VLSI
Question
Bank

Section 2:

Digital Circuits and Systems:

1) Make a 5:1 MUX USING 4 2:1 MUXs..

2) How many MUXs would be required to make any general n:1 MUX and hence 7:1 MUX?

3) Counter decoding.. Where a counter circuit (using gates and FFs is given) and you are required to

decode the circuit and probably tell the mod number of that counter..

4) Simple problem statements were given and we were required to make an appropriate digital

circuit just by solving K-maps. (2 such questions)

5) A combinational circuit was given and we were asked to minimize the number of gates in that

circuit.

6) Two ways to generate a clock signal.. (Interview)

7) Binary to Gray and Gray to Binary Conversion..using FFs.. (Interview)

8) 2 ways to convert Gray back into binary (tricky).. (Interview)

9) SOP to POS.. (Interview)

Cadence Paper

HADRWARE SECTION:

Q1) consider a numeric system with base 20,which is a simple extension to the hexadecimal system such that the letter A,B,C,D,E,F,G,H,I and J stand for the decimal number 10,11,12,1,14,15,16,17,18and 19 respectively. Find the value IA in above system?
ans:562 in the octal system

2). If a signed integer is stored in a 8 bit memory element ,which of the following is true?
ans (a) the minimum value of the integer possible is -128

Q3).where do the data transfer takes place between in DMA ?
ans : I/O port and memory not controlled by CPU

Q4). a module ...18.... Counter requires a minimum of
ans ...7... no of flipflop.

Q5).assume the following convention for storing real numbers of size 10 bits in mantissa exponent form:
Mantissa :6 MSB bits, stored in 2,s complement
Exponent:4 LSB ,stored in excess 4 form.
Under the above rule what is the the decimal number corresponding to 1,110,11,0111?

Q6). consider the circuit
A X
B
What is the Boolean expression for X?

Q7).consider the circuit below (all flip-flop are negative edge triggered toggle flip-flops)
clr clr clr
T Q T Q T Q
The above circuit is for a asynchronous counter of the type: -
a) modulo 3
b) modulo 5
c) modulo 6
d) none of the above

Q8) A 4-bit ring counter is mutually loaded with 0100 After how many clock pulses will come back to the initial state?
a) log 24
b) 4
c) 24
d)none

Q9)Correct values of the control signal B and C for undergraded transmission of input signal A(0 or 1) to output are:
a) B=0 C=0
b) B=0 C=1
c) B=1 C=0
d) D=1 C=1

Q10). Which of the following is minimized expression of the function
f=XYZ+YZ+XZ
a) XY+Y+X
b) Z
c) Y
d) None of the

Software:

Q1).
0
0
1 1 1 1
S is the start as well as end symbol. What type of strings are generated by this machine?

Q2.)Given In order : abcdefgh
preorder :.....
find deorder?

Q3) let a number be 0100 which is input to 4bit ring counter. Find after how many clock pulse this sequence is repeated?
Ans: 16

Q4)for 1=1to n do
{if(i is odd)
for j=1 to n do
s1
else s2
end}
end.
Find the order :6n2{n/2*n.. + n/2}
S1 s2?

Q5)if to a procedure x is passed by reference and y passed by value find the value of x and y :-
p(var x, int y)
begin
x=x+2;
y=y+3;

where the function is invoked by x=3 and y=2, p(x,y)

Q6)Relocatuble code generated by using which type of addressing?
a)relative b)direct c)absolute d).....?

DCM datasystems (programming section):

It contain four sections MCQs on networking ,OS,applications 10 Qs each section major stress on programming section -ve marking is there do them correctly directly run these programmes to get the solutions.In interview they ask these same Qs to solve and few Qs on 'C' refer kerningan and rtchie chapter two.like if(1&2).. else ... it will go to if or else ,if(1&&2) .... Else .... It go to if or else etc.refer the
The points I have written in the programmes.

Give the output of the following programs.

1).

```
#include<stdio.h>
#include<string.h>
void main()
{
int a=1,b=4,c;
for(c=0;c<2;c++)
{
switch(a)
{
case 1:
if (b!=4)
break;
else continue;
case 2 :
a=b;
b++;
break;
}
}
printf("output is %d %d\n",a,b);
}
```

2).

```
 #include<stdio.h>
#include<string.h>
int recsum(int var)
{ static int sum=0;
if(var==3)
return(sum);
```

```
else{ sum=sum+7;
var++;
recsum(var);
}
sum=sum-var;
return sum;
}
void main()
{
int a,b;
a=recsum(1);
b=recsum(0);
printf("output is %d %d\n",a,b);
}
```

3)

```
#include<stdio.h>
#include<string.h>
//#define formula(a) dcm(a*a)
#define max(i,j) ((i)>(j) ? (i):(j))
/*void main()
{ int i=2,j=3,k;
k=max(i++,j++);
// k=((i++)>(j++) ? (i++):(j++));
printf("\n%d",k);
k=i+1>j+2;
printf("\n%d",k);
} */
/*dcm(int var)
{
printf("\noutput is %d %d",var++,var++);
return 0;
}
void main()
{
int a,first=3;
//a=first+1*first+1;=>7 note this expression is replaced in the macro
//macro should be defined as dcm((a)*(a))
// printf("\n%d",a);
formula(first+1);
formula(first++);
} */
```

4)

```
#include<stdio.h>
#include<string.h>
char* function (char *tptr)
```

```
{ char* dcms="Data systems";
tptr=dcms;
return (tptr);
}
void main()
{
char ptr[]="DCM Delhi";
function(ptr);
printf("value is %s %d\n",ptr ,sizeof(ptr));
}
```

5)

```
#include<stdio.h>
#include<string.h>
int dcm(const char*p)
{ char *q;
q=p;
while(*q++);//if *++q is given then it will give 1 less than as before
// printf("%d",q-p);//the above ++ is incrementing the pointer because
// ++ and * both are urinary operator they have right to left associativity
//++ is calculated first the * is calculated
getch();
return(q-p-1);
}
void main()
{
char *str1="DCMODATA\0SYSTEMS";
printf("\n%d %s\n",dcm(str1),str1);
printf("%d %s\n",dcm(str1+8),str1+8);
}
```

6)

```
#include<stdio.h>
#include<string.h>
#include<conio.h>
int dcm(int a,int b,int c)
{
int d;
d=((a>>(b+1-c))&(~(~0<<c)));
printf("%x %x %x %x",~0,(~0<<c)|2,~(~0<<c),7&5);
//keep in mind that ~bit wise not operator so ~0 is 1111 1111 1111 1111
//or ffff <<c will cause --1111 1000 not again gives --0000 0111 forms the
//mask
getch();
return d;
}
void main()
```

```
{ int value;
value=dcm(0xf7,4,3);
printf("%d \n",value);
}
```

7)

```
#include<stdio.h>
#include<string.h>
int a=65;
char func(char c)
{
int b;
b=c;
printf("%c",b);
getch();
return(a);
}
char dcm(char a)
{
printf("%c",a); //
//*c=*c+2;
return(a);
}
void main()
{
char a='B';
printf("\n%d",a);
a=func(a);
printf("%c",a);
dcm(a);
printf("%d",a);
}
```

8)

```
#include<stdio.h>
#include<string.h>
int nbits;
void dcm(short num,unsigned mark)
{
int b,count;
for(count=1;count<=nbits;count++)
{
b=(num&mark)?1:0;
printf(" %x",b);
if(count%4==0)
printf(" ");
}
}
void main()
```

```
{
unsigned mark;
nbits=8*sizeof(short);
mark=0x1<<(nbits-1);
dcm(32767,mark);
dcm(-32768,mark);
}
```

COSMIC CIRCUITS

1) One question on maximum clock frequency of a digital circuit.

2) An op-amp circuit was given with positive as well as negative feedback resistors but the value of these was unknown and we were asked to choose the appropriate value so that the circuit is stable.

3) Level Shifter circuit using two MOS and two capacitors. (This was a tricky question)

4) Controlled Voltage/Current Sources- Choose which controlled source does the circuit represent.

5) Output impedance of n cascaded Common Gate amplifiers.

6) Square wave of1KHz frequency passed through an LPF of cut off frequency 2KHz. Find the maximum amplitude of the output.

7) MOSFET with an inductor connected in its drain and then the inductor to VDD. Source was grounded and a square wave was applied at the input and we had to plot the output waveform. Output was taken across the drain and the source..

8) One question based on ADCs and DACs. I didn't even understand the question!

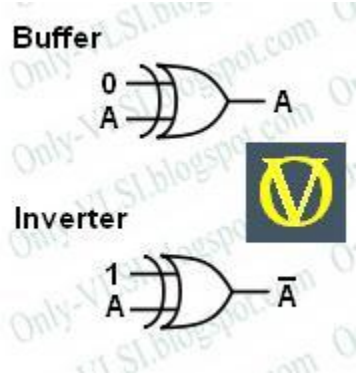
9) One simple question on clampers..

10) 2 questions from LIC where we had to find the T/F of the op-amp based circuit.

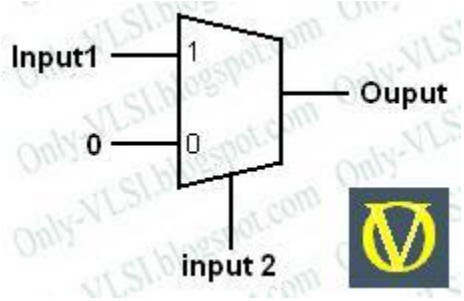
11) One question on circuit resistance noise, where we had to find the component values such that the total circuit noise is less than a certain value..

Digital Design Interview Questions - All in 1

1. How do you convert a XOR gate into a buffer and a inverter (Use only one XOR gate for each)?
Answer



2. Implement an 2-input AND gate using a 2x1 mux.
Answer



3. What is a multiplexer?
Answer

A multiplexer is a combinational circuit which selects one of many input signals and directs to the only output.

4. What is a ring counter?
Answer

A ring counter is a type of counter composed of a circular shift register. The output of the last shift register is fed to the input of the first register. For example, in a 4-register counter, with initial register values of 1100, the repeating pattern is: 1100, 0110, 0011, 1001, 1100, so on.

5. Compare and Contrast Synchronous and Asynchronous reset.
Answer

Synchronous reset logic will synthesize to smaller flip-flops, particularly if the reset is gated with the logic generating the d-input. But in such a case, the combinational logic gate count grows, so the overall gate count savings may not be that significant. The clock works as a filter for small reset glitches; however, if these glitches occur near the active clock edge, the Flip-flop could go metastable. In some designs, the reset must be generated by a set of internal conditions. A synchronous reset is recommended for these types of designs because it will filter the logic equation glitches between clock.
Problem with synchronous resets is that the synthesis tool cannot easily distinguish the reset signal from any other data signal. Synchronous resets may need a pulse stretcher to guarantee a reset pulse width wide enough to ensure reset is present during an active edge of the clock, if you have a gated clock to save power, the clock may be disabled coincident with the assertion of reset. Only an asynchronous reset will work in this situation, as the reset might be removed prior to the resumption of the clock. Designs that are pushing the limit for data path timing, can not afford to have added gates and additional net delays in the data path due to logic inserted to handle synchronous resets.

Asynchronous reset: The major problem with asynchronous resets is the reset release, also called reset removal. Using an asynchronous reset, the designer is guaranteed not to have the reset added to the data path. Another advantage favoring asynchronous resets is that the circuit can be reset with or without a clock present. Ensure that the release of the reset can occur within one clock period else if the release of the reset occurred on or near a clock edge then flip-flops may go into metastable state.

6. What is a Johnson counter?
Answer

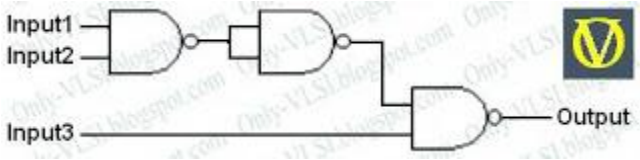
Johnson counter connects the complement of the output of the last shift register to its input and circulates a stream of ones followed by zeros around the ring. For example, in a 4-register counter, the repeating pattern is: 0000, 1000, 1100, 1110, 1111, 0111, 0011, 0001, so on.

7. An assembly line has 3 fail safe sensors and one emergency shutdown switch.The line should keep moving unless any of the following conditions arise:
(1) If the emergency switch is pressed
(2) If the senor1 and sensor2 are activated at the same time.
(3) If sensor 2 and sensor3 are activated at the same time.
(4) If all the sensors are activated at the same time
Suppose a combinational circuit for above case is to be implemented only with NAND Gates. How many minimum number of 2 input NAND gates are required?
Answer

8. In a 4-bit Johnson counter How many unused states are present?
Answer

4-bit Johnson counter: 0000, 1000, 1100, 1110, 1111, 0111, 0011, 0001, 0000.
8 unused states are present.

9. Design a 3 input NAND gate using minimum number of 2 input NAND gates.
Answer



10. How can you convert a JK flip-flop to a D flip-flop?
Answer

11. What are the differences between a flip-flop and a latch?
Answer

Flip-flops are edge-sensitive devices where as latches are level sensitive devices.
Flip-flops are immune to glitches where are latches are sensitive to glitches.
Latches require less number of gates (and hence less power) than flip-flops.
Latches are faster than flip-flops.

12. What is the difference between Mealy and Moore FSM?
Answer

Mealy FSM uses only input actions, i.e. output depends on input and state. The use of a Mealy FSM leads often to a reduction of the number of states.
Moore FSM uses only entry actions, i.e. output depends only on the state. The advantage of the Moore model is a simplification of the behavior.

13. What are various types of state encoding techniques? Explain them.
Answer

One-Hot encoding: Each state is represented by a bit flip-flop). If there are four states then it requires four bits

(four flip-flops) to represent the current state. The valid state values are 1000, 0100, 0010, and 0001. If the value is 0100, then it means second state is the current state.

One-Cold encoding: Same as one-hot encoding except that '0' is the valid value. If there are four states then it requires four bits (four flip-flops) to represent the current state. The valid state values are 0111, 1011, 1101, and 1110.

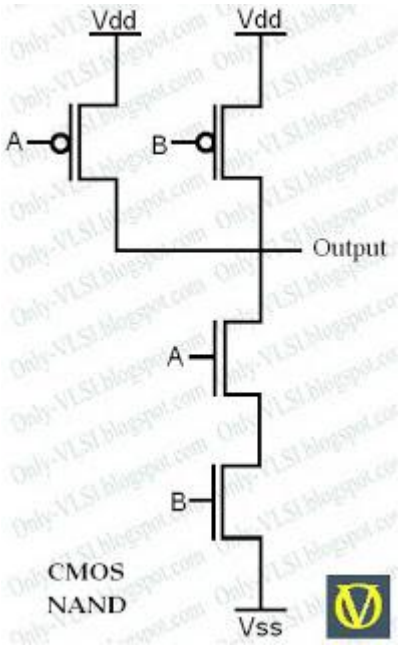
Binary encoding: Each state is represented by a binary code. A FSM having '2 power N' states requires only N flip-flops.

Gray encoding: Each state is represented by a Gray code. A FSM having '2 power N' states requires only N flip-flops.

14. Define Clock Skew , Negative Clock Skew, Positive Clock Skew.
Answer

Clock skew is a phenomenon in synchronous circuits in which the clock signal (sent from the clock circuit) arrives at different components at different times. This can be caused by many different things, such as wire-interconnect length, temperature variations, variation in intermediate devices, capacitive coupling, material imperfections, and differences in input capacitance on the clock inputs of devices using the clock.
There are two types of clock skew: negative skew and positive skew. Positive skew occurs when the clock reaches the receiving register later than it reaches the register sending data to the receiving register. Negative skew is the opposite: the receiving register gets the clock earlier than the sending register.

15. Give the transistor level circuit of a CMOS NAND gate.
Answer



16. Design a 4-bit comparator circuit.
Answer

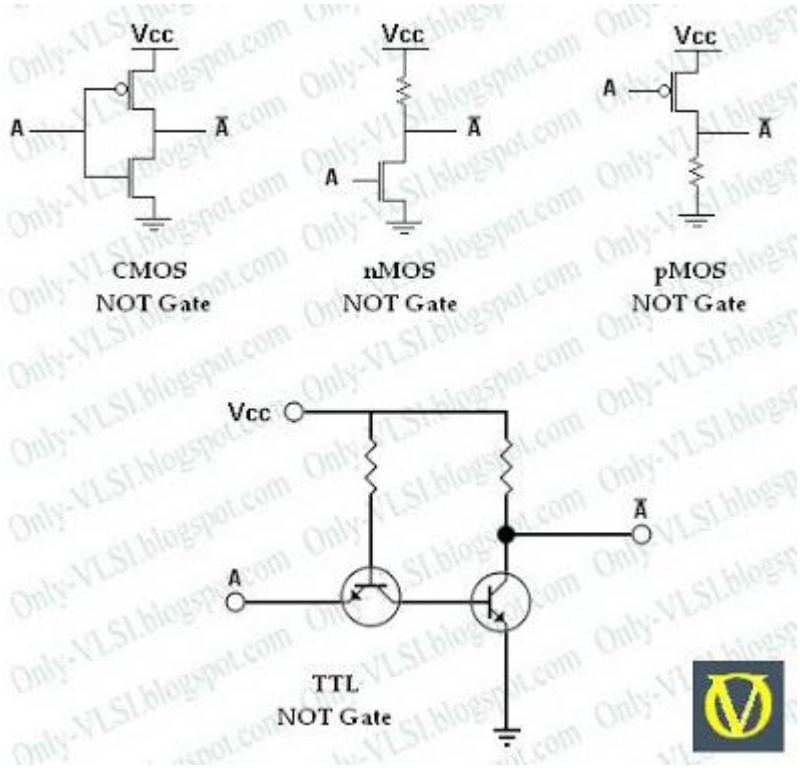
17. Design a Transmission Gate based XOR. Now, how do you convert it to XNOR (without inverting the output)?
Answer

18. Define Metastability.
Answer

If there are setup and hold time violations in any sequential circuit, it enters a state where its output is unpredictable, this state is known as metastable state or quasi stable state, at the end of metastable state, the flip-flop settles down to either logic high or logic low. This whole process is known as metastability.

19. Compare and contrast between 1's complement and 2's complement notation.
Answer

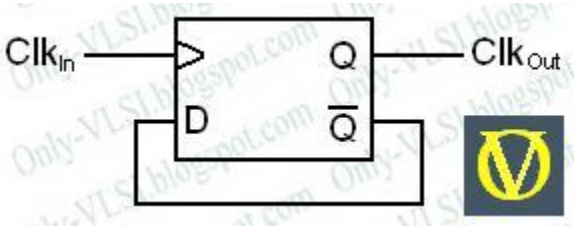
20. Give the transistor level circuit of CMOS, nMOS, pMOS, and TTL inverter gate.
Answer



21. What are set up time and hold time constraints?
Answer

Set up time is the amount of time before the clock edge that the input signal needs to be stable to guarantee it is accepted properly on the clock edge.
Hold time is the amount of time after the clock edge that same input signal has to be held before changing it to make sure it is sensed properly at the clock edge.
Whenever there are setup and hold time violations in any flip-flop, it enters a state where its output is unpredictable, which is known as as metastable state or quasi stable state. At the end of metastable state, the flip-flop settles down to either logic high or logic low. This whole process is known as metastability.

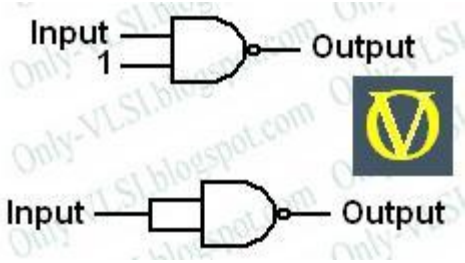
22. Give a circuit to divide frequency of clock cycle by two.
Answer



23. Design a divide-by-3 sequential circuit with 50% duty circle.
Answer

24. Explain different types of adder circuits.
Answer

25. Give two ways of converting a two input NAND gate to an inverter.
Answer



26. Draw a Transmission Gate-based D-Latch.
Answer

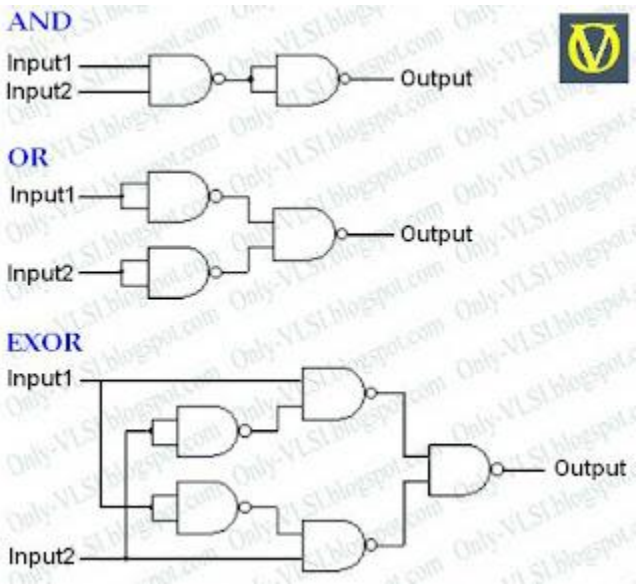
27. Design a FSM which detects the sequence 10101 from a serial line without overlapping.
Answer

28. Design a FSM which detects the sequence 10101 from a serial line with overlapping.
Answer

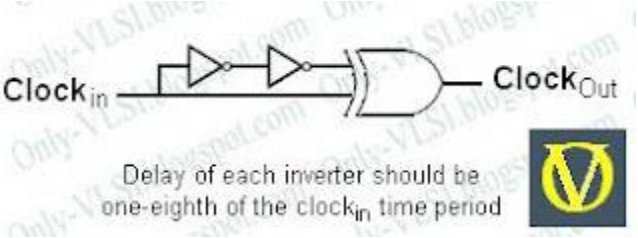
29. Give the design of 8x1 multiplexer using 2x1 multiplexers.
Answer

30. Design a counter which counts from 1 to 10 (Resets to 1, after 10).
Answer

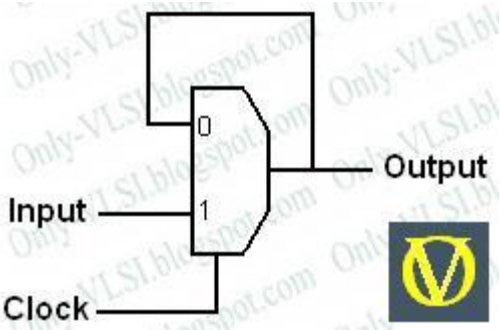
31. Design 2 input AND, OR, and EXOR gates using 2 input NAND gate.
Answer



32. Design a circuit which doubles the frequency of a given input clock signal.
Answer



33. Implement a D-latch using 2x1 multiplexer(s).
Answer



34. Give the excitation table of a JK flip-flop.
Answer

35. Give the Binary, Hexadecimal, BCD, and Excess-3 code for decimal 14.
Answer

14:
Binary: 1110
Hexadecimal: E
BCD: 0001 0100
Excess-3: 10001

36. What is race condition?
Answer

37. Give 1's and 2's complement of 19.
Answer

19: 10011
1's complement: 01100
2's complement: 01101

38. Design a 3:6 decoder.
Answer

39. If A*B=C and C*A=B then, what is the Boolean operator * ?
Answer

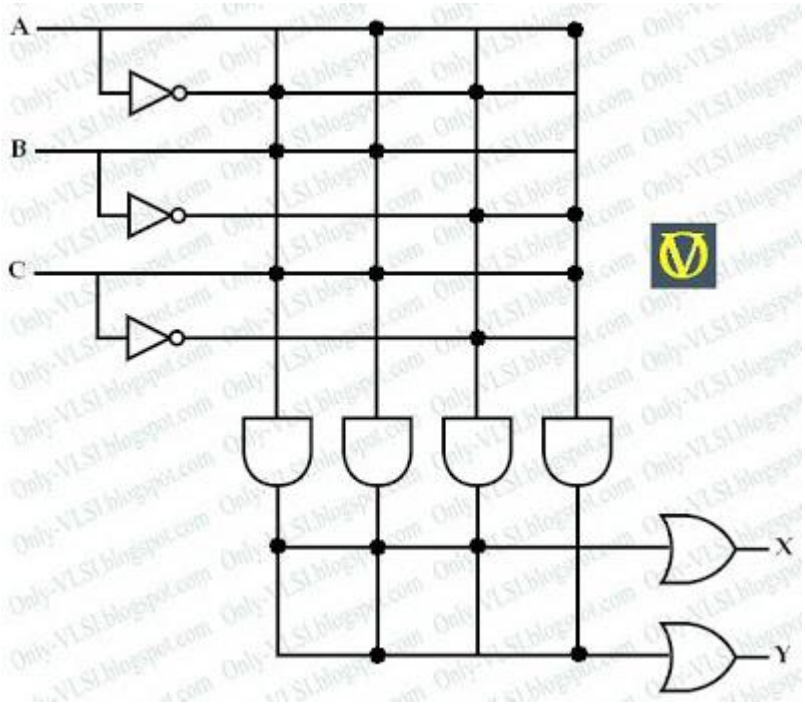
* is Exclusive-OR.

40. Design a 3 bit Gray Counter.
Answer

41. Expand the following: PLA, PAL, CPLD, FPGA.
Answer

PLA - Programmable Logic Array
PAL - Programmable Array Logic
CPLD - Complex Programmable Logic Device
FPGA - Field-Programmable Gate Array

42. Implement the functions: X = A'BC + ABC + A'B'C' and Y = ABC + AB'C using a PLA.
Answer



43. What are PLA and PAL? Give the differences between them.
Answer

Programmable Logic Array is a programmable device used to implement combinational logic circuits. The PLA has a set of programmable AND planes, which link to a set of programmable OR planes, which can then be conditionally complemented to produce an output.
PAL is programmable array logic, like PLA, it also has a wide, programmable AND plane. Unlike a PLA, the OR plane is fixed, limiting the number of terms that can be ORed together.
Due to fixed OR plane PAL allows extra space, which is used for other basic logic devices, such as multiplexers, exclusive-ORs, and latches. Most importantly, clocked elements, typically flip-flops, could be included in PALs. PALs are also extremely fast.

44. What is LUT?
Answer

LUT - Look-Up Table. An n-bit look-up table can be implemented with a multiplexer whose select lines are the inputs of the LUT and whose inputs are constants. An n-bit LUT can encode any n-input Boolean function by modeling such functions as truth tables. This is an efficient way of encoding Boolean logic functions, and LUTs with 4-6 bits of input are in fact the key component of modern FPGAs.

45. What is the significance of FPGAs in modern day electronics? (Applications of FPGA.)
Answer

- ASIC prototyping: Due to high cost of ASIC chips, the logic of the application is first verified by dumping HDL code in a FPGA. This helps for faster and cheaper testing. Once the logic is verified then they are made into ASICs.
- Very useful in applications that can make use of the massive parallelism offered by their architecture. Example: code breaking, in particular brute-force attack, of cryptographic algorithms.
- FPGAs are sued for computational kernels such as FFT or Convolution instead of a microprocessor.
- Applications include digital signal processing, software-defined radio, aerospace and defense systems, medical imaging, computer vision, speech recognition, cryptography, bio-informatics, computer hardware emulation and a growing range of other areas.

46. What are the differences between CPLD and FPGA.
Answer

47. Compare and contrast FPGA and ASIC digital designing.
Answer

Click here.

48. Give True or False.
(a) CPLD consumes less power per gate when compared to FPGA.
(b) CPLD has more complexity than FPGA
(c) FPGA design is slower than corresponding ASIC design.
(d) FPGA can be used to verify the design before making a ASIC.
(e) PALs have programmable OR plane.
(f) FPGA designs are cheaper than corresponding ASIC, irrespective of design complexity.
Answer

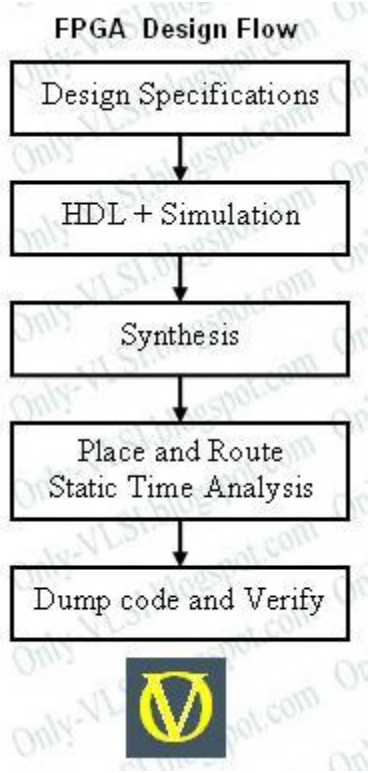
(a) False
(b) False
(c) True
(d) True

(e) False
(f) False

49. Arrange the following in the increasing order of their complexity: FPGA,PLA,CPLD,PAL.
Answer

Increasing order of complexity: PLA, PAL, CPLD, FPGA.

50. Give the FPGA digital design cycle.
Answer



51. What is DeMorgan's theorem?
Answer

For N variables, DeMorgan's theorems are expressed in the following formulas:
(ABC..N)' = A' + B' + C' + ... + N' -- The complement of the product is equivalent to the sum of the complements.
(A + B + C + ... + N)' = A'B'C'...N' -- The complement of the sum is equivalent to the product of the complements.
This relationship so induced is called DeMorgan's duality.

52. F'(A, B, C, D) = C'D + ABC' + ABCD + D. Express F in Product of Sum form.
Answer

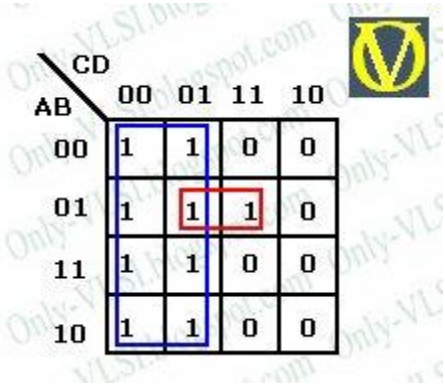
Complementing both sides and applying DeMorgan's Theorem:
F(A, B, C, D) = (C + D')(A' + B' + C)(A' + B' + C' + D')(D')

53. How many squares/cells will be present in the k-map of F(A, B, C)?
Answer

F(A, B, C) has three variables/inputs.
Therefore, number of squares/cells in k-map of F = $2^{(\text{Number of variables})} = 2^3 = 8$.

54. Simplify F(A, B, C, D) = S (0, 1, 4, 5, 7, 8, 9, 12, 13)
Answer

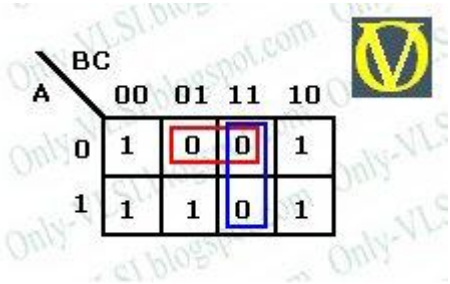
The four variable k-map of the given expression is:



The grouping is also shown in the diagram. Hence we get,
F(A, B, C, D) = C' + A'BD

55. Simplify F(A, B, C) = S (0, 2, 4, 5, 6) into Product of Sums.
Answer

The three variable k-map of the given expression is:



The 0's are grouped to get the F'.
F' = A'C + BC

Complementing both sides and using DeMorgan's theorem we get F,
F = (A + C')(B' + C')

56. The simplified expression obtained by using k-map method is unique. True or False. Explain your answer.
Answer

57. Give the characteristic tables of RS, JK, D and T flip-flops.
Answer

RS flip-flop.

S R Q(t+1)
0 0 Q(t)
0 1 0
1 0 1
1 1 ?

JK flip-flop

J K Q(t+1)
0 0 Q(t)
0 1 0
1 0 1
1 1 Q'(t)

D flip-flop

D	Q(t+1)
0	0
1	1

T flip-flop

T	Q(t+1)
0	Q(t)
1	Q'(t)

58. Give excitation tables of RS, JK, D and T flip-flops.
Answer

RS flip-flop.

Q(t)	Q(t+1)	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

JK flip-flop

Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

D flip-flop

Q(t)	Q(t+1)	D
0	0	0
0	1	1
1	0	0
1	1	1

T flip-flop

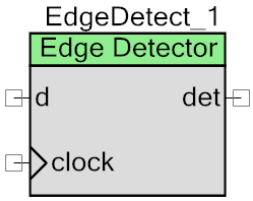
Q(t)	Q(t+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

59. Design a BCD counter with JK flip-flops
Answer

60. Design a counter with the following binary sequence 0, 1, 9, 3, 2, 8, 4 and repeat. Use T flip-flops.
Answer

http://www.scribd.com/doc/36004714/texasinst

Edge Detector
1.0



Features

- Detects Rising Edge, Falling Edge, or Either Edge

General Description

The Edge Detector component samples the connected signal and produces a pulse when the selected edge occurs.

When to Use an Edge Detector

Use the Edge Detector when a circuit needs to respond to a state change on a signal.

Input/Output Connections

This section describes the various input and output connections for the Edge Detector.

d – Input

The signal connected to the d input is the signal that will be sampled for an edge.

clock – Input

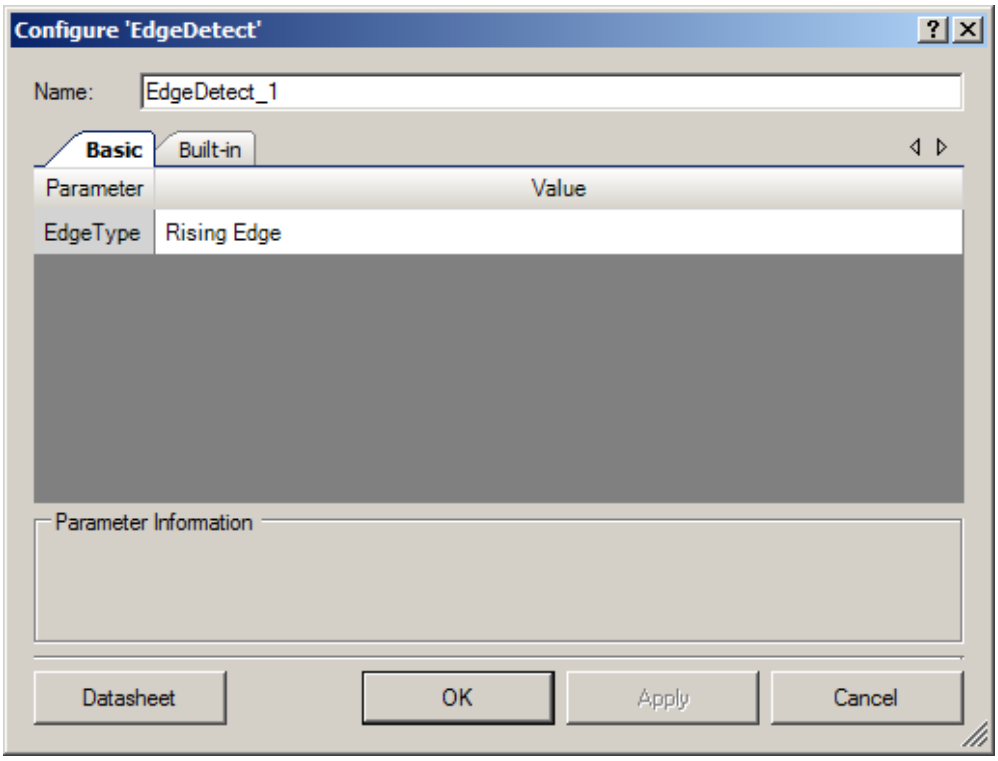
The clock input determines how often the **d** input will be sampled.

det – Output

The det output pulses high when an edge is detected on the **d** input.

Component Parameters

Drag an Edge Detector onto your design and double-click it to open the **Configure** dialog.



The Edge Detector provides the following parameters.

EdgeType

This parameter determines what type of edge to detect. The value must be **Rising Edge**, **Falling Edge**, or **Either Edge**. The default is **Rising Edge**.

Functional Description

The Edge Detector stores the state of the signal at the last rising clock edge, and compares it to the current value of the signal. If the state change matches the edge type selected in the customizer, the **det** terminal will go high until the next rising clock edge. This means that the resulting pulse from an edge may be shorter than one clock cycle, but it will never be longer.

Figure 1. Rising Edge Schematic

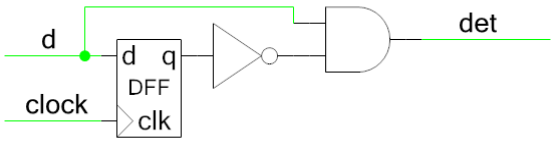
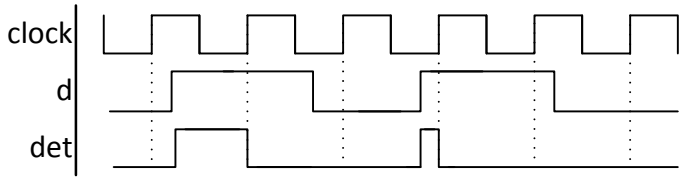


Figure 1 displays a logical representation of the implementation for the **Rising Edge** configuration**.** Figure 2 provides a sample waveform to illustrate the functionality.

Figure 2. Rising Edge Waveform



As seen in Figure 2, the **det** output will go high as soon as a **rising** edge is detected on the **d** input. The **det** output is cleared on the next rising clock edge.

Figure 3. Falling Edge Schematic

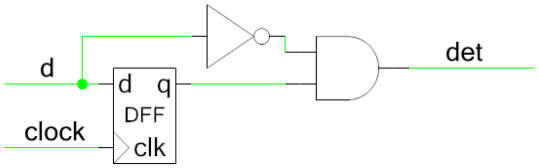
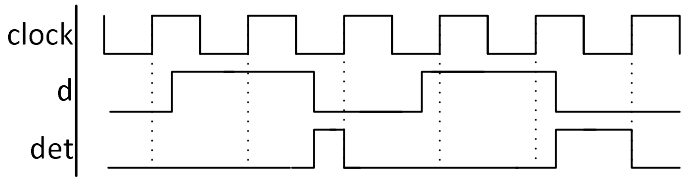


Figure 3 displays a logical representation of the implementation for the **Falling Edge** configuration**.** Figure 4 provides a sample waveform to illustrate the functionality.

Figure 4. Falling Edge Waveform



As seen in Figure 4, the **det** output will go high as soon as a **falling** edge is detected on the **d** input. The **det** output is cleared on the next rising clock edge.

Figure 5. Either Edge Schematic

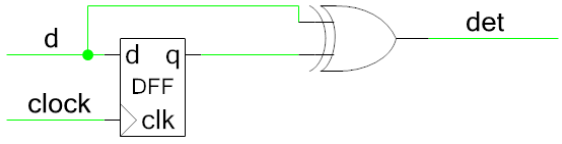
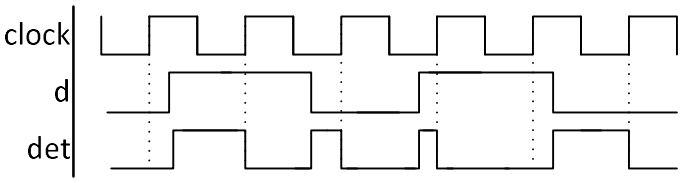


Figure 5 displays a logical representation of the implementation for the **Falling Edge** configuration**.** Figure 6 provides a sample waveform to illustrate the functionality.

Figure 6. Either Edge Waveform



As seen in Figure 6, the **det** output will go high as soon as **any** edge is detected on the **d** input. The **det** output is cleared on the next rising clock edge.

Resources

Configuration	Resource Type					
	Datapath Cells	Macrocells	Status Cells	Control Cells	DMA Channels	Interrupts
Edge Detector	–	1	–	–	–	–

MISRA Compliance

This section describes the MISRA-C:2004 compliance and deviations for the component. There are two types of deviations defined: project deviations – deviations that are applicable for all PSoC Creator components and specific deviations – deviations that are applicable only for this component. This section provides information on component specific deviations. The project deviations are described in the MISRA Compliance section of the *System Reference Guide* along with information on the MISRA compliance verification environment.

The Edge Detector component does not have any C source code APIs.

Component Changes

Version 1.0 is the first release of the Edge Detector Component.

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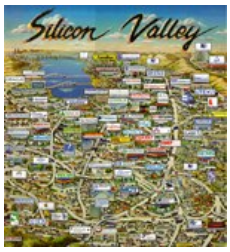
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HITEQUEST
technical interview Q & A for high-tech professionals



=Electronics Hardware Questions=

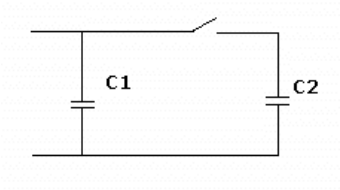
Interview Questions
Electronics Hardware
Computer Software
Intel screening questions
Quick Thinking Tests
General questions
Phone screening questions
Submit your Q or A

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- Two capacitors are connected in parallel through a switch. C1= 1uF, C2= 0.25uF.
  Initially the switch is open, C1 is charged to 10V. What happens if we close the switch? No losses in wires and capacitors.



Answers & follow ups

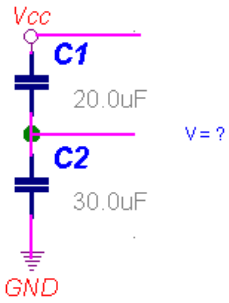
- You have 2 switches to control the light in the long corridor. You want to be able to turn the light on entering the corridor and turn it off at the other end. Do the wiring circuit.

Answers & follow ups

- There are 3 switches that can turn on and off a light in the room. How to connect them?

Answers & follow ups

- What will be the voltage level between the two capacitors? The Vcc = 10v DC.
  Sent by Tanh, VLSI engineer

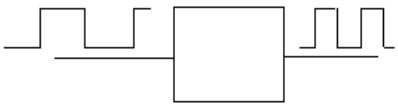


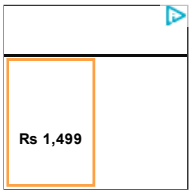
Answers & follow ups

- Suppose, you work on a specification for a system with some digital parameters. The spec table has has Min,Typ and Max colomns for each parameter. In what column would you put a Setup time and a Hold time?

Answers & follow ups

- Design a simple circuit based on combinational logic to double the output frequency.





Do you know how much you are worth on the market?
If you stayed with the same company for a lengthy period of time, there is a big chance you are under-earning.
More...

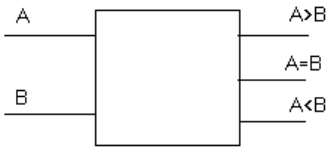
Verilog
IDE for SystemVerilog, e language, VHDL & Verilog. Get a free trial !

Answers & follow ups

- 8bit ADC with parallel output converts an input signal into digital numbers. You have to come up with an idea of a circuit , that finds MAX of every 10 numbers at the output of the ADC.

Answers & follow ups

- Implement a comparator that compares two 2-bit numbers A and B. The comparator should have 3 outputs: A > B, A < B, A = B. Do it two ways:
  - using combinational logic;
  - using multiplexers. Write HDL code for your schematic at RTL and gate level.



Answers & follow ups

- You have a 8 bit ADC clocking data out every 1mS. Design a system that will sort the output data and keep the statistics of how often each binary number appears at the output of ADC.

Answers & follow ups

- What types of flip-flops do you know?

Answers & follow ups

- Implement D- latch from
  - RS flip flop;
  - multiplexer.

Answers & follow ups

- How to convert D-latch into JK-latch and JK-latch into D-latch?

Answers & follow ups

- There are two counters to 16, built from negedge D- FF . The first circuit is synchronous and the second one is "ripple" (cascading). Which circuit has less propagation delay?

Answers & follow ups

- What is the difference between a flip-flop and a latch?
  Write an HDL code for their behavioral models.

Answers & follow ups

- Describe the operation of a DAC. What are the most important parameters of a DAC? Do we really need both INL and DNL to estimate linearity?

Answers & follow ups

- Compare briefly all types of ADC that you know .

Answers & follow ups

- How will the output signal of an ideal integrator look like after
  - a positive pulse is applied to the input;
  - a series of 10 positive pulses ?

Answers & follow ups

- How to design a divide-by-3 counter with equal duty cycle ?
  question from Anonymous

  Answers & follow ups

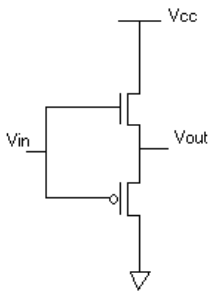
- For an 8-bit flash A/D converter with an input range from 0V to 2.55V, describe what happens when the input voltage changes from 1.27V to 1.28V

  Answers & follow ups

- Your system has CPU, ALU and two 8bit registers. There is no external memory. Can you swap the contence of the registers ?

  Answers & follow ups

- We swapped 2 transistors in CMOS inverter (put n-transistor at the top and p-transistor at the bottom). Can this circuit work as a non-inverting buffer?
  (By E.Martovetsky,design eng)

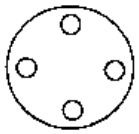


  Answers & follow ups

- Convert D-latch into divider by 2.
  What is the max clock frequency the circuit can handle ?
  T_setup= 6nS
  T_hold = 2nS
  T_propagation = 10nS

  Hint from Hitequest

- The circle can rotate clockwise and back. Use minimum hardware to build a circuit to indicate the direction of rotating.



  Hint from Hitequest

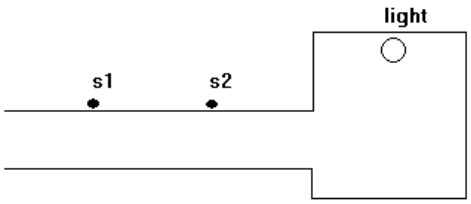
- Provide 2-dimentional plot of how the output of digital circuit will look like, if on axis X we sweep the clock period, while on axis Y we sweep the data setup time (Tclk vs Tsetup).

- For chip design/test/product engineers :
  An IC device draws higher current when temperature gets:
  - higher?
  - lower?

  Hint from Hitequest

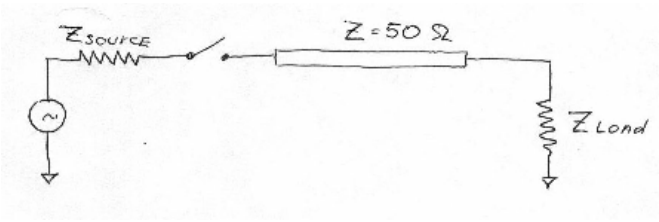
- To enter the office people have to pass through the corridor. Once someone gets into the office the light turns on. It goes off when noone is

present in the room. There are two registration sensors in the corridor. Build a state machine diagram and design a circuit to control the light.



- A voltage source with internal impedance Z_source = 50 OHm is connected to a transmission line with Z = 50 OHm. Z_load is also 50 OHm.
  The voltage source generates a single voltage step 1V.
  What will be the voltage level on the load:

  a) 2V , because the reflected signal will be in-phase with the incident signal;
  b) 0.33V , because the voltage is devided between Z_source , Z_load and Z_transm.line;
  c) 0.5V , because the voltage is devided between Z_source and Z_load.



Hint from Hitequest

- Draw a transistor schematic of NOR gate,it's layout and a cross section of the layout.
  This question is quite popular on interviews.

- The silicon of a new device has memory leak. When all "0" are written into RAM, it reads back all "0" whithout any problem. When all "1" are written, only 80% of memory cells are read back correctly. What can be possibly the problem with the RAM?
  M.Altshuler, product engineer.

- Draw a CMOS inverter. Why does CMOS technology dominate in VLSI manufacturing?
  L.Backer, DFT engineer

- Design a FIFO 1 byte wide and 13 words deep. The FIFO is interfacing 2 blocks with different clocks. On the rising edge of clk the FIFO stores data and increments wptr. On the rising edge of clkb the data is put on the b-output,the rptr points to the next data to be read.
  If the FIFO is empty, the b-output data is not valid. When the FIFO is full the existing data should not be overriden.
  When rst_N is asserted, the FIFO pointers are asynchronously reset.

```
module fifo1 (full,empty,clk,clkb,ain,bout,rst_N)
output [7:0] bout;
input [7:0] ain;
input clk,clkb,rst_N;
output empty, full;
reg [3:0] wptr, rptr;
```

```
...

endmodule
```

Hint from Hitequest

- What does CMOS stand for? VLSI? ASIC?
  This was in the series of quick questions in the interview at Analog Devices. We use these abbreviations daily, but not everyone remembers what they stand for.

Hint from Hitequest

- Design a COMBINATIONAL circuit that can divide the clock frequency by 2.

Answers & follow ups

- Design a 2bit up/down counter with clear using gates. (No verilog or vhdl)

- We have a circular wheel with half painted black and the other half painted white. There are 2 censors mounted 45 degree apart at the surface of this wheel(not touching the wheel) which give a "1" for black and "0" for white passing under them. Design a circuit to detect which way the wheel is moving. Can not assume any fixed position for start.

- We have a FIFO which clocks data in at 100mhz and clocks data out at 80mhz. On the input there are only 80 data bits in any order during each 100 clocks. In other words, a 100 input clock will carry only 80 data bits, and the other twenty clocks carry no data (data is scattered in any order). How big the FIFO needs to be to avoid data over/under-run.

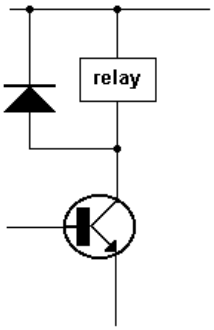
Follow Ups

- Instead of specifying SETUP and HOLD time, can we just specify a SETUP time for '1' and a SETUP time for '0'?

Follow ups

- Here some hardware digital design specific questions, offered by Suhas:
  (1) When will you use a latch and a flipflop in a sequential design?
  (2) Design a 1-bit fulladder using a decoder and 2 "or" gates?
  (3) You have a circuit operating at 20 MHz and 5 volt supply. What would you do to reduce the power consumption in the circuit- reduce the operating frequency of 20Mhz or reduce the power supply of 5Volts and why?
  (4) In a nmos transistor, how does the current flows from drain to source in saturation region when the channel is pinched off?
  (5) In a SRAM circuit, how do you design the precharge and how do you size it?
  (6) In a PLL, what elements(like XOR gates or Flipflops) can be used to design the phase detector?
  (7) While synthesis of a design using synopsys design compiler, why do you specify input and output delays?
  (8) What difference do you see in the timing reports for a propogated clock and an ideal clock?
  (9) What is timeborrowing related to Static timing anaylsis in Primetime?

- What is the purpose of a diode next to relay on schematics?



Answers & follow ups

- Design a combinational circuit, that multiplies an input decimal digit represented in BCD by 5. The output is to be represented in BCD. Show that the outputs can be obtained from the input lines without using any logic gates.

- I got this question on the interview to the company that makes flash memory controllers. They let me take it home and think for a few days. A.M.
  There is a system with 4 flash memory banks. When data comes to the system, it will be randomly sent to one of the 4 banks. The system has a feature called Native Command Queuing (NCQ) which allows the host to send multiple commands at a time.
  The number of commands sent at a time is called the queue depth. With a queue depth of 1, only 1 of 4 memory banks will be active.
  With a queue depth of 2, we expect 2 of 4 flash memory banks to be active except in the case when both commands are to the same bank.
  The question is, what is the expected number of flash memory banks that would be active, given queue depths of:
  2, 4, 8, and 16 ?

Answers & follow ups

- R.K
  How do you synchronize different CLK Speeds in a system?

- R.K
  Two N-Channel MOSFETs are connected in series and are connected between Vcc and GND. Will turning-on the top MOSFET turn on the BOTTOM one? If yes, Why?
  What is a solution to this problem?

Definition of Set-up, Hold and Propagation in Flip-Flops

Figure 1 shows a basic diagram of a D Flip-Flop. Flip-Flops are very common elements in synchronous designs where clock signal provides the timing to various elements and clock domains. **click here if you don't see pictures**

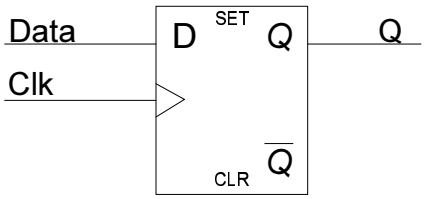


Figure Error! Bookmark not defined.**: D Flip-Flop**

Setup time and hold time describe the timing requirements on the D input of a Flip-Flop with respect to the Clk input. Setup and hold time define a window of time which the D input must be valid and stable in order to assure valid data on the Q output.

Setup Time (Tsu) – Setup time is the time that the D input must be valid before the Flip-Flop samples.

Hold Time (Th) – Hold time is the time that D input must be maintained valid after the Flip-Flop samples.

Propagation Delay (Tpd) – Propagation delay is the time that takes to the sampled D input to propagate to the Q output.

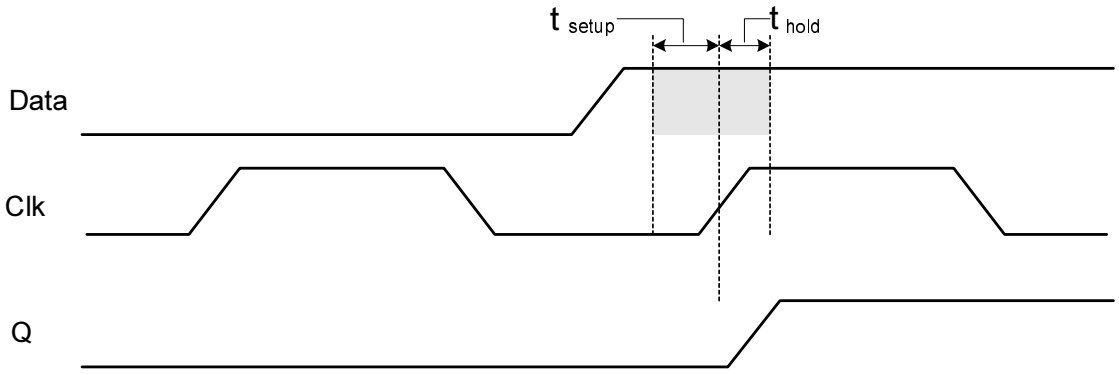


Figure Error! Bookmark not defined.**: Timing Diagram**

Simple Encryption System

The question is to design minimal hardware system, which encrypts 8-bit parallel data. A synchronized clock is provided to this system as well. The output-encrypted data should be at the same rate as the input data but no necessarily with the same phase.

The solution is presented in figure 1. **click here if you don't see pictures**

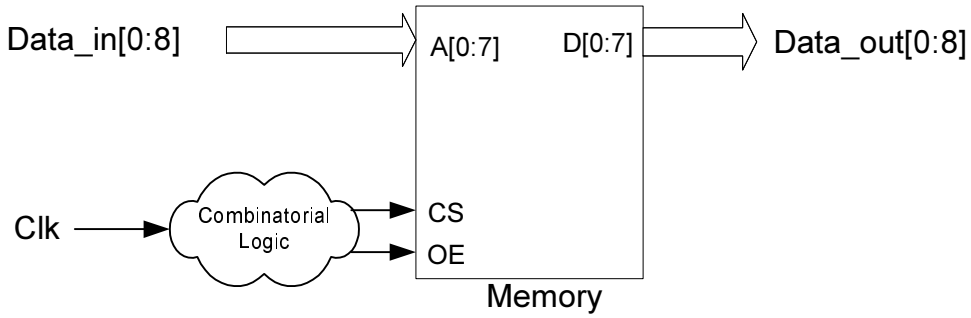


Figure Error! Bookmark not defined.**:**
Block Diagram of Encryption System

The encryption system is centered around a memory device that perform a LUT (Look-Up Table) conversion. This memory functionality can be achieved by using a PROM, EPROM, FLASH and etc. The device contains an encryption code, which may be burned into the device with an external programmer. In encryption operation, the data_in is an address pointer into a memory cell and the combinatorial logic generates the control signals. This creates a read access from the memory. Then the memory device goes to the appropriate address and outputs the associate data. This data represent the data_in after encryption.

Address	Data
01	56
02	12
03	65
04	91
⋮	⋮
FF	03

Table Error! Bookmark not defined.**:**
Example of Memory Content

Pulse Duration Extender

The question is to design a black box that receive a signal input (pulse) and multiply the duration of it by five.

Note: the longer pulse can be transmitted at any time. The length of the longer pulse may not be accurate.
click here if you don't see pictures

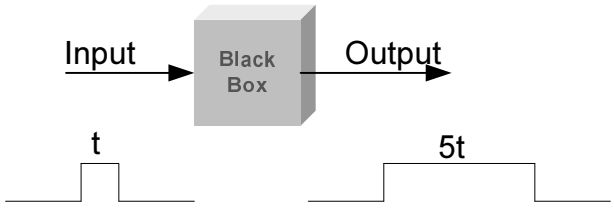


Figure Error! Bookmark not defined.**:**
Pulse Duration Extender

Figure 2 shows a general block diagram of the solution.

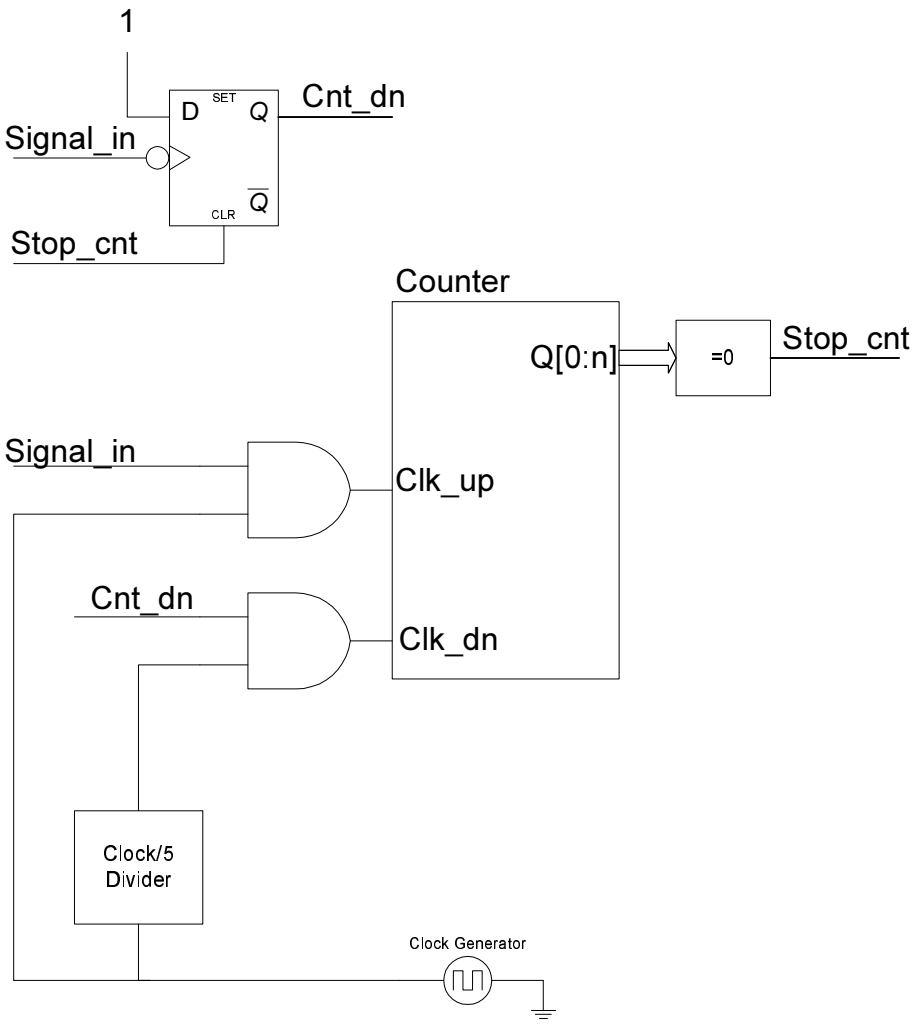


Figure Error! Bookmark not defined.**: General Block Diagram of Duration Extender**

After reset (or power-up), the counter, and Cnt_dn are de-asserted. When the Signal_in set to high (externally), the counter starts receiving clocks and counts up. The clock frequency must by higher then the Signal_in, could be about 100-1000 times to achieve good resolution on the output. When the Signal_in set to low, the counter stop counting up and the Cnt_dn set to high. Now the Clk_dn pin on the counter receives the clocks and start counting down.

When the counter stop counting up and start counting down, the value its' holds represents the number of "Clock Generator" ticks that happened when the Signal_in was high. The idea now is to count down but with slower clock, in this solution we are using a clock divider to divide the "Clock Generator" by 5. Therefore the counter will counter five times slower.

When the counter reaches zero. It means that it finishes to count down and the time passed was five times longer then the Signal_in duration. Stop_cnt create a pulse, which reset the Cnt_dn and put the system in its idle state.

The output of the system can be the Cnt_dn signal.

Note: The above describe general block diagram and general concept. The details are not mention.

Odd Number Clock Divider

This question is really common. Design a clock divider that divides by odd number. The following answer shows how to design a divider by 3 which is asymmetrical. The trivial way is to use a state-machine concept; therefore the answer explains state-machine design step-by-step, from functional specifications towards a complete detailed design and implementation.

A functional description of the asymmetrical divider is shown in Figure 1. **click here if you don't see pictures**

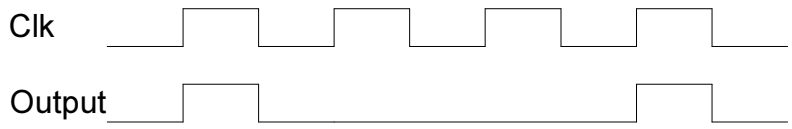


Figure Error! Bookmark not defined.**:**
Functional Description of the Divider

The first step is to draw a state diagram that describes the logical behavior of the circuit. Figure 2 introduces the state diagram of the divider. We can easily see that the divider consist of 3 states which means 2 Flip-Flops. Each step is done every clock cycle.

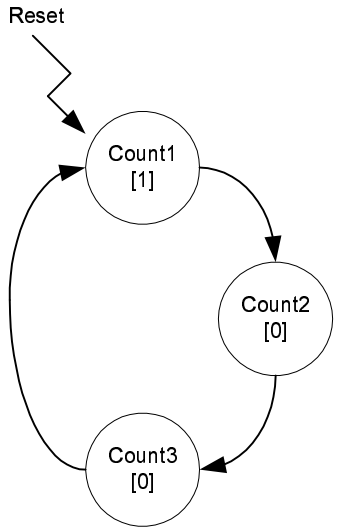


Figure Error! Bookmark not defined.**:**
State diagram of the Divider

We name the state with a unique name and define the outputs in the square brackets. Whenever the state-machine is in Count1, the output shall be 1. Whenever the state-machine is in Count2 or Count3, the output shall be 0.

After obtaining the state diagram it is possible to describe the design with a symbolic state transition table. In this step we put all the information we gathered as shown in the following table.

Present State	Next Step	Output
Count1	*Count2*	1
Count2	*Count3*	0
Count3	*Count1*	0

Table Error! Bookmark not defined.**:**
Symbolic State Transition Table

The next step is to go into details. We have 2 Flip-Flops and one output. This information is entered into an encoded state transition table. The functions can be extracted from a Karnaugh map, or in this case, use the table as a truth table.

Present State		Next Step		Output
Q0	***Q1***	***D0***	***D1***	
0	0	0	1	1
0	1	1	0	0
1	0	0	0	0

Table Error! Bookmark not defined.**:**
Encoded State Transition Table

We can write the functions as:

- D0 = Q1
- D1 = NOT(Q0+Q1)
- OUT = D1

The implement of the divider by 3 is shown in Figure 3. The output can be connected to Q1 pin.

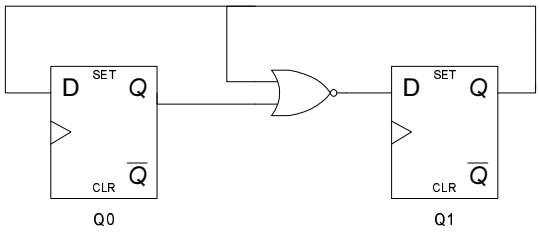


Figure Error! Bookmark not defined.**:**
Implementation of the Divider

Comments and suggestions: interview@hardware-guru.com

Digital One Shot

This "one-shot" shall produce a single output pulse for any long pulse in the input. The length of the output pulse shall be one clock cycle. Assume that the input pulse is as you see in the following figure. **click here if you don't see pictures**

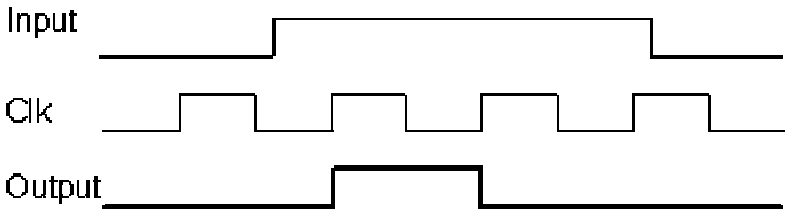


Figure Error! Bookmark not defined.**:**
One Shot Timing Diagram

The answer is showed in Figure 2. It's based on two flip-flops, which create a delay on the signal input. Then the result of the outputs (*Q0, Q1*) are logically AND, and output the result.

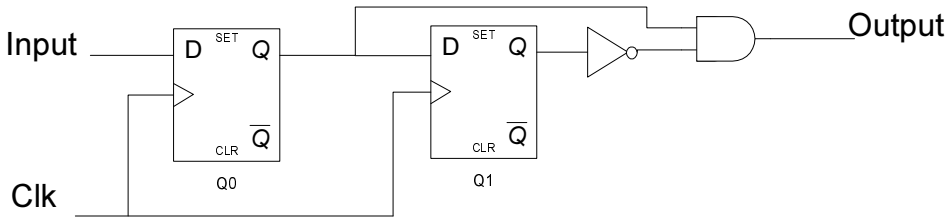


Figure Error! Bookmark not defined.**:**
One Shot Schematics

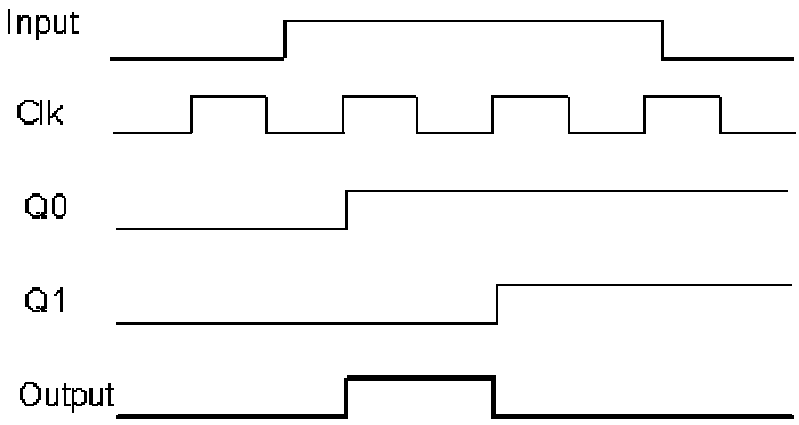


Figure Error! Bookmark not defined.**:**
One Shot Detailed Timing Diagram

This is a simplified design and thus has some problems (hint: asynchronous input). Please write to us with your improvement ideas and we will update the entire solution with your inputs.

Comments and suggestions: *interview@hardware-guru.com*

The following are some of the questions I was asked in my interviews. The questions of course, depend on the position *you* are being interviewed and also on *your* Resume. So if you find any questions *not* relevant to your Resume, you can safely ignore them. Also, these questions are limited to VLSI Design, Computer Architeture and some basic Programming. If you are looking for something in Analog, RF etc, this is NOT the place.

Okay alright...that makes sense...now lets get going...

VLSI Design:

1) Explain why & how a MOSFET works

2) Draw Vds-Ids curve for a MOSFET. Now, show how this curve changes (a) with increasing Vgs (b) with increasing transistor width (c) considering Channel Length Modulation

3) Explain the various MOSFET Capacitances & their significance

4) Draw a CMOS Inverter. Explain its transfer characteristics

5) Explain sizing of the inverter

6)How do you size NMOS and PMOS transistors to increase the threshold voltage?

7) What is Noise Margin? Explain the procedure to determine Noise Margin

8) Give the expression for CMOS switching power dissipation

9) What is Body Effect?

10) Describe the various effects of scaling

11) Give the expression for calculating Delay in CMOS circuit

12) What happens to delay if you increase load capacitance?

13) What happens to delay if we include a resistance at the output of a CMOS circuit?

14) What are the limitations in increasing the power supply to reduce delay?

15) How does Resistance of the metal lines vary with increasing thickness and increasing length?

16) You have three adjacent parallel metal lines. Two out of phase signals pass through the outer two metal lines. Draw the waveforms in the center metal line due to interference. Now, draw the signals if the signals in outer metal lines are in phase with each other

17) What happens if we increase the number of contacts or via from one metal layer to the next?

18) Draw a transistor level two input NAND gate. Explain its sizing (a) considering Vth (b) for equal rise and fall times

19) Let A & B be two inputs of the NAND gate. Say signal A arrives at the NAND gate later than signal B. To optimize delay, of the two series NMOS inputs A & B, which one would you place near the output?

20) Draw the stick diagram of a NOR gate. Optimize it

21) For CMOS logic, give the various techniques you know to minimize power consumption

22) What is Charge Sharing? Explain the Charge Sharing problem while sampling data from a Bus

23) Why do we gradually increase the size of inverters in buffer design? Why not give the output of a circuit to one large inverter?

24) In the design of a large inverter, why do we prefer to connect small transistors in parallel (thus increasing effective width) rather than lay out one transistor with large width?

25) Given a layout, draw its transistor level circuit. (I was given a 3 input AND gate and a 2 input Multiplexer. You can expect any simple 2 or 3 input gates)

26) Give the logic expression for an AOI gate. Draw its transistor level equivalent. Draw its stick diagram

27) Why don't we use just one NMOS or PMOS transistor as a transmission gate?

28) For a NMOS transistor acting as a pass transistor, say the gate is connected to VDD, give the output for a square pulse input going from 0 to VDD

29) Draw a 6-T SRAM Cell and explain the Read and Write operations

30) Draw the Differential Sense Amplifier and explain its working. Any idea how to size this circuit? (Consider Channel Length Modulation)

31) What happens if we use an Inverter instead of the Differential Sense Amplifier?

32) Draw the SRAM Write Circuitry

33) Approximately, what were the sizes of your transistors in the SRAM cell? How did you arrive at those sizes?

34) How does the size of PMOS Pull Up transistors (for bit & bit- lines) affect SRAM's performance?

35) What's the critical path in a SRAM?

36) Draw the timing diagram for a SRAM Read. What happens if we delay the enabling of Clock signal?

37) Give a big picture of the entire SRAM Layout showing your placements of SRAM Cells, Row Decoders, Column Decoders, Read Circuit, Write Circuit and Buffers

38) In a SRAM layout, which metal layers would you prefer for Word Lines and Bit Lines? Why?

39) How can you model a SRAM at RTL Level?

40) What's the difference between Testing & Verification?

41) For an AND-OR implementation of a two input Mux, how do you test for Stuck-At-0 and Stuck-At-1 faults at the internal nodes? (You can expect a circuit with some redundant logic)

42) What is Latch Up? Explain Latch Up with cross section of a CMOS Inverter. How do you avoid Latch Up?

Digital Design:

1) Give two ways of converting a two input NAND gate to an inverter

2) Given a circuit, draw its exact timing response. (I was given a Pseudo Random Signal Generator; you can expect any sequential ckt)

3) What are set up time & hold time constraints? What do they signify? Which one is critical for estimating maximum clock frequency of a circuit?

4) Give a circuit to divide frequency of clock cycle by two

5) Design a divide-by-3 sequential circuit with 50% duty circle. (Hint: Double the Clock)

6) Suppose you have a combinational circuit between two registers driven by a clock. What will you do if the delay of the combinational circuit is greater than your clock signal? (You can't resize the combinational circuit transistors)

7) The answer to the above question is breaking the combinational circuit and pipelining it. What will be affected if you do this?

8) What are the different Adder circuits you studied?

9) Give the truth table for a Half Adder. Give a gate level implementation of the same.

10) Draw a Transmission Gate-based D-Latch.

11) Design a Transmission Gate based XOR. Now, how do you convert it to XNOR? (Without inverting the output)

12) How do you detect if two 8-bit signals are same?

13) How do you detect a sequence of "1101" arriving serially from a signal line?

14) Design any FSM in VHDL or Verilog.

Computer Architecture:

1) What is pipelining?

2) What are the five stages in a DLX pipeline?

3) For a pipeline with 'n' stages, what's the ideal throughput? What prevents us from achieving this ideal throughput?

4) What are the different hazards? How do you avoid them?

5) Instead of just 5-8 pipe stages why not have, say, a pipeline with 50 pipe stages?

6) What are Branch Prediction and Branch Target Buffers?

7) How do you handle precise exceptions or interrupts?

8) What is a cache?

9) What's the difference between Write-Through and Write-Back Caches? Explain advantages and disadvantages of each.

10) Cache Size is 64KB, Block size is 32B and the cache is Two-Way Set Associative. For a 32-bit physical address, give the division between Block Offset, Index and Tag.

11) What is Virtual Memory?

12) What is Cache Coherency?

13) What is MESI?

14) What is a Snooping cache?

15) What are the components in a Microprocessor?

16) What is ACBF(Hex) divided by 16?

17) Convert 65(Hex) to Binary

18) Convert a number to its two's compliment and back

19) The CPU is busy but you want to stop and do some other task. How do you do it?

C/C++, Perl & Unix:

1) How would you decide weather to use C, C++ or Perl for a particular project?

2) What are pointers? Why do we use them?

3) What are the benefits of having Global & Local Variables?

4) What is 'malloc'? Why do we need to use it?

5) Write a C program to compare two arrays and write the common elements in another array

6) Write a function in C to accept two integers and return the bigger integer

7) What are the advantages of C over Perl and vice versa?

8) What does '@' and '&' mean in Perl?

9) What is a 'Package' in Perl?

10) What are Perl Regular Expressions?

11) Perl Regular Expressions are greedy. What does that mean?

12) What are Associative arrays in Perl?

13) Suppose a Perl variable has your name stored in it. Now, how can you define an array by the name? (i.e., you have $a="Adarsh"; now you want @Adarsh=[.....])

14) Write a Perl script to parse a particular txt file and output to another file in a desired format. (You can expect the file to have some data arranged rows & columns)

15) Suppose you have the outputs of a test program in some big test file. In Perl, how can you test if all the outputs match a particular string?

16) What are Data Abstraction and Data Encapsulation?

17) Explain Friend Functions and Polymorphism with examples

18) Commands for changing directory, making directory, going up one directory, knowing the file permissions and changing file permissions.

19) How do you search for a particular string in all the text files in current directory from command line?

20) How do you sort a file alphabetically from command line?

Other Simple Questions:

1) What is j to the power j?

2) What is Normal Distribution? Where is the Mean and Median on the graph for Normal Distribution?

3) Draw a simple RC-Low pass circuit.

Some General Questions:

1) Tell me something about yourself and your interests

2) Tell me something about some problems you faced in a project and how did you handle it?

3) Give one instance where you were criticised by your Professor

4) Where do you see yourself five years from now?

5) What salary are you expecting?

6) Any Questions for me regarding the position or the company?

7) Finally, does this position sound interesting? :-)

Frequently Asked Interview Questions

I gathered these questions from several emails, sent to me by students who attended on-site interviews at varios different companies. I shall try to add more of them in near future.

1. What is the difference between a latch and a flip flop. For the same input, how
would the output look for a latch and for a flip-flop.

2. Finite state machines:
(2.1)Design a state-machine (or draw a state-diagram) to give an output '1' when the # of A's are even
and # of B's are odd. The input is in the form of a serial-stream (one-bit per clock cycle). The inputs could be of the type A, B or C. At any given clock cycle, the output is a '1', provided the # of A's are even and # of B's are odd. At any given clock cycle, the output is a '0', if the above condition is not satisfied.

(2.2). To detect the sequence "abca" when the inputs can be a b c d.

3. minimize a boolean expression.

4. Draw transistor level nand gate.

5. Draw the cross-section of a CMOS inverter.

6. Deriving the vectors for the stuck at 0 and stuck at 1 faults.

7. Given a boolean expression he asked me to implement just with muxes but nothing else.

8. Draw Id Vds curves for mosfets and explain different regions.

9. Given the transfer characteristics of a black box draw the circuit for the black box.

10. Given a circuit and its inputs draw the outputs exact to the timing.

11. Given an inverter with a particular timing derive an inverter using the previous one but with the required timing other than the previous one.

12. Change the rise time and fall time of a given circuit by not changing the transistor sizes but by using current mirrors.

13. Some problems on clamping diodes.

These are some of the questions asked by Microsoft.
(I feel that these type of questions are asked even in Electrical Engineering interviews. Make sure you browse them.)

1. Given a rectangular (cuboidal for the puritans) cake with a rectangular piece removed (any size or orientation), how would you cut the remainder of the cake into two equal halves with one straight cut of a knife ?

2. You're given an array containing both positive and negative integers and required to find the sub-array with the largest sum (O(N) a la KBL). Write a routine in C for the above.

3. Given an array of size N in which every number is between 1 and N, determine if there are any duplicates in it. You are allowed to destroy the array if you like.

4. Write a routine to draw a circle (x ** 2 + y ** 2 = r ** 2) without making use of any floating point computations at all.

5. Given only putchar (no sprintf, itoa, etc.) write a routine putlon the prints out an unsigned long in decimal.

6. Give a one-line C expression to test whether a number is a power of 2. [No loops allowed - it's a simple test.]

7. Given an array of characters which form a sentence of words, give an efficient algorithm to reverse the order of the words (no characters) in it.

8. How many points are there on the globe where by walking one mile south, one mile east and one mile north you reach the place where you started.

9. Give a very good method to count the number of ones in a 32 bit number. (caution: looping through testing each bit is not a solution)

10. What are the different ways to say, the value of x can be either a 0 or a 1. Apparently the if then else solution has a jump when written

```
out in assembly.
if (x == 0)
  y=0
else
  y =x
```

There is a logical, arithmetic and a datastructure soln to the above problem.

Logic design:

1. Draw the transistor level CMOS # input NAND or NOR gate.
After drawing it lot of qestions on that ckt will be asked.

2. Transistor sizing for given rise time and fall time. How do you size it for equal rise and fall time.

3. Given a function whose inputs are dependent on its outputs. Design a sequential circuit.

4. Design a finite state machine to give a modulo 3 counter when x=0 and modulo 4 counter when x=1.

5. Given a boolean equation minimize it.

6. Given a boolean equation draw the transistor level minimum transistor circuit.

7. What is the function of a D-flipflop, whose inverted outputs are connected to its input ?

8. What will you do if you want to drive a large capacitance ?

Layout related questions:

1. asked me to layout the 3 input nand gate.

2. Later he asked me to modify it to consume as much less space as we can.

3. He also asked me about the transistor sizing.

---

1. He asked me to draw the cross section of an inverter and asked me to show all the capacitances on it and reason for those capacitances.

2. Describe the latchup effect.

3. More about the tristate buffers.

3. What will be the voltage at the output node of a triostate buffer in its high impedence state. He gave a waveform for the input and asked me to draw the output waveform for that.

4. Posed a lot of questions on charge sharing problems and keeper circuits.

5. Asked me to draw the I d Vds curves for mosfet. Asked me to explain the regions and some couses for that curve like channel width modulation.

6. He asked me about the electron migration effect and methods to avoid it.

7. Asked me to draw the dynamic logic of a particular gate and then posed lots of tricky questions from the previous discussion.

8. He asked me to draw the 6 transistor contemporary sram cell and asked me to explain how the reading and writing is done in it.

9. Something about trip point.

Computer Architecture Questions:

1. Explain what is DMA?
2. what is pipelining?
3. what are superscalar machines and vliw machines?
4. what is cache?
5. what is cache coherency and how is it eliminated?
6. what is write back and write through caches?
7. what are different pipelining hazards and how are they eliminated.
8. what are different stages of a pipe?
9. eplain more about branch prediction in controlling the control hazards
10. Give examples of data hazards with pseudo codes.
11. Caluculating the number of sets given its way and size in a cache?
12. How is a block found in a cache?
13. scoreboard analysis.
14. What is miss penalty and give your own ideas to eliminate it.
15. How do you improve the cache performance.
16. Different addressing modes.
17. Computer arithmetic with two's complements.
18. About hardware and software interrupts.
19. What is bus contention and how do you eliminate it.

20. What is aliasing?
21) What is the difference between a latch and a flip flop?
22) What is the race around condition? How can it be overcome?
23) What is the purpose of cache? How is it used?
24) What are the types of memory management?

Introduction :

A fresh graduate faces some tough questions in his first job interview. The questions themselves are simple but require practical and innovative approach to solve them. I started collecting some questions from my own experience and from my friends. Answers to most questions are not given. Spend some time to solve these and let me know if you have some more interesting ones.

Please do not send me emails asking for solutions. You are not supposed to answer these questions in 10 seconds like some university multiple choice questions. Some questions may have more than correct answers and some may not even have correct answer :)

What matters is your approach to solution and understanding of basic hardware design principles.

Recently added questions

Q. Design a logic which mimics a infinite width register. It takes input serially 1 bit at a time. Output is asserted high when this register holds a value which is divisible by 5.

For example:

Input	Sequence	Value	Output
1	1	1	0
0	10	2	0
1	101	5	1
0	1010	10	1
1	10101	21	0

(Hint: Use a FSM to create this)

Q.Design a block which has 3 inputs as followed.

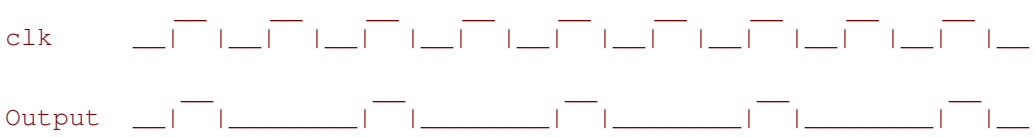
1. system clock of pretty high freq
2. asynch clock input P
3. asynch clock input Q

P and Q clocks have 50% duty cycle each. Their frequencies are close enough and they have phase difference. Design the block to generate these outputs.

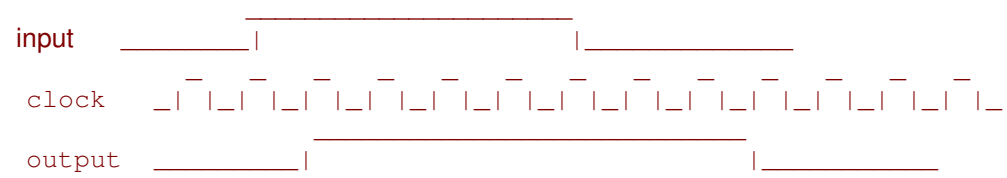
1. PeqQ : goes high if periods of P and Q are same
2. PleQ : goes high if P's period is less than that of Q.
3. PgrQ : goes high if P's period is greater than that of Q.

Q. What's the difference between a latch and a flip-flop? Write Verilog RTL code for each. (This is one of the most common questions but still some EE's don't know how to explain it correctly!)

Q. Design a black box whose input clock and output relationship as shown in diagram.



Q. Design a digital circuit to delay the negative edge of the input signal by 2 clock cycles.

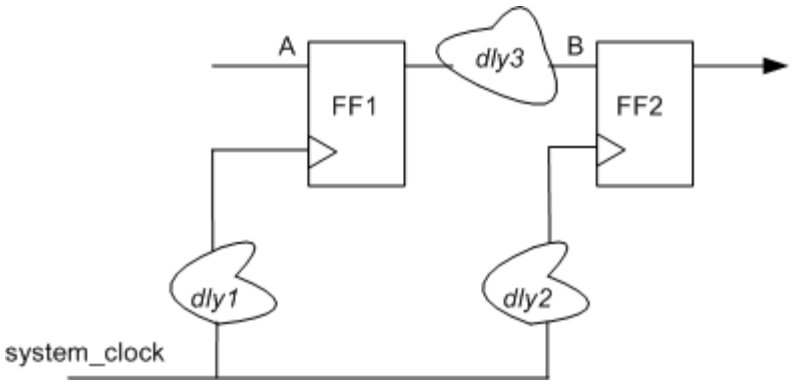


Q. Design a Pattern matching block

- Output is asserted if pattern "101" is detected in last 4 inputs.
- How will you modify this design if it is required to detect same "101" pattern anywhere in last 8 samples?

Questions:

Q1.

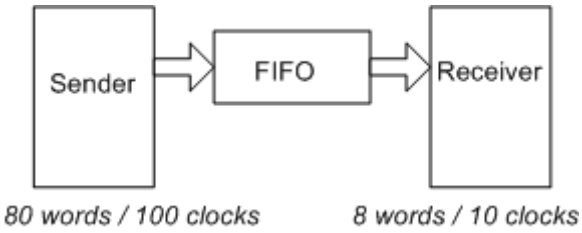


The digital circuit is shown with logic delay (dly3) and two clock buffer delays (dly1, dly2).

- How will you fix setup timing violations occurring at pin B?
- How will you fix hold violations occurring at pin B?

(Hint: Change the values of three delays to get desired effect)

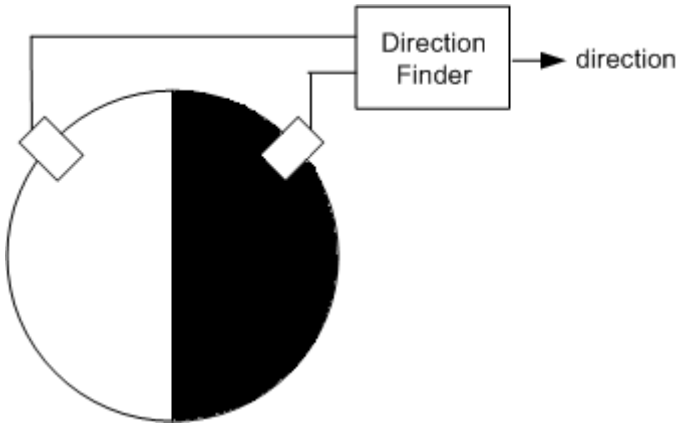
Q2.



Sender sends data at the rate of 80 words / 100 clocks
Receiver can consume at the rate of 8 words / 10 clocks

Calculate the depth of FIFO so that no data is dropped.
Assumptions: There is no feedback or handshake mechanism. Occurrence of data in that time period is guaranteed but exact place in those clock cycles is indeterminate.

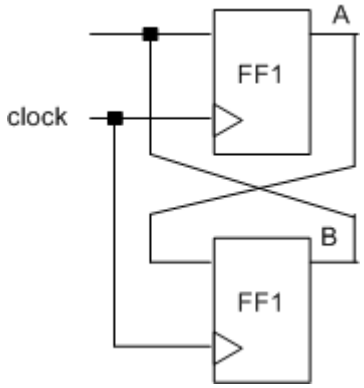
Q3



Optical sensors A and B are positioned at 90 degrees to each other as shown in Figure. Half od the disc is white and remaining is black. When black portion is under sensor it generates logic 0 and logic 1 when white portion is under sensor.

Design Direction finder block using digital components (flip flops and gates) to indicate speed. Logic 0 for clockwise and Logic 1 for counter clockwise.

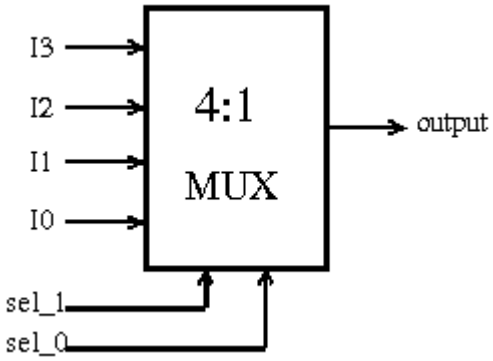
Q4



Will this design work satisfactorily?
Assumptions: thold = tsetup = tclock_out = tclock_skew = 1ns.
After reset A = 0, B = 1

---

1. Design a 4:1 mux in Verilog.



- Multiple styles of coding. e.g.
  Using **if-else** statements

```
if(sel_1 == 0 && sel_0 == 0) output = I0;
else if(sel_1 == 0 && sel_0 == 1) output = I1;
else if(sel_1 == 1 && sel_0 == 0) output = I2;
else if(sel_1 == 1 && sel_0 == 1) output = I3;
```

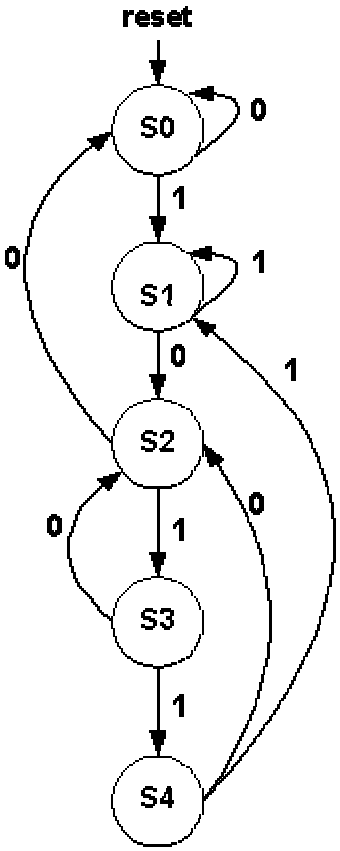
  Using **case** statement

```
case ({sel_1, sel_0})
    00 : output = I0;
    01 : output = I1;
    10 : output = I2;
    11 : output = I3;
    default : output = I0;
endcase
```

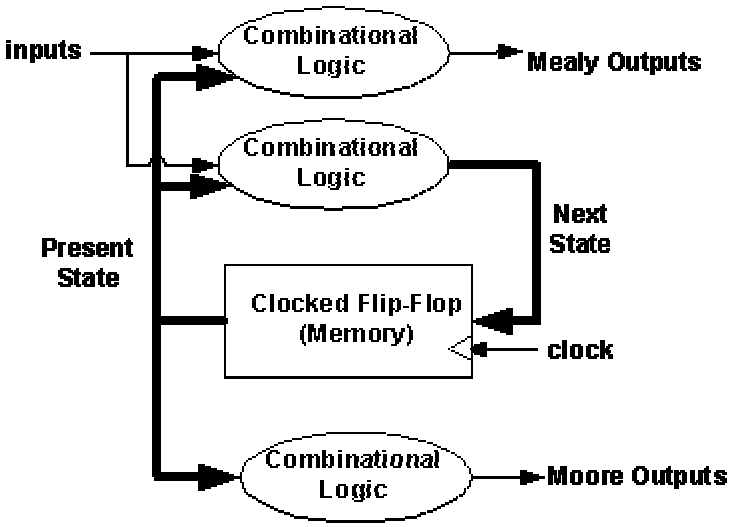
- What are the advantages / disadvantages of each coding style shown above?
- How Synthesis tool will give result for above codes?
- What happens if default statement is removed in case statement?
- What happens if combination 11 and default statement is removed? (Hint Latch inference)
  (Comments : Though this questions looks simple and out of text books, the answers to supporting questions can come only after some experience / experimentation.)

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2. Design a FSM (Finite State Machine) to detect a sequence 10110.



State Transition Diagram



FSM Block Diagram

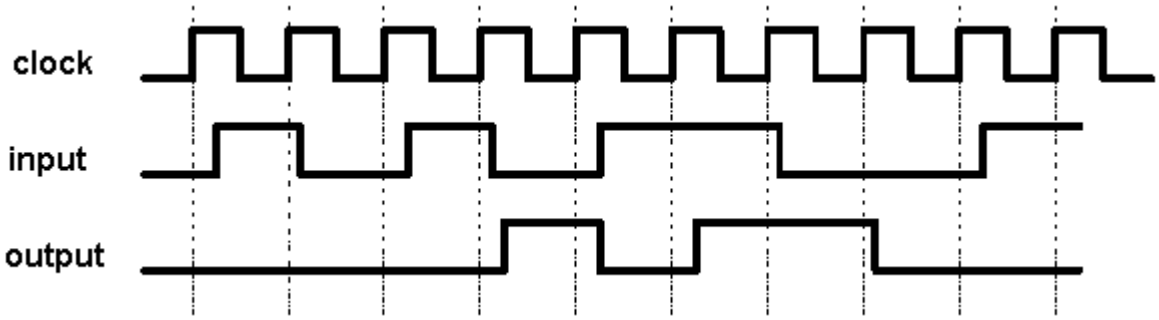
- Have a good approach to solve the design problem.
- Know the difference between Mealy, Moore, 1-Hot type of state encoding.
- Each state should have output transitions for all combinations of inputs.
- All states make transition to appropriate states and not to default if sequence is broken. e.g. S3 makes transition to S2 in example shown.
- Take help of FSM block diagram to write Verilog code.

---

3. One more sequence detector:

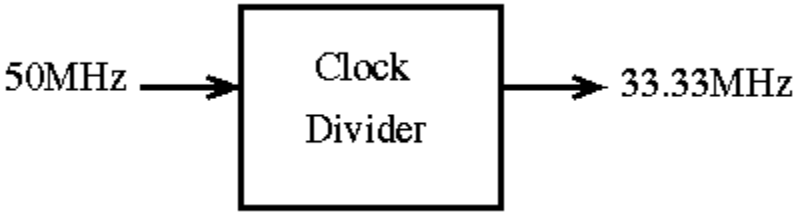
Design a FSM (Finite State Machine) to detect more than one "1"s in last 3 samples.
For example: If the input sampled at clock edges is 0 1 0 1 0 1 1 0 0 1
then output should be 0 0 0 1 0 1 1 1 0 0 as shown in timing diagram.

And yes, you have to design this FSM using not more than 4 states!!



---

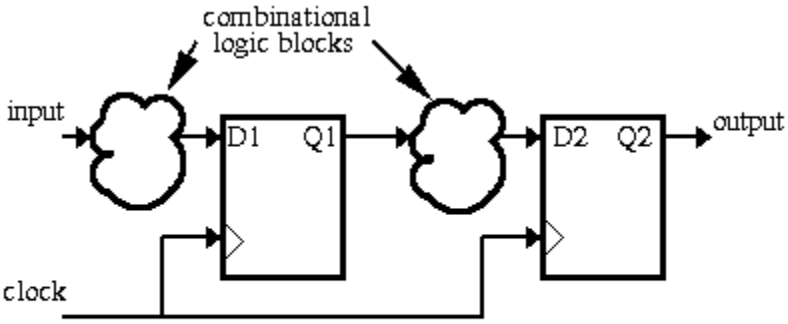
4. Design a state machine to divide the clock by 3/2.



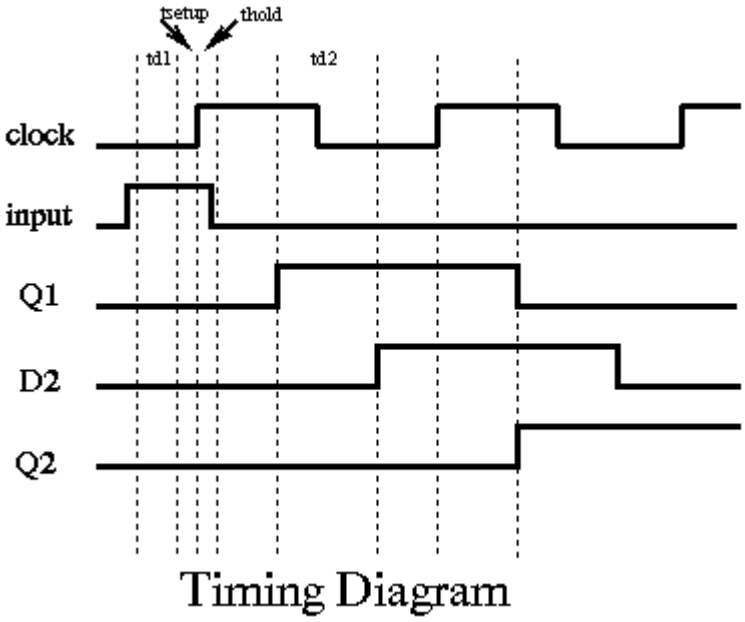
(Hint: 2 FSMs working on posedge and negedge)

---

5. Draw timing diagrams for following circuit.



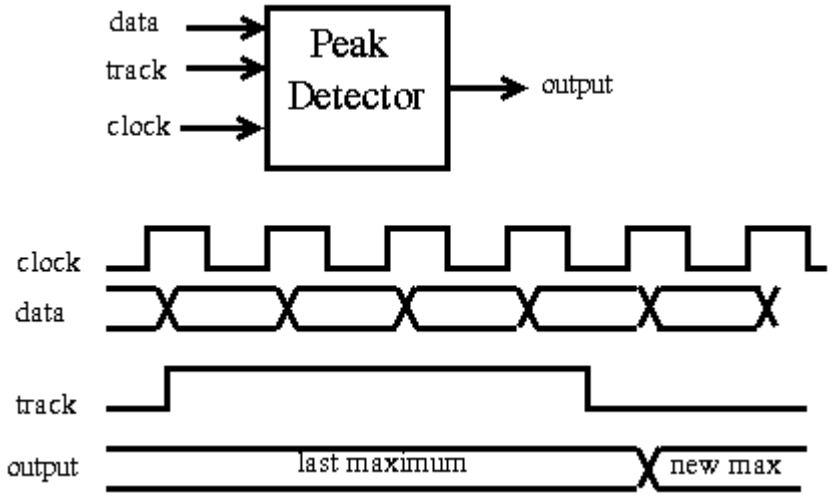
Schematic of two cascaded flip–flops



- What is the maximum frequency at which this circuit can operate?
- What is the minimum width of input pulse and position?
- Problem can be given interesting twist by specifying all delays in min and max types.

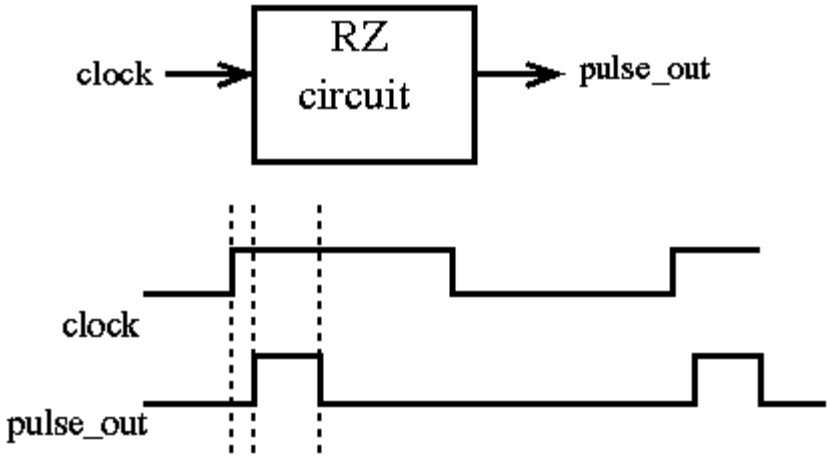
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6. Design a Digital Peak Detector in Verilog.



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7. Design a RZ (return to zero)circuit. Design a clock to pulse circuit in Verilog / hardware gates.



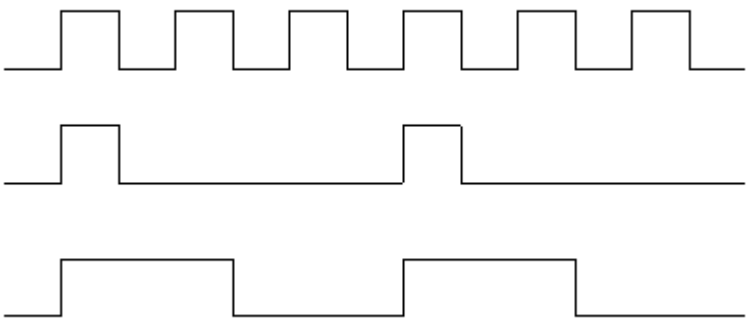
---

8. Miscellaneous Basic Verilog Questions:

- What is the difference between Behavior modeling and RTL modeling?
- What is the benefit of using Behavior modeling style over RTL modeling?
- What is the difference between blocking assignments and non-blocking assignments ?
- How do you implement the bi-directional ports in Verilog HDL
- How to model inertial and transport delay using Verilog?
- How to synchronize control signals and data between two different clock domains?

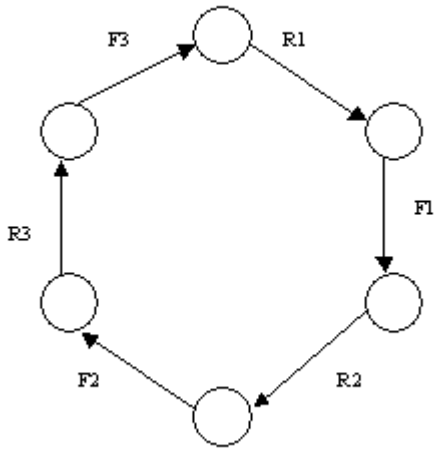
Q: how to design a divide-by-3 counter with equal duty cycle ?

Here is one of the solutions...
Start with the timing diagram. It shows an input clock, an output of a regular divide-by-3 counter and an output of divide-by-3 counter with 50% duty cycle.



It is obvious from the diagram, that we have to use both rising and falling edges of the

input clock.
The next drawing is a state diagram.

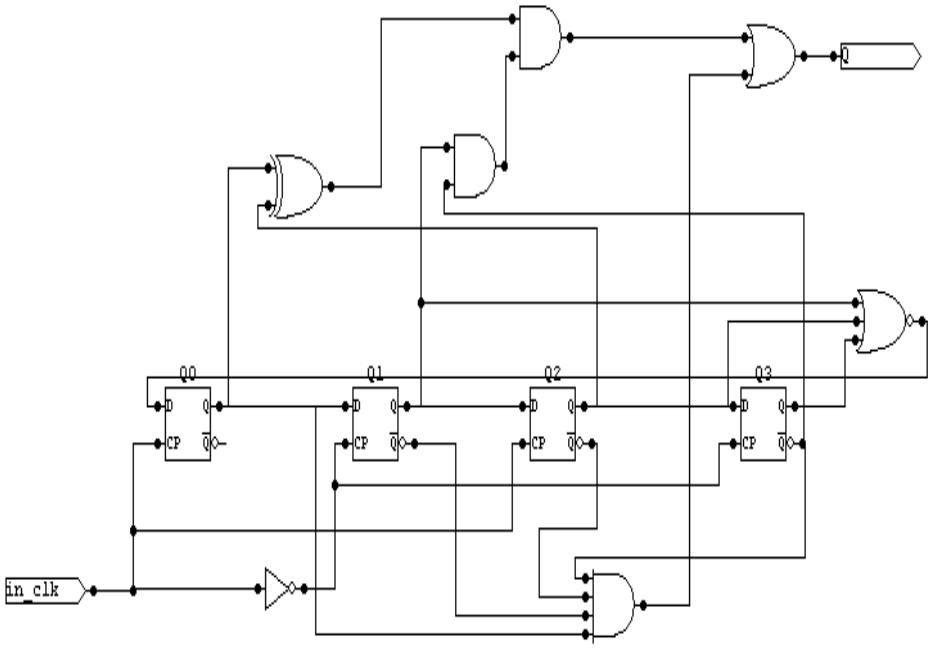


On this diagram R - is a rising edge of input clock, and F - is a falling edge.
How many FF do we need to implement 6 states? At least 3. In this example I am going to use 4 D-type FF just to simplify the design.
Now, look at the table below. Q0 ... Q3 are the outputs of FFs. Q - is the output of the devider with 50% duty cycle. In the first raw the outputs are in the initial state: 0000. In the second raw - the data after the first rising edge and so on.The status of the FFs' ouputs is changing on every rising or falling edge of the input clock according to the information on D-inputs. So, D-inputs should have the data before the clock edge.

in_clk	Q	Q0	Q1	Q2	Q3	D0	D1	D2	D3
R1	1	1	0	0	0	1	1	0	0
F1	1	1	1	0	0	0	1	1	0
R2	1	0	1	1	0	0	0	1	1
F2	0	0	0	1	1	0	0	0	1
R3	0	0	0	0	1	0	0	0	0
F3	0	0	0	0	0	1	0	0	0

These equations are resulting from the table analysis:
D1 = Q0
D2 = Q1
D3 = Q2
D0 = (Q1+Q2+Q3)'
Q = Q0*Q1'*Q2'*Q3'+Q0*Q1*Q2'*Q3'+Q0'*Q1*Q2*Q3' =
Q1*Q3'(Q0*Q2'+Q0'*Q2)+Q0*Q1'*Q2'*Q3'

Now it is the time for the circuit diagram:



Din contributed to this article. Your comments are welcome at eclub@hitequest.com
We will gladly post your design as well.

Q:I swapped transistors in CMOS inverter (put n-transistor at the top and p-transistor at the bottom).
Can I use this circuit as a noninverting buffer?

Not really. High input level can't open n-transistor,because it's source has the same high potential (Vdd) as a gate. By the same reason low level will not open p-transistor.

Further discussion:

```
> > Dear Mr.Martovetsky,
First off, I'd like to thank you and Hitequest for hosting an excellent
website
which has been very useful to me.Great work!I have a question about
 the solution about the noninverting buffer.


I didn't quite understand this because I thought the source and drain
were
 interchangeable depending on which one is at a higher potential.In the
case of
 NMOSFET, if one of the 2 terminals is tied to VDD,then, doesn't that
become the
 drain since it is at a higher potential? I'd really appreciate if you
could
```

enlighten me on this whenever you get time.Thank you and have a great day:-)

Thanks,Sriram

I still am not clear with one thing.Perhaps,I should rephrase my question:
Consider the swapped circuit with NMOS on top and PMOS below.One terminal of
the NMOS is connected to Vdd(this becomes the drain due to its higher potential).The gate is connected to Vdd as well.So shouldn't it act like a pass
transistor and conduct current? there is a voltage difference between source
and drain since drain is connected to Vdd,right?

I'm really curious to know the answer.Thanks for your time and have a great
day:-)

Thanks Sriram

> > OK, think about it this way: what does the current between drain and
> > source depend on? (Vgs - Vt) - any book at the first page,right?
> > If you put nmos transistor at the top, it's Vs = Vdd/2. You can barely
> > open it applying Vg=Vdd. In order to open it you need to apply Vg
> > greater than Vdd.
> > Same story with pmos tarnsistor.
> > Does it make sense?
> > Take care.
> > alex

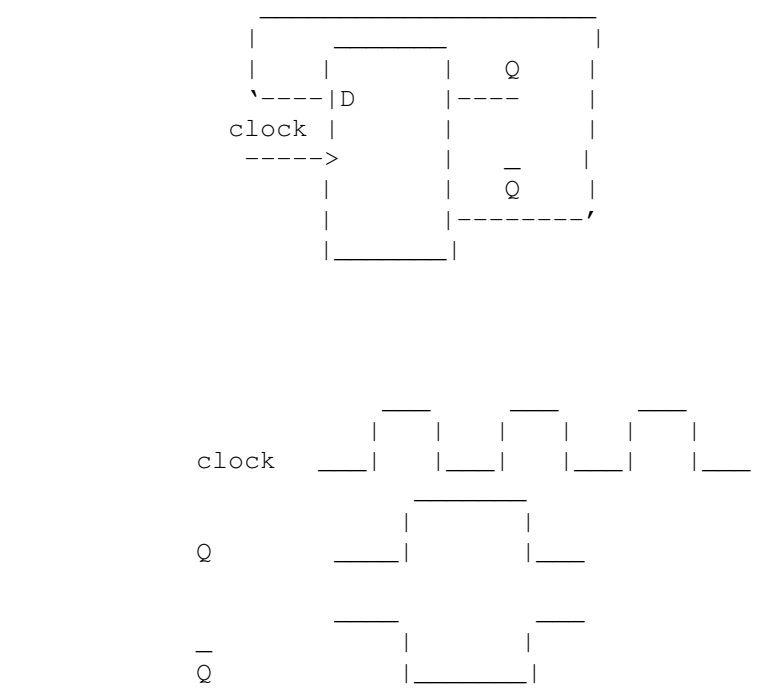
> Dear Mr.Paikin,
> Once again,thanks for responding immediately.I don't know what I'm missing but
> how is the Vs of the NMOS equal to vdd/2.I thought the source of the NMOS is
> basically the output node of the new arrangement and its source voltage is
> indeterminate at the beginning.Look forward to your reply.I appreciate your
> help.
>
> Cheers
> Sriram

---
hello Sriram,
if both transistors are closed,the output is in the tristate,which is Vdd/2 (think about these transistors like of 2 large value resistors).If
you apply "1" to n_mos transistor, it is trying to open ,and it's Vs

would be close to Vdd.But it can't happen,because your Vg is not high enough.
alex

Q: Convert D-latch into divider by 2.
   What is the max clock frequency the circuit can handle ?
                T_setup= 6nS
                T_hold = 2nS
                T_propagation = 10nS

A:



Any system with clock should meet setup and hold time conditions.
Besides since there is a feedback from !Q to D, we should take care of D input timing: the data on D input should not change while clock is high!
Otherwise the results are unpredictable.
To meet these conditions:

t_clock_high <= T_prop

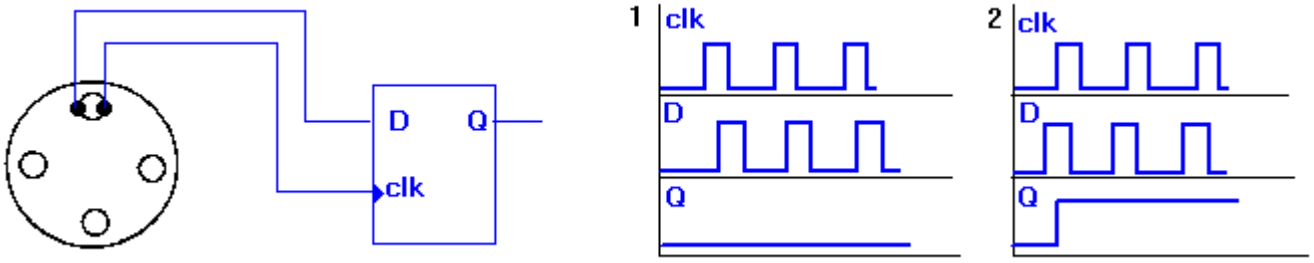
t_clock_low >= T_setup

T_hold <= T_prop

For example if we take t_clock_high= t_clock_low = 6nS
Then clock period = 12nS,i.e max Freq = 80MHz

Q: The circle can rotate clockwise and back. Use minimum hardware to build
a circuit to indicate the direction of rotating.

A: 2 sensors are required to find out the direction of rotating.
They are placed like at the drawing. One of them is connected to the data input of D flip-flop,
and a second one - to the clock input. If the circle rotates the way clock sensor sees the light
first while D input (second sensor) is zero - the output of the flip-flop equals zero, and if D input
sensor "fires" first - the output of the flip-flop becomes high.




```
module fifo1 (full,empty,clk,clkb,ain,bout,rst_N)
output [7:0] bout;
input [7:0] ain;
input clk,clkb,rst_N;
output empty, full;
reg [3:0] wptr, rptr;
...

endmodule
```

wptr -->

0
1
2
3
4
5
6
7
8

<-- rptr

9
10
11
12

Multiple clocks add complexity to this design. We need to define conditions for Empty and Full signals, take care of WR and RD pointers. Here is one of the solutions.

Empty and Full flags:
assign empty=((wptr == rptr) && (w_flag == r_flag);
assign full=((wptr == rptr) && (w_flag == ~r_flag);

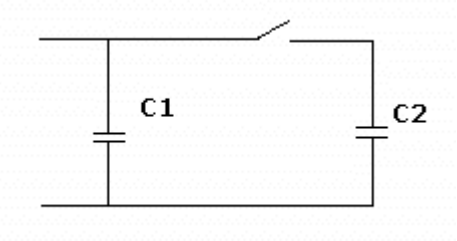
where w_flag is set when wptr =12 (end of FIFO). After that wptr is reset to 0. The same for r_flag and rptr.

Pointer handling:
if (wptr == 12) {w_flag,wptr} <= {~w_flag,4'b0000};
else wptr <= wptr+1;
if (rptr == 12) {r_flag,rptr} <= {~r_flag,4'b0000};
else rptr <= rptr+1;

Q:What does CMOS stand for? VLSI?

A:VLSI - means a Very Large-Scale Integration.
CMOS stands for Complementary Metal-Oxide Semiconductor technology... Don't stop here, draw a complementary transistor pair and tell them how it works!

Q: Two capacitors are connected in parallel through a switch. C1= 1uF, C2=0.25uF. Initially switch is open,C1 is charged to 10V. What happens if we close the switch? No loss in the wires and capacitors.



A:
Since no loss in the circuit the charge remains the same:
U1C1 + U2C2 = U3(C1+C2)
U3 = (U1C1+U2C2)/(C1+C2) = (10*1 + 0*0.25)/1+0.25 = 8
U3= 8V

Q: You work on a specification of a system with some digital parameters. Each parameter has Min,Typ and Max colomns.
What colomn would you put setup and hold time?

A: put SETUP time into the Min colomn, put HOLD time into the Min colomn too.
Example:
usually the data must be set at least (minimum) X nS before clock and being held at least Y nS after the clock. You need to specify Min setup and Min hold time.

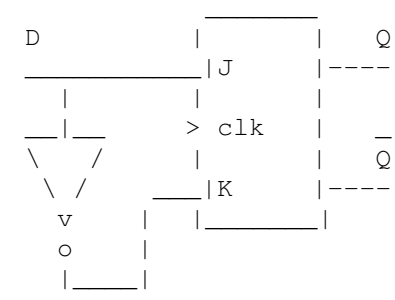
Q:How to convert D-latch into JK-latch and JK-latch into D-latch?

Compare the truth tables of D-latch and JK-latch accordingly:

clk	D	Q
+	0	0
+	1	1

clk	J	K	Q
+	0	0	hold
+	0	1	0
+	1	0	1
+	1	1	switch to opposite

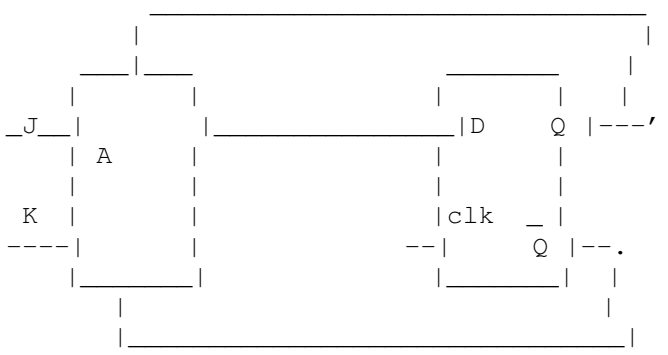
Using these truth tables it is easy to convert JK-latch into D-latch.



To convert D-latch into JK-latch some extra logic is required.
The following table shows the relation between J,K and D

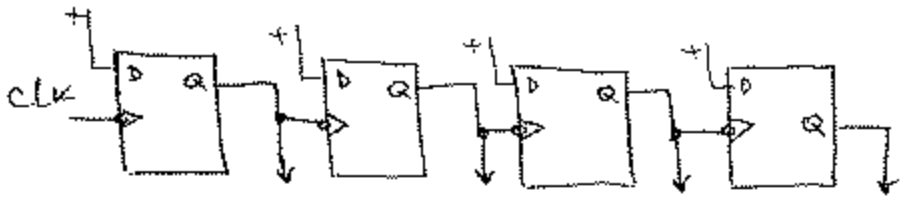
J	K	D

```
0       0       Q
0       1       0
1       0       1
1       1       !Q
```

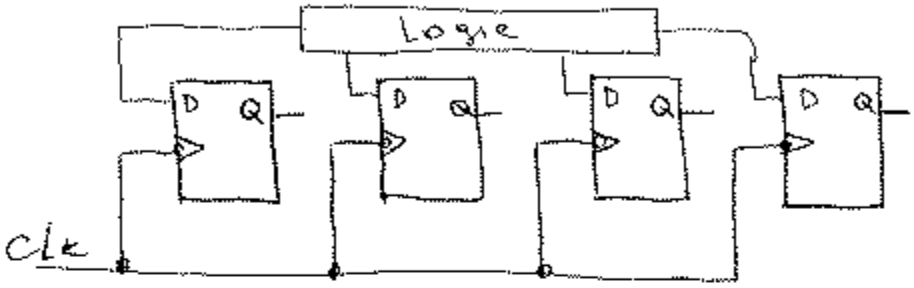



```
Looking at the drawing and the table it is not a problem to implement
block A.
Probably the easiest way is to use a MUX.
J and K are control signals and 1,0,Q,!Q are data inputs.
```

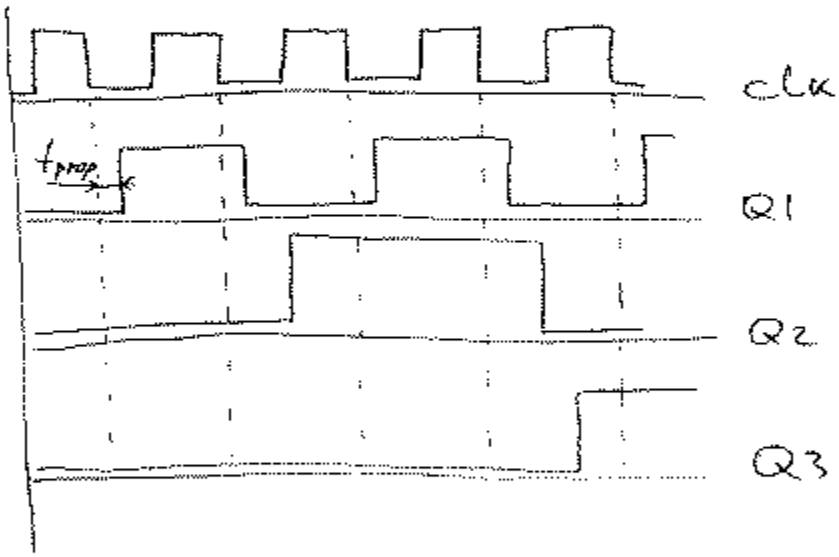
Q:You have two counters to 16 built from negedge D- FF . First circuit is synchronous and second is "ripple" (cascading). Which circuit has less delay?



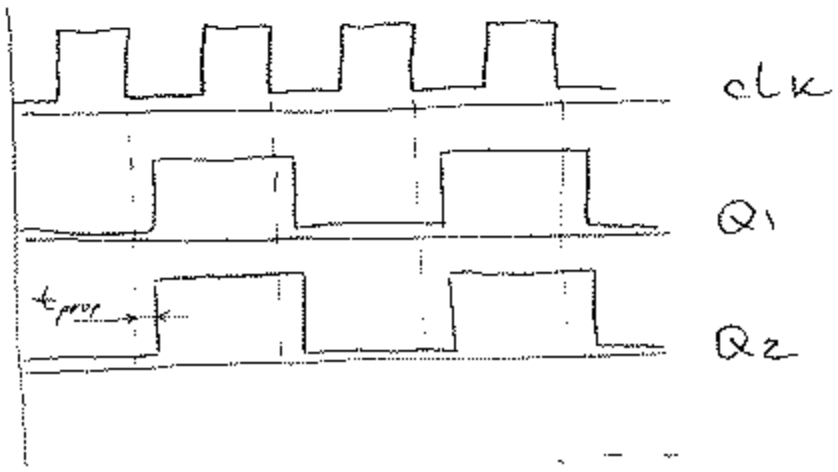
1



2



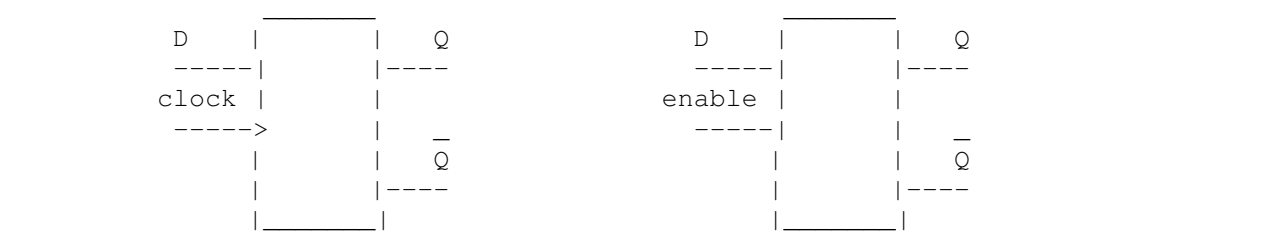
3



4

1 - is ripple counter;
2 - synchronous.
Both consist of 4 FF, synchronous counter also has some logic to control it's operation. From diagram 3 (for ripple) and 4 (for synchronous) it is seen that propagation delay of ripple counter is 4* t_prop , while synchronous counter has only 1* t_prop.

Q:What is the difference between flip-flop and latch?



The example shows D-latch and D-FF.
The simplest form of data storage is latch.It's output responds immidiately
to changes at the input and the input state will be remembered, or "latched" onto.
While "enable" input is active the input of the latch is transparant to the output,
once "enable" is disactivated the output remains locked.
Flip flops use clock as a control input. The transition in output Q occurs only at the edge
of the clock pulse. Input data must present T_setup time before clock edge and remain T_hold time after.

* RESET input, while it is not shown, is present in most FF.

```
module DFF (Q,_Q,D,clk,rst);
output Q,_Q;
input D,clk,rst;
reg Q,_Q;

always @(posedge clk or posedge rst)
 begin
  if (rst) Q  <= 0;
  else     Q  <= D;
  _Q <= !Q;
 end

endmodule

module DLatch (Q,_Q,D,en,rst);
output Q,_Q;
input D,en,rst;
reg Q,_Q;

always @(en or D or posedge rst)
```

```
 begin
  if (rst)      Q  <= 0;
  else if (en) Q  <= D;
  _Q <= !Q;
 end

endmodule
```

Here are the most typical interview questions about your personality.
Nevertheless a proper answer to these questions is just as important, as a technical qualification.

So, tell me something about yourself?
Remember, this question is totally job-related. It is designed for the interviewer to hear you talk and see how you express yourself. Don't talk long about your marital status, your hobbies or go through your whole resume. Instead, you need to summarize your response and talk about key accomplishments in your career: " These are the things I am good at ..., these are the things I can do for the company ..." You can briefly show your professional identity, and what you are looking for (professionally).

Name 3 positive and 3 negative qualities of yours.
Positive qualities: get along with people, high motivation to learn, never give up facing a problem , responsibility
(at least not switching jobs in the middle of the project) etc.
Negative: Hey, nobody is perfect. You have to give something.But don't go too far, you don't want to look as a bad person either.

Where do you see yourself professionally, in 3-5 years?
Some people talk about their professional development.
It is a good idea to tell if you want to take any particular classes or explore any particular areas in the professional environment.

What is the most exciting and boring part of your job?
There is no 100% exciting job. Every job has it's boring part.

Do you prefer team work or individual?

Be carefull. If you say "individual" - it may sound like you are not a team player, if you say you prefer team work - it means you can't work alone.

What is the most significant responsibility you have ever had in your life?
Matovolwa Peter, email: matovolwap@scitz.com

You are assigned to work on an important project containing 7 design modules and you are short of time.
By the end of the month all you can do is either to have 3 modules accomplished or to have all 7 modules started in parallel but not finished.
What strategy will you choose?
You definitely want to have at least part of your work finished.You can demonstrate it, explain your problems and ask some more time to complete the project.

Why do you want to leave your present job?
Do not blame your company,your job,managers. The interviewer may think you are not getting along with your supervisor or coworkers. In this case you are not someone they would want to hire. Instead, you can say it is a time for you to move on, try yourself in the new area, etc

What do you know about our company?
You have to do some homework before the interview. At least look at the company's WEB page.

What particularly would you like to work on?
Some people say:"What ever you want me to do!" In most cases this answer will not be appreciated, especially in start up companies.
It may sound like you have no any other interest in this job but money.

Tell us about one of the technical problems you had to solve recently
Bill Benson, technical recruter from Silicon Valley, says this question is quite frequent on interviews. Don't miss this chance ! Tell how good you are in solving technical problems.

What is the earth?
Sometimes people ask strange questions to see your reaction .Just stay cool and bring your sense of humor.
A friend of mine who really likes talking answered to the above question:

"How much time do you have?"

The next few questions are reproduced from www.microsearchsf.com with kind permission of Janice Schooler Litvin, executive search consultant

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Tell me about a conflict you encountered and how you handled it.
HINT : This is one of the toughest interview questions of all. It's sort of a trick question, as a matter of fact. Never speak negatively about anyone. The ability to successfully resolve conflicts is important for all members of an IS team.. It may be the most important factor if you're working in a service environment, such as a large consulting firm that deals with outside clients. The answer you give here could go a long way toward getting you a job offer. Managers want to see that you are mature and unselfish. The answer should involve proof of your maturity level. They are looking for your ability to handle conflict. Compromise and working it out without external intervention are the keys. A disgruntled person is not going to be productive, and tends to bring down coworkers' morale as well.

What changes have you made in your life that you are most proud of?
HINT : This tells the manager more about your ability to take control of your life. It illustrates your leadership potential, and suggests just how promotable you might be. After all, if he produces a star, he looks good.

What are your salary requirements?
HINT : The use of the word "offer" is critical. It's a subliminal message that an actual job offer is what you are discussing, not just your salary needs in general.
A : "Salary is not my primary consideration. Of course, I have to pay the bills. I'd be open to any reasonable offer." Pause and maintain direct eye contact, even if it seems like forever. Do not be the first one to flinch. Do not over-talk. Be prepared for a long silence. Let the manager be the first to present a figure. It will give you power and control.
If forced to give a specific number, never give a broad range -- you will usually be offered the low end. Instead, be as precise as possible: "I'd be open to something in the low-fifties (or mid-forties, high-seventies, whatever)." Giving such a specific number presumes you've researched

the local job market and know what people with your skills are making.

Are you interviewing at any other companies?
HINT : You want the manager to know that you're extremely interested in his opportunity, but are keeping your options open.
A : "Yes, Mr. X, but at this point XYZ is my first choice."

Remember, all of these interview questions have more than one appropriate answer.
If you are feeling nervous about an upcoming interview, keep in mind that the hiring manager gets just as excited about a potentially strong candidate as the candidate does about him or her.
Strong, qualified, motivated technical people are very hard to find. Be direct, but think before you speak, and you will surely get an offer.

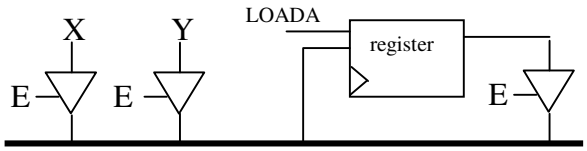
MT1 practice questions. These questions aren't to test your reasoning capabilities, just your knowledge of the basics.

1. 3 Bit Up Counter (Moore FSM Design, Setup/Hold Time, Logic Simplification)
   1a) Build a 3 bit up counter Moore machine using only AND, OR, NOT, and XOR gates. Simplify all logic. (Use the fewest number of gates possible.)

   1b) Will the circuit have any problems if:
   2ns < T_{pINV} < 4ns
   3ns < T_{pAND} < 6ns
   3ns < T_{pOR} < 5ns
   3ns < T_{pXOR} < 4ns
   3ns < Tcko < 8ns
   Tsetup = 4ns
   Thold = 6ns
   CLK = 40 MHz

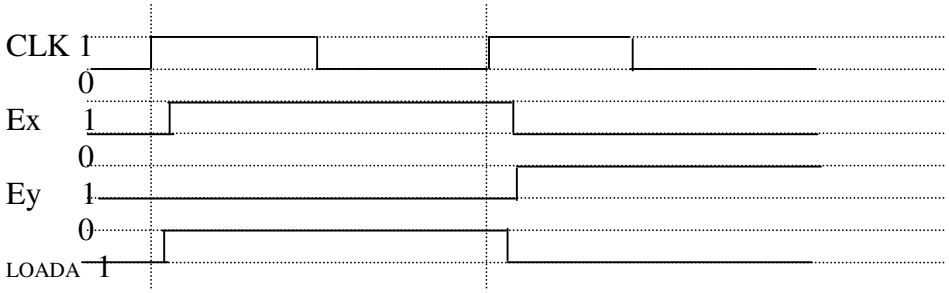
2. Timing problem… (All questions use homework 5, problem 2)
   2a) (HW5, 2b) Without doing any math, do you think increasing T_{DELAY} will increase $(T1\text{-}T0)_{MAX}$ or decrease it.
   2b) (HW5, 2c) Without doing any math, do you think increasing T_{DELAY} will increase $(T1\text{-}T0)_{MIN}$ or decrease it.
   2c) What is the formula for finding a setup time violation on HW #5's 2b? Use 'T' for the period.
   2d) What is the formula for finding a setup time violation on HW #5's 2c? Use 'T' for the period.

3. BUFEs and timing



(Note: E_{XAZ} means the time it takes the output of E_X to go from active to tristated. The turn-off time.)

The controller sends the following signals to the above datapath:



Are there any problems when:

3a) E_{XAZ} = 10ns, E_{YZA} = 5ns, T_{setup} = 10ns, T_{hold} = 4ns, Period = 30ns?

3b) E_{XAZ} = 9ns, E_{YZA} = 10ns, T_{setup} = 10ns, T_{hold} = 10ns, Period = 30ns?

3c) E_{XAZ} = 10ns, E_{YZA} = 12ns, T_{setup} = 10ns, T_{hold} = 4ns, Period = 20ns?

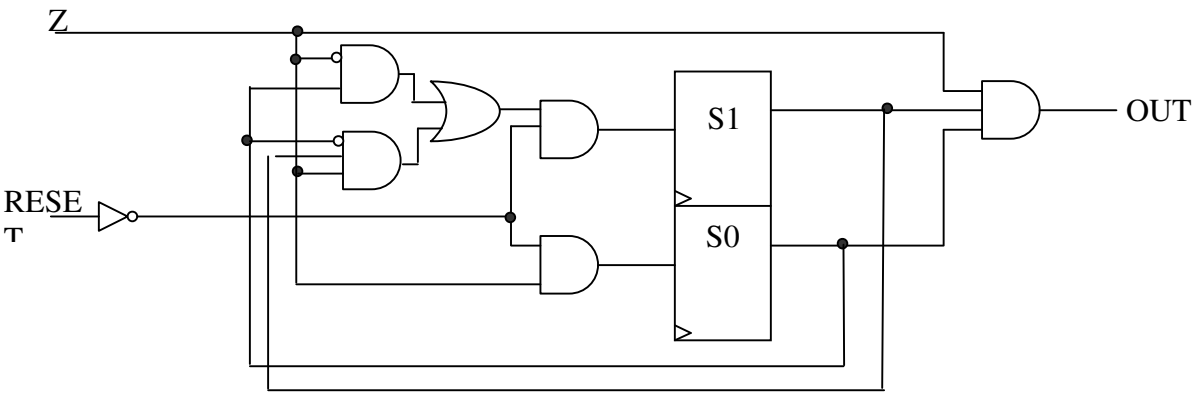
3d) E_{XAZ} = 14ns, E_{YZA} = 6ns, T_{setup} = 12ns, T_{hold} = 8ns, Period = 25ns?

4. Clock divider (FSM Design, Counters, Clocks and Glitches)

Design a clock divider which runs on a 200MHz clock and produces a 1MHz clock. Make the duty cycle of the 1MHz clock 50%. Should you design this to be a Moore or a Mealy machine?

5. FSM Analysis

What does this circuit do? (Is it a Mealy or Moore? Where are the NSD, OD & State FFs.)



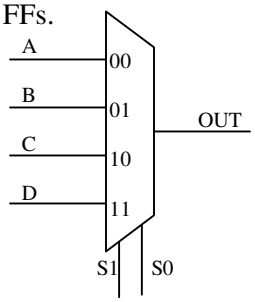
6. STD & STT
   Draw a Moore STD for a circuit that does the same thing as the circuit in problem 5 and then make it's STT.

7. Gate Delay Timing Diagram (Cross Coupled Gates, Gate Delays)
   It's too much to try create a problem like the one on last year's MT. Try to do last year's (and if you can get it, the year before's timing problem might be good practice).

8. Toggle Flip Flops (FSM Design)
   Design a 2 Bit Down Counter that is implemented with T FFs.

9. JK Flip Flops (FSM Design)
   Design a 2 Bit Up Counter that is implemented with JK FFs.

10. Use 3 BUFTs and 3 BUFEs to build a 4-1 MUX.



Also review asserted high, asserted low, & equation simplfication.

Solutions:

1. a) STT:

PS2	PS1	PS0	NS2	NS1	NS0
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0

Equations:

$NS0 = \overline{PS0}$

$NS1 = PS1 \oplus PS0$

$NS2 = PS2\ \overline{PS0} + PS2\ \overline{PS1} + \overline{PS2}\ PS1\ PS0$

b) Any problems?

Check for hold time violations (Can value from flip-flop output propagate back to the flip-flop inputs so quickly that the hold time is violated?):

> Find the shortest propagation path: Looks like NS0 (XOR gates have bigger delay than inverters).

$Tckp_{min} + Tinv_{min} \stackrel{?}{>} T_{hold}$

3ns + 2ns $\stackrel{?}{>}$ 6ns This isn't true → hold time violation possible.

Check for setup time violations (Can value from flip-flop output propagate back to the flip-flop inputs so slowly that the next clock's setup time is violated?)

> Find the longest propagation path: Looks like NS2.

$Tcko_{max} + Tinv_{max} + Tand_{max} + Tor_{max} + Tsetup \stackrel{?}{<} Period$ (1/40MHz = 25ns)

8ns + 4ns + 6ns + 5ns + 4ns < 25ns False → setup time violation possible.

2. 2a) By increasing the delay, you're delaying the change in D1's value. Which means that you can increase T1-T0.

2b) The delay won't affect the minimum T1-T0.

2c) Tckomax + INVTpmax + Tsumax < T – (T1-T0).

2d) Tckomax + Tdelaymax + Tsumax < T – (T1-T0).

3. 3a) Bus conflict (X & Y both on bus at same time).

3b) Hold time violation. Turn-off time (E_{XAZ}) is less than Thold of flip-flop.

3c) Setup time violation. E_{YZA} + Tsetup > Period.

3d) Setup time violation, hold time violation & bus conflict.

4. This should be a Moore machine with two states and no logic on the output – the output should come directly from the state bits. Because... any combinational logic, no matter how simple may cause glitches. It should change states when a counter counts up to 100. Since the output of this circuit is a clock, we have to make especially sure that there aren't glitches on the output. Moore is safer than a Mealy in this case. Why?

5. It's a Mealy pattern detector (1011).

Equations:

$NS1 = \overline{R}\,(\,PS0\,\overline{Z} + PS1\,\overline{PS0\,Z}\,)$

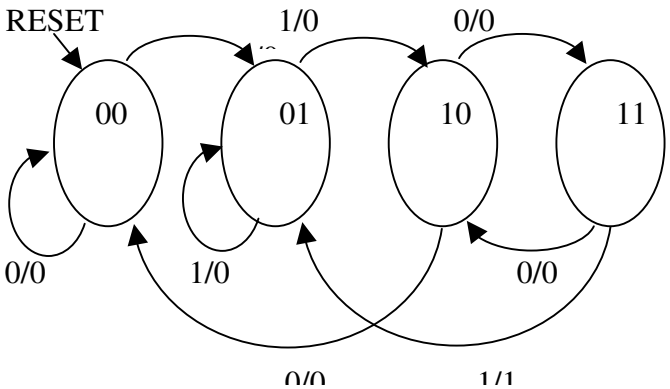
$NS0 = \overline{Z}\,\overline{R}$

$OUT = Z\,PS1\,PS0$

STT:

RESET	PS1	PS0	Z	NS1	NS0
OUT					
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	0	1
0	1	0	0	0	0
0	1	0	1	1	1
0	1	1	0	1	0
0	1	1	1	0	1
1	0	0	0	0	0
1	0	0	1	0	0
1	0	1	0	0	0
1	0	1	1	0	0
1	1	0	0	0	0

STD:



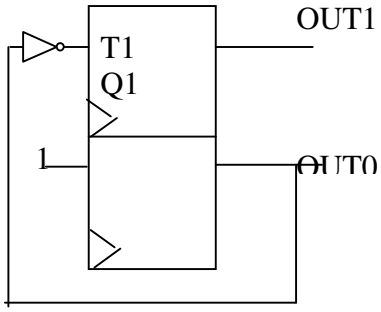
6. The Moore STD has 5 states. 1 is output in the 5th state.
On the STT, make sure that the output is the same for the same PS even though the input changes.

7. ------

8. STT:

PS1	PS0	Bookkeeping T1	Bookkeeping T0	NS1	NS0
0	0	1	1	1	1
0	1	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	0

Equations found for T1 and T0:



9. STT:

Equations:

PS1	PS0	Bookkeeping J1K1	Bookkeeping J0K0	NS1	NS0
0	0	0X	1X	0	1
0	1	1X	X1	1	0
1	0	X0	1X	1	1
1	1	X1	X1	0	0

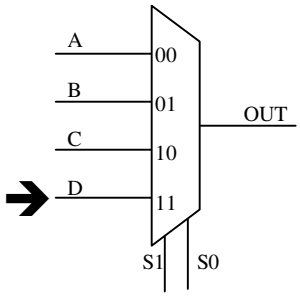
J_1 = PS0 $\overline{K_1}$ = PS0

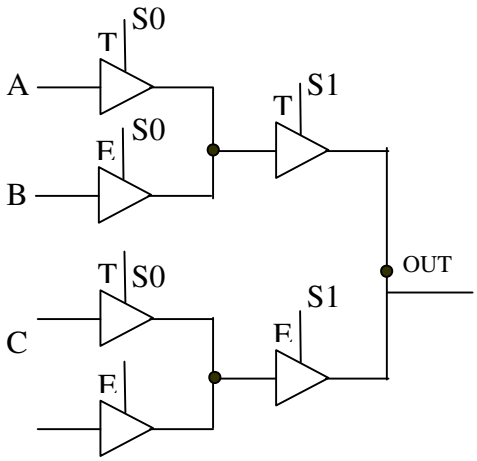
J_0 = PS0

K_0 = PS0

X means “don’t care”.

10.





NVIDIA

1)Divisible by 5 FSM..

2)FIFO Buffer..

Certain writing frequency, certain reading frequency and there were wait or idle states between any two

successive reads and we were asked to find the minimum size of the FIFO buffer required to prevent any

loss of data.

3)Questions from Chapter 10, Sedra and Smith where one is supposed to realize an arbitrary boolean

expression using NMOS and PMOS..and also vice versa i.e. Boolean expression from any given NMOS or

PMOS network or to derive NMOS network from given PMOS network or vice-versa..

4)2 ways to make an inverter out of a NAND gate..which one would be better in terms of power

dissipation and switching speed..

5)Setup and Hold time..

6)Delay Theory.. (2-3 questions)

7)Pipelining..

8)Number of bits required for the Cache Memory.. (You may refer to the book by Patterson).

9) o, t, t, f, f, s, s, e, n, ___

What should be in this blank?

10)Changing the clock frequency would affect: Hold Time or Setup time or both or none?

11)A program to find factorial of a number was implemented with and without using recursion. Using

which of the above two methods would you be able to find the factorial of a larger number and why?

Odd Number Divide By Counters With 50% Outputs and Synchronous Clocks

Prepared by: Cleon Petty and Paul Shockman
Product Applications
ON Semiconductor

ON Semiconductor
Formerly a Division of Motorola
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APPLICATION NOTE

The application inquiries handled by the Product Applications gives opportunities to solve customer needs with new ideas and learn of ways the customer has used our devices in new applications. A couple of these calls lead to techniques of designing odd number counters with synchronous clocks and 50% outputs.

The first technique requires a differential clock, that has a 50% duty cycle, a extra Flip Flop, and a gate to allow Odd integers, such as 3, 5, 7, 9, to have 50% duty cycle outputs and a synchronous clock. The frequency of operations is limited by Tpd of the driving FF, Setup, and Hold of the extra FF, and the times cannot exceed one half on the incoming clock cycle time.

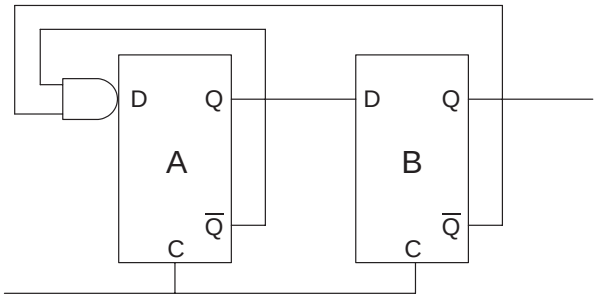
The design begins with producing a odd number counter (Divide By 3 for this discussion) by any means one wishes and add a flip flop, and a couple of gates to produce the desired function. Karnaugh maps usually produce counters that are lockup immune.

Example:

Specify, Divide By 3,
50% duty cycle on the output
Synchronous clocking
50% duty cycle clock in

Using D type Flop flips and karnaugh maps we find;
Ad = A*B* and Bd = A
(Note: * indicates BAR function)

Figure 1 shows schematic and timing of such a design.



Divide By 3

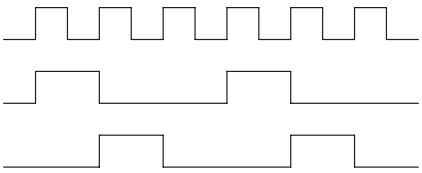


Figure 1.

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Using the technique, we add a gate on the clock to get differential Clock and Clock bar, a flip flop that triggers on the Clock Bar rising edge (Clock Neg.) to shift the output of ”B” by 90 degrees and a gate to AND/OR two FF output to produce the 50% output. We get Figure 2, a Divide By 3 that clocks synchronously with 50% output duty cycle.

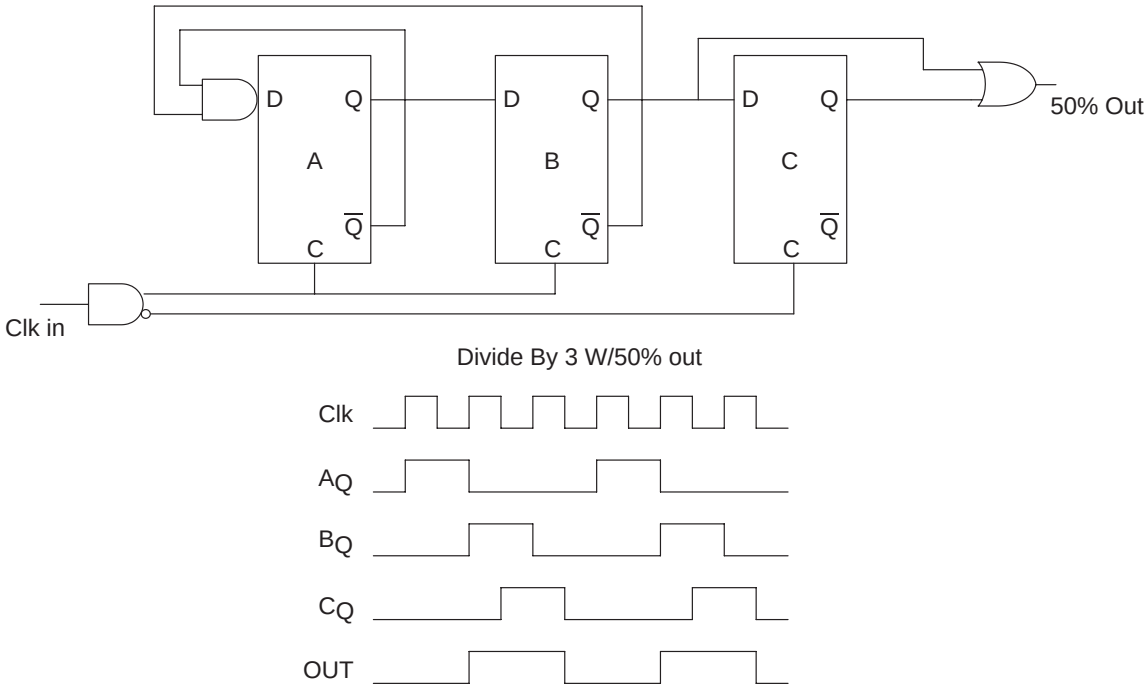


Figure 2.

The Max frequency of the configuration (figure 2) is calculated as Clock input freq./2 = Tpd of FF ”B” + Setup of ”C” + Hold of ”C”.

Example:

Tpd = 1Ns, Setup = !NS and Hold time = 0Ns.

with these numbers the Max Frequency the configuration can expect is; Cycle time = 2*(1 + 1)Ns or 4 Ns that converts to 250MHZ.

The Method is usable on other divide by ”N” counters as well by using the same methodology. The use of different types of Flip Flops (J,K, S,R, Toggle, ETC.) may produce fewer components. The type logic used may also dictate configuration. The configuration should always be checked for lockup conditions before the design is committed to a production.

Example:

A Divide By 3 design has all possible states shown in chart 1 but uses only the states shown in chart 2 leaving the states 2,3,4,5, & 7 for possible lockup.

Chart 1

	A	B	C
0	0	0	0
1	1	0	0
2	0	1	0
3	1	1	0
4	0	0	1
5	1	0	1
6	0	1	1
7	1	1	1

Chart 2

	A	B	C
0	0	0	0
1	1	0	0
6	0	1	1

We need to know that the counter will go into the flow, shown in chart 2, if it happens to come up in one of the unused states at powerup or for any other reason. Figure 3 shows the resulting flow chart of the analysis of the Divide By 3 counter of Figure 2. There is no state that the counter can begin in that doesn't lead to the desired flow after one clock cycle.

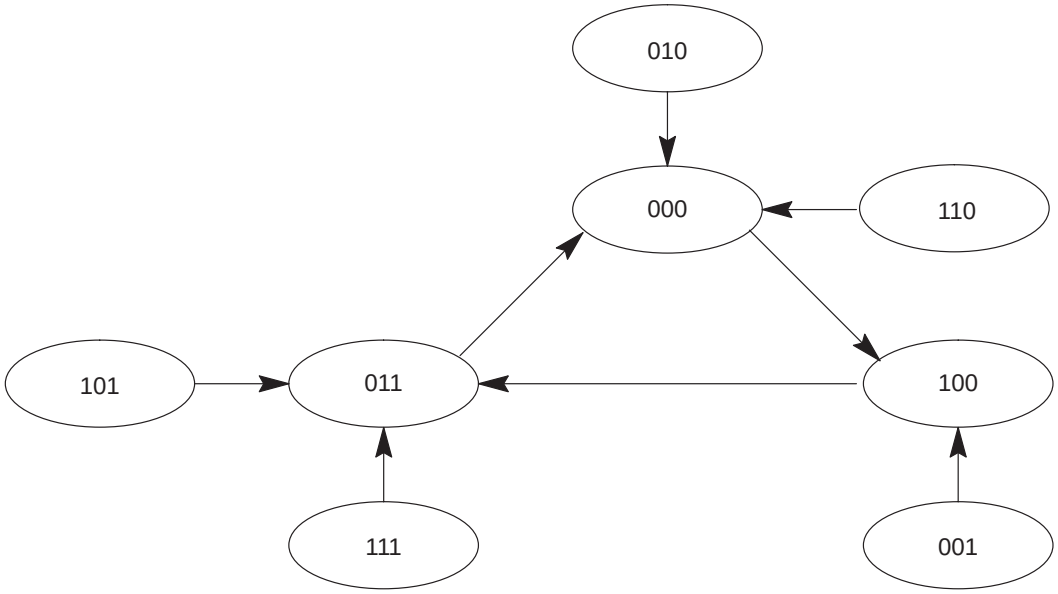


Figure 3.

Observation shows that FF "C" follows FF "B" by a half a clock cycle and will never be able to lockup making the analysis of the Divide By 3 sufficient to assure the whole configuration will have no lockup flow. So; only the 1 1 state of the divide by three needed to be confirmed.

The method is extendible to other odd larger divide by "N" numbers by following the same design flow.

a) Design a stable UP or Down divide by "N" counter

b) Make the Clock input a 50% duty cycle differential signal

c) Add a FF to follow one of the FF's in the counter by 1/2 clock cycle

d) OR/AND the shifted FF with the one that is driving it to obtain the desired 50% output

Example:

Design a 50% Divide By 9

Use "D" type FF's, other types may give smaller component count

Karnaugh maps yield:

Ad = A*B* Bd = A*B + AB*

Cd = ABC* + CB* + A*C Dd = ABC

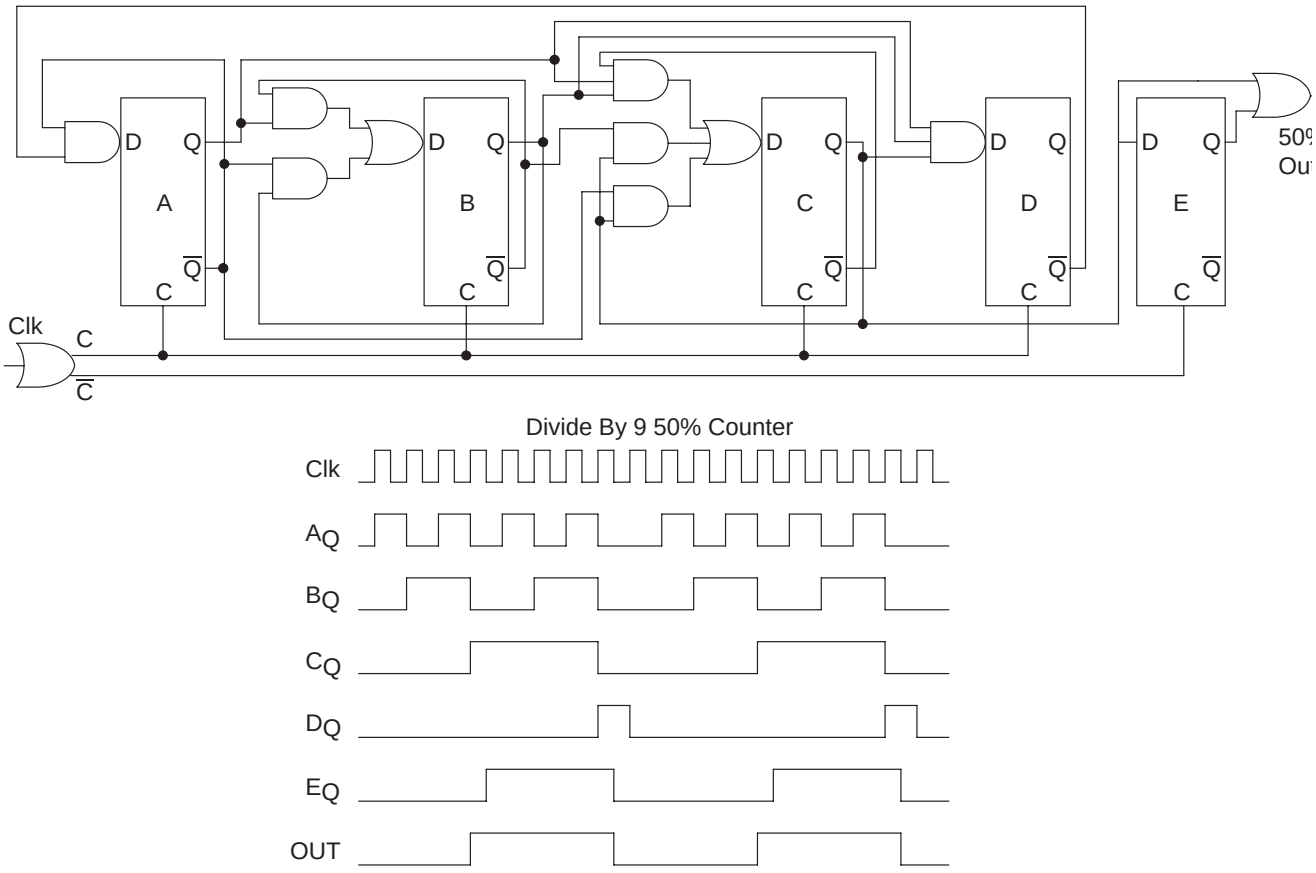


Figure 4.

Choosing to use ”C” as the flip flop to delay by a 1/2 clock cycle is necessary to accomplish the 50% output required when ”ANDed” with ”E”.

Another Synchronous 50% counter for Divide By 6, 10, 12, 14, 18, etc. can be realized by the additions of a J K FF and some gates. Other types of FF’s may be used.

Take the before mentioned Divide By 3 add a J K and a divide by 6, 50% duty cycle, synchronous counter is realized as shown in Figure 5.

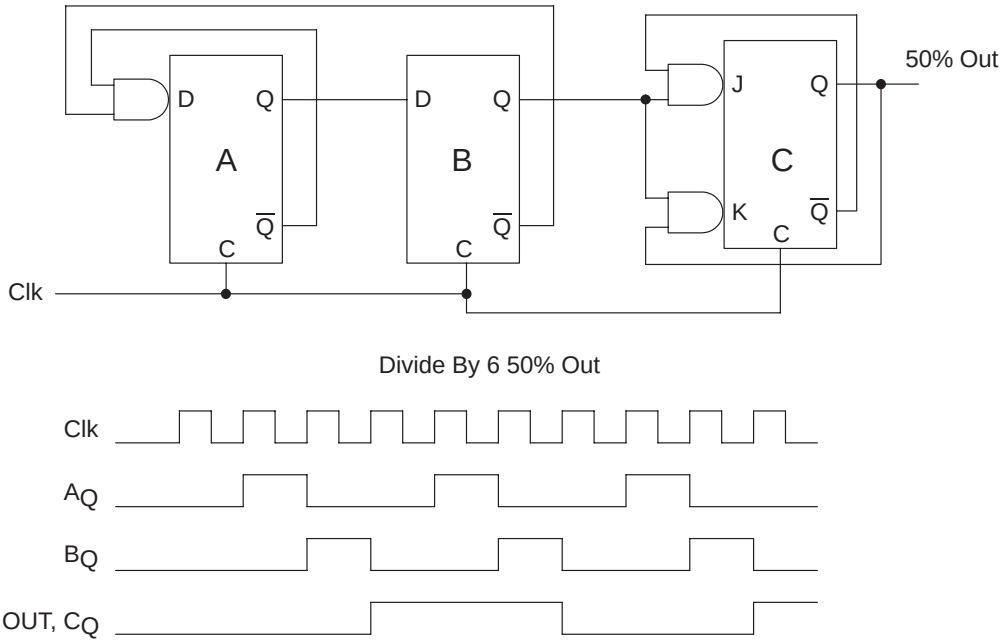


Figure 5.

Of course, there are better ways to realize a Divide By 6 but it does demonstrate how the method works. Note this configuration does not require a 50% input clock duty cycle and it is synchronous. This type of configuration could be useful in a clock generating PLL chip where a Divide By 3 and Divide By 6 are needed to synchronize two signals as shown in figure 6.

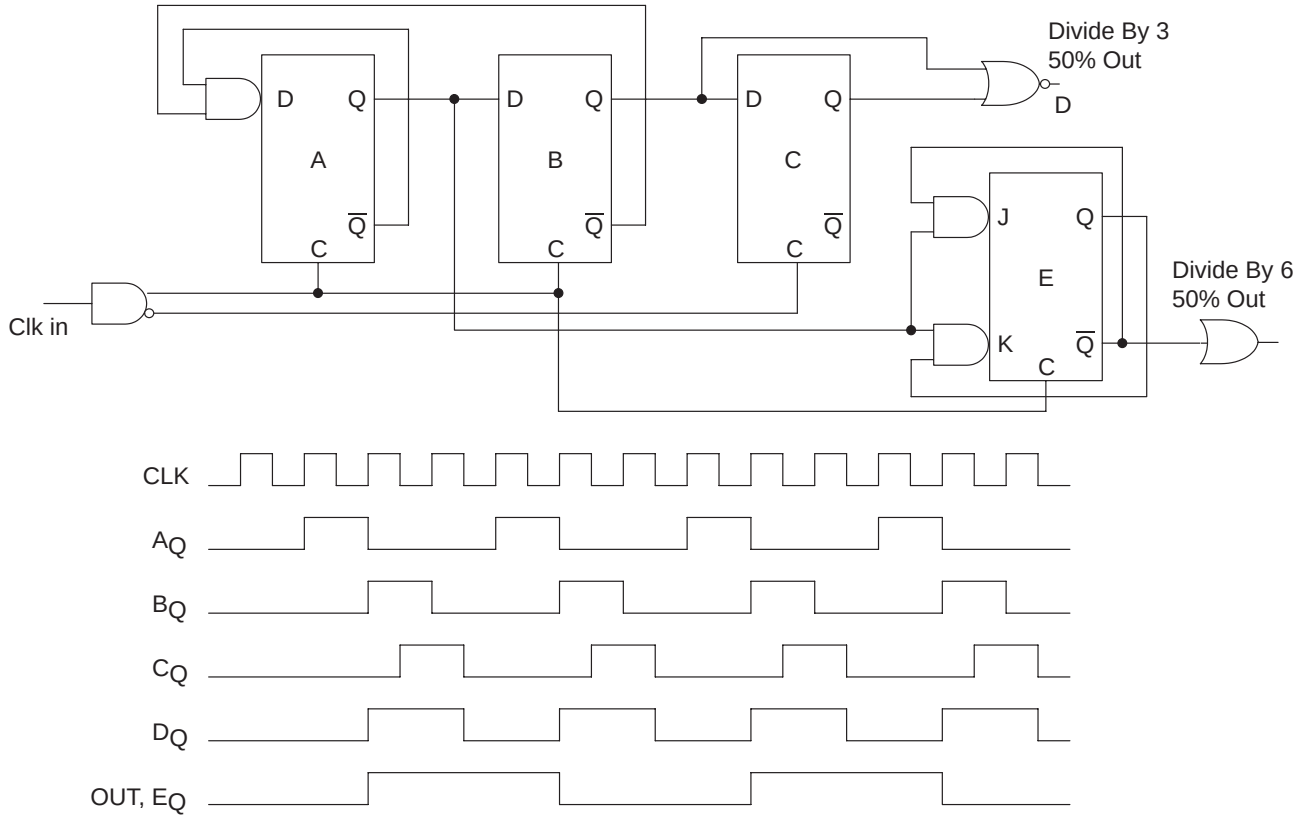


Figure 6.

Notice FF ”A” was chosen as the FF to drive FF ”E” in order to align the positive edges of the clock, Divide By 3, and divide by 6. The overall skew of the output could be better matched if all the same type of FF and gates are used.

We already know the Divide By 3 is lockup immune, following flow chart Figure 7 shows that the addition of the J K does not change that situation for the Divide By 6.

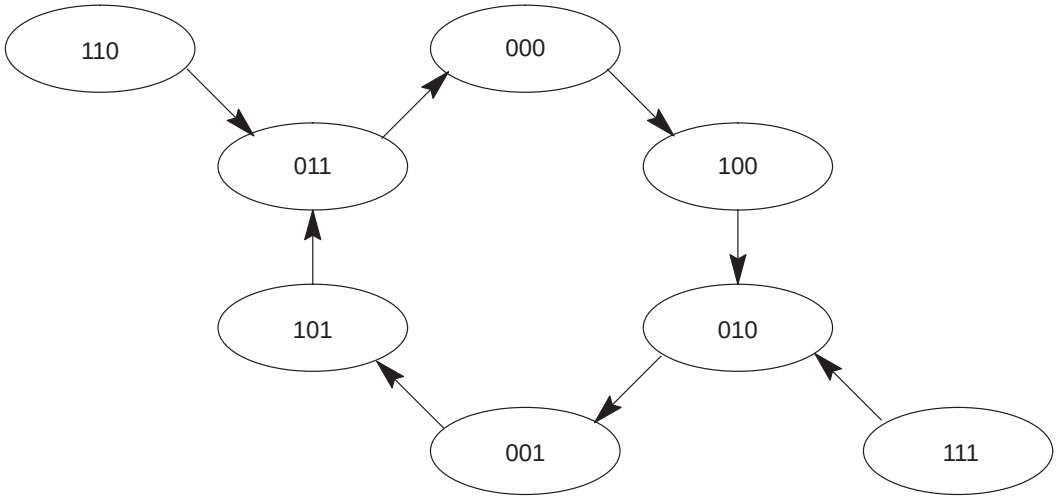


Figure 7. Divide by 6 Flow Chart

The flow shows no lockup, but if one observes that the J K is a sort of toggle device it is obvious that it can’t lock up the counter.

The J K may need bigger input AND gates to accomplish larger divide numbers. As an example, pick a Divide By 12 and use J K type FF’s to do the function.

Maps show:

Ja = 1 JB = AC* Jc = AB

Ka = 1 Kb = A Kc = A

Figure 8 shows the implementation.

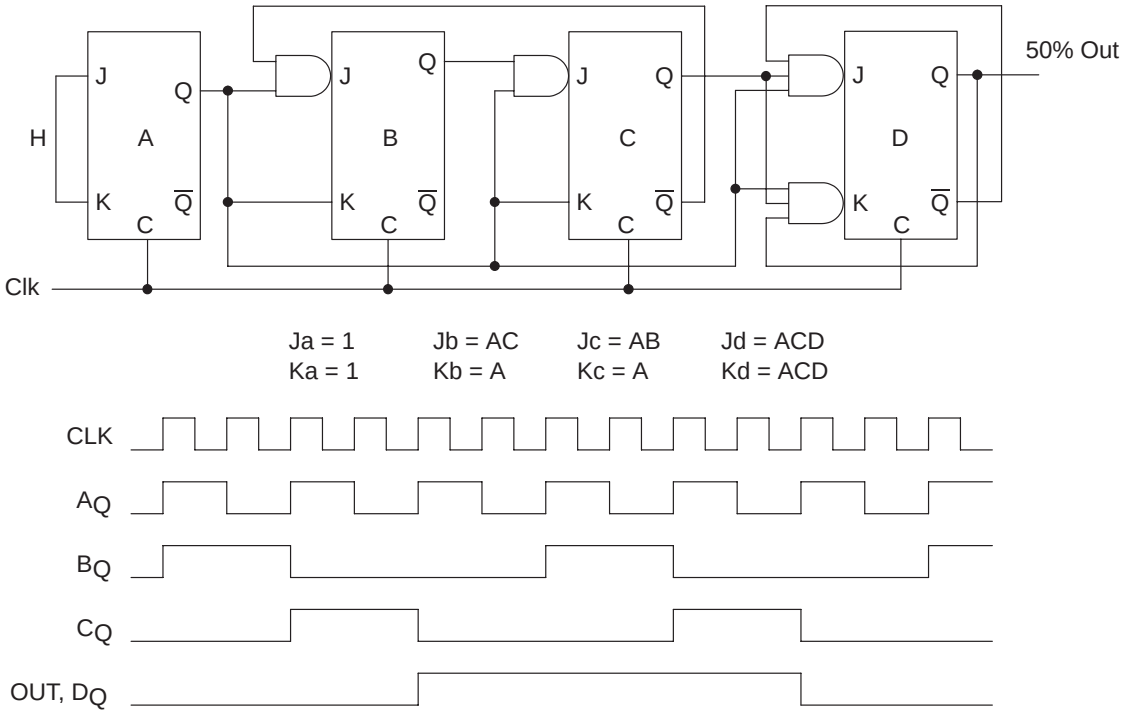


Figure 8. Synchronous Divide By 12

	A	B	C	D
0	0	0	0	0
1	1	0	0	0
2	0	1	0	0
3	1	1	0	0
4	0	0	1	0
5	1	0	1	0*
6	0	0	0	1
7	1	0	0	1
8	0	1	0	1
9	1	1	0	1
10	0	0	1	1
11	1	0	1	1*

The truth table shows that the FF ”D” must change state at 5 and 13

Examination of the truth table shows that the FF ”D” must decode a 5 and a 13 in order to make the desire 50% function.

The inputs to the ”D” FF are J = ACD* and K = ACD and requires 3 input AND gates. For larger counters the inputs on the AND gates will need to increase to reach the desired configuration; However for the single digit integers such as 3, 5, 7, & 9 to realize 6, 10, 14, & 18 a fan in of three is max.

The methods are expandable. A little observation, thinking, and logic typing will allow the designer to minimize the component count and skew on this type of counter.

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22. Companies Interview Questions

Intel

1. Why power stripes routed in the top metal layers?
2. Why do you use alternate routing approach HVH/VHV (Horizontal-Vertical-Horizontal/ Vertical-Horizontal-Vertical)?
3. What are several factors to improve propagation delay of standard cell?
4. How do you compute net delay (interconnect delay) / decode RC values present in tech file?
5. What are various ways of timing optimization in synthesis tools?
6. What would you do in order not to use certain cells from the library?
7. How delays are characterized using WLM (Wire Load Model)?
8. What are various techniques to resolve congestion/noise?
9. Let's say there enough routing resources available, timing is fine, can you increase clock buffers in clock network? If so will there be any impact on other parameters?
10. How do you optimize skew/insertion delays in CTS (Clock Tree Synthesis)?
11. What are pros/cons of latch/FF (Flip Flop)?

12. How you go about fixing timing violations for latch- latch paths?

13. As an engineer, let's say your manager comes to you and asks for next project die size estimation/projection, giving data on RTL size, performance requirements. How do you go about the figuring out and come up with die size considering physical aspects?

14. How will you design inserting voltage island scheme between macro pins crossing core and are at different power wells? What is the optimal resource solution?

15. What are various formal verification issues you faced and how did you resolve?

16. How do you calculate maximum frequency given setup, hold, clock and clock skew?

17. What are effects of meta-stability?

ST Microelectronics

1 What are the challenges you faced in place and route, FV (Formal Verification), ECO (Engineering Change Order) areas?
2 How long the design cycle for your designs?
3 What part are your areas of interest in physical design?
4 Explain ECO (Engineering Change Order) methodology.
5 Explain CTS (Clock Tree Synthesis) flow.

Answer: Clock Tree Synthesis

6 What kind of routing issues you faced?

7 **How does STA (Static Timing Analysis) in OCV (On Chip Variation) conditions done? How do you set OCV (On Chip Variation) in IC compiler? How is timing correlation done before and after place and route?**
8 If there are too many pins of the logic cells in one place within core, what kind of issues would you face and how will you resolve?
9 Define hash/ @array in perl.
10 Using TCL (Tool Command Language, Tickle) how do you set variables?
11 What is ICC (IC Compiler) command for setting derate factor/ command to perform physical synthesis?
12 What are nanoroute options for search and repair?
13 What were your design skew/insertion delay targets?
14 How is IR drop analysis done? What are various statistics available in reports?
15 Explain pin density/ cell density issues, hotspots?
16 How will you relate routing grid with manufacturing grid and judge if the routing grid is set correctly?
17 What is the command for setting multi cycle path?
18 If hold violation exists in design, is it OK to sign off design? If not, why?

Texas Instruments (TI)

1 How are timing constraints developed?
2 Explain timing closure flow/methodology/issues/fixes.
3 Explain SDF (Standard Delay Format) back annotation/ SPEF (Standard Parasitic Exchange Format) timing correlation flow.
4 Given a timing path in multi-mode multi-corner, how is STA (Static Timing Analysis) performed in order to meet timing in both modes and corners, how are PVT (Process-Voltage-Temperature)/derate factors decided and set in the Primetime flow?
5 With respect to clock gate, what are various issues you faced at various stages in the physical design flow?
6 What are synthesis strategies to optimize timing?
7 Explain ECO (Engineering Change Order) implementation flow. Given post routed database and functional fixes, how will you take it to implement ECO (Engineering Change Order) and what physical and functional checks you need to perform?

Qualcomm

3 In building the timing constraints, do you need to constrain all IO (Input-Output) ports?
4 Can a single port have multi-clocked? How do you set delays for such ports?
5 How is scan DEF (Design Exchange Format) generated?
6 What is purpose of lockup latch in scan chain?
7 How do you set inter clock uncertainty?
8 In DC (Design Compiler), how do you constrain clocks, IO (Input-Output) ports, maxcap, max tran?
9 What are differences in clock constraints from pre CTS (Clock Tree Synthesis) to post CTS (Clock Tree Synthesis)?
10 How is clock gating done?

Answer: Clock Gating

11 what constraints you add in CTS (Clock Tree Synthesis) for clock gates?
12 What is tradeoff between dynamic power (current) and leakage power (current)?

Answer:
Leakage Power Trends
Dynamic Power

13 How do you reduce standby (leakage) power?

Answer: Low Power Design Techniques

14 Explain top level pin placement flow? What are parameters to decide?
15 Given block level netlists, timing constraints, libraries, macro LEFs (Layout Exchange Format/Library Exchange Format), how will you start floor planning?
16 With net length of 1000um how will you compute RC values, using equations/tech file info?
17 What do noise reports represent?
18 What does glitch reports contain?
19 What are CTS (Clock Tree Synthesis) steps in IC compiler?
20 What do clock constraints file contain?
21 How to analyze clock tree reports?
22 What do IR drop Voltagestorm reports represent?
23 Where /when do you use DeCAP (Decoupling Capacitor) cells?
24 What are various power reduction techniques?

Answer: Low Power Design Techniques

Hughes Networks

3 What is setup/hold? What are setup and hold time impacts on timing? How will you fix setup and hold violations?
4 Explain function of Muxed FF (Multiplexed Flip Flop) /scan FF (Scal Flip Flop).
5 What are tested in DFT (Design for Testability)?
6 In equivalence checking, how do you handle scanen signal?
7 In terms of CMOS (Complimentary Metal Oxide Semiconductor), explain physical parameters that affect the propagation delay?
8 Why is power signal routed in top metal layers?

Top Metal Layers has Less Resistance and High Current Density.

Avago Technologies (former HP group)

1 How do you minimize clock skew/ balance clock tree?
2 Given 11 minterms and asked to derive the logic function.
3 Given C1= 10pf, C2=1pf connected in series with a switch in between, at t=0 switch is open and one end having 5v and other end zero voltage; compute the voltage across C2 when the switch is closed?
4 Explain the modes of operation of CMOS (Complimentary Metal Oxide Semiconductor) inverter? Show IO (Input-Output) characteristics curve.
5 Implement a ring oscillator.

6 How to slow down ring oscillator?

Hynix Semiconductor

1 How do you optimize power at various stages in the physical design flow?
2 What timing optimization strategies you employ in pre-layout /post-layout stages?
3 What are process technology challenges in physical design?
4 Design divide by 2, divide by 3, and divide by 1.5 counters. Draw timing diagrams.
5 What are multi-cycle paths, false paths? How to resolve multi-cycle and false paths?
6 Given a flop to flop path with combo delay in between and output of the second flop fed back to combo logic. Which path is fastest path to have hold violation and how will you resolve?
7 What are RTL (Register Transfer Level) coding styles to adapt to yield optimal backend design?
8 Draw timing diagrams to represent the propagation delay, set up, hold, recovery, removal, minimum pulse width.

Review Questions/Answers

Low Power Circuits and Systems



Ajit Pal

Professor

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Review Questions of Lec-1

Q1. Why low power has become an important issue in the present day VLSI circuit realization?

Q2. How reliability of a VLSI circuit is related to its power dissipation?

Q3. How environment is affected by the power dissipation of VLSI circuits?

Q4. Why leakage power dissipation has become an important issue in deep submicron technology?

Q5. Distinguish between energy and power dissipation of VLSI circuits. Which one is more important for portable systems?

Answer to Questions of Lec-1

Q1. Why low power has become an important issue in the present day VLSI circuit realization?

- **Ans: In deep submicron technology the power has become as one of the most important issue because of:**
  - **Increasing transistor count; the number of transistors is getting doubled in every 18 months based on Moore,s Law**
  - **Higher speed of operation; the power dissipation is proportional to the clock frequency**
  - **Greater device leakage currents; In nanometer technology the leakage component becomes a significant percentage of the total power and the leakage current increases at a faster rate than dynamic power in technology generations**

Answer to Questions of Lec-1

Q2. How reliability of a VLSI circuit is related to its power dissipation?

Ans: It has been observed that every 10ºC rise in temperature roughly doubles the failure rate because various failure mechanism such as silicon interconnect fatigue, electromigration diffusion, junction diffusion and thermal runaway starts occurring as temperature increases.

Q3. How environment is affected by the power dissipation of VLSI circuits?

- **Ans: According to an estimate of the U.S. Environmental Protection Agency (EPA), 80% of the power consumption by office equipment are due to computing equipment and a large part from unused equipment. Moreover, the power is dissipated mostly in the form of heat. The cooling techniques, such as AC transfers the heat to the environment.**

Answer to Questions of Lec-1

Q4. Why leakage power dissipation has become an important issue in deep submicron technology?

Ans: In deep submicron technology the leakage component becomes a significant percentage of the total power and the leakage current increases at a faster rate than dynamic power in new technology generations. That is why the leakage power has become an important issue.

Q5. Distinguish between energy and power dissipation of VLSI circuits. Which one is more important for portable systems?

Ans: Power (P) is the power dissipation in Watts at different instances of time. On the other has energy (E) refers to the energy consumed in Joule over a period of time (E = P*t).

Review Questions of Lec-2

Q1. What are the commonly used conducting layers used in IC fabrication?

Q2. Show the basic structure of a MOS transistor.

Q3. What is the latch up problem that arises in bulk CMOS technology? How is it overcome?

Q3. Distinguish between the bulk CMOS technology with the SoI technology fabrications.

Q4. What are the benefits of SOI technology relative to conventional bulk CMOS technology?

Answer to Questions of Lec-2

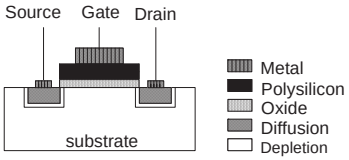
Q1. What are the commonly used conducting layers used in IC fabrication?

Ans: Fabrication involves fabrication of patterned layers of the three conducting materials: *metal*, *poly-silicon* and *diffusion by using* a series of photolithographic techniques and chemical processes involving oxidation of silicon, diffusion of impurities into the silicon and deposition and etching of aluminum or polysilicon on the silicon to provide interconnection.

Answer to Questions of Lec-2

Q2. Show the basic structure of a MOS transistor.

➢Ans: The basic structure of a MOS transistor is given below. On a lightly doped substrate of silicon two islands of diffusion regions called as *source* and *drain*, of opposite polarity of that of the substrate, are created. Between these two regions, a thin insulating layer of silicon dioxide is formed and on top of this a conducting material made of poly-silicon or metal called *gate* is deposited.



Answer to Questions of Lec-2

Q3. What is the latch up problem that arises in bulk CMOS technology?

Ans: The latch-up is an inherent problem in both n-well as well as p-well based CMOS circuits. The phenomenon is caused by the parasitic bipolar transistors formed in the bulk of silicon as shown in the figure for the n-well process. *Latch-up* can be defined as the formation of a low-impedance path between the power supply and ground rails through the parasitic npn and pnp bipolar transistors. As shown the BJTs are cross-coupled to form the structure of a silicon-controlled-rectifier (SCR) providing a short-circuit path between the power rail and ground. Leakage current through the parasitic resistors can cause one transistor to turn on, which in turn turns on the other transistor due to positive feedback and leading to heavy current flow and consequent device failure.

Answer to Questions of Lec-2

Q4. How the latch up problem can be overcome?

Ans: There are several approaches to reduce the tendency of Latch-up. Some of the important techniques are mentioned below:

- **Use guard ring around p- and/or n-well with frequent contacts to the rings**
- **To reduce the gain product B1XB2**
- **Moving the n-well and the n+ source/drain further apart**
- **Buried n+ layer in well to reduce gain of Q1**
- **Higher substrate doping level to reduce R-sub**
- **Reduce R-well by making low resistance contact to GND**

Answer to Questions of Lec-2

Q5. Distinguish between the bulk CMOS technology with the Sol technology fabrications.

Ans: In bulk CMOS technology, a lightly doped p-type or n-type substrate is used to fabricate MOS transistors. On the other hand, an insulator can be used as a substrate to fabricate MOS transistors

Answer to Questions of Lec-2

Q6. What are the benefits of SOI technology relative to conventional bulk CMOS technology?

- **Ans: Benefits of SOI technology relative to conventional silicon (bulk CMOS):**
  - **Lowers parasitic capacitance due to isolation from the bulk silicon, which improves power consumption and thus high speed performance.**
  - **Reduced short channel effects**
  - **Better sub-threshold slope.**
  - **No Latch up due to BOX (buried oxide).**
  - **Lower Threshold voltage.**
  - **Reduction in junction depth leads to low leakage current.**
  - **Higher Device density.**

Review Questions of Lec-3



Q1. What are the basic assumptions of the fluid model?

Q2. Explain the function of a MOS transistor in the saturation mode using fluid model.

Q3. Explain the function of a MOS transistor in the nonsaturation mode using the fluid model.

Q4. Explain the three modes of operation of a MOS transistors.

Q5. Explain the linear region of the I-V characteristic of an nMOS transistor using the fluid model.

Answer to the Questions of Lec-3



Q1. What are the basic assumptions of the fluid model?

Ans: There are two basic assumptions as follows:

(a) Electrical charge is considered as fluid, which can move from one place to another depending on the difference in their level, of one from the other, just like a fluid.

(b) Electrical potentials can be mapped into the geometry of a container, in which the fluid can move around.

Q2. Explain the function of a MOS transistor in the nonsaturation mode using the fluid model.

Ans: Gate voltage higher than the threshold voltage and the drain voltage is slightly higher than source voltage. In such a situation, as the drain voltage is increased the slope of the fluid flowing out increases indicating linear increase in the flow of current.

Answer to the Questions of Lec-3



Q3. Explain the function of a MOS transistor in the saturation mode using fluid model.

Ans: The saturation mode corresponds to the drain voltage is much higher than source voltage. In such a situation the slope of the fluid cannot increase representing constant flow of fluid.

Q4. Explain the three modes of operation of a MOS transistors.

Ans: The three modes are:

(a) Accumulation mode when Vgs is much less than Vt

(b) Depletion mode when Vgs is equal to Vt

(c) Inversion mode when Vgs is greater than Vt

Answer to the Questions of Lec-3



Q5. What are the three regions of operation of a MOS transistor?

Ans: The three regions are:

***Cut-off region*: This is essentially the accumulation mode, where there is no effective flow of current between the source and drain.**

***Non-saturated region:* This is the *active, linear* or *week inversion* region, where the drain current is dependent on both the gate and drain voltages.**

***Saturated region:* This is the *strong inversion* region, where the drain current is independent of the drain-to-source voltage but depends on the gate voltage.**

Review Questions of Lec-4



Q1. What is the threshold voltage of a MOS transistor? How it varies with the body bias?

Q2. The following parameters are given for an nMOS process: t_{ox} 500Å, $N_A = 1x10^{16}cm^{-3}$, $N_D = 1x10^{20}cm^{-3}$, $N_{OX} = 2x10^{10}cm^{-1}$. (i) Calculate Vt for an unimplanted transistor, (ii) what type and what concentration must be implanted to achieve Vt = +1.5V and Vt = -2.0V?

Q3. What is channel length modulation effect? How the voltage current characteristics are affected because of this effect?

Q4. What is body effect? How does it influences the threshold voltage of a MOS transistor?

Q5. What is transconductance of a MOS transistor? Explain its role in the operation of the transistor.

Answer to the Questions of Lec-4



Q1. What is the threshold voltage of a MOS transistor? How it varies with the body bias?

Ans: One of the parameters that characterizes the switching behavior of a MOS transistor is its threshold voltage V_t. This can be defined as the gate voltage at which a MOS transistor begins to conduct.

Q2. The following parameters are given for an nMOS process: t_{ox} 500Å, $N_A = 1x10^{16}cm^{-3}$, $N_D = 1x10^{20}cm^{-3}$, $N_{OX} = 2x10^{10}cm^{-1}$. (i) Calculate Vt for an unimplanted transistor, (ii) what type and what concentration must be implanted to achieve Vt = +1.5V and Vt = -2.0V?

Answer to the Questions of Lec-4

Q3. What is channel length modulation effect? How the voltage current characteristics are affected because of this effect?

Ans: It is assumed that channel length remains constant as the drain voltage is increased appreciably beyond the on set of saturation. As a consequence, the drain current remains constant in the saturation region. In practice, however the channel length shortens as the drain voltage is increased. For long channel lengths, say more than 5 μm, this variation of length is relatively very small compared to the total length and is of little consequence. However, as the device sizes are scaled down, the variation of length becomes more and more predominant and should be taken into consideration. As a consequence, the drain current increases with the increase in drain voltage even in the saturation region.

Answer to the Questions of Lec-4

Q4. What is body effect? How does it influences the threshold voltage of a MOS transistor?

Ans: All MOS transistors are usually fabricated on a common substrate and substrate (body) voltage of all devices is normally constant. However, as we shall see in subsequent chapters, when circuits are realized using a number of MOS devices, several devices are connected in series. This results in different source potentials for different devices. It may be noted that the threshold voltage V_t is not constant with respect to the voltage difference between the substrate and the source of the MOS transistor. This is known as the substrate-bias effect or *body effect*. Increasing the V_{sb} causes the channel to be depleted of charge carries and this leads to increase in the threshold voltage.

Answer to the Questions of Lec-4

Q5. What is transconductance of a MOS transistor? Explain its role in the operation of the transistor.

Ans: Trans-conductance is represented by the change in drain current for change in gate voltage for constant value of drain voltage. This parameter is somewhat similar to β, the current gain of bipolar junction transistors. The following equation shows the dependence of on various parameters. As MOS transistors are voltage controlled devices, this parameter plays an important role in identifying the efficiency of the MOS transistor.

$$g_m = \frac{\mu_n \varepsilon_{ox} \varepsilon_0}{D} \frac{W}{L} (V_{gs} - V_t)$$

Review Questions of Lec-5

Q1. Explain the behaviour of a nMOS transistor as a switch.

Q2. Explain the behaviour of a pMOS transistor as a switch.

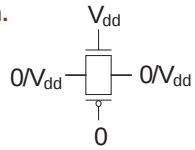
Q3. How one nMOS and one pMOS transistor are combined to behave like an ideal switch.

Q4. The input of a lightly loaded transmission gate is slowly changes from HIGH level to LOW level. How the currents through the two transistors vary?

Q5. How its ON-resistance of a transmission gate changes as the input varies from 0 V to Vdd, when the output has a light capacitive load.

Answer to the Questions of Lec-5

Q3. How one nMOS and one pMOS transistor are combined to behave like an ideal switch.



Ans: To overcome the limitation of either of the transistors, one pMOS and one nMOS transistor can be connected in parallel with complementary inputs at their gates. In this case we can get both LOW and HIGH levels of good quality at the output. The low level passes through the nMOS switch and HIGH level passes through the pMOS switch without any degradation as shown in the figure.

Answer to the Questions of Lec-5

Q4. The input of a lightly loaded transmission gate is slowly changes from HIGH level to LOW level. How the currents through the two transistors vary?

Ans: Another situation is the operation of the transmission gate when the output is lightly loaded (smaller load capacitance). In this case, the output closely follows the input. In this case the transistors operate in three regions depending on the input voltage as follows:

Region I: nMOS non-saturated, pMOS cut-OFF

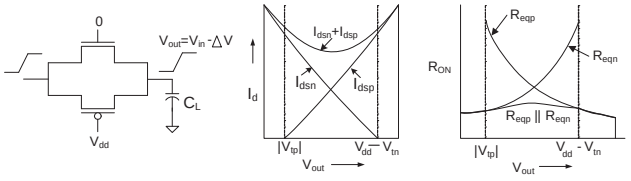
Region II: nMOS non-saturated, pMOS non-saturated

Region III: nMOS cut off, pMOS non-saturated

Answer to the Questions of Lec-5

Q5. How its ON-resistance of a transmission gate changes as the input varies from 0 V to Vdd, when the output has a light capacitive load.

Ans: The variation of ON resistance is shown in the figure. The parallel resistance remains more or less constant.



Review Questions of Lec-6

Q1. Draw the ideal characteristics of a CMOS inverter and compare it with the actual characteristics.

Q2. What is noise margin? Find out the noise margin from the actual characteristics of the inverter.

Q3. Compare the characteristics of the different types of MOS inverters in terms of noise margin and power dissipation.

Answer to the Questions of Lec-6

Q1. Draw the ideal characteristics of a CMOS inverter and compare it with the actual characteristics.

Ans: The ideal and actual characteristics are given below. In the ideal characteristics, the output voltage is Vdd for input voltage from o to Vdd/2 and 0 for input voltage from Vdd/2 to Vdd. This is not true in case of the actual characteristics as shown below.



Answer to the Questions of Lec-6

Q2. What is noise margin? Find out the noise margin from the actual characteristics of the inverter.

Ans: An important parameter called *noise margin* is associated with the input-output voltage characteristics of a gate. It is defined as the allowable noise voltage on the input of a gate so that the output is not affected. The deviations in logic levels from the ideal values, which are restored as the signal propagates to the output, can be obtained from the DC characteristic curves. The logic levels at the input and output are given by

The noise margins are:

$$NM_L = |V_{IL} - V_{OL}|$$

$$NM_H = |V_{OH} - V_{IH}|$$

logic 0 input : $0 \le V_{in} \le V_{IL}$

logic 1 input : $V_{IH} \le V_{in} \le V_{dd}$

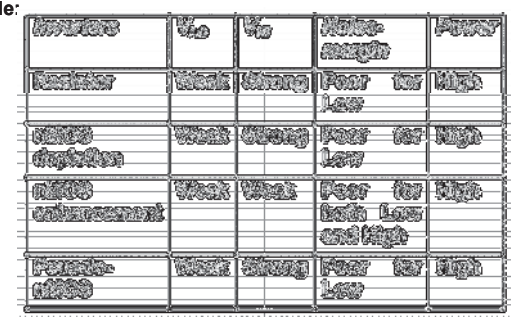
logic 0 output : $0 \le V_o \le V_{OL}$

logic 1 output : $V_{OH} \le V_o \le V_{dd}$

Answer to the Questions of Lec-6

Q3. Compare the characteristics of the different types of MOS inverters in terms of noise margin and power dissipation.

Ans: Various characteristic parameters are compared in the following table:



Review Questions of Lec-7

Q1. What is the inversion voltage of an inverter? Find out the inversion voltage of a CMOS inverter.

Q2. How the inversion voltage is affected by the relative sizes of the nMOS and pMOS transistors of the CMOS inverter?

Q3. Find out the noise margin of a CMOS inverter.

Q4. How the noise margin is affected by voltage scaling?

Q5. What is the lower limit of supply voltage of a CMOS inverter. What happens if the supply voltage is further reduced?

Answer to Questions of Lec-7

Q1. What is the inversion voltage of an inverter? Find out the inversion voltage of a CMOS inverter.

Ans: The inversion voltage Vinv is defined as the voltage at which the output voltage Vo is equal to the input voltage Vin. For a CMOS inverter it can be expressed in terms of the threshold voltages of the MOS transistors and other parameters.

$$V_{inv} = \frac{V_{dd} + V_{tp} + V_{tn}\sqrt{\frac{\beta_n}{\beta_p}}}{1+\sqrt{\frac{\beta_n}{\beta_p}}}$$

For $\beta_n = \beta_p$ and $V_{tn} = -V_{tp}$, we get $V_{inv} = V_{dd}/2$

Answer to Questions of Lec-7

Q2. How the inversion voltage is affected by the relative sizes of the nMOS and pMOS transistors of the CMOS inverter?

Ans: In a CMOS process $\frac{K_n}{K_p} = \frac{\mu_n}{\mu_p} \approx 2.5$

To make one may choose $\left(\frac{W}{L}\right)_p = 2.5\left(\frac{W}{L}\right)_n$,

to get Vinv = Vdd/2

Answer to Questions of Lec-7

Q3. Find out the noise margin of a CMOS inverter.

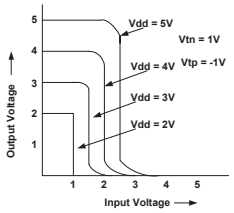
Ans: For a symmetric inverter $V_{IH} + V_{IL} = V_{dd}$

Noise Margin $NM_L = V_{IL} - V_{OL} = V_{IL}$

and $NM_H = V_{OH} - V_{IH} = V_{dd} - V_{IH} = V_{IL}$

Q4. How the noise margin is affected by voltage scaling?

Ans: As the supply voltage is reduced, the margin also decreases as shown in the figure.

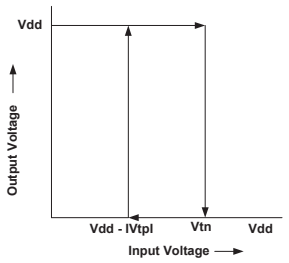


Answer to Questions of Lec-7

Q5. What is the lower limit of supply voltage of a CMOS inverter. What happens if the supply voltage is further reduced?

Ans: The lower limit of the supply voltage depends on the sum of the threshold voltages of the nMOS and the pMOS transistors.

Vdd = Vtn +|Vtp|. As the supply voltage is reduced further, it leads to hysteresis in the transfer characteristics.



Review Questions of Lec-8

Q1. What is sheet resistance? Find out the expression of the resistance of rectangular sheet in terms of sheet resistance.

Q2. Find out the capacitance of a MOS capacitor.

Q3. Find out the expression of delay time of a CMOS inverter.

Q4. What are the various ways to reduce the delay time of a CMOS inverter?

Q5. Explain the commonly used technique to estimate the delay time of a CMOS inverter.

Answer to the Questions of Lec-8

- **Q1. What is sheet resistance? Find out the expression of the resistance of rectangular sheet in terms of sheet resistance.**

Ans: The sheet resistance is defined as the resistance per unit area of a sheet of material. Consider a rectangular sheet of material with Resistivity = ρ, Width = W, Thickness = t and Length = L.

Then, the resistance between the two ends is

$$R_{AB} = \frac{\rho L}{t.W} = \frac{\rho L}{A} ohm$$ **For L = W** $$R_S = \frac{\rho}{t} = R\, ohm$$

Where, R_S is defined as the sheet resistance

Answer to the Questions of Lec-8

Q2. Find out the capacitance of a MOS capacitor.

Ans: The capacitance of a parallel plate capacitor is given by

$$Co = \frac{\varepsilon_0 \varepsilon_{ins} A}{D} \text{ Farads}$$

Where A is the area of the plates and D is the thickness of the insulator between the plates.

Answer to the Questions of Lec-8

Q3. Find out the expression of delay time of a CMOS inverter.

Ans: The delay time td is given by the expression

$$t_d = \left[\frac{L_n}{K_n W_n} + \frac{L_p}{K_p W_p}\right] \frac{C_L}{V_{dd}\left(1 - \frac{V_t}{V_{dd}}\right)^2}$$

Where C is the load capacitance, Vdd is the supply voltage and Vt is the threshold voltages of the MOS transistors.

Answer to the Questions of Lec-8

Q4. What are the various ways to reduce the delay time of a CMOS inverter?

Ans: Various ways for reducing the delay time are given below:

(a) The width of the MOS transistors can be increased to reduce the delay. This is known as gate sizing, which will be discussed later in more details.

(b) The load capacitance can be reduced to reduce delay. This is achieved by using transistors of smaller and smaller dimensions as provided by future generation technologies.

(c) Delay can also be reduced by increasing the supply voltage Vdd and/or reducing the threshold voltage Vt of the MOS transistors.

Answer to the Questions of Lec-8

Q5. Explain the commonly used technique to estimate the delay time of a CMOS inverter.

Ans: As the delay time of an inverter is very small, it cannot be measured accurately using conventional method with the help of an oscilloscope. Delay is usually measured by realizing a ring oscillator using a large number of inverters, say 101. Then the frequency of oscillation is measured using the expression $f = \frac{1}{2nt_d}$ **where n is the number of inverters and td is the delay time.**

Review Questions of Lec-9

Q1. Justify the reason for not recommending more than 4 pass transistors to use in series in realizing logic circuits.
Q2. Draw the schematic diagram of an inverting super-buffer and explain its operation.
Q3. Give the schematic diagram of a Bi-CMOS inverter. Explain its operation. . Compare the switching characteristics of a BiCMOS inverter with respect to that for static CMOS for different fan out conditions.
Q4. Prove that the delay of a series of pass transistors can be reduced from quardratic dependence to linear dependence on the number of transistors in series by inserting buffers at suitable intervals.
Q5. Design a scaled chain of inverters such that the delay time between the logic gate (C_g = 100 fF) and a load capacitance 2 pF in minimized. Find out the number of stages and stage ratio.

Answer to Questions of Lec-9

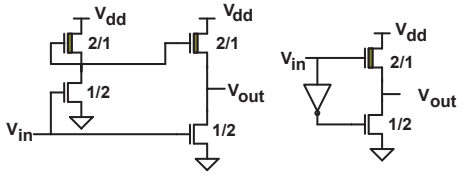
Q1. Explain the basic concept of super buffer.

Ans: There are situations when a large load capacitance such as, long buffers, off-chip capacitive load, or I/O buffer are to be driven by a gate. In such cases, the delay can be very high if driven by a standard gate. Limitations of driving by a simple nMOS inverter is the asymmetric drive capability of pull-up and pull-down devices (ratioed logic). Moreover, when the pull-down transistor is ON, the pull-up transistor also remains ON. So, the pull-down transistor should also sink the current of the pull-up device. Although this limitation is overcome in CMOS circuits, there is asymmetry in drive capability of pull-up and pull-down devices having the same minimum size.

Answer to Questions of Lec-9

Q2. Draw the schematic diagram of an inverting and non-inverting super-buffers and explain its operation.

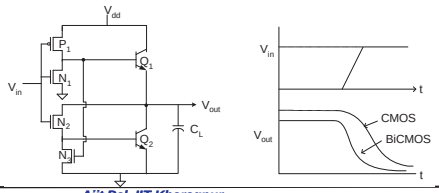
Ans: The schematic diagrams of the super buffers are given below. The average of the saturation currents for Vds = 5V and linear current for Vds = 2.5V is approximately 4.4 βpu for the standard inverter. On the other hand the, for the super-buffer, the average current is 19.06 βpu, which is 4 times that of standard inverter. Thus, the pull-up device is capable of sourcing about four times the current of the standard nMOS inverter.



Answer to Questions of Lec-9

Q3. Give the schematic diagram of a Bi-CMOS inverter. Explain its operation. . Compare the switching characteristics of a BiCMOS inverter with respect to that for static CMOS for different fan out conditions.

Ans: The schematic diagram of a BiCMOS inverter is given below. Higher current drive capability of bipolar NPN transistors is used in realizing bi-CMOS inverters. The delays of CMOS and BiCMOS inverters are compared for different fan-outs. It may be noted that for fan-out of 1 or 2, CMOS provides smaller delay compared to BiCMOS. However, as fan-out increases further, BiCMOS performs better and better.



Answer to Questions of Lec-9

Q4. Design a scaled chain of inverters such that the delay time between the logic gate (C_g = 100 fF) and a load capacitance 2 pF in minimized. Find out the number of stages and stage ratio.

Ans: It may be noted that a MOS transistor of unit length (2λ) has gate capacitance proportional to its width (W), which may be multiple of λ. With the increase of the width, the current driving capability is increased. But, this in turn, also increases the gate capacitance. As a consequence the delay in driving a load capacitance C_L by a transistor of gate capacitance Cg is given by the relationship $\tau.C_L / Cg$. So, delay in driving by a single stage is $2000/100\ \tau = 200\ \tau$. With n stages, where n dependent on the stage ratio, the delay is

$$t_{\min} = nf\tau = e\tau \ln\left[\frac{C_L}{C_g}\right] \qquad \text{or} \quad \ln y = n \ln f \qquad \text{or} \quad n = \frac{\ln y}{\ln f}$$

Review Questions of Lec-10

Q1. How the transfer characteristic of a CMOS NAND gate is affected with increase in fan-in?

Q2. How the transfer characteristic of a CMOS NOR gate is affected with increase in fan-in?

Q3 How switching characteristic of a CMOS NAND gate is affected with increase in fan-in?

Q4. How switching characteristic of a CMOS NOR gate is affected with increase in fan-in?

Q5. How noise margin of a CMOS NAND/NOR gate is affected with increase in fan-in?

Answer to Questions of Lec-10

Q1. How the transfer characteristic of a CMOS NAND gate is affected with increase in fan-in?

Ans: Transfer characteristic does not remain symmetric with increase in fan-in of the NAND gate. The inversion voltage moves towards right with the increase in fan-in.

Q2. How the transfer characteristic of a CMOS NOR gate is affected with increase in fan-in?

Ans: In case of NOR gate the transfer characteristic also does not remain symmetric and the inversion voltage moves towards left with the increase in fan-in.

Answer to Questions of Lec-10

Q3 How switching characteristic of a CMOS NAND gate is affected with increase in fan-in?

Ans: When the load capacitance is relatively large, the fall time increases linearly with the increase in fan-in and the rise time is not affected much.

Q4. How switching characteristic of a CMOS NOR gate is affected with increase in fan-in?

Ans: When the load capacitance is relatively large, the rise time increases linearly with the increase in fan-in and the fall time is not affected much. For the same area, NAND gates are superior to NOR gates in switching characteristics because of higher mobility of electrons compared holes. For the same delay, NAND gates require smaller area than NOR gates

Answer to Questions of Lec-10

Q5. How noise margin of a CMOS NAND/NOR gate is affected with increase in fan-in?

Ans: Because of the change in the inversion voltage, the noise margin is affected with the increase in fan-in. For equal fan-in, noise margin is better for NAND gates compared to NOR gates. We may conclude that for equal area design NAND gates are faster and better alternative to NOR gates

Review Questions of Lec-11

Q1. For a complex/compound CMOS logic gate, how do you realize the pull-up and the pull-down networks?

Q2. Give the two possible topologies AND-OR-INVERT AND-OR-INVERT (AOI) and OR-AND-INVERT (OAI) to realize CMOS logic gate. Explain with an example.

Q3. Give the AOI and OAI realizations for the sum and carry functions of a full adder.

Q4. How do you realize pseudo nMOS logic circuits. Compare its advantage and disadvantages with respect to standard static CMOS circuits.

Answer to Questions of Lec-11

Q1. For a complex/compound CMOS logic gate, how do you realize the pull-up and the pull-down networks?

- **Ans: A CMOS logic gate consists of a nMOS pull-down network and a pMOS pull-up network. The nMOS network is connected between the output and the ground, whereas the pull-up network is connected between the output and the power supply. The nMOS network corresponds to the complement of the function either in sum-of-product or product-of-sum forms and the pMOS network is dual of the nMOS network .**

Answer to Questions of Lec-11

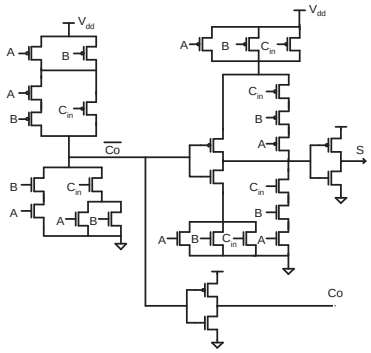
Q2. Give the two possible topologies AND-OR-INVERT AND-OR-INVERT (AOI) and OR-AND-INVERT (OAI) to realize CMOS logic gate. Explain with an example.

Ans: The AND-OR-INVERT network corresponds to the realization of the nMOS network in sum-of-product form. Where as the OR-AND-INVERT network corresponds to the realization of the nMOS network in product-of-sum form. In both the cases, the pMOS network is dual of the nMOS network .

Answer to Questions of Lec-11

Q3. Give the AOI and OAI realizations for the sum and carry functions of a full adder.

Ans: AOI form of realization is shown in the figure.



Answer to Questions of Lec-11

Q4. How do you realize pseudo nMOS logic circuits. Compare its advantage and disadvantages with respect to standard static CMOS circuits.

Ans: In the pseudo-nMOS realization, the pMOS network of the static CMOS realization is replaced by a single pMOS transistor with its gate connected to GND. An n-input pseudo nMOS requires n+1 transistors compared to 2n transistors of the corresponding static CMOS gates. This leads to substantial reduction in area and delay in pseudo nMOS realization. As the pMOS transistor is always ON, it leads to static power dissipation when the output is LOW.

Review Questions of Lec-12

Q1. In what way relay logic circuits differ from pass transistor logic circuits? Why the output of a pass transistor circuit is not used as a control signal for the next stage?

Q2. What are the advantages and limitations of pass transistor logic circuits? How the limitations are overcome?

Q3. Why is it necessary to insert a buffer after not more than four pass transistors in cascade?

Q4. Why is it necessary to have swing restoration logic in pass transistor logic circuits? Explain its operation.

Q5. What is the 'sneak path' problem of pass transistor logic circuits? How sneak path is avoided in Universal Logic Module (ULM) based realization of pass transistor network. Illustrate with an example.

Answer to Questions of Lec-12

Q1. In what way relay logic circuits differ from pass transistor logic circuits? Why the output of a pass transistor circuit is not used as a control signal for the next stage?

Ans: Logic functions can be realized using pass transistors in a manner similar to relay contact networks. However, there are some basic differences as mentioned below:

(a) In relay logic, output is considered to be '1' when there is some voltage passing through the relay logic. Absence of voltage is considered to be '0'. On the other hand, is case of pass transistor logic it is essential to provide both charging and discharging path for the output load capacitance.

(b) There is no voltage drop in the relay logic, but there is some voltage drop across the pass transistor network.

(c) Pass transistor logic is faster than relay logic.

Answer to Questions of Lec-12

Q2. What are the advantages and limitations of pass transistor logic circuits? How the limitations are overcome?

Ans: Pass transistor realization is ratioless, i.e. there is no need to have L:W ration in the realization. All the transistors can be of minimum dimension. Lower area due to smaller number of transistors in pass transistor realization compared to static CMOS realization. Pass transistor realization also has lesser power dissipation because there is no static power and short-circuit power dissipation in pass transistor circuits. The limitations are (a) Higher delay in long chain of pass transistors (b) Multi-threshold Voltage drop (Vout = Vdd – Vtn) (c) Complementary control signals and (d) Possibility of sneak path because of the presence of path to Vdd and GND.

Answer to Questions of Lec-12

Q3. Why is it necessary to insert a buffer after not more than four pass transistors in cascade?

Ans: When a signal is steered through several stages of pass transistors, the delay can be considerable. For n stages of pass transistors, the delay is given by the relationship $\tau = CR_{eq}\frac{n(n+1)}{2}$

To overcome the problem of long delay, buffers should be inserted after every three or four pass transistor stages.

Answer to Questions of Lec-12

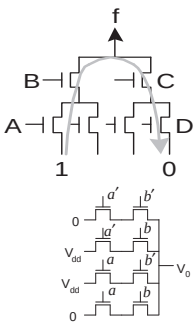
Q4. Why is it necessary to have swing restoration logic in pass transistor logic circuits? Explain its operation.

Ans: In order to avoid the voltage drop at the output (Vout = Vdd – Vtn) , it is necessary to use additional hardware known swing restoration logic at the gate output. At the output of the swing restoration logic there is rail to rail voltage swing. The swing restoration can be done using a pMOS transistor with its gate connected to GND.

Answer to Questions of Lec-12

Q5. What is the 'sneak path' problem of pass transistor logic circuits? How sneak path is avoided in Universal Logic Module (ULM) based realization of pass transistor network. Illustrate with an example.

Ans: As shown in the figure, the output is connected to both '1' (Vdd) and '0' (GND). The output attains some intermediate Value between Vdd and GND. The MUX based realization allows connection of the output to only one input, which can be either 0 or 1. Multiplexer realization of $f = a'b + ab'$ Is shown in the figure.



Review Questions Lec-13

Q1. What is Shanon's Expansion Theorem? How can be used realize pass transistor circuit for a given Boolean function?

Q2. Obtain an ROBDD for the Boolean function F = ∑(3, 7, 9, 11, 12, 13, 14,15). Realize the function using a CPL circuit.

Q3. Compare the area, in terms of the number of transistors, for the three different implementations of a full adder using (i) static CMOS, (ii) domino CMOS, and (iii) complementary pass transistor logic (CPL).

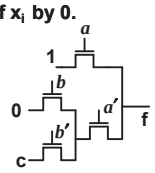
Answer to Questions of Lec-13

Q1. What is Shanon's Expansion Theorem? How can be used realize pass transistor circuit for a given Boolean function?

Ans: According to Shanon's expansion theorem, a Boolean function can be expanded around a variable x_i to represent the function as $f = x_i f_{xi} + x_i' f_{xi'}$, where f_{xi} and $f_{xi'}$ are the positive and negative cofactors of f, respectively. A positive Shannon cofactor of function f with respect to variable x_i is defined as that function with all instances of x_i replaced by 1. A negative Shannon cofactor is the same, but replaces all instances of x_i by 0.

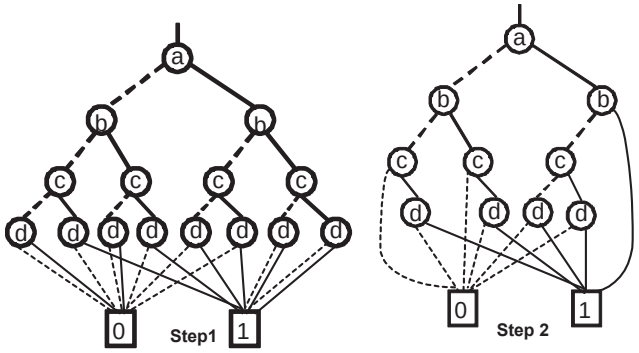
This is illustrated by the following example:

$$f = a.1 + a'(b.0 + b'.c)$$



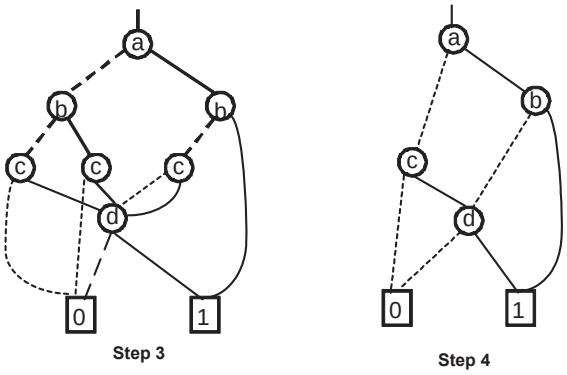
Answer to Questions of Lec-13

Q2. Obtain an ROBDD for the Boolean function F = ∑(3, 7, 9, 11, 12, 13, 14,15). Realize the function using a CPL circuit.



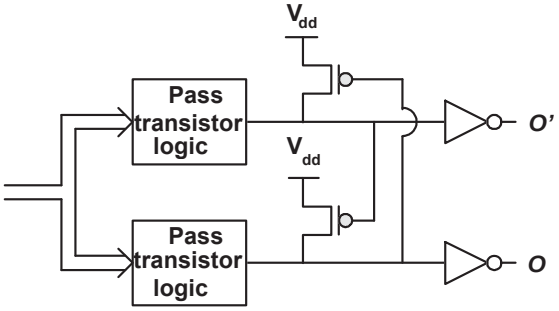
Answer to Questions of Lec-13

Ans 2:



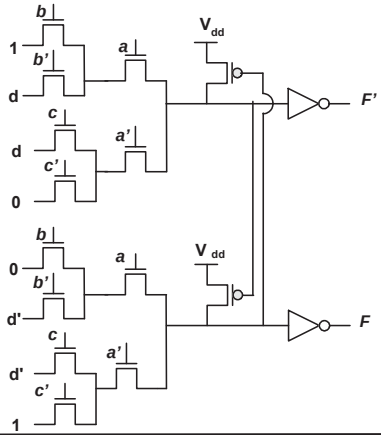
Answer to Questions of Lec-13

Ans 2:



Answer to Questions of Lec-13

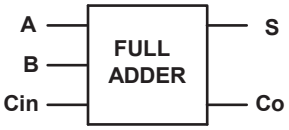
Ans 2:



Answer to Questions of Lec-13

Q3. Compare the area, in terms of the number of transistors, for the three different implementations of a full adder using (i) static CMOS, (ii) domino CMOS, and (iii) complementary pass transistor logic (CPL).

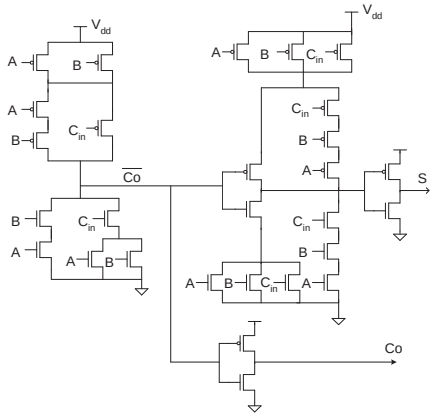
Ans: The full adder block diagram is given below:



Answer to Questions of Lec-13

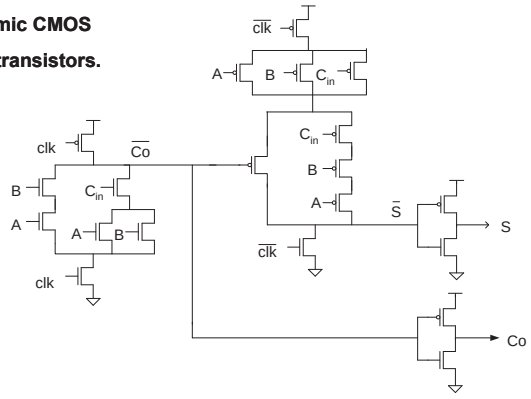
The full adder Realization using Static CMOS is given here.

It requires 28 transistors.



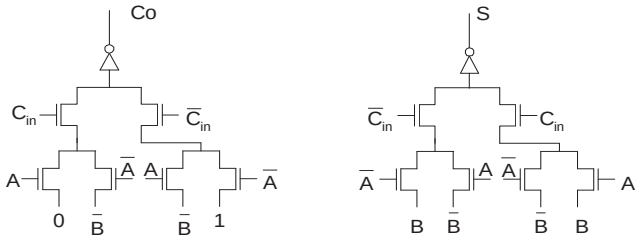
Answer to Questions of Lec-13

The realization using NORA dynamic CMOS requires 20 transistors.



Answer to Questions of Lec-13

The pass transistor realization of the full adder is given below. It requires 16 (8+8) transistors.



Review Questions of Lec-14

Q1. What are the key characteristics of MOS dynamic circuits?

Q2. Explain the basic operation of a 2-phase dynamic circuit?

Q3. How 2-phase clocks can be generated using inverters?

Q4. What makes dynamic CMOS circuits faster than static CMOS circuits?

Q5. Compare the sources of power dissipation between static CMOS and dynamic CMOS circuits?

Answer to Questions of Lec-14

Q1. What are the key characteristics of MOS dynamic circuits?

Ans: The advantage of low power of static CMOS circuits and smaller chip area of nMOS circuits are combined in dynamic circuits leading to circuits of smaller area and lower power dissipation. Smaller area due to lesser number of transistors (n+2) compared to static CMOS realization requiring 2n transistors to realize a n-variable function. Dynamic CMOS circuits have Lower static power dissipation because of smaller capacitance. There is no short circuit power dissipation and no glitching power dissipation. Dynamic CMOS circuits are also faster because the capacitance is about half that of the static CMOS circuits.

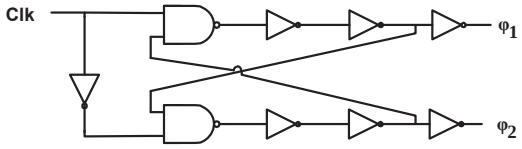
Answer to Questions of Lec-14

Q2. Explain the basic operation of a 2-phase dynamic circuit?

Ans: The operation of the circuit can be explained using precharge logic in which the output is precharged to HIGH level during $\phi 2$ clock and the output is evaluated during $\phi 1$ clock.

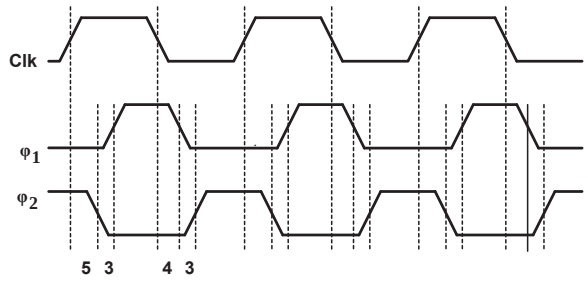
Q3. How 2-phase clocks can be generated using inverters?

Ans: As shown below, two phase clock can generated using inverters. The timing diagram is given in the next slide.



Answer to Questions of Lec-14

Timing diagram of the two-phase clock generated from a single-phase clock.



Answer to Questions of Lec-14

Q4. What makes dynamic CMOS circuits faster than static CMOS circuits?

Ans: As MOS dynamic circuits require lesser number of transistors and lesser capacitance is to be driven by it. This makes MOS dynamic circuits faster.

Q5. Compare the sources of power dissipation between static CMOS and dynamic CMOS circuits?

Ans: In both the cases there is switching power and leakage power dissipations. However, the short circuit and glitching power dissipations, which are present in static CMOS circuits, are not present in dynamic CMOS circuits.

Review Questions of Lec-15

Q1. What is charge leakage problem of dynamic CMOS circuits? How is it overcome?

Q2. What is charge sharing problem? How can it be overcome?

Q3. Explain the clock skew problem of dynamic CMOS circuits?

Q4. How clock skew problem is overcome in in domino CMOS circuits?

Q5. How clock skew problem is overcome in in NORA CMOS circuits?

Answer to Questions of Lec-15

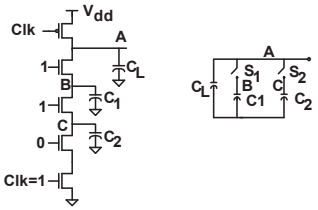
Q1. What is charge leakage problem of dynamic CMOS circuits? How is it overcome?

Ans: The source-drain diffusions form parasitic diodes with the substrate. There is reverse bias leakage current .The current is in the range 0.1nA to 0.5nA per device at room temperature and the current doubles for every 10°C increase in temperature . This leads to slow but steady discharge of the charge on the capacitor, which represent information. This needs to be compensated by refreshing the charge at regular interval.

Answer to Questions of Lec-15

Q2. What is charge sharing problem of dynamic CMOS circuits?

Ans: The charge sharing problem is illustrated in the following diagram



Before the switches are closed, the charge on C_L is given by QA = V_{dd} C_L and charges at node B and C are Q_B = 0 and Q_C = 0

After the switches are closed, there will be redistribution of charges based of charge conservation principle, and the voltage V_A at node A is given by VA, which is less than Vdd.

$$C_L V_{dd} = (C_L + C_1 + C_2) V_{A.} \qquad V_A = \frac{C_L}{(C_L + C_1 + C_2)} V_{dd}$$

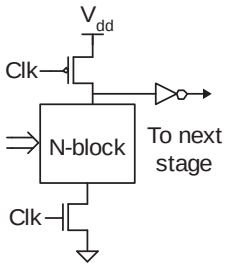
Answer to Questions of Lec-15

Q3. Explain the clock skew problem of dynamic CMOS circuits?

Ans: Clock skew problem arises because of delay due to resistance and parasitic capacitances associated with the wire that carry the clock pulse and this delay is approximately proportional to the square of the length of the wire. When the clock signal reaches a later stage before its preceding stage, the precharge phase of the preceding stage overlaps with the evaluation phase of the later stage, which may lead to premature discharge of the load capacitor and incorrect output during evaluation phase.

Answer to Questions of Lec-15

Q4. How clock skew problem is overcome in in domino CMOS circuits?

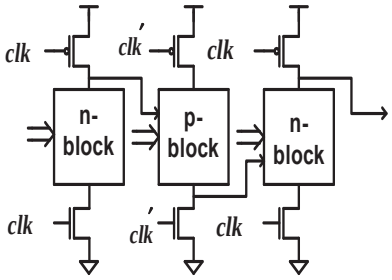


Ans: In domino CMOS circuits the problem is overcome by adding an inverter as shown in the diagram. It consists of two distinct components: The first component is a conventional dynamic CMOS gate and the second component is a static inverting CMOS buffer . During precharge phase, the output of the dynamic gate is high, but the output of the inverter is LOW. As a consequence it cannot drive an nMOS transistor ON. So, the clock skew problem is overcome.

Answer to Questions of Lec-15

Q5. How clock skew problem is overcome in in NORA CMOS circuits?

Ans: The problem can be overcome using NORA logic, nMOS and pMOS transistor networks are alternatively used. The output of an nMOS block is HIGH during precharge, which cannot turn a pMOS transistor ON. Similarly, the output of an pMOS block is LOW during precharge, which cannot turn a nMOS transistor ON.



Review Questions of Lec-16

Q1. Distinguish between Mealy and Moore machines.

Q2. A sequence detector produces a '1' for each occurrence of the input sequence '1001' at its input.

(a) Draw the state-transition diagram of the FSM realizing the sequence detector.

(b) Obtain state table from the state transition diagram.

(c) Realize the FSM using D FFs and a PLA.

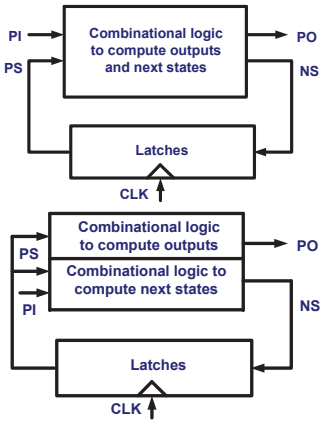
Q3. How can you realize a set of Boolean functions using a ROM.

Q4. How the limitations of a ROM-based realization is overcome in a PLA-based realization.

Answer to Questions of Lec-16

Q1. Distinguish between Mealy and Moore machines.

Ans: In a Mealy machine the outputs are dependent on the inputs and present state. The Output transition function is represented by Z = λ(S,X). Where as in a Moore machine the outputs are dependent only on present state. The output transition function is represented by Z = λ(S)



Answer to Questions of Lec-16

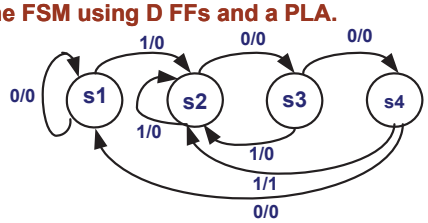
Q2. A sequence detector produces a '1' for each occurrence of the input sequence '1001' at its input.

(a) Draw the state-transition diagram of the FSM realizing the sequence detector.

(b) Obtain state table from the state transition diagram.

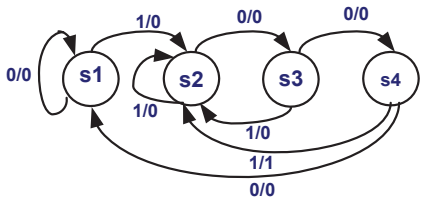
(c) Realize the FSM using D FFs and a PLA.

Ans: 2(a)



Answer to Questions of Lec-16

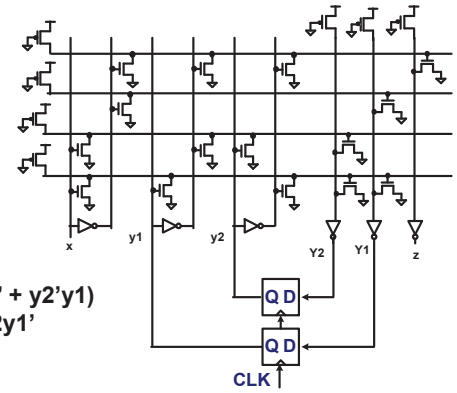
Ans: 2(b) The state table from the state transition diagram is given below:



		Y2Y1, z	
PS	y2y1	x =0	X = 1
S1	0 0	00,0	01,0
S2	01	10,0	01,0
S3	1 0	11,0	01,0
S4	1 1	00,0	01,1

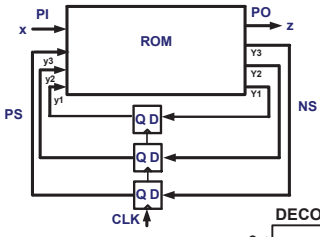
Answer to Questions of Lec-16

Ans: 2(c) Realization of the FSM using D FFs and a PLA is shown in the diagram.



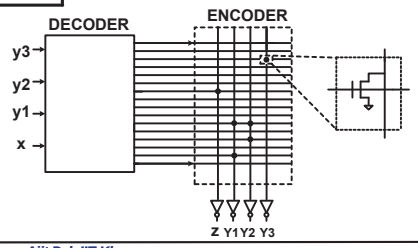
Y2 = x'(y2y1' + y2'y1)
Y1 = x + x'y2y1'
z = xy2y1

Answer to Questions of Lec-16



Q3. How can you realize a set of Boolean functions using a ROM?

Ans: ROMs can be used to realize a set of Boolean functions as show In the diagram. Encoder part of the ROM is realized according to the functions.



Answer to Questions of Lec-16

Q4. How the limitations of a ROM-based realization is overcome in a PLA-based realization.

Ans: In a ROM, the encoder part is only programmable and use of ROMs to realize Boolean functions is wasteful in many situations because there is no cross-connect for a significant part . This wastage can be overcome by using Programmable Logic array (PLA), which requires much lesser chip area.

Review Questions of Lec-17

Q1. Sketch the schematic diagram of a SRAM memory cell along with sense amplifier and data write circuitry.

Q2. Explain how read and write operations are performed in a SRAM.

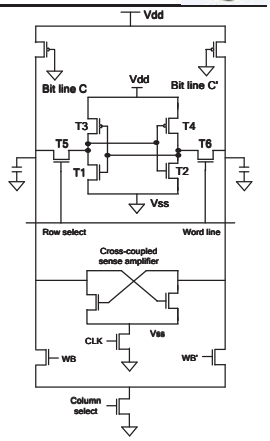
Q3. In what way the DRAMs differ from SRAMs?

Q4. Explain the read and write operations for a one-transistor DRAM cell.

Answer to Questions of Lec-17

Q1. Sketch the schematic diagram of a SRAM memory cell along with sense amplifier and data write circuitry.

Ans: The schematic diagram of a SRAM memory along with the sense amplifier and data write circuitry



Answer to Questions of Lec-17

Q2. Explain how read and write operations are performed.

Ans: Steps of READ operation:

1. **Precharge and equalization circuit is activated to precharge the bus lines to Vdd/2**
2. **The word line is activated connecting the cell to B and B' lines. As a consequence, a voltage is developed between B and B'.**
3. **Once adequate voltage is developed between B and B', the sense amplifier amplifies the signals to rail to rail using regenerative action. The output is then directed to the chip I/O pin by the column decoder.**

Steps of write operation:

1. The data available on the chip I/O pin re directed to the B and B' lines. By activating the word line signal, the data is transferred to the memory cell

Answer to Questions of Lec-17

Q3. In what way the DRAMs differ from SRAMs?

Ans: Both SRAMS and DRAMs are volatile in nature, i.e. information is lost if power line is removed. However, SRAMs provide high switching speed, good noise margin but require larger chip area than DRAMs.

Q4. Explain the read and write operations for a one-transistor DRAM cell.

Ans: A significant improvement in the DRAM evolution was to realize 1-T DRAM cell. One additional capacitor is explicitly fabricated for storage purpose. To store '1', it is charged to Vdd-Vt and to store '0' it is discharged to 0V. Read operation is destructive. Sense amplifier is needed for reading. Read operation is followed by restoration operation.

Review Questions of Lec-18

Q1. How charge sharing leads to power dissipation?

Q2. What is short circuit power dissipation? On what parameters does it depend?

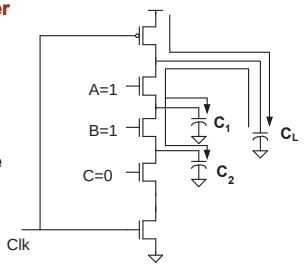
Q3. Justify the statement; "there is no short circuit power dissipation in a static CMOS circuit if Vdd < (Vtn + !Vtp!)"

Q4. What is glitching power dissipation? How can it be minimized?

Answer to Questions of Lec-18

Q1. How charge sharing leads to power dissipation?

Ans: In case of dynamic gates, power dissipation takes place due to the phenomenon of charge sharing even when the output is not 0 at the time evaluation. At the time of precharge, the output is charged to Vdd, but at the time of evaluation, the output decreases because of the of the sharing of charge by the two capacitors C1 and C2. In the next precharge phase a power dissipation equal to P takes place.



Answer to Questions of Lec-18

After charge sharing, the new voltage level is Vnew. Because of the conservation of charge, we get

$$(C_1 + C_2 + C_L).V_{new} = V_{dd}C_L$$

$$V_{new} = \frac{C_L V_{dd}}{C_1 + C_2 + C_3}$$

$$\Delta V = V_{dd} - \frac{C_L}{C_1 + C_2 + C_L}.V_{dd} \qquad \frac{C_1 + C_2}{C_1 + C_2 + C_L}.V_{dd}$$

$$P = C_L.V_{dd}.\Delta V = \frac{C_L.(C_1 + C_2)}{(C_1 + C_2 + C_L)}V_{dd}^2$$

Answer to Questions of Lec-18

Q2. What is short circuit power dissipation? On what parameters does it depend?

Ans: As input changes slowly, power dissipation takes place even when there is no load or parasitic capacitor. When the input is greater than Vtn and less than (Vdd – Vtp), both the nMOS and pMOS transistors are ON. The supply voltage is now shorted to GND through the two transistors. This leads to the short circuit power dissipation.

Answer to Questions of Lec-18

Q3. Justify the statement; "there is no short circuit power dissipation in a static CMOS circuit if Vdd < (Vtn + !Vtp!)".

Ans: When Vdd < (Vtn + !Vtp!), only one thansistor can turn on at a time. Since bothe transistors cannot turn on simultaneously, there is no short circuit power dissipation.

Q4. What is glitching power dissipation? How can it be minimized?

Ans: Because of finite delay of the gates used to realize Boolean functions, different signals cannot reach the inputs of a gate simultaneously. This leads to spurious transitions at the output before it settles down to its final value. The spurious transitions leads to charging and discharging of the outputs causing glitching power dissipation. It can be minimized by having balanced realization having same delay at the inputs.

Review Questions of Lec-19

Q1. List various sources of leakage currents.

Q2. Why leakage power is an important issue in deep submicron technology?

Q3. What is band-to-band tunneling current?

Q4. What is body effect?

Q5. What is subthreshold leakage current? Briefly discuss various mechanisms responsible for this leakage current?

Answer to Questions of Lec-19

Q1. List various sources of leakage currents.

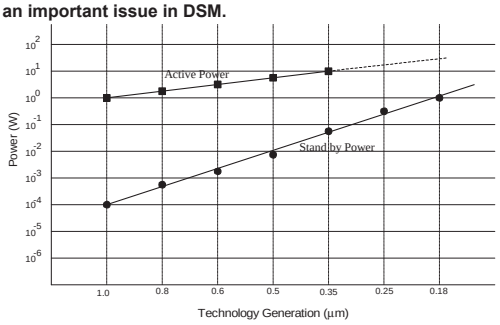
Ans: Various sources of leakage currents are listed below:

- **I_1= Reverse-bias p-n junction diode leakage current**
- **I_2 = Band-to-band tunneling current**
- **I_3 = Subthreshold leakage current**
- **I_4 = Gate Oxide tunneling current**
- **I_5 = Gate current due to hot-carrier injection**
- **I_6 = Channel punch-through**
- **I_7 = Gate induced drain-leakage current**

Answer to Questions of Lec-19

Q2. Why leakage power is an important issue in deep submicron technology?

Ans: In deep submicron technology, the leakage component is a significant % of total power as shown in the diagram. Moreover, the leakage current is increasing at a faster rate than dynamic power. As a consequence, it has become an important issue in DSM.



Answer to Questions of Lec-19

Q3. What is band-to-band tunneling current?

Ans: When both n regions and p regions are heavily doped, a high electric field across a reverse biased p-n junction causes significant current to flow through the junction due to tunneling of electrons from the valence bond of the p-region to the conduction band of n-region. This is known as band-to-band tunneling.

Q4. What is body effect?

Ans: As a negative voltage is applied to the substrate with respect to the source, the well-to-source junction the device is reverse biased and bulk depletion region is widened. This leads to increase the threshold voltage. This effect is known as body effect.

Answer to Questions of Lec-19

Q5. What is subthreshold leakage current? Briefly discuss various mechanisms responsible for this leakage current?

Ans: The subthreshold leakage current in CMOS circuits is due to carrier diffusion between the source and the drain regions of the transistor in weak inversion, when the gate voltage is below V_t. The behavior of an MOS transistor in the subthreshold operating region is similar to a bipolar device, and the subthreshold current exhibits an exponential dependence on the gate voltage. The amount of the subthreshold current may become significant when the gate-to-source voltage is smaller than, but very close to the threshold voltage of the device.

Answer to Questions of Lec-20

Q1. Explain the basic concepts of supply voltage scaling.

Q2. As you move to a new process technology with a scaling factor S = 1.4, how the drain current, power density, delay and energy requirement changes for the constant field scaling?

Q3. Distinguish between constant field and constant voltage feature size scaling? Compare their advantages and disadvantages.

Q4. Compare the constant field and constant voltage scaling approaches in terms of area, delay, energy and power density parameters.

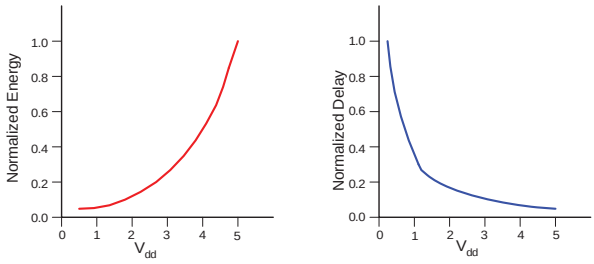
Q5. Explain how *parallelism* can be used to achieve low power instead of high performance in realizing digital circuits.

Q6.Explain how multicore architecture provides low power compared to the single core architecture of the same performance.

Answer to Questions of Lec-20

Q1. Explain the basic concepts of supply voltage scaling.

Ans: Power dissipation is proportional the square of the supply voltage. So, a factor of two reduction in supply voltage yields a factor of four decrease in energy. But, as the supply voltage is reduced, delay increases as shown in the diagram. So, the challenge is to scale down the supply voltage without compromise in performance.



Answer to Questions of Lec-20

Q2. As you move to a new process technology with a scaling factor S = 1.4, how the drain current, power dissipation, power density, delay and energy requirement changes for the constant field scaling?

Ans: Drain current reduces by a factor of S. Although power dissipation decreases by a factor of S^2 , the power density remains the same. The delay decreases by a factor of S and the energy decreases by a factor of S^3.

Quality	Before Scaling	After Scaling
Gate Capacitance	C_g	$C_g' = C_g / S$
Drain Current	I_D	$I_D' = I_D / S$
Power Dissipation	P	$P' = P / S^2$
Power Density	P/ Area	$P'/Area' = (P/Area)$
Delay	t_d	$t_d' = t_d / S$
Energy	$E = P.t_d$	$E' = \frac{P}{S^2} \times \frac{t_d}{S} = \frac{Pt_d}{S^3} = \frac{1}{S^3}E$

Answer to Questions of Lec-20

Q3. Distinguish between constant field and constant voltage feature size scaling? Compare their advantages and disadvantages.

Ans: In this approach the magnitude of all the internal electric fields within the device are preserved, while the dimensions are scaled down by a factor of S. This requires that all potentials must be scaled down by the same factor. Accordingly, supply as well as threshold voltages are scaled down proportionately. But, in constant-voltage scaling, all the device dimensions are scaled down by a factor of S just like constant-voltage scaling, supply voltage and threshold voltages are not scaled.

Answer to Questions of Lec-20

Q4. Compare the constant field and constant voltage scaling approaches in terms of area, delay, energy and power density parameters.

Quality	Cons field Scaling	Constant Voltage Scaling
Gate Capacitance	$C_g' = C_g / S$	$C_g' = C_g / S$
Drain Current	$I_D' = I_D / S$	$I_D' = I_D.S$
Power Dissipation	$P' = P / S^2$	$P' = P.S$
Power Density	$P'/Area' = (P/Area)$	$P'/Area' = S^3 P/Area$
Delay	$t_d' = t_d / S$	$t_d' = t_d / S^2$
Energy	$E' = \frac{P}{S^2} \times \frac{t_d}{S} = \frac{Pt_d}{S^3} = \frac{1}{S^3}E$	$E' = E / S$

Answer to Questions of Lec-20

Q5. Explain how *parallelism* can be used to achieve low power instead of high performance in realizing digital circuits.

Ans: Traditionally, parallelism is used to improve performance at the expense of larger power dissipation. But, instead of trying to improve performance, the power dissipation can be reduced by scaling down the supply voltage such that the performance remains unaltered.

Q6.Explain how multicore architecture provides low power compared to the single core architecture of the same performance.

Ans: The idea behind the parallelism for low-power can be extended to multi-core architecture. The clock frequency can be reduced with commensurate scaling of the supply voltage as the number of cores is increased from one to more than one while maintaining the same throughput.

Review Questions of Lec-21



Q1. In what situation pipelining can be implemented?

Q2. Explain how pipelining can be used to achieve low power instead of high performance in realizing digital circuits.

Q3. How clock frequency, speed up, throughput and power dissipation changes for a pipelined implementation with k stages with respect to non-pipelined implementation?

Q4. How can you combine sizing and supply voltage scaling to realize low power circuits?

Q5. Explain with an example how pipelining and parallelism can be combined to realize low power circuits?

Answer to Questions of Lec-21



Q1. In what situation pipelining can be implemented?

Ans: A task can be pipelined when it can be divided into more than one independent subtasks, which can be executed in a overlapped manner.

Q2. Explain how pipelining can be used to achieve low power instead of high performance in realizing digital circuits.

Ans: In a conventional pipelined implementation, the subtasks are executed in a overlapped manner at a faster rate. Instead of that, the subtasks can be executed at the same rate as the original task but with reduced supply voltage. This is how the pipelined the implementation will require lower power.

Answer to Questions of Lec-21



Q3. How clock frequency, speed up, throughput and power dissipation changes for a pipelined implementation with k stages with respect to non-pipelined implementation?

Ans: Clock frequency = kf, speedup = $S_k = \frac{n.k}{k+(n-1)}$

where n is the number of tasks executed using k-stage pipeline, and power dissipation = $1/k^2$.

Q4. How can you combine sizing and supply voltage scaling to realize low power circuits?

Ans: It can be done in three steps (a) Upsize gates on the critical path to reduce delay of the circuit (b) Scale down the supply voltage to equalize with the original delay (c) Upsize gates on non-critical paths selectively without exceeding the critical path delay.

Answer to Questions of Lec-21



Q5. Explain with an example how pipelining and parallelism can be combined to realize low power circuits?

Ans: Here, more than one parallel structure is used and each structure is pipelined. Both power supply and frequency of operation are reduced to achieve substantial overall reduction in power dissipation.

Review Questions of Lec-22



Q1. Explain the basic concept of multi level voltage scaling.

Q2. What is the impact of multiple supply voltages on the distribution of path delays of a circuit with respect to that for single supply voltage?

Q3. List and explain three important issues in the context of multiple supply voltage scaling?

Q4. What problem arises when a signal passes from low voltage domain to high voltage domain? How this problem is overcome?

Q5. Explain the design decision for the placement of converters in the voltage scaling interfaces.

Answer to Questions of Lec-22



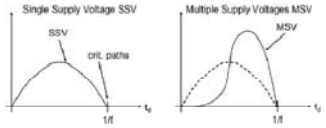
Q1. Explain the basic concept of multi level voltage scaling.

Ans: This is an extension of SVS where two or few fixed voltage domains are used in different parts of a circuit,. As we know, high Vdd gates have less delay, but higher dynamic and static power dissipation and low Vdd gates have larger delay but lesser power dissipation. Voltage islands can be generated at different levels of granularity, such as macro level and standard cell level. The slack of the off-critical path can be utilized for allocation of macro modules of low-Vdd to off-critical-path macro modules. Total power dissipation can be reduced without degrading the overall circuit performance.

Answer to Questions of Lec-22

Q2. What is the impact of multiple supply voltages on the distribution of path delays of a circuit with respect to that for single supply voltage?

Ans: Path delay for different paths in a circuit for single supply voltage is shown. The graph of a Gaussian is a characteristic symmetric "bell curve" shape that quickly falls off towards plus/minus infinity plus/minus infinity. However, when multiple supply voltages are used, the path delay distribution is not Gaussian because modules having smaller delays are assigned with smaller supply voltage and their delay increases.



Answer to Questions of Lec-22

Q3. List and explain the important issues in the context of multiple supply voltage scaling?

Ans: Important issues in the context of MVS are listed below:

(a) Voltage Scaling Interfaces
(b) Converter Placement
(c) Floor planning, Routing and Placement
(d) Multiple Supply Voltages
(e) Static Timing Analysis
(f) Power up and Power down Sequencing
(g) Clock distribution

Answer to Questions of Lec-22

Q4. What problem arises when a signal passes from low voltage domain to high voltage domain? How this problem is overcome?

Ans: A high-level output from the low-Vdd domain has output VddL, which may turn on both nMOS and pMOS transistors of the high-Vdd domain inverter resulting in short circuit between VddH to GND. A level converter needs to be inserted to avoid this static power consumption

Answer to Questions of Lec-22

Q5. Explain the design decision for the placement of converters in the voltage scaling interfaces.

Ans: One important design decision in the voltage scaling interfaces is the placement of converters . As the high-to-low level converters use low-Vdd voltage rail, it is a appropriate to place them in the receiving or destination domain. It is also recommended to place the low-to-high level converters in the receiving domain. As the low-to-high level converters require both low and high-Vdd supply rails, at least one of the supply rails needs to be routed from one domain to the other.

Review Questions of Lec-23

Q1. For a workload of 50%, explain how reduction in power dissipation takes place for DVS and DVFS?

Q2. With the help of a schematic diagram, explain how the dynamic voltage and frequency scaling technique is used to achieve low power dissipation of a processor.

Q3. In what way adaptive voltage scaling differs from dynamic voltage scaling?

Q4. Show various building blocks required for the implementation of adaptive voltage scaling system.

Answer to Questions of Lec-23

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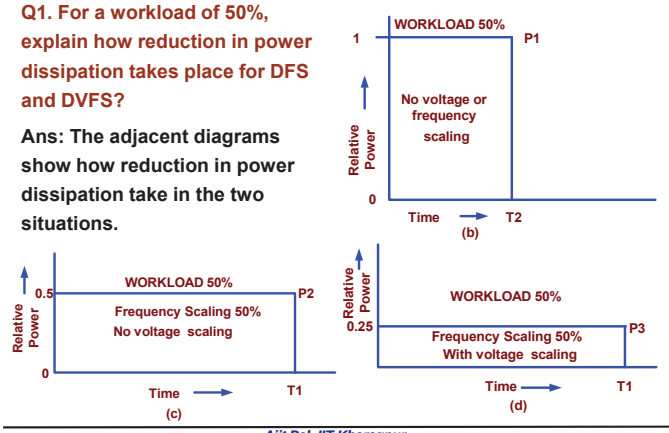
Q3. In what way adaptive voltage scaling differs from dynamic voltage scaling?

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Answer to Questions of Lec-23

Q1. For a workload of 50%, explain how reduction in power dissipation takes place for DFS and DVFS?

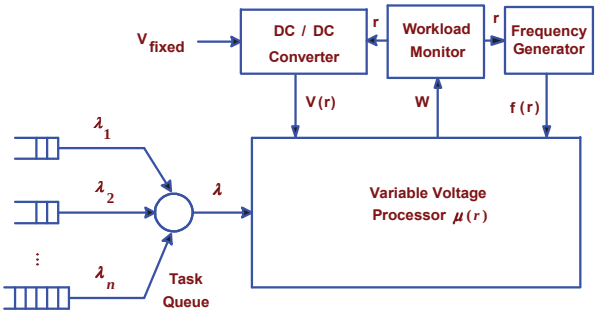
Ans: The adjacent diagrams show how reduction in power dissipation take in the two situations.



Answer to Questions of Lec-23

Q2. With the help of a schematic diagram, explain how the dynamic voltage and frequency scaling technique is used to achieve low power dissipation of a processor.

Ans: The diagram shows how both voltage and frequency are controlled for different predicted workload.



Answer to Questions of Lec-23

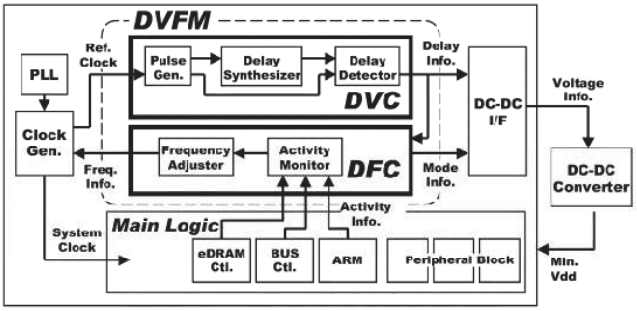
Q3. In what way adaptive voltage scaling differs from dynamic voltage scaling?

Ans: The DVFS approach is open-loop in nature. Voltage-frequency pairs are determined at design time keeping sufficient margin for guaranteed operation across the entire range of best and worst case process, voltage and temperature (PVT) conditions. As the design needs to be very conservative for successful operation, the actual benefit obtained is lesser than actually possible. A better alternative that can overcome this limitation is the Adaptive Voltage Scaling (AVS) where a close-loop feedback system is implemented between the voltage scaling power supply and delay sensing performance monitor at execution time.

Answer to Questions of Lec-23

Q4. Show various building blocks required for the implementation of adaptive voltage scaling system.

Ans: The schematic diagram is given below. The operation of different building blocks are as follows:



Answer to Questions of Lec-23

The on-chip monitor not only checks the actual voltage developed, but also detects whether the silicon is slow, typical or fast and the effect of temperature on the surrounding silicon. The DVC emulates the critical path characteristic of the system by using a delay synthesizer and controls the dynamic supply voltage (0.9 to 1.6V at 5mV step in real time). The DFC adjusts the clock frequency at the required minimum value by monitoring the system activity (20MHz to 120MHz). The voltage and frequency are predicted according to the performance monitoring of the system. The DVFM system can track the required performance with a high level of accuracy over the full range of temperature and process deviations.

Review Questions of Lec-24

Q1. Explain the basic concept of hardware-software tradeoff with an example for low power .

Q2. Distinguish between superscalar and VLIW architecture

Q3. Discuss why VLIW architecture allows lower power dissipation compared to Superscalar architecture.

Q4. In the context of Transmeta's Crusoe processor explain the role of Code Morphing Software.

Q5. How caching can be used to achieve lower power dissipation in VLIW architecture?

Answer to Questions of Lec-24



Q1. Explain the basic concept of hardware-software tradeoff with an example for low power .

Ans: It is well known that the same functionality can be either realized by hardware or by software or by a combination of both. The hardware-based approach has the following characteristic:

- Faster
- Costlier
- Consumes more power

On the other hand the software-based approachhas the following characteristics:

- Cheaper
- Slower
- Consumes lesser power

Answer to Questions of Lec-24



Q2. Distinguish between superscalar and VLIW architecture.

Ans: Both superscalar and VLIW architectures implement a form of parallelism called instruction-level parallelism within a single processor. A superscalar processor executes more than one instructions during a clock cycle by simultaneously dispatching multiple instructions to more than one functional units on the processor. In a superscalar CPU the dispatcher reads instructions from memory and decides which ones can be run in parallel, dispatching them to two or more functional units contained inside a single CPU. So, in this case ILP is implemented by hardware.

On the other hand, a compiler identifies that can be executed in parallel and generates long instructions having multiple operations meant for different functional units. Therefore, in this case ILP is implemented by software.

Answer to Questions of Lec-24



Q3. Discuss why VLIW architecture allows lower power dissipation compared to Superscalar architecture.

Ans: As parallelism is identified by software at compile time, the VLIW incurs lower power dissipation.

Q4. In the context of Transmeta's Crusoe processor explain the role of Code Morphing Software.

Ans: The Code Morphing software mediates between x86 software and the VLIW engine. It is fundamentally a dynamic translation system. A program that translates instructions from one instruction set architecture to another instruction set architecture. Here, x86 code is compiled into VLIW code of the Cruosoe processor. Code Morphing software insulates x86 programs from the hardware engine's native instruction set. The code morphing software is the only program that is written directly for the VLIW processor.

Answer to Questions of Lec-24



Q5. How caching can be used to achieve lower power dissipation in VLIW architecture?

Ans: In real-life applications it is very common to execute a block of code many times over and over after it has been translated once. A separate memory space is used to store the translation cache and the code morphing software. It allows reuse the translated code by making use of locality of reference property. Caching translations provide excellent opportunity of reuse in many real-life applications.

Review Questions of Lec-25



Q1. Explain the basic concept of bus encoding to reduce switched capacitance.

Q2. Find out the switching activity of a modulo-7 counter using binary and Gray codes for state encoding

Q3. How gray coding helps to reduce power dissipation for fetching instructions from main memory?

Q4. How reduction in power dissipation is achieved by dividing a 64-bit bus into eight 8-bit buses.

Q5. How T0 encoding achieves almost zero transition on a bus.

Answer to Questions of Lec-25



Q1. Explain the basic concept of bus encoding to reduce switched capacitance.

Ans: Switching activity can be reduced by coding the address bit before sending over the bus. This is done introducing sample to sample correlation such that total number of bit transitions is reduced. Similarly, communicating data bits in an appropriately coded form can reduce the switching activity.

Answer to Questions of Lec-25

Q2. Find out the switching activity of a modulo-7 counter using binary and Gray codes for state encoding.

Ans: The reduction in the number of bit transitions for the two types of coding is given below:

Binary code	Transitions	Gray code	Transitions
000		000	
001	1	001	1
010	2	011	1
011	1	010	1
100	3	110	1
101	1	111	1
110	2	101	1
	2		2
Total	12		8

Answer to Questions of Lec-25

Q3. How gray coding helps to reduce power dissipation for fetching instructions from main memory?

Ans: A gray code sequence is a set of numbers in which adjacent numbers have only one bit difference. On the other hand, the number of transitions vary from 1 to n (n/2 on the average) as shown in the adjacent table. The power dissipations of the bus driver decreases because of the reduction of switching activity.

Decimal Value	Binary Code	Gray Code
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

Answer to Questions of Lec-25

Q4. How T0 encoding achieves almost zero transition on a bus.

Ans: In T0 encoding, after sending the first address, the same address is sent for infinite streams of consecutive addresses. The receiver side is informed about it by sending an additional bit known as increment (INC) bit. However, if the address is not consecutive, then the actual address is sent. The T0 code provides, zero transition property for infinite streams of consecutive addresses.

Review Questions of Lec-26

1. **Explain the basic concept of clock gating to reduce power dissipation in a digital circuit.**
2. **What are the three levels of clock gating granularity? Compare their pros and cons.**
3. **Explain the important issues related to clock gating in the clock tree.**
4. **Explain with an example, how power dissipation in a combinational circuit can be reduced by using operand isolation.**

Answer to Questions of Lec-26

1. **Explain the basic concept of clock gating to reduce power dissipation in a digital circuit.**

Ans: It has been observed that a major component of processor power is the clock power (50% of the total power). So, there is scope for large reduction of power dissipation by using suitable technique to remove a large number of unnecessary transitions. Such transitions can be suppressed without affecting functionality. One of the most successful and commonly used low power technique is clock gating.

Answer to Questions of Lec-26

2. What are the three levels of clock gating granularity? Compare their pros and cons.

Ans: There are three levels of granularity:

- **Module-level clock gating: Large reduction in power but there is limited opportunity.**
- **Register-level clock gating: There is more opportunity compared to module level clock gating, but lesser reduction of power.**
- **Cell-level clock gating: Provides many more opportunities and it lends itself to automated insertion and can result in massively clock gated designs.**

Answer to Questions of Lec-26

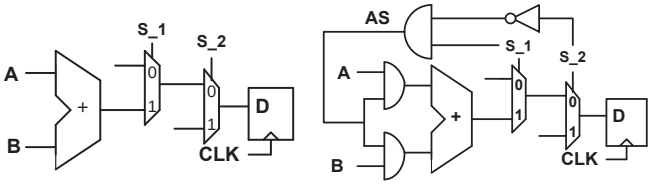
3. Explain the important issues related to clock gating in the clock tree.

Ans: Clock tree amounts to a significant portion of the total dynamic power consumption. It leads to a tradeoff between the physical capacitance that is prevented from switching and the number of unnecessary transitions. Disabling at higher up in the tree prevents larger capacitance from switching, but the gating condition is satisfied fewer times. Having multiple gating points along a path may be beneficial.

4. Explain with an example, how power dissipation in a combinational circuit can be reduced by using operand isolation.

Answer to Questions of Lec-26

Ans: Operand isolation is a technique for power reduction in the combinational part of the circuit. Here the basic concept is to 'shut-off' logic blocks when they do not perform any useful computation. Shutting-off is done by not allowing the inputs to toggle in clock cycles when the output of the block is not used. In the following example, the output of the adder is loaded into the latch only when S_1 is 1 and S_2 is 0. So, input lines of the adder may be gated based on this condition, as shown in the diagram.



Review Questions of Lec-27

Q1. What is the basic concept of clock gated FSM.

Q2. How state encoding can be used to reduce power dissipation in an FSM? Explain with an example.

Q3. How power dissipation is reduced by partitioning an FSM> Explain with an example.

Q4. A sequence detector produces a '1' for each occurrence of the input sequence '1001' at its input.

(a) Draw the state-transition diagram of the FSM realizing the sequence detector.

(b) Obtain state table from the state transition diagram.

(c) Realize the FSM using D FFs and a PLA.

Answer to Questions of Lec-27

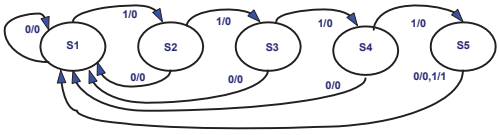
Q1. What is the basic concept of clock gated FSM.

Ans: There are conditions when the next state and output values do not change (idle condition). Clocking the circuit during this idle condition leads to unnecessary wastage of power. The clock can be stopped, if the idle conditions can be detected. This saves power both in the combinational circuit as well as the registers/latches.

Answer to Questions of Lec-27

Q2. How state encoding can be used to reduce power dissipation in an FSM? Explain with an example.

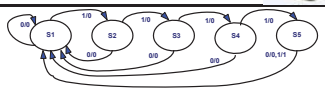
Ans: In the state assignment phase of an FSM, each state is given a unique code. It has been observed that states assignment strongly influences the complexity of its combinational logic part used to realize the FSM. Traditionally state assignment has been used to optimize the area and delay of the circuit. It can also be used to reduce switching activity for the reduction of the dynamic power. This is illustrated with the help of the following example:



Answer to Questions of Lec-27

State	Encoding
S1	000
S2	111
S3	001
S4	110
S5	101

State	Encoding
S1	000
S2	001
S3	011
S4	010
S5	100



Transitions	Assignment-1	Assignment-2
S1→S1	0.0	0.0
S1→S2	1.5	0.5
S2→S1	1.5	0.5
S2→S3	1.0	0.5
S3→S1	0.5	1.0
S3→S4	1.5	0.5
S4→S1	1.0	0.5
S4→S5	1.0	1.0
S5→S1	2.0	1.0
Total	10.0	5.5

Answer to Questions of Lec-27

Q3. How power dissipation is reduced by partitioning an FSM. Explain with an example.

Ans: The idea is to decompose a large FSM into a several smaller FSMs with smaller number of state registers and combinational blocks. Out of all the FSMs, only the active FSMs receive clock and switching inputs, and the others are idle and consume no dynamic power. This is the basic concept of reducing dynamic power by partitioning an FSM.

Q4. A sequence detector produces a '1' for each occurrence of the input sequence '1001' at its input.

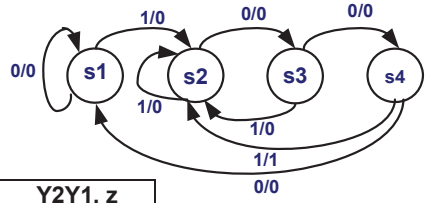
(a) Draw the state-transition diagram of the FSM realizing the sequence detector.

(b) Obtain state table from the state transition diagram.

(c) Realize the FSM using D FFs and a PLA.

Answer to Questions of Lec-27

Ans: 4(a) The state transition diagram is given below:

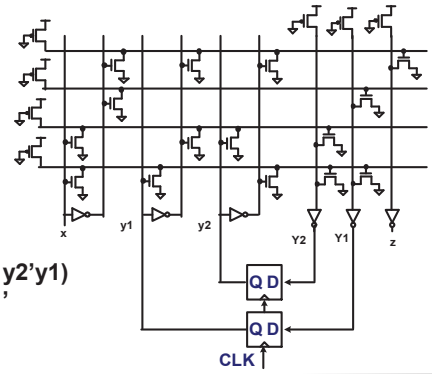


		Y2Y1, z	
PS	y2y1	x =0	X = 1
S1	0 0	00,0	01,0
S2	01	10,0	01,0
S3	1 0	11,0	01,0
S4	1 1	00,0	01,1

Ans: 4(b) The state table is given in the adjacent diagram

Answer to Questions of Lec-27

Ans: 4(c) The FSM using D FFs and a PLA to realize the combinational circuit is given below:.



Y2 = x'(y2y1' + y2'y1)
Y1 = x + x'y2y1'
z = xy2y1

Review Questions of Lec-28

Q1. In what situation the use of sign-magnitude form of number representation instead of 2's complement form is beneficial in terms of power dissipation? Illustrate with an example.

Q2. Explain how the ordering of input signal does affect the dynamic power dissipation on a bus? Illustrate with an example.

Q3. What is glitching power dissipation? How can it be minimized?

Q4. How is it possible to reduce power dissipation by duplicating a resource rather than using it twice?

Answer to Questions of Lec-28

Q1. In what situation the use of sign-magnitude form of number representation instead of 2's complement form is beneficial in terms of power dissipation?

Ans: In most of the signal processing applications, 2's complement is typically chosen to represent numbers. Sign extension causes MSB sign-bits to switch when a signal transitions from +ve to – ve or vice versa. 2's complement can result in significant switching activity when the signals being processed switch frequently around zero. Switching in MSBs can be minimized by using sign-magnitude (S-M) representation.

Answer to Questions of Lec-28

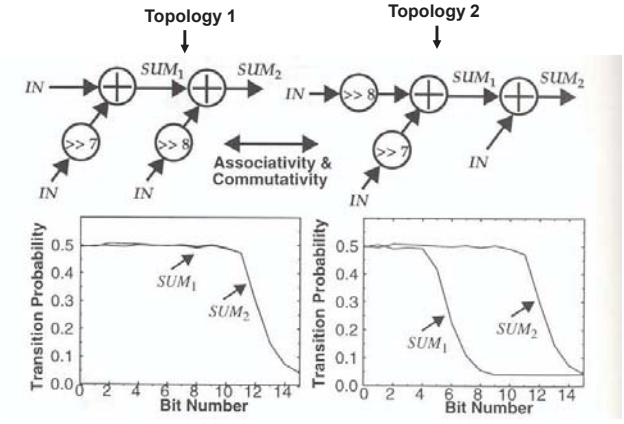
Q2. Explain how the ordering of input signal does affect the dynamic power dissipation on a bus? Illustrate with an example.

Ans: Let us consider two alternative topologies to implement the required 2 additions

Topology #1
- SUM1 = IN + (IN>>7)
- SUM2 = SUM1 + (IN>>8)

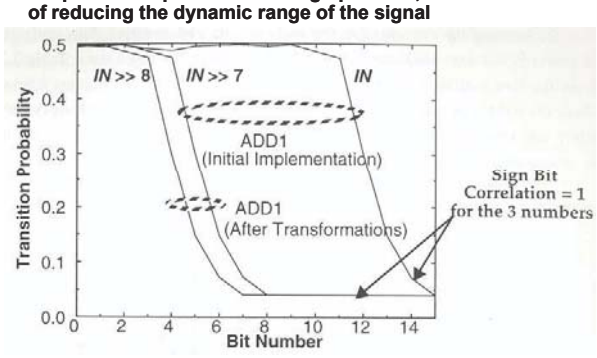
Topology #2
- SUM1 = (IN>>7) + (IN>>8)
- SUM2 = SUM1 + IN

Answer to Questions of Lec-28



Answer to Questions of Lec-28

Shift operation represent a scaling operation, which has the effect of reducing the dynamic range of the signal

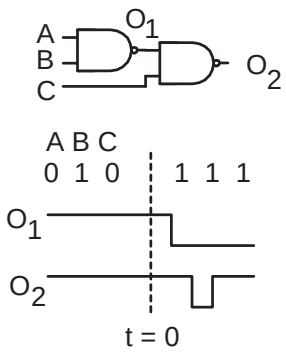


- Transition Probability for 3 signals – IN, IN>>7, IN>>8

Answer to Questions of Lec-28

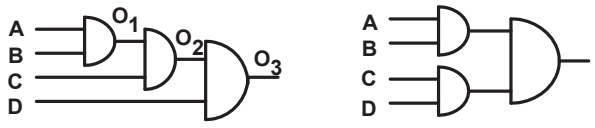
Q3. What is glitching power dissipation? How can it be minimized?

Ans: In digital circuits glitch is an undesired transition that occurs before the signal settles to its intended value. In other words, glitch is an electrical pulse of short duration that is usually the result of a fault or design error. As shown in the adjacent diagram, there is some delay at the output O_1, which results in a glitch at output O_2. As there is some capacitance associated with the output O_2, it leads to switching power dissipation. This switching power dissipation arising out of a glitch is known as glitching power dissipation.



Answer to Questions of Lec-28

These "Extra" transitions can be minimized by
- Balancing all signal paths
- Reducing logic depth



Realization of A.B.C.D in cascaded form where there is possibility of glitch.

Balanced realization of the same function with lesser possibility of glitch

Answer to Questions of Lec-28

Q4. What are the potential logic styles for the realization of low power high performance CMOS circuits?

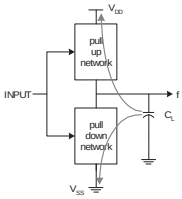
Ans: Potential Logic Styles are:
- Static CMOS Logic
- Dynamic CMOS Logic
- Pass-Transistor Logic (PTL)

Ref: D. Samanta, Ajit Pal, *Logic Styles for High Performance and Low Power*, Proceedings of the 12th International Workshop on Logic and Synthesis, 2003 (IWLS-2003), pp. 355-362, May 2003

Answer to Questions of Lec-28

Static CMOS Logic

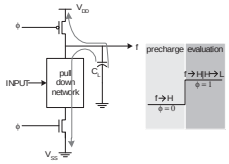
- Advantages
  - Ease of fabrication
  - Good noise margin
  - Robust
  - Lower switching activity
  - Good input/output decoupling
  - No charge sharing problem
  - Availability of matured logic synthesis tools and techniques



- Disadvantages
  - Larger number of transistors (larger chip area and delay)
  - Spurious transitions (glitch) due to finite propagation delays leading to extra power dissipation and incorrect operation
  - Short circuit power dissipation
  - Weak output driving capability
  - Large number of standard cells requiring substantial engineering effort for technology mapping

Answer to Questions of Lec-28

Dynamic CMOS Logic



- Advantages
  - Combines the advantages of low power of static CMOS and low chip area of pseudo-nMOS
  - Reduced number of transistors compared to static CMOS (n+2 versus 2n)
  - Faster than static CMOS logic
  - No short circuit power dissipation
  - No spurious transition and glitching power dissipation
- Disadvantages
  - Higher switching activity
  - Not as robust as static CMOS logic
  - Clock skew problem in cascaded realization
  - Suffers from charge sharing problem
  - Mature synthesis tool not available

Answer to Questions of Lec-28

Pass-Transistor Logic

- Advantages
  - Lower area due to smaller number of transistors and smaller input loads
  - Ratio-less PTL allows minimum dimension transistors and hence makes area efficient circuit realization
  - No short circuit current leading to lower power dissipation
- Disadvantages
  - Increased delay due to long chain pf pass-transistors
  - Multi-threshold voltage drop
  - Dual-rail logic to provide all signals in complementary form
  - There is possibility of sneak path

Review Questions of Lec-29

Q1. Why leakage power is an important in the deep sub micron technology?

Q2. What challenges we face in using threshold voltage scaling to minimize leakage power dissipation?

Q3. Explain how transistor staking can be used to reduce leakage power dissipation.

Q4. Distinguish between standby and runtime leakage power. Why runtime leakage power is becoming important in the present day context?

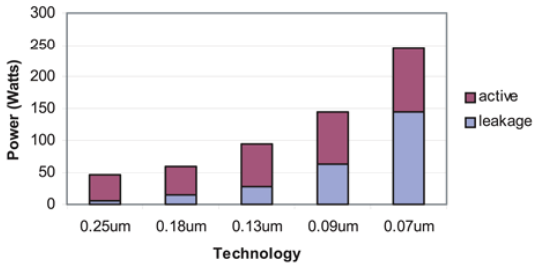
Q5. Compare between VTCMOS and MTCMOS for leakage power reduction.

Q6. Give the advantages and limitations of MTCMOS approach.

Answer to Questions of Lec-29

Q1. Why leakage power is an important in the deep sub micron technology?

Ans: As shown in the diagram, the leakage power component is increasing at a higher rate compared to dynamic power as we move towards deep sub micron technology. So, it has become an important issue.

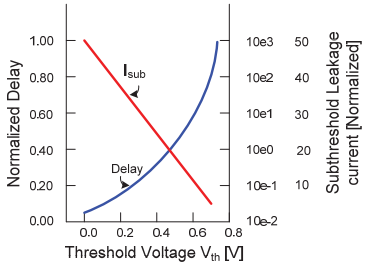


Source: Microprocessor power consumption, Intel

Answer to Questions of Lec-29

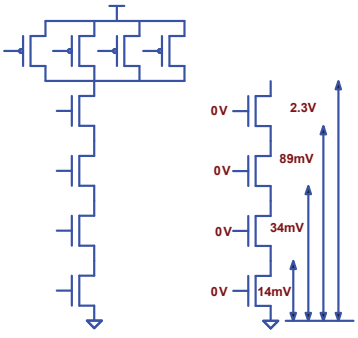
Q2. What challenges we face in using threshold voltage scaling to minimize leakage power dissipation?

Ans: As supply voltage is scaled down to reduce power dissipation, the threshold voltage is also scaled down to maintain performance. As we do so, the subthreshold leakage current increases leading to high power dissipation. So, the challenge is to maintain performance with lower power dissipation.



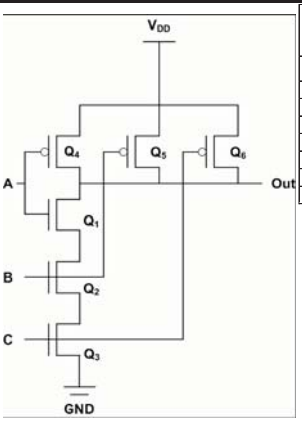
Answer to Questions of Lec-29

Q3. Explain how transistor staking can be used to reduce leakage power dissipation.



Ans: When more than one transistor is in series in a CMOS circuit, the leakage current has strong dependence on the number of turned off transistor. This is known as stacking effect. The highest leakage current is 99 times that of the lowest leakage current

Answer to Questions of Lec-29



State (ABC)	Leakage Current (*nA*)	Leaking Transistors
000	0.095	*Q*1, *Q*2, *Q*3
001	0.195	*Q*1,*Q*2
010	0.195	*Q*1,*Q*3
011	1.874	*Q*1
100	0.184	*Q*2,*Q*3
101	1.220	*Q*2
110	1.140	*Q*3
111	9.410	*Q*4,*Q*5,*Q*6

A suitable input combination is used such that the reduction in leakage current due to stacking effect is maximized.

Answer to Questions of Lec-29

Q4. Distinguish between standby and runtime leakage power. Why runtime leakage power is becoming important in the present day context?

Ans: Standby leakage power dissipation takes place when the circuit is not in use, i.e. inputs do not change and clock is not applied. On the other hand, runtime leakage power dissipation takes place when the circuit is being used.

Answer to Questions of Lec-29

Q5. Compare between VTCMOS and MTCMOS for leakage power reduction.

Ans: In case of VTCMOS, basic principle is to adjust threshold voltage by changing substrate bias. Transistors initially have low V_{th} during normal operation and substrate bias is altered using substrate bias control circuit. The threshold is increased by using reverse body bias when the circuit is not in use. Effective in reducing leakage power dissipation in standby mode and it involved additional area and higher circuit complexity. So, it is a post-silicon approach.

On the other hand, in case of MTCMOS approach MOS transistors of multiple threshold voltages are fabricated in which a power gating transistor is inserted in the stack between the logic transistors and either power or ground, thus creating a virtual supply rail or a virtual ground rail, respectively. The logic block contains all low-Vth transistors for fastest switching speeds while the switch transistors, header and footer, are built using high-Vth transistors to minimize the leakage power dissipation. So, it is a pre-silicon approach.

Review Questions of Lec-30

Q1. Explain how standby power is reduced in MTCMOS technique. Discuss its advantage and disadvantages.

Q2. Explain how multiple threshold voltage transistors can be used to realize low voltage high performance circuits requiring low power (i) only in the standby mode, (ii) both in standby and active mode.

Q3. Compare the advantage and disadvantages of fine-grained and coarse-grained power gating approaches.

Q4. Distinguish between local and global power gating and compare their advantage and disadvantages.

Q5. Why is it necessary to isolate a signal as it goes from one voltage domain to another voltage domain? Explain how is it implemented?

Q6. Explain different approaches of state retention.

Answer to Questions of Lec-30

Q1. Compare and contrast clock gating versus power gating approaches.

Ans: Clock gating minimizes dynamic power by stopping unnecessary transitions, but power gating minimizes leakage power by inserting a high-Vt transistor in series with the low-vt logic blocks.

Q2. Compare the advantage and disadvantages of fine-grained and coarse-grained power gating approaches.

Ans: In case of fine-grained approach the switch is placed locally inside each standard cell and the switch must be designed to supply worst case current so that it does not impact performance. In this approach, the area overhead of each cell is significant (2X-4X). On the other hand, in case of coarse-grained approach, a block of gates has its power switched by a collection of switched cells. In this case the sizing is very difficult, but it has significantly less area overhead than that of fine grain approach. It is a preferred approach because of lesser area overhead.

Answer to Questions of Lec-30

Q3. Distinguish between local and global power gating and compare their advantage and disadvantages.

Ans: Global power gating refers to a logical topology in which multiple switches are connected to one or more blocks of logic, and a single virtual ground is shared in common among all the power gated logic blocks. This topology is effective for large blocks (coarse-grained) in which all the logic is power gated, but is less effective for physical design reasons, when the logic blocks are small. It does not apply when there are many different power gated blocks, each controlled by different sleep enable signals.

On the other hand, local power gating refers to a logical topology in which each switch singularly gates its own virtual ground connected to its own group of logic. This arrangement results in multiple segmented virtual grounds for a single sleep domain.

Answer to Questions of Lec-30

Q4. Why is it necessary to isolate a signal as it goes from one voltage domain to another voltage domain? Explain how is it implemented?

Ans: There is no guarantee that the power gated blocks will fully charge or discharge. As a consequence, the outputs of powered down blocks may result in crowbar currents in the powered up blocks. To overcome this problem, it is necessary to isolate a power gated block from a non power gated block.

Answer to Questions of Lec-30

Q5. Explain different approaches of state retention.

Ans: Given a power switching fabric and an isolation strategy, it is possible to power gate a block of logic, but unless a retention strategy is employed, all state information is lost when the block is powered down. To resume its operation on power up, the block must either have its state restored from an external source or build up its state from the reset condition. In either case, the time and power required can be significant. The following three approached can be used

- A software approach based on reading and writing registers
- A scan-based approach based on the re-use of scan chains to store state off chip
- A register-based approach that uses retention registers

Review Questions of Lec-31

Q1. How supply voltage scaling leads to run time leakage power reduction?

Q2. How can you combine power gating with dynamic voltage scaling to reduce power dissipation?

Q3. Explain the basic concept of dual Vt assignment for reduction of leakage power.

Q4. Distinguish between delay-constrained and energy-constrained dual Vt assignment approaches.

Q5. Explain how the threshold voltage can be dynamically adjusted to reduce leakage power dissipation.

Answer to Questions of Lec-31

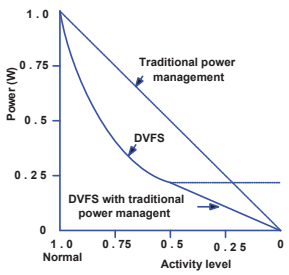
Q1. How supply voltage scaling leads to run time leakage power reduction.

Ans: Supply voltage reduction not only leads to the reduction of dynamic power, it also leads to the reduction of leakage power. The subthreshold leakage due to GIDL and DIBL decreases as supply voltage is scaled down. It has also been demonstrated that the supply voltage scaling impacts in the orders of V^3 and V^4 on subthreshold leakage and gate leakage, respectively.

Answer to Questions of Lec-31

Q2. How can you combine power gating with dynamic voltage scaling to reduce power dissipation?

Ans: As the supply voltage along the frequency is lowered using the DVFS, the supply voltage hits the lower limit and the curve flattens out and the supply voltage cannot be further lowered. At this point it is more efficient to switch over to traditional power management, i.e. the supply voltage is turned on and off depending on the performance requirement. This is how one can combine the DVFS and traditional power management approach as shown in the diagram.



Answer to Questions of Lec-31

Q3. Explain the basic concept of dual Vt assignment for the reduction of leakage power.

Ans: This is based on the observation that all gates are not on the critical path when the circuit is represented with the help of a directed acyclic graph (DAG). So, gates on the critical path can be realized using Low-V_{th} transistors for high performance and the gates on the noncritical path are realized using high-V_{th} transistors to reduce leakage power. This is the basic concept of dualVt assignment.

Answer to Questions of Lec-31



Q4. Distinguish between delay-constrained and energy-constrained dual Vt assignment approaches.

Ans: In delay-constrained approach, no compromise is made on performance. So, dual-Vt assignment is done such that there is no performance degradation. On the other hand, in energy constrained approach some compromise in performance is made, say 10% to 15%, to achieve larger reduction in the leakage power.

Answer to Questions of Lec-31



Q5. Explain how the threshold voltage can be dynamically adjusted to reduce leakage power dissipation.

Ans: Just like dynamic the Vdd scaling scheme, a dynamic Vth scheme (DVTS) can be used to reduce runtime leakage power in sub-100-nm generations, where leakage power is significant portion of the total power at runtime. When the workload is less than the maximum, the processor is operated at lower clock frequency. Instead of reducing the supply voltage, the DVTS hardware raises the threshold voltage using reverse body biasing to reduce runtime leakage power. Just enough throughput is delivered for the current workload by dynamically adjusting the Vth in an optimal manner to maximize leakage power reduction.

A simpler scheme is called Vth-hopping which dynamically switches between only two threshold voltages; Low-Vt and High-Vt as the frequency controller generates either FCLK or FCLK/2, respectively

Review Questions of Lec-32



Q1. Distinguish between conventional charging (used in static CMOS circuits) and adiabatic charging of a load capacitance.

Q2. Explain how dynamic power dissipation is minimized using adiabatic switching?

Q3. Prove that the charging of a capacitor C in *n* steps to a voltage Vdd instead of a conventional single-step charging reduces the power dissipation by a factor of *n*.

Q4. Realize a 2-input OR/NOR gate using positive feedback adiabatic logic (PFAL) circuit. Explain its operation.

Answer to Questions of Lec-32



Q1. Distinguish between conventional charging (used in static CMOS circuits) and adiabatic charging of a load capacitance.

Ans: Switching power dissipation in static CMOS circuit with capacitive load C_L has a lower limit of $C_L V_{dd}^2/2$. On the other hand, charge moves efficiently from power supply to the load capacitance by using slow, constant-current charging. Reversing the current source will cause the energy to flow from the load capacitance back into the power supply. The power supply must be so designed to retrieve the energy fed back to it. Adiabatic-switching circuits require non-constant, non-standard power supply with time-varying voltage. This supply is called "Pulsed-power supplies".

Answer to Questions of Lec-32



Q2. Explain how dynamic power dissipation is minimized using adiabatic switching?

Ans: In adiabatic switching, a time-dependent current source, I(t) is used to charge the capacitance C through a resistor R. For T>2RC, the dissipated energy is smaller than the conventional lower limit $C_L V_{dd}^2/2$. The dissipation can be made arbitrarily small by extending the charging time T. As the dissipated energy is proportional to R, a smaller R results in a lower dissipation unlike conventional case.

Answer to Questions of Lec-32



Q3. Prove that the charging of a capacitor C in *n* steps to a voltage Vdd instead of a conventional single-step charging reduces the power dissipation by a factor of *n*.

Ans: Energy dissipation for single step charging $E = C_L V^2 / 2$.

Energy dissipation in each of n step charging is equal to $Estep = C_L V^2 / 2n^2$ .The total energy dissipation in N step charging equal to $N.Estep = N. (C_L V^2 / 2n^2) = C_L V^2 / 2n$. Therefore, $Estep / E = 1/n$. Therefore, the power dissipation reduces by a factor of n.

Answer to Questions of Lec-32

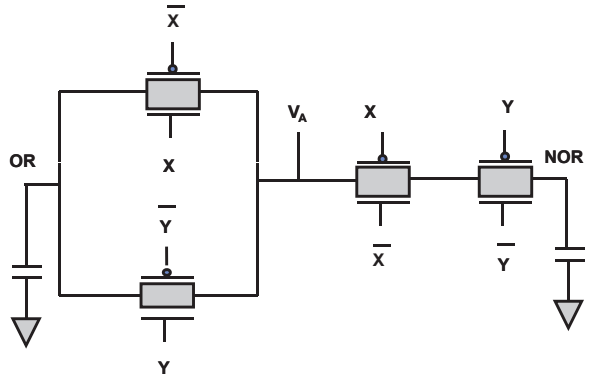
Q4. Realize a 2-input OR/NOR gate using positive feedback adiabatic logic (PFAL) circuit. Explain its operation.

Ans: Perform the following steps:

1. Replace each of the PMOS and NMOS devices in the pull-up and pull-down networks with T-gates.
2. Use expanded pull-up network to drive the true output. Use expanded pull-down network to drive the complementary output.
3. Both networks in the transformed circuit are used to charge and discharge the load capacitance.
4. Replace DC Vdd by a pulsed power supply with varying voltage to allow adiabatic operation.

The realization is shown in the next slide

Answer to Questions of Lec-32



Review Questions of Lec-33

Q1. Write a short note on "Battery-driven system design".

Q2. Explain *rate capacity effect for* rechargeable batteries.

Q3 Explain *recovery effect for* rechargeable batteries.

Q3. What is the 'non-increasing profile effect' of a battery?

Q4. Explain the basic steps of battery aware task scheduling. How does it improves the lifetime of a Battery?

Q5. Why is the reverse body biasing important to extend the battery life in the present day context?

Answer to Questions of Lec-33

Q1. Write a short note on "Battery-driven system design".

Ans: In recent years there large proliferation of portable computing and communication equipment, such as laptops, palmtops, cell-phones, etc. and the growth rate of these portable equipment is very high. The complexity of these devices is also increasing with time, leading to larger energy consumption. As these devices are battery operated, battery life is of primary concern. Unfortunately, the battery technology has not kept up with the energy requirement of the portable equipment. Moreover, the commercial success of these products depend on weight, cost and battery life. Low power design methodology is very important to make these battery-operated devices commercially viable.

Answer to Questions of Lec-33

Q2. Explain *rate capacity effect for* rechargeable batteries.

Ans: Dependency between the actual capacity and the magnitude of the discharge current depends on the availability of active region. When discharge rate is high, surface of the cathode gets coated with insoluble compound. This prevents access to many active areas and consequent reduction of actual capacity of the battery. As a result, a higher rate of discharge leads to a lower available capacity. This is known as Rate Capacity effect.

Q3 Explain *recovery effect for* rechargeable batteries.

Ans: Availability of charge carriers D\depends of the concentration of positively charged ions near the cathode. When heavy current is drawn, rate at which positively charged ions consumed at the cathode is more than supplied. This improves as the battery is kept idle for some duration. As a consequence, the battery voltage recovers in idle periods.

Answer to Questions of Lec-33

Q3. What is the 'non-increasing profile effect' of a battery?

Ans: It has been experimentally verified that if the tasks consuming higher power are scheduled first followed by tasks with decreasing power consumption, then energy available in the battery is larger compared to other schedules. This is known as non-increasing profile effect.

Answer to Questions of Lec-33



Q4. Explain the basic steps of battery aware task scheduling. How does it improves the lifetime of a Battery?

Ans: There are three steps. In the first step, an early deadline first (EDF) based schedule is made, provided the task dependencies are not violated. In the second step, the task schedule is modified by scheduling the tasks in the non increasing order of the current loads provided the deadlines and the task dependencies are not violated. In the third step, starting from the last task, the slack obtained at the end of the task is utilized to get the optimal pair of supply voltage and the body bias voltage. This procedure is followed until either there is no more slack, or further scaling down not possible.

Answer to Questions of Lec-33



Q5. Why is the reverse body biasing important to extend the battery life in the present day context?

Ans: With the advancement of technology, as the process technology further gets lower, the energy due to static power becomes more significant, and the algorithm using RBB to reduce the leakage current provides larger saving in power dissipation.

Review Questions of Lec-34



Q1. What is the limitation of contemporary CAD tools?

Q2. Give a summary of the benefits and impacts of the different low power techniques.

Q3. What is provided by UPF?

Q4. What are the key features of Eclypse, the low-power CAD tool of Synopsis?

Answer to Questions of Lec-34



Q1. What is the limitation of contemporary CAD tools?

Ans: In RTL coding there is no provision to use Multi-Vt , Multi-Vdd, Body biasing and power gating in RTL synthesis. So, the static power reduction techniques cannot be used. As supply voltage and the operating frequency are also not handled at the RTL level, the dynamic power can be reduced primarily by reducing the switching activity α. Commonly used techniques in RTL synthesis to reduce α are:

- Bus encoding
- Clock gating
- FSM state assignment

Q2. Give a summary of the benefits and impacts of the different low power techniques.

Answer to Questions of Lec-34



Ans: The benefits and impacts of different low power techniques is summarized in the following table. These techniques can significantly reduce power consumption in deep submicron chips. However, these techniques traditionally require ad-hoc, time-consuming, risk-prone, and manual verification and implementation approaches, unless automated using CAD tools.

Technique	Dynamic Power Benefit	Static Power Benefit	Design Impact	Verification Impact	Implementation Impact
Clock Gating	Large	Small	Small	Small	Small
Multi-Vdd	Large	Small	Little	Low	Medium
DVFS	Large	Small	Medium	Large	Medium
Multi-Vt	Small	Large	Medium	Small	Medium
Power Gating	Small	Very Large	Medium	Large	Medium

Answer to Questions of Lec-34



Q3. What is provided by UPF?

Ans: UPF provides the ability for electronic systems to be designed with power as a key consideration early in the process. It accomplishes this through the ability to allow the specification of implementation-relevant power information early in the design process — RTL (register transfer level) or earlier. UPF provides a consistent format to specify power-aware design information that cannot be specified in HDL code or when it is undesirable to directly specify within the HDL logic, as doing so would tie the logic specification directly to a constrained power implementation.

Answer to Questions of Lec-34



Q4. What are the key features of Eclypse, the low-power CAD tool of Synopsis?

Ans: Eclypse provides a comprehensive approach – power-aware tools at all levels of design hierarchy starting from early architectural and system-level analysis to verification, RTL synthesis, test, physical implementation and sign-off. This supports the Accellera Unified Power format – an open industry standard to specify power intent and it is backed by the popular "Low Power Methodology Manual" (LPMM).

Review Questions of Lec-35



Q1. How parameter variations impact on circuits and microarchitecture of present day VLSI circuits?

Q2. Why current or future technologies result in two-sided constraints?

Q3. What are the basic approaches for variation tolerant design?

Q4. Explain how you can achieve low power single-Vt circuits using judicious use of sizing?

Q5. Explain how you can use adaptive body biasing to improve yield.

Answer to Questions of Lec-35



Q1. How parameter variations impact on yield of present day VLSI circuits?

- **Ans: Fluctuations are attributed to the manufacturing process (e.g., drifts in L_{eff}, T_{ox}, V_t, or N_{cheff}), which affect circuit yield. For example, with in die variation in L_{eff} can be as high as 50%. 30% delay variation and 20X leakage variation between fast and slow dies have been reported for 0.18µ CMOS process. Low leakage chips with too low frequency must be discarded and high frequency chips with too high leakage must also be discarded. This lwads to reduction in yield.**

Answer to Questions of Lec-35



Q2. Why current or future technologies result in two-sided constraints?

Ans: Lower channel length leads to smaller threshold voltage and larger power dissipation and higher channel length leads to higher threshold voltage and longer delay. This leads to two-sided constraints for current and future technologies.

Q3. What are the basic approaches for variation tolerant design?

Ans: Basic approaches are:

(a) To reduce the sources of variations, (b) Reduce the effects of variation at the time of design and (c) Reduce effects of variation after fabrication (post-silicon) such as reverse body biasing.

Answer to Questions of Lec-35



Q4. Explain how you can achieve low power single-Vt circuits using judicious use of sizing?

Ans: This can be done in three steps:

Step-I: Upsize gates on the critical path to reduce delay of the entire circuit.

Step II: Increase threshold voltage of the MOSFETs of the entire circuit to minimize leakage power

Step III: Downsize gates on the non critical path to reduce area and dynamic power

Answer to Questions of Lec-35



Q5. Explain how you can use adaptive body biasing to improve yield.

Ans: Starting with Vt lower than the target voltage, adaptive body biasing can be used to match the mean Vt of all the die samples. The die samples that require larger body bias to match its mean Vt to the target Vt end with higher within-die variation.

Questions on Logic Synthesis

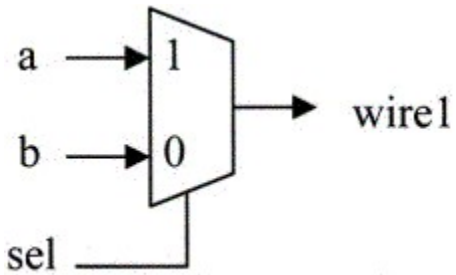
What logic is inferred when there are multiple assign statements targeting the same wire?

It is illegal to specify multiple assign statements to the same wire in a synthesizable code that will become an output port of the module. The synthesis tools give a syntax error that a net is being driven by more than one source.
However, it is legal to drive a three-state wire by multiple assign statements.

What do conditional assignments get inferred into?

Conditionals in a continuous assignment are specified through the "?:" operator. Conditionals get inferred into a multiplexor. For example, the following is the code for a simple multiplexor

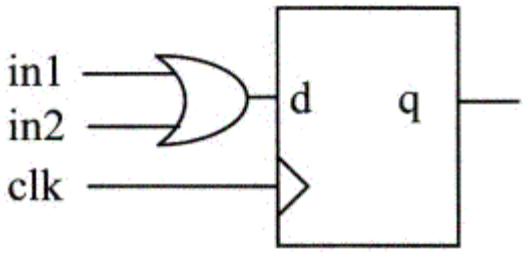
assign wire1 = (sel==1'b1) ? a : b;



What value is inferred when multiple procedural assignments made to the same reg variable in an always block?

When there are multiple nonblocking assignments made to the same reg variable in a sequential always block, then the last assignment is picked up for logic synthesis. For example

```
always @ (posedge clk) begin
out <= in1^in2;
out <= in1 &in2;
out <= in1|in2;
```



In the example just shown, it is the OR logic that is the last assignment. Hence, the logic synthesized was indeed the OR gate. Had the last assignment been the "&" operator, it would have synthesized an AND gate.

1) What is minimum and maximum frequency of dcm in spartan-3 series fpga?

Spartan series dcm's have a minimum frequency of 24 MHZ and a maximum of 248

2)Tell me some of constraints you used and their purpose during your design?

There are lot of constraints and will vary for tool to tool ,I am listing some of Xilinx constraints
a) Translate on and Translate off: the Verilog code between Translate on and Translate off is ignored for synthesis.
b) CLOCK_SIGNAL: is a synthesis constraint. In the case where a clock signal goes through combinatorial logic before being connected to the clock input of a flip-flop, XST cannot identify what input pin or internal net is the real clock signal. This constraint allows you to define the clock net.
c) XOR_COLLAPSE: is synthesis constraint. It controls whether cascaded XORs should be collapsed into a single XOR.
For more constraints detailed description refer to constraint guide.

3) Suppose for a piece of code equivalent gate count is 600 and for another code equivalent gate count is 50,000 will the size of bitmap change?in other words will size of bitmap change if gate count changes?

The size of bitmap is irrespective of resource utilization, it is always the same,for Spartan xc3s5000 it is 1.56MB and will never change.

4) What are different types of FPGA programming modes?what are you currently using ?how to change from one to another?

Before powering on the FPGA, configuration data is stored externally in a PROM or some other nonvolatile medium either on or off the board. After applying power, the configuration data is written to the FPGA using any of five different modes: Master Parallel, Slave Parallel, Master Serial, Slave Serial, and Boundary Scan (JTAG). The Master and Slave Parallel modes

Mode selecting pins can be set to select the mode, refer data sheet for further details.

5) Tell me some of features of FPGA you are currently using?

I am taking example of xc3s5000 to answering the question .

Very low cost, high-performance logic solution for
high-volume, consumer-oriented applications
- Densities as high as 74,880 logic cells
- Up to 784 I/O pins
- 622 Mb/s data transfer rate per I/O
- 18 single-ended signal standards
- 6 differential I/O standards including LVDS, RSDS
- Termination by Digitally Controlled Impedance
- Signal swing ranging from 1.14V to 3.45V
- Double Data Rate (DDR) support
• Logic resources
- Abundant logic cells with shift register capability
- Wide multiplexers
- Fast look-ahead carry logic
- Dedicated 18 x 18 multipliers
- Up to 1,872 Kbits of total block RAM
- Up to 520 Kbits of total distributed RAM
• Digital Clock Manager (up to four DCMs)
- Clock skew elimination
• Eight global clock lines and abundant routing

6) What is gate count of your project?

Well mine was 3.2 million, I don't know yours.!

7) Can you list out some of synthesizable and non synthesizable constructs?

not synthesizable->>>>
initial
ignored for synthesis.
delays
ignored for synthesis.
events
not supported.
real
Real data type not supported.
time
Time data type not supported.
force and release
Force and release of data types not supported.

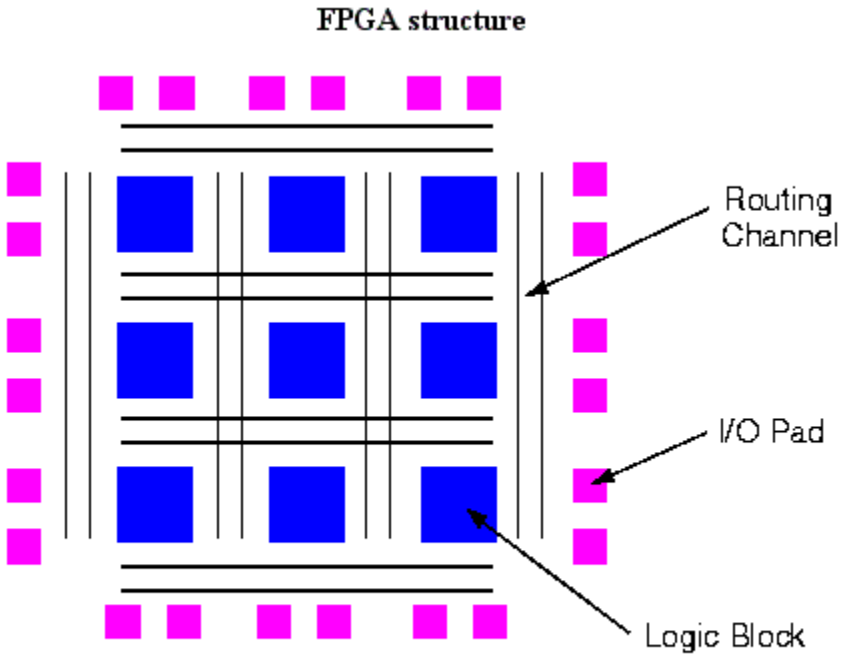
fork join
Use nonblocking assignments to get same effect.
user defined primitives
Only gate level primitives are supported.

synthesizable constructs->>
assign,for loop,Gate Level Primitives,repeat with constant value...

8)Can you explain what struck at zero means?

These stuck-at problems will appear in ASIC. Some times, the nodes will permanently tie to 1 or 0 because of some fault in manufacturing/layout. To avoid that, we need to provide testability in RTL. If it is permanently 1 it is called stuck-at-1 If it is permanently 0 it is called stuck-at-0.

9) Can you draw general structure of fpga?



10) Difference between FPGA and CPLD?

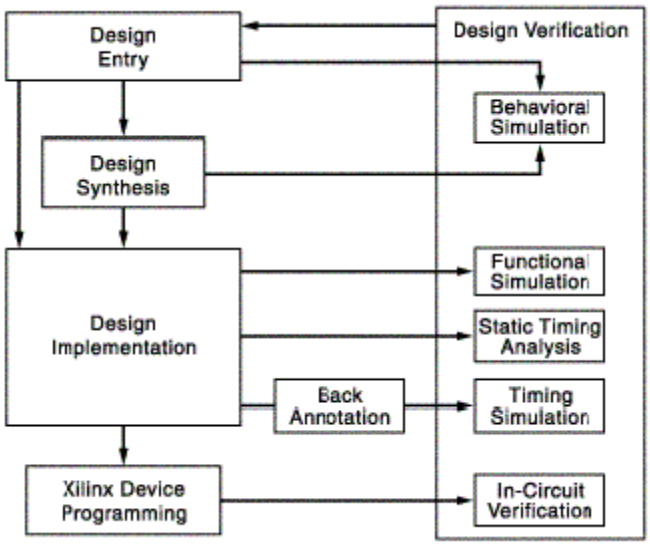
FPGA:
a)SRAM based technology.
b)Segmented connection between elements.
c)Usually used for complex logic circuits.
d)Must be reprogrammed once the power is off.
e)Costly

CPLD:
a)Flash or EPROM based technology.
b)Continuous connection between elements.
c)Usually used for simpler or moderately complex logic circuits.
d)Need not be reprogrammed once the power is off.
e)Cheaper

11) What are dcm's?why they are used?

Digital clock manager (DCM) is a fully digital control system that uses feedback to maintain clock signal characteristics with a high degree of precision despite normal variations in operating temperature and voltage.
That is clock output of DCM is stable over wide range of temperature and voltage , and also skew associated with DCM is minimal and all phases of input clock can be obtained . The output of DCM coming form global buffer can handle more load.

12) FPGA design flow?



Also,Please refer to presentation section synthesis ppt on this site.

13)what is slice,clb,lut?

I am taking example of xc3s500 to answer this question

The Configurable Logic Blocks (CLBs) constitute the main logic resource for implementing synchronous as well as combinatorial circuits.
CLB are configurable logic blocks and can be configured to combo,ram or rom depending on coding style

CLB consist of 4 slices and each slice consist of two 4-input LUT (look up table) F-LUT and G-LUT.

14) Can a clb configured as ram?

YES.

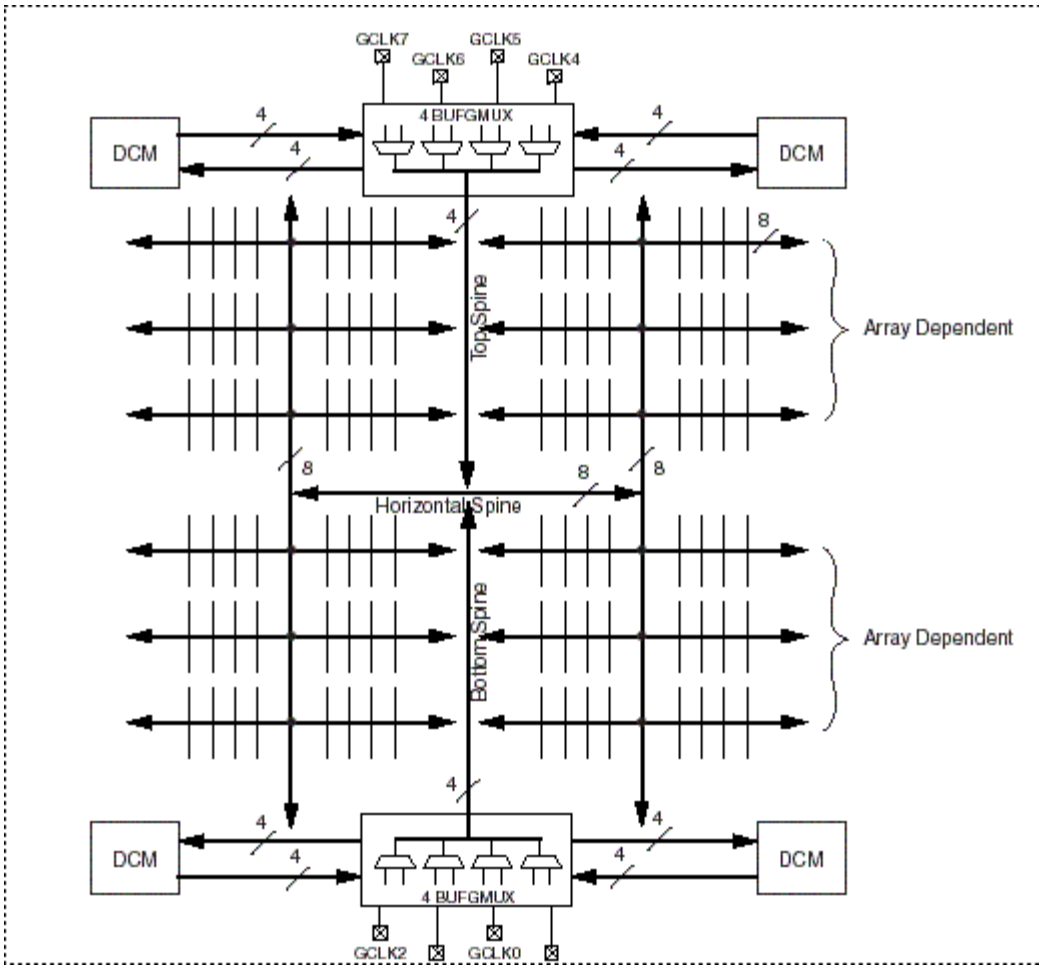
The memory assignment is a clocked behavioral assignment, Reads from the memory are asynchronous, And all the address lines are shared by the read and write statements.

15)What is purpose of a constraint file what is its extension?

The UCF file is an ASCII file specifying constraints on the logical design. You create this file and enter your constraints in the file with a text editor. You can also use the Xilinx Constraints Editor to create constraints within a UCF(extention) file. These constraints affect how the logical design is implemented in the target device. You can use the file to override constraints specified during design entry.

16) What is FPGA you are currently using and some of main reasons for choosing it?

17) Draw a rough diagram of how clock is routed through out FPGA?



18) How many global buffers are there in your current fpga,what is their significance?

There are 8 of them in xc3s5000
An external clock source enters the FPGA using a Global Clock Input Buffer (IBUFG), which directly accesses the global clock network or an Input Buffer (IBUF). Clock signals within the FPGA drive a global clock net using a Global Clock Multiplexer Buffer (BUFGMUX). The global clock net connects directly to the CLKIN input.

19) What is frequency of operation and equivalent gate count of u r project?

20)Tell me some of timing constraints you have used?

21)Why is map-timing option used?

Timing-driven packing and placement is recommended to improve design performance, timing, and packing for highly utilized designs.

22)What are different types of timing verifications?

Dynamic timing:
a. The design is simulated in full timing mode.
b. Not all possibilities tested as it is dependent on the input test vectors.
c. Simulations in full timing mode are slow and require a lot of memory.
d. Best method to check asynchronous interfaces or interfaces between different timing domains.
Static timing:
a. The delays over all paths are added up.
b. All possibilities, including false paths, verified without the need for test vectors.
c. Much faster than simulations, hours as opposed to days.
d. Not good with asynchronous interfaces or interfaces between different timing domains.

23) Compare PLL & DLL ?

PLL:
PLLs have disadvantages that make their use in high-speed designs problematic, particularly when both high performance and high reliability are required.
The PLL voltage-controlled oscillator (VCO) is the greatest source of problems. Variations in temperature, supply voltage, and manufacturing process affect the stability and operating performance of PLLs.

DLLs, however, are immune to these problems. A DLL in its simplest form inserts a variable delay line between the external clock and the internal clock. The clock tree distributes the clock to all registers and then back to the feedback pin of the DLL.
The control circuit of the DLL adjusts the delays so that the rising edges of the feedback clock align with the input clock. Once the edges of the clocks are aligned, the DLL is locked, and both the input buffer delay and the clock skew are reduced to zero.
Advantages:
· precision
· stability
· power management
· noise sensitivity
· jitter performance.

24) Given two ASICs. one has setup violation and the other has hold violation. how can they be made to work together without modifying the design?

Slow the clock down on the one with setup violations..
And add redundant logic in the path where you have hold violations.

25)Suggest some ways to increase clock frequency?

· Check critical path and optimize it.
· Add more timing constraints (over constrain).

· pipeline the architecture to the max possible extent keeping in mind latency req's.

26)What is the purpose of DRC?

DRC is used to check whether the particular schematic and corresponding layout(especially the mask sets involved) cater to a pre-defined rule set depending on the technology used to design. They are parameters set aside by the concerned semiconductor manufacturer with respect to how the masks should be placed , connected , routed keeping in mind that variations in the fab process does not effect normal functionality. It usually denotes the minimum allowable configuration.

27)What is LVs and why do we do that. What is the difference between LVS and DRC?

The layout must be drawn according to certain strict design rules. DRC helps in layout of the designs by checking if the layout is abide by those rules.
After the layout is complete we extract the netlist. LVS compares the netlist extracted from the layout with the schematic to ensure that the layout is an identical match to the cell schematic.

28)What is DFT ?

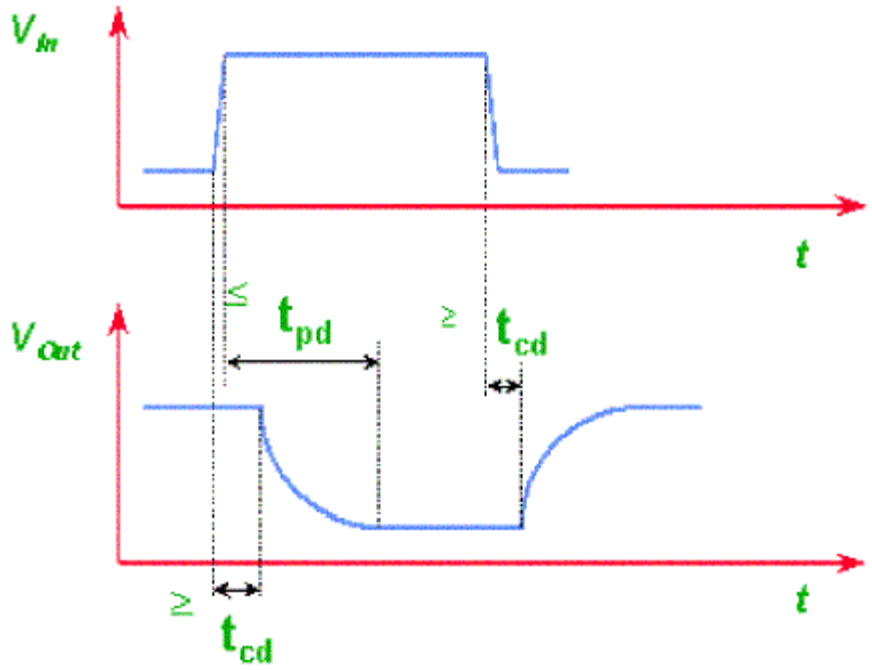
DFT means design for testability. 'Design for Test or Testability' - a methodology that ensures a design works properly after manufacturing, which later facilitates the failure analysis and false product/piece detection
Other than the functional logic,you need to add some DFT logic in your design.This will help you in testing the chip for manufacturing defects after it come from fab. Scan,MBIST,LBIST,IDDQ testing etc are all part of this. (this is a hot field and with lots of opportunities)

29) There are two major FPGA companies: Xilinx and Altera. Xilinx tends to promote its hard processor cores and Altera tends to promote its soft processor cores. What is the difference between a hard processor core and a soft processor core?

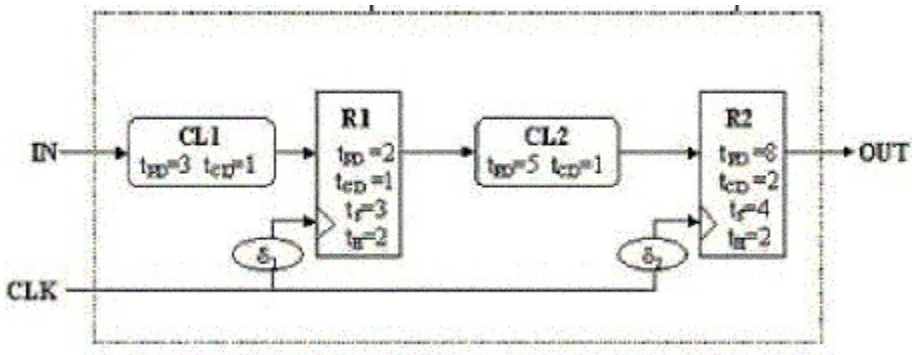
A hard processor core is a pre-designed block that is embedded onto the device. In the Xilinx Virtex II-Pro, some of the logic blocks have been removed, and the space that was used for these logic blocks is used to implement a processor. The Altera Nios, on the other hand, is a design that can be compiled to the normal FPGA logic.

30)What is the significance of contamination delay in sequential circuit timing?

Look at the figure below. tcd is the contamination delay.



Contamination delay tells you if you meet the hold time of a flip flop. To understand this better please look at the sequential circuit below.



The contamination delay of the data path in a sequential circuit is critical for the hold time at the flip flop where it is exiting, in this case R2.
mathematically, th(R2) <= tcd(R1) + tcd(CL2)
Contamination delay is also called tmin and Propagation delay is also called tmax in many data sheets.

31)When are DFT and Formal verification used?

DFT:
· manufacturing defects like stuck at "0" or "1".
· test for set of rules followed during the initial design stage.

Formal verification:

· Verification of the operation of the design, i.e, to see if the design follows spec.
· gate netlist == RTL ?
· using mathematics and statistical analysis to check for equivalence.

32)What is Synthesis?

Synthesis is the stage in the design flow which is concerned with translating your Verilog code into gates - and that's putting it very simply! First of all, the Verilog must be written in a particular way for the synthesis tool that you are using. Of course, a synthesis tool doesn't actually produce gates - it will output a netlist of the design that you have synthesised that represents the chip which can be fabricated through an ASIC or FPGA vendor.

33)We need to sample an input or output something at different rates, but I need to vary the rate? What's a clean way to do this?

Many, many problems have this sort of variable rate requirement, yet we are usually constrained with a constant clock frequency. One trick is to implement a digital NCO (Numerically Controlled Oscillator). An NCO is actually very simple and, while it is most naturally understood as hardware, it also can be constructed in software. The NCO, quite simply, is an accumulator where you keep adding a fixed value on every clock (e.g. at a constant clock frequency). When the NCO "wraps", you sample your input or do your action. By adjusting the value added to the accumulator each clock, you finely tune the AVERAGE frequency of that wrap event. Now - you may have realized that the wrapping event may have lots of jitter on it. True, but you may use the wrap to increment yet another counter where each additional Divide-by-2 bit reduces this jitter. The DDS is a related technique. I have two examples showing both an NCOs and a DDS in my File Archive. This is tricky to grasp at first, but tremendously powerful once you have it in your bag of tricks. NCOs also relate to digital PLLs, Timing Recovery, TDMA and other "variable rate" phenomena

TEXAS INSTRUMENTS: TECHNICAL TEST

Date: 20th December 2003

1. For a CMOS inverter, the transition slope of Vout vs Vin DC characteristics can be increased (steeper transition) by:

a. Increasing W/L of PMOS transistor
b. Increasing W/L of NMOS transistor
c. Increasing W/L of both transistors by the same factor
d. Decreasing W/L of both transistor by the same factor

Ans: c

2. Minimum number of 2-input NAND gates that will be required to implement the function: Y = AB + CD + EF is

a. 4
b. 5
c. 6
d. 7

ans: c

3. Consider a two-level memory hierarchy system M1 & M2. M1 is accessed first and on miss M2 is accessed. The access of M1 is 2 nanoseconds and the miss penalty (the time to get the data from M2 in case of a miss) is 100 nanoseconds. The probability that a valid data is found in M1 is 0.97. The average memory access time is:

a. 4.94 nanoseconds
b. 3.06 nanoseconds
c. 5.00 nanoseconds
d. 5.06 nanoseconds

ans: a

4. Interrupt latency is the time elapsed between:

a. Occurrence of an interrupt and its detection by the CPU
b. Assertion of an interrupt and the start of the associated ISR
c. Assertion of an interrupt and the completion of the associated ISR
d. Start and completion of associated ISR

Ans: d (not confirmed)

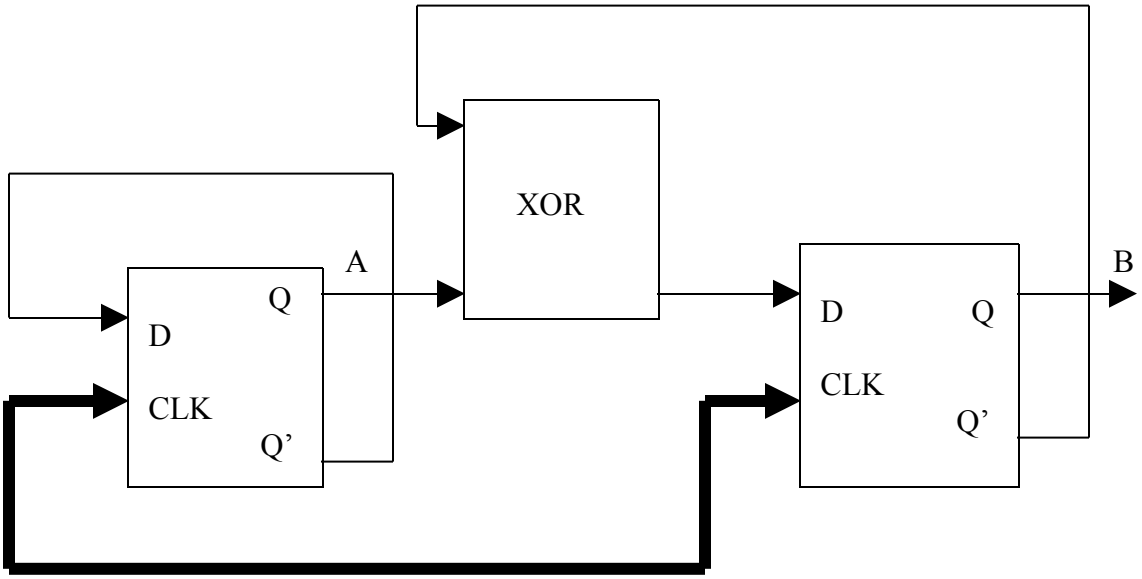
5. Which of the following is true for the function (A.B + A'.C + B.C)

a. This function can glitch and can be further reduced
b. This function can neither glitch nor can be further reduced
c. This function can glitch and cannot be further reduced
d. This function cannot glitch but can be further reduced

Ans: c **This can be reduced further using K-map, don't know abt glich, but it should not glitch**

6. For the two flip-flop configuration below, what is the relationship of the output at B to the clock frequency?

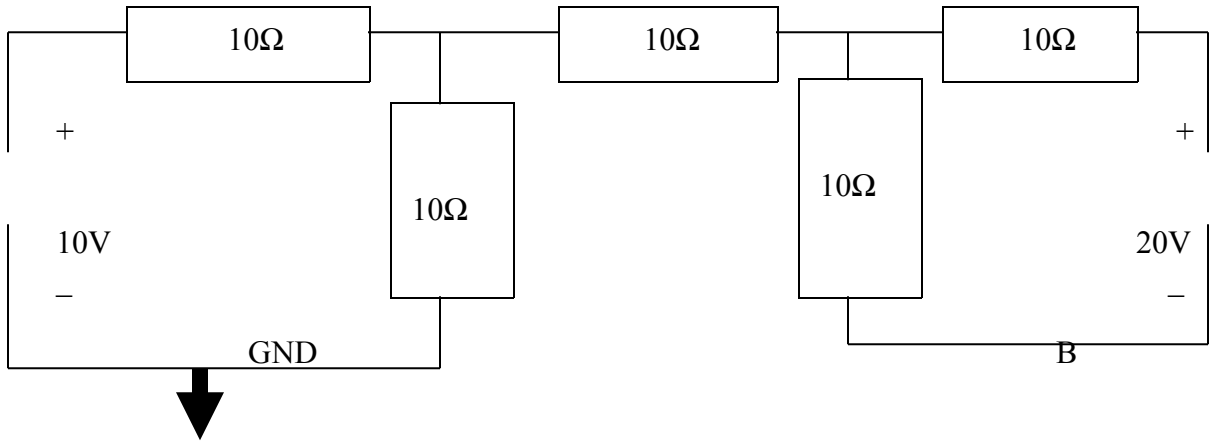
a. Output frequency is 1/4th the clock frequency, with 50% duty cycle
b. Output frequency is 1/3rd the clock frequency, with 50% duty cycle
c. Output frequency is 1/4th the clock frequency, with 25% duty cycle
d. Output frequency is equal to the clock frequency



Ans: a

7. The voltage on Node B is:

a. 0
b. 10
c. –10
d. –5



Ans: d

8. A CPU supports 250 instructions. Each instruction op-code has these fields:

- The instruction type (one among 250)
- A conditional register specification
- 3 register operands
- Addressing mode specification for both source operands

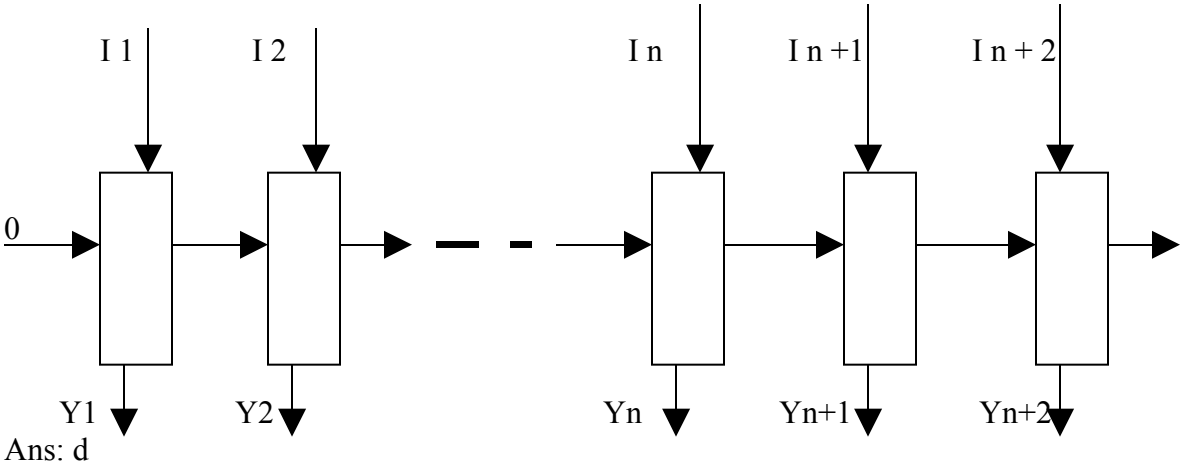
The CPU has 16 registers and supports 5 addressing modes. What is the instruction op-code length in bits?

a. 32
b. 24
c. 30
d. 36

ans: don't know

9. In the iterative network shown, the output Yn of any stage N is 1 if the total number of 1s at the inputs starting from the first stage to the Nth stage is odd. (Each identical box in the iterative network has two inputs and two outputs). The optimal logic structure for the box consists of:

a. One AND gate and one NOR gate
b. One NOR gate and one NAND gate
c. Two XNOR gates
d. One XOR gate



Ans: d

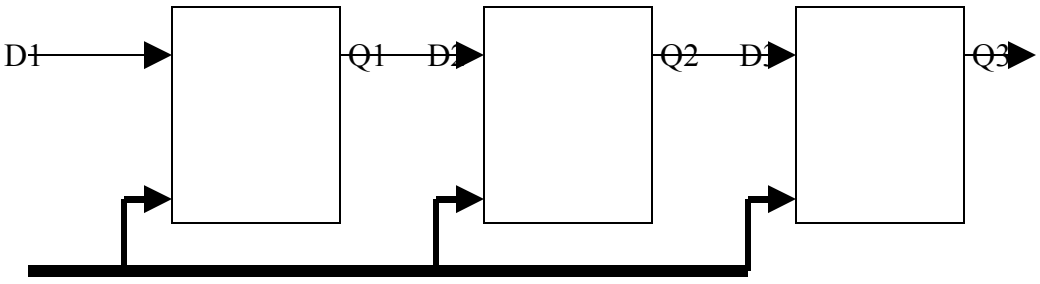
10. Consider a circuit with N logic nets. If each net can be stuck-at either values 0 and 1, in how many ways can the circuit be faulty such that only one net in it can be faulty, and such that up-to all nets in it can be faulty?

a. 2 and 2N
b. N and 2^N
c. 2N and 3^N-1
d. 2N and 3N

ans: 2N and 2^N (no match) see it .

sorry , no idea abt this

11. In the circuit shown, all the flip-flops are identical. If the set-up time is 2 ns, clock->Q delay is 3 ns and hold time is 1 ns, what is the maximum frequency of operation for the circuit?



a. 200 MHz
b. 333 MHz
c. 250 MHz
d. None of the above

Ans: a

12. Which of the following statements is/are true?

I. Combinational circuits may have feedback, sequential circuits do not.
II. Combinational circuits have a 'memory-less' property, sequential circuits do not.
III. Both combinational and sequential circuits must be controlled by an external clock.

a. I only
b. II and III only
c. I and II only
d. II only

Ans: d

13. Consider an alternate binary number representation scheme, wherein the number of ones M, in a word of N bits, is always the same. This scheme is called the M-out-of-N coding scheme. If M=N/2, and N=8, what is the efficiency of this coding scheme as against the regular binary number representation scheme? (As a hint, consider that the number of unique words represent able in the latter representation with N bits is 2^N. Hence the efficiency is 100%)

a. Close to 30%

b. Close to 50%
c. Close to 70%
d. Close to 100%

Ans: a

14. A CPU supports 4 interrupts- I1, I2, I3 and I4. It supports priority of interrupts. Nested interrupts are allowed if later interrupt is higher priority than previous one. During a certain period of time, we observe the following sequence of entry into and exit from the interrupt service routine:
I1-start---I2-start---I2-end---I4-start---I3-start---I3-end---I4-end---I1-end
From this sequence, what can we infer about the interrupt routines?

a. I3 > I4 > I2 > I1
b. I4 > I3 > I2 > I1
c. I2 > I1; I3 > I4 > I1
d. I2 > I1, I3 > I4 > I2 > I1

Ans: c

15. I decide to build myself a small electric kettle to boil my cup of tea. I need 200 ml of water for my cup of tea. Assuming that typical tap water temperature is 25 C and I want the water boiling in exactly one minute, then what is the wattage required for the heating element?
[Assume: Boiling point of water is 100 C, 1 Calorie (heat required to change 1 gm of water by 1 C)= 4 joules, 1 ml of water weighs 1 gm.]

a. Data given is insufficient
b. 800 W
c. 300 W
d. 1000 W
e. 250 W

ans: d

16. The athletics team from REC Trichy is traveling by train. The train slows down, (but does not halt) at a small wayside station that has a 100 mts long platform. The sprinter (who can run 100 mts in 10 sec) decides to jump down and get a newspaper and some idlis. He jumps out just as his compartment enters the platform and spends 5 secs buying his newspaper that is at the point where he jumped out. He then sprints along the platform to buy idlis that is another 50 mts. He spends another 5 secs buying the idlis. He is now just 50 mts from the other end of the platform where the train is moving out. He begins running in the direction of the train and the only other open door in his train is located 50 mts behind the door from where he jumped. At what(uniform) speed should the train be traveled if he just misses jumping into the open door at the very edge of the platform? Make the following assumptions

- He always runs at his peak speed uniformly
- The train travels at uniform speed
- He does not wait (other than for the idlis & newspaper) or run baclwards

a. Data given is insufficient
b. 4 m/s
c. 5 m/s

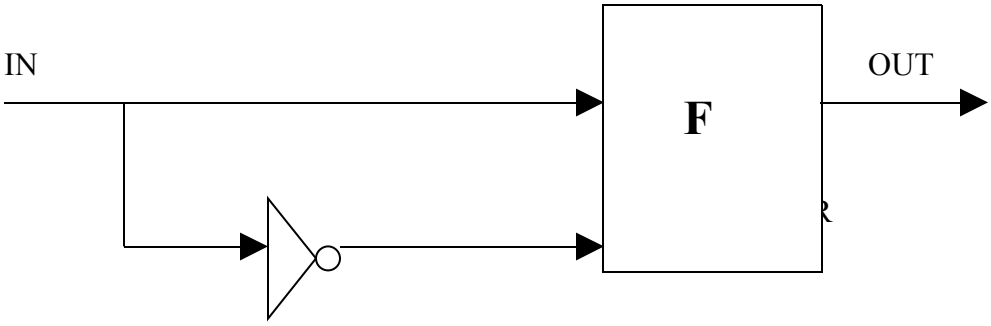
d. 7.5 m/s
e. 10 m/s

ans: c

17. State which of the following gate combinations does not form a universal logic set:
a. 2-input AND + 2-input OR
b. 2-to-1 multiplexer
c. 2-input XOR + inverter
d. 3-input NAND

ans: a

18. For the circuit shown below, what should the function F be, so that it produces an output of the same frequency (function F1), and an output of double the frequency (function F2).

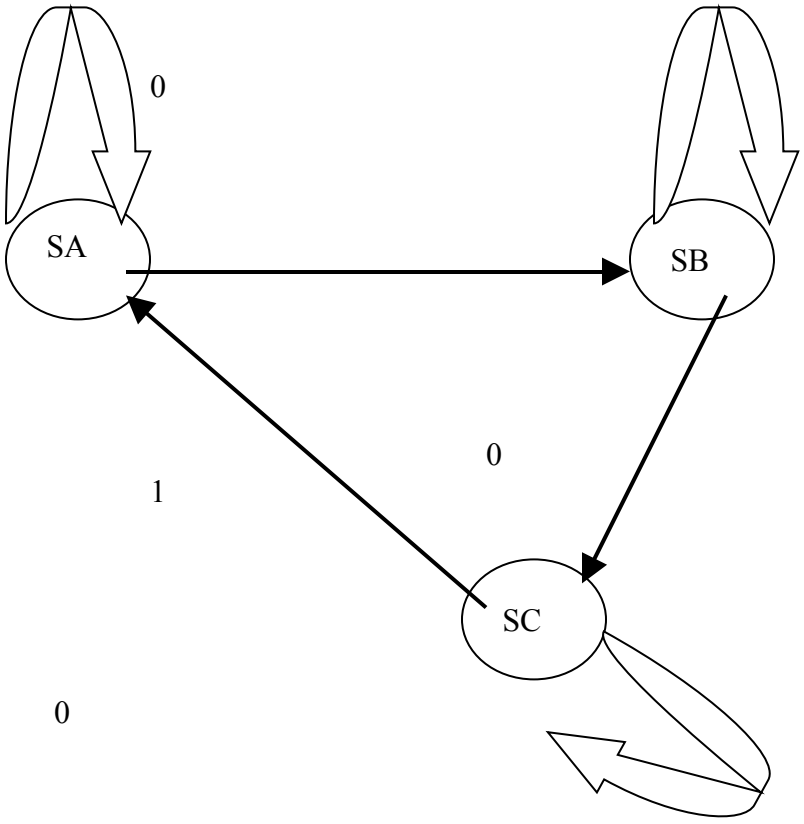


a. F1= NOR gate and F2= OR gate
b. F1=NAND gate and F2= AND gate
c. F1=AND gate and F2=XOR gate
d. None of the above

Ans: c

19. The FSM (finite state machine) below starts in state Sa, which is the reset state, and detects a particular sequence of inputs leading it to state Sc. FSMs have a few characteristics. An autonomous FSM has no inputs. For a Moore FSM, the output depends on the present state alone. For a Mealy FSM, the output depends on the present state as well as the inputs. Which of the statements best describes the FSM below?

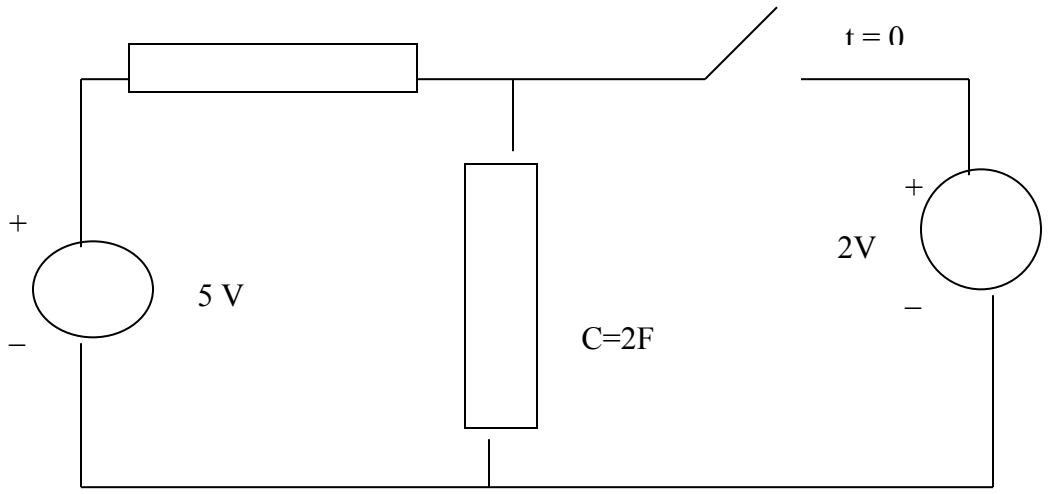
a. It has two states and is autonomous
b. The information available is insufficient
c. It is a Mealy machine with three states
d. It is a Moor machine with three states



Ans :d

20. In the circuit given below, the switch is opened at time t=0. Voltage across the capacitor at t=infinity is:

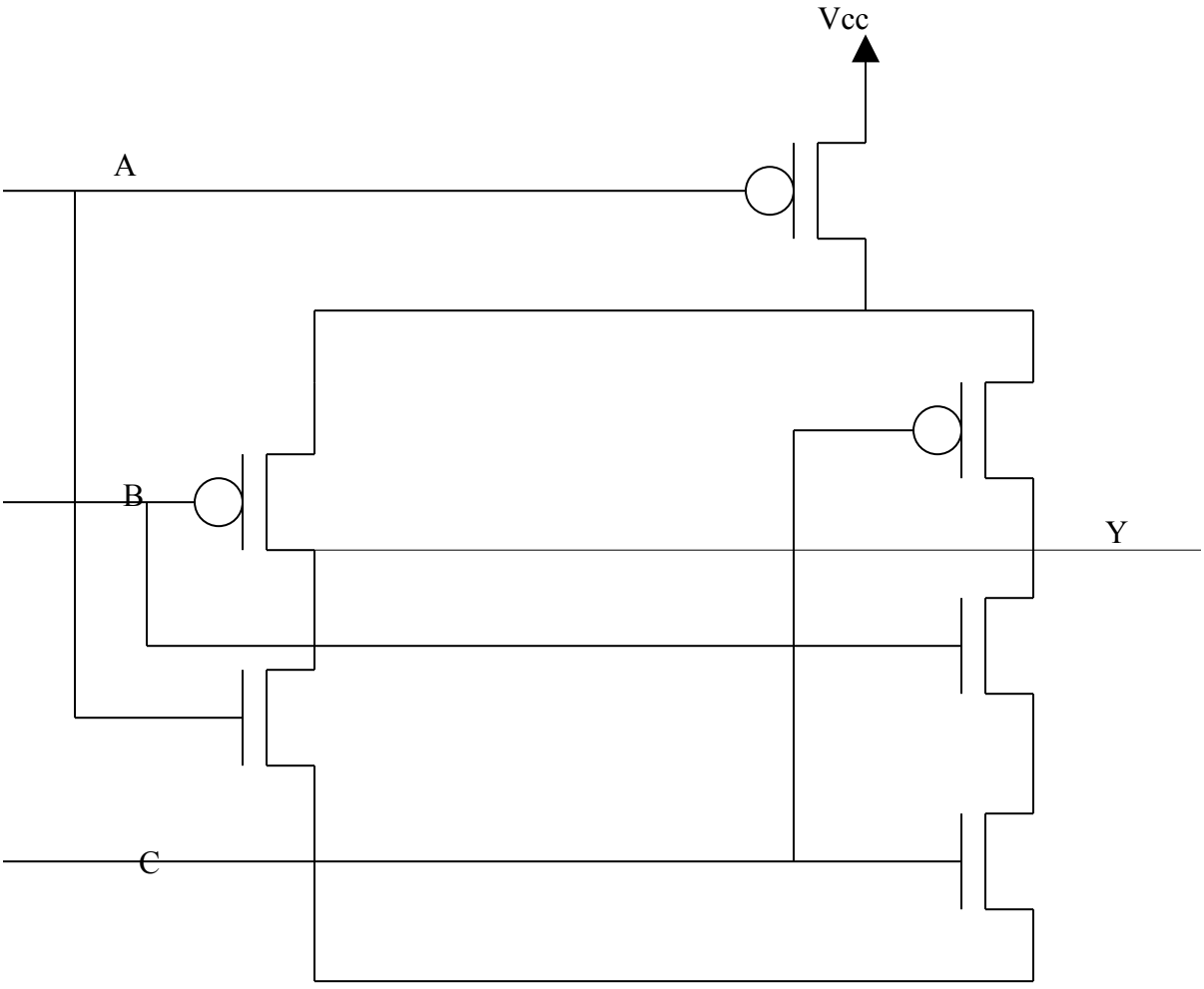
a. 2V
b. 3V
c. 5V
d. 7V



Ans: c

21. What is the functionality represented by the following circuit?

a. y= ! (b+ac)
b. y= ! (a+bc)
c. y= ! (a(b+c))
d. y= ! (a+b+c)



Ans: b

22. The value (0xdeadbeef) needs to stored at address 0x400. Which of the below ways will the memory look like in a big endian machine:

	0x403	0x402	0x401	0x400
a.	be	ef	de	ad
b.	ef	be	ad	de
c.	fe	eb	da	ed
d.	ed	da	eb	fe

ans: don't know

ans should be (b), just check with some CS guy, little endian is Intel type, Big-endian is perhaps Motorola type

23. In a given CPU-memory sub-system, all accesses to the memory take two cycles. Accesses to memories in two consecutive cycles can therefore result in incorrect data transfer. Which of the following access mechanisms guarantees correct data transfer?

a. A read operation followed by a write operation in the next cycle.
b. A write operation followed by a read operation in the next cycle.
c. A NOP between every successive reads & writes
d. None of the above

Ans: c(not confirm)
I'm also not sure.

24. An architecture saves 4 control registers automatically on function entry (and restores them on function return). Save of each registers costs 1 cycle (so does restore). How many cycles are spent in these tasks (save and restore) while running the following un-optimized code with n=5:

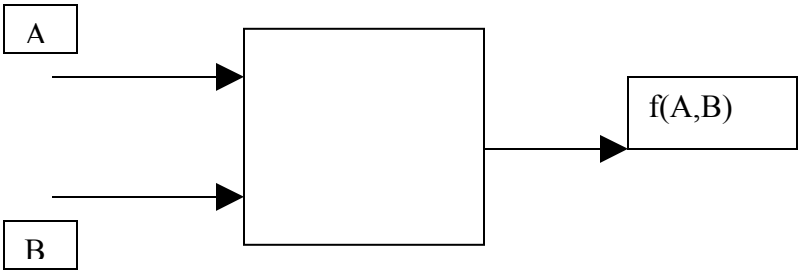
```
Void fib(int n)
{
        if((n==0) || (n==1)) return 1;
        return(fib(n-1) + fib(n-2));
}
```

a. 120
b. 80
c. 125
d. 128

ans: a

25. The maximum number of unique Boolean functions F(A,B), realizable for a two input (A,B) and single output (Z) circuit is:

a. 2
b. 6
c. 8
d. None of the above

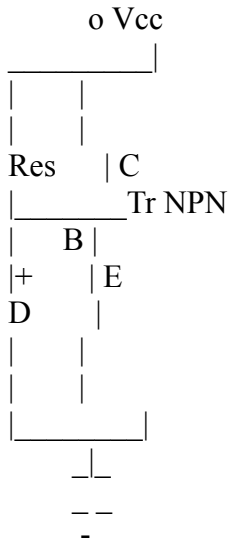


Ans: 2*(2*2)=16 ie d

paper of TI 1999

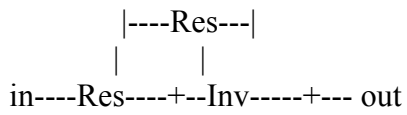
Hard ware part only. There was one part of reasoning and there was separate paper for software persons.

1.



Find the current I delivered by the battery.

2.



What is the given circuit

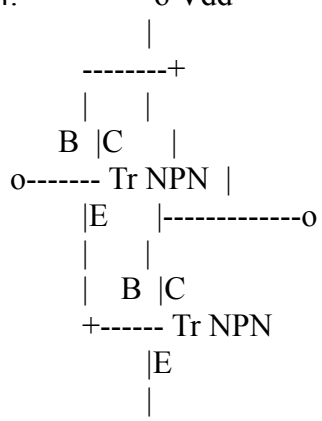
a) Latch b)Amplifier c)Schmitt trigger. d)

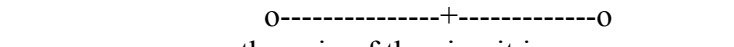
3. The total no of flip flop required for N stage sequential circuit

N N-1 N

a)2 b)2 c) Log N d) 2 -1

4.

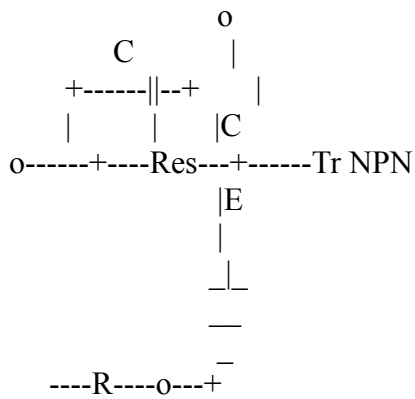




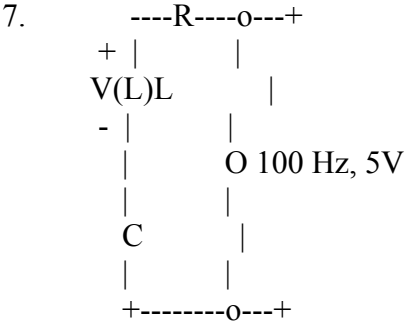
the gain of the circuit is
a) beta square b)beta + 1 c) (beta+1) ka square d)

5. If the o/p and i/p are related by y=k(x square) and i/p is a sum of 2 waveforms then the modulation scheme is
a) FM b)AM c)PM and d)None
Ans. B

6.Function of C in the circuit below is
a) Improve switching b)dc coupling c) ac coupling d) None

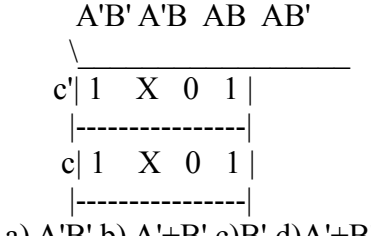


7.



if the ckt is at resonance and V(L)= (constant) V (given)
the value of V(R) and V(C) is
a)100V,5V b)-100V,5V c)5V,5V
(Use V(L)=5 /_100 and V(C)=5/_-100, V(R)=5V

8. Minimize the K-map



a) A'B' b) A'+B' c)B' d)A'+B'+C'

9. IF the rate of removal of elements in a queue containing N elements is proportional to the no of elements already existing in the queue at that

instant then the no. of elements----
a)decrease linearly b)Exponetialy decrease b) Logarithmcally

10. One question on CMOS ckt.
11. Two question on OP-AMP.

THIS IS TI 1999 jadavpur for ECE students.for cs another paper is ^M
>given^M
>^M
>1.two transistors are connected Vbe is 0.7volts .this is simple ckt.one ^M
>transistor is diode equivalent. & asked the o/p across the 2 nd transistor.^M
>2.simple k map ans is Bbar.^M
>3.^M
>^M
> Emitter^M
>---R-------transistorbase| --^M
> | ---^M
> collector^M
> in above capacitor is connected parallel with resistance ^M
>r.capacitor is not shown^M
> in fig.capacitor is used for in this ckt:^M
>^M
>^M
> ans:a.speedupb.active bypass c.decoupling^M
> 4.^M
>^M
> -----R------I----------o/p^M
> |___R____ |^M
> in above r is resistence.I is cmos inverter.^M
> then ckt is used for:^M
>^M
>^M
> a.schmitt trigger b.latch c.inverter ^M
>d.amplifier^M
>^M
>^M
> 5.simple amplifier ckt openloop gain of amplifier is 4.V in ^M
>=1v.asked for V x?^M
> amplifdier + is connected to base. - is connected to i/p in between ^M
>5k is connected.^M
> from o/p feedback connected to - of amplifier with 15k.this is ckt.^M
>^M
>^M
> 6.resistence inductot cap are serially connected to ac voltage 5 ^M
>volts.voltage across^M

> inductor is given.R I C values are given & asked for^M
> voltages across resistence & capacitor.^M
> 7.^M
> ___ R_____^M
> | |^M
> ---R------OPAMP ----------^M

> |---^M
> R1 R1 is for wjhat i mean what is the purpose of R1.^M
> |^M
>^M
> ground^M
>^M
>^M
> 8.asked for Vo at the o/p.it is like simple cmos realization that is n ^M
>block is above^M
> & p block is below.Vdd is 3 volts at supply.V threshold 5 volts.^M
> 9.2 d ffs are connected in asyncro manner .clock 10 MEGAHZ.gate delay ^M
>is 1 nanosec.^M
> A B are the two given D FFs.asked for AB output is:^M
>^M
>^M
> a.updown^M
> b.up c. updown glitching like that (take care abt glitching word)^M
>^M
> 10.^M
>^M
>^M
> ----------------| subtractor|---------o/p^M
> |___HPF____|^M

>^M
> the ckt is LPF ,HPF or APF ?^M
>^M
> 11.in a queue at the no of elements removed is proportional to no of ^M
>elements in^M
> the queue.then no of elements in the queue:^M
> a.increases decreases exp or linearly(so these are the 4 options given ^M
>choose 1 option)^M
> 12.with 2 i/p AND gates u have to form a 8 i/p AND gate.which is the ^M
>fastest in the^M
> following implementations.^M
> ans we think ((AB)(CD))((EF)(GH))^M
> 13.with howmany 2:1 MUX u can for 8:1 MUX.answer is 7.^M
> 14. there are n states then ffs used are log n.^M
> 15.cube each side has r units resistence then the resistence across ^M

>diagonal of cube.^M
> 16.op amp connections asked for o/p^M
> the answer is (1+1/n)(v2-v1).check it out.practise this type of model.^M
> 17.^M
> _____ supply^M
> ---|__ ___|^M
> Ii >_____ |___ Tranistot^M
> > _____Vo^M
> > _____Vo^M
> |^M
> |^M
> R |^M
> | | Io^M
> ground.^M
>^M
>^M
>^M
>^M
> asked for Io/Ii=? transistor gain is beta.^M
>^M
>^M
> a.(1+beta)square b.1+beta c. beta^M
>^M
>^M
> 18.y=kxsquare. this is transfer function of a block with i/p x & o/p ^M
>y.if i/p is^M
> sum of a & b then o/p is :--^M
>^M
> a. AM b.FM c. PM^M
> 19.^M
> ------MULTIPLIER--- |^M

> | |^M
> _____R__|__OPAMP_____Vo^M
> ---^M
> |^M
> ground.^M
> v in = -Ez then o/p Vo =?^M
> answer is squareroot of -Ez.multiplier i/ps are a & b then ^M
>its o/p^M
> is a.b;^M

Here is Texas paper for you.
in this paper there was 20 questions as follows in 60 minutes .
second part consists of 36 que. in 30 minutes all questions are
diagramatical.(figurs)..

1. if a 5-stage pipe-line is flushed and then we have to execute 5 and
12
instructions respectively then no. of cycles will be
a. 5 and 12
b. 6 and 13
c. 9 and 16
d.none

2. k-map

ab
---
c 1 x 0 0
1 x 0 x

solve it

a. A.B
B. ~A
C. ~B
D. A+B

3.CHAR A[10][15] AND INT B[10][15] IS DEFINED
WHAT'S THE ADDRESS OF A[3][4] AND B[3][4]
IF ADDRESS OD A IS OX1000 AND B IS 0X2000

A. 0X1030 AND 0X20C3
B. OX1031 AND OX20C4
AND SOME OTHERS..

4. int f(int *a)
{
int b=5;
a=&b;
}

main()
{
int i;
printf("\n %d",i);
f(&i);
printf("\n %d",i);
}

what's the output .

1.10,5
2,10,10
c.5,5
d. none

5. main()
{
int i;
fork();
fork();
fork();
printf("----");
}

how many times the printf will be executed .
a.3
b. 6
c.5
d. 8

6.
void f(int i)
{
int j;
for (j=0;j<16;j++)
{
if (i & (0x8000>>j))
printf("1");
else
printf("0");
}
}
what's the purpose of the program

a. its output is hex representation of i
b. bcd
c. binary
d. decimal

7.#define f(a,b) a+b
#define g(a,b) a*b

main()
{

```
int m;
m=2*f(3,g(4,5));
printf("\n m is %d",m);
}
```

what's the value of m
a.70
b.50
c.26
d. 69

8.

```
main()
{
char a[10];
strcpy(a,"\0");
if (a==NULL)
printf("\a is null");
else
printf("\n a is not null");}
```

what happens with it .
a. compile time error.
b. run-time error.
c. a is null
d. a is not null.

9. char a[5]="hello"

a. in array we can't do the operation .
b. size of a is too large
c. size of a is too small
d. nothing wrong with it .

10. local variables can be store by compiler
a. in register or heap
b. in register or stack
c .in stack or heap .
d. global memory.

11. average and worst time complexity in a sorted binary tree is

12. a tree is given and ask to find its meaning (parse-tree)
(_expression tree)
ans. ((a+b)-(c*d)) (not confirmed)
13. convert 40.xxxx into binary .

14. global variable conflicts due to multiple file occurance
is resolved during
a. compile-time
b. run-time
c. link-time
d. load-time

15.
two program is given of factorial.
one with recursion and one without recursion .
question was which program won't run for very big no. input because
of stack overfow .
a. i only (ans.)
b. ii only
c. i& ii both .
c. none

16.
struct a
{
int a;
char b;
int c;
}

union b
{
char a;
int b;
int c;
};
which is correct .
a. size of a is always diff. form size of b.(ans.)
b. size of a is always same form size of b.
c. we can't say anything because of not-homogeneous (not in ordered)
d. size of a can be same if ...

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one with recursion and one without recursion .
question was which program won't run for very big no. input because
of stack overfow .
a. i only (ans.)
b. ii only
c. i& ii both .
c. none

16.
struct a
{
int a;
char b;
int c;
}

union b
{
char a;
int b;
int c;
};
which is correct .
a. size of a is always diff. form size of b.(ans.)
b. size of a is always same form size of b.
c. we can't say anything because of not-homogeneous (not in ordered)
d. size of a can be same if ...

bye..
p.sreenivasa rao

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> ATTACHMENT part 5 application/msword name=texas.doc

> ATTACHMENT part 6 application/octet-stream name=texas

Date:
Wed, 30 Dec 1998 19:30:34 +0500
From:
PVSAK Viswanadham Add to Address Book
Subject:
TI
Organization:
Computer Science Dept., Indian Institute of Technology, Kharagpur
To:
bkup

for TI aptitude test consist of all pictorial questions. ie in each question he will give 8 diagrams and ask to find the 9'th diagram in that sequence.
You go through RS Agarwal. These aptitude questins are very easy. Just pratice them. In RS Agarwal gothrough SERIES chapter. It is suffient. There are 35 aptitude questions. First 25 are very easy. Do these questions

in just 15 or 20 minutes. Because last questions are
very touch.

TECHNICAL TEST:
---

1)3 flipflops are connected so that after 0 to 5 count occured
next number is zero. So what is the counter?
Ans: mod 6 counter

2)simplication of some boolean _expression which is simple.
Boolean _Expression is A+A'B.
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find out postorder sequence.

4)Some question on value of a static variable.

5) Given an interger in binary form,find the number of ones in
that number without counting each bit.(This questin is not
multiple choice question. This question carries more
marks. So please take care for this question.)

6) 1-way set associative memory is called-----
a)direct b)something c)1-way set associative 4)something
Ans: c

7)Fastest IPC mechanism is

a)shared memory b)pipes c)named pipes d)semaphores
Ans:c

8)Some page references are given. You are asked to implement
it with Least Frequently Used algorithm.

9)Some diagram is given.
Iam describinmg the diagram. A 2*1 MUX is given. The inputs
are A,B. Output is C. C and A are tied together. What is
the diagram.?

Ans:Latch.

****

This paper is for Electrical & Electronics students. There is

separate test for computer Science Students. There are 20 questions.

1)Some circuit is given. Iam describing the circuit.
A resistor R & a capacitor C are connected in parallel.
To this circuit another circuit which is having a capacitor of capacity 2C & an impedence Z, is connected in series.
You are asked to find out the value of Z? Note that 2C & Z are connected in series.
a)Z=2C
b)Z=2L
c)Z=L/2
d)Z=2R

2)Some circuit which consist of only resistors R is given.
This is a repetative circuit. U have to find the effctive resistance of the entire circuit.
A)Rin=R
B)Rin=(5+sqrt(3))/7
C)Rin=(19+sqrt(3))/8
D)None.

3)Two wave forms are given. You are asked to write the cirsuit to get B(second wave form) from A(first wave form).

4)#define SUM(a,b) a+b

```
main()
{
a=2;
b=3;
x=SUM(a,b)*2;
printf("x=%d\n",x);
}
```
Ans:8.

5)
```
number(int i)
{
number++;
printf("%d\n",number);
}

main()
{
static int i=0;
number(i);
```

}
Ans: I don't know.

6)Some circuit is given. I can't describe the circuit. There are
3 resistors,3 capacitors & one inverter.. The question is
What is the value of the frequency such that the circuit oscillates.
A)f=RC
B)f=sqrt(3)/(Pi*R*C)
C)f=1/(Pi*R*C)
D)something
Ans:I don't know the answer.

7)Question on flipflop. So gothrough all flipflops.

8)There are 5 questions on Nmos & Pmos circuits.

****
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****

(1) The fastest memory is
(i) DRAM, (ii) ROM, (iii) SRAM, (iv) Main memory
Ans : SRAM

(2) Programing exceptions are
(i) Asynchronous, (ii) Synchronous, (iii) None
Ans : Asynchronous

(3) DSP which architecture is used
(i) MIMD, (ii) SIMD, (iii) Nueman, (iv) Harvard Architecture
Ans : Harvard Architecture

(4) C prog. for searching for an element in linked list

(5) main()
{
unsigned char i;
int sum;

for(i=0; i<300; i++)
sum+ = i;
printf("\nSum = %d\n", sum);
}

Ans : infinite loop

(6) void fn(int *p)
{
static int val = 100;
p = &val;
}

main()
{
int i=10;
printf("i=%d\n", i);
fn(&i);
printf("i=%d\n", i);
}

Ans : i=10 i=10

(7) int a[10[15];
char b[10[15];
(a) location g a[3][4], if base location g a[0][0] is ox1000
(b) location g b[3][4], if base location g b[0][0] is ox2000
int taken 32 bits and char taken 8 bits.

Ans : (a) ox10C4 (b) ox2031

(8) Implement OR gate function with 2*1 MUX

Ans : A _____
--------|2*1 MUX |
B | |--------o/p
--------| |
| -----------
|_____|C

B=C

(9) Implement 4*1 MUX with 2*1 MUXES

(10) Swapping without using a temporary variables. (2 methods)

(i) x = x+y;
y = x-y;
x = x-y;

(ii) x = x^y;

```
y = x^y;
x = x^y;
```

(11) Count no of 1's in a word without using bit by bit.
(This question carries more marks. It is not a multiple choice question.)

(12) Code 1 :

```
for(i=0; i<1000; i++)
for(j=0; j<100; j++)
x = y;
```

Code 2 :

```
for(i=0; i<100; i++)
for(j=0; j<1000; j++)
x = y;
```

Which code will execute faster
(i) Code 1 and Code 2 are of same speed,
(ii) Code 1,
(iii) Code 2,
(iv) None.

Ans : Code 2

(13)

```
main()
{
int a[10] = {1, 2, 3, ...., 10}, i, x=10, temp;

for(i=0; i temp = a[i];
a[i] = a[x-i-1];
a[x-i-1] = temp;
}
```

(i) All contents of array a are reversed
(ii) Only some portions are altered
(iii) Remains same
(iv) None

Ans : (iii)

(14) An array is stored in row major order. The memory capacity is 30 MB. And in unix system demand paging is used. Which one will give more page faults?

```
#define V_L_I 10000
int i, j, array[V_L_I][V_L_I];
```

Code 1 :
array[i][j] = 1;

Code 1 :
for(j=0; j for(i=0; i array[i][j] = 1;

Ans : Code 2

(15) In C which parameter passing technique is used?
(i) call by value,
(ii) call by reference,
(iii) both

Ans : call by value

(16) A circuit is given with 2 exclusive OR gates whose boolean
_expression will be y = '(AB) + AB
(' indicates bar)

(17) main()
{
int i = 1;
fork();
fork();
printf("\ni = %d\n", i+1);
}

Ans : 4 printfs will occur and i = 2

(18) Compute the complexity of Binary search.
Ans : O(lg n) (Answer in detail. This is not a multiple choice question.
It carries more marks.)

(19) Write _expression for the tree graph :
Ans : ((a-b) + c*d)/x

(20) # define MAX(a, b) a>b ? a:b
main()
{
int m, n;
m = 3 + MAX(2, 3);
n = 2 * MAX(3, 2);
printf("m = %d, n = %d\n", m, n)
}

Ans : m=2, n=3
paper of texas instruments.
Technical + aptitude + interview.
* Questions on c
here i am not strictly following syntax it is just to show what was asked.
1 #define sum(a,b) a+b
value of sum(2,3)*2 Ans:8

2. a=5,b=6
swap(&a,b);
This function is written to swap a and b
find value of a and b . Ans 6,6

3.
```
function()
{
static int a=0;
a=a+1;
return a;
}
main()
{
function();
function();
function();
printf a;
}
```
final value of a ? Ans : a=3. static initializes once.
4.Write two prog. to swap a & b without using temp variable.
5.
```
unsigned char i;
int sum=0;
for(i=0;i<300;i++)
sum=sum+i;
printf(sum);
```

Ans:Program will held in infinite loop b/c i can not exceed 255.

6.
five questions on MOSFETS.
four were having single mosfets.
questions were simple.as i told u in Gwalior.
just on the funda that it will conduct if Vg-Vs > Vt .

In one question output at drain was to be calculated while o/p was initially charged to 5v and to the gate 5v were applied.

In one question output at drain was to be calculated while o/p was initially charged to 5v and the gate was shorted to drain.
7.
Clear the concept of settling time , hold time and other times. 3 ques on that.
like values of various delays were given and max frequency at which the circuit can work
hint : 1/sum of all delays . In our case ans was 200 Mhz.

8. An input and output waveform was given and circuit was to be designed with the use of one
delay.
Ans : exor gate in which second input is first input with a delay.

9. A question to determine sequence of counter. Don't get puzzled it was a tough question.

10.The output and input of a inverter is connected by three RC stages in between of each stage two amplifiers with poles at imaginary axis were connected.
Hint : The poles at imaginary axis will create extra 180 phase shift thus the circuit will oscillate and calculate the frequency of operation.

11. A series of infinite connected rc circuit and overall input resistance is calculated.(question of 12 class)

```
        TECHNICAL TEST:
        -------------------------

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to get B(second wave form) from A(first wave form).

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```
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  printf("x=%d\n",x);
  }
```
Ans:8.

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```
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{
number++;
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```

```
}

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Ans: I don't know.

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(5)
```
main()
{
        unsigned char   i;
        int     sum;

        for(i=0; i<300; i++)
                sum+ = i;
        printf("\nSum = %d\n", sum);
}
```

Ans : infinite loop

(6)
```
void fn(int *p)
        {
                static int val = 100;
                p = &val;
```

```
        }

main()
{
        int i=10;
        printf("i=%d\n", i);
        fn(&i);
        printf("i=%d\n", i);
}
```

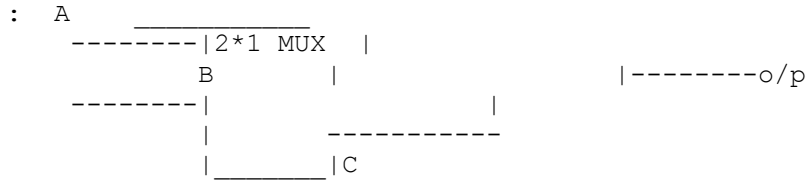
Ans : i=10 i=10

(7) int a[10[15];
char b[10[15];
(a) location g a[3][4], if base location g a[0][0] is ox1000
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int taken 32 bits and char taken 8 bits.

Ans : (a) ox10C4 (b) ox2031

(8) Implement OR gate function with 2*1 MUX

Ans :



B=C

(9) Implement 4*1 MUX with 2*1 MUXES

(10) Swapping without using a temporary variables. (2 methods)

(i)
```
x = x+y;
y = x-y;
x = x-y;
```

(ii)
```
x = x^y;
 y = x^y;
 x = x^y;
```

(11) Count no of 1's in a word without using bit by bit.

>^M
> THIS IS TI 1999 jadavpur for ECE students.for cs another paper is ^M
>given^M
>^M
>1.two transistors are connected Vbe is 0.7volts .this is simple ckt.one ^M

>transistor is diode equivalent. & asked the o/p across the 2 nd transistor.^M
>2.simple k map ans is Bbar.^M
>3.^M
>^M
>   Emitter^M
>---R-------transistorbase| --^M
> | ---^M
>   collector^M
>   in above capacitor is connected parallel with resistance ^M
>r.capacitor is not shown^M
>   in fig.capacitor is used for in this ckt:^M
>^M
>^M
>   ans:a.speedupb.active bypass c.decoupling^M
>   4.^M
>^M
>   -----R------I----------o/p^M
> |___R____ |^M
>   in above r is resistence.I is cmos inverter.^M
>   then ckt is used for:^M
>^M
>^M
>   a.schmitt trigger b.latch c.inverter ^M
>d.amplifier^M
>^M
>^M
>   5.simple amplifier ckt openloop gain of amplifier is 4.V in ^M
>=1v.asked for V x?^M
>   amplifdier + is connected to base. - is connected to i/p in between ^M
>5k is connected.^M
>   from o/p feedback connected to - of amplifier with 15k.this is ckt.^M
>^M
>^M
>   6.resistence inductot cap are serially connected to ac voltage 5 ^M
>volts.voltage across^M
>   inductor is given.R I C values are given & asked for^M
>   voltages across resistence & capacitor.^M
>   7.^M
>   ___ R_____^M
> | |^M
>   ---R------OPAMP ----------^M

                  > |---^M
>   R1   R1 is for wjhat i mean what is the purpose of R1.^M
> |^M
>^M
>   ground^M
>^M
>^M

>   8.asked for Vo at the o/p.it is like simple cmos realization that is n ^M
>block is above^M
>   & p block is below.Vdd is 3 volts at supply.V threshold 5 volts.^M
>   9.2 d ffs are connected in asyncro manner .clock 10 MEGAHZ.gate delay ^M
>is 1 nanosec.^M
>   A B are the two given D FFs.asked for AB output is:^M
>^M
>^M
>   a.updown^M
>   b.up c. updown glitching like that (take care abt glitching word)^M
>^M
>   10.^M
>^M
>^M
>   ----------------| subtractor|---------o/p^M
> |___HPF____|^M

                        >^M
>   the ckt is LPF ,HPF or APF ?^M
>^M
>   11.in a queue at the no of elements removed is proportional to no of ^M
>elements in^M
>   the queue.then no of elements in the queue:^M
>   a.increases decreases exp or linearly(so these are the 4 options given ^M
>choose 1 option)^M
>   12.with 2 i/p AND gates u have to form a 8 i/p AND gate.which is the ^M
>fastest in the^M
>   following implementations.^M
>   ans we think ((AB)(CD))((EF)(GH))^M
>   13.with howmany 2:1 MUX u can for   8:1 MUX.answer is 7.^M
>   14. there are n states then ffs used are log n.^M
>   15.cube each side has r units resistence then the resistence across ^M
>diagonal of cube.^M
>   16.op amp connections asked for o/p^M
>   the answer is (1+1/n)(v2-v1).check it out.practise this type of model.^M
>   17.^M
>   _____________ supply^M
>   ---|__   ___|^M
> Ii   >________ |___   Tranistot^M
>   > _______Vo^M
>   > _______Vo^M
> |^M
> |^M
>   R |^M
> | | Io^M
>   ground.^M
>^M
>^M
>^M

>^M
>   asked for Io/Ii=? transistor gain is beta.^M
>^M
>^M
>   a.(1+beta)square b.1+beta c. beta^M
>^M
>^M
>   18.y=kxsquare. this is transfer function of a block with i/p x & o/p ^M
>y.if i/p is^M
>   sum of a & b then o/p is :--^M
>^M
>   a. AM b.FM c. PM^M
>   19.^M
>   ------MULTIPLIER--- |^M

                       > | |^M
>   _____R__|__OPAMP_____________________Vo^M
>   ---^M
> |^M
>   ground.^M
>   v in = -Ez then o/p Vo =?^M
>   answer is squareroot of -Ez.multiplier i/ps are a & b then ^M
>its o/p^M
>   is a.b;^M

Here is Texas paper for you.
in this paper there was 20 questions as follows in 60 minutes .
second part consists of 36 que. in 30 minutes all questions are diagramatical.(figurs)..

1. if a 5-stage pipe-line is flushed and then we have to execute 5 and
12
instructions respectively then no. of cycles will be
a. 5 and 12
b. 6 and 13
c. 9 and 16
d.none

2. k-map

ab
---
c   1   x 0 0
       1   x 0 x

solve it

a. A.B
B. ~A
C. ~B
D. A+B

3.CHAR A[10][15] AND INT B[10][15] IS DEFINED
WHAT'S THE ADDRESS OF A[3][4] AND B[3][4]

IF ADDRESS OD A IS OX1000 AND B IS 0X2000

A. 0X1030 AND 0X20C3
B. OX1031 AND OX20C4
AND SOME OTHERS..

```
4. int f(int *a)
{
int b=5;
a=&b;
}

main()
{
int i;
printf("\n %d",i);
f(&i);
printf("\n %d",i);
}
```

what's the output .

1.10,5
2,10,10
c.5,5
d. none

```
5. main()
{
int i;
fork();
fork();
fork();
printf("----");
}
```

how many times the printf will be executed .
a.3
b. 6
c.5
d. 8

```
6.
void f(int i)
{
int j;
for (j=0;j<16;j++)
{
if (i & (0x8000>>j))
printf("1");
else
printf("0");
}
}
```
what's the purpose of the program

a. its output is hex representation of i

b. bcd
c. binary
d. decimal

7.#define f(a,b) a+b
#define g(a,b) a*b

```
main()
{

int m;
m=2*f(3,g(4,5));
printf("\n m is %d",m);
}
```

what's the value of m
a.70
b.50
c.26
d. 69

8.

```
main()
{
char a[10];
strcpy(a,"\0");
if (a==NULL)
printf("\a is null");
else
printf("\n a is not null");}
```

what happens with it .
a. compile time error.
b. run-time error.
c. a is null
d. a is not null.

9. char a[5]="hello"

a. in array we can't do the operation .
b. size of a is too large
c. size of a is too small
d. nothing wrong with it .

10. local variables can be store by compiler
a. in register or heap
b. in register or stack
c .in stack or heap .
d. global memory.

11. average and worst time complexity in a sorted binary tree is

12. a tree is given and ask to find its meaning (parse-tree)
(expression tree)
ans. ((a+b)-(c*d)) (not confirmed)
13. convert 40.xxxx into binary .

14. global variable conflicts due to multiple file occurance
is resolved during
a. compile-time
b. run-time
c. link-time
d. load-time

15.
two program is given of factorial.
one with recursion and one without recursion .
question was which program won't run for very big no. input
because
of stack overfow .
a. i only (ans.)
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c. i& ii both .
c. none

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struct a
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int c;
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which is correct .
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c. we can't say anything because of not-homogeneous (not in ordered)
d. size of a can be same if ...

Hi friends

here is the full paper of TI India IITB 20/07/2001.
WITH SOME ANSWER

the paper had 4 sections in which we have to attend
2 sections
i attempted digtal and analog

and in apti, there was 75 Qs in 60min. be prepare
for
it also, apti was very tough. Apti was more
mathematical stuff, requires a
lot of thinking and very high speed. ex. coding,
reading compreh.(RC),
calculations, relationship, travelling problem
(given some cities and
journey conditions were given) etc etc...

Best of luck
Regards
Vijay Mathur

DIGITAL

1. nand gate is
a) associative &cumulative b)cumulative but not associative
c)not cumulative but associative d)not cumultive and associative

ANS. b

2. which imp has les delay
a) (a xor b) xor (c xor d)
b) (((a xor b) xor c) xor d)
(think on the situation when input a, b, c, d are comes in ascending (i.e. a frist then b then c and then d) and desending (opposite order)

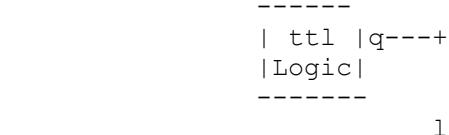
3) one inverter cmos circuit was given with A variable and enable B signal

ANS. tristate inverter with B as enable

4)a logic cell which dertermines(op =1) for odd no.. of 1s in the given seq is

ANS. one xor gate

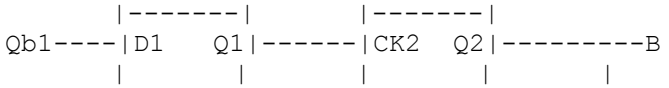
5)circuit

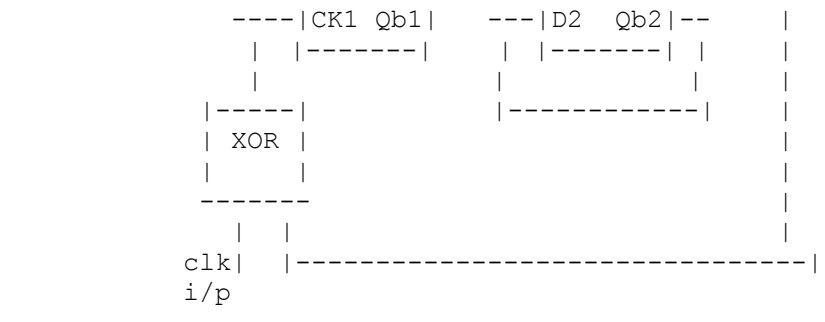


led should glow when q=0 and off when q=1
the choices are
a.ckt will funct as given
b.it wont funct as given
c.q cant drive ttl inverter
d.non of these
ANS: b

6)n nets are givenin how many ways can we model each
of the stuck at fault in n nets in single(one at a time) and w....
ANS. 2n and (3^n)-1

7) circiut with 2 d ffs was given





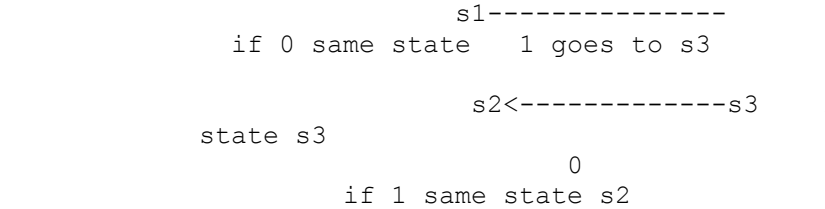
whaT is the relation between B and clk i/p?

Ans. b is 1/3 of clk i/p with 50% duty cycle.

8)3 dffs was given with common clk
setup time 3ns
hold time 1ns
clk to q delay 2ns find the maximum frequency of operation

ANS: 200MHz

9)fsm question there states given ques what is the
machne called



ans: MOORE M/C

10. consider a adder and multiplier
question ios some what like can both implemented in
same fsm or cant

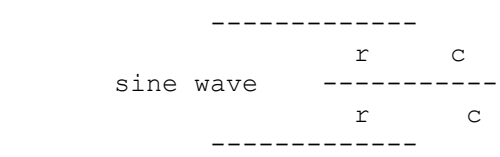
ANS: Yes they can implemented in same machine.

these are the 10 ques asked in digitl section and we are in a
situation to attend 2 section i attented analog part
and i am giving the ruogh idea of analog section

****
Analog part 10questions

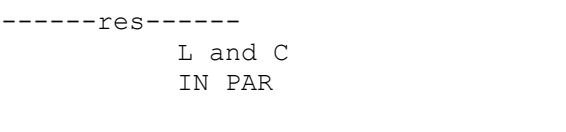
1. input is sine wave
what is the op in the middle of serially connected 2

rc parallel ckts



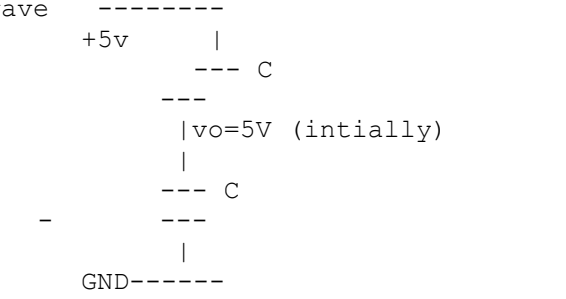
a. cos wave
b.sine wave with 0 phase shift
etc
ANS: b

2.square wave as input given to the ckt



ANS: SINE WAVE.

3.in 5v wave

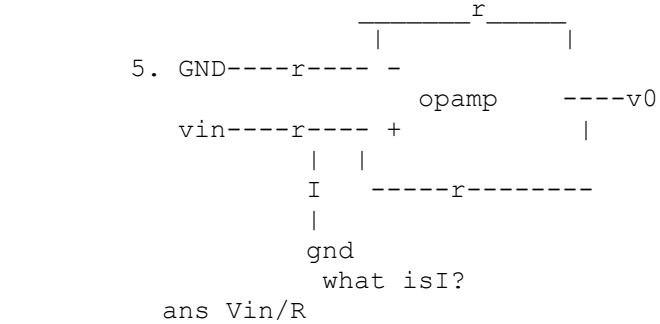


to the circuit in which two cap are connected in series
what is the op at the vo at time T
intially it was at 5v

ans: may be 5V. (CHECK IT MAY BE 2.5 V)

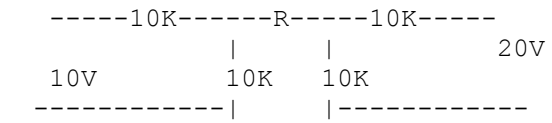
4.darlinton pair was given with beta as ct gain of each trans
what is the overall gain
ans: (beta+1)^2

5.



what isI?
ans Vin/R

6.

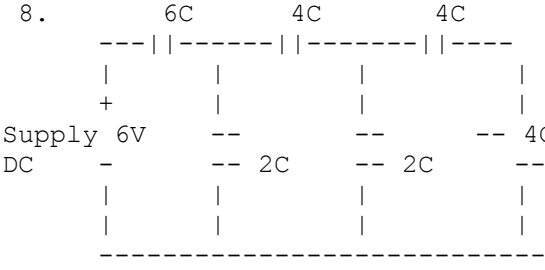


7. IF THE INPUT SGL IS 95khz and it is sampled at 120samples per sec the at what freq wil the fft opt fundemental freq will come

totally there were 10 questions in analog ssection also

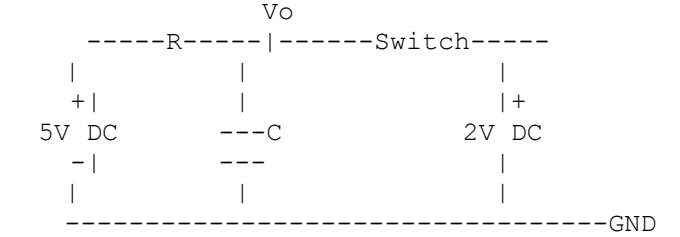
the numbering is not right

8.



find out Vo=? (caps vaule may be changed).
ANS was 0.75V but here i think cap. values changed, so calculate urself for currect answer.

9.



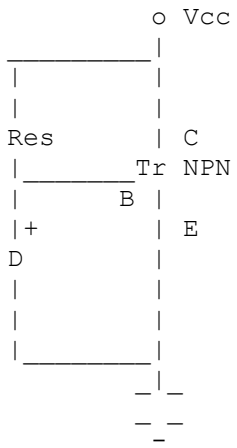
Switch is open at t=0, what is the value of Vo at t=infinity. Ans. 5V

10. one Qs on current mirror. there are n current mirrors are connected in series, you have to find out
the condition for whitch current mirrors will be in linear region.
(a)Vt
(b)Vt+ deta V
(c)n*Vt+ (n-1)* delta V
(d)n(Vt+delta V)

ANS: c

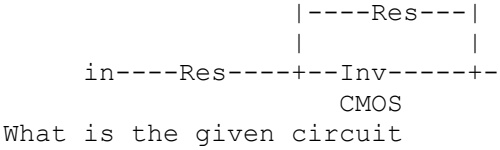
paper of TI 1999
Hard ware part only. There was one part of reasoning and there was separate paper for software persons.

1.



Find the current I delivered by the battery.

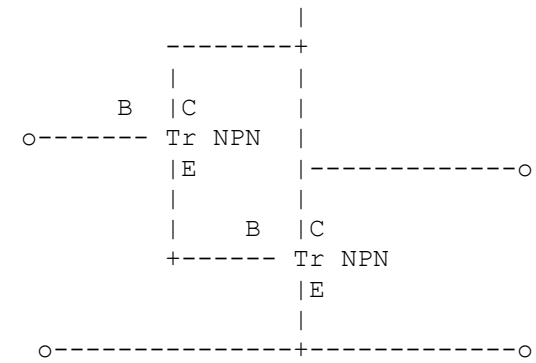
2.



What is the given circuit
a) Latch b)Amplifier c)Schmitt trigger. d)

3. The total no of flip flop required for N stage sequential circuit
N N-1 N
a)2 b)2 c) Log N d) 2 -1

4.

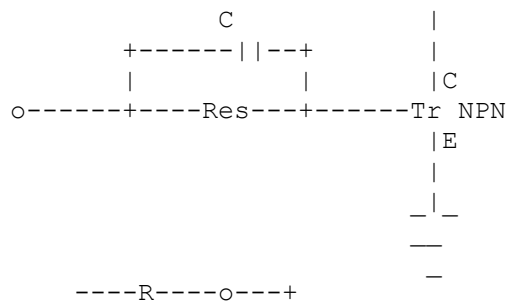


the gain of the circuit is
a) beta square b)beta + 1 c) (beta+1) ka square d)

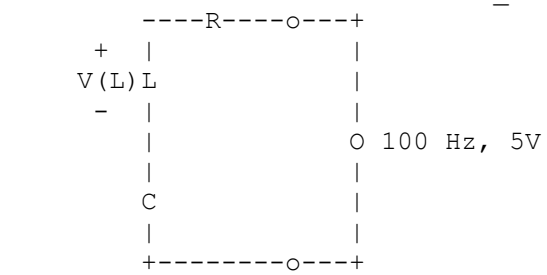
5. If the o/p and i/p are related by y=k(x square) and i/p is a sum of 2 waveforms
then the modulation scheme is
a) FM b)AM c)PM and d)None
Ans. B

6.Function of C in the circuit below is
a) Improve switching b)dc coupling c) ac coupling d) None

o



7.



if the ckt is at resonance and V(L)= (constant) V (given)
the value of V(R) and V(C) is
a)100V,5V b)-100V,5V c)5V,5V
   (Use V(L)=5 /_100 and V(C)=5/_-100, V(R)=5V

8. Minimize the K-map

	A'B'	A'B	AB	AB'
c'	1	X	0	1
c	1	X	0	1

a) A'B' b) A'+B' c)B' d)A'+B'+C'

9. IF the rate of removal of elements in a queue containing N elements is
proportional to the no of elements already existing in the queue at that
instant then the no. of elements----
a)decrease linearly b)Exponetialy decrease b) Logarithmcally

10. One question on CMOS ckt.
11. Two question on OP-AMP.

THIS IS TI 1999 jadavpur for ECE students.for cs another paper is ^M
>given^M
>^M
>1.two transistors are connected Vbe is 0.7volts .this is simple
ckt.one ^M
>transistor is diode equivalent. & asked the o/p across the 2 nd
transistor.^M
>2.simple k map ans is Bbar.^M
>3.^M
>^M
> Emitter^M
>---R-------transistorbase| --^M
> | ---^M
> collector^M

> in above capacitor is connected parallel with resistance ^M
>r.capacitor is not shown^M
> in fig.capacitor is used for in this ckt:^M
>^M
>^M
> ans:a.speedupb.active bypass c.decoupling^M
> 4.^M
>^M
> -----R------I----------o/p^M
> |___R____ |^M
> in above r is resistence.I is cmos inverter.^M
> then ckt is used for:^M
>^M
>^M
> a.schmitt trigger b.latch c.inverter ^M
>d.amplifier^M
>^M
>^M
> 5.simple amplifier ckt openloop gain of amplifier is 4.V in ^M
>=1v.asked for V x?^M
> amplifdier + is connected to base. - is connected to i/p in between ^M
>5k is connected.^M
> from o/p feedback connected to - of amplifier with 15k.this is ckt.^M
>^M
>^M
> 6.resistence inductot cap are serially connected to ac voltage 5 ^M
>volts.voltage across^M
> inductor is given.R I C values are given & asked for^M
> voltages across resistence & capacitor.^M
> 7.^M
> ___ R_____^M
> | |^M
> ---R------OPAMP ----------^M

> |---^M
> R1 R1 is for wjhat i mean what is the purpose of R1.^M
> |^M
>^M
> ground^M
>^M
>^M
> 8.asked for Vo at the o/p.it is like simple cmos realization that is
n ^M
>block is above^M
> & p block is below.Vdd is 3 volts at supply.V threshold 5 volts.^M
> 9.2 d ffs are connected in asyncro manner .clock 10 MEGAHZ.gate delay
^M
>is 1 nanosec.^M
> A B are the two given D FFs.asked for AB output is:^M
>^M
>^M
> a.updown^M
> b.up c. updown glitching like that (take care abt glitching word)^M
>^M
> 10.^M
>^M
>^M

> ----------------| subtractor|---------o/p^M
> |___HPF____|^M

>^M
> the ckt is LPF ,HPF or APF ?^M
>^M
> 11.in a queue at the no of elements removed is proportional to no of
^M
>elements in^M
> the queue.then no of elements in the queue:^M
> a.increases decreases exp or linearly(so these are the 4 options
given ^M
>choose 1 option)^M
> 12.with 2 i/p AND gates u have to form a 8 i/p AND gate.which is the
^M
>fastest in the^M
> following implementations.^M
> ans we think ((AB)(CD))((EF)(GH))^M
> 13.with howmany 2:1 MUX u can for 8:1 MUX.answer is 7.^M
> 14. there are n states then ffs used are log n.^M
> 15.cube each side has r units resistence then the resistence across ^M
>diagonal of cube.^M
> 16.op amp connections asked for o/p^M
> the answer is (1+1/n)(v2-v1).check it out.practise this type of
model.^M
> 17.^M
> _____________ supply^M
> ---|__ ___|^M
> Ii >________ |___ Tranistot^M
> > _______Vo^M
> > _______Vo^M
> |^M
> |^M
> R |^M
> | | Io^M
> ground.^M
>^M
>^M
>^M
>^M
> asked for Io/Ii=? transistor gain is beta.^M
>^M
>^M
> a.(1+beta)square b.1+beta c. beta^M
>^M
>^M
> 18.y=kxsquare. this is transfer function of a block with i/p x & o/p
^M
>y.if i/p is^M
> sum of a & b then o/p is :--^M
>^M
> a. AM b.FM c. PM^M
> 19.^M
> ------MULTIPLIER--- |^M

> | |^M
> _____R__|__OPAMP_____________________Vo^M

> ---^M
> |^M
> ground.^M
> v in = -Ez then o/p Vo =?^M
> answer is squareroot of -Ez.multiplier i/ps are a & b then ^M
>its o/p^M
> is a.b;^M

Here is Texas paper for you.
in this paper there was 20 questions as follows in 60 minutes .
second part consists of 36 que. in 30 minutes all questions are
diagramatical.(figurs)..

1. if a 5-stage pipe-line is flushed and then we have to execute 5 and
12
instructions respectively then no. of cycles will be
a. 5 and 12
b. 6 and 13
c. 9 and 16
d.none

2. k-map

ab
----
c 1 x 0 0
1 x 0 x

solve it

a. A.B
B. ~A
C. ~B
D. A+B

3.CHAR A[10][15] AND INT B[10][15] IS DEFINED
WHAT'S THE ADDRESS OF A[3][4] AND B[3][4]
IF ADDRESS OD A IS OX1000 AND B IS 0X2000

A. 0X1030 AND 0X20C3
B. OX1031 AND OX20C4
AND SOME OTHERS..

4. int f(int *a)
{
int b=5;
a=&b;
}

main()
{
int i;
printf("\n %d",i);
f(&i);
printf("\n %d",i);
}

what's the output .

1.10,5
2,10,10
c.5,5
d. none

```
5. main()
{
int i;
fork();
fork();
fork();
printf("----");
}
```

how many times the printf will be executed .
a.3
b. 6
c.5
d. 8

```
6.
void f(int i)
{
int j;
for (j=0;j<16;j++)
{
if (i & (0x8000>>j))
printf("1");
else
printf("0");
}
}
```

what's the purpose of the program

a. its output is hex representation of i
b. bcd
c. binary
d. decimal

```
7.#define f(a,b) a+b
#define g(a,b) a*b

main()
{

int m;
m=2*f(3,g(4,5));
printf("\n m is %d",m);
}
```

what's the value of m
a.70
b.50
c.26
d. 69

8.

```
main()
{
char a[10];
strcpy(a,"\0");
if (a==NULL)
printf("\a is null");
else
printf("\n a is not null");}
```

what happens with it .
a. compile time error.
b. run-time error.
c. a is null
d. a is not null.

9. char a[5]="hello"

a. in array we can't do the operation .
b. size of a is too large
c. size of a is too small
d. nothing wrong with it .

10. local variables can be store by compiler
a. in register or heap
b. in register or stack
c .in stack or heap .
d. global memory.

11. average and worst time complexity in a sorted binary tree is

12. a tree is given and ask to find its meaning (parse-tree)
(_expression tree)
ans. ((a+b)-(c*d)) (not confirmed)
13. convert 40.xxxx into binary .

14. global variable conflicts due to multiple file occurance
is resolved during
a. compile-time
b. run-time
c. link-time
d. load-time

15.
two program is given of factorial.
one with recursion and one without recursion .
question was which program won't run for very big no. input because
of stack overfow .
a. i only (ans.)
b. ii only
c. i& ii both .
c. none

16.

```
struct a
{
```

```
int a;
char b;
int c;
}

union b
{
char a;
int b;
int c;
};
```
which is correct .
a. size of a is always diff. form size of b.(ans.)
b. size of a is always same form size of b.
c. we can't say anything because of not-homogeneous (not in ordered)
d. size of a can be same if ...

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bye..
p.sreenivasa rao

_____

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> ATTACHMENT part 6 application/octet-stream name=texas

Date:
Wed, 30 Dec 1998 19:30:34 +0500
From:
PVSAK Viswanadham Add to Address Book
Subject:
TI
Organization:
Computer Science Dept., Indian Institute of Technology, Kharagpur
To:
bkup

for TI aptitude test consist of all pictorial questions. ie in each question he will give 8 diagrams and ask to find the 9'th diagram in that sequence.
You go through RS Agarwal. These aptitude questins are very easy. Just pratice them. In RS Agarwal gothrough SERIES chapter. It is suffient. There are 35 aptitude questions. First 25 are very easy. Do these questions in just 15 or 20 minutes. Because last questions are very touch.

TECHNICAL TEST:
---

1)3 flipflops are connected so that after 0 to 5 count occured next number is zero. So what is the counter?
Ans: mod 6 counter

2)simplication of some boolean _expression which is simple.
Boolean _Expression is A+A'B.
Ans:A+B

3)Given inorder sequence and preorder sequence and asked to find out postorder sequence.

4)Some question on value of a static variable.

5) Given an interger in binary form,find the number of ones in that number without counting each bit.(This questin is not multiple choice question. This question carries more marks. So please take care for this question.)

6) 1-way set associative memory is called-----
a)direct b)something c)1-way set associative 4)something
Ans: c

7)Fastest IPC mechanism is

a)shared memory b)pipes c)named pipes d)semaphores
Ans:c

8)Some page references are given. You are asked to implement it with Least Frequently Used algorithm.

9)Some diagram is given.
Iam describinmg the diagram. A 2*1 MUX is given. The inputs are A,B. Output is C. C and A are tied together. What is the diagram.?

Ans:Latch.

****

This paper is for Electrical & Electronics students. There is separate test for computer Science Students. There are 20 questions.

1)Some circuit is given. Iam describing the circuit.
A resistor R & a capacitor C are connected in parallel.
To this circuit another circuit which is having a capacitor of capacity 2C & an impedence Z, is connected in series.
You are asked to find out the value of Z? Note that 2C & Z are connected in series.
a)Z=2C
b)Z=2L
c)Z=L/2
d)Z=2R

2)Some circuit which consist of only resistors R is given.
This is a repetative circuit. U have to find the effctive resistance of the entire circuit.
A)Rin=R
B)Rin=(5+sqrt(3))/7
C)Rin=(19+sqrt(3))/8
D)None.

3)Two wave forms are given. You are asked to write the cirsuit to get B(second wave form) from A(first wave form).

4)#define SUM(a,b) a+b

```
main()
{
a=2;
b=3;
x=SUM(a,b)*2;
printf("x=%d\n",x);
}
```
Ans:8.

5)
```
number(int i)
{
number++;
printf("%d\n",number);
```

```
}

main()
{
static int i=0;
number(i);
}
```
Ans: I don't know.

6)Some circuit is given. I can't describe the circuit. There are
3 resistors,3 capacitors & one inverter.. The question is
What is the value of the frequency such that the circuit oscillates.
A)f=RC
B)f=sqrt(3)/(Pi*R*C)
C)f=1/(Pi*R*C)
D)something
Ans:I don't know the answer.

7)Question on flipflop. So gothrough all flipflops.

8)There are 5 questions on Nmos & Pmos circuits.

****
This Paper is for Computer Science Students. THis paper is
very easy. You can definitely do it in one hour.
****

(1) The fastest memory is
(i) DRAM, (ii) ROM, (iii) SRAM, (iv) Main memory
Ans : SRAM

(2) Programing exceptions are
(i) Asynchronous, (ii) Synchronous, (iii) None
Ans : Asynchronous

(3) DSP which architecture is used
(i) MIMD, (ii) SIMD, (iii) Nueman, (iv) Harvard Architecture
Ans : Harvard Architecture

(4) C prog. for searching for an element in linked list

```
(5) main()
{
unsigned char i;
int sum;

for(i=0; i<300; i++)
sum+ = i;
printf("\nSum = %d\n", sum);
}
```

Ans : infinite loop

```
(6) void fn(int *p)
{
static int val = 100;
p = &val;
```

```
}

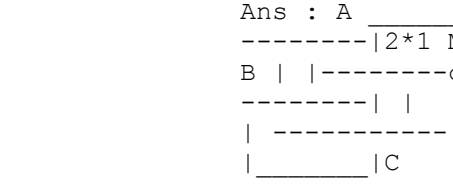
main()
{
int i=10;
printf("i=%d\n", i);
fn(&i);
printf("i=%d\n", i);
}
```

Ans : i=10 i=10

(7) int a[10[15];
char b[10[15];
(a) location g a[3][4], if base location g a[0][0] is ox1000
(b) location g b[3][4], if base location g b[0][0] is ox2000
int taken 32 bits and char taken 8 bits.

Ans : (a) ox10C4 (b) ox2031

(8) Implement OR gate function with 2*1 MUX

Ans : A


B=C

(9) Implement 4*1 MUX with 2*1 MUXES

(10) Swapping without using a temporary variables. (2 methods)

```
(i) x = x+y;
y = x-y;
x = x-y;

(ii) x = x^y;
y = x^y;
x = x^y;
```

(11) Count no of 1's in a word without using bit by bit.
(This question carries more marks. It is not a multiple choice question.)

```
(12) Code 1 :
for(i=0; i<1000; i++)
for(j=0; j<100; j++)
x = y;

Code 2 :
for(i=0; i<100; i++)
for(j=0; j<1000; j++)
x = y;
```

Which code will execute faster
(i) Code 1 and Code 2 are of same speed,
(ii) Code 1,
(iii) Code 2,
(iv) None.

Ans : Code 2

(13) main()
{
int a[10] = {1, 2, 3, ..., 10}, i, x=10, temp;

for(i=0; i temp = a[i];
a[i] = a[x-i-1];
a[x-i-1] = temp;
}
(i) All contents of array a are reversed
(ii) Only some portions are altered
(iii) Remains same
(iv) None

Ans : (iii)

(14) An array is stored in row major order. The memory capacity is 30 MB. And in unix system demand paging is used. Which one will give more page faults?

#define V_L_I 10000
int i, j, array[V_L_I][V_L_I];

Code 1 :
array[i][j] = 1;

Code 1 :
for(j=0; j for(i=0; i array[i][j] = 1;

Ans : Code 2

(15) In C which parameter passing technique is used?
(i) call by value,
(ii) call by reference,
(iii) both

Ans : call by value

(16) A circuit is given with 2 exclusive OR gates whose boolean _expression will be y = '(AB) + AB
(' indicates bar)

(17) main()
{
int i = 1;
fork();
fork();
printf("\ni = %d\n", i+1);
}

Ans : 4 printfs will occur and i = 2

(18) Compute the complexity of Binary search.
Ans : O(lg n) (Answer in detail. This is not a multiple choice question.
It carries more marks.)

(19) Write _expression for the tree graph :
Ans : ((a-b) + c*d)/x

(20) # define MAX(a, b) a>b ? a:b
```
main()
{
int m, n;
m = 3 + MAX(2, 3);
n = 2 * MAX(3, 2);
printf("m = %d, n = %d\n", m, n)
}
```

Ans : m=2, n=3
paper of texas instruments.
Technical + aptitude + interview.
* Questions on c
here i am not strictly following syntax it is just to show what was asked.
1 #define sum(a,b) a+b
value of sum(2,3)*2 Ans:8

2. a=5,b=6
swap(&a,b);
This function is written to swap a and b
find value of a and b . Ans 6,6

3.
```
function()
{
static int a=0;
a=a+1;
return a;
}
main()
{
function();
function();
function();
printf a;
}
```
final value of a ? Ans : a=3. static initializes once.
4.Write two prog. to swap a & b without using temp variable.
5.
```
unsigned char i;
int sum=0;
for(i=0;i<300;i++)
sum=sum+i;
printf(sum);
```

Ans:Program will held in infinite loop b/c i can not exceed 255.

6.
five questions on MOSFETS.
four were having single mosfets.
questions were simple.as i told u in Gwalior.
just on the funda that it will conduct if Vg-Vs > Vt .

In one question output at drain was to be calculated while o/p was initially
charged to 5v and to the gate 5v were applied.

In one question output at drain was to be calculated while o/p was initially
charged to 5v and the gate was shorted to drain.
7.
Clear the concept of settling time , hold time and other times. 3 ques on that.
like values of various delays were given and max frequency at which the circuit can work
hint : 1/sum of all delays . In our case ans was 200 Mhz.

8. An input and output waveform was given and circuit was to be designed with the use of one
delay.
Ans : exor gate in which second input is first input with a delay.

9. A question to determine sequence of counter. Don't get puzzled it was a
tough question.

10.The output and input of a inverter is connected by three RC stages in between of each stage two amplifiers with poles at imaginary axis were connected.
Hint : The poles at imaginary axis will create extra 180 phase shift thus the
circuit will oscillate and calculate the frequency of operation.

11. A series of infinite connected rc circuit and overall input resistance is
calculated.(question of 12 class)

1: given an expression tree and asked us to write the in fix of that expression

four choices

2:
global variables in different files are

a:at compiletime
b) loading time
c) linking time
d)execution time

3)size of(int)

a) always 2 bytes
b) depends on compiler that is being used
c) always 32 bits
d) can't tell

4)which one will over flow given two programs
2

prog 1:

```
main()
{
int fact;
long int x;
fact=factoral(x);

}
```

prog2:

```
main()
{
        int fact=0
for(i=1;i<=n;i++)
        fact=fact*i;

}
```

```
int factorial(long int x)
{

if(x>1) return(x*factorial(x-1);
}
```

a) program 1;
b) program 2;
c) both 1 &2
d) none

}

5) variables of fuction call are allocated in
a) registers and stack
b) registers and heap
c) stack and heap
d)

6)

avg and worst case time of sorted binary tree

7) data structure used for proority queue
a) linked list b) double linkedd list c)array d) tree

8)

```
 main(){
char str[5]="hello";
if(str==NULL) printf("string null");
else printf("string not null");
}
```

what is out put of the program?
a) string is null b) string is not null c) error in program d) it executes but print nothing

9)there are 0ne 5 pipe line and another 12 pipe line sates are there and flushed time taken to execute five instructions

a) 10,17
b) 9,16
c)25,144
d)

10)

for hashing which is best on terms of buckets
a)100 b)50 c)21 d)32 ans 32

11)

```
void f(int value){
for (i=0;i<16;i++){
if(value &0x8000>>1) printf("1")
else printf("0");
}
}
```

what is printed?
a) bineray value of argument b)bcd value c) hex value d) octal value

12)

```
void f(int *p){
static val=100;
val=&p;
}
main(){
int a=10;
printf("%d ",a);
f(&a);
printf("%d ",a);
}
```

what will be out put?
a)10,10

13)

```
struck a{
int x;
float y;
char c[10];
}
union b{
int x;
float y;
char c[10];
}
```

which is true?

a) size of(a)!=sizeof(b);
b)
c)
d)

14)

```
# define f(a,b) a+b
#defiune g(c,d) c*d
```

find valueof f(4,g(5,6))
a)26 b)51 c) d)

15)

find avg access time of cache
a)tc*h+(1-h)*tm b)tcH+tmH

c) d) tc is time to access cache tm is time to access when miss occure

16)

```
main()
{
char a[10]="hello";
strcpy(a,'\0');
printf("%s",a);
}
```

out put of the program?
a) string is null b) string is not null c) program error d)

17)

simplyfy k map

1 x x 0
1 x 0 1

18)

```
int f(int a)
{
a=+b;

//some stuff


}

main()
{
x=fn(a);
y=&fn;
```

what are x & y types
a) x is int y is pointer to afunction which takes integer value

19) char a[5][15];
int b[5][15];
address of a 0x1000 and b is 0x2000 find address of a[3][4] and b[3][4]
assume char is 8 bits and int is 32 bits

a) b) c) d)

there are 20 questions all in techinical paper and 36 questions in appititude test

in appititude thay have given all diagrams and asked to find what comes next

thay are quite easy and i hope if u practice r.s aggraval u can do it easily

for tecnical thay have given 1 hr for 20 questions and for not technical thay have given only 40 min and 36 questions

this is the paper i have right now

Here is Texas paper for you.
in this paper there was 20 questions as follows in 60 minutes .
second part consists of 36 que. in 30 minutes all questions are diagramatical.(figurs)..

1. if a 5-stage pipe-line is flushed and then we have to execute 5 and
12
instructions respectively then no. of cycles will be
a. 5 and 12
b. 6 and 13
c. 9 and 16
d.none

2. k-map

ab
---
c 1 x 0 0
   1 x 0 x

solve it

a. A.B
B. ~A
C. ~B
D. A+B

3.CHAR A[10][15] AND INT B[10][15] IS DEFINED
WHAT'S THE ADDRESS OF A[3][4] AND B[3][4]
IF ADDRESS OD A IS OX1000 AND B IS 0X2000

A. 0X1030 AND 0X20C3
B. OX1031 AND OX20C4
AND SOME OTHERS..

```
4. int f(int *a)
{
int b=5;
a=&b;
}

main()
{
int i;
printf("\n %d",i);
f(&i);
printf("\n %d",i);
}
```

what's the output .

1.10,5
2,10,10
c.5,5
d. none

```
5. main()
{
int i;
fork();
fork();
fork();
printf("----");
}
```

how many times the printf will be executed .
a.3
b. 6
c.5
d. 8

```
6.
void f(int i)
{
int j;
for (j=0;j<16;j++)
{
if (i & (0x8000>>j))
printf("1");
else
printf("0");
}
}
```

what's the purpose of the program

a. its output is hex representation of i
b. bcd

c. binary
d. decimal

7.
```
#define f(a,b) a+b
#define g(a,b) a*b

main()
{

int m;
m=2*f(3,g(4,5));
printf("\n m is %d",m);
}
```

what's the value of m
a.70
b.50
c.26
d. 69

8.
```
main()
{
char a[10];
strcpy(a,"\0");
if (a==NULL)
printf("\a is null");
else
printf("\n a is not null");}
```

what happens with it .
a. compile time error.
b. run-time error.
c. a is null
d. a is not null.

9. char a[5]="hello"

a. in array we can't do the operation .
b. size of a is too large
c. size of a is too small
d. nothing wrong with it .

10. local variables can be store by compiler
a. in register or heap
b. in register or stack
c .in stack or heap .
d. global memory.

11. average and worst time complexity in a sorted binary tree is

12. a tree is given and ask to find its meaning (parse-tree)
(expression tree)
ans. ((a+b)-(c*d)) (not confirmed)
13. convert 40.xxxx into binary .

14. global variable conflicts due to multiple file occurance

is resolved during
a. compile-time
b. run-time
c. link-time
d. load-time

15.
two program is given of factorial.
one with recursion and one without recursion .
question was which program won't run for very big no. input
because
of stack overfow .
a. i only (ans.)
b. ii only
c. i& ii both .
c. none

16.
struct a
{
int a;
char b;
int c;
}

union b
{
char a;
int b;
int c;
};
which is correct .
a. size of a is always diff. form size of b.(ans.)
b. size of a is always same form size of b.
c. we can't say anything because of not-homogeneous (not in ordered)
d. size of a can be same if ...

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{
int b=5;
a=&b;
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main()
{
int i;
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f(&i);
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}

what's the output .

1.10,5
2,10,10
c.5,5
d. none

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{
int i;
fork();
fork();
fork();
printf("----");
}

how many times the printf will be executed .
a.3
b. 6
c.5
d. 8

6.
void f(int i)
{
int j;

```
for (j=0;j<16;j++)
{
if (i & (0x8000>>j))
printf("1");
else
printf("0");
}
}
```

what's the purpose of the program

a. its output is hex representation of i
b. bcd
c. binary
d. decimal

```
7.#define f(a,b) a+b
#define g(a,b) a*b

main()
{

int m;
m=2*f(3,g(4,5));
printf("\n m is %d",m);
}
```

what's the value of m
a.70
b.50
c.26
d. 69

```
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```

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c. we can't say anything because of not-homogeneous (not in ordered)
d. size of a can be same if ...

DIGITAL DESIGN

1.Using a 2:1 Mux realize the following
a) NOT gate b) AND gate c) OR gate d) Ex-OR gate e) Ex-NOR gate
f) NAND gate g) NOR gate h) Latch i) FlipFlop

Answer:
For these kind of questions always use Shannon's Expansion.
hint : Use Shannon's Expansion , get expression in the form of Mux equation
muxout = sel_bar * Input0 + sel*Input1.

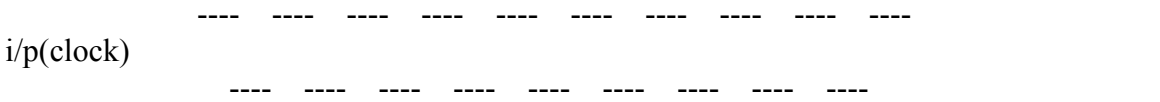
Ex: Realize a 2-i/p AND gate using a 2:1 mux.

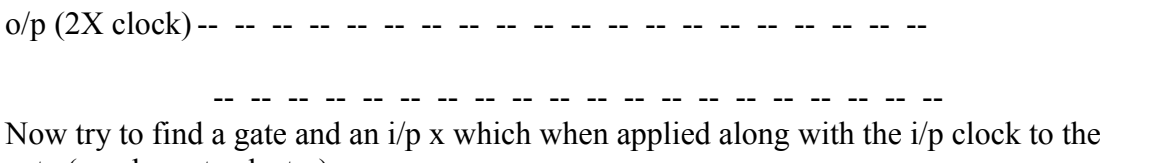
AND gate: Y = A*B.
= A*B + ~A*'0'

Now select A as Mux control signal and Input0 is '0' (ground potential/electrical equivalent of logic '0').
Input1 is 'B'.

2.Using Combo logic Multiply Clock by two (freq of clock at o/p = 2* freq at i/p).

Answer:
For these kind of questions, first draw the i/p and o/p waveforms, then try to add one or more waveforms which applied to a gate (or a combination of gates) will give the o/p waveform.

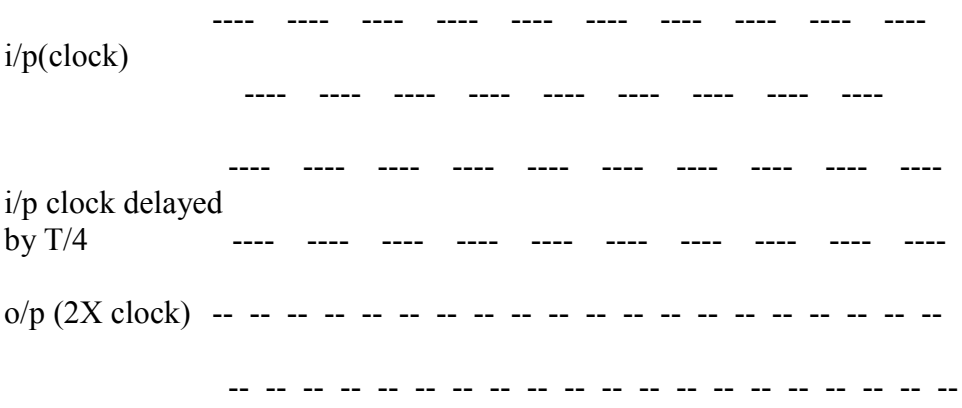




Now try to find a gate and an i/p x which when applied along with the i/p clock to the gate (combo gate cluster)

this is purely based on systematic approach... develop it...
you should be able to find that if the i/p clock is delayed by T/4 (where T is the period of the clock) and this applied to Ex-OR gate along with the actual clock would give the 2xclock.

Dont worry about the delay element for T/4, that would not be difficult, you can add a buffer.
Now try to get 3X clock using combo logic only. (you may need more than two i/ps ;)).



3.Realize a transistor level circuit for
Y = { [(ABC+Abar)bar] * (AB + Bbar) }

4.Given/using a Positive Trigger as input generate Square wave.

5. Question on Static Hazards
AND gate 1 has two i/ps A , sel
AND gate 2 has two i/ps B,sel_bar
output of these AND gates are given as i/p to ex-or gate
Tand = Tex-or= 2ns, Tinv (used for sel_bar) = 1ns

find Glitch width and draw the hazard-free circuit
hint: See switching theory book by Kohavi

6.Draw FSM for "0101" sequence detector and code it in Verilog/VHDL.
How many FFs are needed?

7.Given a 8 bit number how would you check whether it is a palindrome or not???

8.Two FFs are cascaded with combo logic in between (Q of FF1 to D of FF2)
Tcombo,min = 1ns and Tcombo,max = 3ns
Tsetup = Thold = 2ns, Tclk = 10ns, Tclock-to-Q = 2ns
check for Setup and hold time violations.

9.What is Synchronizer used for ? draw the ciruit and comment on sizing of Txs.
hint : see DIGITAL INTEGRATED CIRCUITS book by Rabaey.

10.Draw Tx level ckt for Y= AB + AC + BD + CD.

11.What is RACE condition ? How to avoid it?

12.Using D FF and combo logic realize T FF.

13.Using D FF and COMBO logic realize JK FF.

14.What are the advantages and disadvanteages of Dynamic Logic ?

15.Draw NAND and NOR equivalents of CMOS inverter for equal rise and fall times.
hint : see DIGITAL INTEGRATED CIRCUITS by Rabaey.

16. Is it possible to have negative setup and hold times ? Explain.

17.A 7 bit ring counter has initial state 0100010 after how many clock cycles it will return to
initial state?

18. Which device is fast BJT or MOS? Why ?

19. A 4 bit shift register has _____ number of states.

20. What is Mealy FSM and Moore FSM? Which one is fast?

21.Give adv and disadv of Mealy and Moore FSMs? Give examples of applications of both.

22.Swap two 8-bit registers without using another register.
hint : use boolean logic

23.Realize a two i/p AND gate using Ex-OR gate .
hint: don't waste time , come 2 a conclusion , ... ya u r right.... :-)

24.Describe an FSM to detect three successive coin tosses that result in Heads.

25.In what cases do you need to double clock a signal before presneting it to a Synchronous state
machine?

26.You have a driver that drives a long signal and connects to an i/p device. At the i/p there is
either overshoot or undershoot or signal threshold violations. What can be done to correct this problem?

27.What is a Silicon Compiler and a Memory Compiler used for?

28.To realize a 4x4 multiplier using ROM, what is the size of ROM needed?

29.In a system there are two modules A and B. A is operating at 25 MHz and B at 25 KHz From module A if a pulse of width equal to width of clock (1/25 Micro seconds) is sent, How
ensure that the pulse will be correctly received at module B without using handshaking or Buffers like FIFO?

30.A D FF has its D i/p from a MUX. MUX input0 is connected to external i/p and MUXi input1 is connected to output of D FF (Q) through combo block(i.e: feedback of o/p to i/p
thru combo block). If Mux delay is 0 ns and
Tsetup = 3ns, Thold = 2ns , TClock-to-Q = 1ns
What is the max frequency of the circuit with and without feedbak?

31.Why PMOS Tx is made 2.5 times wider than NMOS ?

32.If PMOS and NMOS Txs are interchanged in a CMOS inverter, what does it work like?

33.Draw Ids-Vds curve of a MOSFET with
a)increasing VGS, b) increasing W, c) considering Channel Length modulation

34.Why MOSFET goes into saturation and what type of current flows (drift/diffusion) at saturation?
(or)
If channel is pinched of how current flows from source to drain ?

35.List variuos Capacitances in a MOS device and their approximate values in Linear , saturaiton and cut-off regions.

36.Explain VTC of a CMOS inverter .what is the effect of channel length modulation in VTC ?

37.How to increase gain of a CMOS inverter in transition region ?On what factors does it depend?

38. What is Noise Margin, Noise Immunity? differentiate.

39.What is regenerative property of a CMOS inverter? explain with graphs.

40.What is Switching/logic threshold of a CMOS inverter ? How to change it?

41.How to measure Noise Margin?

42.What is Body effect?

43.What is CMOS latchup ? how to avoid it?

44.What is Electromigration ? How to avoid it ?

45.What is ESD ? How to avoid it?

46.What is Ground Bounce ? How to avoid it?

47.Why don't you use a NMOS/PMOS as a TG?

48.What is Full scaling and constant voltage scaling ?

49.Why scaling is done?

50. If a technology is scaled by 30 % (VDD also), how the following change
a) Cox,Cg b) Power c) Area d) Delay.

51.GIve the Expression for Elmore delay and penfield Rubenstein delay models.

52.Why NAND logic is preferred in CMOS ?

53.What happpens if we increase number of contacts and vias from one metal layer to another?

54.Draw a 2 i/p NAND gate and explain sizing regarding Vth and rise/fall times.

55.What are limitations in increasing Vdd to reduce intrinsic dcelay?

56.What happens to delay if we include a resistence at the o/p of a cmos ckt?

57.What is crosstalk ? On what factors does it depend?

58.What are various kinds of power dissipation in CMOS circuits?

59.What are the disadvantages of scaling?

60.You have three adjacent parallel metal lines.Two out of phase signals pass through outer
lines.Draw the signal in central metal line due to interference. repeat for inphase signals in the outer lines.

61.What happens if we increase no: of contacts or vias from one metal layer to another?

62.Draw Tx level ckt for a 2-i/p NAND gate and explain sizing considering
a) Logic threshold b) equal rise and fall times.

63. Why is it preferred to have logic threshold at Vdd/2 ?

64.What is Self-loading ?

65.Let A and B are inputs to a two i/p NAND gate, which signal should be close to the output
a) if signal A arrives later than signal B,
b) if signal B has higher switching activity than signal A,

66.Why fan-in of gates is resricted to 4 ?What is done to have large fan-in ?

67.Draw Stick diagram of a NOR gate and optimize it.

68.Give various methods used for reducing power in CMOS ciruits.

69.What is charge sharing ? Explain charge sharing while sampling data from a bus.

70.When driving a large capacitive load why do we use a chain of inverters with progressive
increase in size, instead of having a large buffer?

71.Explain difference between normal Buffers and Clock buffers.

72.Mention algorithms used for CLOCK distribution.

73.While laying out a large(wide) Transistor , why do we connect small transistors in parallel
rather than laying out a Tx with large width?

74.Why don't we use NMOS or PMOS as a switch?

75.Draw 6T SRAM cell . Explain read and write operation.
which one takes more time read/write ? why?

76.Draw a Differntial Sense amplifier and expalin its operation.

77.Draw a Cross coupled Snese amp and expalin its operation.

78.What is a double stage Differential sense Amplifier? what is it needed for?

79.Comment on sizing of Access Tx used in 6T SRAM cell.

80.Which one is fast NAND/ NOR ROM ?Give applications of each?

81.In memory design interconnect delay becomes critical , How is it reduced?

82.How does size of a PMOS pull up Tx affect performance of a 6T SRAM cell?

83.Explain sizing of variuos Txs used in SRAM cell.

84.What is critical path in SRAM?

85.In SRAM which metal layers would you prefer for word and bit lines?why?

86.How do you model SRAM in RTL ?

87.For an AND-OR implementation of a 2:1 Mux, how would you check for stuck-at-faults at
internal nodes?

88. Mention algorithms used for Stuck-at-fault analysis.

89.What is the differnce between testing and verification?

90.What Kind of circuit is this
A and B are inputs to an AND gate
AND gate output goes to one i/p of OR gate
The other i/p of OR gate comes from a Ex-OR gate
inputs to the Ex-OR gate are C and the output of the OR gate
(final output fedback to i/p)
combo/sequential?
synchronous/asynchronous?

91.Realize the boolean function
Y= A'B'C +A'BC+ABC+ABC'+AB'C
a) using 2-i/p and 3-i/p NAND gate,
b) using 2-i/p and 3-i/p NOR gate
c) using AOI gate
d) using inverter

92.What is the importance of SCAN in a digital system?

93. A Ex-OR B = C, Prove that
a) B Ex-OR C = A,
b) A Ex-OR B Ex-OR C = 0.

94.Construct a test pattern that can detect stuck-at-1 fault in the ckt given below
NAND gate NAND1 has two i/ps C and D
NAND gate NAND2 has two i/ps A and Y
AND gate has o/ps of NAND gates NAND1 and NAND2 as i/ps
and its o/p is Y (this is fedback to i/p of NAND gate NAND2)

95.In an Op-Amp ckt i/p offest is 5mv, Voltage gain = 10,000, Vsat = +/- 15v.Find o/p voltage.

96.Draw P-n/w for the function Y = ((AB+C) D)'.

97.Realize JK FF using D FF and MUX.

98.Realize the function Y= A + BC' + BC (A + B) using 2:1 Mux.

99.For the circuit given below
D FF "DFF1" has its D i/p,D1, connected to o/p of Ex-OR "Ex-OR1"gate.
D FF "DFF2" has its D i/p,D2, connected to o/p of Ex-OR gate "Ex-OR1".
i/ps of Ex-OR gate "Ex-OR1" are o/ps of "DFF1" and "DFF2" (Q1 and Q2)
CLK i/p of "DFF1" is connected directly to clock signal and CLK i/p of "DFF2" is connected to inverted clock signal (clcok signal goes to DFF2 thru inverter).
What is the realtion between input and output frequencies?

100.Design a Synchronous ckt for the following clock waveform
CLK ---> thrice the CLK period ---> half the period of i/p

101.What are setup and hold times of a FF? What happens if we don't consider them when
designing a digital circuit?

102.Two D FFs, "DFF1" and "DFF2" are cascaded, if Tsetup = Thold = 2ns and Twire = 0ns.What is the max Clock frequency for the ckt ? If DFF2 is negative edge triggered D FF
then what is the maximum clock frequency?

103.What is a FIFO buffer ? What is a FIFO buffer used for ?Give example.

104.How can you make sure that Glitches does not occur in a circuit at logic level?

105.What is the function of a D FF whose Complemented o/p (Qbar) is connected to it's input,D. What is the max clock frequency that can be used for it?

106.What happens if Setup violation occurs ? what happens if Hold violation occurs? Can a circuit have both setup and hold violations? Is it possible to have Setup and hold violations together on the same path?

107. Which one will have less switching activity ?
a) Tree real;ization or b) chain realization .

108.Two D FFs,DFF1 and DFF2 are cscaded and clock arrives late at the clcok input of DFF2.

What happens if the delay (in path from clock signal to clk i/p of DFF2) is large?How can this problem be solved?

109.Design a divide-by-3 sequential circuit with 50% duty cycle.

110.Draw the circuit of a TG based Latch.

111. _____

i/p ----------Buffer----------o/p
In the above circuit, what is the purpose of the buffer.(Note that o/p is fedback to i/p)? Is it redundant /necessary to have a buffer?

112.What is the o/p of the ciruit given below
2-i/p Ex-OR "Ex-OR1" has its i/ps tied to X,
2-i/p Ex-OR "Ex-OR2" has one of it's i/p connected to o/p of "Ex-OR1"
and the other i/p connected to X.
2-i/p Ex-OR "Ex-OR3" has one of it's i/p connected to o/p of "Ex-OR2"
and the other i/p connected to X.
What is the o/p of the circuit(o/p of "Ex-OR3").

113.Given a Circular disk with a sector of 45 degrees painted in blue. Two sensors are given and they can detect change in color. Design a circuit with minimum number of gates to detect the direction of the disk when it is rotated.

114.Given two transparent latches, realize a positive edge triggered D FF using minimum number of gates.

115.How many 2:1 Muxes are needed to realize a 16:1 Mux?

116.What is metastability? Why it occurs ? How to avoid it?

117.Convert a 2-i/p NAND gate to an inverter in two different ways.

118.Realize a T FF using 2:1 Muxes and few gates.

119.Realize D FF from RS latch (not Flip Flop).

120.What is the difference between EEPROM and Flash Memory?

121.Define Clock skew. What are the causes for it ? How Positive skew effects the system?

122.Define Clock jitter and differentiate skew and jitter.How clock jitter effects the system?

123.Which one is good Synchronous reset or Asynchronous reset?

124.Describe an FSM to detect the string "abca" if i/ps are a,b,c,d. Code it in verilog/VHDL.

125.Change rise and fall times of a CMOS inverter without changing W/L ratios.
hint: rise and fall time depend on current drive available.

126.What are setup and hold times? what do they signify ? which one is critical for estimating maximum clock frequency?

127.Suppose you have a combo ckt b/w two registers driven by a clock.If the delay of Combo ckt is larger than the clock period, then how would you overcome the problem?

128.The answer to the above question is break the combo ckt (functionality of combo into simple functions) and pipeline the combo block.What is the penalty in doing so?

129.Draw the ckts of TG based D latch and D FlipFlop(positive edge triggered).
how would you reduce load on the clock signal? what is the penalty in doing so?

130.Realize Ex-OR using TGs and modify to Ex-NOR gate (without complementing o/p).

131.Design an FSM to give modulo-3 counter when input X=0 and modulo-4 counter when input
X=1.

132.What is clock feedthrough?

133.Given a Clock signal, generate nonoverlapping clcoks (clock and clock_bar) using Combo logic.

134. What happens to VTC of a CMOS inverter, if supply voltage is reduced?

135.What are the limitations on reducing Vdd from delay point of view and from noise point of view?

136.Design a logic circuit using AOI configuration sich that if input a=1, output Y = AB+CD
else Y=DE + CF.

137.What is charge sharing? how to avoid it?

138.Design a ckt that clips every alternate clock pulse.

139.If A ? B = C and A?C = B, then what is the operator "?".

140.Dynamic circuits with feedback are called _____?

141.Design a circuit to count No: of ones in a 7-bit binary number (data comes in parallel).
(do not do it bit by bit)

142.Generate a square wave using Mux.

143.Draw CMOS ckt for a Tri-state Buffer.Realize a 2:1 Mux using Tri-state Buffer.
COMPUTER ORGANIZATION:

Hi folks,
I thought, Computer organization is required for a VLSI design engineer.Intel,amd,....do processor design and expect you to have "what is what" knowledge, you may not be doing the architecture development but nothing wrong in knowing "what is what "......

these are the Questions I have collected from my frens (and personal experience).

1.What is a Cache? What is it used for? What is the principle behind it?

2.what should be the size of a cache -- large/small?

3. What is a cache hit and cache hit ratio?

4. what are the various mappings used in Cache?
(direct, assosciative , set-assosciative)

5.What are the stages of a 5 stage DLX pipeline?

6. What are bubbles in a pipeline ?

7. What are HAZARDS in a pipelined system?

8. What is the ideal throughput of a N stage pipeline system? What prevents from achieving the
ideal throughput ? Is it better to have a 5 stage pipeline or 20 stage pipeline?

9.Expand TLB. what is it used for?

10. Name some Bus standards u know. Compare them.

11.Explain purpose of cache in a single Processor system and a double processor system with a
separate cache for each processor.

12.Explain difference between "Write through" and "Write back" caches.

13.What is MESI ?

14.What is Snooping?

15.Swap two 8-bit registers without using any other register.

16.Differentiate Overflow and Carry flag.

17.Differntiate Superscalar and VLIW processors.

18.What is MicroProgram control and Hardwired control?

19.What is Von-Numan architecture and Harvard architecture ?
Which one is used for MicroProcessor and which one forDigital signal Processor? Why?

20.What is Branch Prediction and BTB?

21.What is virtual memory?

22.What is cache Cohorency?

23.Differntiate MicroProcessor and MicroController.
Ans: In addition to all arithmetic and logic elements of a general purpose microprocessor, the microcontroller usually also integrates additional elements such as read-only and read-write memory, and input/output interfaces.

24.Processor is busy , but you want to perform some task . How will you do that?
Ans: Interrupts (Interrupts are used to pause execution of processor's program service a routine and then continue with the program)

25.What is ACBF (hex number) divided by 16 , give Quotient and remainder?

26.Given cache size is 64KB , Block size is 32B and the cache is two-way set assosciative.
For a 32-bit physical address, give the division between block offset, index and tag.

27.Differentiate RISC and CISC. Is RISC always fast?

28. How is a DSP different from a GPP?

Ans:The essential difference between a DSP and a microprocessor is that a DSP processor has features designed to support high-performance, repetitive, numerically intensive tasks. In contrast, general-purpose processors or microcontrollers (GPPs/MCUs for short) are either not specialized for a specific kind of applications (in the case of general-purpose processors), or they are designed for control-oriented applications (in the case of microcontrollers). Features that accelerate performance in DSP applications include:

* Single-cycle multiply-accumulate capability; high-performance DSPs often have two multipliers that enable two multiply-accumulate operations per instruction cycle; some DSP have four or more multipliers
* Specialized addressing modes, for example, pre- and post-modification of address pointers, circular addressing, and bit-reversed addressing
* Most DSPs provide various configurations of on-chip memory and peripherals tailored for DSP applications. DSPs generally feature multiple-access memory architectures that enable DSPs to complete several accesses to memory in a single instruction cycle
* Specialized execution control. Usually, DSP processors provide a loop instruction that allows tight loops to be repeated without spending any instruction cycles for updating and testing the loop counter or for jumping back to the top of the loop
* DSP processors are known for their irregular instruction sets, which generally allow several operations to be encoded in a single instruction. For example, a processor that uses 32-bit instructions may encode two additions, two multiplications, and four 16-bit data moves into a single instruction. In general, DSP processor instruction sets allow a data move to be performed in parallel with an arithmetic operation. GPPs/MCUs, in contrast, usually specify a single operation per instruction

While the above differences traditionally distinguish DSPs from GPPs/MCUs, in practice it is not important what kind of processor you choose. What is really important is to choose the processor that is best suited for your application; if a GPP/MCU is better suited for your DSP application than a DSP processor, the processor of choice is the GPP/MCU. It is also worth noting that the difference between DSPs and GPPs/MCUs is fading: many GPPs/MCUs now include DSP features, and DSPs are increasingly adding microcontroller features.

1. What is the difference between a latch and a flip flop. For the same input, how
would the output look for a latch and for a flip-flop.

2. Finite state machines:
(2.1)Design a state-machine (or draw a state-diagram) to give an output '1' when the # of A's are even
and # of B's are odd. The input is in the form of a serial-stream (one-bit per clock cycle). The inputs could be of the type A, B or C. At any given

clock cycle, the output is a '1', provided the # of A's are even and # of B's are odd. At any given clock cycle, the output is a '0', if the above condition is not satisfied.

(2.2). To detect the sequence "abca" when the inputs can be a b c d.

3. minimize a boolean expression.

4. Draw transistor level nand gate.

5. Draw the cross-section of a CMOS inverter.

6. Deriving the vectors for the stuck at 0 and stuck at 1 faults.

7. Given a boolean expression he asked me to implement just with muxes but nothing else.

8. Draw Id Vds curves for mosfets and explain different regions.

9. Given the transfer characteristics of a black box draw the circuit for the black box.

10. Given a circuit and its inputs draw the outputs exact to the timing.

11. Given an inverter with a particular timing derive an inverter using the previous one but with the required timing other than the previous one.

12. Change the rise time and fall time of a given circuit by not changing the transistor sizes but by using current mirrors.

13. Some problems on clamping diodes.

These are some of the questions asked by Microsoft.
(I feel that these type of questions are asked even in Electrical Engineering interviews. Make sure you browse them.)

1. Given a rectangular (cuboidal for the puritans) cake with a rectangular piece removed (any size or orientation), how would you cut the remainder of the cake into two equal halves with one straight cut of a knife ?

2. You're given an array containing both positive and negative integers and required to find the sub-array with the largest sum (O(N) a la KBL). Write a routine in C for the above.

3. Given an array of size N in which every number is between 1 and N, determine if there are any duplicates in it. You are allowed to destroy the array if you like.

4. Write a routine to draw a circle (x ** 2 + y ** 2 = r ** 2) without making use of any floating point computations at all.

5. Given only putchar (no sprintf, itoa, etc.) write a routine putlon the prints out an unsigned long in decimal.

6. Give a one-line C expression to test whether a number is a power of 2. [No loops allowed - it's a simple test.]

7. Given an array of characters which form a sentence of words, give an efficient algorithm to reverse the order of the words (no characters) in it.

8. How many points are there on the globe where by walking one mile south, one mile east and one mile north you reach the place where you started.

9. Give a very good method to count the number of ones in a 32 bit number. (caution: looping through testing each bit is not a solution)

10. What are the different ways to say, the value of x can be either a 0 or a 1. Apparently the if then else solution has a jump when written

```
out in assembly.
if (x == 0)
  y=0
else
  y =x
```

There is a logical, arithmetic and a datastructure soln to the above problem.

Logic design:

1. Draw the transistor level CMOS #input NAND or NOR gate. After drawing it lot of qestions on that ckt will be asked.

2. Transistor sizing for given rise time and fall time. How do you size it for equal rise and fall time.

3. Given a function whose inputs are dependent on its outputs. Design a sequential circuit.

4. Design a finite state machine to give a modulo 3 counter when x=0 and modulo 4 counter when x=1.

5. Given a boolean equation minimize it.

6. Given a boolean equation draw the transistor level minimum transistor circuit.

7. What is the function of a D-flipflop, whose inverted outputs are connected to its input ?

8. What will you do if you want to drive a large capacitance ?

Layout related questions:

1. asked me to layout the 3 input nand gate.

2. Later he asked me to modify it to consume as much less space as we can.

3. He also asked me about the transistor sizing.

---

1. He asked me to draw the cross section of an inverter and asked me to show all the capacitances on it and reason for those capacitances.

2. Describe the latchup effect.

3. More about the tristate buffers.

3. What will be the voltage at the output node of a triostate buffer in its high impedence state. He gave a waveform for the input and asked me to draw the output waveform for that.

4. Posed a lot of questions on charge sharing problems and keeper circuits.

5. Asked me to draw the Id Vds curves for mosfet. Asked me to explain the regions and some couses for that curve like channel width modulation.

6. He asked me about the electron migration effect and methods to avoid it.

7. Asked me to draw the dynamic logic of a particular gate and then posed lots of tricky questions from the previous discussion.

8. He asked me to draw the 6 transistor contemporary sram cell and asked me to explain how the reading and writing is done in it.

9. Something about trip point.

Computer Architecture Questions:

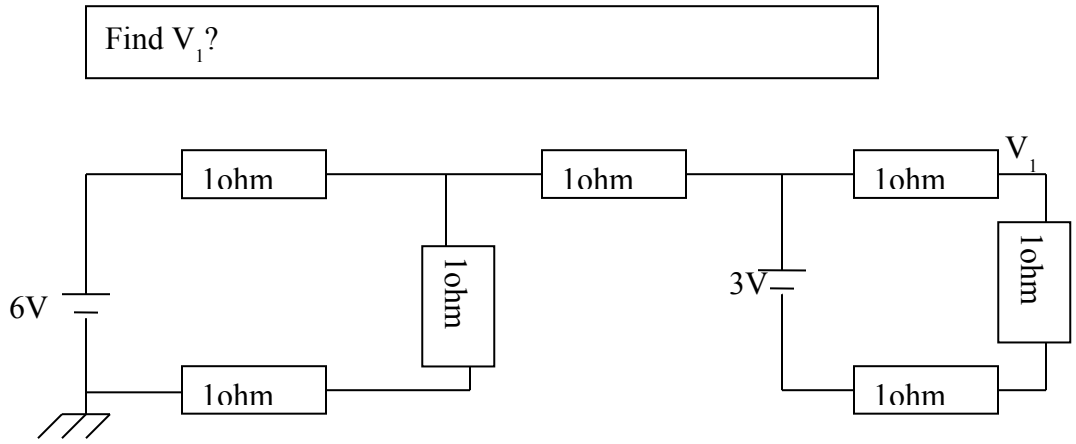
1. Explain what is DMA?
2. what is pipelining?
3. what are superscalar machines and vliw machines?
4. what is cache?
5. what is cache coherency and how is it eliminated?
6. what is write back and write through caches?
7. what are different pipelining hazards and how are they eliminated.
8. what are different stages of a pipe?
9. eplain more about branch prediction in controlling the control hazards
10. Give examples of data hazards with pseudo codes.
11. Caluculating the number of sets given its way and size in a cache?
12. How is a block found in a cache?

13. scoreboard analysis.
14. What is miss penalty and give your own ideas to eliminate it.
15. How do you improve the cache performance.
16. Different addressing modes.
17. Computer arithmetic with two's complements.
18. About hardware and software interrupts.
19. What is bus contention and how do you eliminate it.
20. What is aliasing?
21) What is the difference between a latch and a flip flop?
22) What is the race around condition? How can it be overcome?
23) What is the purpose of cache? How is it used?
24) What are the types of memory management?

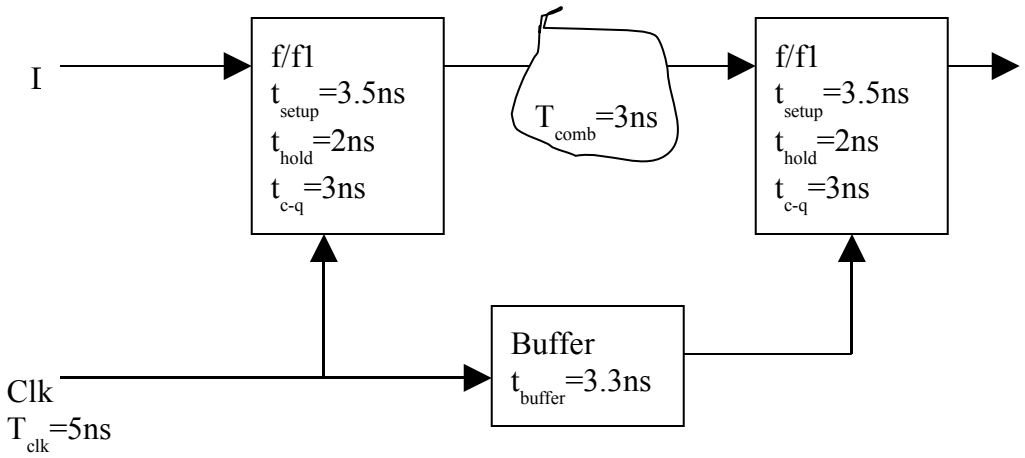
VLSI Interview questions

CISCO:

1)



2) Some star to delta conversion networks for finding the R
3) What shud we do to reduce latch up -----
3) How to reduce short channel effects – substrate is to heavily doped
4) Some mental ability q's
5) Convert a mux to an OR gate
6) Design a 2X1 mux using half adders
7) Some clk skew q
8) Some simple ckt which has 2 voltage sources in series to it and a current source u had to find the I through resistor which is a easy one to solve
9) Two latches constructed using muxes are cascaded such that it acts like a master slave flipflop and u shud mention wether it is +ve edge triggered or –ve edge triggered..
10) Some stuck at fault in a ckt and u shud mention the test vector for it.
11) Some k map simplification….
12) Given a boolean eq. and u shud design the ckt using min no. of nmos and pmos for that go for pseudo nmos technique.
13) Given the below ckt and u shud tell wether the clk period is enough or not and what problems that the ckt will faces (I m not able to remember the correct q and diagram)



Interview questions:
Some basic inverter q's
Latch up q's
Timing violation q's

Freescale:
1. How to design AND Gate using one pMOS and one nMOS.

2. Design a flip flop using MUX.

3.Design a divide by 3 synchronous circuit.

4. Positive edge detector circuit.

5. A simple combinational circuit was asked to be simplified.

6. Design a two bit comparator with and without using MUX.

7. A transistor circuit is given.find out the output voltage given Vbe and Vce. This is a simple one.

8. Design a square wave generator which takes only one positive edge trigger.

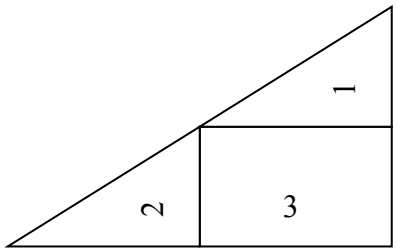
9. A question on maximum frequency of operation of a circuit. the setup time, hold time of the flip flops are given.

10. What is the purpose of the impedence matching between the load and source?
ans: To avoid the reflection of the power.

ITTIAM:

Written Test (Apti) :

1. Probability of 0 → 1 is p1 and 1 → 0 is p2. If 00 is xferred what is the prob of receiving at least one of them is 0. Ans 1 – (p1*p1)
2. Triangle 1: width is 5, height is 2.
   Triangle 2: width is 8, height is 3
   Rectangle 3: width is 5, height is 3
   What's the total area?
   a) 32
   b) 32.5
   c) 33
   d) both a and b



2. Something like this:

6471p + 3245q = 263452
3245p + 6471q = 236231

a) 1.5 <= p <= 2
b) 2 <= p <= 2.5 etc.

3. Speed downstream is 72 Kmph, level is 63 Kmph, upstream is 54 Kmph. A person travels A to B in 4 hrs and returns in 4 hrs 40 min. Distance from A to B is
   a) 203 Km
   b) 273 Km
   c) 302 Km
   d) Data insufficient
4. Something like this:
   A said "B didn't do it"
   B said "I didn't do it"
   C said "A did it"
   D said "B lies"
   Who is true?
5. Speed uphill=53miles/hr; speed downhill=70 miles/hr; speed on flat road=63miles/sec. It takes 4 hr to travel from town A to B and 4hrs 40min to travel from B to A. Find the distance between the towns.

6. A has n+1 coins and B has n coins. Both of them together toss all their coins. What is the probability that A gets more no of heads than B.

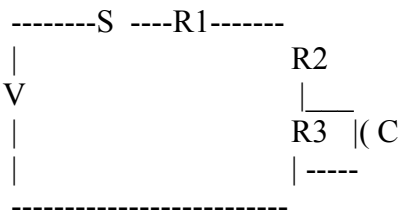
Interview

1. How to construct 4x1 mux using 2x1 mux only.
2. How to find out contents of PC at any point in the code. Ans Using CALL and reading top of stack.
3. x = (x +1) % 2 in the body of big loop. Optimize this to single operation. Initially x = 0. Ans x = not (x)
4. How to make a monostable (one shot) multivibrator using flipflops.
5. If the clock and D input of a D flipflop are shoted and clock connected to this circuit, how will it respond?
6. Some opamp circuit with several voltage and current sources connected through resistor dividers, find output.
7. Basic DSP theory: What is the frequency domain representation of (1) sinewave (2) cosine wave (3) the combination of sine and cosine waves. Given the output of (3)above, how will you find the input? Draw and show how it looks like.
8. If a LPF and HPF are connected in series, how will they respond under different cases of their cutoff frequencies (example if f1 < f2, what will happen)?
9. Interface an 8 bit µP with two 8Kx8 RAM chips. What would you do if A0, A1 are interchanged in h/w for only one memory chip. What'd you do in case of PROMs in case of RAMs?

DSP Paper

There are 3 sections ee, dsp and cse each with 20 q?s u have to attempt any one section only . Here I am sending the dsp section which I took , in other section the first and last 4 q?s were same as dsp.

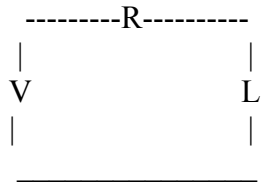
1



instantaneous Voltage across R2 when switch S is closed :
a.V*R2/(R1+R2)
b.V*R2/(R1+R2+R3)
c.0
d.V

Ans V*R2/(R1+R2+R3)------is what I wrote

2



as freq increases which of the following increases

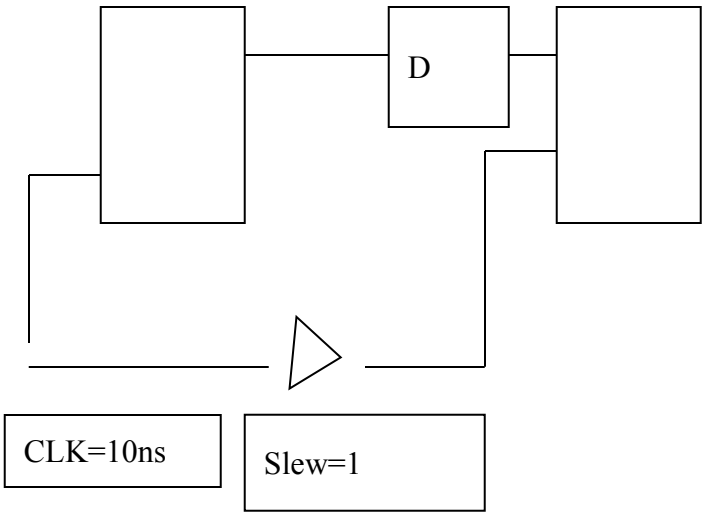
ans : Z and V(L)

3 q? on setup time and dealy

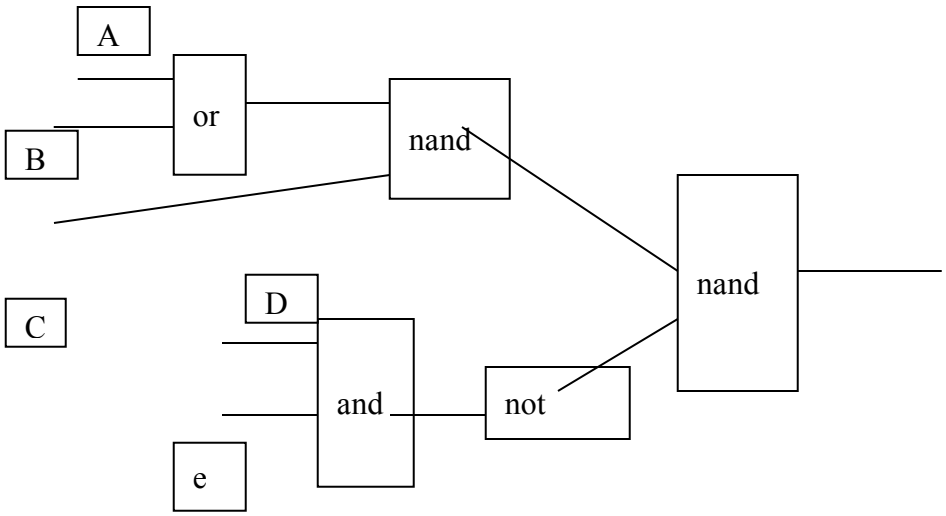
diagram below given not very clear\

clock period is 10ns ,
setuptime of each ff is 2 ns
clk2Q delay is 3ns
Slew dew to inverter is 1ns
Wht is the max allowable dealy of block D hold time=0

Ans 10-2-3-1=4ns



4 o/p of the following gate



ans : (a+b)c+de

5 SER=10^-4 the BER of a QPSK
a =SER
b <=SER
c>=SER
d =SER/2
ans >=SER

6 for 62db of PCM System what is the no of bits =10

7 for a 4 level pipeline processor the no of machine cycles required for executing 4 and (someno I don't rember) with initially pipeline flushed

ans = 4+3 and …+3
u add there for initial latency

8 An ideal LPF is
a causal
b non causal
c non stable
d none
ans: non causal ,

9 impluse func and white noise have same
a *magnitude and phase response*
b magnitude response
c phase response
d none
ans magnitude and phase response

10
y(t)=y(t-1)+0.1x(n) is what typr of filter
ans : IIR LPF

11 a signal s(t)=sin(omega*t) is sampled at fs, then the resulting signal spectrum is periodic depends on:
a *Omega/fs*
b omega *fs
c omega
d fs
ans omega/fs

12 if 2 gaussian func of mean m1 and m2 are added the wht is the resulting PDF
a guassian func with mean m1+m2
b guassian func with mean m1+m2/2
c uniform with mean m1+m2
d rayelig with mean m1+m2

13 u(t)+ sumof(deltafunc(n-k)){n=- infinitive to+ infinitive) is equaltent to
I hope I am made this clear it is a simple one
Ans u(-t-1) +2u(t)

14 if the probable of drawing an even no is p the wht is the probailty of drawing odd no in 2nd chance given 1st draw resulted in even one
a.p
b.(1-p)
c.p(1-p)
d.p/(1-p)
ans p(1-p)

15 no of multi required to mutli 2 upper triangular matrixes
a p(p+1)/2
b (p-1)(p+1)/2
c.summation (i=1 to p) i(i+1)/2
d…..
ans:c

16 a c program given something like this
unsigned short int i, j=1;
for (i=0,i<10,i++)
if(i&j)
printf("ITTIAM");

how many time is Ittiam printed .
ans 5 since bitwise and therefore only odd no will result in true if condtion

17 some c program abt function concerned with pointer and local variable easy one
ans 25

18 f=100 khz fs=125khz
o/p of filter with cutoff 150khzs
ans 25 and 100 khzs

19 stack in a processor is used for
a function call
b unlimted function call
c local variable
d something
I am not very sure if it is function call or unlimited function call , since function call also be done with shadow registers but only to a certain depth but most processor don't use shadow registers . at the same time depth of stack is also limited

20 x(F) is a signal whose freq response is asymmetric i.e H(F)=H(-f)
then it can be concluded that x(t) is
a real
b real or complex
c comples
d none
ans complex

here is the apti paper it also has 20 q?s and 30 min

1 some q? on some no is appended with 7(on right of units place) multiplied by 5 then result is similar to intial no with 7 on the left most (most significant digit)find the 3rd digit
ans =2 (?)

2 OTTSSFF?N
a.T
b.E
c.N
d…
ans: I wrote E

3 abcdefghij
a=no of zeros in the no
b= no of ones in the no

c= no of twos
so on
wht is the sum of digts
a.10
b.55
c.9
d….
ans 9(?)

4 ¼ of a no +2/3 of another no =3/8 of sum
wht is their ratio
ans: 3:7 (7:3)

5,6,7,8 9 four q? on some gre type analytical it was abt some 4 family runs a 4 restaurant name of husband (jai, jayesh, Parikh,bipin), wife(beena, chand, preethi, sangeetha) and their familyname (joshi,natwar,sahni,….)give , some hint and who runs which hotel(Indian court, American court,….) asked

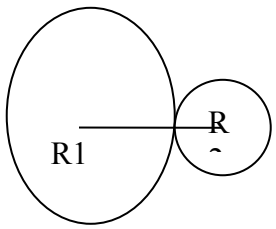
ans D D D C in that order in our paper :ANSWER THIS Q (I think this was an important question)

10 If in a test 1 mark is for correct answer ,what negative mark should be kept for nullifying the correct answers…
a.1/4
b.1/2
c.3/4
d…
ans=1/4

11 A wins B by 28 meters or (some) seconds(time) the A is ahead of B

Ans 4 min 20 sec

12 Given 2 circles of radius R1 & R2. how many rotations will the smaller circle have to make a full revolution around the circle with radius R1.



ans (r1+r2)/r1

13 An equlateral triangle and its circumcircle..what is the probability that a line drawn inside this circle is longer than the side of the equilateral triangle

ans =1/3(?)

14 given wt 1,3,9,27 how much max can u weigh
ans 40

15 43 players play some knock out game . how many games should be conducted to declare a winner
ans 42

15A man traveling at a speed of….misses a train by 7 mins…if he travels at a speed …..how far should he travel to catch the train
ans 6 km

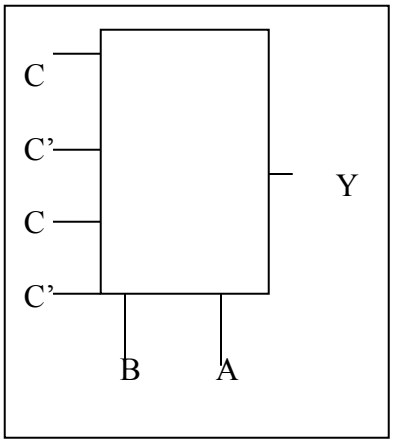
16 2 traingle made form circles fiven

0	0000
00	000
000	00
0000	0

how many min circles have to be removed to get some thing
ans 3

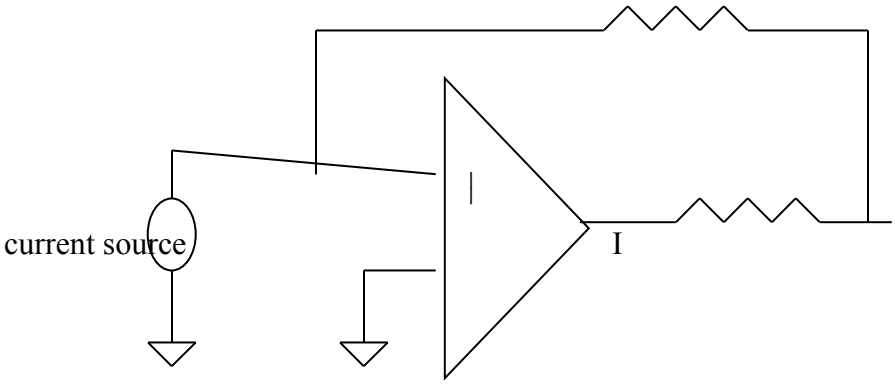
17 a boy has trasfered 100 galss from one palce to another the owner puts a condition tht if he delivers safely he gets 3 paise for each glassand he would forfeit 9 paise for every broken glass. He loses some glass and gets Rs.2.40/- wht are the no of broken glass
ans 5

in EE paper:
some questions which I saw:
5. what is the expression for o/p?
a.AC
b.A'C+AC'
c.C
d.B'C+BC'

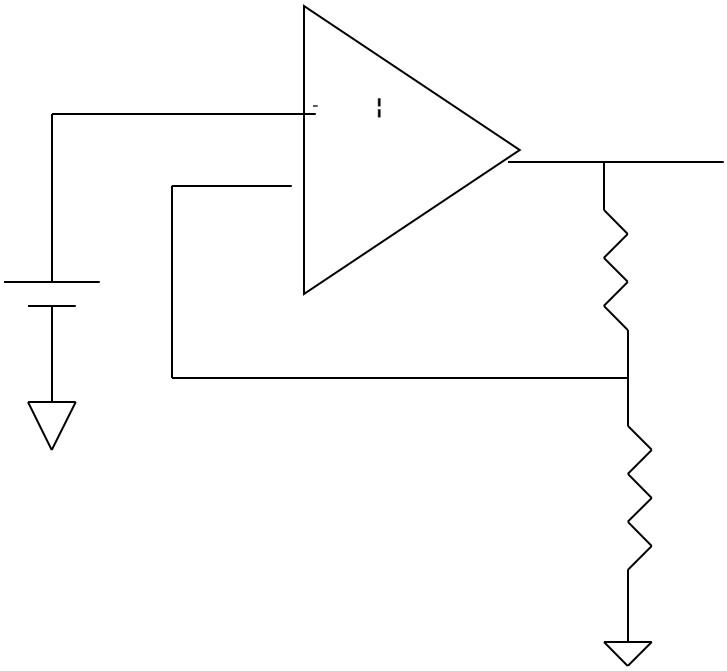


ans:A'C+AC'

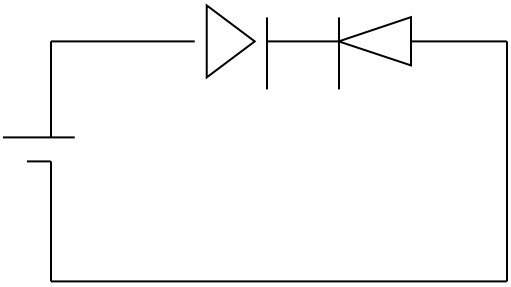
6.what is the current I in the ckt assuming ideal opamp as shown: resistance values given



7.what does the ckt below work as:
something similar to this : I think it was Schmitt trigger ckt (check it out)



8.what is the current flowing in the ckt:
a.Is
b.0
c…
d….



9.some problem on writing laplace transform of the given ckt
a.E/s[….]
b.Es[…]
c….
d….

Hi …… These are the questions,in interview for ITTIAM. I have written the answers I gave. There might be better solns ☺

Q. Give the time and frequency domain representation of a sine wave.
A. Sketch a sine wave for time domain rep. For freq domain rep, it is 1/2j [delta(f-fc)-delta(f+fc)]. Phase of the sine wave is 90 deg. (lag)

Q. Repeat the above for a cosine wave – Phase is zero deg.

Q. Repeat the above for sum of sine and cos waves.

Q. Given large amount of data of ones and zeros, how wud u compress it using huffman coding.
A. Use Run Length coding and code the run lengths using Huffman coding

Q. Any alternate way for the abov prob.
A. Consider three bits at a time. That can give rise to eight possible symbols. Then code these symbols using Huffman Coding.

Q. Will Huffman coding be always advantageous.
A. No, If the prob of occurance of symbols are equal then there is no advantage

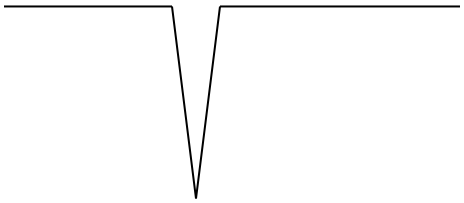
Q. When is Huffman coding optimum
A. When prob of symbols are powers of ½ it can be optimum

Q. why
A. Use the equation, bits reqd = – log p. If p is powers of ½ then no of bits will be an integer and hence it will be optimum.

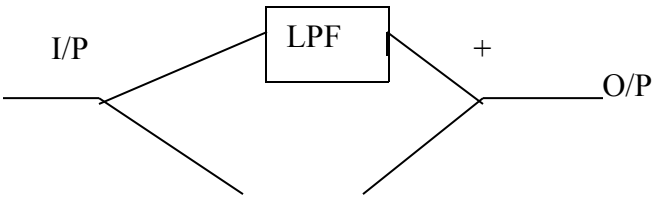
Q. Is there any other method of optimal entropy coding
A. Arithmetic Coding

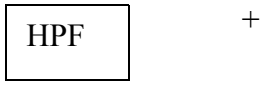
Q. Why is it optimal
A. Since group of symbols are coded, fractional bit rate can be used.

Q. What is a notch filter
A. Filter with response



Q. Given Low pass and High pass filters how to realise above filter
A.





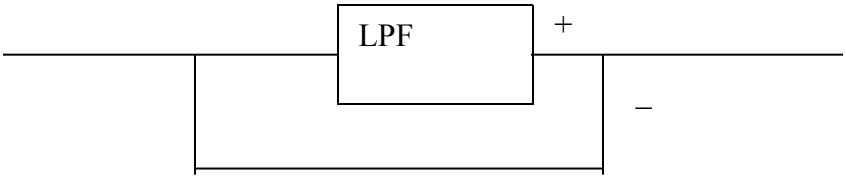
Choose LPF with cut off slightly below the notch freq.
HPF with cut off slightly above the notch freq.

Q. How to design filter with gain at single freq
A. Put LPF and HPF in series. Choose Cut off of LPF slightly above the notch freq.
Choose cut off of HPF slightly below notch freq.

Q. Given an RC-Ckt, Low pass output is obtained across which component and why
A. Across C. Because impedance of C inc as freq reduces, so voltage across it inc, as freq reduces.

Q. Given two LTI systems in cascade what is the resultant gain and phase
A. Resultant Gain is product of the two gains, Resultant Phase is sum of the two phases.

Q. what is the O/P of the following system



A. All the low freq components will be attenuated and high freq components will be inverted

Q. How will the phase of LPF affect the above ckt.
A. Low frequencies get cancelled only when phase shift is integer multiples of 2pi.

Q. Two LTI sys are in cascade. Impulse response of first system is h1(n). Second system is described by the diff equ. Y(n)=x(n-1)+x(n), what is the overall response of the cascade
A. overall response is h1(n-1) + h1(n)

Q. I have a database of 5 faces. Given a test image of some face, what is the simplest way to recognize it.
A. Correlation

Q. If the test face is taken in totally diff lighting conditions wud correlation work, How wud o solve the prob
A. I gave solns like removal of DC Component, Histogram equalisation and then correlation etc…….he was not convinced.

Q. Is an image zero mean signal
A. No

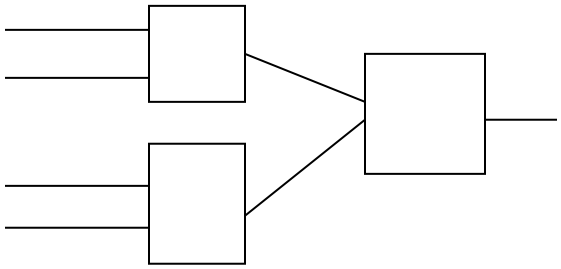
Q. P.T. sum of 2 odd nos is even

Q. Is sum of two prime nos prime. If Yes Prove, If not Prove

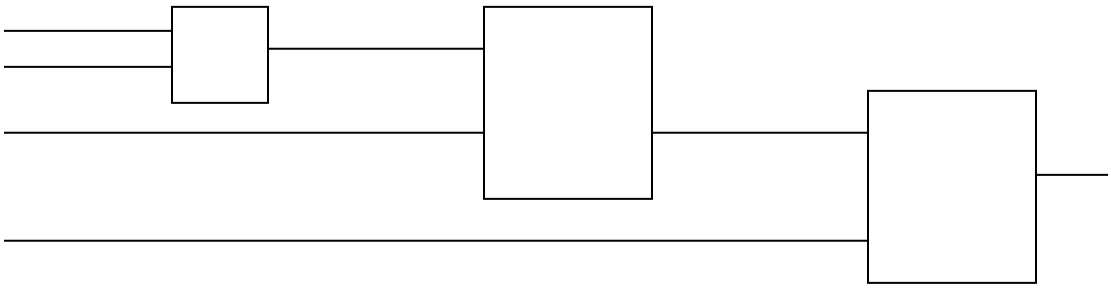
A. Same proof for both of the above

Q. Give diff configurations to realise 4 - I/P AND gate using 2 – I/P AND gates
A.

Config 1:



Config 2:



Q. Given a black box consisting of one of the above config, how wud u detect as to which is the config.
A. In Config 1, delay is same for all i/ps. In Config 2 delay for i/p 1110 is less than for 1011.

There were other small questions which I do not remember. All were very very basic. Just stay cool and u can answer everything.

ITTIAM paper 2004 (EE section only)

1) If the probability of 0 being received as 1 is p_1 and that of 1 being received as 0 is p_2. What is the prob of receiving at least one 1 if two consecutive zeroes are sent?
→$1-(1-p_1)^2$

2) For symbols a, r, and p having a prob of occurrence as 0.4, 0.2, and 0.2 and t being another symbol, what is the length of code for 'a' in huffman binary coding?
→ 1

3) For probability of 0.25,0.25,0.5, find the entropy.
→3/2

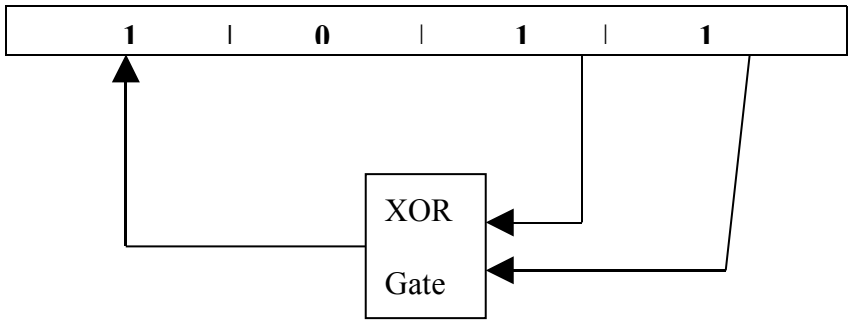
4) How many multiplications will be required for multiplying two p×p upper triangular matrices.
→ sigma p^3 (I am not sure of the answer)

5) How many cycles are required for a N,M convolution given that each addition, mult and mac requires one cycle?
→ I ticked (N-M+1)M/2 (not sure of the answer)

6) For 62db of SNR, what it the channel capacity? (think they meant per unit BW)
→ use $C=B \log_2(1+SNR)$

7) One question on odd-parity detector. ('what is this' type)

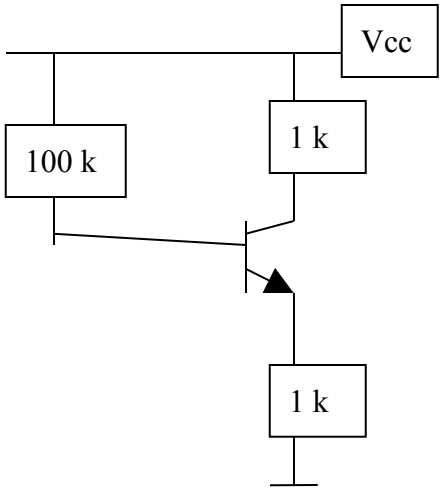
8) One question on bistable multivibrator using 741. (again 'what it this' type)



9) The above is a 4 bit shift reg. The feedback path has an XOR gate. Tell the value of the reg after two shift right.

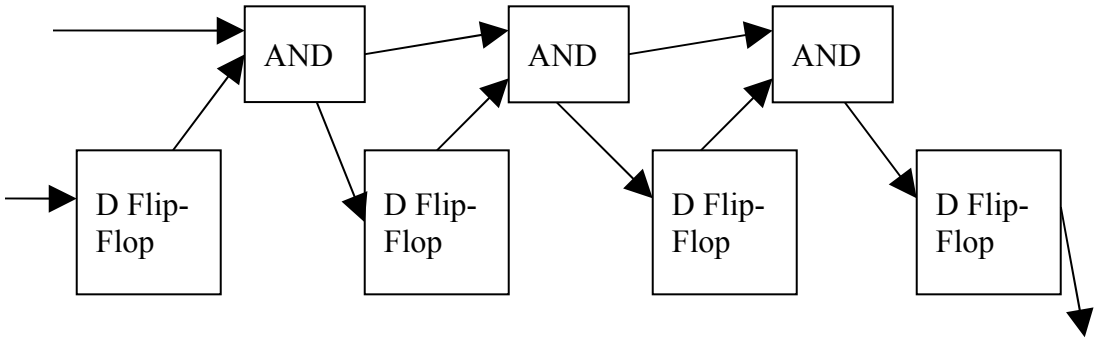
10) A simple C program. What will be the output of i&j where i=10, j=20.
→ 0

11) For Vcc =20 V and beta =100 find Ie for V_{be}= 0.6 V. Take Ic = Ie.
→9.6 mA



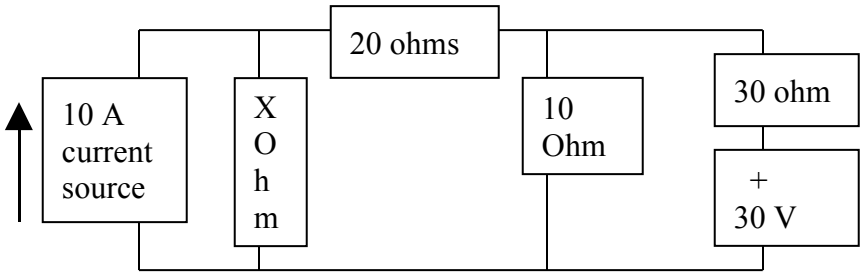
12) The propagation delay of each AND gate is 10 ns. What could be the max clock frequency.

→ 10^8 Hz

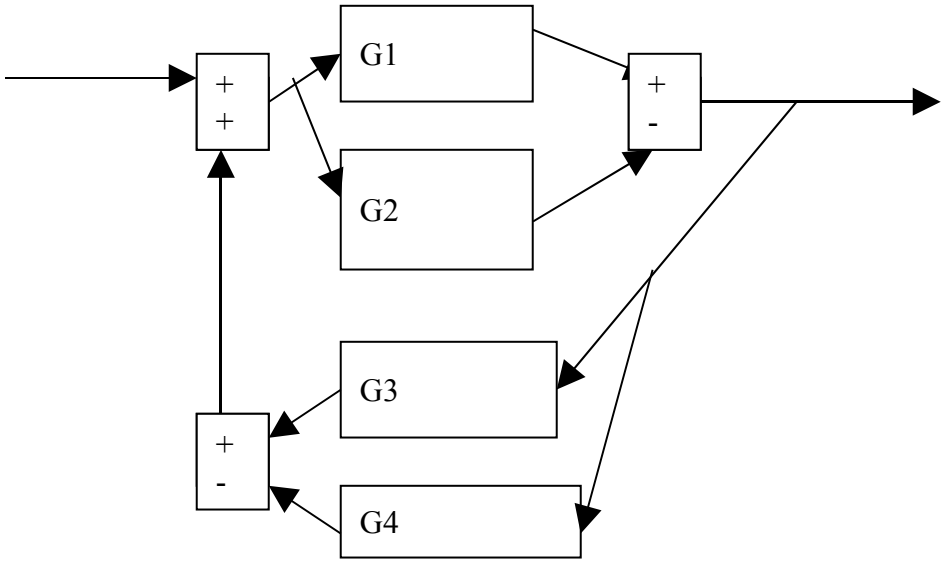


13) The current across 20 ohm res is

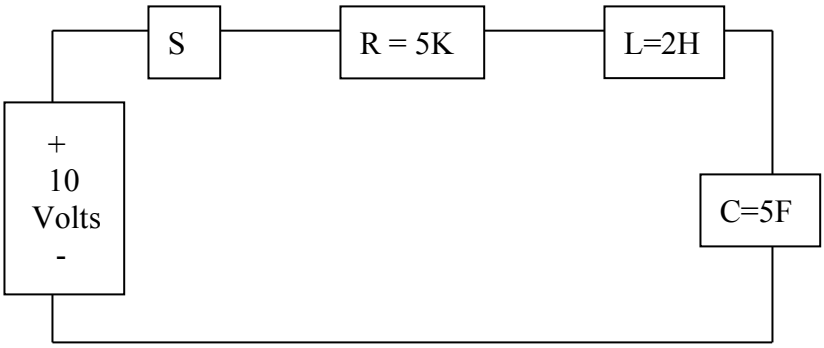
→0.5



14) Block diagram (find the Transfer Function):



15) Find I for t=0 when the switch S is closed. Also dI/dt.
→ 0A, 5A/s



16) There was a question about the causality and non-linearity of a system given its difference equation.

Ittiam interveiw questions

1. make a 4x1 mux using 2x1 multiplexers
2. make a 2-input Or gate using 2x1 mux
ans. make A & B as the input of Mux & B as the select signal
3. In the z-plane there is a zero at intersection of unit circle and x-axis & there are two poles somewhere inside the unit circle Then what can u tell about the fourier transform of the signal??
ans. the FT will definitely be zero at origin
4. A signal s1(t) is passed through a LPF to get s2(t) and the s2(t) is

subtracted from s1(t) to get s3(t) =s1(t)-s2(t)
How will s3(t) relate with s1(t)??
ans. It may or may not be the high pass filtered version of s1(t)
depending on the phase of LPF (he was not convinced with the answer)
5. what is difference between A-law & mu-law ??
ans. ??

Some questions of Ittiam:

'ABCDEFGHIJ' is a number.
A is the number of 0s.
B is the number of 1s etc
J is the number of 9s
What is the number

Some question on logic was there. like finding whose husband is john or whose wife is mary type.

in general other aptitude questions were peace

(interview) what is the algorithm to find if p is a prime. Why do u need to test only upto sqrt(p) factors.

Technical test and interview):
DSP: y(n) = a y(n-1) + b x(n); what is the condition for system to be stable ? why?
find region of convergence of 1 +2 z inverse
if y1(n)= h1(n)*x(n) and y(n)= ay1(n)+by1(n-1). Find impulse response of over all sysemin terms of h1(n) only.

Huffman coding.(given set of symbols and probabilities construct code)
ASked the algorithm in general to generate such an optimal code.(there is something like a tree diagram given in every digi compression book).

asked to give output of RC circuit with const voltage source. then with const current source. Asked why in 2nd case the voltage cannot goto infinity(could be capacitor break down or that the voltage across current source can't exceed a certain value)

Construct four input and gate using 3 two input and gates in two ways.
If a black box with one of them is given how will u find which configuration it is. U have only black box and nothing to compare that with. u can give any input and see output.
How many min number of input combinations do u need

Pseudo code for matrix transpose. Should be optimal and swapping alos should be optimal. I gave sswapping using arithmetic operators. But they wanted that using logical operators. jus replace the arithmetic operations with xor. how many computations do u need for the getting the transpose of n*n matrix. Why

Whats the probability that u pick two red balls out of a bag of two red balls and 3 black balls? they tried to confuse. But i held on . They were seeing it in a different manner but finally landed up with the same answer i gave. Soif ur sure don't give up

Intel:

Paper I

1. Find Voltage across R and C in the following circuits.

a. In a given RC circuit find the voltage across C and R?

b. In a given CR circuit find the voltage across R and C ?;

2. For the given _expression Y=A'B'C+A'BC+AB'C+ABC+ABC' realize using the following

a. 2 input and 3input NAND gate

b. 2 input and 3 input NOR gate

c. AND,OR, INVERTER.

d. INVERTER;

3. What is the importance of scan in digital system.;

4.Given A XOR B =C, such that prove the following

a. B XOR C =A

b. A XOR BXOR C=0;

5. Construct an input test pattern that can detect the result E stuck at 1 in the ckt below

NAND (A,B)->E, NAND(C,D)->F

AND(E,F)->A.

6. In a given opamp ckt input offcet is 5mv,volatage gain =10,000,vsat=+-15v

such that find the output voltage .

7. Draw the p side equation of the circuit.(I am not sulre about it)

8. Make a JK FF using a D FF and 4->1 MUX.

9.Use 2->1 MUX to implement the following _expression

Y=A+BC'+BC(A+B).

10.For the following ckt what is the relation between fin and fout.?

the D FF use +ve edge triggered and have a intial value is 0

CLK->two DFFs with complementing (i.e one DFF have CLK and other one have

Complement of it),inputs of DFF is same and output of DFFs is given to NOR

Gate and output of NOR gate is feedback to the two DFFs.

11. Design a asyncronous circuit for the following clk waveforms.

CLK->thrice the CLK period->half the period of input.

12. What is the setup time and hold time parameters of the FF, what happens if we are not consider it in designing the digital ckt.

13. Given two DFF A,B ones output is the input of other and have the common clock.

Fmax if A and B are +ve edge triggered, if A is+ve edge triggered ,B is -ve edge triggered what is the Fmax relation to previous Fmax relation…

14. What are the FIFOS .? give some use of FIFOS in design.

Paper II

1. What is FIFO ? where it is used?
2. what is set-up and hold time?
3. Two +ive triggered FFs are connected in series and if the maximum frequency that can operate this circuit is Fmax. Now assume other circuit that has +ive trigger FF followed by –ive trigger FF than what would be maximum frequency in terms of the Fmax that the circuit can work?
4. layout of gates were shown and u have to identify the gates (NAND & NOR gates)
5. make a JK FF using a mux(4:1) and a FF.

6. the waveform of clk, i/p and o/p were shown and u have to make a seqential circuit that should satisfy the required waveform.
7. resistor is connected in series with capacitor and the input is dc voltage. Draw the waveform across the capacitor and resistor.
8. two FFs, one is –ive triggered and other is +ive triggered are connected in parallel. The 2 i/p NAND gate is has the i/ps from the q_out of both the FFs and the output of the NAND gate is connected with the I/p of both FFs . Find the frequency of the output of the NAND gate w.r.t clk.

Interview questions (face to face discussion)

1. Draw the circuit for inverter. How does it work?
2. If the pmos and nmos is changed in the inveretr, how does it behave?
3. Design flow for ASICs and FPGA. what are the difference between the ASICs and FPGA?where do u use ASIC and where u use FPGA?
4. What is floorplanning?
5. What do u mean by technology file used in the synthesis or optimization for the circuit (netlist)? What is the difference in the technology files used for the ASICs and FPGAs based designing?
6. Using a FF and gates. Make a memory (i.e include RD, WR etc.)
7. If the setup & hold time gets violated than what u 'll do to remove it?
8. What is clock skew? How u 'll minimize it?
9. What is clock tree? How it looks like? Concept behind that.
10. What about the Vdd and Gnd lines ? does one Vdd and Gnd pins will be sufficient for the chip. What will be the effect of using single Vdd and Gnd pins in the chip?
11. What is voltage refernce circuit? What is bandgap? How does it work?
12. what is FIFO? How does it work? Draw the circuit of FIFO of 1-bit and 4memory location deep? What would happen if memory is full and again u try to write in FIFO? What u 'll do to overcome this problem? Which one would be more easier to implement :- either dropping the packet, when the FIFO is full or pushing the data of FIFO every time. And why ?

The following questions are used for screening the candidates during the prescreening interview. The questions apply mostly to fresh college grads pursuing an engineering career at Intel.

COMPUTER ARCHITECTURE QUESTIONS

1. For a single computer processor computer system, what is the purpose of a processor cache and describe its operation?

2. Explain the operation considering a two processor computer system with a cache for each processor.

What are the main issues associated with multiprocessor caches and how might you solve it?

3. Explain the difference between write through and write back cache.

4. Are you familiar with the term MESI?

5. Are you familiar with the term snooping?

STATE MACHINE QUESTIONS

1. Describe a finite state machine that will detect three consecutive coin tosses (of one coin) that results in heads.

2. In what cases do you need to double clock a signal before presenting it to a synchronous state machine?

3. Design a FSM that will assert output when more than one ‘1’ is recieved in last three samples. Do not use more then 4 states.

SIGNAL LINE QUESTIONS

1. You have a driver that drives a long signal & connects to an input device. At the input device there is either overshoot,

undershoot or signal threshold violations, what can be done to correct this problem?

VALIDATION QUESTIONS:

What are the total number of lines written in C/C++? What is the most complicated/valuable program written in C/C++?

What compiler was used?

Have you studied busses? What types?

Have you studied pipelining? List the 5 stages of a 5 stage pipeline. Assuming 1 clock per stage, what is the latency of an

instruction in a 5 stage machine? What is the throughput of this machine ?

How many bit combinations are there in a byte?

What is the difference between = and == in C?

Are you familiar with VHDL and/or Verilog?

MEMORY, I/O, CLOCK AND POWER QUESTIONS

1. What types of CMOS memories have you designed? What were their size? Speed? Configuration Process technology?

2. What work have you done on full chip Clock and Power distribution? What process technology and budgets were used?

3. What types of I/O have you designed? What were their size? Speed? Configuration? Voltage requirements?

Process technology? What package was used and how did you model the package/system?

What parasitic effects were considered?

4. What types of high speed CMOS circuits have you designed?

5. What transistor level design tools are you proficient with? What types of designs were they used on?

6. What products have you designed which have entered high volume production?

What was your role in the silicon evaluation/product ramp? What tools did you use?

7. If not into production, how far did you follow the design and why did not you see it into production?

nVIDIA:

```
1-6 are multiple choice questions)

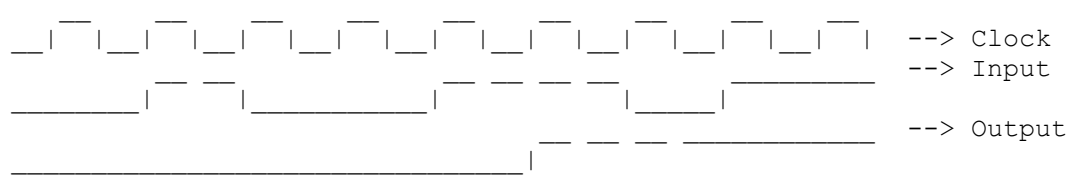
1. The max. value and min. value of 16-bit 2's complement (hex, dec,
binary)?
2. The max.,min. value of 16-bit 1's complement (hex, dec, binary)?
3. max. no. of logic functions for n-variables? ans: 2^2^n
4. about physical and virtual address, which is greater?
5. TLB (Translation Lookahead Buffer) is used for ?
   options:
   1. L1 cache misses
   2. L2 cache miss
   3. some thing page miss (not remembered exactly)

6. Minimum no. of P and N Mos transistors required to implement The
logic
   Function Y= !(A | B & C) using CMOS
   1. 1 p & 3 n
   2. 3 p & 1 n
```

3. 3 p & 3 n

7.One shot digital circuit

8.Noise eliminator (both -ve and +ve pulses of one clock cycle duration)



Observe the one clock delay in output.

Design a state machine for this operation.

9. Implement the following 2 functions using only 2x1 MUX without gates. U can use 1 or 0 for inputs.

a) Y = AB+not(C) b) Y = A xor B

10. Write a program in C or C++ to implement Stack and its functions such as
isEmpty, isFull, Push, Pop, Init etc. (I think based on the question given,
array implementation is sufficient)

11. Question on clock frequency required for given Sequential ckt.
Given 2 F/Fs, 3 delays (2 delays for clock dly1, dly2 and 1 delay dly3 of
combinational circuit) given setup, hold and propagation times

Ans: T+(dly2-dly1) >= Tpd + dly3 + Tsetup
Tpd + dly3 > Thold + (dly2-dly1)

Others:

1. A Positive logic NAND gate will be equuivalent to a '-'ive logic ---------- gate.

a) NAND
b) EX-NOR
c) NOR
d) OR

SANDISK:

SANDISK IIT BOMBAY PAPER, 26th DECEMBER, 2005

Written Test 45 mins

1) No. of universal logic gates reqd to implement EXOR

a) 4 NAND
b) 4 NOR
c) 5 NAND
d) 5 NOR

2) Using (A AND Bbar), we can implement
   a) only AND
   b) only OR
   c) any logic function
   d) none

3) A –V to +V pulse voltage source is connected to a RC series ckt. Draw the waveforms of voltage across R, voltage across C, and current in the circuit.

4) Draw the capacitance vs voltage characteristics of MOSFET and MOS cap, and point their differences in the HF region

5) Arrange an underdamped, critically damped, and overdamped system in order of phase margins

6) Find the voltage gain of a transconductance amplifier of transconducatnce gm, with Vi at +ve terminal, C btwn –ve term and gnd, and R between –ve term and output

7) Considering MOS caps Cgs and Cgd,
   a) Cgs>Cgd in cut-off region
   b) Cgd>Cgs in saturation region
   c) Cgd=Cgs in triode region
   d) None

8) Draw the waveform of "A" from the verilog code

```
Always(@clk)
Begin
A=0;
#5 A=1;
end;
```

9) Draw a NORbased latch, calculate its setup time if delay of each gate is td

10) A 1V dc source is connected to the source of an NMOS, a 0.1 nf cap is connected to the drain, and a 5V single pulse of duration 1 us is applied to the gate. To act as an integrator,
   a) W/L >>1
   b) W/L<<1

c) W/L=1
d) Cant be said from the given data

Interview 1st round :

Questions from the written test which I could not answer correctly, transfer characteristics of a CMOS inverter, implementation of an FSM given a state diagram, and a riddle :-given only a 3 l and a 5 l bottle, and nothing else, how would u measure 4 l water?

Interview 2nd round :

What are the issues if the duty cycle of the clock in a digital ckt is changed from 50%?
What are the different tests you would do to verify your verilog code?
How would your friends describe you?
What is the greatest risk you have taken so far in life?
What are the differences between academics and industry?

Paper II
1 simple current mirror question.

2 to generate non-overlapping clock.(see Rabaey page 339)

3 question on Verilog synthesis

4 always@(posedge clk)
begin
a=0;
#5 a=1;
end
what is the output waveform of a?

5 question on differential amplifier gain with $(w/l)_1=2*(w/l)_2$

6 V=vin1 – vin2 (vin1 and vin2 are two input voltages of 1 stage diff. amplifier).Now V is varied from -5 to +5 then draw the output voltage vs V.(Vdd=+5 Vss= -5).

7 two simple question on charging of capacitor with constant current source.

8 draw the VTC of buffer (PMOS and NMOS are interchanged in inverter)

9 what should be the ratio of $(W/L)_{PMOS}$ / $(W/L)_{NMOS}$ for switching threshold of Vdd/2.Given Kn/Kp=2.8.

10 there is 2 input CMOS NAND gate .inputs A and B changes from 0 to Vdd. but A goes to Vdd after B(after some delay). which input should be closer to Vout.

11 what are the benefits of finger layout----less junction capacitance etc

Others couldn't recall ..

Two rounds of interview-- HR and technical.
Nearly 45 mins for technical (Device, digital and mostly analog).
HR also of 45 mins.

Paper III

Q1) why noise margin in invertor calculated when slope becomes -1

Q2) one question on OTA acting as HPF (resistance with -ve f/b) and a capacitance at vin-
ans: gm(1+rsc)/gm+sc

Q3) question on verilog synthesis

Q4) draw c-v w/f for mos capacitance and mosfet

Q5) an ideal current pulse source charging a capacitance what wud be voltage across it

Q6) 3 step response given wat wud be the relative phase margin

ST Micro:

There were two papers
1. separate for the hardware (electonics people(VLSI))
2. and other for the embedded software design(both for electronics and comp).

1. which conversion is not possible
   a. float to int
   b. int to float
   c. char to float
   d. all are possible

2. threads have which thing in common
   a. register set
   b. data section
   c. thread id
   d. ...

3. one que like

```
main()
{
int x=5, y;
y= x*x++ * ++x ;

// print x and y
}
```

4. A CPU has four group of instruction set A, B, C, D
   CPI of A = 1
   CPI of B=3
   Cpi of c =2
   cpi of d= 4
   the cpu access 20% of A, 30% of b, 30% of C and 20 % of D
   what will be the average CPI.

   (this que was repeated in section 2 & 3)

5 . a question on hit ratio n effective memory access time.

6.

```
main()
{
int a=10,b=5
while ( --b>=0 && ++a)
{
--b;
++a;
}
print (a);
print (b);
}
```

7.

```
main()
{
char i;

for (i=0; i<=255; i++)
{
printf("%c", i);
}
}
```

8. One question on controls systems
   to find the transfer function.

poles n zeroes were given in a graph

9. One question on sampling theorem,
highest frequency of a signal is f, and it is sampled at fs(>= 2f). what is the frequency range of a bandpass signal whose spectrum looks exactly like the original signal.

10.One on the signal to noise ratio
There are N bits to represent each sample and total number of levels is M. If the amplitude range is reduced by half then SNR will be reduced by:

11. A cellular network operator is operating at 9.6Kbps. If he want to transmitt the audio quality of 44.1KHz with 16 bits for each sample how much bandwidth should be increased.

12. calculating the checksum for the bits to be transmitted given the frame-11000101 and generator is1100.

13. calculating the no of bits required for the error detection & the error correction for the given codeword set.
codeword a:
0000
0001
0011
1111
codeword b:
101111
.
.
.
.
110101

14. options were given to choose as which was an example of multitasking.
a:multiple remote users accessing a server
b:user working on spreadsheet, downloading some matter from internet
c:multiple programs resident in memory

15. CA in CSMA/ CA stands for
a. collision approval
b. collision avoidance
c. critical access

16. in a triangle, without changing the angle, if we double the sides,then new area will be

17. there is a pipe having dia 6mm, then how many pipes having 1mm dia wiill be needed to provide same amount of water.

18. in which of the folwng schemes after page replacement the entered page will enter in the same memory location as of the replaced one

a. direct mapping
b. n-set associative
c. associative
d. none of them

19. belady anamoly is related to.
ans. page replacement algos

20.which one uses cache mechanism
ans TLB

21.what will happen in following code..
signal(mutex)
critical section
wait(mutex)

ans. violation of mutual exclusion

22.an RLC ckt was given, fuctioning of ckt to be determined.
a: will act like FM
b: PM
c:AM
d: none of the above

23.

```
int i=0;
switch(i)
{
case 1: printf("hi");
case 0: printf("zero");
case 2: printf("world");
}
```

24.which one is the declaration of static string
a: static string
b: 'static string'
c: "static string"
d:char sting[30]

25.a que on file handling in c
a: file cant be opened
b:msg.txt is copied to msg
c:only first string be copied
d:

26. which of the fuction will store a 100 char string in X
a: fread(x,100,....)
b. fread(100,x,.....)
c.gets(x)
d.read(x)

27. which of the following data type will occupy the same memory irrespective of the compiler.
a.int
b.double
c.char
d.float

TI:

TEXAS INSTRUMENTS: TECHNICAL TEST

Date: 20th December 2003

1. For a CMOS inverter, the transition slope of Vout vs Vin DC characteristics can be increased (steeper transition) by:

a. Increasing W/L of PMOS transistor
b. Increasing W/L of NMOS transistor
c. Increasing W/L of both transistors by the same factor
d. Decreasing W/L of both transistor by the same factor

Ans: c

2. Minimum number of 2-input NAND gates that will be required to implement the function: Y = AB + CD + EF is

a. 4
b. 5
c. 6
d. 7

ans: c

3. Consider a two-level memory hierarchy system M1 & M2. M1 is accessed first and on miss M2 is accessed. The access of M1 is 2 nanoseconds and the miss penalty (the time to get the data from M2 in case of a miss) is 100 nanoseconds. The probability that a valid data is found in M1 is 0.97. The average memory access time is:

a. 4.94 nanoseconds
b. 3.06 nanoseconds

VLSI interview questions and answers, VLSI FAQs

Next Page »

VLSI interview questions and answers for freshers and experienced candidates. Also find VLSI online practice tests to fight written tests and certification exams on VLSI. In this section we have covered almost all VLSI questions that might be asked during an interview.

What are the steps required to solve setup and Hold violations in VLSI?

There are few steps that has to be performed to solved the setup and hold violations in VLSI. The steps are as follows:

- The optimization and restructuring of the logic between the flops are carried way. This way the logics are combined and it helps in solving this problem.

- There is way to modify the flip-flops that offer lesser setup delay and provide faster services to setup a device. Modifying the launch-flop to have a better hold on the clock pin, which provides CK->Q that makes the launch-flop to be fast and helps in fixing the setup violations.

- The network of the clock can be modified to reduce the delay or slowing down of the clock that captures the action of the flip-flop.

- There can be added delay/buffer that allows less delay to the function that is used.

Latest topics

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- Data Structure - Part 2
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- Embedded Systems for freshers
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- Arrays
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- Stack & Queue
- Trees
- SOA
- VC++
- PERL
- OOPS - Part 1
- OOPS - Part 2
- Inheritance & Polymorphism - OOPS

What are the different ways in which antenna violation can be prevented?

Antenna violation occurs during the process of plasma etching in which the charges generating from one metal strip to another gets accumlated at a single place. The longer the strip the more the charges gets accumulated. The prevention can be done by following method:

- Creating a jogging the metal line, that consists of atleast one metal above the protected layer.

- There is a requirement to jog the metal that is above the metal getting the etching effect. This is due to the fact that if a metal gets the etching then the other metal gets disconnected if the prevention measures are not taken.

- There is a way to prevent it by adding the reverse Diodes at the gates that are used in the circuits.

What is the function of tie-high and tie-low cells?

Tie-high and tie-low are used to connect the transistors of the gate by using either the power or the ground. The gates are connected using the power or ground then it can be turned off and on due to the power bounce from the ground. The cells are used to stop the bouncing and easy from of the current from one cell to another. These cells are required Vdd that connects to the tie-high cell as there is a power supply that is high and tie-low gets connected to Vss. This connection gets established and the transistors function properly without the need of any ground bounce occurring in any cell.

What is the main function of metastability in VSDL?

Metastability is an unknown state that is given as neither

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one or zero. It is used in designing the system that violates the setup or hole time requirements. The setup time requirement need the data to be stable before the clock-edge and the hold time requires the data to be stable after the clock edge has passed. There are potential violation that can lead to setup and hold violations as well. The data that is produced in this is totally asynchronous and clocked synchronous. This provide a way to setup the state through which it can be known that the violations that are occuring in the system and a proper design can be provided by the use of several other functions.

What are the steps involved in preventing the metastability?

Metastability is the unknown state and it prevents the violations using the following steps:

1. proper synchronizers are used that can be two stage or three stage whenever the data comes from the asynchronous domain. This helps in recovering the metastable state event.

2. The synchronizers are used in between cross-clocking domains. This reduces the metastability by removing the delay that is caused by the data element that are coming and taking time to get removed from the surface of metal.

3. Use of faster flip-flops that allow the transaction to be more faster and it removes the delay time between the one component to another component. It uses a narrower metastable window that makes the delay happen but faster flip-flops help in making the process faster and reduce the time delay as well.

What are the different design constraints occur in the Synthesis phase?

The steps that are involved in which the design constraint occurs are:
1. first the creation of the clock with the frequency and the duty cycle gets created. This clock helps in maintaining the flow and synchronizing various devices that are used.

2. Define the transition time according the requirement on the input ports.

3. The load values are specified for the output ports that are mapped with the input ports.

4. Setting of the delay values for both the input and output ports. The delay includes the input and output delay.

5. Specify the case-settings to report the correct time that are matched with the specific paths

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used in VLSI?

There are three types of skew that are used in VLSI. The skew are used in clock to reduce the delay or to understand the process accordingly. The skew are as follows:

Local skew:
This contain the difference between the launching flip-flop and the destination flip-flop. This defines a time path between the two.

Global skew:
Defines the difference between the earliest component reaching the flip flow and the the latest arriving at the flip flow with the same clock domain. In this delays are not measured and the clock is provided the same.

Useful skew:
Defines the delay in capturing a flip flop paths that helps in setting up the environment with specific requirement for the launch and capture of the timing path. The hold requirement in this case has to be met for the design purpose.

What are the changes that are provided to meet design power targets?

To meet the design power target there should be a process to design with Multi-VDD designs, this area requires high performance, and also the high VDD that requires low-performance. This is used to create the voltage group that allow the appropriate level-shifter to shift and placed in cross-voltage domains. There is a design with the multiple threshold voltages that require high performance when the Vt becomes low. This have lots of current leakage that makes the Vt cell to lower the performance. The reduction can be performed in the leakage power as the clock in this consume more power, so placing of an optimal clock controls the module and allow it to be given more power. Clock tree allow the switching to take place when the clock buffers are used by the clock gating cells and reduce the switching by the power reduction.

What are the different measures that are required to achieve the design for better yield?

To achieve better yeild then there should be reduction in maufacturability flaws. The circuit perfomance has to be high that reduces the parametric yield. This reduction is due to process variations The measures that can be taken are:
- Creation of powerful runset files that consists of spacing and shorting rules. This also consists of all the permissions that has to be given to the user.
- Check the areas where the design is having lithographic issues, that consists of sharp cuts.
- Use of redundant vias to reduce the breakage of the

current and the barrier.
- Optimal placing of the de-coupling capacitances can be done so that there is a reduction in power-surges.

What is the difference between the mealy and moore state machine?

- Moore model consists of the machine that have an entry action and the output depends only on the state of the machine, whereas mealy model only uses Input Actions and the output depends on the state and also on the previous inputs that are provided during the program.

- Moore models are used to design the hardware systems, whereas both hardware and software systems can be designed using the mealy model.

- Mealy machine's output depend on the state and input, whereas the output of the moore machine depends only on the state as the program is written in the state only.

- Mealy machine is having the output by the combination of both input and the state and the change the state of state variables also have some delay when the change in the signal takes place, whereas in Moore machine doesn't have glitches and its ouput is dependent only on states not on the input signal level.

What is the difference between Synchronous and Asynchronous reset?

- Synchronous reset is the logic that will synthesize to smaller flip-flops. In this the clock works as a filter providing the small reset glitches but the glitches occur on the active clock edge, whereas the asynchronous reset is also known as reset release or reset removal. The designer is responsible of added the reset to the data paths.

- The synchronous reset is used for all the types of design

that are used to filter the logic glitches provided between the clocks. Whereas, the circuit can be reset with or without the clock present.

- Synchronous reset doesn't allow the synthesis tool to be used easily and it distinguishes the reset signal from other data signal. The release of the reset can occur only when the clock is having its initial period. If the release happens near the clock edge then the flip-flops can be metastable.

What are the different design techniques required to create a Layout for Digital Circuits?

The different design techniques to create the Layout for digital circuits are as follows:

- Digital design consists of the standard cells and represent the height that is required for the layout. The layout depends on the size of the transistor. It also consists of the specification for Vdd and GND metal paths that has to be maintained uniformly.

- Use of metal in one direction only to apply the metal directly. The metal can be used and displayed in any direction.

- Placing of the substrate that place where it shows all the empty spaces of the layout where there is resistances.

- Use of fingered transistors allows the design to be more easy and it is easy to maintain a symmetry as well.

Write a program to explain the comparator?

To make a comparator there is a requirement to use multiplexer that is having one input and many outputs. This allows the choosing of the maximum numbers that are

required to design the comparator. The implementation of the 2 bit comparator can be done using the law of tigotomy that states that A > B, A < B, A = B (Law of trigotomy). The comparator can be implemented using:
combinational logic circuits or multiplexers that uses the HDL language to write the schematic at RTL and gate level.
Behavioral model of comparator represented like:

```
module comp0 (y1,y2,y3,a,b);
input [1:0] a,b;
output y1,y2,y3;
wire y1,y2,y3;
assign y1= (a >b)? 1:0;
assign y2= (b >a)? 1:0;
assign y3= (a==b)? 1:0;
endmodule
```

What is the function of chain reordering?

The optimization technique that is used makes it difficult for the chain ordering system to route due to the congestion caused by the placement of the cells. There are tool available that automate the reordering of the chain to reduce the congestion that is produced at the first stage. It increases the problem of the chain system and this also allow the overcoming of the buffers that have to be inserted into the scan path. The increase of the hold time in the chain reordering can cause great amount of delay. Chain reordering allows the cell to be come in the ordered format while using the different clock domains. It is used to reduce the time delay caused by random generation of the element and the placement of it.

What are the steps involved in designing an optimal pad ring?

- To make the design for an optimal pad ring there is a requirement for the corner-pads that comes across all the

corners of the pad-ring. It is used to give power continuity and keep the resistance low.

- It requires the pad ring that is to fulfil the power domains that is common for all the ground across all the domains.

- It requires the pad ring to contain simultaneous switching noise system that place the transfer cell pads in cross power domains for different pad length.

- Drive strength is been seen to check the current requirements and the timings to make the power pads.

- Choose a no-connection pad that is used to fill the pad-frame when there is no requirement for the inputs to be given. This consumes less power when there is no input given at a particular time.

- Checking of the oscillators pads take place that uses the synchronous circuits to make the clock data synchronize with the existing one.

What is the function of enhancement mode transistor?

The enhancement mode transistors are also called as field effect transistors as they rely on the electric filed to control the shape and conductivity of the channel. This consists of one type of charge carrier in a semiconductor material environment. This also uses the unipolar transistors to differentiate themselves with the single-carrier type operation transistors that consists of the bipolar junction transistor. The uses of field effect transistor is to physical implementation of the semiconductor materials that is compared with the bipolar transistors. It provides with the majority of the charge carrier devices. The devices that consists of active channels to make the charge carriers pass through. It consists of the concept of drain and the source.

What is the purpose of having Depletion mode Device?

Depletion modes are used in MOSFET it is a device that remains ON at zero gate-source voltage. This device consists of load resistors that are used in the logic circuits. This types are used in N-type depletion-load devices that allow the threshold voltages to be taken and use of -3 V to +3V is done. The drain is more positive in this comparison of PMOS where the polarities gets reversed. The mode is usually determined by the sign of threshold voltage for N-type channel. Depletion mode is the positive one and used in many technologies to represent the actual logic circuit. It defines the logic family that is dependent on the silicon VLSI. This consists of pull-down switches and loads for pull-ups.

What is the difference between NMOS and PMOS technologies?

- PMOS consists of metal oxide semiconductor that is made on the n-type substrates and consists of active careers named as holes. These holes are used for migration purpose of the charges between the p-type and the drain. Whereas, NMOS consists of the metal oxide semiconductor and they are made on p-type substrates. It consists of electrons as their carriers and migration happens between the n-type source and drain.

- On applying the high voltage on the logic gates NMOS will be conducted and will get activated, whereas PMOS require low voltage to be activated.

- NMOS are faster than PMOS as the carriers that NMOS uses are electrons that travels faster than holes. The speed is twice as fast as holes.

- PMOS are more immune to noice than NMOS.

What is the difference between CMOS and Bipolar technologies?

- CMOS technology allows the power dissipation to be low and it gives more power output, whereas bipolar takes lots of power to run the system and the ciricutary require lots of power to get activated.

- CMOS technology provides high input impedance that is low drive current that allow more current to be flown in the cirucit and keep the circuit in a good position, whereas it provides high drive current means more input impedance.

- CMOS technology provides scalable threshold voltage more in comparison to the Bipolar technology that provides low threshold voltage.

- CMOS technology provides high noise margin, packing density whereas Bipolory technology allows to have low noise margin so that to reduce the high volues and give low packing density of the components.

What are the different classification of the timing control?

There are different classification in which the timing control data is divided and they are:

1. Delay based timing control: this is based on timing control that allows to manage the component such that the delay can be notified and wherever it is required it can be given. The delays that are based on this are as:
- Regular delay control: that controls the delay on the regular basis.
- Intra-assignment delay control: that controls the internal delays.
- Zero delay control

2. Events based timing control: this is based on the events

that are performed when an event happens or a trigger is set on an event that takes place. It includes
- Regular event control
- Named event control
- Event OR control

3. Level sensitive timing control: this is based on the levels that are given like 0 level or 1 level that is being given or shown and the data is being modified according the levels that are being set. When a level changes the timing control also changes.

Next Page »

Data Structures Arrays - Answers to Arrays related interview questions

Data Structures Arrays - In this series, we have covered all about Arrays and answered the questions that might be asked during an interview.

Linked list interview questions and answers - freshers, experienced

Data Structures Linked list - In this series, we have covered all about Linked list and answered the questions that might be asked during an interview.

Data Structures Sorting - Answers to Sorting related interview questions

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about Sorting and answered the questions that might be asked during an interview.

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Zoran Corporation

Posted by **VLSI_Rules** at **6:05 PM 4 comments:**
Labels: **backend**, **chip**, **cmos**, **Companies**, **design**, **fabless**, **frontend**, **hyderabad**, **india**, **List**, **logic**, **of**, **physical**, **semiconductor**, **verilog**, **vhdl**, **vlsi**, **world**

Monday, November 17, 2008

CMOS Interview Questions

1. Explain why & how a MOSFET works
2. Draw Vds-Ids curve for a MOSFET. Now, show how this curve changes (a) with increasing Vgs (b) with increasing transistor width (c) considering Channel Length Modulation
3. Explain the various MOSFET Capacitances & their significance
4. Draw a CMOS Inverter. Explain its transfer characteristics
5. Explain sizing of the inverter
6. How do you size NMOS and PMOS transistors to increase the threshold voltage?
7. What is Noise Margin? Explain the procedure to determine Noise Margin
8. Give the expression for CMOS switching power dissipation
9. What is Body Effect?
10. Describe the various effects of scaling
11. Give the expression for calculating Delay in CMOS circuit
12. What happens to delay if you increase load capacitance?
13. What happens to delay if we include a resistance at the output of a CMOS circuit?
14. What are the limitations in increasing the power supply to reduce delay?
15. How does Resistance of the metal lines vary with increasing thickness and increasing length?
16. You have three adjacent parallel metal lines. Two out of phase signals pass through the outer two metal lines. Draw the waveforms in the center metal line due to interference. Now, draw the signals if the signals in outer metal lines are in phase with each other
17. What happens if we increase the number of contacts or via from one metal layer to the next?
18. Draw a transistor level two input NAND gate. Explain its sizing (a) considering Vth (b) for equal rise and fall times
19. Let A & B be two inputs of the NAND gate. Say signal A arrives at the NAND gate later than signal B. To optimize delay, of the two series NMOS inputs A & B, which one would you place near the output?
20. Draw the stick diagram of a NOR gate. Optimize it
21. For CMOS logic, give the various techniques you know to minimize power consumption

22. What is Charge Sharing? Explain the Charge Sharing problem while sampling data from a Bus
23. Why do we gradually increase the size of inverters in buffer design? Why not give the output of a circuit to one large inverter?
24. In the design of a large inverter, why do we prefer to connect small transistors in parallel (thus increasing effective width) rather than lay out one transistor with large width?
25. Given a layout, draw its transistor level circuit. (I was given a 3 input AND gate and a 2 input Multiplexer. You can expect any simple 2 or 3 input gates)
26. Give the logic expression for an AOI gate. Draw its transistor level equivalent. Draw its stick diagram
27. Why don't we use just one NMOS or PMOS transistor as a transmission gate?
28. For a NMOS transistor acting as a pass transistor, say the gate is connected to VDD, give the output for a square pulse input going from 0 to VDD
29. Draw a 6-T SRAM Cell and explain the Read and Write operations
30. Draw the Differential Sense Amplifier and explain its working. Any idea how to size this circuit? (Consider Channel Length Modulation)
31. What happens if we use an Inverter instead of the Differential Sense Amplifier?
32. Draw the SRAM Write Circuitry
33. Approximately, what were the sizes of your transistors in the SRAM cell? How did you arrive at those sizes?
34. How does the size of PMOS Pull Up transistors (for bit & bit-lines) affect SRAM's performance?
35. What's the critical path in a SRAM?
36. Draw the timing diagram for a SRAM Read. What happens if we delay the enabling of Clock signal?
37. Give a big picture of the entire SRAM Layout showing your placements of SRAM Cells, Row Decoders, Column Decoders, Read Circuit, Write Circuit and Buffers
38. In a SRAM layout, which metal layers would you prefer for Word Lines and Bit Lines? Why?
39. How can you model a SRAM at RTL Level?
40. What's the difference between Testing & Verification?
41. For an AND-OR implementation of a two input Mux, how do you test for Stuck-At-0 and Stuck-At-1 faults at the internal nodes? (You can expect a circuit with some redundant logic)
42. What is Latch Up? Explain Latch Up with cross section of a CMOS Inverter. How do you avoid Latch Up?

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1. Give two ways of converting a two input NAND gate to an inverter
2. Given a circuit, draw its exact timing response. (I was given a Pseudo Random Signal Generator; you can expect any sequential ckt)
3. What are set up time & hold time constraints? What do they signify? Which one is critical for estimating maximum clock frequency of a circuit?

4. Give a circuit to divide frequency of clock cycle by two
5. Design a divide-by-3 sequential circuit with 50% duty circle. (Hint: Double the Clock)
6. Suppose you have a combinational circuit between two registers driven by a clock. What will you do if the delay of the combinational circuit is greater than your clock signal? (You can't resize the combinational circuit transistors)
7. The answer to the above question is breaking the combinational circuit and pipelining it. What will be affected if you do this?
8. What are the different Adder circuits you studied?
9. Give the truth table for a Half Adder. Give a gate level implementation of the same.
10. Draw a Transmission Gate-based D-Latch.
11. Design a Transmission Gate based XOR. Now, how do you convert it to XNOR? (Without inverting the output)
12. How do you detect if two 8-bit signals are same?
13. How do you detect a sequence of "1101" arriving serially from a signal line?
14. Design any FSM in VHDL or Verilog.
15. Explain RC circuit's charging and discharging.
16. Explain the working of a binary counter.
17. Describe how you would reverse a singly linked list.

Posted by **VLSI_Rules** at **10:43 AM No comments:**
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FPGA Interview Questions

1) What is minimum and maximum frequency of dcm in spartan-3 series fpga?

Spartan series dcm's have a minimum frequency of 24 MHZ and a maximum of 248

2)Tell me some of constraints you used and their purpose during your design?

There are lot of constraints and will vary for tool to tool ,I am listing some of Xilinx constraints
a) Translate on and Translate off: the Verilog code between Translate on and Translate off is ignored for synthesis.
b) CLOCK_SIGNAL: is a synthesis constraint. In the case where a clock signal goes through combinatorial logic before being connected to the clock input of a flip-flop, XST cannot identify what input pin or internal net is the real clock signal. This constraint allows you to define the clock net.
c) XOR_COLLAPSE: is synthesis constraint. It controls whether cascaded XORs should be collapsed into a single XOR.
For more constraints detailed description refer to constraint guide.

3) Suppose for a piece of code equivalent gate count is 600 and for

another code equivalent gate count is 50,000 will the size of bitmap change?in other words will size of bitmap change it gate count change?

The size of bitmap is irrespective of resource utilization, it is always the same,for Spartan xc3s5000 it is 1.56MB and will never change.

4) What are different types of FPGA programming modes?what are you currently using ?how to change from one to another?

Before powering on the FPGA, configuration data is stored externally in a PROM or some other nonvolatile medium either on or off the board. After applying power, the configuration data is written to the FPGA using any of five different modes: Master Parallel, Slave Parallel, Master Serial, Slave Serial, and Boundary Scan (JTAG). The Master and Slave Parallel modes
Mode selecting pins can be set to select the mode, refer data sheet for further details.

5) Tell me some of features of FPGA you are currently using?

I am taking example of xc3s5000 to answering the question .

Very low cost, high-performance logic solution for
high-volume, consumer-oriented applications
- Densities as high as 74,880 logic cells
- Up to 784 I/O pins
- 622 Mb/s data transfer rate per I/O
- 18 single-ended signal standards
- 6 differential I/O standards including LVDS, RSDS
- Termination by Digitally Controlled Impedance
- Signal swing ranging from 1.14V to 3.45V
- Double Data Rate (DDR) support
• Logic resources
- Abundant logic cells with shift register capability
- Wide multiplexers
- Fast look-ahead carry logic
- Dedicated 18 x 18 multipliers
- Up to 1,872 Kbits of total block RAM
- Up to 520 Kbits of total distributed RAM
• Digital Clock Manager (up to four DCMs)
- Clock skew elimination
• Eight global clock lines and abundant routing

6) What is gate count of your project?

Well mine was 3.2 million, I don’t know yours.!

7) Can you list out some of synthesizable and non synthesizable constructs?

not synthesizable->>>>
initial
ignored for synthesis.
delays
ignored for synthesis.
events
not supported.
real
Real data type not supported.
time
Time data type not supported.
force and release
Force and release of data types not supported.
fork join
Use nonblocking assignments to get same effect.
user defined primitives
Only gate level primitives are supported.

synthesizable constructs->>
assign,for loop,Gate Level Primitives,repeat with constant value...

8)Can you explain what struck at zero means?

These stuck-at problems will appear in ASIC. Some times, the nodes will permanently tie to 1 or 0 because of some fault. To avoid that, we need to provide testability in RTL. If it is permanently 1 it is called stuck-at-1 If it is permanently 0 it is called stuck-at-0.

9) Can you draw general structure of fpga?

10) Difference between FPGA and CPLD?

FPGA:
a)SRAM based technology.
b)Segmented connection between elements.
c)Usually used for complex logic circuits.
d)Must be reprogrammed once the power is off.
e)Costly

CPLD:
a)Flash or EPROM based technology.
b)Continuous connection between elements.
c)Usually used for simpler or moderately complex logic circuits.
d)Need not be reprogrammed once the power is off.
e)Cheaper

11) What are dcm's?why they are used?

Digital clock manager (DCM) is a fully digital control system that

uses feedback to maintain clock signal characteristics with a high degree of precision despite normal variations in operating temperature and voltage.
That is clock output of DCM is stable over wide range of temperature and voltage , and also skew associated with DCM is minimal and all phases of input clock can be obtained . The output of DCM coming form global buffer can handle more load.

12) FPGA design flow?

13)what is slice,clb,lut?

I am taking example of xc3s500 to answer this question

The Configurable Logic Blocks (CLBs) constitute the main logic resource for implementing synchronous as well as combinatorial circuits.
CLB are configurable logic blocks and can be configured to combo,ram or rom depending on coding style
CLB consist of 4 slices and each slice consist of two 4-input LUT (look up table) F-LUT and G-LUT.

14) Can a clb configured as ram?

YES.

The memory assignment is a clocked behavioral assignment, Reads from the memory are asynchronous, And all the address lines are shared by the read and write statements.

15)What is purpose of a constraint file what is its extension?

The UCF file is an ASCII file specifying constraints on the logical design. You create this file and enter your constraints in the file with a text editor. You can also use the Xilinx Constraints Editor to create constraints within a UCF(extention) file. These constraints affect how the logical design is implemented in the target device. You can use the file to override constraints specified during design entry.

16) What is FPGA you are currently using and some of main reasons for choosing it?

17) Draw a rough diagram of how clock is routed through out FPGA?

18) How many global buffers are there in your current fpga,what is their significance?

There are 8 of them in xc3s5000
An external clock source enters the FPGA using a Global Clock Input Buffer (IBUFG), which directly accesses the global clock network or an Input Buffer (IBUF). Clock signals within the FPGA drive a global clock net using a Global Clock Multiplexer Buffer (BUFGMUX). The global clock net connects directly to the CLKIN input.

19) What is frequency of operation and equivalent gate count of u r project?

20)Tell me some of timing constraints you have used?

21)Why is map-timing option used?

Timing-driven packing and placement is recommended to improve design performance, timing, and packing for highly utilized designs.

22)What are different types of timing verifications?

Dynamic timing:
a. The design is simulated in full timing mode.
b. Not all possibilities tested as it is dependent on the input test vectors.
c. Simulations in full timing mode are slow and require a lot of memory.
d. Best method to check asynchronous interfaces or interfaces between different timing domains.
Static timing:
a. The delays over all paths are added up.
b. All possibilities, including false paths, verified without the need for test vectors.
c. Much faster than simulations, hours as opposed to days.
d. Not good with asynchronous interfaces or interfaces between different timing domains.

23) Compare PLL & DLL ?

PLL:
PLLs have disadvantages that make their use in high-speed designs problematic, particularly when both high performance and high reliability are required.
The PLL voltage-controlled oscillator (VCO) is the greatest source of problems. Variations in temperature, supply voltage, and manufacturing process affect the stability and operating performance of PLLs.

DLLs, however, are immune to these problems. A DLL in its simplest form inserts a variable delay line between the external clock and the internal clock. The clock tree distributes the clock to all registers and then back to the feedback pin of the DLL.

The control circuit of the DLL adjusts the delays so that the rising edges of the feedback clock align with the input clock. Once the edges of the clocks are aligned, the DLL is locked, and both the input buffer delay and the clock skew are reduced to zero.
Advantages:
· precision
· stability
· power management
· noise sensitivity
· jitter performance.

24) Given two ASICs. one has setup violation and the other has hold violation. how can they be made to work together without modifying the design?

Slow the clock down on the one with setup violations..
And add redundant logic in the path where you have hold violations.

25)Suggest some ways to increase clock frequency?

· Check critical path and optimize it.
· Add more timing constraints (over constrain).
· pipeline the architecture to the max possible extent keeping in mind latency req's.

26)What is the purpose of DRC?

DRC is used to check whether the particular schematic and corresponding layout(especially the mask sets involved) cater to a pre-defined rule set depending on the technology used to design. They are parameters set aside by the concerned semiconductor manufacturer with respect to how the masks should be placed , connected , routed keeping in mind that variations in the fab process does not effect normal functionality. It usually denotes the minimum allowable configuration.

27)What is LVs and why do we do that. What is the difference between LVS and DRC?

The layout must be drawn according to certain strict design rules. DRC helps in layout of the designs by checking if the layout is abide by those rules.
After the layout is complete we extract the netlist. LVS compares the netlist extracted from the layout with the schematic to ensure that the layout is an identical match to the cell schematic.

28)What is DFT ?

DFT means design for testability. 'Design for Test or Testability' - a

methodology that ensures a design works properly after manufacturing, which later facilitates the failure analysis and false product/piece detection
Other than the functional logic,you need to add some DFT logic in your design.This will help you in testing the chip for manufacturing defects after it come from fab. Scan,MBIST,LBIST,IDDQ testing etc are all part of this. (this is a hot field and with lots of opportunities)

29) There are two major FPGA companies: Xilinx and Altera. Xilinx tends to promote its hard processor cores and Altera tends to promote its soft processor cores. What is the difference between a hard processor core and a soft processor core?

A hard processor core is a pre-designed block that is embedded onto the device. In the Xilinx Virtex II-Pro, some of the logic blocks have been removed, and the space that was used for these logic blocks is used to implement a processor. The Altera Nios, on the other hand, is a design that can be compiled to the normal FPGA logic.

30)What is the significance of contamination delay in sequential circuit timing?

31)When are DFT and Formal verification used?

DFT:
· manufacturing defects like stuck at "0" or "1".
· test for set of rules followed during the initial design stage.

Formal verification:
· Verification of the operation of the design, i.e, to see if the design follows spec.
· gate netlist == RTL ?
· using mathematics and statistical analysis to check for equivalence.

32)What is Synthesis?

Synthesis is the stage in the design flow which is concerned with translating your Verilog code into gates - and that's putting it very simply! First of all, the Verilog must be written in a particular way for the synthesis tool that you are using. Of course, a synthesis tool doesn't actually produce gates - it will output a netlist of the design that you have synthesised that represents the chip which can be fabricated through an ASIC or FPGA vendor.

33)We need to sample an input or output something at different rates, but I need to vary the rate? What's a clean way to do this?

Many, many problems have this sort of variable rate requirement, yet we are usually constrained with a constant clock frequency. One trick

is to implement a digital NCO (Numerically Controlled Oscillator). An NCO is actually very simple and, while it is most naturally understood as hardware, it also can be constructed in software. The NCO, quite simply, is an accumulator where you keep adding a fixed value on every clock (e.g. at a constant clock frequency). When the NCO "wraps", you sample your input or do your action. By adjusting the value added to the accumulator each clock, you finely tune the AVERAGE frequency of that wrap event. Now - you may have realized that the wrapping event may have lots of jitter on it. True, but you may use the wrap to increment yet another counter where each additional Divide-by-2 bit reduces this jitter. The DDS is a related technique. I have two examples showing both an NCOs and a DDS in my File Archive. This is tricky to grasp at first, but tremendously powerful once you have it in your bag of tricks. NCOs also relate to digital PLLs, Timing Recovery, TDMA and other "variable rate" phenomena.

Posted by **VLSI_Rules** at **10:42 AM 2 comments:**
Labels: **asic**, **chip**, **cmos**, **combinational**, **design**, **digital**, **fifo**, **flip**, **flop**, **fpga**, **fsm**, **interview**, **latch**, **questions**, **RTL**, **sequential**, **synchronous**, **verilog**, **vhdl**, **vlsi**

VHDL Interview Questions

What is the difference between using direct instntiations and component ones except that you need to declare the component ?

What is the use of BLOCKS ?

What is the use of PROCEDURES?

What is the usage of using more then one architecture in an entity?

What is a D-latch? Write the VHDL Code for it?

Implement D flip-flop with a couple of latches? Write a VHDL Code for a D flip-flop?

Differences between Signals and Variables in VHDL? If the same code is written using Signals and Variables what does it synthesize to?

Differences between functions and Procedures in VHDL?

Explain the concept of a Clock Divider Circuit? Write a VHDL code for the same?

What you would use in RTL a 'boolean' type or a 'std_logic' type and why.

What are/may be the implications of using an 'integer' type in RTL.

A timing path fails: what are your options?

What are VHDL structures, give an example to exploit them

What is grey coding, any example where they are used

Discuss Async interfaces

Metastability

Synopsys unwanted latch

Verilog blocking vs non-blocking

VHDL variables: example where you have to use them

What is pipelining and how it may improve the performance

What are multicycle paths.

What are false paths

What are Async counters, what are advantages of using these over sync counters. and what are the disadvantages

Sensitivity List:
How does it matter.What will happen
if you dont include a signal in the sensitivity list
and use/read it inside the process

How you will implement a C language pointer in VHDL

What is Design For Test and why it is done.

What is clock gating? How and why it is done.
Low Power: discuss how it may be done

Discuss disadvantages/challenges of shrinking technology

What is pipelining, how may it affect the performance of a design
What is the difference between transport delays and inertial delays in VHDL
What determines the max frequency a digital design may work on
Why thold(hold time) is not included in the calculation for the above.
What will happen if output of an inverter is shorted to its input
What is noise margin.
Why are p-mos larger than n-mos in CMOS design.
Draw DC curve of inverter and Re-Draw it if pmos and nmos are equal.
What is Latch-up
How can an Inverter work as an amplifier

Design a state machine which divides the input frequency of a clock by 3.

Why does a pass gate requires two transistors(1 N and 1 P type) Can we use a
single transistor N or P type in a pass gate? If not why? and if yes then in what conditions?

Why CMOS why not N-MOS or P-MOS logic, when we know that the number
of gates required in CMOS are grater than in n-mos or p-mos logic.

How much is the max fan out of a typical CMOS gate. Or alternatively,

discuss the limiting factors.

What are dynamic logic gates? What are their advantages over conventional logic gates

Design a digital circuit to delay the negative edge of the input signal by 2 clock cycles

What is the relation between binary encoding and grey(or gray) encoding.

Write a vhdl function to implement a length independent grey code counter.
alternatively, discuss the logic to do that.

How you will constraint a combinational logic path through your design
in dc_shell.

Make a T Flip Flop using a D Flip Flop

How you will make a Nand Gate function like an inverter.

Design a state machine to detect a '1101' pattern in a stream.
Detect both, overlapping and non overlapping patterns.

What are MISRs, example usage?

Posted by **VLSI_Rules** at **10:41 AM No comments:**
Labels: **asic**, **chip**, **cmos**, **combinational**, **design**, **digital**, **fifo**, **flip**, **flop**, **fsm**, **interview**, **latch**, **questions**, **RTL**, **sequential**, **skew**, **synchronous**, **verilog**, **vhdl**, **vlsi**

Verilog Interview Questions

1. What is the difference between Behavior modeling and RTL modeling?
2. What is the benefit of using Behavior modeling style over RTL modeling?

3. What is the difference between blocking assignments and non-blocking assignments ?
4. How do you implement the bi-directional ports in Verilog HDL
5. How to model inertial and transport delay using Verilog?
6. How to synchronize control signals and data between two different clock domains?
7. Create 4 bit multiplier using a ROM and what will be the size of the ROM. How can you realize it when the outputs are specified.
8. How can you swap 2 integers a and b, without using a 3rd variable
9. Which one is preferred? 1's complement or 2's complement and why?
10. Which one is preferred in FSM design? Mealy or Moore? Why?
11. Which one is preferred in design entry? RTL coding or Schematic? Why?
12. Design a 2 input OR gate using a 2:1 mux.
13. Design a 2 input AND gate using a 2 input XOR gate.
14. Design a hardware to implement following equations without using multipliers or dividers.
a. out = 7x + 8y;
b. out = .78x + .17y;
15. Design Gray counter to count 6.
16. Design XOR gate using just NAND gates.
17. Create "AND" gate using a 2:1 multiplexer. (Create all other gates too.)
18. How are blocking and non-blocking statements executed?
19. How do you model a synchronous and asynchronous reset in Verilog?
20. What happens if there is connecting wires width mismatch?
21. What are different options that can be used with $display statement in Verilog?
22. Give the precedence order of the operators in Verilog.
23. Should we include all the inputs of a combinational circuit in the sensitivity list? Give reason.
24. Give 10 commonly used Verilog keywords.
25. Is it possible to optimize a Verilog code such that we can achieve low power design?
26. Which is updated first: signal or variable?

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Basic Digital Interview Questions

What is the function of a D flip-flop, whose inverted output is connected to its input ?

Design a circuit to divide input frequency by 2.

Design a divide-by-3 sequential circuit with 50% duty cycle.

Design a divide-by-5 sequential circuit with 50% duty cycle.

What are the different types of adder implementations ?

Draw a Transmission Gate-based D-Latch.

Give the truth table for a Half Adder. Give a gate level implementation of it.

Design an XOR gate from 2:1 MUX and a NOT gate

What is the difference between a LATCH and a FLIP-FLOP ?

* Latch is a level sensitive device while flip-flop is an edge sensitive device.
* Latch is sensitive to glitches on enable pin, whereas flip-flop is immune to glitches.
* Latches take less gates (also less power) to implement than flip-flops.
* Latches are faster than flip-flops.

Design a D Flip-Flop from two latches.

Design a 2 bit counter using D Flip-Flop.

What are the two types of delays in any digital system ?

Design a Transparent Latch using a 2:1 Mux.

Design a 4:1 Mux using 2:1 Muxes and some combo logic.

What is metastable state ? How does it occur ?

What is metastability ?

Design a 3:8 decoder

Design a FSM to detect sequence "101" in input sequence.

Convert NAND gate into Inverter, in two different ways.

Design a D and T flip flop using 2:1 mux; use of other components not allowed, just the mux.

Design a divide by two counter using D-Latch.

Design D Latch from SR flip-flop.

Define Clock Skew , Negative Clock Skew, Positive Clock Skew.

What is Race Condition ?

Design a 4 bit Gray Counter.

Design 4-bit Synchronous counter, Asynchronous counter.

Design a 16 byte Asynchronous FIFO.

What is the difference between an EEPROM and a FLASH ?

What is the difference between a NAND-based Flash and a NOR-based Flash ?

You are given a 100 MHz clock. Design a 33.3 MHz clock with and without 50&37; duty cycle.

Design a Read on Reset System ?

Which one is superior: Asynchronous Reset or Synchronous Reset ? Explain.

Design a State machine for Traffic Control at a Four point Junction.

What are FIFO's? Can you draw the block diagram of FIFO? Could

you modify it to make it asynchronous FIFO ?

How can you generate random sequences in digital circuits?

Posted by **VLSI_Rules** at **10:39 AM 1 comment:**
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Physical Design Interview Questions

Companywise ASIC/VLSI Interview Questions

Below questions are asked for senior position in Physical Design domain. The questions are also related to Static Timing Analysis and Synthesis. Answers to some questions are given as link. Remaining questions will be answered in coming blogs.

Common introductory questions every interviewer asks are:

* Discuss about the projects worked in the previous company.
* What are physical design flows, various activities you are involved?
* Design complexity, capacity, frequency, process technologies, block size you handled.

Intel

* Why power stripes routed in the top metal layers?

The resistivity of top metal layers are less and hence less IR drop is seen in power distribution network. If power stripes are routed in lower metal layers this will use good amount of lower routing resources and therefore it can create routing congestion.

* Why do you use alternate routing approach HVH/VHV (Horizontal-Vertical-Horizontal/ Vertical-Horizontal-Vertical)?

Answer:

This approach allows routability of the design and better usage of routing resources.

* What are several factors to improve propagation delay of standard cell?

Answer:

Improve the input transition to the cell under consideration by up

sizing the driver.
Reduce the load seen by the cell under consideration, either by placement refinement or buffering.
If allowed increase the drive strength or replace with LVT (low threshold voltage) cell.

* How do you compute net delay (interconnect delay) / decode RC values present in tech file?
* What are various ways of timing optimization in synthesis tools?

Answer:

Logic optimization: buffer sizing, cell sizing, level adjustment, dummy buffering etc.

Less number of logics between Flip Flops speedup the design.

Optimize drive strength of the cell , so it is capable of driving more load and hence reducing the cell delay.

Better selection of design ware component (select timing optimized design ware components).

Use LVT (Low threshold voltage) and SVT (standard threshold voltage) cells if allowed.

* What would you do in order to not use certain cells from the library?

Answer:

Set don't use attribute on those library cells.

* How delays are characterized using WLM (Wire Load Model)?

Answer:

For a given wireload model the delay are estimated based on the number of fanout of the cell driving the net.

Fanout vs net length is tabulated in WLMs.

Values of unit resistance R and unit capacitance C are given in technology file.

Net length varies based on the fanout number.

Once the net length is known delay can be calculated; Sometimes it is again tabulated.

* What are various techniques to resolve congestion/noise?

Answer:

Routing and placement congestion all depend upon the connectivity in the netlist , a better floor plan can reduce the congestion.

Noise can be reduced by optimizing the overlap of nets in the design.

* Let's say there enough routing resources available, timing is fine, can you increase clock buffers in clock network? If so will there be any impact on other parameters?

Answer:

No. You should not increase clock buffers in the clock network. Increase in clock buffers cause more area , more power. When everything is fine why you want to touch clock tree??

* How do you optimize skew/insertion delays in CTS (Clock Tree Synthesis)?

Answer:

Better skew targets and insertion delay values provided while building the clocks.

Choose appropriate tree structure – either based on clock buffers or clock inverters or mix of clock buffers or clock inverters.

For multi clock domain, group the clocks while building the clock tree so that skew is balanced across the clocks. (Inter clock skew analysis).

* What are pros/cons of latch/FF (Flip Flop)?

* How you go about fixing timing violations for latch- latch paths?
* As an engineer, let's say your manager comes to you and asks for next project die size estimation/projection, giving data on RTL size, performance requirements. How do you go about the figuring out and come up with die size considering physical aspects?
* How will you design inserting voltage island scheme between macro pins crossing core and are at different power wells? What is the optimal resource solution?
* What are various formal verification issues you faced and how did you resolve?
* How do you calculate maximum frequency given setup, hold, clock and clock skew?

* What are effects of metastability?

* Consider a timing path crossing from fast clock domain to slow clock domain. How do you design synchronizer circuit without knowing the source clock frequency?
* How to solve cross clock timing path?
* How to determine the depth of FIFO/ size of the FIFO?

STmicroelectronics

* What are the challenges you faced in place and route, FV (Formal Verification), ECO (Engineering Change Order) areas?
* How long the design cycle for your designs?
* What part are your areas of interest in physical design?
* Explain ECO (Engineering Change Order) methodology.
* Explain CTS (Clock Tree Synthesis) flow.

* What kind of routing issues you faced?
* How does STA (Static Timing Analysis) in OCV (On Chip Variation) conditions done? How do you set OCV (On Chip Variation) in IC compiler? How is timing correlation done before and after place and route?

* If there are too many pins of the logic cells in one place within core, what kind of issues would you face and how will you resolve?
* Define hash/ @array in perl.
* Using TCL (Tool Command Language, Tickle) how do you set variables?
* What is ICC (IC Compiler) command for setting derate factor/ command to perform physical synthesis?
* What are nanoroute options for search and repair?
* What were your design skew/insertion delay targets?
* How is IR drop analysis done? What are various statistics available in reports?
* Explain pin density/ cell density issues, hotspots?
* How will you relate routing grid with manufacturing grid and judge if the routing grid is set correctly?
* What is the command for setting multi cycle path?
* If hold violation exists in design, is it OK to sign off design? If not, why?

Texas Instruments (TI)

* How are timing constraints developed?

* Explain timing closure flow/methodology/issues/fixes.
* Explain SDF (Standard Delay Format) back annotation/ SPEF (Standard Parasitic Exchange Format) timing correlation flow.
* Given a timing path in multi-mode multi-corner, how is STA (Static Timing Analysis) performed in order to meet timing in both modes and corners, how are PVT (Process-Voltage-Temperature)/derate factors decided and set in the Primetime flow?
* With respect to clock gate, what are various issues you faced at various stages in the physical design flow?
* What are synthesis strategies to optimize timing?
* Explain ECO (Engineering Change Order) implementation flow. Given post routed database and functional fixes, how will you take it to implement ECO (Engineering Change Order) and what physical and functional checks you need to perform?

Qualcomm

* In building the timing constraints, do you need to constrain all IO (Input-Output) ports?
* Can a single port have multi-clocked? How do you set delays for such ports?
* How is scan DEF (Design Exchange Format) generated?
* What is purpose of lockup latch in scan chain?
* Explain short circuit current.

* What are pros/cons of using low Vt, high Vt cells?

* How do you set inter clock uncertainty?

Answer:

set_clock_uncertainty –from clock1 -to clock2

* In DC (Design Compiler), how do you constrain clocks, IO (Input-Output) ports, maxcap, max tran?
* What are differences in clock constraints from pre CTS (Clock Tree Synthesis) to post CTS (Clock Tree Synthesis)?

Answer:

Difference in clock uncertainty values; Clocks are propagated in post CTS.

In post CTS clock latency constraint is modified to model clock jitter.

* How is clock gating done?

* What constraints you add in CTS (Clock Tree Synthesis) for clock gates?

Answer:

Make the clock gating cells as through pins.

* What is trade off between dynamic power (current) and leakage power (current)?

Answer:

* How do you reduce standby (leakage) power?

* Explain top level pin placement flow? What are parameters to decide?
* Given block level netlists, timing constraints, libraries, macro LEFs (Layout Exchange Format/Library Exchange Format), how will you start floor planning?
* With net length of 1000um how will you compute RC values, using equations/tech file info?
* What do noise reports represent?
* What does glitch reports contain?
* What are CTS (Clock Tree Synthesis) steps in IC compiler?
* What do clock constraints file contain?
* How to analyze clock tree reports?
* What do IR drop Voltagestorm reports represent?
* Where /when do you use DCAP (Decoupling Capacitor) cells?
* What are various power reduction techniques?

Hughes Networks

* What is setup/hold? What are setup and hold time impacts on timing? How will you fix setup and hold violations?
* Explain function of Muxed FF (Multiplexed Flip Flop) /scan FF (Scal Flip Flop).
* What are tested in DFT (Design for Testability)?
* In equivalence checking, how do you handle scanen signal?
* In terms of CMOS (Complimentary Metal Oxide Semiconductor), explain physical parameters that affect the propagation delay?
* What are power dissipation components? How do you reduce them?

* How delay affected by PVT (Process-Voltage-Temperature)?

* Why is power signal routed in top metal layers?

Avago Technologies (former HP group)

* How do you minimize clock skew/ balance clock tree?

* Given 11 minterms and asked to derive the logic function.
* Given C1= 10pf, C2=1pf connected in series with a switch in between, at t=0 switch is open and one end having 5v and other end zero voltage; compute the voltage across C2 when the switch is closed?
* Explain the modes of operation of CMOS (Complimentary Metal Oxide Semiconductor) inverter? Show IO (Input-Output) characteristics curve.
* Implement a ring oscillator.
* How to slow down ring oscillator?

Hynix Semiconductor

* How do you optimize power at various stages in the physical design flow?
* What timing optimization strategies you employ in pre-layout /post-layout stages?
* What are process technology challenges in physical design?
* Design divide by 2, divide by 3, and divide by 1.5 counters. Draw timing diagrams.
* What are multi-cycle paths, false paths? How to resolve multi-cycle and false paths?
* Given a flop to flop path with combo delay in between and output of the second flop fed back to combo logic. Which path is fastest path to have hold violation and how will you resolve?
* What are RTL (Register Transfer Level) coding styles to adapt to yield optimal backend design?
* Draw timing diagrams to represent the propagation delay, set up, hold, recovery, removal, minimum pulse width.

About Contributor

ASIC_diehard has more than 5 years of experience in physical design, timing, netlist to GDS flows of Integrated Circuit development. ASIC_diehard's fields of interest are backend design, place and route, timing closure, process technologies.

Readers are encouraged to discuss answers to these questions. Just click on the 'post a comment' option below and put your comments there. Alternatively you can send your answers/discussions to my mail id: shavakmm@gmail.com
Physical Design Objective Type of Questions and Answers

* 1) Chip utilization depends on ___.

a. Only on standard cells
b. Standard cells and macros

c. Only on macros
d. Standard cells macros and IO pads

* 2) In Soft blockages ____ cells are placed.

a. Only sequential cells
b. No cells
c. Only Buffers and Inverters
d. Any cells

* 3) Why we have to remove scan chains before placement?

a. Because scan chains are group of flip flop
b. It does not have timing critical path
c. It is series of flip flop connected in FIFO
d. None

* 4) Delay between shortest path and longest path in the clock is called ____.

a. Useful skew
b. Local skew
c. Global skew
d. Slack

* 5) Cross talk can be avoided by ___.

a. Decreasing the spacing between the metal layers
b. Shielding the nets
c. Using lower metal layers
d. Using long nets

* 6) Prerouting means routing of _____.

a. Clock nets
b. Signal nets
c. IO nets
d. PG nets

* 7) Which of the following metal layer has Maximum resistance?

a. Metal1
b. Metal2
c. Metal3
d. Metal4

* 8) What is the goal of CTS?

a. Minimum IR Drop
b. Minimum EM

c. Minimum Skew
d. Minimum Slack

* 9) Usually Hold is fixed ___.

a. Before Placement
b. After Placement
c. Before CTS
d. After CTS

* 10) To achieve better timing ____ cells are placed in the critical path.

a. HVT
b. LVT
c. RVT
d. SVT

* 11) Leakage power is inversely proportional to ___.

a. Frequency
b. Load Capacitance
c. Supply voltage
d. Threshold Voltage

* 12) Filler cells are added ___.

a. Before Placement of std cells
b. After Placement of Std Cells
c. Before Floor planning
d. Before Detail Routing

* 13) Search and Repair is used for ___.

a. Reducing IR Drop
b. Reducing DRC
c. Reducing EM violations
d. None

* 14) Maximum current density of a metal is available in ___.

a. .lib
b. .v
c. .tf
d. .sdc

* 15) More IR drop is due to ___.

a. Increase in metal width
b. Increase in metal length

c. Decrease in metal length
d. Lot of metal layers

* 16) The minimum height and width a cell can occupy in the design is called as ___.

a. Unit Tile cell
b. Multi heighten cell
c. LVT cell
d. HVT cell

* 17) CRPR stands for ___.

a. Cell Convergence Pessimism Removal
b. Cell Convergence Preset Removal
c. Clock Convergence Pessimism Removal
d. Clock Convergence Preset Removal

* 18) In OCV timing check, for setup time, ___.

a. Max delay is used for launch path and Min delay for capture path
b. Min delay is used for launch path and Max delay for capture path
c. Both Max delay is used for launch and Capture path
d. Both Min delay is used for both Capture and Launch paths

* 19) "Total metal area and(or) perimeter of conducting layer / gate to gate area" is called ___.

a. Utilization
b. Aspect Ratio
c. OCV
d. Antenna Ratio

* 20) The Solution for Antenna effect is ___.

a. Diode insertion
b. Shielding
c. Buffer insertion
d. Double spacing

* 21) To avoid cross talk, the shielded net is usually connected to ___.

a. VDD
b. VSS
c. Both VDD and VSS
d. Clock

* 22) If the data is faster than the clock in Reg to Reg path ___ violation may come.

a. Setup
b. Hold
c. Both
d. None

* 23) Hold violations are preferred to fix ___.

a. Before placement
b. After placement
c. Before CTS
d. After CTS

* 24) Which of the following is not present in SDC ___?

a. Max tran
b. Max cap
c. Max fanout
d. Max current density

* 25) Timing sanity check means (with respect to PD)___.

a. Checking timing of routed design with out net delays
b. Checking Timing of placed design with net delays
c. Checking Timing of unplaced design without net delays
d. Checking Timing of routed design with net delays

* 26) Which of the following is having highest priority at final stage (post routed) of the design ___?

a. Setup violation
b. Hold violation
c. Skew
d. None

* 27) Which of the following is best suited for CTS?

a. CLKBUF
b. BUF
c. INV
d. CLKINV

* 28) Max voltage drop will be there at(with out macros) ___.

a. Left and Right sides
b. Bottom and Top sides
c. Middle
d. None

* 29) Which of the following is preferred while placing macros ___?

a. Macros placed center of the die
b. Macros placed left and right side of die
c. Macros placed bottom and top sides of die
d. Macros placed based on connectivity of the I/O

* 30) Routing congestion can be avoided by ___.

a. placing cells closer
b. Placing cells at corners
c. Distributing cells
d. None

* 31) Pitch of the wire is ___.

a. Min width
b. Min spacing
c. Min width - min spacing
d. Min width + min spacing

* 32) In Physical Design following step is not there ___.

a. Floorplaning
b. Placement
c. Design Synthesis
d. CTS

* 33) In technology file if 7 metals are there then which metals you will use for power?

a. Metal1 and metal2
b. Metal3 and metal4
c. Metal5 and metal6
d. Metal6 and metal7

* 34) If metal6 and metal7 are used for the power in 7 metal layer process design then which metals you will use for clock ?

a. Metal1 and metal2
b. Metal3 and metal4
c. Metal4 and metal5
d. Metal6 and metal7

* 35) In a reg to reg timing path Tclocktoq delay is 0.5ns and TCombo delay is 5ns and Tsetup is 0.5ns then the clock period should be ___.

a. 1ns
b. 3ns

c. 5ns
d. 6ns

* 36) Difference between Clock buff/inverters and normal buff/inverters is __.

a. Clock buff/inverters are faster than normal buff/inverters
b. Clock buff/inverters are slower than normal buff/inverters
c. Clock buff/inverters are having equal rise and fall times with high drive strengths compare to normal buff/inverters
d. Normal buff/inverters are having equal rise and fall times with high drive strengths compare to Clock buff/inverters.

* 37) Which configuration is more preferred during floorplaning ?

a. Double back with flipped rows
b. Double back with non flipped rows
c. With channel spacing between rows and no double back
d. With channel spacing between rows and double back

* 38) What is the effect of high drive strength buffer when added in long net ?

a. Delay on the net increases
b. Capacitance on the net increases
c. Delay on the net decreases
d. Resistance on the net increases.

* 39) Delay of a cell depends on which factors ?

a. Output transition and input load
b. Input transition and Output load
c. Input transition and Output transition
d. Input load and Output Load.

* 40) After the final routing the violations in the design ___.

a. There can be no setup, no hold violations
b. There can be only setup violation but no hold
c. There can be only hold violation not Setup violation
d. There can be both violations.

* 41) Utilisation of the chip after placement optimisation will be ___.

a. Constant
b. Decrease
c. Increase
d. None of the above

* 42) What is routing congestion in the design?

a. Ratio of required routing tracks to available routing tracks
b. Ratio of available routing tracks to required routing tracks
c. Depends on the routing layers available
d. None of the above

* 43) What are preroutes in your design?

a. Power routing
b. Signal routing
c. Power and Signal routing
d. None of the above.

* 44) Clock tree doesn't contain following cell ___.

a. Clock buffer
b. Clock Inverter
c. AOI cell
d. None of the above

* Answers:

1)b
2)c
3)b
4)c
5)b
6)d
7)a
8)c
9)d
10)b
11)d
12)d
13)b
14)c
15)b
16)a
17)c
18)a
19)d
20)a
21)b
22)b
23)d
24)d
25)c
26)b
27)a
28)c

29)d
30)c
31)d
32)c
33)d
34)c
35)d
36)c
37)a
38)c
39)b
40)d
41)c
42)a
43)a
44)c

Backend (Physical Design) Interview Questions and Answers

* Below are the sequence of questions asked for a physical design engineer.

In which field are you interested?

* Answer to this question depends on your interest, expertise and to the requirement for which you have been interviewed.

* Well..the candidate gave answer: Low power design

Can you talk about low power techniques?
How low power and latest 90nm/65nm technologies are related?

Do you know about input vector controlled method of leakage reduction?

* Leakage current of a gate is dependant on its inputs also. Hence find the set of inputs which gives least leakage. By applyig this minimum leakage vector to a circuit it is possible to decrease the leakage current of the circuit when it is in the standby mode. This method is known as input vector controlled method of leakage reduction.

How can you reduce dynamic power?

* -Reduce switching activity by designing good RTL
* -Clock gating
* -Architectural improvements

* -Reduce supply voltage
* -Use multiple voltage domains-Multi vdd

What are the vectors of dynamic power?

* Voltage and Current

How will you do power planning?

If you have both IR drop and congestion how will you fix it?

* -Spread macros
* -Spread standard cells
* -Increase strap width
* -Increase number of straps
* -Use proper blockage

Is increasing power line width and providing more number of straps are the only solution to IR drop?

* -Spread macros
* -Spread standard cells
* -Use proper blockage

In a reg to reg path if you have setup problem where will you insert buffer-near to launching flop or capture flop? Why?

* (buffers are inserted for fixing fanout voilations and hence they reduce setup voilation; otherwise we try to fix setup voilation with the sizing of cells; now just assume that you must insert buffer !)

* Near to capture path.

* Because there may be other paths passing through or originating from the flop nearer to lauch flop. Hence buffer insertion may affect other paths also. It may improve all those paths or degarde. If all those paths have voilation then you may insert buffer nearer to launch flop provided it improves slack.

How will you decide best floorplan?

What is the most challenging task you handled?
What is the most challenging job in P&R flow?

* -It may be power planning- because you found more IR drop
* -It may be low power target-because you had more dynamic and

leakage power
* -It may be macro placement-because it had more connection with standard cells or macros
* -It may be CTS-because you needed to handle multiple clocks and clock domain crossings
* -It may be timing-because sizing cells in ECO flow is not meeting timing
* -It may be library preparation-because you found some inconsistancy in libraries.
* -It may be DRC-because you faced thousands of voilations

How will you synthesize clock tree?

* -Single clock-normal synthesis and optimization
* -Multiple clocks-Synthesis each clock seperately
* -Multiple clocks with domain crossing-Synthesis each clock seperately and balance the skew

How many clocks were there in this project?

* -It is specific to your project
* -More the clocks more challenging !

How did you handle all those clocks?

* -Multiple clocks-->synthesize seperately-->balance the skew-->optimize the clock tree

Are they come from seperate external resources or PLL?

* -If it is from seperate clock sources (i.e.asynchronous; from different pads or pins) then balancing skew between these clock sources becomes challenging.

* -If it is from PLL (i.e.synchronous) then skew balancing is comparatively easy.

Why buffers are used in clock tree?

* To balance skew (i.e. flop to flop delay)

What is cross talk?

* Switching of the signal in one net can interfere neigbouring net due

to cross coupling capacitance.This affect is known as cros talk. Cross talk may lead setup or hold voilation.

How can you avoid cross talk?

* -Double spacing=>more spacing=>less capacitance=>less cross talk
* -Multiple vias=>less resistance=>less RC delay
* -Shielding=> constant cross coupling capacitance =>known value of crosstalk
* -Buffer insertion=>boost the victim strength

How shielding avoids crosstalk problem? What exactly happens there?

* -High frequency noise (or glitch)is coupled to VSS (or VDD) since shilded layers are connected to either VDD or VSS.

* Coupling capacitance remains constant with VDD or VSS.

How spacing helps in reducing crosstalk noise?

* width is more=>more spacing between two conductors=>cross coupling capacitance is less=>less cross talk

Why double spacing and multiple vias are used related to clock?

* Why clock?-- because it is the one signal which chages it state regularly and more compared to any other signal. If any other signal switches fast then also we can use double space.

* Double spacing=>width is more=>capacitance is less=>less cross talk

* Multiple vias=>resistance in parellel=>less resistance=>less RC delay

How buffer can be used in victim to avoid crosstalk?

* Buffer increase victims signal strength; buffers break the net length=>victims are more tolerant to coupled signal from aggressor.

Physical Design Questions and Answers

* I am getting several emails requesting answers to the questions

posted in this blog. But it is very difficult to provide detailed answer to all questions in my available spare time. Hence i decided to give "short and sweet" one line answers to the questions so that readers can immediately benefited. Detailed answers will be posted in later stage.I have given answers to some of the physical design questions here. Enjoy !

What parameters (or aspects) differentiate Chip Design and Block level design?

* Chip design has I/O pads; block design has pins.

* Chip design uses all metal layes available; block design may not use all metal layers.

* Chip is generally rectangular in shape; blocks can be rectangular, rectilinear.

* Chip design requires several packaging; block design ends in a macro.

How do you place macros in a full chip design?

* First check flylines i.e. check net connections from macro to macro and macro to standard cells.

* If there is more connection from macro to macro place those macros nearer to each other preferably nearer to core boundaries.

* If input pin is connected to macro better to place nearer to that pin or pad.

* If macro has more connection to standard cells spread the macros inside core.

* Avoid criscross placement of macros.

* Use soft or hard blockages to guide placement engine.

Differentiate between a Hierarchical Design and flat design?

* Hierarchial design has blocks, subblocks in an hierarchy; Flattened design has no subblocks and it has only leaf cells.

* Hierarchical design takes more run time; Flattened design takes less run time.

Which is more complicated when u have a 48 MHz and 500 MHz clock design?

* 500 MHz; because it is more constrained (i.e.lesser clock period) than 48 MHz design.

Name few tools which you used for physical verification?

* Herculis from Synopsys, Caliber from Mentor Graphics.

What are the input files will you give for primetime correlation?

* Netlist, Technology library, Constraints, SPEF or SDF file.

If the routing congestion exists between two macros, then what will you do?

* Provide soft or hard blockage

How will you decide the die size?

* By checking the total area of the design you can decide die size.

If lengthy metal layer is connected to diffusion and poly, then which one will affect by antenna problem?

* Poly

If the full chip design is routed by 7 layer metal, why macros are designed using 5LM instead of using 7LM?

* Because top two metal layers are required for global routing in chip design. If top metal layers are also used in block level it will create routing blockage.

In your project what is die size, number of metal layers, technology, foundry, number of clocks?

* Die size: tell in mm eg. 1mm x 1mm ; remeber 1mm=1000micron which is a big size !!

* Metal layers: See your tech file. generally for 90nm it is 7 to 9.

* Technology: Again look into tech files.

* Foundry:Again look into tech files; eg. TSMC, IBM, ARTISAN etc

* Clocks: Look into your design and SDC file !

How many macros in your design?

* You know it well as you have designed it ! A SoC (System On Chip) design may have 100 macros also !!!!

What is each macro size and number of standard cell count?

* Depends on your design.

What are the input needs for your design?

* For synthesis: RTL, Technology library, Standard cell library, Constraints

* For Physical design: Netlist, Technology library, Constraints, Standard cell library

What is SDC constraint file contains?

* Clock definitions

* Timing exception-multicycle path, false path

* Input and Output delays

How did you do power planning? How to calculate core ring width, macro ring width and strap or trunk width? How to find number of power pad and IO power pads? How the width of metal and number of straps calculated for power and ground?

* Get the total core power consumption; get the metal layer current density value from the tech file; Divide total power by number sides of the chip; Divide the obtained value from the current density to get core power ring width. Then calculate number of straps using some more equations. Will be explained in detail later.

How to find total chip power?

* Total chip power=standard cell power consumption,Macro power consumption pad power consumption.

What are the problems faced related to timing?

* Prelayout: Setup, Max transition, max capacitance

* Post layout: Hold

How did you resolve the setup and hold problem?

* Setup: upsize the cells

* Hold: insert buffers

In which layer do you prefer for clock routing and why?

* Next lower layer to the top two metal layers(global routing layers). Because it has less resistance hence less RC delay.

If in your design has reset pin, then it'll affect input pin or output pin or both?

* Output pin.

During power analysis, if you are facing IR drop problem, then how did you avoid?

* Increase power metal layer width.

* Go for higher metal layer.

* Spread macros or standard cells.

* Provide more straps.

Define antenna problem and how did you resolve these problem?

* Increased net length can accumulate more charges while manufacturing of the device due to ionisation process. If this net is connected to gate of the MOSFET it can damage dielectric property of the gate and gate may conduct causing damage to the MOSFET. This

is antenna problem.

* Decrease the length of the net by providing more vias and layer jumping.

* Insert antenna diode.

How delays vary with different PVT conditions? Show the graph.

* P increase->dealy increase

* P decrease->delay decrease

* V increase->delay decrease

* V decrease->delay increase

* T increase->delay increase

* T decrease->delay decrease

Explain the flow of physical design and inputs and outputs for each step in flow.

What is cell delay and net delay?

* Gate delay

* Transistors within a gate take a finite time to switch. This means that a change on the input of a gate takes a finite time to cause a change on the output.[Magma]

* Gate delay =function of(i/p transition time, Cnet+Cpin).

* Cell delay is also same as Gate delay.

* Cell delay

* For any gate it is measured between 50% of input transition to the corresponding 50% of output transition.

* Intrinsic delay

* Intrinsic delay is the delay internal to the gate. Input pin of the cell to output pin of the cell.

* It is defined as the delay between an input and output pair of a cell, when a near zero slew is applied to the input pin and the output does not see any load condition.It is predominantly caused by the internal capacitance associated with its transistor.

* This delay is largely independent of the size of the transistors forming the gate because increasing size of transistors increase internal capacitors.

* Net Delay (or wire delay)

* The difference between the time a signal is first applied to the net and the time it reaches other devices connected to that net.

* It is due to the finite resistance and capacitance of the net.It is also known as wire delay.

* Wire delay =fn(Rnet , Cnet+Cpin)

What are delay models and what is the difference between them?

* Linear Delay Model (LDM)

* Non Linear Delay Model (NLDM)

What is wire load model?

* Wire load model is NLDM which has estimated R and C of the net.

Why higher metal layers are preferred for Vdd and Vss?

* Because it has less resistance and hence leads to less IR drop.

What is logic optimization and give some methods of logic

optimization.

* Upsizing

* Downsizing

* Buffer insertion

* Buffer relocation

* Dummy buffer placement

What is the significance of negative slack?

* negative slack==> there is setup voilation==> deisgn can fail

What is signal integrity? How it affects Timing?

* IR drop, Electro Migration (EM), Crosstalk, Ground bounce are signal integrity issues.

* If Idrop is more==>delay increases.

* crosstalk==>there can be setup as well as hold voilation.

What is IR drop? How to avoid? How it affects timing?

* There is a resistance associated with each metal layer. This resistance consumes power causing voltage drop i.e.IR drop.

* If IR drop is more==>delay increases.

What is EM and it effects?

* Due to high current flow in the metal atoms of the metal can displaced from its origial place. When it happens in larger amount the metal can open or bulging of metal layer can happen. This effect is known as Electro Migration.

* Affects: Either short or open of the signal line or power line.

What are types of routing?

* Global Routing

* Track Assignment

* Detail Routing

What is latency? Give the types?

* Source Latency

* It is known as source latency also. It is defined as "the delay from the clock origin point to the clock definition point in the design".

* Delay from clock source to beginning of clock tree (i.e. clock definition point).

* The time a clock signal takes to propagate from its ideal waveform origin point to the clock definition point in the design.

* Network latency

* It is also known as Insertion delay or Network latency. It is defined as "the delay from the clock definition point to the clock pin of the register".

* The time clock signal (rise or fall) takes to propagate from the clock definition point to a register clock pin.

What is track assignment?

* Second stage of the routing wherein particular metal tracks (or layers) are assigned to the signal nets.

What is congestion?

* If the number of routing tracks available for routing is less than the required tracks then it is known as congestion.

Whether congestion is related to placement or routing?

* Routing

What are clock trees?

* Distribution of clock from the clock source to the sync pin of the registers.

What are clock tree types?

* H tree, Balanced tree, X tree, Clustering tree, Fish bone

What is cloning and buffering?

* Cloning is a method of optimization that decreases the load of a heavily loaded cell by replicating the cell.

* Buffering is a method of optimization that is used to insert beffers in high fanout nets to decrease the dealy.

What is the difference between soft macro and hard macro?

* What is the difference between hard macro, firm macro and soft macro?

or

* What are IPs?

* Hard macro, firm macro and soft macro are all known as IP (Intellectual property). They are optimized for power, area and performance. They can be purchased and used in your ASIC or FPGA design implementation flow. Soft macro is flexible for all type of ASIC implementation. Hard macro can be used in pure ASIC design flow, not in FPGA flow. Before bying any IP it is very important to evaluate its advantages and disadvantages over each other, hardware compatibility such as I/O standards with your design blocks, reusability for other designs.

Soft macros

* Soft macros are in synthesizable RTL.

* Soft macros are more flexible than firm or hard macros.

* Soft macros are not specific to any manufacturing process.

* Soft macros have the disadvantage of being somewhat unpredictable in terms of performance, timing, area, or power.

* Soft macros carry greater IP protection risks because RTL source code is more portable and therefore, less easily protected than either a netlist or physical layout data.

* From the physical design perspective, soft macro is any cell that has been placed and routed in a placement and routing tool such as Astro. (This is the definition given in Astro Rail user manual !)

* Soft macros are editable and can contain standard cells, hard macros, or other soft macros.

Firm macros

* Firm macros are in netlist format.

* Firm macros are optimized for performance/area/power using a specific fabrication technology.

* Firm macros are more flexible and portable than hard macros.

* Firm macros are predictive of performance and area than soft macros.

Hard macro

* Hard macros are generally in the form of hardware IPs (or we termed it as hardwre IPs !).

* Hard macos are targeted for specific IC manufacturing technology.

* Hard macros are block level designs which are silicon tested and proved.

* Hard macros have been optimized for power or area or timing.

* In physical design you can only access pins of hard macros unlike soft macros which allows us to manipulate in different way.

* You have freedom to move, rotate, flip but you can't touch anything inside hard macros.

* Very common example of hard macro is memory. It can be any

design which carries dedicated single functionality (in general).. for example it can be a MP4 decoder.

* Be aware of features and characteristics of hard macro before you use it in your design... other than power, timing and area you also should know pin properties like sync pin, I/O standards etc

* LEF, GDS2 file format allows easy usage of macros in different tools.

From the physical design (backend) perspective:

* Hard macro is a block that is generated in a methodology other than place and route (i.e. using full custom design methodology) and is brought into the physical design database (eg. Milkyway in Synopsys; Volcano in Magma) as a GDS2 file.

Synthesis and placement of macros in modern SoC designs are challenging. EDA tools employ different algorithms accomplish this task along with the target of power and area. There are several research papers available on these subjects. Some of them can be downloaded from the given link below.

What is difference between normal buffer and clock buffer?

Answer:

Clock net is one of the High Fanout Net(HFN)s. The clock buffers are designed with some special property like high drive strength and less delay. Clock buffers have equal rise and fall time. This prevents duty cycle of clock signal from changing when it passes through a chain of clock buffers.

Normal buffers are designed with W/L ratio such that sum of rise time and fall time is minimum. They too are designed for higher drive strength.
What is difference between HFN synthesis and CTS?

Answer:

HFNs are synthesized in front end also.... but at that moment no placement information of standard cells are available... hence

backend tool collapses synthesized HFNs. It resenthesizes HFNs based on placement information and appropriately inserts buffer. Target of this synthesis is to meet delay requirements i.e. setup and hold.

For clock no synthesis is carried out in front end (why.....????..because no placement information of flip-flops ! So synthesis won't meet true skew targets !!) ... in backend clock tree synthesis tries to meet "skew" targets...It inserts clock buffers (which have equal rise and fall time, unlike normal buffers !)... There is no skew information for any HFNs.
Is it possible to have a zero skew in the design?

Answer:

Theoretically it is possible....!

Practically it is impossible....!!

Practically we cant reduce any delay to zero.... delay will exist... hence we try to make skew "equal" (or same) rather than "zero".....now with this optimization all flops get the clock edge with same delay relative to each other.... so virtually we can say they are having "zero skew " or skew is "balanced".
Physical Design Interview Questions

Below are the important interview questions for VLSI physical design aspirants. Interview starts with flow of physical design and goes on.....on....on..... I am trying to make your life easy..... let me prepare answers to all these if soft form.... as soon as it happens those answers will be posted in coming blogs.

*
What parameters (or aspects) differentiate Chip Design & Block level design??
*
How do you place macros in a full chip design?
*
Differentiate between a Hierarchical Design and flat design?
*
Which is more complicated when u have a 48 MHz and 500 MHz clock design?
*
Name few tools which you used for physical verification?
*
What are the input files will you give for primetime correlation?
*
What are the algorithms used while routing? Will it optimize wire

length?
*
How will you decide the Pin location in block level design?
*
If the routing congestion exists between two macros, then what will you do?
*
How will you place the macros?
*
How will you decide the die size?
*
If lengthy metal layer is connected to diffusion and poly, then which one will affect by antenna problem?
*
If the full chip design is routed by 7 layer metal, why macros are designed using 5LM instead of using 7LM?
*
In your project what is die size, number of metal layers, technology, foundry, number of clocks?
*
How many macros in your design?
*
What is each macro size and no. of standard cell count?
*
How did u handle the Clock in your design?
*
What are the Input needs for your design?
*
What is SDC constraint file contains?
*
How did you do power planning?
*
How to find total chip power?
*
How to calculate core ring width, macro ring width and strap or trunk width?
*
How to find number of power pad and IO power pads?
*
What are the problems faced related to timing?
*
How did u resolve the setup and hold problem?
*
If in your design 10000 and more numbers of problems come, then what you will do?
*
In which layer do you prefer for clock routing and why?
*
If in your design has reset pin, then it'll affect input pin or output pin or both?

*
During power analysis, if you are facing IR drop problem, then how did u avoid?
*
Define antenna problem and how did u resolve these problem?
*
How delays vary with different PVT conditions? Show the graph.
*
Explain the flow of physical design and inputs and outputs for each step in flow.
*
What is cell delay and net delay?
*
What are delay models and what is the difference between them?
*
What is wire load model?
*
What does SDC constraints has?
*
Why higher metal layers are preferred for Vdd and Vss?
*
What is logic optimization and give some methods of logic optimization.
*
What is the significance of negative slack?
*
What is signal integrity? How it affects Timing?
*
What is IR drop? How to avoid .how it affects timing?
*
What is EM and it effects?
*
What is floor plan and power plan?
*
What are types of routing?
*
What is a grid .why we need and different types of grids?
*
What is core and how u will decide w/h ratio for core?
*
What is effective utilization and chip utilization?
*
What is latency? Give the types?
*
How the width of metal and number of straps calculated for power and ground?
*
What is negative slack ? How it affects timing?
*
What is track assignment?

*
What is grided and gridless routing?
*
What is a macro and standard cell?
*
What is congestion?
*
Whether congestion is related to placement or routing?
*
What are clock trees?
*
What are clock tree types?
*
Which layer is used for clock routing and why?
*
What is cloning and buffering?
*
What are placement blockages?
*
How slow and fast transition at inputs effect timing for gates?
*
What is antenna effect?
*
What are DFM issues?
*
What is .lib, LEF, DEF, .tf?
*
What is the difference between synthesis and simulation?
*
What is metal density, metal slotting rule?
*
What is OPC, PSM?
*
Why clock is not synthesized in DC?
*
What are high-Vt and low-Vt cells?
*
What corner cells contains?
*
What is the difference between core filler cells and metal fillers?
*
How to decide number of pads in chip level design?
*
What is tie-high and tie-low cells and where it is used
*
What is LEF?
*
What is DEF?
*
What are the steps involved in designing an optimal pad ring?

* What are the steps that you have done in the design flow?
* What are the issues in floor plan?
* How can you estimate area of block?
* How much aspect ratio should be kept (or have you kept) and what is the utilization?
* How to calculate core ring and stripe widths?
* What if hot spot found in some area of block? How you tackle this?
* After adding stripes also if you have hot spot what to do?
* What is threshold voltage? How it affect timing?
* What is content of lib, lef, sdc?
* What is meant my 9 track, 12 track standard cells?
* What is scan chain? What if scan chain not detached and reordered? Is it compulsory?
* What is setup and hold? Why there are ? What if setup and hold violates?
* In a circuit, for reg to reg path ...Tclktoq is 50 ps, Tcombo 50ps, Tsetup 50ps, tskew is 100ps. Then what is the maximum operating frequency?
* How R and C values are affecting time?
* How ohm (R), fared (C) is related to second (T)?
* What is transition? What if transition time is more?
* What is difference between normal buffer and clock buffer?
* What is antenna effect? How it is avoided?
* What is ESD?
* What is cross talk? How can you avoid?
* How double spacing will avoid cross talk?
* What is difference between HFN synthesis and CTS?
* What is hold problem? How can you avoid it?
* For an iteration we have 0.5ns of insertion delay and 0.1 skew and for other iteration 0.29ns insertion delay and 0.25 skew for the same circuit then which one you will select? Why?
* What is partial floor plan?

Posted by **VLSI_Rules** at **10:37 AM 4 comments:**
Labels: **analysis**, **asic**, **backend**, **buffer**, **chip**, **clock**, **cmos**, **delay**, **design**, **layout**, **optimization**, **physical**, **routing**, **sta**, **synthesis**, **timing**, **tree**, **vlsi**

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VLSI Interview Questions

For Any answers you may contact Aviral Mittal at avimit@yahoo.com
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1(a). What you would use in RTL a 'boolean' type or a 'std_logic' type and why.
1(b). What are/may be the implications of using an 'integer' type in RTL.
2). What you would use a RAM or a regfile discuss:
RAM: Low Power, Low Area
REGFILE: better timing.
3). A timing path fails: what are your options?
Ans: a). Look for parallelism in RTL.
b). Look for small RAMs which might be synthesized
c). Look for better placement
d). Look for Pipelining opportunity
e). Look for moving logic before the Reg
f). Look for replicating the drivers to reduce load if the delay is caused by loading
g). '< or >' are better than -, when comparators are used in RTL
h). Look for if..elsif..elseif..elsif
i). Use One Hot instead of Binary coded State Registers
4). Any example you can provide in which verilog might be a prob
5). What are VHDL structures, give an example to exploit them
6). What is grey coding, any example where they are used
7). Discuss Async interfaces
8). Metastability
9). Synopsys unwanted latch
10). Verilog blocking vs non-blocking
11). VHDL variables: example where you have to use them
12). What is pipelining and how it may improve the performance
13). What are multicycle paths.
14). What are false paths
15). What are Async counters, what are advantages of using these over sync counters. and what are the disadvantages
16). q_in : IN std_logic
variable q0 : std_logic
variable q1 : std_logic
variable q2 : std_logic
q_out : OUT std_logic;

```
if(rising_edge(clk)) then
  q0 := q_in;
  q1 := q0;
  q2 := q1;
  q_out <= q2;
endif;
```

what will be the result.

17). Sensitivity List:
How does it matter.What will happen
if you dont include a signal in the sensitivity list
and use/read it inside the process

18). process(a_sig,b_sig)

```
begin
if(a_sig = '1' and b_sig = '0') then
  sig_out <= '1';
else
  sig_out <= '0';
end if;
end process

process(c_sig,d_sig)
begin
if(c_sig = '1' and d_sig = '0') then
  sig_out <= '1';
else
  sig_out <= '0';
end if;
end process
```

Any thing wrong with the above code
Ans: Same signal being driven in two different processes

19). While writing a state machine in RTL
it is generally recommended that the seq and combi logics
should be written in different processes
Even if its not a state machine, it is recommended
that seq and combi logics should be written in
different processes
Any example you may give to support this recommendation

Ans: Sometime its desired to use the 'next_RegValue' which can
only be accessed in recommended method, not in the
method where you combine seq and combi parts into one
seq process.

20). Following are two methods to determine if 'a_sig' is greater than 'b_sig'
Which one would you use and why?

```
signal a_sig : std_logic_vector(n downto 0);
signal b_sig : std_logic_vector(n downto 0);

process(a_sig,bsig)
  variable c_sig : std_logic_vector(n downto 0);
begin
  c_sig := a_sig-bsig;
  a_is_smaller <= c_sig(c_sig'high);
```

```
end process;

process(a_sig,bsig)
  variable c_sig : std_logic_vector(n downto 0);
begin
  if(asig < bsig) then
    a_is_smaller <= '1';
  else
    a_is_smaller <= '0';
  end if;
end process;
```

21). How you will implement a C language pointer in VHDL

22). A certain chip(silicon) fails hold test and another fails a setup test. Which one MAY still be used how and why

23). What is Design For Test and why it is done.

24). What is clock gating? How and why it is done.

25). Low Power: discuss how it may be done

Ans : a) Clock Gating
b) Reducing the frequency of operation using pipelining
c) Shutting the power down
d) Different voltage domains
e) Reducing number of transitions, eg. using grey coding
f) Bus Invert coding: invert bus if hamming distance is greater than 1/2 word size
g) Async design techniques

26). Discuss disadvantages/challenges of shrinking technology

Ans : Leakage Power

27). Simulation Problem. Register going 'X' even when RTL has reset pin defining the state of register at power up reset. And such a reset has been applied in simulation.

28). Generally dc_shell tries to optimise the path with worst violation. Is there any thing that you can do to make it work on more paths parallely

Ans :Use group_path may be with 'critical_range' on that group

29). What is pipelining, how may it affect the performance of a design

30). What is the difference between transport delays and inertial delays in VHDL

31). What determines the max frequency a digital design may work on. Why thold(hold time) is not included in the calculation for the above.

32). What will happen if output of an inverter is shorted to its input

33). What is noise margin.

34). Why are p-mos larger than n-mos in CMOS design

35). Draw DC curve of inverter and Re-Draw it if pmos and nmos are equal

36). What is Latch-up

37). How can an Inverter work as an amplifier

38). Design a state machine to implement a edge detector That is.
The output of this state machine is a pulse of logic '1' of duration one clock when ever there is a negative edge on an input signal. Given that the frequency at which the negative edge is appearing on the input signal is low as compared to the clock of the state machine

39). Design a state machine which divides the input frequency of a clock by 3.
Given that the phase change in the output due to propogation delay in of the flip flop is acceptable up to a delay offered by a single flip flop only.

40). Why does a pass gate requires two transistors(1 N and 1 P type) Can we use a single transistor N or P type in a pass gate? If not why? and if yes then in what conditions?

41). Why CMOS why not N-MOS or P-MOS logic, when we know that the number of gates required in CMOS are grater than in n-mos or p-mos logic.

42). How much is the max fan out of a typical CMOS gate. Or alternatively, discuss the limiting factors.

43). What are dynamic logic gates? What are their advantages over conventional logic gates

44). Design a digital circuit to delay the negative edge of the input signal by 2 clock cycles

45). What is the relation between binary encoding and grey(or gray) encoding.

46). Write a vhdl function to implement a length independent grey code counter. alternatively, discuss the logic to do that.

47). How you will constraint a combinational logic path through your design in dc_shell.

48). Make a T Flip Flop using a D Flip Flop

49). How you will make a Nand Gate function like an inverter.

50). Design a state machine to detect a '1101' pattern in a stream.
Detect both, overlapping and non overlapping patterns.

51). From a CMOS logic gate IC, the 40xx series, Take an inverter IC, power it up, and leave the input of the inverter floating. Comment upon the ouput of this inverter.
Similarly take a TTL logic gate IC, the 74xx series. Take an inverter IC, power it up, and leave the input of the inverter floating. Comment upon the output of this inverter.

52). What are LFSRs, example usage?

53). What are MISRs, example usage?

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VLSI Made Easy

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VLSI Interview Questions

VLSI FAQs

1. What is metastability?

When setup or hold window is violated in an flip flop then signal attains a unpredictable value or state known as metastability.

2. What is MTBF? What it signifies?

- MTBF-Mean Time Before Failure
- Average time to next failure

3. How chance of metastable state failure can be reduced?

- Lowering clock frequency
- Lowering data speed
- Using faster flip flop

4. What are the advantages of using synchronous reset ?

- No metastability problem with synchronous reset (provided recovery and removal time for reset is taken care).
- Simulation of synchronous reset is easy.

5. What are the disadvantages of using synchronous reset ?

- Synchronous reset is slow.
- Implementation of synchronous reset requires more number of gates compared to asynchronous reset design.
- An active clock is essential for a synchronous reset design. Hence you can expect more power consumption.

6. What are the advantages of using asynchronous reset ?

- Implementation of asynchronous reset requires less number of gates compared to synchronous reset design.
- Asynchronous reset is fast.
- Clocking scheme is not necessary for an asynchronous design. Hence design consumes less power. Asynchronous design style is also one of the latest design options to achieve low power. Design community is scrathing their head over asynchronous design possibilities.

7. What are the disadvantages of using asynchronous reset ?

- Metastability problems are main concerns of asynchronous reset scheme (design).
- Static timing analysis and DFT becomes difficult due to asynchronous reset.

8. What are the 3 fundamental operating conditions that determine the delay characteristics of gate? How operating conditions affect gate delay?

- Process
- Voltage
- Temperature

9. Is verilog/VHDL is a concurrent or sequential language?

- Verilog and VHDL both are concurrent languages.
- Any hardware descriptive language is concurrent in nature.

10. In a system with insufficient hold time, will slowing down the clock frequency help?

- No.
- Making data path slower can help hold time but it may result in setup violation.

11. In a system with insufficient setup time, will slowing down the clock frequency help?

- Yes.
- Making data path faster can also help setup time but it may result in hold violation.

Physical Design Objective Type of Questions and Answers

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About Me

Gopi Premala

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- **1) Chip utilization depends on ___.**

a. Only on standard cells b. Standard cells and macros c. Only on macros d. Standard cells macros and IO pads

- **2) In Soft blockages _____ cells are placed.**

a. Only sequential cells b. No cells c. Only Buffers and Inverters d. Any cells

- **3) Why we have to remove scan chains before placement?**

a. Because scan chains are group of flip flop b. It does not have timing critical path c. It is series of flip flop connected in FIFO d. None

- **4) Delay between shortest path and longest path in the clock is called _____.**

a. Useful skew b. Local skew c. Global skew d. Slack

- **5) Cross talk can be avoided by ___.**

a. Decreasing the spacing between the metal layers b. Shielding the nets c. Using lower metal layers d. Using long nets

- **6) Prerouting means routing of _____.**

a. Clock nets b. Signal nets c. IO nets d. PG nets

- **7) Which of the following metal layer has Maximum resistance?**

a. Metal1 b. Metal2 c. Metal3 d. Metal4

- **8) What is the goal of CTS?**

a. Minimum IR Drop b. Minimum EM c. Minimum Skew d. Minimum Slack

- **9) Usually Hold is fixed ___.**

a. Before Placement b. After Placement c. Before CTS d. After CTS

- **10) To achieve better timing _____ cells are placed in the critical path.**

a. HVT b. LVT c. RVT d. SVT

- **11) Leakage power is inversely proportional to ___.**

a. Frequency b. Load Capacitance c. Supply voltage d. Threshold Voltage

- **12) Filler cells are added ___.**

a. Before Placement of std cells b. After Placement of Std Cells c. Before Floor planning d. Before Detail Routing

- **13) Search and Repair is used for ___.**

a. Reducing IR Drop b. Reducing DRC c. Reducing EM violations d. None

- **14) Maximum current density of a metal is available in ___.**

a. .lib b. .v c. .tf d. .sdc

- **15) More IR drop is due to ___.**

a. Increase in metal width b. Increase in metal length c. Decrease in metal length d. Lot of metal layers

- **16) The minimum height and width a cell can occupy in the design is called as ___.**

a. Unit Tile cell b. Multi heighten cell c. LVT cell d. HVT cell

- **17) CRPR stands for ___.**

a. Cell Convergence Pessimism Removal b. Cell Convergence Preset Removal c. Clock Convergence Pessimism Removal d. Clock Convergence Preset Removal

- **18) In OCV timing check, for setup time, ___.**

a. Max delay is used for launch path and Min delay for capture path b. Min delay is used for launch path and Max delay for capture path c. Both Max delay is used for launch and Capture path d. Both Min delay is used for both Capture and Launch paths

- **19) "Total metal area and(or) perimeter of conducting layer / gate to gate area" is called ___.**

a. Utilization b. Aspect Ratio c. OCV d. Antenna Ratio

- **20) The Solution for Antenna effect is ___.**

a. Diode insertion b. Shielding c. Buffer insertion d. Double spacing

- **21) To avoid cross talk, the shielded net is usually connected to ___.**

a. VDD b. VSS c. Both VDD and VSS d. Clock

- **22) If the data is faster than the clock in Reg to Reg path ___ violation may come.**

a. Setup b. Hold c. Both d. None

- **23) Hold violations are preferred to fix ___.**

a. Before placement b. After placement c. Before CTS d. After CTS

- **24) Which of the following is not present in SDC ___?**

a. Max tran b. Max cap c. Max fanout d. Max current density

- **25) Timing sanity check means (with respect to PD)___.**

a. Checking timing of routed design with out net delays b. Checking Timing of placed design with net delays c. Checking Timing of unplaced design without net delays d. Checking Timing of routed design with net delays

- **26) Which of the following is having highest priority at final stage (post routed) of the design ___?**

a. Setup violation b. Hold violation c. Skew d. None

- **27) Which of the following is best suited for CTS?**

a. CLKBUF b. BUF c. INV d. CLKINV

- **28) Max voltage drop will be there at(with out macros) ___.**

a. Left and Right sides b. Bottom and Top sides c. Middle d. None

- **29) Which of the following is preferred while placing macros ___?**

a. Macros placed center of the die b. Macros placed left and right side of die c. Macros placed bottom and top sides of die d. Macros placed based on connectivity of the I/O

- **30) Routing congestion can be avoided by ___.**

a. placing cells closer b. Placing cells at corners c. Distributing cells d. None

- **31) Pitch of the wire is ___.**

a. Min width b. Min spacing c. Min width - min spacing d. Min width + min spacing

- **32) In Physical Design following step is not there ___.**

a. Floorplaning b. Placement c. Design Synthesis d. CTS

- **33) In technology file if 7 metals are there then which metals you will use for power?**

a. Metal1 and metal2 b. Metal3 and metal4 c. Metal5 and metal6 d. Metal6 and metal7

- **34) If metal6 and metal7 are used for the power in 7 metal layer process design then which metals you will use for clock ?**

a. Metal1 and metal2 b. Metal3 and metal4 c. Metal4 and metal5 d. Metal6 and metal7

- **35) In a reg to reg timing path Tclocktoq delay is 0.5ns and TCombodelay is 5ns and Tsetup is 0.5ns then the clock period should be ___.**

a. 1ns b. 3ns c. 5ns d. 6ns

- **36) Difference between Clock buff/inverters and normal buff/inverters is __.**

a. Clock buff/inverters are faster than normal buff/inverters b. Clock buff/inverters are slower than normal buff/inverters c. Clock buff/inverters are having equal rise and fall times with high drive strengths compare to normal buff/inverters d. Normal buff/inverters are having equal rise and fall times with high drive strengths compare to Clock buff/inverters.

- **37) Which configuration is more preferred during floorplaning ?**

a. Double back with flipped rows b. Double back with non flipped rows c. With channel spacing between rows and no double back d. With channel spacing between rows and double back

- **38) What is the effect of high drive strength buffer when added in long net?**

a. Delay on the net increases b. Capacitance on the net increases c. Delay on the net decreases d. Resistance on the net increases.

- **39) Delay of a cell depends on which factors ?**

a. Output transition and input load b. Input transition and Output load c. Input transition and Output transition d. Input load and Output Load.

- **40) After the final routing the violations in the design ___.**

a. There can be no setup, no hold violations b. There can be only setup violation but no hold c. There can be only hold violation not Setup violation d. There can be both violations.

- **41) Utilisation of the chip after placement optimisation will be ___.**

a. Constant b. Decrease c. Increase d. None of the above

- **42) What is routing congestion in the design?**

a. Ratio of required routing tracks to available routing tracks b. Ratio of available routing tracks to required routing tracks c. Depends on the routing layers available d. None of the above

- **43) What are preroutes in your design?**

a. Power routing b. Signal routing c. Power and Signal routing d. None of the above.

- **44) Clock tree doesn't contain following cell ___.**

a. Clock buffer b. Clock Inverter c. AOI cell d. None of the above

- **Answers:**

1)b 2)c 3)b 4)c 5)b 6)d 7)a 8)c 9)d 10)b 11)d 12)d 13)b 14)c 15)b 16)a 17)c 18)a 19)d 20)a 21)b 22)b 23)d 24)d 25)c 26)b 27)a 28)c 29)d 30)c 31)d 32)c 33)d 34)c 35)d 36)c 37)a 38)c 39)b 40)d 41)c 42)a 43)a 44)c

CMOS Design Interview Questions

Below are the important VLSI CMOS interview questions. This set of interview questions may be updated in future. Answers will be posted one by one as and when i prepare them ! Readers are encouraged to post answers in comment section. Here we go.....

· Draw Vds-Ids curve for an MOSFET. How it varies with a) increasing Vgs b)velocity saturation c)Channel length modulation d)W/L ratio.
· What is body effect? Write mathematical expression? Is it due to parallel or serial connection of MOSFETs?
· What is latch-up in CMOS design and what are the ways to prevent it?
· What is Noise Margin? Explain with the help of Inverter.
· What happens to delay if you increase load capacitance?
· Give the various techniques you know to minimize power consumption for CMOS logic?
· What happens when the PMOS and NMOS are interchanged with one another in an inverter?
· What is body effect?
· Why is NAND gate preferred over NOR gate for fabrication?
· What is Noise Margin? Explain the procedure to determine Noise Margin
· Explain sizing of the inverter?
· How do you size NMOS and PMOS transistors to increase the threshold voltage?

- What happens to delay if we include a resistance at the output of a CMOS circuit?
- What are the limitations in increasing the power supply to reduce delay?
- How does Resistance of the metal lines vary with increasing thickness and increasing length?
- What is Charge Sharing? Explain the Charge Sharing problem while sampling data from a Bus?
- Why do we gradually increase the size of inverters in buffer design? Why not give the output of a circuit to one large inverter?
- Give the expression for CMOS switching power dissipation?
- Why is the substrate in NMOS connected to ground and in PMOS to VDD?
- What is the fundamental difference between a MOSFET and BJT ?
- Which transistor has higher gain- BJT or MOS and why?
- Why PMOS and NMOS are sized equally in a Transmission Gates?
- What is metastability? When/why it will occur? What are the different ways to avoid this?
- Explain zener breakdown and avalanche breakdown?

* What happens if Vds is increased over saturation?

- In the I-V characteristics curve, why is the saturation curve flat or constant?
- What happens if a resistor is added in series with the drain in a CMOS transistor?
- What are the different regions of operation in a CMOS transistor?
- What are the effects of the output characteristics for a change in the beta (β) value?
- What is the effect of body bias?
- What is hot electron effect and how can it be eliminated?
- What is channel length modulation?
- What is the effect of temperature on threshold voltage?
- What is the effect of temperature on mobility?
- What is the effect of gate voltage on mobility?
- What are the different types of scaling?
- What is stage ratio?
- What is charge sharing on a bus?
- What is electron migration and how can it be eliminated?
- Can both PMOS and NMOS transistors pass good 1 and good 0? Explain.
- Why is only NMOS used in pass transistor logic?
- What are the different methodologies used to reduce the charge sharing in dynamic logic?
- What are setup and hold time violations? How can they be eliminated?
- Explain the operation of basic SRAM and DRAM.
- Which ones take more time in SRAM: Read operation or Write operation? Why?
- What is meant by clock race?
- What is meant by single phase and double phase clocking?
- If given a choice between NAND and NOR gates, which one would you pick? Explain.
- Explain the origin of the various capacitances in the CMOS transistor and the physical reasoning behind it.
- Why should the number of CMOS transistors that are connected in series be reduced?
- What is charge sharing between bus and memory element?
- What is crosstalk and how can it be avoided?
- Realize an XOR gate using NAND gate.
- What are the advantages and disadvantages of Bi-CMOS process?
- Draw an XOR gate with using minimum number of transistors and explain the operation.
- What are the critical parameters in a latch and flip-flop?
- What is the significance of sense amplifier in an SRAM?
- Explain Domino logic.
- What are the advantages of depletion mode devices over the enhancement mode devices?
- How can the rise and fall times in an inverter be equated?
- What is meant by leakage current?
- Realize an OR gate using NAND gate.
- Realize an NAND gate using a 2:1 multiplexer.
- Realize an NOR gate using a 2:1 multiplexer.
- Draw the layout of a simple inverter.
- What are the substrates of PMOS and NMOS transistors connected to and explain the results if the connections are interchanged with the other.
- What are repeaters?
- What is tunneling problem?
- What is meant by negative biased instability and how can it be avoided?
- What is Elmore delay algorithm?
- What is meant by metastability?
- What is the effect of Vdd on delay?
- What is the effect of delay, rise and fall times with increase in load capacitance?
- What is the value of mobility of electrons?
- What is value of mobility of holes?
- Give insights of an inverter. Draw Layout. Explain the working.

* Give insights of a 2 input NOR gate. Draw Layout. Explain the working.

- Give insights of a 2 input NAND gate. Draw layout. Explain the working?
- Implement F= not (AB+CD) using CMOS gates.
- What is a pass gate. Explain the working?
- Why do we need both PMOS and NMOS transistors to implement a pass gate?
- What does the above code synthesize to?
- Draw cross section of a PMOS transistor.
- Draw cross section of an NMOS transistor.
- What is a D-latch?
- Implement D flip-flop with a couple of latches?
- Implement a 2 input AND gate using transmission gate?

· Explain various adders and difference between them?
· How can you construct both PMOS and NMOS on a single substrate?
· What happens when the gate oxide is very thin?
· What is SPICE?
· What are the differences between IRSIM and SPICE?
· What are the differences between netlist of HSPICE and Spectre?
· Implement F = AB+C using CMOS gates?
· What is hot electron effect?
· Define threshold voltage?
· List out the factors affecting power consumption on a chip?
· What r the phenomenon which come into play when the devices are scaled to the sub-micron lengths?
· What is clock feed through?
· Implement an Inverter using a single transistor?
· What is Fowler-Nordheim Tunneling?
· Which gate is normally preferred while implementing circuits using CMOS logic, NAND or NOR? Why?
· Draw the Differential Sense Amplifier and explain its working. How to size this circuit?
· What happens if we use an Inverter instead of the Differential Sense Amplifier?
· Draw the SRAM Write Circuitry
· How did you arrive at sizes of transistor in SRAM?
· How does the size of PMOS pull up transistors for bit and bitbar lines affect SRAM's performance?
· What is the critical path in a SRAM?
· Draw the timing diagram for a SRAM Read. What happens if we delay the enabling of Clock signal?
· Give a big picture of the entire SRAM layout showing placements of SRAM cells, row decoders, column decoders, read circuit, write circuit and buffers.
· In a SRAM layout, which metal layers would you prefer for Word Lines and Bit Lines? Why?

Design For Test-DFT

In scan chains if some flip flops are +ve edge triggered and remaining flip flops are -ve edge triggered how it behaves?
Answer:
For designs with both positive and negative clocked flops, the scan insertion tool will always route the scan chain so that the negative clocked flops come before the positive edge flops in the chain. This avoids the need of lockup latch.
For the same clock domain the negedge flops will always capture the data just captured into the posedge flops on the posedge of the clock.
For the multiple clock domains, it all depends upon how the clock trees are balanced. If the clock domains are completely asynchronous, ATPG has to mask the receiving flops.

What you mean by scan chain reordering?
Answer1:
Based on timing and congestion the tool optimally places standard cells. While doing so, if scan chains are detached, it can break the chain ordering (which is done by a scan insertion tool like DFT compiler from Synopsis and can reorder to optimize it.... it maintains the number of flops in a chain.
Answer2:
During placement, the optimization may make the scan chain difficult to route due to congestion. Hence the tool will re-order the chain to reduce congestion.
This sometimes increases hold time problems in the chain. To overcome these buffers may have to be inserted into the scan path. It may not be able to maintain the scan chain length exactly. It cannot swap cell from different clock domains.
Because of scan chain reordering patterns generated earlier is of no use. But this is not a problem as ATPG can be redone by reading the new net list.

what are the differences between SIMULATION and SYNTHESIS

Simulation <= verify your design.

synthesis <= Check for your timing

Simulation is used to verify the functionality of the circuit.. a)Functional Simulation:study of ckt's operation independent of timing parameters and gate delays. b) Timing Simulation :study including estimated delays, verify setup,hold and other timing requirements of devices like flip flops are met.

Synthesis:One of the foremost in back end steps where by synthesizing is nothing but converting VHDL or VERILOG description to a set of primitives(equations as in CPLD) or components(as in FPGA'S)to fit into the target technology.Basically the synthesis tools convert the design description into equations or components

Can u tell me the differences between latches & flipflops?

There are 2 types of circuits:

1. Combinational

2. Sequential

Latches and flipflops both come under the category of "sequential circuits", whose output depends not only on the current inputs, but also on previous inputs and outputs. Difference: Latches are level-sensitive, whereas, FF are edge sensitive. By edge sensitive, I mean O/p changes only when there is a clock transition.(from 1 to 0, or from 0 to 1)

Example: In a flipflop, inputs have arrived on the input lines at time= 2 seconds. But, output won't change immediately. At time = 3 seconds, clock transition takes place. After that, O/P will change.

Flip-flops are of 2 types:

1.Positive edge triggered

2. negative edge triggered

1)fllipflops take twice the nymber of gates as latches

2) so automatically delay is more for flipflops

3)power consumption is also more

latch does not have a clock signal, whereas a flip-flop always does.

What is slack?

The slack is the time delay difference from the expected delay(1/clock) to the actual delay in a particular path.

Slack may be +ve or -ve.

Equivalence between VHDL and C?

There is concept of understanding in C there is structure.Based upon requirement structure provide facility to store collection of different data types.

In VHDL we have direct access to memory so instead of using pointer in C (and member of structure) we can write interface store data in memory and access it.

RTL and Behavioral

Register transfer language means there should be data flow between two registers and logic is in between them for end registers data should flow.

Behavioral means how hardware behave determine the exact way it works we write using HDL syntax.For complex projects it is better mixed approach or more behavioral is used.

VHDL QUESTIONS

1. What is the difference between using direct instantiations and component ones except that you need to declare the component?
2. What is the use of BLOCKS?
3. What is the use of PROCEDURES?
4. What is the usage of using more then one architecture in an entity?
5. What is a D-latch? Write the VHDL Code for it?
6. Implement D flip-flop with a couple of latches? Write a VHDL Code for a D flip-flop?
7. Differences between Signals and Variables in VHDL? If the same code is written using Signals and Variables what does it synthesize to?
8. Differences between functions and Procedures in VHDL?
9. Explain the concept of a Clock Divider Circuit? Write a VHDL code for the same?

Digital Design interview questions:

1. Give two ways of converting a two input NAND gate to an inverter
2. Given a circuit, draw its exact timing response. (I was given a Pseudo Random Signal Generator; you can expect any sequential ckt)
3. What are set up time & hold time constraints? What do they signify? Which one is critical for estimating maximum clock frequency of a circuit?
4. Give a circuit to divide frequency of clock cycle by two
5. Design a divide-by-3 sequential circuit with 50% duty circle. (Hint: Double the Clock)
6. Suppose you have a combinational circuit between two registers driven by a clock. What will you do if the delay of the combinational circuit is greater than your clock signal? (You can't resize the combinational circuit transistors)
7. The answer to the above question is breaking the combinational circuit and pipelining it. What will be affected if you do this?
8. What are the different Adder circuits you studied?
9. Give the truth table for a Half Adder. Give a gate level implementation of the same.
10. Draw a Transmission Gate-based D-Latch.
11. Design a Transmission Gate based XOR. Now, how do you convert it to XNOR? (Without inverting the output)
12. How do you detect if two 8-bit signals are same?
13. How do you detect a sequence of "1101" arriving serially from a signal line?
14. Design any FSM in VHDL or Verilog.

Intel interview questions

The following questions are used for screening the candidates during the first interview. The questions apply mostly to fresh college grads pursuing an engineering career at Intel.

1. Have you studied buses? What types?

2. Have you studied pipelining? List the 5 stages of a 5 stage pipeline. Assuming 1 clock per stage, what is the latency of an instruction in a 5 stage machine? What is the throughput of this machine ?

3. How many bit combinations are there in a byte?

4. For a single computer processor computer system, what is the purpose of a processor cache and describe its operation?

5. Explain the operation considering a two processor computer system with a cache for each processor.

6. What are the main issues associated with multiprocessor caches and how might you solve them?

7. Explain the difference between write through and write back cache.

8. Are you familiar with the term MESI?
9. Are you familiar with the term snooping?
10. Describe a finite state machine that will detect three consecutive coin tosses (of one coin) that results in heads.
11. In what cases do you need to double clock a signal before presenting it to a synchronous state machine?
12. You have a driver that drives a long signal & connects to an input device. At the input device there is either overshoot, undershoot or signal threshold violations, what can be done to correct this problem?
13. What are the total number of lines written by you in C/C++? What is the most complicated/valuable program written in C/C++?
14. What compiler was used?
15. What is the difference between = and == in C?
16. Are you familiar with VHDL and/or Verilog?
17. What types of CMOS memories have you designed? What were their size? Speed?
18. What work have you done on full chip Clock and Power distribution? What process technology and budgets were used?
19. What types of I/O have you designed? What were their size? Speed? Configuration? Voltage requirements?
20. Process technology? What package was used and how did you model the package/system? What parasitic effects were considered?
21. What types of high speed CMOS circuits have you designed?
22. What transistor level design tools are you proficient with? What types of designs were they used on?
23. What products have you designed which have entered high volume production?
24. What was your role in the silicon evaluation/product ramp? What tools did you use?
25. If not into production, how far did you follow the design and why did not you see it into production?

VLSI Design Interview questions

1. Explain why & how a MOSFET works
2. Draw Vds-Ids curve for a MOSFET. Now, show how this curve changes (a) with increasing Vgs (b) with increasing transistor width (c) considering Channel Length Modulation
3. Explain the various MOSFET Capacitances & their significance
4. Draw a CMOS Inverter. Explain its transfer characteristics
5. Explain sizing of the inverter
6. How do you size NMOS and PMOS transistors to increase the threshold voltage?
7. What is Noise Margin? Explain the procedure to determine Noise Margin
8. Give the expression for CMOS switching power dissipation
9. What is Body Effect?
10. Describe the various effects of scaling
11. Give the expression for calculating Delay in CMOS circuit
12. What happens to delay if you increase load capacitance?
13. What happens to delay if we include a resistance at the output of a CMOS circuit?
14. What are the limitations in increasing the power supply to reduce delay?
15. How does Resistance of the metal lines vary with increasing thickness and increasing length?
16. You have three adjacent parallel metal lines. Two out of phase signals pass through the outer two metal lines. Draw the waveforms in the center metal line due to interference. Now, draw the signals if the signals in outer metal lines are in phase with each other
17. What happens if we increase the number of contacts or via from one metal layer to the next?
18. Draw a transistor level two input NAND gate. Explain its sizing (a) considering Vth (b) for equal rise and fall times
19. Let A & B be two inputs of the NAND gate. Say signal A arrives at the NAND gate later than signal B. To optimize delay, of the two series NMOS inputs A & B, which one would you place near the output?
20. Draw the stick diagram of a NOR gate. Optimize it
21. For CMOS logic, give the various techniques you know to minimize power consumption
22. What is Charge Sharing? Explain the Charge Sharing problem while sampling data from a Bus
23. Why do we gradually increase the size of inverters in buffer design? Why not give the output of a circuit to one large inverter?
24. In the design of a large inverter, why do we prefer to connect small transistors in parallel (thus increasing effective width) rather than lay out one transistor with large width?
25. Given a layout, draw its transistor level circuit. (I was given a 3 input AND gate and a 2 input Multiplexer. You can expect any simple 2 or 3 input gates)
26. Give the logic expression for an AOI gate. Draw its transistor level equivalent. Draw its stick diagram
27. Why don't we use just one NMOS or PMOS transistor as a transmission gate?
28. For a NMOS transistor acting as a pass transistor, say the gate is connected to VDD, give the output for a square pulse input going from 0 to VDD

29. Draw a 6-T SRAM Cell and explain the Read and Write operations
30. Draw the Differential Sense Amplifier and explain its working. Any idea how to size this circuit? (Consider Channel Length Modulation)
31. What happens if we use an Inverter instead of the Differential Sense Amplifier?
32. Draw the SRAM Write Circuitry
33. Approximately, what were the sizes of your transistors in the SRAM cell? How did you arrive at those sizes?
34. How does the size of PMOS Pull Up transistors (for bit & bit- lines) affect SRAM's performance?
35. What's the critical path in a SRAM?
36. Draw the timing diagram for a SRAM Read. What happens if we delay the enabling of Clock signal?
37. Give a big picture of the entire SRAM Layout showing your placements of SRAM Cells, Row Decoders, Column Decoders, Read Circuit, Write Circuit and Buffers
38. In a SRAM layout, which metal layers would you prefer for Word Lines and Bit Lines? Why?
39. How can you model a SRAM at RTL Level?
40. Whatï¿½s the difference between Testing & Verification?
41. For an AND-OR implementation of a two input Mux, how do you test for Stuck-At-0 and Stuck-At-1 faults at the internal nodes? (You can expect a circuit with some redundant logic)
42. What is Latch Up? Explain Latch Up with cross section of a CMOS Inverter. How do you avoid Latch Up?

Verilog Interview Questions

- What is the difference between $display and $monitor and $write and $strobe?
- What is the difference between code-compiled simulator and normal simulator?
- What is the difference between wire and reg?
- What is the difference between blocking and non-blocking assignments?
- What is the significance Timescale directivbe?
- What is the difference between bit wise, unary and logical operators?
- What is the difference between task and function?
- What is the difference between casex, casez and case statements?
- Which one preferred-casex or casez?
- For what is defparam used?
- What is the difference between "= =" and "= = =" ?
- What is a compiler directive like 'include' and 'ifdef'?
- Write a verilog code to swap contents of two registers with and without a temporary register?
- What is the difference between inter statement and intra statement delay?
- What is delta simulation time?
- What is difference between Verilog full case and parallel case?
- What you mean by inferring latches?
- How to avoid latches in your design?
- Why latches are not preferred in synthesized design?
- How blocking and non blocking statements get executed?
- Which will be updated first: is it variable or signal?
- What is sensitivity list?
- If you miss sensitivity list what happens?
- In a pure combinational circuit is it necessary to mention all the inputs in sensitivity disk? If yes, why? If not, why?
- In a pure sequential circuit is it necessary to mention all the inputs in sensitivity disk? If yes, why? If not, why?
- What is general structure of Verilog code you follow?
- What are the difference between Verilog and VHDL?
- What are system tasks?
- List some of system tasks and what are their purposes?
- What are the enhancements in Verilog 2001?
- Write a Verilog code for synchronous and asynchronous reset?
- What is pli? why is it used?
- What is file I/O?
- What is difference between freeze deposit and force?
- Will case always infer priority register? If yes how? Give an example.
- What are inertial and transport delays ?
- What does `timescale 1 ns/ 1 ps' signify in a verilog code?
- How to generate sine wav using verilog coding style?
- How do you implement the bi-directional ports in Verilog HDL?
- How to write FSM is verilog?
- What is verilog case (1)?
- What are Different types of Verilog simulators available?
- What is Constrained-Random Verification ?
- How can you model a SRAM at RTL Level?

Physical Design Questions and Answers

- I am getting several emails requesting answers to the questions posted in this blog. But it is very difficult to provide detailed answer to all questions in my available spare time. Hence i decided to give "short and sweet" one line answers to the questions so that readers can immediately benefited. Detailed answers will be posted in later stage.I have given answers to some of the physical design questions here. Enjoy !

What parameters (or aspects) differentiate Chip Design and Block level design?

- Chip design has I/O pads; block design has pins.
- Chip design uses all metal layes available; block design may not use all metal layers.
- Chip is generally rectangular in shape; blocks can be rectangular, rectilinear.
- Chip design requires several packaging; block design ends in a macro.

How do you place macros in a full chip design?

- First check flylines i.e. check net connections from macro to macro and macro to standard cells.
- If there is more connection from macro to macro place those macros nearer to each other preferably nearer to core boundaries.
- If input pin is connected to macro better to place nearer to that pin or pad.
- If macro has more connection to standard cells spread the macros inside core.
- Avoid criscross placement of macros.
- Use soft or hard blockages to guide placement engine.

Differentiate between a Hierarchical Design and flat design?

- Hierarchial design has blocks, subblocks in an hierarchy; Flattened design has no subblocks and it has only leaf cells.
- Hierarchical design takes more run time; Flattened design takes less run time.

Which is more complicated when u have a 48 MHz and 500 MHz clock design?

- 500 MHz; because it is more constrained (i.e.lesser clock period) than 48 MHz design.

Name few tools which you used for physical verification?

- Herculis from Synopsys, Caliber from Mentor Graphics.

What are the input files will you give for primetime correlation?

- Netlist, Technology library, Constraints, SPEF or SDF file.

If the routing congestion exists between two macros, then what will you do?

- Provide soft or hard blockage

How will you decide the die size?

- By checking the total area of the design you can decide die size.

If lengthy metal layer is connected to diffusion and poly, then which one will affect by antenna problem?

- Poly

If the full chip design is routed by 7 layer metal, why macros are designed using 5LM instead of using 7LM?

- Because top two metal layers are required for global routing in chip design. If top metal layers are also used in block level it will create routing blockage.

In your project what is die size, number of metal layers, technology, foundry, number of clocks?

- Die size: tell in mm eg. 1mm x 1mm ; remeber 1mm=1000micron which is a big size !!
- Metal layers: See your tech file. generally for 90nm it is 7 to 9.
- Technology: Again look into tech files.
- Foundry:Again look into tech files; eg. TSMC, IBM, ARTISAN etc
- Clocks: Look into your design and SDC file !

How many macros in your design?

- You know it well as you have designed it ! A SoC (System On Chip) design may have 100 macros also !!!!

What is each macro size and number of standard cell count?

- Depends on your design.

What are the input needs for your design?

- For synthesis: RTL, Technology library, Standard cell library, Constraints
- For Physical design: Netlist, Technology library, Constraints, Standard cell library

What is SDC constraint file contains?

- Clock definitions
- Timing exception-multicycle path, false path
- Input and Output delays

How did you do power planning?
How to calculate core ring width, macro ring width and strap or trunk width?
How to find number of power pad and IO power pads?
How the width of metal and number of straps calculated for power and ground?

- Get the total core power consumption; get the metal layer current density value from the tech file; Divide total power by number sides of the chip; Divide the obtained value from the current density to get core power ring width. Then calculate number of straps using some more equations. Will be explained in detail later.

How to find total chip power?

- Total chip power=standard cell power consumption,Macro power consumption pad power consumption.

What are the problems faced related to timing?

- Prelayout: Setup, Max transition, max capacitance
- Post layout: Hold

How did you resolve the setup and hold problem?

- Setup: upsize the cells
- Hold: insert buffers

In which layer do you prefer for clock routing and why?

- Next lower layer to the top two metal layers(global routing layers). Because it has less resistance hence less RC delay.

If in your design has reset pin, then it'll affect input pin or output pin or both?

- Output pin.

During power analysis, if you are facing IR drop problem, then how did you avoid?

- Increase power metal layer width.
- Go for higher metal layer.
- Spread macros or standard cells.
- Provide more straps.

Define antenna problem and how did you resolve these problem?

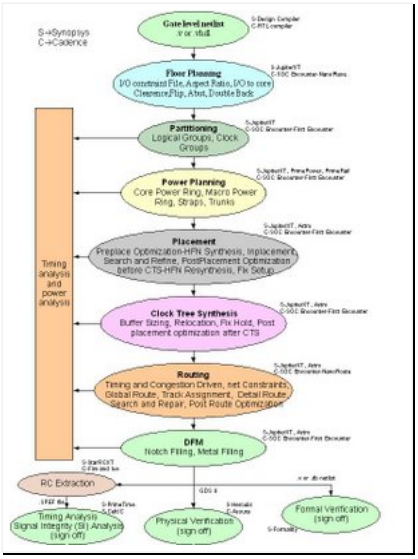
- Increased net length can accumulate more charges while manufacturing of the device due to ionisation process. If this net is connected to gate of the MOSFET it can damage dielectric property of the gate and gate may conduct causing damage to the MOSFET. This is antenna problem.
- Decrease the length of the net by providing more vias and layer jumping.
- Insert antenna diode.

How delays vary with different PVT conditions? Show the graph.

- P increase->dealy increase
- P decrease->delay decrease
- V increase->delay decrease
- V decrease->delay increase
- T increase->delay increase
- T decrease->delay decrease

Explain the flow of physical design and inputs and outputs for each step in flow.

Physical Design Flow



The physical design flow is generally explained in the Figure (1.). In each section of the flow EDA tools available from the two main EDA companies-Synopsys and Cadence is also listed. In each and every step of the flow timing and power analysis can be carried out. If timing and power requirements are not met then either the whole flow has to be re-exercised or going back one or two steps and optimizing the design or incremental optimization may meet the requirements

What is cell delay and net delay?

- **Gate delay**
- Transistors within a gate take a finite time to switch. This means that a change on the input of a gate takes a finite time to cause a change on the output.[Magma]
- Gate delay =function of(i/p transition time, Cnet+Cpin).

- Cell delay is also same as Gate delay.
- **Cell delay**
- For any gate it is measured between 50% of input transition to the corresponding 50% of output transition.
- Intrinsic delay
- Intrinsic delay is the delay internal to the gate. Input pin of the cell to output pin of the cell.
- It is defined as the delay between an input and output pair of a cell, when a near zero slew is applied to the input pin and the output does not see any load condition.It is predominantly caused by the internal capacitance associated with its transistor.
- This delay is largely independent of the size of the transistors forming the gate because increasing size of transistors increase internal capacitors.
- **Net Delay (or wire delay)**
- The difference between the time a signal is first applied to the net and the time it reaches other devices connected to that net.
- It is due to the finite resistance and capacitance of the net.It is also known as wire delay.
- Wire delay =fn(Rnet , Cnet+Cpin)

What are delay models and what is the difference between them?

- Linear Delay Model (LDM)
- Non Linear Delay Model (NLDM)

What is wire load model?

- Wire load model is NLDM which has estimated R and C of the net.

Why higher metal layers are preferred for Vdd and Vss?

- Because it has less resistance and hence leads to less IR drop.

What is logic optimization and give some methods of logic optimization.

- Upsizing
- Downsizing
- Buffer insertion
- Buffer relocation
- Dummy buffer placement

What is the significance of negative slack?

- negative slack==> there is setup voilation==> deisgn can fail

What is signal integrity? How it affects Timing?

- IR drop, Electro Migration (EM), Crosstalk, Ground bounce are signal integrity issues.
- If Idrop is more==>delay increases.
- crosstalk==>there can be setup as well as hold voilation.

What is IR drop? How to avoid? How it affects timing?

- There is a resistance associated with each metal layer. This resistance consumes power causing voltage drop i.e.IR drop.
- If IR drop is more==>delay increases.

What is EM and it effects?

- Due to high current flow in the metal atoms of the metal can displaced from its origial place. When it happens in larger amount the metal can open or bulging of metal layer can happen. This effect is known as Electro Migration.
- Affects: Either short or open of the signal line or power line.

What are types of routing?

- Global Routing
- Track Assignment
- Detail Routing

What is latency? Give the types?

- **Source Latency**
- It is known as source latency also. It is defined as "the delay from the clock origin point to the clock definition point in the design".
- Delay from clock source to beginning of clock tree (i.e. clock definition point).
- The time a clock signal takes to propagate from its ideal waveform origin point to the clock definition point in the design.
- **Network latency**
- It is also known as Insertion delay or Network latency. It is defined as "the delay from the clock definition point to the clock pin of the register".
- The time clock signal (rise or fall) takes to propagate from the clock definition point to a register clock pin.

What is track assignment?

- Second stage of the routing wherein particular metal tracks (or layers) are assigned to the signal nets.

What is congestion?

- If the number of routing tracks available for routing is less than the required tracks then it is known as congestion.

Whether congestion is related to placement or routing?

- Routing

What are clock trees?

- Distribution of clock from the clock source to the sync pin of the registers.

What are clock tree types?

- H tree, Balanced tree, X tree, Clustering tree, Fish bone

What is cloning and buffering?

- Cloning is a method of optimization that decreases the load of a heavily loaded cell by replicating the cell.
- Buffering is a method of optimization that is used to insert beffers in high fanout nets to decrease the dealy.

ASIC

Different Types of Delays in ASIC or VLSI design

- **Source Delay/Latency**
- **Network Delay/Latency**
- **Insertion Delay**
- **Transition Delay/Slew: Rise time, fall time**
- **Path Delay**
- **Net delay, wire delay, interconnect delay**
- **Propagation Delay**
- **Phase Delay**
- **Cell Delay**
- **Intrinsic Delay**
- **Extrinsic Delay**
- **Input Delay**
- **Output Delay**
- **Exit Delay**
- **Latency (Pre/post CTS)**
- **Uncertainty (Pre/Post CTS)**
- **Unateness: Positive unateness, negative unateness**
- **Jitter: PLL jitter, clock jitter**

Gate delay

- Transistors within a gate take a finite time to switch. This means that a change on the input of a gate takes a finite time to cause a change on the output.[Magma]
- Gate delay =function of(i/p transition time, Cnet+Cpin).
- Cell delay is also same as Gate delay.

Source Delay (or Source Latency)

- It is known as source latency also. It is defined as "the delay from the clock origin point to the clock definition point in the design".
- Delay from clock source to beginning of clock tree (i.e. clock definition point).
- The time a clock signal takes to propagate from its ideal waveform origin point to the clock definition point in the design.

Network Delay(latency)

- It is also known as Insertion delay or Network latency. It is defined as "the delay from the clock definition point to the clock pin of the register".
- The time clock signal (rise or fall) takes to propagate from the clock definition point to a register clock pin.

Insertion delay

- The delay from the clock definition point to the clock pin of the register.

Transition delay

- It is also known as "Slew". It is defined as the time taken to change the state of the signal. Time taken for the transition from logic 0 to logic 1 and vice versa . or Time taken by the input signal to rise from 10%(20%) to the 90% (80%) and vice versa.
- Transition is the time it takes for the pin to change state.

Slew

- Rate of change of logic.See Transition delay.
- Slew rate is the speed of transition measured in volt / ns.

Rise Time

- Rise time is the difference between the time when the signal crosses a low threshold to the time when the signal crosses the high threshold. It can be absolute or percent.
- Low and high thresholds are fixed voltage levels around the mid voltage level or it can be either 10% and 90% respectively or 20% and 80% respectively. The percent levels are converted to absolute voltage levels at the time of measurement by calculating percentages from the difference between the starting voltage level and the final settled voltage level.

Fall Time

- Fall time is the difference between the time when the signal crosses a high threshold to the time when the signal crosses the low threshold.
- The low and high thresholds are fixed voltage levels around the mid voltage level or it can be either 10% and 90% respectively or 20% and 80% respectively. The percent levels are converted to absolute voltage levels at the time of measurement by calculating percentages from the difference between the starting voltage level and the final settled voltage level.
- For an ideal square wave with 50% duty cycle, the rise time will be 0.For a symmetric triangular wave, this is reduced to just 50%.
- The rise/fall definition is set on the meter to 10% and 90% based on the linear power in Watts. These points translate into the -10 dB and -0.5 dB points in log mode (10 log 0.1) and (10 log 0.9). The rise/fall time values of 10% and 90% are calculated based on an algorithm, which looks at the mean power above and below the 50% points of the rise/fall times
- **Path delay**
- Path delay is also known as pin to pin delay. It is the delay from the input pin of the cell to the output pin of the cell.

Net Delay (or wire delay)

- The difference between the time a signal is first applied to the net and the time it reaches other devices connected to that net.
- It is due to the finite resistance and capacitance of the net.It is also known as wire delay.
- Wire delay =fn(Rnet , Cnet+Cpin)

Propagation delay

- For any gate it is measured between 50% of input transition to the corresponding 50% of output transition.
- This is the time required for a signal to propagate through a gate or net. For gates it is the time it takes for a event at the gate input to affect the gate output.
- For net it is the delay between the time a signal is first applied to the net and the time it reaches other devices connected to that net.
- It is taken as the average of rise time and fall time i.e. Tpd= (Tphl+Tplh)/2.

Phase delay

- Same as insertion delay

Cell delay

- For any gate it is measured between 50% of input transition to the corresponding 50% of output transition.

Intrinsic delay

- Intrinsic delay is the delay internal to the gate. Input pin of the cell to output pin of the cell.
- It is defined as the delay between an input and output pair of a cell, when a near zero slew is applied to the input pin and the output does not see any load condition.It is predominantly caused by the internal capacitance associated with its transistor.
- This delay is largely independent of the size of the transistors forming the gate because increasing size of transistors increase internal capacitors.

Extrinsic delay

- Same as wire delay, net delay, interconnect delay, flight time.
- Extrinsic delay is the delay effect that associated to with interconnect. output pin of the cell to the input pin of the next cell.

Input delay

- Input delay is the time at which the data arrives at the input pin of the block from external circuit with respect to reference clock.

Output delay

- Output delay is time required by the external circuit before which the data has to arrive at the output pin of the block with respect to reference clock.

Exit delay

- It is defined as the delay in the longest path (critical path) between clock pad input and an output. It determines the maximum operating frequency of the design.

Latency (pre/post cts)

- Latency is the summation of the Source latency and the Network latency. Pre CTS estimated latency will be considered during the synthesis and after CTS propagated latency is considered.

Uncertainty (pre/post cts)

- Uncertainty is the amount of skew and the variation in the arrival clock edge. Pre CTS uncertainty is clock skew and

clock Jitter. After CTS we can have some margin of skew + Jitter.

Unateness

- A function is said to be unate if the rise transition on the positive unate input variable causes the ouput to rise or no change and vice versa.
- Negative unateness means cell output logic is inverted version of input logic. eg. In inverter having input A and output Y, Y is -ve unate w.r.to A. Positive unate means cell output logic is same as that of input.
- These +ve ad -ve unateness are constraints defined in library file and are defined for output pin w.r.to some input pin.
- A clock signal is positive unate if a rising edge at the clock source can only cause a rising edge at the register clock pin, and a falling edge at the clock source can only cause a falling edge at the register clock pin.
- A clock signal is negative unate if a rising edge at the clock source can only cause a falling edge at the register clock pin, and a falling edge at the clock source can only cause a rising edge at the register clock pin. In other words, the clock signal is inverted.
- A clock signal is not unate if the clock sense is ambiguous as a result of non-unate timing arcs in the clock path. For example, a clock that passes through an XOR gate is not unate because there are nonunate arcs in the gate. The clock sense could be either positive or negative, depending on the state of the other input to the XOR gate.

Jitter

- The short-term variations of a signal with respect to its ideal position in time.
- Jitter is the variation of the clock period from edge to edge. It can varry +/- jitter value.
- From cycle to cycle the period and duty cycle can change slightly due to the clock generation circuitry. This can be modeled by adding uncertainty regions around the rising and falling edges of the clock waveform.

Sources of Jitter Common sources of jitter include:

- Internal circuitry of the phase-locked loop (PLL)
- Random thermal noise from a crystal
- Other resonating devices
- Random mechanical noise from crystal vibration
- Signal transmitters
- Traces and cables
- Connectors
- Receivers

Skew

- The difference in the arrival of clock signal at the clock pin of different flops.
- Two types of skews are defined: Local skew and Global skew.

Local skew

- The difference in the arrival of clock signal at the clock pin of related flops.

Global skew

- The difference in the arrival of clock signal at the clock pin of non related flops.
- Skew can be positive or negative.
- When data and clock are routed in same direction then it is **Positive skew.**
- When data and clock are routed in opposite then it is **negative skew.**

Recovery Time

- Recovery specifies the minimum time that an asynchronous control input pin must be held stable after being de-asserted and before the next clock (active-edge) transition.
- Recovery time specifies the time the inactive edge of the asynchronous signal has to arrive before the closing edge of the clock.
- Recovery time is the minimum length of time an asynchronous control signal (eg.preset) must be stable before the next active clock edge. The recovery slack time calculation is similar to the clock setup slack time calculation, but it applies asynchronous control signals.

Equation 1:

- Recovery Slack Time = Data Required Time â€“ Data Arrival Time
- Data Arrival Time = Launch Edge + Clock Network Delay to Source Register + Tclkq+ Register to Register Delay
- Data Required Time = Latch Edge + Clock Network Delay to Destination Register =Tsetup

If the asynchronous control is not registered, equations shown in Equation 2 is used to calculate the recovery slack time.
Equation 2:

- Recovery Slack Time = Data Required Time â€“ Data Arrival Time
- Data Arrival Time = Launch Edge + Maximum Input Delay + Port to Register Delay
- Data Required Time = Latch Edge + Clock Network Delay to Destination Register Delay+Tsetup
- If the asynchronous reset signal is from a port (device I/O), you must make an Input Maximum Delay assignment to the asynchronous reset pin to perform recovery analysis on that path.

Removal Time

- Removal specifies the minimum time that an asynchronous control input pin must be held stable before being de-asserted and after the previous clock (active-edge) transition.

- Removal time specifies the length of time the active phase of the asynchronous signal has to be held after the closing edge of clock.
- Removal time is the minimum length of time an asynchronous control signal must be stable after the active clock edge. Calculation is similar to the clock hold slack calculation, but it applies asynchronous control signals. If the asynchronous control is registered, equations shown in Equation 3 is used to calculate the removal slack time.
- If the recovery or removal minimum time requirement is violated, the output of the sequential cell becomes uncertain. The uncertainty can be caused by the value set by the resetbar signal or the value clocked into the sequential cell from the data input.

Equation 3

- Removal Slack Time = Data Arrival Time â€" Data Required Time
- Data Arrival Time = Launch Edge + Clock Network Delay to Source Register + Tclkq of Source Register + Register to Register Delay
- Data Required Time = Latch Edge + Clock Network Delay to Destination Register + Thold
- If the asynchronous control is not registered, equations shown in Equation 4 is used to calculate the removal slack time.

Equation 4

- Removal Slack Time = Data Arrival Time â€" Data Required Time
- Data Arrival Time = Launch Edge + Input Minimum Delay of Pin + Minimum Pin to Register Delay
- Data Required Time = Latch Edge + Clock Network Delay to Destination Register +Thold
- If the asynchronous reset signal is from a device pin, you must specify the Input Minimum Delay constraint to the asynchronous reset pin to perform a removal analysis on this path.

What is the difference between soft macro and hard macro?

- **What is the difference between hard macro, firm macro and soft macro?**

or

- **What are IPs?**
- Hard macro, firm macro and soft macro are all known as IP (Intellectual property). They are optimized for power, area and performance. They can be purchased and used in your ASIC or FPGA design implementation flow. Soft macro is flexible for all type of ASIC implementation. Hard macro can be used in pure ASIC design flow, not in FPGA flow. Before bying any IP it is very important to evaluate its advantages and disadvantages over each other, hardware compatibility such as I/O standards with your design blocks, reusability for other designs.

Soft macros

- Soft macros are in synthesizable RTL.
- Soft macros are more flexible than firm or hard macros.
- Soft macros are not specific to any manufacturing process.
- Soft macros have the disadvantage of being somewhat unpredictable in terms of performance, timing, area, or power.
- Soft macros carry greater IP protection risks because RTL source code is more portable and therefore, less easily protected than either a netlist or physical layout data.
- From the physical design perspective, soft macro is any cell that has been placed and routed in a placement and routing tool such as Astro. (This is the definition given in Astro Rail user manual !)
- Soft macros are editable and can contain standard cells, hard macros, or other soft macros.

Firm macros

- Firm macros are in netlist format.
- Firm macros are optimized for performance/area/power using a specific fabrication technology.
- Firm macros are more flexible and portable than hard macros.
- Firm macros are predictive of performance and area than soft macros.

Hard macro

- Hard macros are generally in the form of hardware IPs (or we termed it as hardwre IPs !).
- Hard macos are targeted for specific IC manufacturing technology.
- Hard macros are block level designs which are silicon tested and proved.
- Hard macros have been optimized for power or area or timing.
- In physical design you can only access pins of hard macros unlike soft macros which allows us to manipulate in different way.
- You have freedom to move, rotate, flip but you can't touch anything inside hard macros.
- Very common example of hard macro is memory. It can be any design which carries dedicated single functionality (in general).. for example it can be a MP4 decoder.
- Be aware of features and characteristics of hard macro before you use it in your design... other than power, timing and area you also should know pin properties like sync pin, I/O standards etc
- LEF, GDS2 file format allows easy usage of macros in different tools.

From the physical design (backend) perspective:

- Hard macro is a block that is generated in a methodology other than place and route (i.e. using full custom design methodology) and is brought into the physical design database (eg. Milkyway in Synopsys; Volcano in Magma) as a GDS2 file.

What is the difference between FPGA and CPLD?

FPGA-Field Programmable Gate Array and CPLD-Complex Programmable Logic Device-- both are programmable logic devices made by the same companies with different characteristics.

"A Complex Programmable Logic Device (CPLD) is a Programmable Logic Device with complexity between that of PALs (Programmable Array Logic) and FPGAs, and architectural features of both. The building block of a CPLD is the macro cell, which contains logic implementing disjunctive normal form expressions and more specialized logic operations".

Architecture

Granularity is the biggest difference between CPLD and FPGA.

FPGA are "fine-grain" devices. That means that they contain hundreds of (up to 100000) of tiny blocks (called as LUT or CLBs etc) of logic with flip-flops, combinational logic and memories.FPGAs offer much higher complexity, up to 150,000 flip-flops and large number of gates available.

CPLDs typically have the equivalent of thousands of logic gates, allowing implementation of moderately complicated data processing devices. PALs typically have a few hundred gate equivalents at most, while FPGAs typically range from tens of thousands to several million.

CPLD are "coarse-grain" devices. They contain relatively few (a few 100's max) large blocks of logic with flip-flops and combinational logic. CPLDs based on AND-OR structure.

CPLD's have a register with associated logic (AND/OR matrix). CPLD's are mostly implemented in control applications and FPGA's in datapath applications. Because of this course grained architecture, the timing is very fixed in CPLDs.

- FPGA are RAM based. They need to be "downloaded" (configured) at each power-up. CPLD are EEPROM based. They are active at power-up i.e. as long as they've been programmed at least once.

FPGA needs boot ROM but CPLD does not. In some systems you might not have enough time to boot up FPGA then you need CPLD+FPGA.

Generally, the CPLD devices are not volatile, because they contain flash or erasable ROM memory in all the cases. The FPGA are volatile in many cases and hence they need a configuration memory for working. There are some FPGAs now which are nonvolatile. This distinction is rapidly becoming less relevant, as several of the latest FPGA products also offer models with embedded configuration memory.

- The characteristic of non-volatility makes the CPLD the device of choice in modern digital designs to perform 'boot loader' functions before handing over control to other devices not having this capability. A good example is where a CPLD is used to load configuration data for an FPGA from non-volatile memory.
- Because of coarse-grain architecture, one block of logic can hold a big equation and hence CPLD have a faster input-to-output timings than FPGA.

Features

- FPGA have special routing resources to implement binary counters,arithmetic functions like adders, comparators and RAM. CPLD don't have special features like this.
- FPGA can contain very large digital designs, while CPLD can contain small designs only.The limited complexity (<500>
- ***Speed***: CPLDs offer a single-chip solution with fast pin-to-pin delays, even for wide input functions. Use CPLDs for small designs, where "instant-on", fast and wide decoding, ultra-low idle power consumption, and design security are important (e.g., in battery-operated equipment).
- ***Security***: In CPLD once programmed, the design can be locked and thus made secure. Since the configuration bitstream must be reloaded every time power is re-applied, design security in FPGA is an issue.
- ***Power***: The high static (idle) power consumption prohibits use of CPLD in battery-operated equipment. FPGA idle power consumption is reasonably low, although it is sharply increasing in the newest families.
- ***Design flexibility***: FPGAs offer more logic flexibility and more sophisticated system features than CPLDs: clock management, on-chip RAM, DSP functions, (multipliers), and even on-chip microprocessors and Multi-Gigabit Transceivers.These benefits and opportunities of dynamic reconfiguration, even in the end-user system, are an important advantage.
- Use FPGAs for larger and more complex designs.
- FPGA is suited for timing circuit becauce they have more registers , but CPLD is suited for control circuit because they have more combinational circuit. At the same time, If you synthesis the same code for FPGA for many times, you will find out that each timing report is different. But it is different in CPLD synthesis, you can get the same result.

As CPLDs and FPGAs become more advanced the differences between the two device types will continue to blur. While this trend may appear to make the two types more difficult to keep apart, the architectural advantage of CPLDs combining low cost, non-volatile configuration, and macro cells with predictable timing characteristics will likely be sufficient to maintain a product differentiation for the foreseeable future.

What is the difference between FPGA and ASIC?

This question is very popular in VLSI fresher interviews. It looks simple but a deeper insight into the subject reveals the fact that there are lot of thinks to be understood !! So here is the answer.

FPGA vs. ASIC

- Difference between ASICs and FPGAs mainly depends on costs, tool availability, performance and design flexibility. They have their own pros and cons but it is designers responsibility to find the advantages of the each and use either FPGA or ASIC for the product. However, recent developments in the FPGA domain are narrowing down the benefits of the ASICs.

FPGA

- **Field Programable Gate Arrays**

FPGA Design Advantages

- ***Faster time-to-market:*** No layout, masks or other manufacturing steps are needed for FPGA design. Readymade FPGA is available and burn your HDL code to FPGA ! Done !!
- **No NRE (Non Recurring Expenses):** This cost is typically associated with an ASIC design. For FPGA this is not there. FPGA tools are cheap. (sometimes its free ! You need to buy FPGA.... thats all !). ASIC youpay huge NRE and tools are expensive. I would say "very expensive"...Its in crores....!!

- ***Simpler design cycle:*** This is due to software that handles much of the routing, placement, and timing. Manual intervention is less.The FPGA design flow eliminates the complex and time-consuming floorplanning, place and route, timing analysis.
- ***More predictable project cycle:*** The FPGA design flow eliminates potential re-spins, wafer capacities, etc of the project since the design logic is already synthesized and verified in FPGA device.
- ***Field Reprogramability:*** A new bitstream (i.e. your program) can be uploaded remotely, instantly. FPGA can be reprogrammed in a snap while an ASIC can take $50,000 and more than 4-6 weeks to make the same changes. FPGA costs start from a couple of dollars to several hundreds or more depending on the hardware features.
- ***Reusability:*** Reusability of FPGA is the main advantage. Prototype of the design can be implemented on FPGA which could be verified for almost accurate results so that it can be implemented on an ASIC. Ifdesign has faults change the HDL code, generate bit stream, program to FPGA and test again.Modern FPGAs are reconfigurable both partially and dynamically.

FPGAs are good for prototyping and limited production.If you are going to make 100-200 boards it isn't worth to make an ASIC.

Generally FPGAs are used for lower speed, lower complexity and lower volume designs.But today's FPGAs even run at 500 MHz with superior performance. With unprecedented logic density increases and a host of other features, such as embedded processors, DSP blocks, clocking, and high-speed serial at ever lower price, FPGAs are suitable for almost any type of design.

- Unlike ASICs, FPGA's have special hardwares such as Block-RAM, DCM modules, MACs, memories and highspeed I/O, embedded CPU etc inbuilt, which can be used to get better performace. Modern FPGAs are packed with features. Advanced FPGAs usually come with phase-locked loops, low-voltage differential signal, clock data recovery, more internal routing, high speed, hardware multipliers for DSPs, memory,programmable I/O, IP cores and microprocessor cores. Remember Power PC (hardcore) and Microblaze (softcore) in Xilinx and ARM (hardcore) and Nios(softcore) in Altera. There are FPGAs available now with built in ADC ! Using all these features designers can build a system on a chip. Now, dou yo really need an ASIC ?
- FPGA sythesis is much more easier than ASIC.
- In FPGA you need not do floor-planning, tool can do it efficiently. In ASIC you have do it.

FPGA Design Disadvantages

- Powe consumption in FPGA is more. You don't have any control over the power optimization. This is where ASIC wins the race !
- You have to use the resources available in the FPGA. Thus FPGA limits the design size.
- Good for low quantity production. As quantity increases cost per product increases compared to the ASIC implementation.

ASIC

- **Application Specific Intergrated Circiut**

ASIC Design Advantages

- ***Cost....cost....cost....Lower unit costs:*** For very high volume designs costs comes out to be very less. Larger volumes of ASIC design proves to be cheaper than implementing design using FPGA.
- ***Speed...speed...speed....ASICs are faster than FPGA:*** ASIC gives design flexibility. This gives enoromous opportunity for speed optimizations.
- ***Low power....Low power....Low power:*** ASIC can be optimized for required low power. There are several low power techniques such as power gating, clock gating, multi vt cell libraries, pipelining etc are available to achieve the power target. This is where FPGA fails badly !!! Can you think of a cell phone which has to be charged for every call.....never.....low power ASICs helps battery live longer life !!
- In ASIC you can implement analog circuit, mixed signal designs. This is generally not possible in FPGA.
- In ASIC DFT (Design For Test) is inserted. In FPGA DFT is not carried out (rather for FPGA no need of DFT !) .

ASIC Design Diadvantages

- ***Time-to-market:*** Some large ASICs can take a year or more to design. A good way to shorten development time is to make prototypes using FPGAs and then switch to an ASIC.
- ***Design Issues:*** In ASIC you should take care of DFM issues, Signal Integrity isuues and many more. In FPGA you don't have all these because ASIC designer takes care of all these. (Don't forget FPGA isan IC and designed by ASIC design enginner !!)
- ***Expensive Tools:*** ASIC design tools are very much expensive. You spend a huge amount of NRE.

Structured ASICS

- Structured ASICs have the bottom metal layers fixed and only the top layers can be designed by the customer.
- Structured ASICs are custom devices that approach the performance of today's Standard Cell ASIC while dramatically simplifying the design complexity.
- Structured ASICs offer designers a set of devices with specific, customizable metal layers along with predefined metal layers, which can contain the underlying pattern of logic cells, memory, and I/O.

FPGA vs. ASIC Design Flow Comparison



ASIC Design Check List

Silicon Process and Library Characteristics

- What exact process are you using?
- How many layers can be used for this design?
- Are the Cross talk Noise constraints, Xtalk Analysis configuration, Cell EM & Wire EM available?

Design Characteristics

- What is the design application?
- Number of cells (placeable objects)?
- Is the design Verilog or VHDL?
- Is the netlist flat or hierarchical?
- Is there RTL available?
- Is there any datapath logic using special datapath tools?
- Is the DFT to be considered?
- Can scan chains be reordered?
- Is memory BIST, boundary scan used on this design?
- Are static timing analysis constraints available in SDC format?

Clock Characteristics

- How many clock domains are in the design?
- What are the clock frequencies?
- Is there a target clock skew, latency or other clock requirements?
- Does the design have a PLL?
- If so, is it used to remove clock latency?
- Is there any I/O cell in the feedback path?
- Is the PLL used for frequency multipliers?
- Are there derived clocks or complex clock generation circuitry?
- Are there any gated clocks?
- If yes, do they use simple gating elements?
- Is the gate clock used for timing or power?
- For gated clocks, can the gating elements be sized for timing?
- Are you muxing in a test clock or using a JTAG clock?
- Available cells for clock tree?
- Are there any special clock repeaters in the library?
- Are there any EM, slew or capacitance limits on these repeaters?
- How many drive strengths are available in the standard buffers and inverters?
- Do any of the buffers have balanced rise and fall delays?
- Any there special requirements for clock distribution?
- Will the clock tree be shielded? If so, what are the shielding requirements?

Floorplan and Package Characteristics

- Target die area?
- Does the area estimate include power/signal routing?
- What gates/mm2 has been assumed?
- Number of routing layers?
- Any special power routing requirements?
- Number of digital I/O pins/pads?
- Number of analog signal pins/pads?

- Number of power/ground pins/pads?
- Total number of pins/pads and Location?
- Will this chip use a wire bond package?
- Will this chip use a flip-chip package?
- If Yes, is it I/O bump pitch? Rows of bumps? Bump allocation?Bump pad layout guide?
- Have you already done floorplanning for this design?
- If yes, is conformance to the existing floorplan required?
- What is the target die size?
- What is the expected utilization?
- Please draw the overall floorplan ?
- Is there an existing floorplan available in DEF?
- What are the number and type of macros (memory, PLL, etc.)?
- Are there any analog blocks in the design?
- What kind of packaging is used? Flipchip?
- Are the I/Os periphery I/O or area I/O?
- How many I/Os?
- Is the design pad limited?
- Power planning and Power analysis for this design?
- Are layout databases available for hard macros ?
- Timing analysis and correlatio?
- Physical verification ?

Data Input

- Library information for new library
- .lib for timing information
- GDSII or LEF for library cells including any RAMs
- RTL in Verilog/VHDL format
- Number of logical blocks in the RTL
- Constraints for the block in SDC
- Floorplan information in DEF
- I/O pin location
- Macro locations

ASIC General

General ASIC questions are posted here. More questions related to different catagories of ASICs can be found at respective sections.

- What are the differences between PALs, PLAs, FPGAs, ASICs and PLDs?
- In system with insufficient hold time, will slowing down the clock help?
- In system with insufficient setup time, will slowing down the clock help?
- Why would a testbench not have pins (port) on it?
- When declaring a flip flop, why would not you declare its output value in the port statement?
- Give 2 advantages of using a script to build a chip?
- A "tri state " bus is directly connected to a set of CMOS input buffers. No other wires or components are attached to the bus wires. Upon observation we can find that under certain conditions, this circuit is consuming considerable power. Why it is so? Is circuit correct? If not, how to correct?
- Is Verilog (or that matter any HDL) is a concurrent or sequential language?
- What is the function of sensitivity list?
- A mealy –type state machine is coded using D-type rising edge flip flops. The reset and clock signals are in the sensitivity list but with one of the next state logic input signals have been left out of the sensitivity list. Explain what happens when the state machine is simulated? Will the state machine be synthesized correctly?
- A moore –type state machine is coded using D-type rising edge flip flops. The reset and clock signals are in the sensitivity list but with one of the next state logic input signals have been left out of the sensitivity list. Explain what happens when the state machine is simulated? Will the state machine be synthesized correctly?
- What type of delay is most like a infinite bandwidth transmission line?
- Define metastability.
- When does metastability occur?
- Give one example of a situation where metastability could occur.
- Give two ways metastability could manifest itself in a state machine.
- What is MTBF?
- Does MTBF give the time until the next failure occurs?
- Give 3 ways in which to reduce the chance of metastable failure.
- Give 2 advantages of using a synchronous reset methodology.
- Give 2 disadvantages of using a synchronous reset methodology.
- Give 2 advantages of using an asynchronous reset methodology.
- Give 2 disadvantages of using an asynchronous reset methodology.
- What are the two most fundamental inputs (files) to the synthesis tool?
- What are two important steps in synthesis? What happens in those steps?
- What are the two major output (files) from the synthesis process?

- Name the fundamental 3 operating consitions that determine (globally) the delay characteristics of CMOS gates. For each how they affect gate delay?
- For a single gate, with global gating conditions held constant , what 3 delay coefficients effect total gate delay? Which is the most sensitive to circuit topology?

FPGA.

What is the difference between FPGA and CPLD?

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- This is what Wiki defines.....!!

Architecture

- Granularity is the biggest difference between CPLD and FPGA.
- FPGA are "fine-grain" devices. That means that they contain hundreds of (up to 100000) of tiny blocks (called as LUT or CLBs etc) of logic with flip-flops, combinational logic and memories.FPGAs offer much higher complexity, up to 150,000 flip-flops and large number of gates available.
- CPLDs typically have the equivalent of thousands of logic gates, allowing implementation of moderately complicated data processing devices. PALs typically have a few hundred gate equivalents at most, while FPGAs typically range from tens of thousands to several million.

CPLD are "coarse-grain" devices. They contain relatively few (a few 100's max) large blocks of logic with flip-flops and combinational logic. CPLDs based on AND-OR structure.

CPLD's have a register with associated logic (AND/OR matrix). CPLD's are mostly implemented in control applications and FPGA's in datapath applications. Because of this course grained architecture, the timing is very fixed in CPLDs.

FPGA are RAM based. They need to be "downloaded" (configured) at each power-up. CPLD are EEPROM based. They are active at power-up i.e. as long as they've been programmed at least once.

FPGA needs boot ROM but CPLD does not. In some systems you might not have enough time to boot up FPGA then you need CPLD+FPGA.

- Generally, the CPLD devices are not volatile, because they contain flash or erasable ROM memory in all the cases. The FPGA are volatile in many cases and hence they need a configuration memory for working. There are some FPGAs now which are nonvolatile. This distinction is rapidly becoming less relevant, as several of the latest FPGA products also offer models with embedded configuration memory.
- The characteristic of non-volatility makes the CPLD the device of choice in modern digital designs to perform 'boot loader' functions before handing over control to other devices not having this capability. A good example is where a CPLD is used to load configuration data for an FPGA from non-volatile memory.
- Because of coarse-grain architecture, one block of logic can hold a big equation and hence CPLD have a faster input-to-output timings than FPGA.

Features

- FPGA have special routing resources to implement binary counters,arithmetic functions like adders, comparators and RAM. CPLD don't have special features like this.
- FPGA can contain very large digital designs, while CPLD can contain small designs only.The limited complexity (<500>
- ***Speed***: CPLDs offer a single-chip solution with fast pin-to-pin delays, even for wide input functions. Use CPLDs for small designs, where "instant-on", fast and wide decoding, ultra-low idle power consumption, and design security are important (e.g., in battery-operated equipment).
- ***Security***: In CPLD once programmed, the design can be locked and thus made secure. Since the configuration bitstream must be reloaded every time power is re-applied, design security in FPGA is an issue.
- ***Power***: The high static (idle) power consumption prohibits use of CPLD in battery-operated equipment. FPGA idle power consumption is reasonably low, although it is sharply increasing in the newest families.

Design flexibility: FPGAs offer more logic flexibility and more sophisticated system features than CPLDs: clock management, on-chip RAM, DSP functions, (multipliers), and even on-chip microprocessors and Multi-Gigabit Transceivers.These benefits and opportunities of dynamic reconfiguration, even in the end-user system, are an important advantage.

- Use FPGAs for larger and more complex designs.
- FPGA is suited for timing circuit becauce they have more registers , but CPLD is suited for control circuit because they have more combinational circuit. At the same time, If you synthesis the same code for FPGA for many times, you will find out that each timing report is different. But it is different in CPLD synthesis, you can get the same result.

As CPLDs and FPGAs become more advanced the differences between the two device types will continue to blur. While this trend may appear to make the two types more difficult to keep apart, the architectural advantage of CPLDs combining low cost, non-volatile configuration, and macro cells with predictable timing characteristics will likely be sufficient to maintain a product differentiation for the foreseeable future.

What is the difference between FPGA and ASIC?

- This question is very popular in VLSI fresher interviews. It looks simple but a deeper insight into the subject reveals the fact that there are lot of thinks to be understood !! So here is the answer.

FPGA vs. ASIC

- Difference between ASICs and FPGAs mainly depends on costs, tool availability, performance and design flexibility. They have their own pros and cons but it is designers responsibility to find the advantages of the each and use either FPGA or ASIC for the product. However, recent developments in the FPGA domain are narrowing down the benefits of the ASICs.

FPGA

- **Field Programable Gate Arrays**

FPGA Design Advantages

- ***Faster time-to-market:*** No layout, masks or other manufacturing steps are needed for FPGA design. Readymade FPGA is available and burn your HDL code to FPGA ! Done !!
- **No NRE (Non Recurring Expenses):** This cost is typically associated with an ASIC design. For FPGA this is not there. FPGA tools are cheap. (sometimes its free ! You need to buy FPGA.... thats all !). ASIC youpay huge NRE and tools are expensive. I would say "very expensive"...Its in crores....!!
- ***Simpler design cycle:*** This is due to software that handles much of the routing, placement, and timing. Manual intervention is less.The FPGA design flow eliminates the complex and time-consuming floorplanning, place and route, timing analysis.
- ***More predictable project cycle:*** The FPGA design flow eliminates potential re-spins, wafer capacities, etc of the project since the design logic is already synthesized and verified in FPGA device.
- ***Field Reprogramability:*** A new bitstream (i.e. your program) can be uploaded remotely, instantly. FPGA can be reprogrammed in a snap while an ASIC can take $50,000 and more than 4-6 weeks to make the same changes. FPGA costs start from a couple of dollars to several hundreds or more depending on the hardware features.

Reusability: Reusability of FPGA is the main advantage. Prototype of the design can be implemented on FPGA which could be verified for almost accurate results so that it can be implemented on an ASIC. Ifdesign has faults change the HDL code, generate bit stream, program to FPGA and test again.Modern FPGAs are reconfigurable both partially and dynamically.

- FPGAs are good for prototyping and limited production.If you are going to make 100-200 boards it isn't worth to make an ASIC.
- Generally FPGAs are used for lower speed, lower complexity and lower volume designs.But today's FPGAs even run at 500 MHz with superior performance. With unprecedented logic density increases and a host of other features, such as embedded processors, DSP blocks, clocking, and high-speed serial at ever lower price, FPGAs are suitable for almost any type of design.
- Unlike ASICs, FPGA's have special hardwares such as Block-RAM, DCM modules, MACs, memories and highspeed I/O, embedded CPU etc inbuilt, which can be used to get better performace. Modern FPGAs are packed with features. Advanced FPGAs usually come with phase-locked loops, low-voltage differential signal, clock data recovery, more internal routing, high speed, hardware multipliers for DSPs, memory,programmable I/O, IP cores and microprocessor cores. Remember Power PC (hardcore) and Microblaze (softcore) in Xilinx and ARM (hardcore) and Nios(softcore) in Altera. There are FPGAs available now with built in ADC ! Using all these features designers can build a system on a chip. Now, dou yo really need an ASIC ?
- FPGA sythesis is much more easier than ASIC.
- In FPGA you need not do floor-planning, tool can do it efficiently. In ASIC you have do it.

FPGA Design Disadvantages

- Powe consumption in FPGA is more. You don't have any control over the power optimization. This is where ASIC wins the race !
- You have to use the resources available in the FPGA. Thus FPGA limits the design size.
- Good for low quantity production. As quantity increases cost per product increases compared to the ASIC implementation.

ASIC

Application Specific Intergrated Circiut

ASIC Design Advantages

Cost....cost....cost....Lower unit costs: For very high volume designs costs comes out to be very less. Larger volumes of ASIC design proves to be cheaper than implementing design using FPGA.

- ***Speed...speed...speed....ASICs are faster than FPGA:*** ASIC gives design flexibility. This gives enoromous opportunity for speed optimizations.
- ***Low power....Low power....Low power:*** ASIC can be optimized for required low power. There are several low power techniques such as power gating, clock gating, multi vt cell libraries, pipelining etc are available to achieve the power target. This is where FPGA fails badly !!! Can you think of a cell phone which has to be charged for every call.....never.....low power ASICs helps battery live longer life !!
- In ASIC you can implement analog circuit, mixed signal designs. This is generally not possible in FPGA.
- In ASIC DFT (Design For Test) is inserted. In FPGA DFT is not carried out (rather for FPGA no need of DFT !) .

ASIC Design Diadvantages

- ***Time-to-market:*** Some large ASICs can take a year or more to design. A good way to shorten development time is to make prototypes using FPGAs and then switch to an ASIC.
- ***Design Issues:*** In ASIC you should take care of DFM issues, Signal Integrity isuues and many more. In FPGA you don't have all these because ASIC designer takes care of all these. (Don't forget FPGA isan IC and designed by ASIC design enginner !!)
- ***Expensive Tools:*** ASIC design tools are very much expensive. You spend a huge amount of NRE.

Structured ASICS

- Structured ASICs have the bottom metal layers fixed and only the top layers can be designed by the customer.
- Structured ASICs are custom devices that approach the performance of today's Standard Cell ASIC while dramatically simplifying the design complexity.
- Structured ASICs offer designers a set of devices with specific, customizable metal layers along with predefined metal

layers, which can contain the underlying pattern of logic cells, memory, and I/O.

FPGA Interview Questions

- What is minimum and maximum frequency of DCM in spartan-3 series FPGA?
- List some of constraints you used and their purpose during your design?
- What is the size of bitmap with changing gate count?
- What are different types of FPGA programming modes? How to change from one to another?
- List out some important features of FPGA.
- List out some of synthesizable and non synthesizable constructs?
- Draw general structure of FPGA?
- What is the difference between FPGA and CPLD?
- What is DCM? Why they are used?
- Draw FPGA design flow. Explain each step. What is input and output from each step?
- What is slice, CLB, LUT?
- Is it possible to configure CLB as RAM?
- What is purpose of a constraint file? What is its extension?
- How you will choose an FPGA?
- How clock is routed through out FPGA?
- What are difference between PLL and DLL ?
- What is soft processor?
- What is hard processor?

4 comments:



navin kumar February 11, 2014 at 12:04 PM

This comment has been removed by the author.

Reply



anagha ghosh April 30, 2014 at 10:52 AM

Good Work Sir !!

Reply



Semicon Technologies April 27, 2016 at 4:23 PM

This is really good questions of vlsi. Thanks for sharing.**VLSI Training Courses in Bangalore**|**VLSI Training Institute in Bangalore**

Reply

Moderator CEO September 15, 2016 at 4:56 AM

I liked your posts, I was wondering why such good content is not monetised are u waiting for adsense approval.I have found another alternative ads http://tinyurl.com/zl83sk9 works fine for my blog http://basicsofvlsi.blogspot.in/

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1. what is the difference between mealy and moore state-machines

2. How to solve setup & Hold violations in the design

To solve setup violation

1. optimizing/restructuring combination logic between the flops.
2. Tweak flops to offer lesser setup delay [DFFX1 -> DFFXx]
3. Tweak launch-flop to have better slew at the clock pin, this
will make CK->Q of launch flop to be fast there by helping fixing
setup violations
4. Play with skew [tweak clock network delay, slow-down clock to
capturing flop and fasten the clock to launch-flop](otherwise called as Useful-skews)

To solve Hold Violations

1. Adding delay/buffer[as buffer offers lesser delay, we go for spl
Delay cells whose functionality Y=A, but with more delay]
2. Making the launch flop clock reaching delayed

3. Also, one can add lockup-latches [in cases where the hold time
requirement is very huge, basically to avoid data slip]

3. What is antenna Violation & ways to prevent it

During the process of plasma etching, charges accumulate along the metal strips. The longer the strips are, the more charges are accumulated. IF a small transistor gate connected to these long metal strips, the gate oxide can be destroyed (large electric field over a very thin electric) , This is called as Antenna violation.

The ways to prevent is , by making jogging the metal line, which is atleast one metal above the layer to be protected. If we want to remove antenna violation in metal2 then need to jog it in metal3 not in metal1. The reason being while we are etching metal2, metal3 layer is not laid out. So the two
pieces of metal2 got disconnected. Only the piece of metal connected to gate have charge to gate. When we laydown metal3, the remaining portion of metal got charge added to metal3. This is called
accumulative antenna effect.
Another way of preventing is adding reverse Diodes at the gates

4. We have multiple instances in RTL(Register Transfer Language), do you do anything special during synthesis stage?

While writing RTL(Register Transfer language),say in verilog or in VHDL language, we dont write the same module functionality again and again, we use a concept called as

instantiation, where in as per the language, the instanciation of a module will behave like the parent module in terms of functionality, where during synthesis stage we need the full code so that the synthesis tool can study the logic , structure and map it to the library cells, so we use a command in synthesis , called as "UNIQUIFY" which will replace the instantiations with the real logic, because once we are in a synthesis stages we have to visualize as real cells and no more modelling just for functionality alone, we need to visualize in-terms of physical world as well.

5. what is tie-high and tie-low cells and where it is used

Tie-high and Tie-Low cells are used to connect the gate of the transistor to either power or ground. In deep sub micron processes, if the gate is connected to power/ground the transistor might be turned on/off due to power or ground bounce. The suggestion from foundry is to use tie cells for this purpose. These cells are part of standard-cell library. The cells which require Vdd, comes and connect to Tie high...(so tie high is a power supply cell)...while the cells which wants Vss connects itself to Tie-low.

6. what is the difference between latches and flip-flops based designs

Latches are level-sensitive and flip-flops are edge sensitive. latch based design and flop based design is that latch allowes time borrowing which a tradition flop does not. That makes latch based design more efficient. But at the same time, latch based design is more complicated and has more issues in min timing (races). Its STA with time borrowing in deep pipelining can be quite complex.

7. What is High-Vt and Low-Vt cells.

Hvt cells are MOS devices with less leakage due to high Vt but they have higher delay than low VT, where as the low Vt cells are devices which have less delay but leakage is high. The thereshold(t) vloltage dictates the transistor switching speed , it matters how much minimum threshold voltage applied can make the transistor switching to active state which results to how fast we can switch the trasistor. disadvantage is it needs to maintain the transistor in a minimum subthreshold voltage level to make ir switch fast so it leads to leakage of current inturn loss of power.

8. What is LEF mean?

LEF is an ASCII data format from Cadence Design inc, to describe a standard cell library. It includes the design rules for routing and the Abstract layout of the cells. LEF file contains the following,

Technology: layer, design rules, via-definitions, metal-capacitance

Site : Site extension

Macros : cell descriptions, cell dimensions, layout of pins and blockages, capacitances

To get further insight to the topic, please check this
http://www.csee.umbc.edu/~cpatel2/links/414/slides/lect03_LEF.pdf

9. what is DEF mean?

DEF is an ASCII data format from Cadence Design inc., to describe Design related information.

10. Steps involved in designing an optimal padring

1. Make sure you have corner-pads, across all the corners of the padring, This is mainly to have the power-continuity as well as the resistance is less .

2. Ensure that the Padring ful-fills the ESD requirement, Identifyh the power-domains, split the domains, Ensure common ground across all the domains.

3. Ensure the padring has ful-filled the SSN(Simultaneous Switching Noise) requirement.

4. Placing Transfer-cell Pads in the cross power-domains, for different height pads, to have rail connectivity.

5. Ensure that the design has sufficient core power-pads.

6. Choose the Drive-strenght of the pads based on the current requirements, timing.

7. Ensure that there is seperate analog ground and power pads.

8. A No-Connection Pad is used to fill out the pad-frame if there is no requirement for I/O's.Extra VDD/GND pads also could be used. Ensure that no Input/output pads are used with un-connected inputs, as they consume power if the inputs float.

9. Ensure that oscillator-pads are used for clock inputs.

10. In-case if the design requirement for source synchronous circuits, make sure that the clock and data pads are of same drive-strength.

11. Breaker-pads are used to break the power-ring, and to isolate the power-structure across the pads.

12. Ensure that the metal-wire connected to the pin can carry sufficient amount of the current, check if more than one metal-layer is necessary to carry the maximum current provided at the pin.

13. In case if required , place pads with capacitance.

related information.

11. What is metastability and steps to prevent it.

Metastability is an unknown state it is neither Zero nor One.Metastability happens for the design systems violating setup or hole time requirements. Setup time is a requirement , that the data has to be stable before the clock-edge and hold time is a requirement , that the data has to be stable after the clock-edge. The potential violation of the setup and hold violation can happen when the data is purely asynchronous and clocked synchronously.

Steps to prevent Metastability.

1. Using proper synchronizers(two-stage or three stage), as soon as the data is coming from the asynchronous domain. Using Synchronizers, recovers from the metastable event.

2. Use synchronizers between cross-clocking domains to reduce the possibility from metastability.

3. Using Faster flip-flops (which has narrower Metastable Window).

12. what is local-skew, global-skew,useful-skew mean?

Local skew : The difference between the clock reaching at the launching flop vs the clock reaching the destination flip-flop of a timing-path.

Global skew : The difference between the earliest reaching flip-flop and latest reaching flip-flop for a same clock-domain.

Useful skew: Useful skew is a concept of delaying the capturing flip-flop clock path, this approach helps in meeting setup requirement with in the launch and capture timing path. But the hold-requirement has to be met for the design.

13. What are the various timing-paths which i should take care in my STA runs?

1. Timing path starting from an input-port and ending at the output port(purely combinational path).

2. Timing path starting from an input-port and ending at the register.

3. Timing path starting from an Register and ending at the output-port.

4. Timing path starting from an register and ending at the register.

14. What are the various components of Leakage-power?

1. sub-threshold leakage

- Subthreshold leakage exponential with V_{gs}

$$I_{ds} = I_{ds0} e^{\frac{V_{gs} - V_t}{n v_T}} \left(1 - e^{\frac{-V_{ds}}{v_T}} \right) \qquad I_{ds0} = \beta v_T^2 e^{1.8}$$

Drain-Induced Barrier Lowering

- Drain voltage also affect V_t

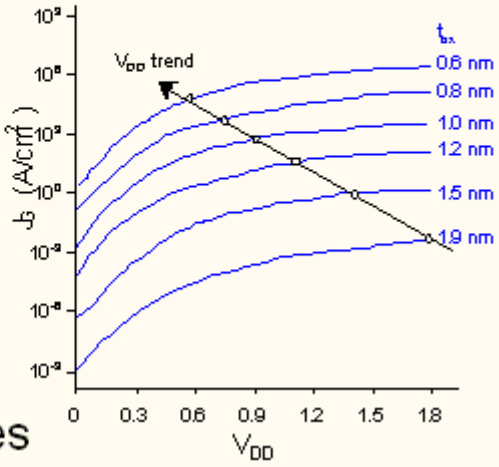
$$V_t' = V_t - \eta V_{ds}$$

- High drain voltage causes subthreshold leakage to increase.

-courtesy Khondker

2. gate leakage

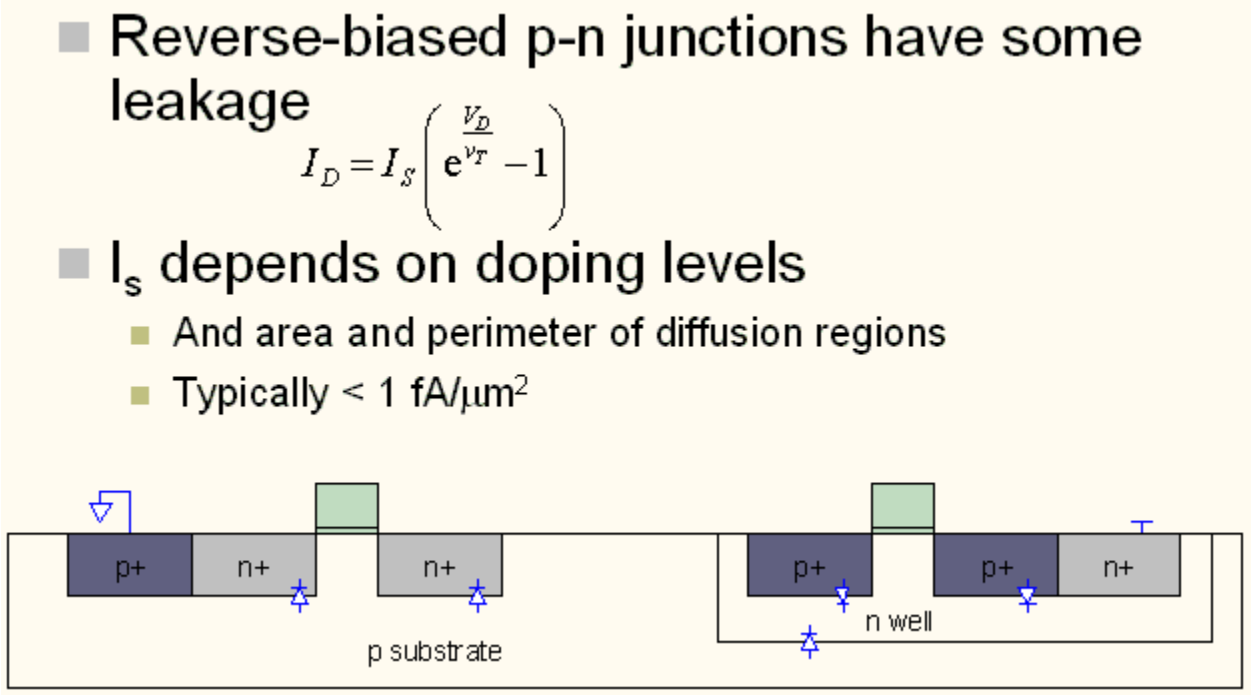
- Carriers may tunnel thorough very thin gate oxides
- Predicted tunneling current (from [Song01])



- Negligible for older processes

-courtesy Khondker

3. reverse biased drain substrate and drain substrate junction band-band tunnelling



15. What are the various yield-losses in the design?

The yield loss in the design is characterized by

1. Functional yield losses, mainly caused by spot defects , especially (shorts & opens)

2. Parametric yield losses, due to process variations.

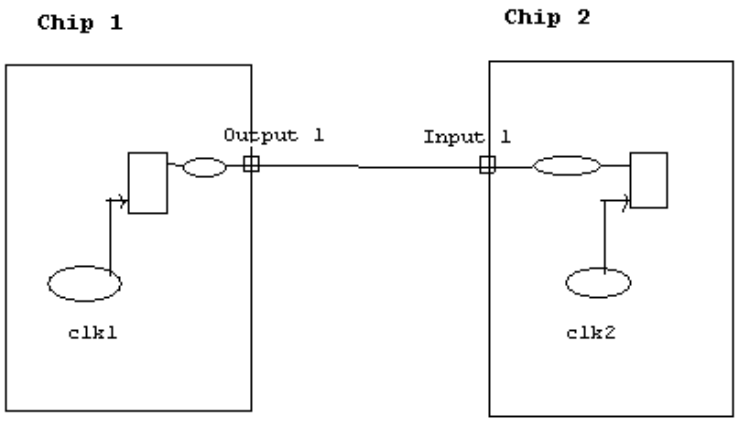
16. what is meant by virtual clock definition and why do i need it?

Virtual clock is mainly used to model the I/O timing specification. Based on what clock the output/input pads are passing the data.

For Further Understanding of the concept.
http://www.vlsichipdesign.com/images/virtual_clock.jpg

attached below..

Concept Behind Virtual-Clocks



Data transfer is from Chip1 ouput1 to Chip2 Input1
we are modelling Chip2. Create a Virtual clock with the
frequency of clk2. Model the pad output, w.r.t. VCLK2, and model
the output delay = (Output1 + Board delay + Input1 to
Flop(chip2).

17. What are the various Variations which impacts timing of the design?

Delay impact of variations

Parameter	*Delay Impa*
BEOL metal (Metal mistrack, thin/thick wires)	-10% → +25%
Environmental (Voltage islands, IR drop, temperature)	±15 %
Device fatigue (NBTI, hot electron effects)	±10%
V_t and T_{ox} device family tracking (Can have multiple V_t and T_{ox} device families)	± 5%
Model/hardware uncertainty (Per cell type)	± 5%
N/P mistrack (Fast rise/slow fall, fast fall/slow rise)	±10%
PLL (Jitter, duty cycle, phase error)	±10%

- **Requires 2^{20} timing runs or [-65%,+80%] guard band!**

[Courtesy I
Chandu

18. What are the various Design constraints used while performing Synthesis for a design?

1. Create the clocks (frequency, duty-cycle).

2. Define the transition-time requirements for the input-ports.3

3. Specify the load values for the output ports

4. For the inputs and the output specify the delay values(input delay and ouput delay), which are

already consumed by the neighbour chip.

5. Specify the case-setting (in case of a mux) to report the timing to a specific paths.

6. Specify the false-paths in the design

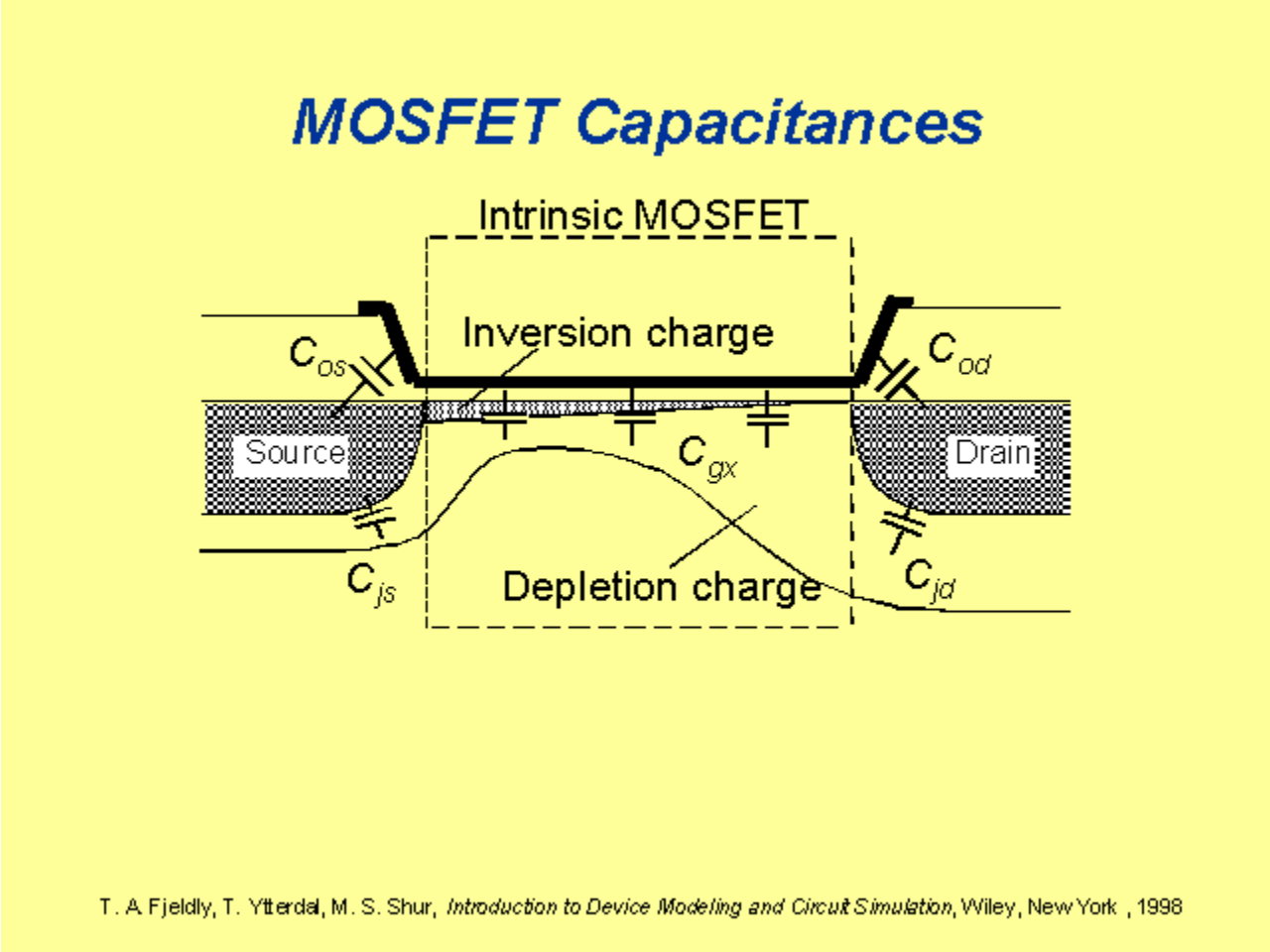
7. Specify the multi-cycle paths in the design.

8. Specify the clock-uncertainity values(w.r.t jitter and the margin values for setup/hold).

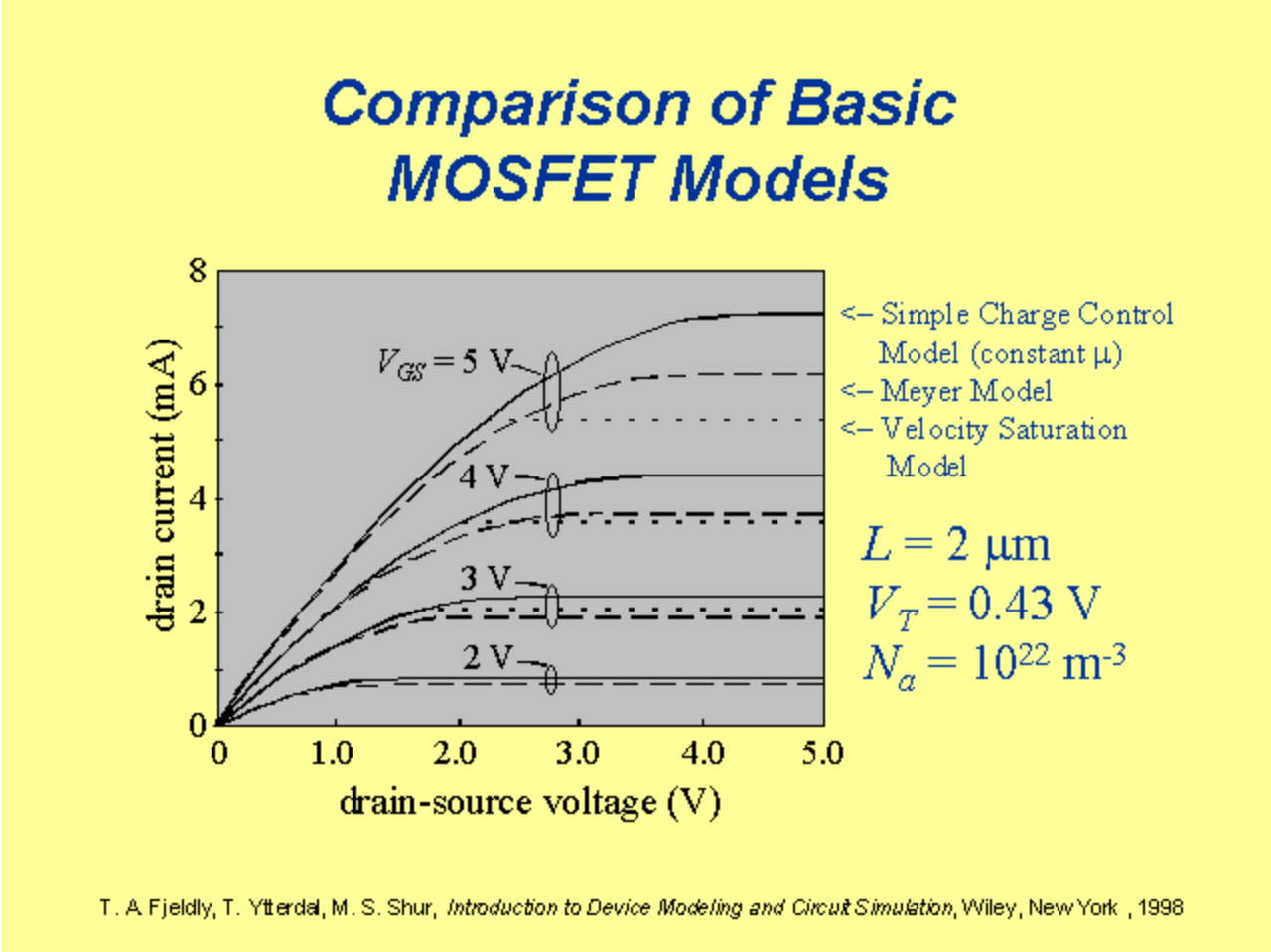
19. Specify few verilog constructs which are not supported by the synthesis tool.

initial, delays, real and time data types, force and release, fork join.

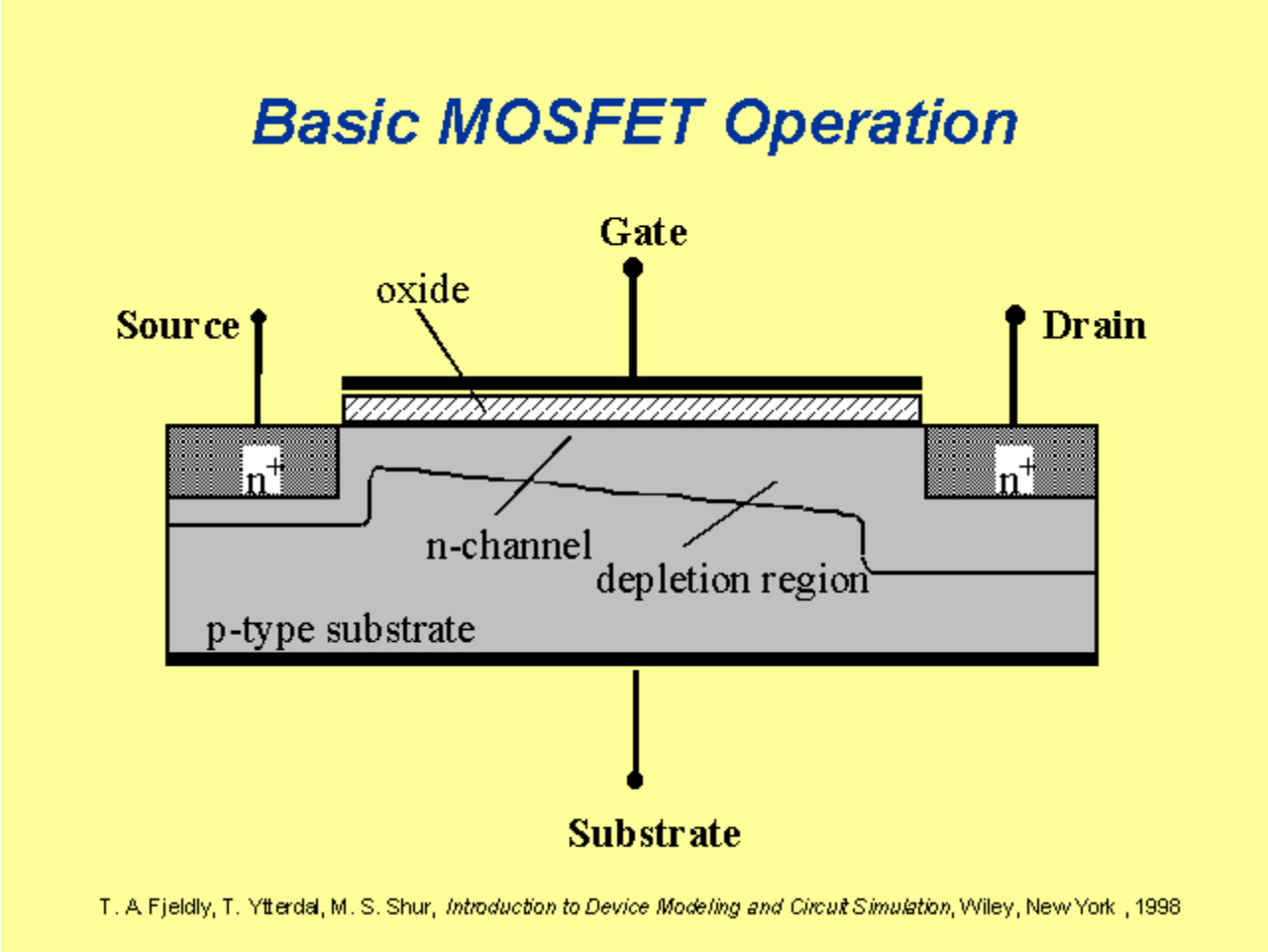
20.what are the various capacitances with an MOSFET?/strong>



21.Vds-Ids curve for an MOSFET, with increasing Vgs.



22. Basic Operation of an MOSFET.



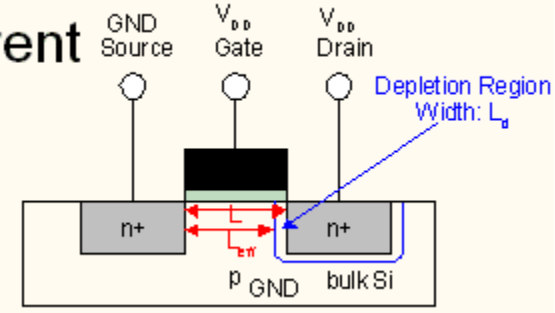
23. What is Channel length Modulation?

Reverse-biased p-n junctions form a *depletion region*

- Region between n and p with no carriers
- Width of depletion L_d region grows with reverse bias
- $L_{eff} = L - L_d$

Shorter L_{eff} gives more current

- I_{ds} increases with V_{ds}
- Even in saturation



-courtesy Khondker

24. what is body effect?

Increase in Vt(threshold voltage) , due to increase in Vs(voltage at source), is called as body effect.

Body Effect Model

$$V_t = V_{t0} + \gamma\left(\sqrt{\phi_s + V_{sb}} - \sqrt{\phi_s}\right)$$

- ϕ_s = *surface potential* at threshold

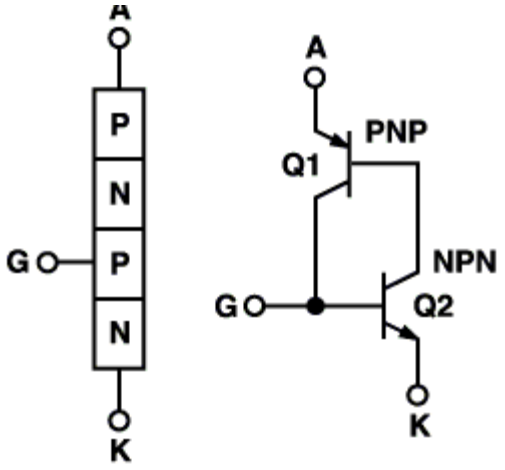
$$\phi_s = 2v_T \ln\frac{N_A}{n_i}$$

  - Depends on doping level N_A
  - And intrinsic carrier concentration n_i
- γ = *body effect coefficient*

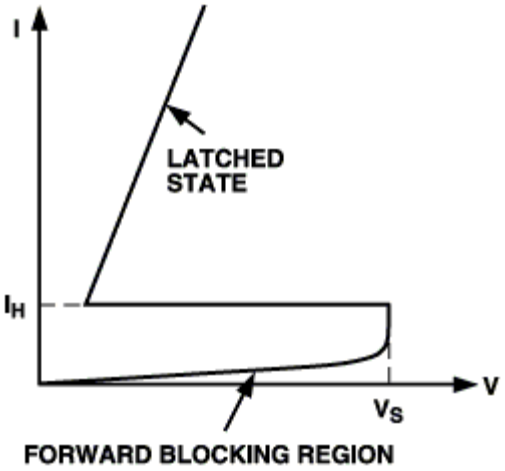
$$\gamma = \frac{t_{ox}}{\varepsilon_{ox}}\sqrt{2q\varepsilon_{si}N_A} = \frac{\sqrt{2q\varepsilon_{si}N_A}}{C_{ox}}$$

25. What is latchup in CMOS design and ways to prevent it?

To best understand the concept behind the latchup, we need to understand the concept behind SCR(Silicon Controlled Rectifiers), and how to model the basic transistor in an SCR structure and on what conditions SCR structures are created in the CMOS design process and its effects and what are the ways used to prevent it in the design-phase.An SCR is an acronym for Silicon Controlled Rectifier. It works similar to a typical diode, but is controlled similar to a bipolar transistor as far as connections go. Connection points are Anode [A], Cathode [K], and Gate [G]. The SCR is made up of two "P-N" junctions with a "Gate" attachment between them. The gate is connected between the two P-N junctions with a current waiting in the forward bias direction [+ to -] and the voltage is above 1-volt. A momentary pulse to the gate will cause the SCR to conduct and current will flow across the device until the value changes.



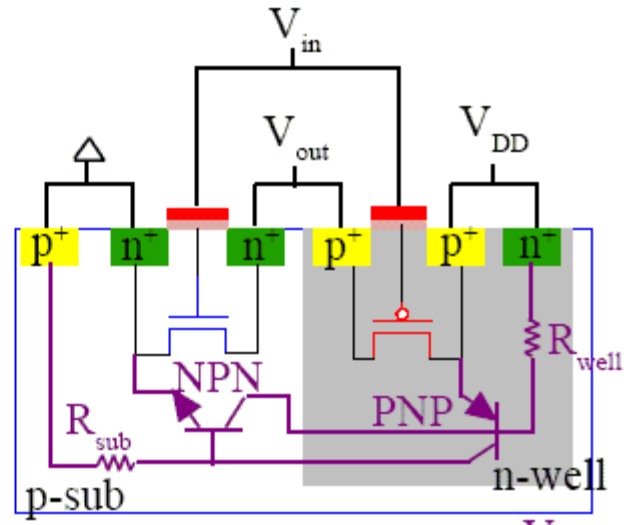
a) Transistor equivalent of an SCR.

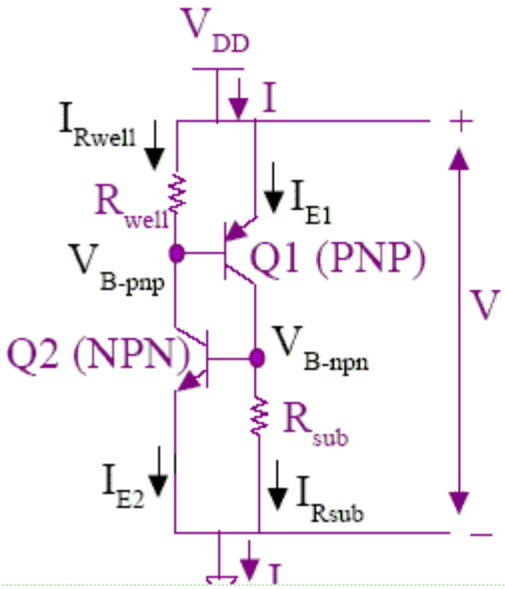


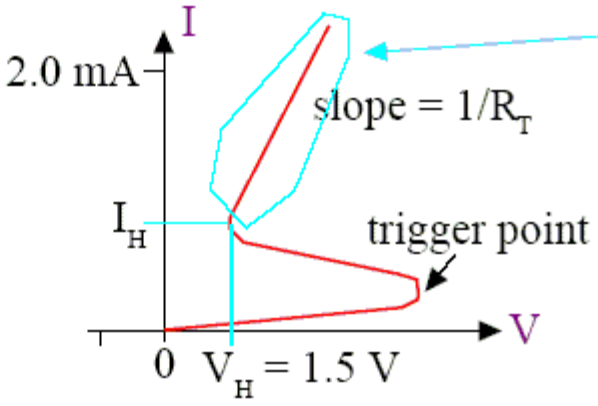
LATCHUP IN CMOS CIRCUITS

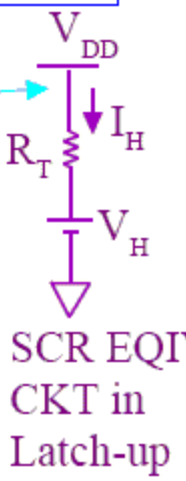
latch-up susceptability proportional to

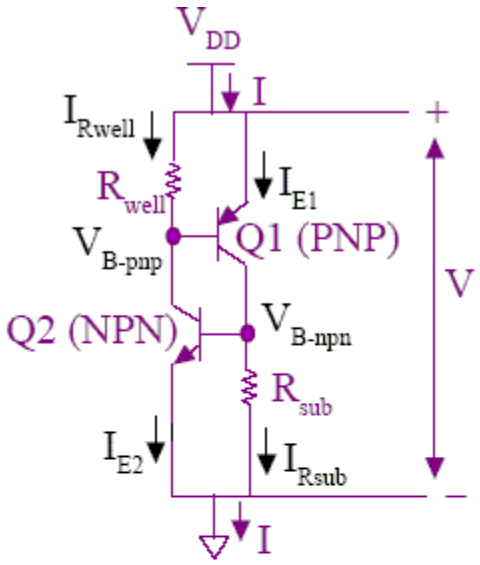
$$\frac{1}{\text{NSUB}} \times \frac{1}{(\text{nMOS} - \text{to} - \text{pMOS spacing})^2}$$

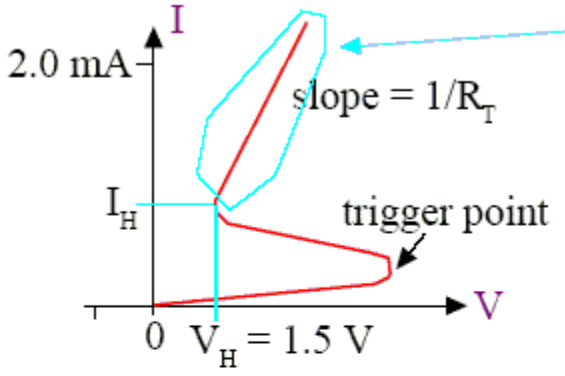


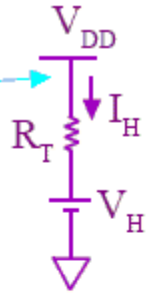












SCR EQIV CKT in Latch-up

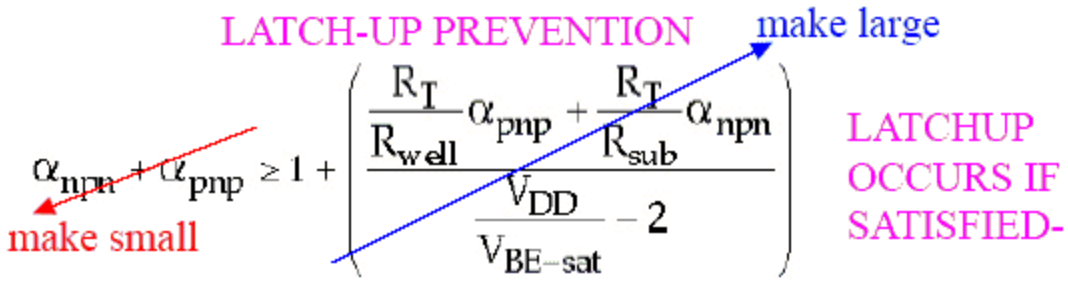
$$I_H = \frac{I_{Rsub}\alpha_{pnp} + I_{Rwell}\alpha_{npn}}{\alpha_{pnp} + \alpha_{npn} - 1}$$

At the on-set of latch-up

$I > I_H = (V_{DD} - V_H)/R_T$

LATCH-UP CONDITION

$$\alpha_{npn} + \alpha_{pnp} \geq 1 + \left(\frac{\frac{R_T}{R_{well}}\alpha_{pnp} + \frac{R_T}{R_{sub}}\alpha_{npn}}{\frac{V_{DD}}{V_{BE-sat}} - 2} \right)$$



TO PREVENT LATCH-UP: Insure Latchup Condition is Violated!

Reduce α_{npn}, α_{pnp}; Reduce R_{sub}, R_{well}

A. Use a latchup resistant process.

Latchup Prevention LAYOUT Guidelines:

B. Use p^+ guard rings connected to GND around nMOS transistors and n^+quard rings connected to V_{DD} around the pMOS transistors to reduce R_{well} and R_{sub} and to weaken BJTs.

C. Place sub, well contacts close to the nMOS, pMOS source connections to supply rails (i.e. GND for nMOS, V_{DD} for pMOS) to reduce R_{well} and R_{sub}.

CONSERVATIVE RULE: One sub contact per source connection to a supply, or GND.

LESS CONSERVATIVE: One sub contact per 5-10 transistors.

D. Layout nMOS, pMOS transistors close to GND, VDD rails, respectively and maintain space between nMOS, pMOS transistors.

25. What are the various design changes you do to meet design power targets?

Design with Multi-VDD designs, Areas which requires high performance, goes with high VDD and areas which needs low-performance are working with low Vdd's, by creating Voltage-islands and making sure that appropriate level-shifters are placed in the cross-voltage domains

Designing with Multi-Vt's(threshold voltages), areas which require high performance, goes with low Vt, but takes lot of leakage current, and areas which require low performance with high Vt cells, which has low leakage numbers, by incorporating this design process, we can reduce the leakage power.

As in the design , clocks consume more amount of power, placing optimal clock-gating cells, in the design and controlling them by the module enable's gives a lot of power-savings.

As clock-tree's always switch making sure that most number of clock-buffers are after the clock-gating cells, this reduces the switching there by power-reduction.

Incorporating Dynamic Voltage & Frequency scaling (DVFS) concepts based on the application , there by reducing the systems voltage and frequency numbers when the application does not require to meet the performance targets.

Ensure the design with IR-Drop analysis and ground-bounce analysis, is with-in the design specification requirement.

Place power-switches, so that the leakage power can be reduced. related information.

27. what is meant by Library Characterizing: ***"Chip designing is all about Modeling the silicon",*** and how well we characterize the silicon, is all the game. So initially let us assume our process technology is say "32nm", for example: Now we need to develop a test-chip, having modules (digital & analog), and study our silicon timings. Now the toughest job is to generate library views(formats specific to each tool understandable formats).There is a bit of timing in accuracy possible in the views across the formats.

28. what is meant by wireload model:

In the synthesis tool, in order to model the wires we use a concept called as "Wireload models", Now the question is what is wireload models: Wireload models are statistical based on models with respect to fanout. say for a particular technology based on our previous chip experience we have a rough estimate we know if a wire goes for "n" number of fanin then we estimate its delay as say "x" delay units. So a model file is created with the fanout numbers and corresponding estimated delay values. This file is used while performing Synthesis to estimate the delay for Wires, and to estimate the delay for cells, technology specific library model files will be available

31. what are the measures in the Design taken for Meeting Signal-integrity targets

As more and more devices are getting packed, results in more congested areas, and coupling capactiances dominating the wire-capacitance, creates SI violations. Let's see now by what are all the measures we can reduce/solve it.

- As clock-tree runs across the whole chip, optimizing the design for SI, is essential route the clock with double-pitch and triple spacing.
- In-case of SI violation, spacing the signal nets reduces cross-talk impacts.
- Shield the nets with power-nets for high frequency signal nets to prevent from SI.
- Enable SI aware routing , so that the tool takes care for SI
- Ensure SI enabled STA runs, and guarantee the design meeting the SI requirements
- Route signals on different layers orthogonal to each other
- Minimize the parallel run-length wires, by inserting buffers.

32. what are the measures taken in the Design achieving better Yield

Better yield could be achieved by reducing the possibility of manufacturability flaws. Guaranting the circuit performance, by reducing parametric yield, with process variations playing a major role is a big-challenge.

- Create more powerful stringent runset files with pessimistic spacing/short rules.

- Check for the areas where the design is prone to lithographic issues, like sharp cuts and try to re-route it.
- For via-reliability issues, use redundant vias, to reduce the chances for via-breakage.
- In order to design for yield-enhancement , design systems, which could have optimal redundancy, like repairable memories.
- Optimal placing of de-coupling capacitances, reduces the power-surges.
- Doubling the width of the non-critical nets, clock-nets can increase the yield parameter.
- Ensure that the poly-orientation are maintained.

32. what are the measures or precautions to be taken in the Design when the chip has both analog and digital portions

Designing for Optimal integration of Analog and Digital

- As today's IC has analog components also inbuilt , some design practices are required for optimal integration.
- Ensure in the floorplanning stage that the analog block and the digital block are not siting close-by, to reduce the noise.
- Ensure that there exists seperate ground for digital and analog ground to reduce the noise.
- Place appropriate guard-rings around the analog-macro's.
- Incorporating in-built DAC-ADC converters, allows us to test the analog portion using digital testers in an analog loop-back fashion.
- Perform techniques like clock-dithering for the digital portion.

33. what are the steps incorporated for Engineering Change Order[ECO]

As more and more complex the IC design is , and with lot of first time application , is more prone to

last minute changes, there should be provision in the design-flow to accomodate the functional and timing bugs. The step to perform this called as Engineering change order(ECO).

- Ensure that the design has spare functional gates well distributed across the layout.
- Ensure that the selection the spare gates, has many flavours of gates and universal gates, so that any functionality could be achieved.

34. what are the steps performed to achieve Lithography friendly Design

Designing for Manufacturability requires validating the design full-filling lithography rules

- Checking the layout confirming the design rules (spacing,trace-width,shorts).
- Check for the less-congested areas and increasing the spacing of the nets.

35. what does synthesis mean

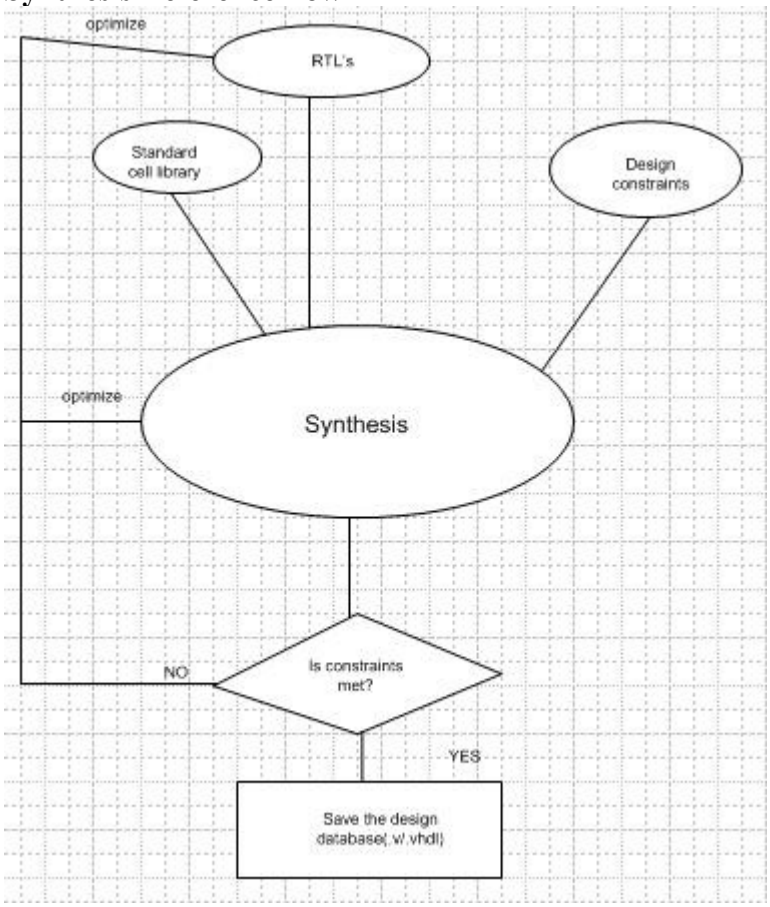
Synthesis is a step of mapping the RTL files (verilog format or vhdl format) to convert it to the technology specific cells..

36. what are the pre-requisties to perform synthesis

1. RTL files
2. Synopsys constraints file, Design constraints file, explaining the priorities of cost functions like area/timing/power
3. Technology specific library files.

34. Explain the Synthesis flow

Synthesis Reference flow



35. What are the various ways to reduce Clock Insertion Delay in the Design

1. Number of Clock sinks
2. Balancing two different clock frequencies
3. Placement of clock sinks.
4. Placement of Clock gating cells
5. Clock tree buffers/inverters drive strength's
6. Clock Transition
7. placement of Clockgating cells and the clock sinks
8. Combinationals cells in the path of clocks (say clock dividers, muxes, clockgates) ...

36. what are the various functional verification methodologies

- TLM(Transaction Level Modelling)
- Linting
- RTL Simulation (Enivronment involving : stimulus generators, monitors, response checkers, transactors)
- Gate level Simulation
- Mixed-signal simulations
- Regression

36. What does formal verification mean?

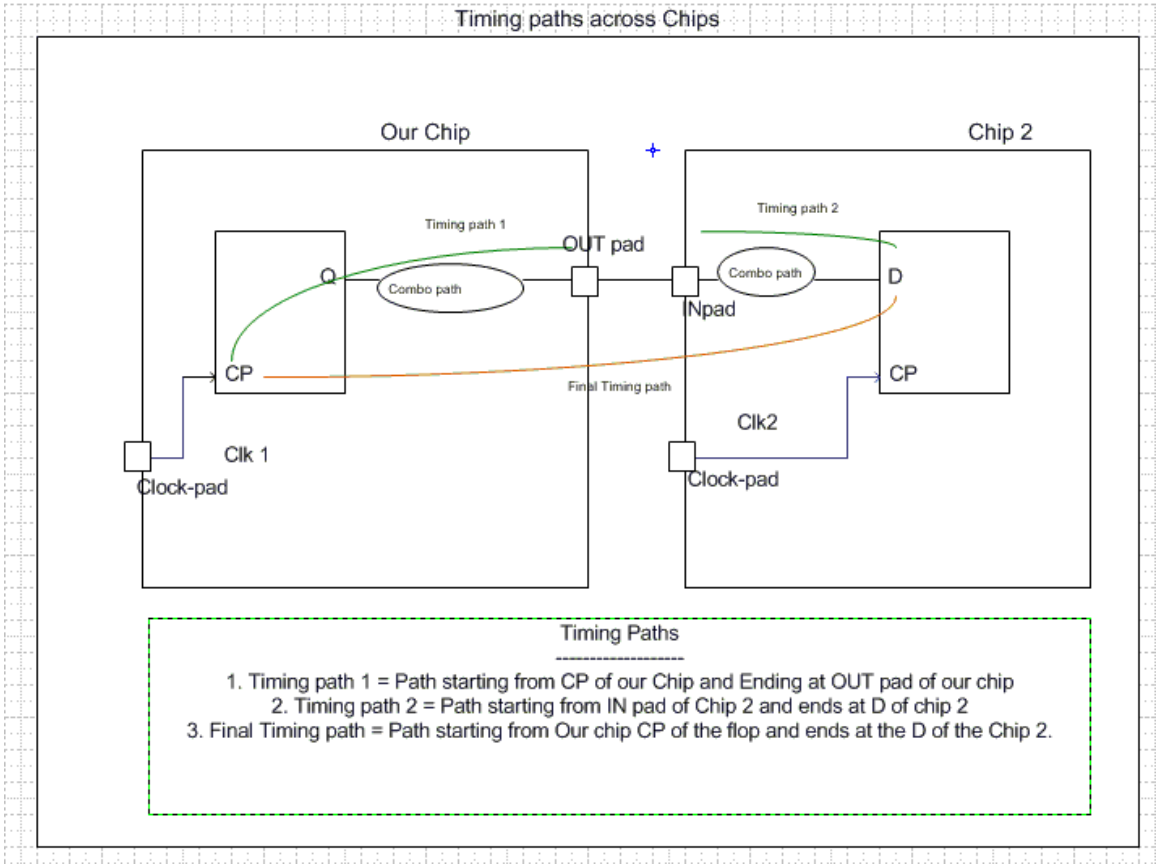
Formal verification uses Mathematical techniquest by prooving the design through assertions or properties. Correctness of the design can be achieved through assertions with out the necessity for simulations. The methods of formal verification are
1. Equivalence checking In this method of checking the designs are compared based on mathematical equations and compared whether they are equal or not .

- Original RTL vs Modified RTL
- RTL vs Netlist
- Golden Netlist vs Modified/Edited Netlist
- Synthesis Netlist vs Place and route Netlist

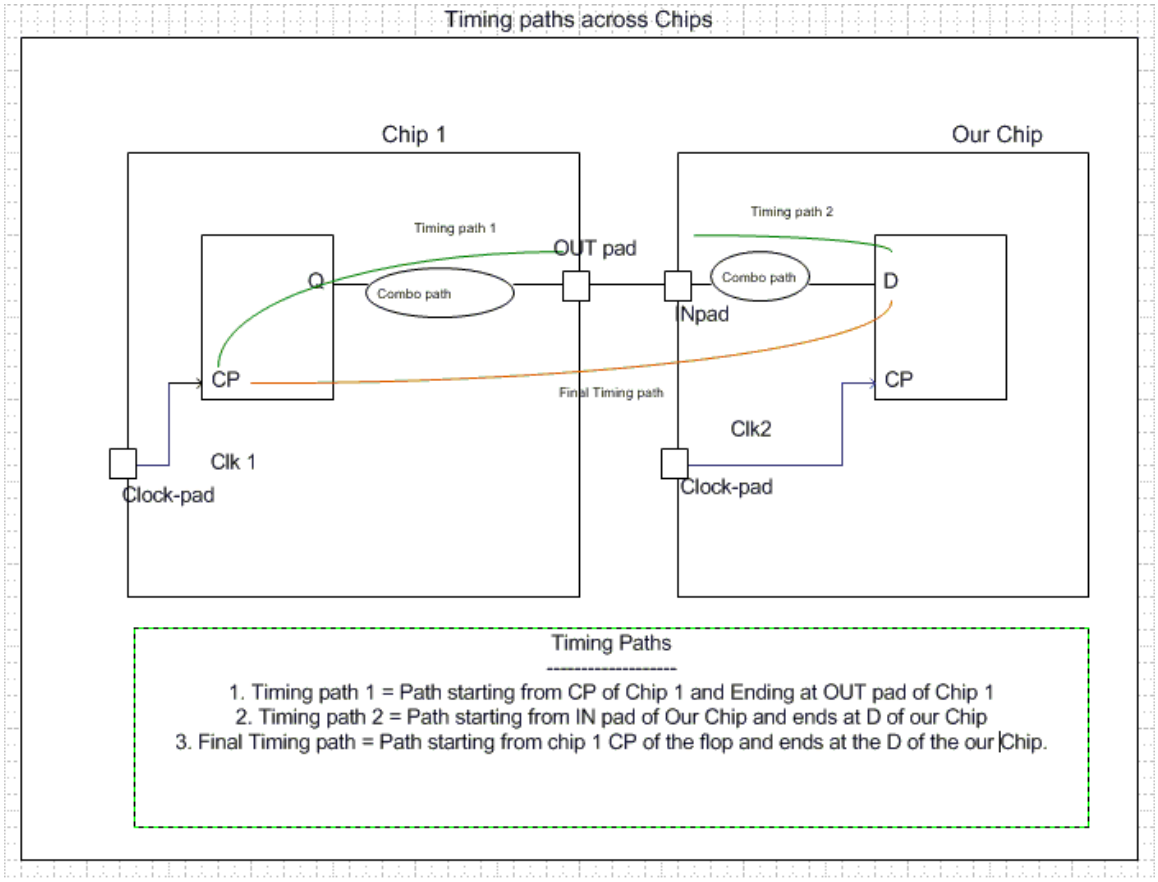
Remember : Formal verification doesnt check for functionality of the RTL code. It will be only checking the equivalence.

2. Model checking Property specification languages like PSL or SVA, are formally analyzed to see if they are always true for a design. This can exhaustively prove if a property is correct, but does tend to suffer from state-space explosion: the time to analyse a design is directly propotional to the amount of states.

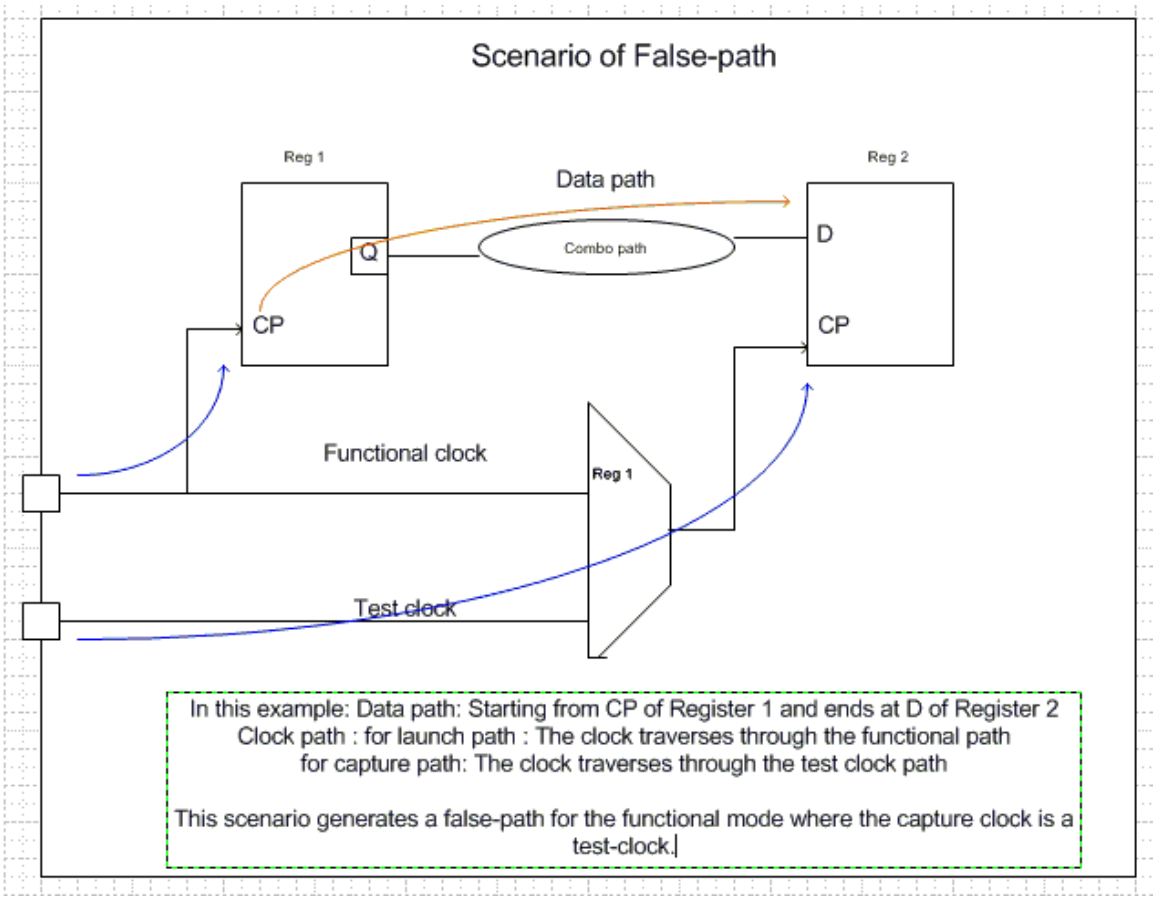
37. How will you time the output paths?



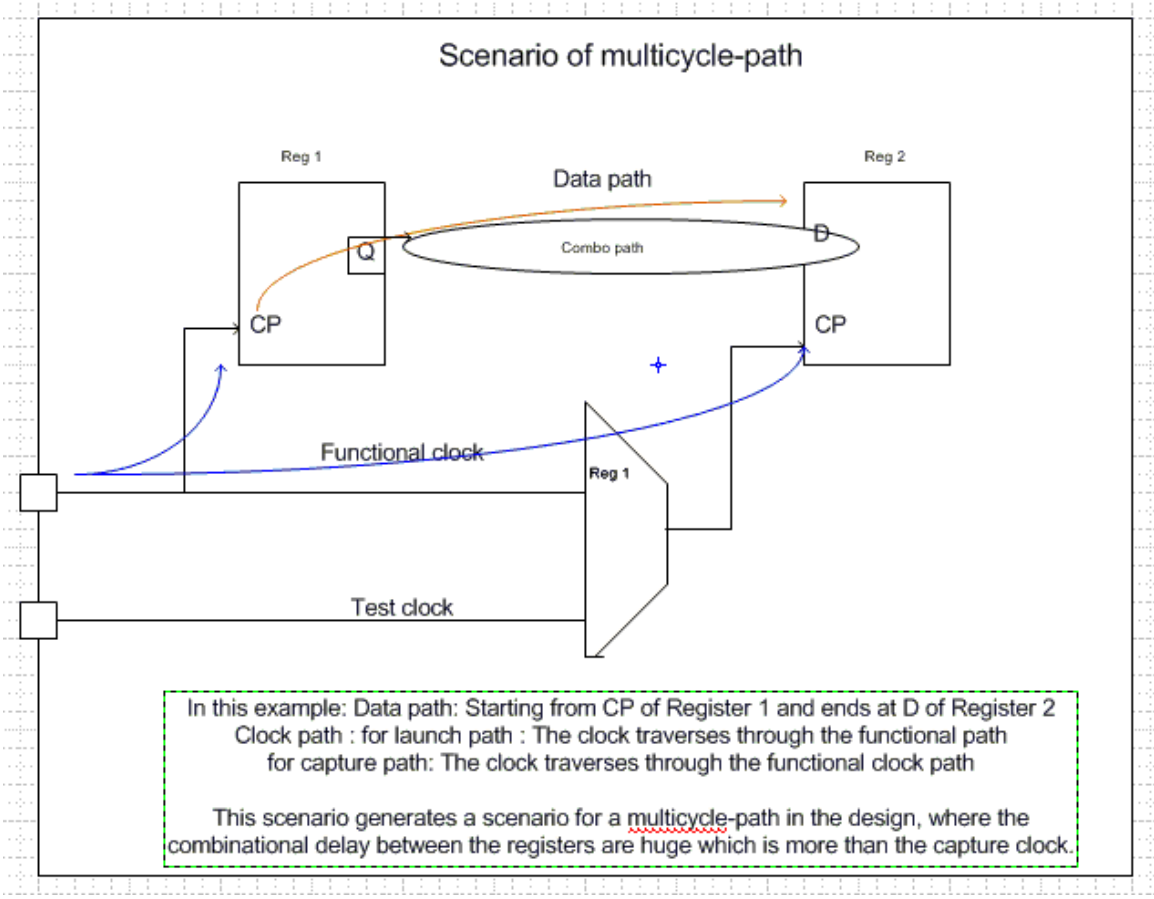
38. How will you time the input paths?

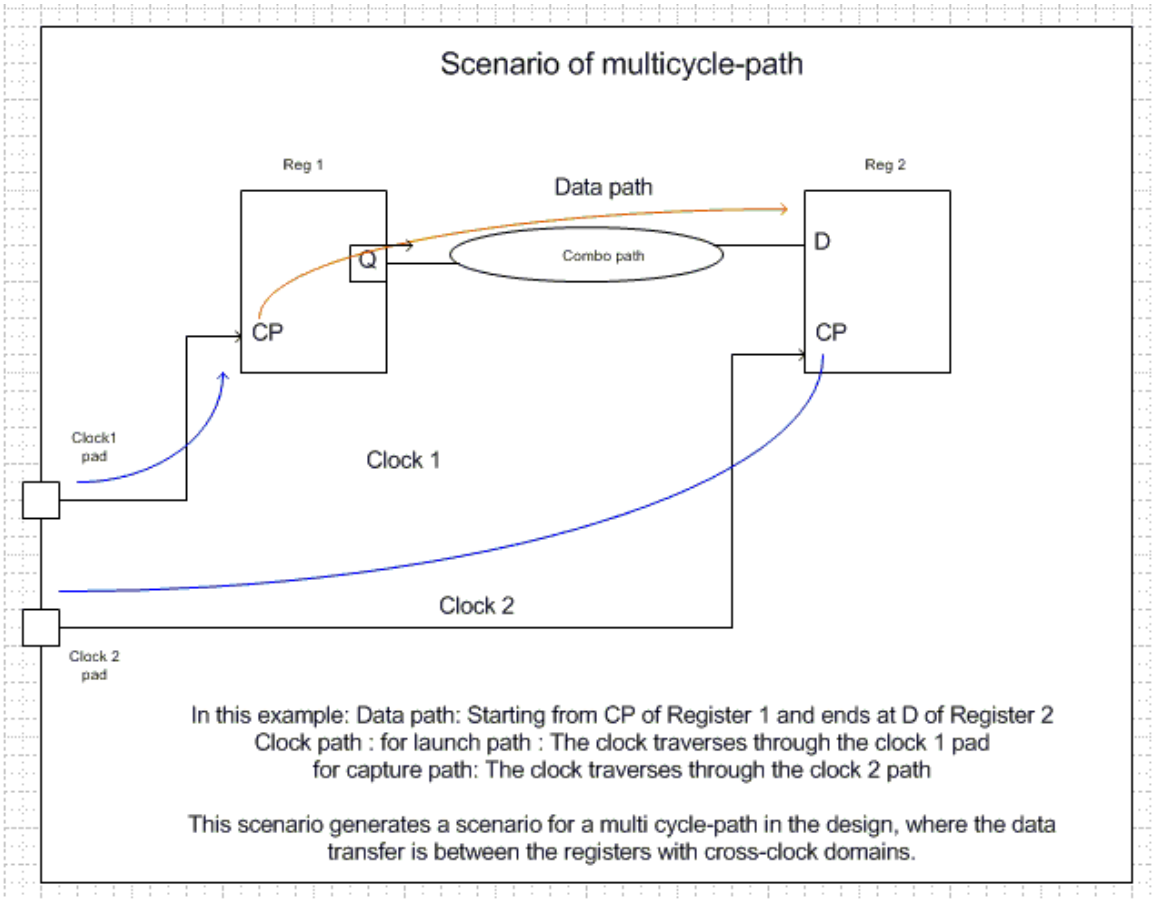


39. what is false path mean in STA and in what scenarios falsepath can come?

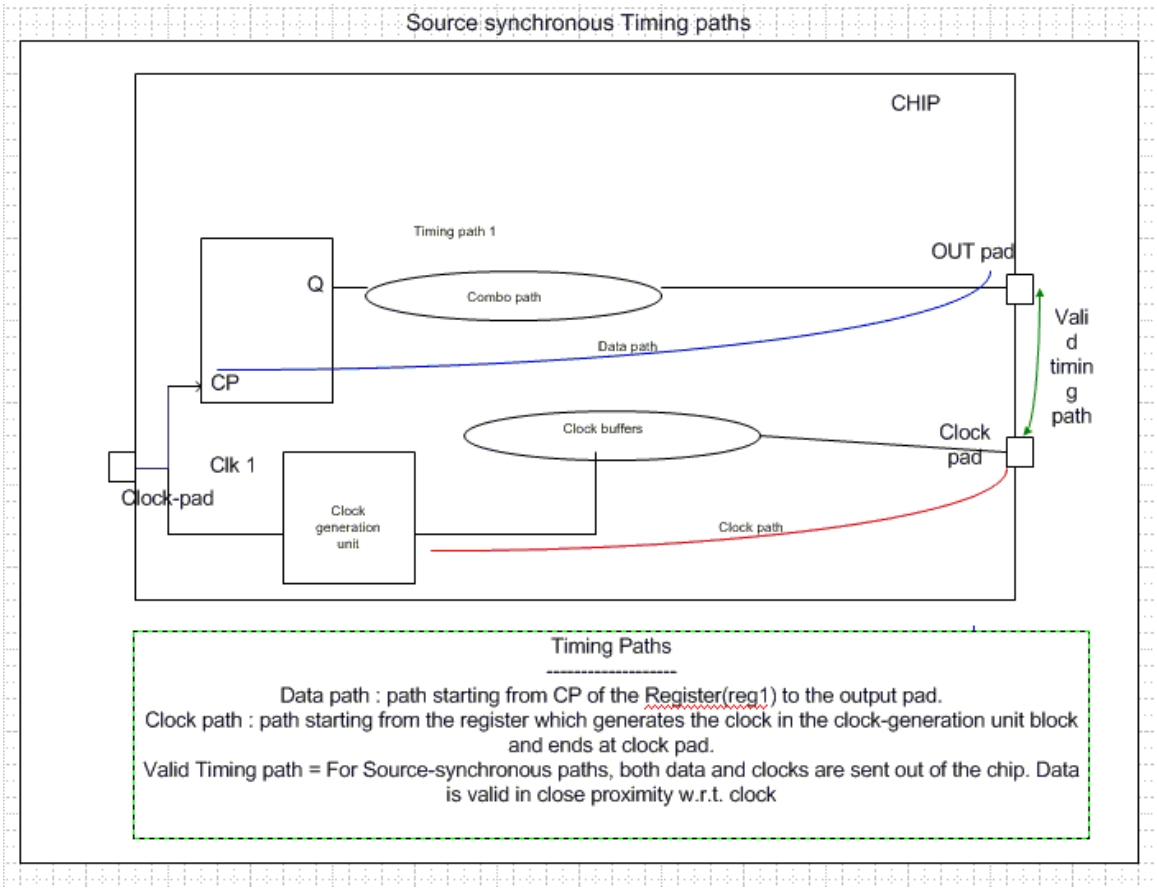


40. what does Multicycle path mean in STA and in what scenarios multicycle paths can come?


Scenario of multicycle-path
Reg 1
Reg 2
Data path
Q
Combo path
D
CP
CP
Functional clock
Reg 1
Test clock
In this example: Data path: Starting from CP of Register 1 and ends at D of Register 2
Clock path : for launch path : The clock traverses through the functional path
for capture path: The clock traverses through the functional clock path
This scenario generates a scenario for a multicycle-path in the design, where the combinational delay between the registers are huge which is more than the capture clock.



41. what are source synchronous paths in STA?



42. assume you have defined latency specified by user both in Master clock and in the Generated clock in STA, how the tool will behave any idea?

If we have defined only Master latency and Generated clock with latency numbers, and the clocks are set to propagated mode after clock-tree, then the Static Timing Analysis Tool, will honour the Generated clock source and Generated clock network latency numbers only and the master clock source and master clock network latencies are ignored.

43. Assume there is a specific requirement to preserve the logic during synthesis, how will do it.

If there is a requirement that some logic needs to be preserved then we can use a command called set_dont_touch or set_dont_design (complete module) and convey the message to the tool not to optimize or smash the logic.

44. We have multiple instances in RTL(Register Transfer Language), do you do anything special during synthesis stage?

While writing RTL(Register Transfer language),say in verilog or in VHDL language, we dont write the same module functionality again and again, we use a concept called as instantiation, where in as per the language, the instanciation of a module will behave like the parent module in terms of functionality, where during synthesis stage we need the full code so that the synthesis tool can study the logic , structure and map it to the library cells, so we use a command in synthesis , called as "UNIQUIFY" which will replace the instantiations with the real logic, because once we are in a synthesis stages we have to visualize as real cells and no more modelling just for functionality alone, we need to visualize in-terms of physical world as well.

45. what do you call an event and when do you call an assertion?

Assertion based Verification Tools, checks whether a statement holds a defined property or not, whereas, Event based Simulators, checks whether there is change in any event, say for every edge of a clock whether there is some activity in a signal or not, in case of an asynchronous designs, checks whether a signal is enabled or not.

1) Explain about setup time and hold time, what will happen if there is setup time and hold tine violation, how to overcome this?

Set up time is the amount of time before the clock edge that the input signal needs to be stable to guarantee it is accepted properly on the clock edge.
Hold time is the amount of time after the clock edge that same input signal has to be held before changing it to make sure it is sensed properly at the clock edge.
Whenever there are setup and hold time violations in any flip-flop, it enters a state where its output is unpredictable: this state is known as metastable state (quasi stable state); at the end of metastable state, the flip-flop settles down to either '1' or '0'. This whole process is known as metastability

2) What is skew, what are problems associated with it and how to minimize it?

In circuit design, clock skew is a phenomenon in synchronous circuits in which the clock signal (sent from the clock circuit) arrives at different components at different times. This is typically due to two causes. The first is a material flaw, which causes a signal to travel faster or slower than expected. The second is distance: if the signal has to travel the entire length of a circuit, it will likely (depending on the circuit's size) arrive at different parts of the circuit at different times. Clock skew can cause harm in two ways. Suppose

that a logic path travels through combinational logic from a source flip-flop to a destination flip-flop. If the destination flip-flop receives the clock tick later than the source flip-flop, and if the logic path delay is short enough, then the data signal might arrive at the destination flip-flop before the clock tick, destroying there the previous data that should have been clocked through. This is called a hold violation because the previous data is not held long enough at the destination flip-flop to be properly clocked through. If the destination flip-flop receives the clock tick earlier than the source flip-flop, then the data signal has that much less time to reach the destination flip-flop before the next clock tick. If it fails to do so, a setup violation occurs, so-called because the new data was not set up and stable before the next clock tick arrived. A hold violation is more serious than a setup violation because it cannot be fixed by increasing the clock period.
Clock skew, if done right, can also benefit a circuit. It can be intentionally introduced to decrease the clock period at which the circuit will operate correctly, and/or to increase the setup or hold safety margins. The optimal set of clock delays is determined by a linear program, in which a setup and a hold constraint appears for each logic path. In this linear program, zero clock skew is merely a feasible point.
Clock skew can be minimized by proper routing of clock signal (clock distribution tree) or putting variable delay buffer so that all clock inputs arrive at the same time

3) What is slack?

'Slack' is the amount of time you have that is measured from when an event 'actually happens' and when it 'must happen'.. The term 'actually happens' can also be taken as being a predicted time for when the event will 'actually happen'.
When something 'must happen' can also be called a 'deadline' so another definition of slack would be the time from when something 'actually happens' (call this Tact) until the deadline (call this Tdead).
Slack = Tdead - Tact.
Negative slack implies that the 'actually happen' time is later than the 'deadline' time...in other words it's too late and a timing violation....you have a timing problem that needs some attention.

4) What is glitch? What causes it (explain with waveform)? How to overcome it?

The following figure shows a gated clock. The gated clock's corresponding timing diagram shows that this implementation can lead to clock glitches, which can cause the flip-flop to clock at the wrong time.

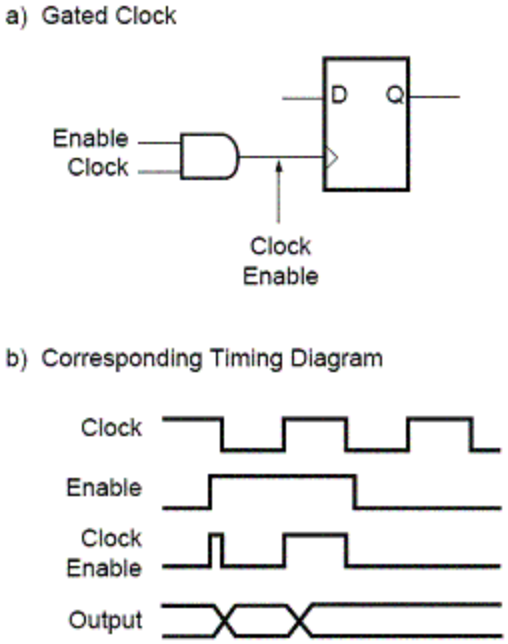
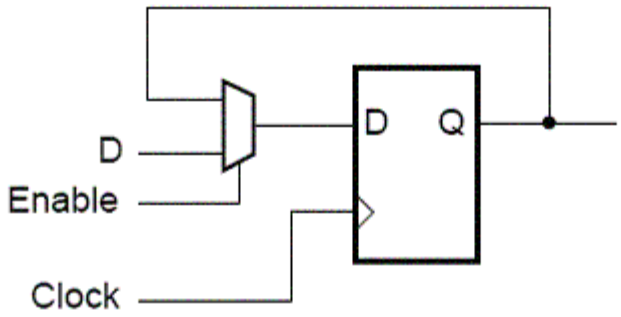


Figure 2-12: **Gated Clock**

The following figure shows a synchronous alternative to the gated clock using a data path. The flip-flop is clocked at every clock cycle and the data path is controlled by an enable. When the enable is Low, the multiplexer feeds the output of the register back on itself. When the enable is High, new data is fed to the flip-flop and the register changes its state

a) Using a Feedback Path



b) Corresponding Timing Diagram

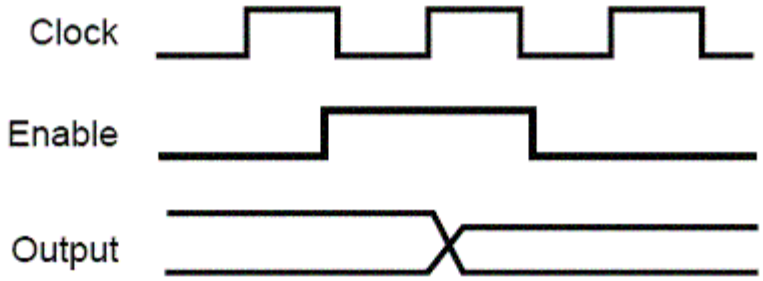


Figure 2-13: **Synchronous Design Using Data Feedback**

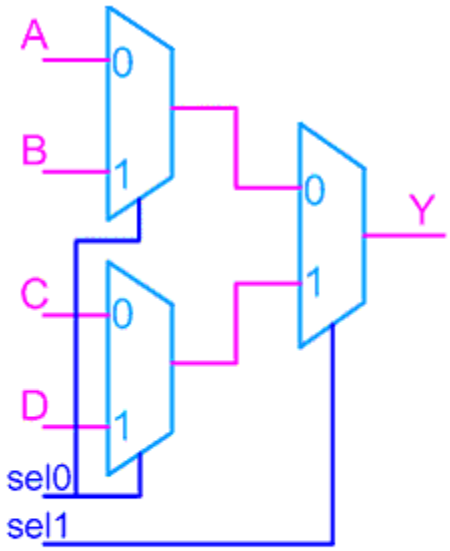
5) Given only two xor gates one must function as buffer and another as inverter?

Tie one of xor gates input to 1 it will act as inverter.
Tie one of xor gates input to 0 it will act as buffer.

6) What is difference between latch and flipflop?

The main difference between latch and FF is that latches are level sensitive while FF are edge sensitive. They both require the use of clock signal and are used in sequential logic. For a latch, the output tracks the input when the clock signal is high, so as long as the clock is logic 1, the output can change if the input also changes. FF on the other hand, will store the input only when there is a rising/falling edge of the clock.

7) Build a 4:1 mux using only 2:1 mux?



Difference between heap and stack?

The Stack is more or less responsible for keeping track of what's executing in our code (or what's been "called"). The Heap is more or less responsible for keeping track of our objects (our data, well... most of it - we'll get to that later.).
Think of the Stack as a series of boxes stacked one on top of the next. We keep track of what's going on in our application by stacking another box on top every time we call a method (called a Frame). We can only use what's in the top box on the stack. When we're done with the top box (the method is done executing) we throw it away and proceed to use the stuff in the previous box on the top of the stack. The Heap is similar except that its purpose is to hold information (not keep track of execution most of the time) so anything in our Heap can be accessed at any time. With the Heap, there are no constraints as to what can be accessed like in the stack. The Heap is like the heap of clean laundry on our bed that we have not taken the time to put away yet - we can grab what we need quickly. The Stack is like the stack of shoe boxes in the closet where we have to take off the top one to get to the one underneath it.

9) Difference between mealy and moore state machine?

A) Mealy and Moore models are the basic models of state machines. A state machine which uses only Entry Actions, so that its output depends on the state, is called a Moore model. A state machine which uses only Input Actions, so that the output depends on the

state and also on inputs, is called a Mealy model. The models selected will influence a design but there are no general indications as to which model is better. Choice of a model depends on the application, execution means (for instance, hardware systems are usually best realized as Moore models) and personal preferences of a designer or programmer

B) Mealy machine has outputs that depend on the state and input (thus, the FSM has the output written on edges)
Moore machine has outputs that depend on state only (thus, the FSM has the output written in the state itself.

Adv and Disadv
In Mealy as the output variable is a function both input and state, changes of state of the state variables will be delayed with respect to changes of signal level in the input variables, there are possibilities of glitches appearing in the output variables. Moore overcomes glitches as output dependent on only states and not the input signal level.
All of the concepts can be applied to Moore-model state machines because any Moore state machine can be implemented as a Mealy state machine, although the converse is not true.
Moore machine: the outputs are properties of states themselves... which means that you get the output after the machine reaches a particular state, or to get some output your machine has to be taken to a state which provides you the output.The outputs are held until you go to some other state Mealy machine:
Mealy machines give you outputs instantly, that is immediately upon receiving input, but the output is not held after that clock cycle.

10) Difference between onehot and binary encoding?

Common classifications used to describe the state encoding of an FSM are Binary (or highly encoded) and One hot.
A binary-encoded FSM design only requires as many flip-flops as are needed to uniquely encode the number of states in the state machine. The actual number of flip-flops required is equal to the ceiling of the log-base-2 of the number of states in the FSM.
A onehot FSM design requires a flip-flop for each state in the design and only one flip-flop (the flip-flop representing the current or "hot" state) is set at a time in a one hot FSM design. For a state machine with 9- 16 states, a binary FSM only requires 4 flip-flops while a onehot FSM requires a flip-flop for each state in the design
FPGA vendors frequently recommend using a onehot state encoding style because flip-flops are plentiful in an FPGA and the combinational logic required to implement a onehot FSM design is typically smaller than most binary encoding styles. Since FPGA performance is typically related to the combinational logic size of the FPGA design, onehot FSMs typically run faster than a binary encoded FSM with larger combinational logic blocks

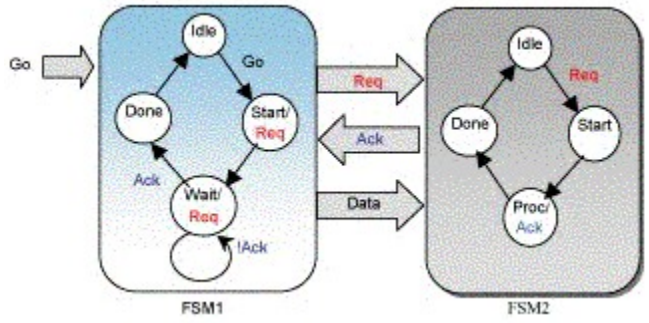
11) What are different ways to synchronize between two clock domains?

Clock Domain Crossing. . .

The following section explains clock domain interfacing

One of the biggest challenges of system-on-chip (SOC) designs is that different blocks operate on independent clocks. Integrating these blocks via the processor bus, memory ports, peripheral busses, and other interfaces can be troublesome because unpredictable behavior can result when the asynchronous interfaces are not properly synchronized

A very common and robust method for synchronizing multiple data signals is a handshake technique as shown in diagram below This is popular because the handshake technique can easily manage changes in clock frequencies, while minimizing latency at the crossing. However, handshake logic is significantly more complex than standard synchronization structures.



FSM1(Transmitter) asserts the req (request) signal, asking the receiver to accept the data on the data bus. FSM2(Receiver) generally a slow module asserts the ack (acknowledge) signal, signifying that it has accepted the data.

it has loop holes: when system Receiver samples the systems Transmitter req line and Transmitter samples system Receiver ack line, they have done it with respect to their internal clock, so there will be setup and hold time violation. To avoid this we go for double or triple stage synchronizers, which increase the MTBF and thus are immune to metastability to a good extent. The figure below shows how this is done.

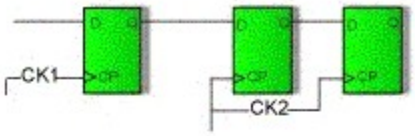


Figure 1a — Single-bit metastability sync

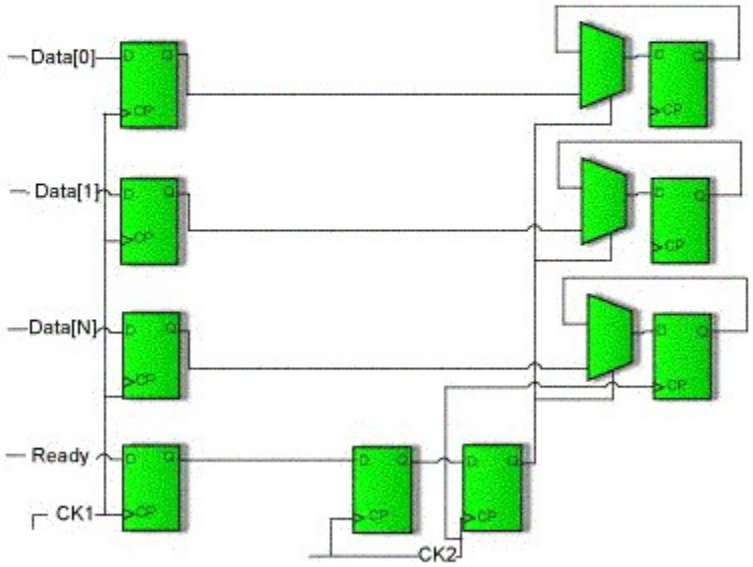


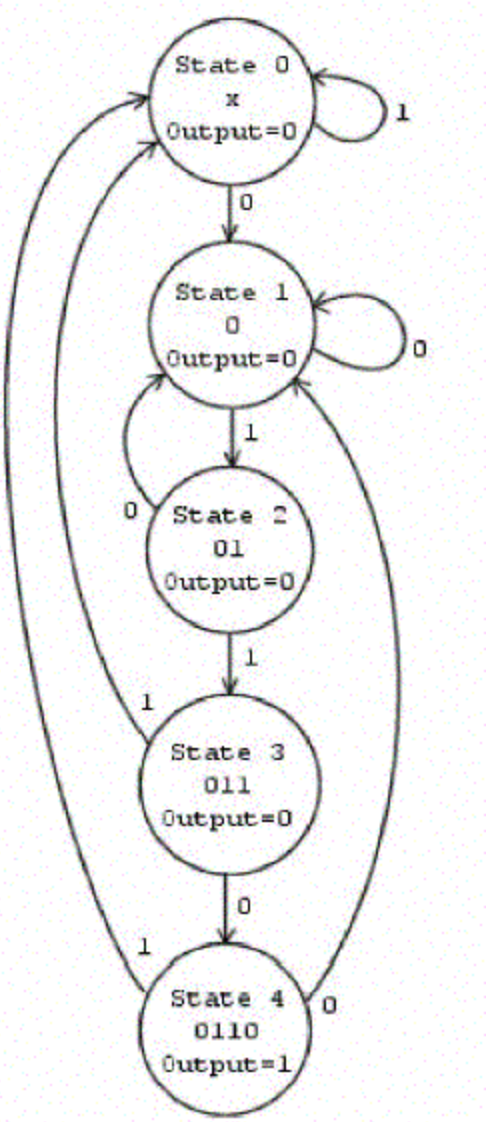
Figure 1b — Multi-bit sync

12) How to calculate maximum operating frequency?

13) How to find out longest path?

You can find answer to this in timing.ppt of presentations section on this site

14) Draw the state diagram to output a "1" for one cycle if the sequence "0110" shows up (the leading 0s cannot be used in more than one sequence)?



15) How to achieve 180 degree exact phase shift?

Never tell using inverter
a) dcm's an inbuilt resource in most of fpga can be configured to get 180 degree phase shift.
b) Bufgds that is differential signaling buffers which are also inbuilt resource of most of FPGA can be used.

16) What is significance of ras and cas in SDRAM?

SDRAM receives its address command in two address words.
It uses a multiplex scheme to save input pins. The first address word is latched into the DRAM chip with the row address strobe (RAS).
Following the RAS command is the column address strobe (CAS) for latching the second

address word.
Shortly after the RAS and CAS strobes, the stored data is valid for reading.

17) Tell some of applications of buffer?

a)They are used to introduce small delays
b)They are used to eliminate cross talk caused due to inter electrode capacitance due to close routing.
c)They are used to support high fanout,eg:bufg

18) Implement an AND gate using mux?

This is the basic question that many interviewers ask. for and gate, give one input as select line,incase if u r giving b as select line, connect one input to logic '0' and other input to a.

19) What will happen if contents of register are shifter left, right?

It is well known that in left shift all bits will be shifted left and LSB will be appended with 0 and in right shift all bits will be shifted right and MSB will be appended with 0 this is a straightforward answer

What is expected is in a left shift value gets Multiplied by 2 eg:consider 0000_1110=14 a left shift will make it 0001_110=28, it the same fashion right shift will Divide the value by 2.

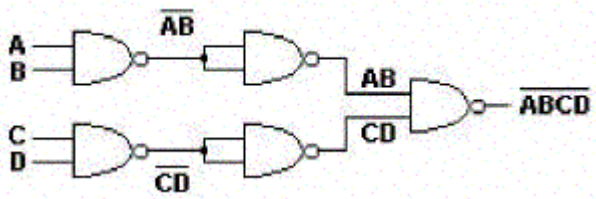
20)Given the following FIFO and rules, how deep does the FIFO need to be to prevent underflow or overflow?

RULES:
1) frequency(clk_A) = frequency(clk_B) / 4
2) period(en_B) = period(clk_A) * 100
3) duty_cycle(en_B) = 25%

Assume clk_B = 100MHz (10ns)
From (1), clk_A = 25MHz (40ns)
From (2), period(en_B) = 40ns * 400 = 4000ns, but we only output for
1000ns,due to (3), so 3000ns of the enable we are doing no output work. Therefore, FIFO size = 3000ns/40ns = 75 entries.

21) Design a four-input NAND gate using only two-input NAND gates.

A:Basically, you can tie the inputs of a NAND gate together to get an inverter, so...



22)Difference between Synchronous and Asynchronous reset.?

Synchronous reset logic will synthesize to smaller flip-flops, particularly if the reset is gated with the logic generating the d-input. But in such a case, the combinational logic gate count grows, so the overall gate count savings may not be that significant.
The clock works as a filter for small reset glitches; however, if these glitches occur near the active clock edge, the Flip-flop could go metastable. In some designs, the reset must be generated by a set of internal conditions. A synchronous reset is recommended for these types of designs because it will filter the logic equation glitches between clock.

Disadvantages of synchronous reset:
Problem with synchronous resets is that the synthesis tool cannot easily distinguish the reset signal from any other data signal.
Synchronous resets may need a pulse stretcher to guarantee a reset pulse width wide enough to ensure reset is present during an active edge of the clock[if you have a gated clock to save power, the clock may be disabled coincident with the assertion of reset. Only an asynchronous reset will work in this situation, as the reset might be removed prior to the resumption of the clock.
Designs that are pushing the limit for data path timing, can not afford to have added gates and additional net delays in the data path due to logic inserted to handle synchronous resets.
Asynchronous reset :
The biggest problem with asynchronous resets is the reset release, also called reset removal. Using an asynchronous reset, the designer is guaranteed not to have the reset added to the data path. Another advantage favoring asynchronous resets is that the circuit can be reset with or without a clock present.
Disadvantages of asynchronous reset: ensure that the release of the reset can occur within one clock period. if the release of the reset occurred on or near a clock edge such that the flip-flops went metastable.

23) Why are most interrupts active low?

This answers why most signals are active low

If you consider the transistor level of a module, active low means the capacitor in the output terminal gets charged or discharged based on low to high and high to low transition respectively. when it goes from high to low it depends on the pull down resistor that pulls it down and it is relatively easy for the output capacitance to discharge rather than charging. hence people prefer using active low signals.

24)Give two ways of converting a two input NAND gate to an inverter?

(a) short the 2 inputs of the nand gate and apply the single input to it.
(b) Connect the output to one of the input and the other to the input signal.

25) What are set up time & hold time constraints? What do they signify? Which one is critical for estimating maximum clock frequency of a circuit?

set up time: - the amount of time the data should be stable before the application of the clock signal, where as the hold time is the amount of time the data should be stable after the application of the clock. Setup time signifies maximum delay constraints; hold time is for minimum delay constraints. Setup time is critical for establishing the maximum clock frequency.

26) Differences between D-Latch and D flip-flop?

D-latch is level sensitive where as flip-flop is edge sensitive. Flip-flops are made up of latches.

27) What is a multiplexer?

Is a combinational circuit that selects binary information from one of many input lines and directs it to a single output line. (2n =>n).

28)How can you convert an SR Flip-flop to a JK Flip-flop?

By giving the feed back we can convert, i.e !Q=>S and Q=>R.Hence the S and R inputs will act as J and K respectively.

29)How can you convert the JK Flip-flop to a D Flip-flop?

By connecting the J input to the K through the inverter.

30)What is Race-around problem?How can you rectify it?

The clock pulse that remains in the 1 state while both J and K are equal to 1 will cause the output to complement again and repeat complementing until the pulse goes back to 0, this is called the race around problem.To avoid this undesirable operation, the clock pulse must have a time duration that is shorter than the propagation delay time of the F-F, this is restrictive so the alternative is master-slave or edge-triggered construction.

31)How do you detect if two 8-bit signals are same?

XOR each bits of A with B (for e.g. A[0] xor B[0]) and so on.the o/p of 8 xor gates are then given as i/p to an 8-i/p nor gate. if o/p is 1 then A=B.

32)7 bit ring counter's initial state is 0100010. After how many clock cycles will it return to the initial state?

6 cycles

33) Convert D-FF into divide by 2. (not latch) What is the max clock frequency the circuit can handle, given the following information?

T_setup= 6nS T_hold = 2nS T_propagation = 10nS

Circuit: Connect Qbar to D and apply the clk at clk of DFF and take the O/P at Q. It gives freq/2. Max. Freq of operation: 1/ (propagation delay+setup time) = 1/16ns = 62.5 MHz

34)Guys this is the basic question asked most frequently. Design all the basic gates(NOT,AND,OR,NAND,NOR,XOR,XNOR) using 2:1 Multiplexer?

Using 2:1 Mux, (2 inputs, 1 output and a select line)
(a) NOT
Give the input at the select line and connect I0 to 1 & I1 to 0. So if A is 1, we will get I1 that is 0 at the O/P.
(b) AND
Give input A at the select line and 0 to I0 and B to I1. O/p is A & B
(c) OR
Give input A at the select line and 1 to I1 and B to I0. O/p will be A | B
(d) NAND
AND + NOT implementations together
(e) NOR
OR + NOT implementations together
(f) XOR
A at the select line B at I0 and ~B at I1. ~B can be obtained from (a) (g) XNOR
A at the select line B at I1 and ~B at I0

35)N number of XNOR gates are connected in series such that the N inputs (A0,A1,A2.....) are given in the following way: A0 & A1 to first XNOR gate and A2 & O/P of First XNOR to second XNOR gate and so on..... Nth XNOR gates output is final output. How does this circuit work? Explain in detail?

If N=Odd, the circuit acts as even parity detector, ie the output will 1 if there are even number of 1's in the N input...This could also be called as odd parity generator since with this additional 1 as output the total number of 1's will be ODD.

If N=Even, just the opposite, it will be Odd parity detector or Even Parity Generator.

36)An assembly line has 3 fail safe sensors and one emergency shutdown switch.The line should keep moving unless any of the following conditions arise:
(i) If the emergency switch is pressed
(ii) If the senor1 and sensor2 are activated at the same time.
(iii) If sensor 2 and sensor3 are activated at the same time.
(iv) If all the sensors are activated at the same time
Suppose a combinational circuit for above case is to be implemented only with NAND Gates. How many minimum number of 2 input NAND gates are required?

No of 2-input NAND Gates required = 6 You can try the whole implementation.

37)Design a circuit that calculates the square of a number? It should not use any multiplier circuits. It should use Multiplexers and other logic?

This is interesting....
1^2=0+1=1
2^2=1+3=4
3^2=4+5=9
4^2=9+7=16
5^2=16+9=25
and so on
See a pattern yet?To get the next square, all you have to do is add the next odd number to the previous square that you found.See how 1,3,5,7 and finally 9 are added.Wouldn't this be a possible solution to your question since it only will use a counter,multiplexer and a couple of adders?It seems it would take n clock cycles to calculate square of n.

38) How will you implement a Full subtractor from a Full adder?

all the bits of subtrahend should be connected to the xor gate. Other input to the xor being one.The input carry bit to the full adder should be made 1. Then the full adder works like a full subtractor

39)A very good interview question... What is difference between setup and hold time. The interviewer was looking for one specific reason , and its really a good answer too..The hint is hold time doesn't depend on clock, why is it so...?

Setup violations are related to two edges of clock, i mean you can vary the clock frequency to correct setup violation. But for hold time, you are only concerned with one edge and does not basically depend on clock frequency.

40)In a 3-bit Johnson's counter what are the unused states?

2(power n)-2n is the one used to find the unused states in johnson counter.
So for a 3-bit counter it is 8-6=2.Unused states=2. the two unused states are 010 and 101

41)The question is to design minimal hardware system, which encrypts 8-bit parallel data. A synchronized clock is provided to this system as well. The output encrypted data should be at the same rate as the input data but no necessarily with the same phase.

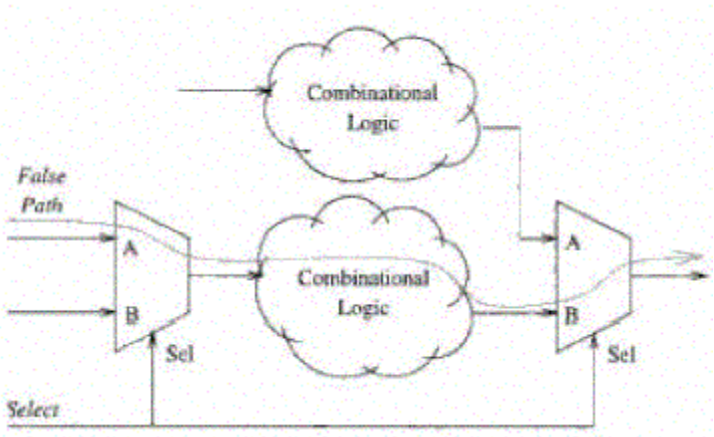
The encryption system is centered around a memory device that perform a LUT (Look-Up Table) conversion. This memory functionality can be achieved by using a PROM, EPROM, FLASH and etc. The device contains an encryption code, which may be burned into the device with an external programmer. In encryption operation, the data_in is an address pointer into a memory cell and the combinatorial logic generates the control signals. This creates a read access from the memory. Then the memory device goes to the appropriate address and outputs the associate data. This data represent the data_in after encryption.

41) What is an LFSR .List a few of its industry applications.?

LFSR is a linear feedback shift register where the input bit is driven by a linear function of the overall shift register value. coming to industrial applications, as far as I know, it is used for encryption and decryption and in BIST(built-in-self-test) based applications..

42)what is false path?how it determine in ckt? what the effect of false path in ckt?

By timing all the paths in the circuit the timing analyzer can determine all the critical paths in the circuit. However, the circuit may have false paths, which are the paths in the circuit which are never exercised during normal circuit operation for any set of inputs. An example of a false path is shown in figure below. The path going from the input A of the first MUX through the combinational logic out through the B input of the second MUS is a false path. This path can never be activated since if the A input of the first MUX is activated, then Sel line will also select the A input of the second MUX.
STA (Static Timing Analysis) tools are able to identify simple false paths; however they are not able to identify all the false paths and sometimes report false paths as critical paths. Removal of false paths makes circuit testable and its timing performance predictable (sometimes faster)

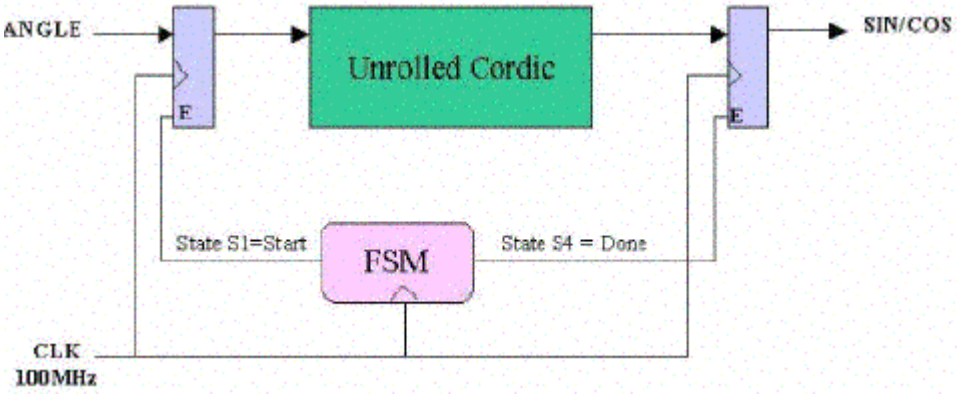


43)Consider two similar processors, one with a clock skew of 100ps and other with a clock skew of 50ps. Which one is likely to have more power? Why?

Clock skew of 50ps is more likely to have clock power. This is because it is likely that low-skew processor has better designed clock tree with more powerful and number of buffers and overheads to make skew better.

44)What are multi-cycle paths?

Multi-cycle paths are paths between registers that take more than one clock cycle to become stable.
For ex. Analyzing the design shown in fig below shows that the output SIN/COS requires 4 clock-cycles after the input ANGLE is latched in. This means that the combinatorial block (the Unrolled Cordic) can take up to 4 clock periods (25MHz) to propagate its result. Place and Route tools are capable of fixing multi-cycle paths problem.



45)You have two counters counting upto 16, built from negedge DFF , First circuit is synchronous and second is "ripple" (cascading), Which circuit has a less propagation delay? Why?

The synchronous counter will have lesser delay as the input to each flop is readily available before the clock edge. Whereas the cascade counter will take long time as the

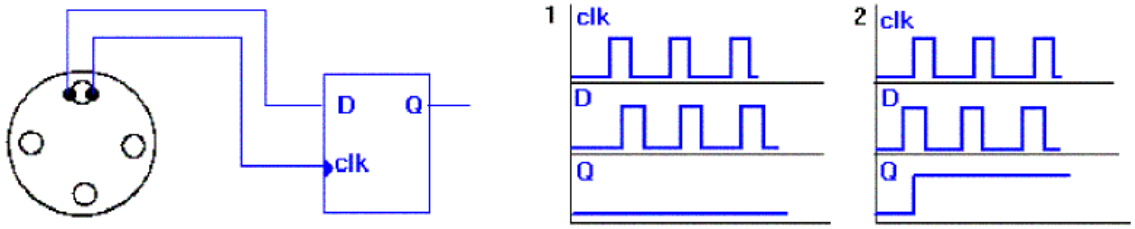
output of one flop is used as clock to the other. So the delay will be propagating. For Eg: 16 state counter = 4 bit counter = 4 Flip flops Let 10ns be the delay of each flop The worst case delay of ripple counter = 10 * 4 = 40ns The delay of synchronous counter = 10ns only.(Delay of 1 flop)

46) what is difference between RAM and FIFO?

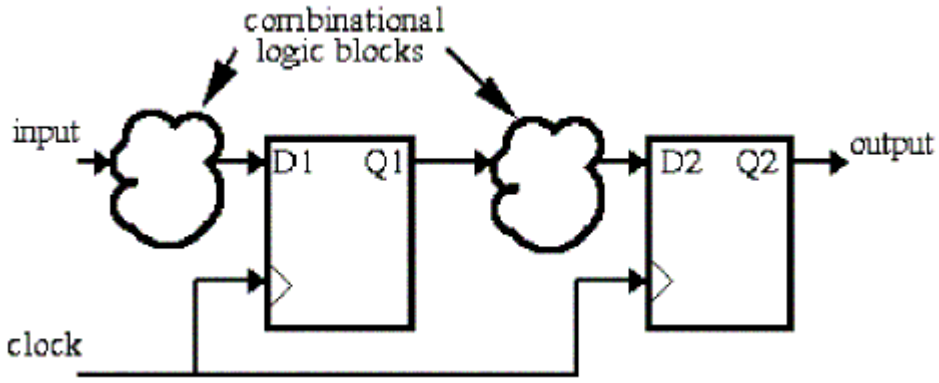
FIFO does not have address lines
Ram is used for storage purpose where as fifo is used for synchronization purpose i.e. when two peripherals are working in different clock domains then we will go for fifo.

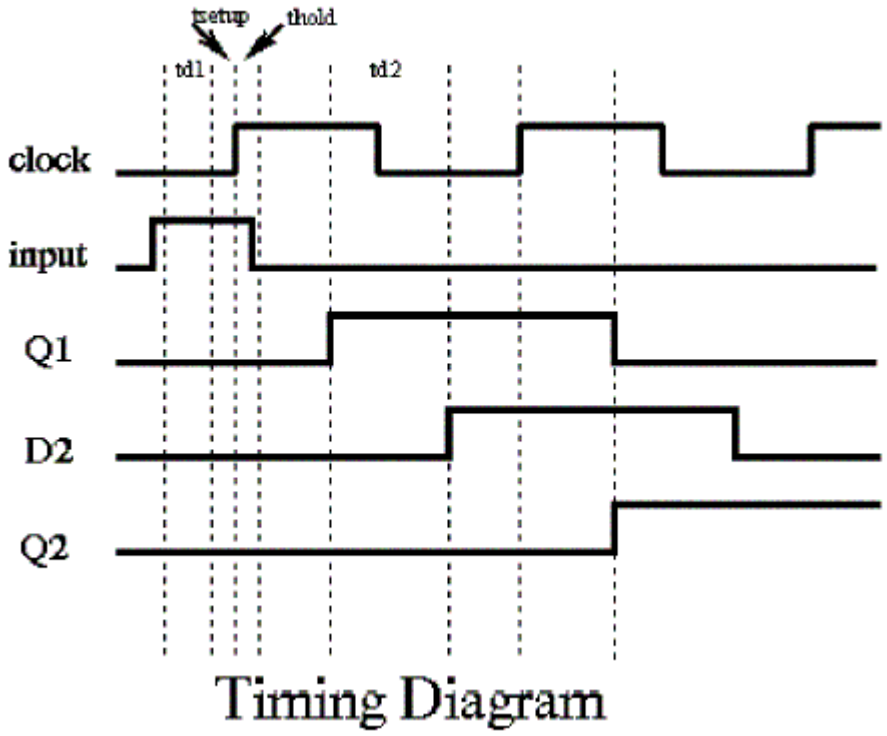
47)The circle can rotate clockwise and back. Use minimum hardware to build a circuit to indicate the direction of rotating.?

2 sensors are required to find out the direction of rotating. They are placed like at the drawing. One of them is connected to the data input of D flip-flop,and a second one - to the clock input. If the circle rotates the way clock sensor sees the light first while D input (second sensor) is zero - the output of the flip-flop equals zero, and if D input sensor "fires" first - the output of the flip-flop becomes high.



48) Draw timing diagrams for following circuit.?





Timing Diagram

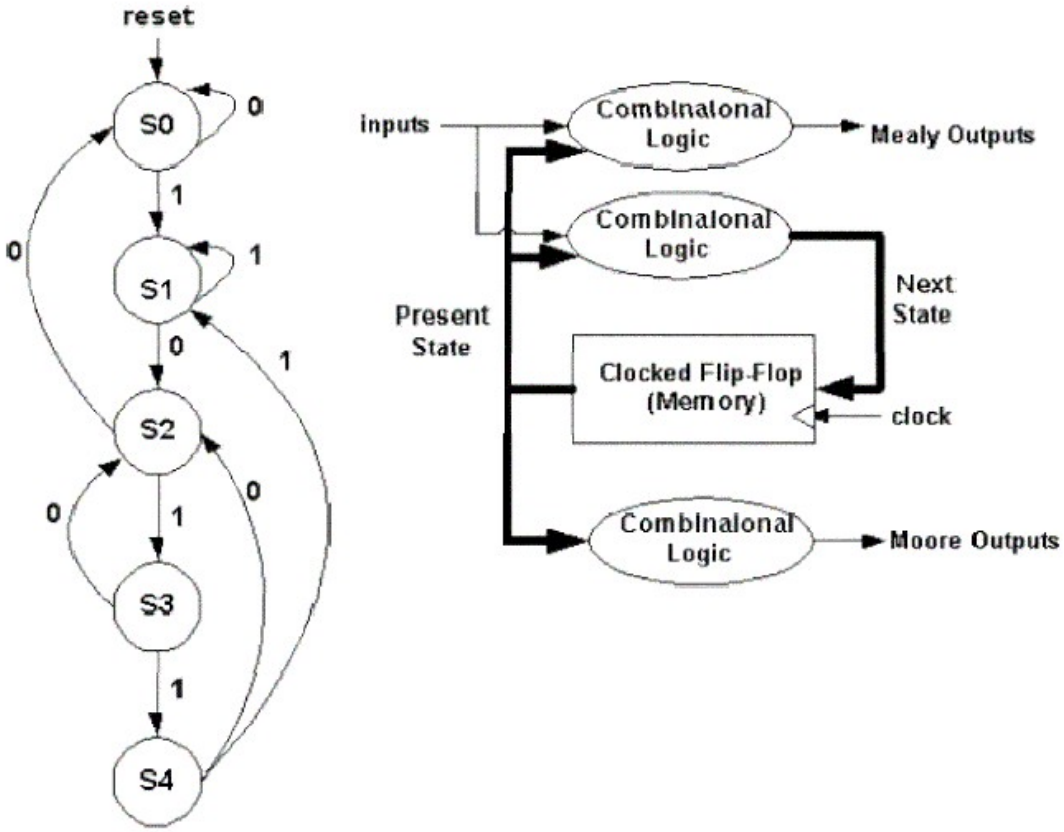
49)Implement the following circuits:
(a) 3 input NAND gate using min no of 2 input NAND Gates
(b) 3 input NOR gate using min no of 2 inpur NOR Gates
(c) 3 input XNOR gate using min no of 2 inpur XNOR Gates
Assuming 3 inputs A,B,C?

3 input NAND:
Connect :
a) A and B to the first NAND gate
b) Output of first Nand gate is given to the two inputs of the second NAND gate (this basically realizes the inverter functionality)
c) Output of second NAND gate is given to the input of the third NAND gate, whose other input is C
((A NAND B) NAND (A NAND B)) NAND C Thus, can be implemented using '3' 2-input NAND gates. I guess this is the minimum number of gates that need to be used.
3 input NOR:
Same as above just interchange NAND with NOR ((A NOR B) NOR (A NOR B)) NOR C
3 input XNOR:
Same as above except the inputs for the second XNOR gate, Output of the first XNOR gate is one of the inputs and connect the second input to ground or logical '0'
((A XNOR B) XNOR 0)) XNOR C

50) Is it possible to reduce clock skew to zero? Explain your answer ?

Even though there are clock layout strategies (H-tree) that can in theory reduce clock skew to zero by having the same path length from each flip-flop from the pll, process variations in R and C across the chip will cause clock skew as well as a pure H-Tree scheme is not practical (consumes too much area).

51)Design a FSM (Finite State Machine) to detect a sequence 10110?

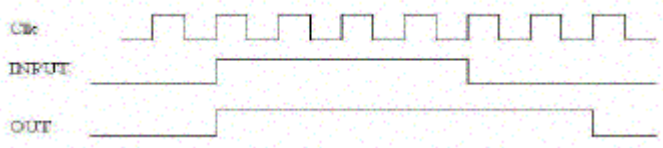


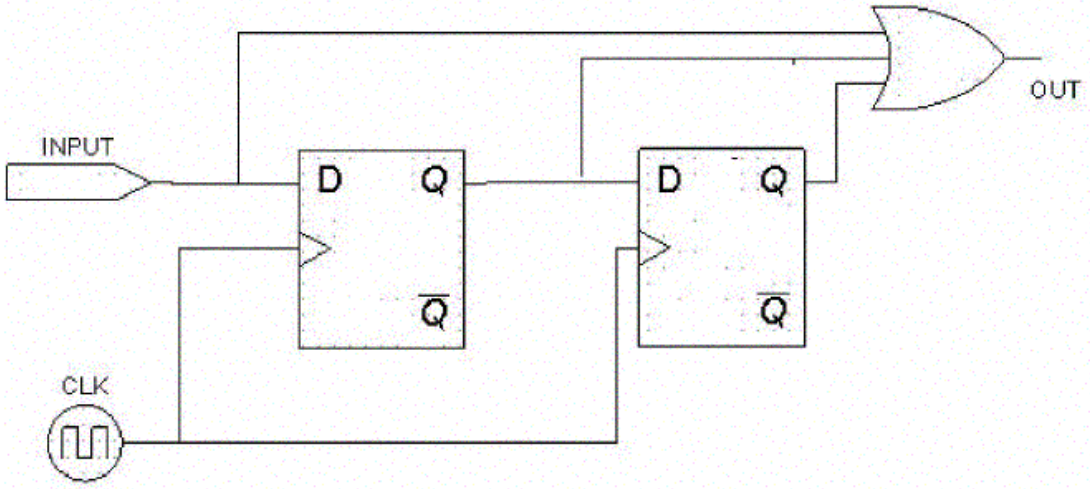
52)Convert D-FF into divide by 2. (not latch)? What is the max clock frequency of the circuit , given the following information?
T_setup= 6nS
T_hold = 2nS
T_propagation = 10nS

Circuit:
Connect Qbar to D and apply the clk at clk of DFF and take the O/P at Q. It gives freq/2.
Max. Freq of operation:
1/ (propagation delay+setup time) = 1/16ns = 62.5 MHz

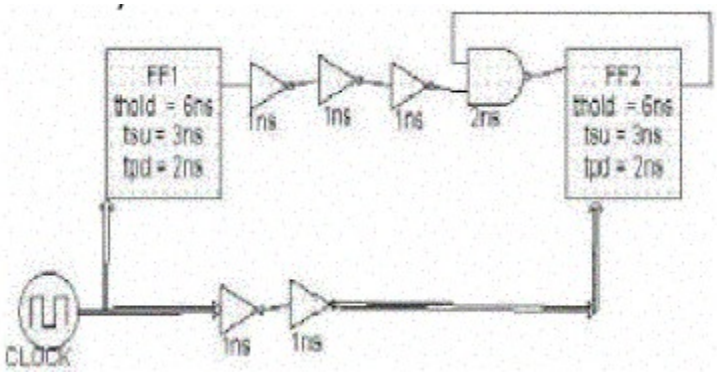
53)Give the circuit to extend the falling edge of the input by 2 clock pulses?The

waveforms are shown in the following figure.



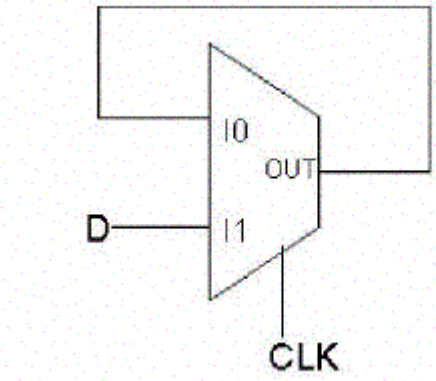


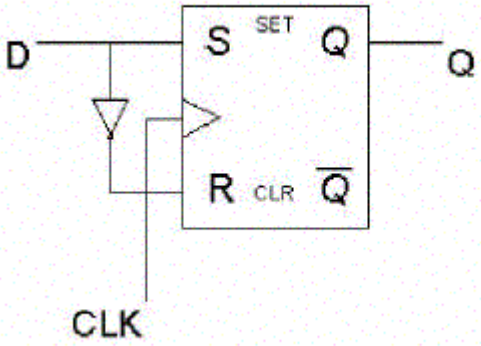
54) For the Circuit Shown below, What is the Maximum Frequency of Operation?Are there any hold time violations for FF2? If yes, how do you modify the circuit to avoid them?



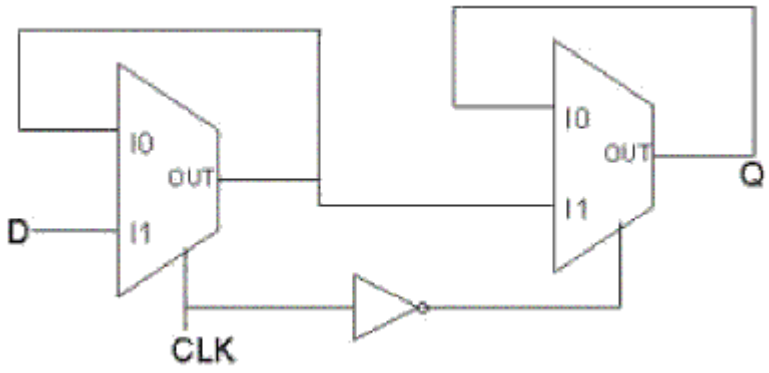
The minumum time period = 3+2+(1+1+1) = 8ns Maximum Frequency = 1/8n= 125MHz.
And there is a hold time violation in the circuit,because of feedback, if you observe, tcq2+AND gate delay is less than thold2,To avoid this we need to use even number of inverters(buffers). Here we need to use 2 inverters each with a delay of 1ns. then the hold time value exactly meets.

55)Design a D-latch using (a) using 2:1 Mux (b) from S-R Latch ?





56)How to implement a Master Slave flip flop using a 2 to 1 mux?

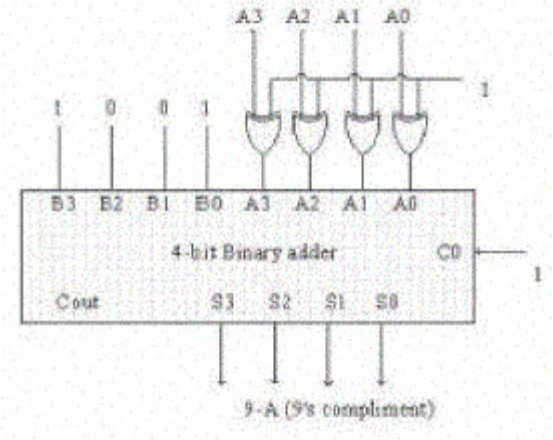


57)how many 2 input xor's are needed to inplement 16 input parity generator ?

It is always n-1 Where n is number of inputs.So 16 input parity generator will require 15 two input xor's .

58)Design a circuit for finding the 9's compliment of a BCD number using 4-bit binary adder and some external logic gates?

9's compliment is nothing but subracting the given no from 9.So using a 4 bit binary adder we can just subract the given binary no from 1001(i.e. 9).Here we can use the 2's compliment method addition.



59) what is Difference between writeback and write through cache?

A caching method in which modifications to data in the cache aren't copied to the cache source until absolutely necessary. Write-back caching is available on many microprocessors , including all Intel processors since the 80486. With these microprocessors, data modifications to data stored in the L1 cache aren't copied to main memory until absolutely necessary. In contrast, a write-through cache performs all write operations in parallel -- data is written to main memory and the L1 cache simultaneously. Write-back caching yields somewhat better performance than write-through caching because it reduces the number of write operations to main memory. With this performance improvement comes a slight risk that data may be lost if the system crashes. A write-back cache is also called a copy-back cache.

60)Difference between Synchronous,Asynchronous & Isynchronous communication?

Sending data encoded into your signal requires that the sender and receiver are both using the same enconding/decoding method, and know where to look in the signal to find data. Asynchronous systems do not send separate information to indicate the encoding or clocking information. The receiver must decide the clocking of the signal on it's own. This means that the receiver must decide where to look in the signal stream to find ones and zeroes, and decide for itself where each individual bit stops and starts. This information is not in the data in the signal sent from transmitting unit.

Synchronous systems negotiate the connection at the data-link level before communication begins. Basic synchronous systems will synchronize two clocks before transmission, and reset their numeric counters for errors etc. More advanced systems may negotiate things like error correction and compression.

Time-dependent. it refers to processes where data must be delivered within certain time constraints. For example, Multimedia stream require an isochronous transport mechanism to ensure that data is delivered as fast as it is displayed and to ensure that the audio is synchronized with the video.

61) What are different ways Multiply & Divide?

Binary Division by Repeated Subtraction

Set quotient to zero
Repeat while dividend is greater than or equal to divisor

- Subtract divisor from dividend
- Add 1 to quotient

End of repeat block
quotient is correct, dividend is remainder
STOP

Binary Division by Shift and Subtract

Basically the reverse of the mutliply by shift and add.

Set quotient to 0
Align leftmost digits in dividend and divisor
Repeat

- **If** that portion of the dividend above the divisor is greater than or equal to the divisor
  - **Then** subtract divisor from that portion of the dividend and
  - Concatentate 1 to the right hand end of the quotient
  - **Else** concatentate 0 to the right hand end of the quotient

- Shift the divisor one place right

Until dividend is less than the divisor
quotient is correct, dividend is remainder
STOP

Binary Multiply - Repeated Shift and Add

Repeated shift and add - starting with a result of 0, shift the second multiplicand to correspond with each 1 in the first multiplicand and add to the result. Shifting each position left is equivalent to multiplying by 2, just as in decimal representation a shift left is equivalent to multiplying by 10.

Set result to 0
Repeat

- Shift 2nd multiplicand left until rightmost digit is lined up with leftmost 1 in first multiplicand
- Add 2nd multiplicand in that position to result
- Remove that 1 from 1st multiplicand

Until 1st multiplicand is zero
Result is correct
STOP

62)What is a SoC (System On Chip), ASIC, "full custom chip", and an FPGA?

There are no precise definitions. Here is my sense of it all. First, 15 years ago, people were unclear on exactly what VLSI meant. Was it 50000 gates? 100000 gates? was is just anything bigger than LSI? My professor simply told me that; VLSI is a level of complexity and integration in a chip that demands Electronic Design Automation tools in order to succeed. In other words, big enough that manually drawing lots of little blue, red and green lines is too much for a human to reasonably do. I think that, likewise, SoC is that level of integration onto a chip that demands more expertise beyond traditional skills of electronics. In other words, pulling off a SoC demands Hardware, Software, and Systems Engineering talent. So, trivially, SoCs aggressively combine HW/SW on a single chip. Maybe more pragmatically, SoC just means that ASIC and Software folks are learning a little bit more about each other's techniques and tools than they did before. Two other interpretations of SoC are 1) a chip that integrates various IP (Intellectual

Property) blocks on it and is thus highly centered with issues like Reuse, and 2) a chip integrating multiple classes of electronic circuitry such as Digital CMOS, mixed-signal digital and analog (e.g. sensors, modulators, A/Ds), DRAM memory, high voltage power, etc.

ASIC stands for “Application Specific Integrated Circuit”. A chip designed for a specific application. Usually, I think people associate ASICs with the Standard Cell design methodology. Standard Cell design and the typical “ASIC flow” usually means that designers are using Hardware Description Languages, Synthesis and a library of primitive cells (e.g. libraries containing AND, NAND, OR, NOR, NOT, FLIP-FLOP, LATCH, ADDER, BUFFER, PAD cells that are wired together (real libraries are not this simple, but you get the idea..). Design usually is NOT done at a transistor level. There is a high reliance on automated tools because the assumption is that the chip is being made for a SPECIFIC APPLICATION where time is of the essence. But, the chip is manufactured from scratch in that no pre-made circuitry is being programmed or reused. ASIC designer may, or may not, even be aware of the locations of various pieces of circuitry on the chip since the tools do much of the construction, placement and wiring of all the little pieces.

Full Custom, in contrast to ASIC (or Standard Cell), means that every geometric feature going onto the chip being designed (think of those pretty chip pictures we have all seen) is controlled, more or less, by the human design. Automated tools are certainly used to wire up different parts of the circuit and maybe even manipulate (repeat, rotate, etc.) sections of the chip. But, the human designer is actively engaged with the physical features of the circuitry. Higher human crafting and less reliance on standard cells takes more time and implies higher NRE costs, but lowers RE costs for standard parts like memories, processors, uarts, etc.

FPGAs, or Field Programmable Gate Arrays are completely designed chips that designers load a programming pattern into to achieve a specific digital function. A bit pattern (almost like a software program) is loaded into the already manufactured device which essentially interconnects lots of available gates to meet the designers purposes. FPGAs are sometimes thought of as a “Sea of Gates” where the designer specifies how they are connected. FPGA designers often use many of the same tools that ASIC designers use, even though the FPGA is inherently more flexible. All these things can be intermixed in hybrid sorts of ways. For example, FPGAs are now available that have microprocessor embedded within them which were designed in a full custom manner, all of which now demands “SoC” types of HW/SW integration skills from the designer.

63)What is "Scan" ?

Scan Insertion and ATPG helps test ASICs (e.g. chips) during manufacture. If you know what JTAG boundary scan is, then Scan is the same idea except that it is done inside the chip instead of on the entire board. Scan tests for defects in the chip's circuitry after it is manufactured (e.g. Scan does not help you test whether your Design *functions* as intended). ASIC designers usually implement the scan themselves and occurs just after synthesis. ATPG (Automated Test Pattern Generation) refers to the creation of "Test

Vectors" that the Scan circuitry enables to be introduced into the chip. Here's a brief summary:

- Scan Insertion is done by a tool and results in all (or most) of your design's flip-flops to be replaced by special "Scan Flip-flops". Scan flops have additional inputs/outputs that allow them to be configured into a "chain" (e.g. a big shift register) when the chip is put into a test mode.
- The Scan flip-flops are connected up into a chain (perhaps multiple chains)
- The ATPG tool, which knows about the scan chain you've created, generates a series of test vectors.
- The ATPG test vectors include both "Stimulus" and "Expected" bit patterns. These bit vectors are shifted into the chip on the scan chains, and the chips reaction to the stimulus is shifted back out again.
- The ATE (Automated Test Equipment) at the chip factory can put the chip into the scan test mode, and apply the test vectors. If any vectors do not match, then the chip is defective and it is thrown away.
- Scan/ATPG tools will strive to maximize the "coverage" of the ATPG vectors. In other words, given some measure of the total number of nodes in the chip that could be faulty (shorted, grounded, "stuck at 1", "stuck at 0"), what percentage of them can be detected with the ATPG vectors? Scan is a good technology and can achive high coverage in the 90% range.
- Scan testing does not solve all test problems. Scan testing typically does not test memories (no flip-flops!), needs a gate-level netlist to work with, and can take a long time to run on the ATE.
- FPGA designers may be unfamiliar with scan since FPGA testing has already been done by the FPGA manufacturer. ASIC designers do not have this luxury and must handle all the manufacturing test details themselves.

1. what is Body effect ?

The threshold voltage of a MOSFET is affected by the voltage which is applied to the back contact. The voltage difference between the source and the bulk, VBS changes the width of the depletion layer and therefore also the voltage across the oxide due to the change of the charge in the depletion region. This results in a difference in threshold voltage which equals the difference in charge in the depletion region divided by the oxide capacitance, yielding:.

$$\Delta V_T = \frac{\sqrt{2\varepsilon_s q N_a}}{C_{OX}} (\sqrt{(2\phi_F + V_{SB})} - \sqrt{2\phi_F})$$

3.What are stansdard Cell's?

In semiconductor design, standard cell methodology is a method of designing Application Specific Integrated Circuits (ASICs) with mostly digital-logic features. Standard cell methodology is an example of design abstraction, whereby a low-level VLSI-layout is encapsulated into an abstract logic representation (such as a NAND gate). Cell-based methodology (the general class that standard-cell belongs to) makes it possible for one designer to focus on the high-level (logical function) aspect of digital-design, while another designer focused on the implementation (physical) aspect. Along with semiconductor manufacturing advances, standard cell methodology was responsible for allowing designers to scale ASICs from comparatively simple single-function ICs (of several thousand gates), to complex multi-million gate devices (SoC).

3.What are Design Rule Check (DRC) and Layout Vs Schematic (LVS) ?

Design Rule Check (DRC) and Layout Vs Schematic (LVS) are verification processes. Reliable device fabrication at modern deep submicrometre (0.13 µm and below) requires strict observance of transistor spacing, metal layer thickness, and power density rules. DRC exhaustively compares the physical netlist against a set of "foundry design rules" (from the foundry operator), then flags any observed violations. LVS is a process that confirms that the layout has the same structure as the associated schematic; this is typically the final step in the layout process. The LVS tool takes as an input a schematic diagram and the extracted view from a layout. It then generates a netlist from each one and compares them. Nodes, ports, and device sizing are all compared. If they are the same, LVS passes and the designer can continue. Note: LVS tends to consider transistor fingers to be the same as an extra-wide transistor. For example, 4 transistors in parallel (each 1 um wide), a 4-finger 1 um transistor, and a 4 um transistor are all seen as the same by the LVS tool. Functionality of .lib files will be taken from spice models and added as an attribute to the .lib file.

4.What is Antenna effect ?

The antenna effect, more formally plasma induced gate oxide damage, is an efffect that can potentially cause yield and reliability problems during the manufacture of MOS integrated circuits[1][2][3][4][5]. Fabs normally supply antenna rules, which are rules that must be obeyed to avoid this problem. A violation of such rules is called an antenna violation. The word antenna is somewhat of a misnomer in this context—the problem is really the collection of charge, not the normal meaning of antenna, which is a device for converting electromagnetic fields to/from electrical currents. Occasionally the phrase antenna effect is used this context[6] but this is less common since there are many effects[7] and the phrase does not make clear which is meant.

4.What are steps involved in Semiconductor device fabrication ?

This is a list of processing techniques that are employed numerous times in a modern electronic device and do not necessarily imply a specific order.

Wafer processing
Wet cleans
Photolithography
Ion implantation (in which dopants are embedded in the wafer creating regions of increased (or decreased) conductivity)
Dry etching
Wet etching
Plasma ashing
Thermal treatments
Rapid thermal anneal
Furnace anneals
Thermal oxidation
Chemical vapor deposition (CVD)
Physical vapor deposition (PVD)
Molecular beam epitaxy (MBE)
Electrochemical Deposition (ECD). See Electroplating
Chemical-mechanical planarization (CMP)
Wafer testing (where the electrical performance is verified)
Wafer backgrinding (to reduce the thickness of the wafer so the resulting chip can be put into a thin device like a smartcard or PCMCIA card.)
Die preparation
Wafer mounting
Die cutting
IC packaging
Die attachment
IC Bonding
Wire bonding
Flip chip
Tab bonding
IC encapsulation
Baking
Plating
Lasermarking
Trim and form
IC testing

5.What is Clock distribution network ?

In a synchronous digital system, the clock signal is used to define a time reference for the movement of data within that system. The clock distribution network distributes the clock

signal(s) from a common point to all the elements that need it. Since this function is vital to the operation of a synchronous system, much attention has been given to the characteristics of these clock signals and the electrical networks used in their distribution. Clock signals are often regarded as simple control signals; however, these signals have some very special characteristics and attributes.
Clock signals are typically loaded with the greatest fanout, travel over the greatest distances, and operate at the highest speeds of any signal, either control or data, within the entire synchronous system. Since the data signals are provided with a temporal reference by the clock signals, the clock waveforms must be particularly clean and sharp. Furthermore, these clock signals are particularly affected by technology scaling (see Moore's law), in that long global interconnect lines become significantly more resistive as line dimensions are decreased. This increased line resistance is one of the primary reasons for the increasing significance of clock distribution on synchronous performance. Finally, the control of any differences and uncertainty in the arrival times of the clock signals can severely limit the maximum performance of the entire system and create catastrophic race conditions in which an incorrect data signal may latch within a register. The clock distribution network often takes a significant fraction of the power consumed by a chip. Furthermore, significant power can be wasted in transitions within blocks, even when their output is not needed. These observations have lead to a power saving technique called clock gating, which involves adding logic gates to the clock distribution tree, so portions of the tree can be turned off when not needed.

5.What is Clock Gating ?

Clock gating is one of the power-saving techniques used on many synchronous circuits including the Pentium 4 processor. To save power, clock gating refers to adding additional logic to a circuit to prune the clock tree, thus disabling portions of the circuitry where flip flops do not change state. Although asynchronous circuits by definition do not have a "clock", the term "perfect clock gating" is used to illustrate how various clock gating techniques are simply approximations of the data-dependent behavior exhibited by asynchronous circuitry, and that as the granularity on which you gate the clock of a synchronous circuit approaches zero, the power consumption of that circuit approaches that of an asynchronous circuit.

6.What is Netlist ?

Netlists are connectivity information and provide nothing more than instances, nets, and perhaps some attributes. If they express much more than this, they are usually considered to be a hardware description language such as Verilog, VHDL, or any one of several specific languages designed for input to simulators.
Most netlists either contain or refer to descriptions of the parts or devices used. Each time a part is used in a netlist, this is called an "instance." Thus, each instance has a "master", or "definition". These definitions will usually list the connections that can be made to that kind of device, and some basic properties of that device. These connection points are called "ports" or "pins", among several other names.
An "instance" could be anything from a vacuum cleaner, microwave oven, or light bulb,

to a resistor, capacitor, or integrated circuit chip.
Instances have "ports". In the case of a vacuum cleaner, these ports would be the three metal prongs in the plug. Each port has a name, and in continuing the vacuum cleaner example, they might be "Neutral", "Live" and "Ground". Usually, each instance will have a unique name, so that if you have two instances of vacuum cleaners, one might be "vac1" and the other "vac2". Besides their names, they might otherwise be identical.
Nets are the "wires" that connect things together in the circuit. There may or may not be any special attributes associated with the nets in a design, depending on the particular language the netlist is written in, and that language's features.
Instance based netlists usually provide a list of the instances used in a design. Along with each instance, either an ordered list of net names are provided, or a list of pairs provided, of an instance port name, along with the net name to which that port is connected. In this kind of description, the list of nets can be gathered from the connection lists, and there is no place to associate particular attributes with the nets themselves. SPICE is perhaps the most famous of instance-based netlists.
Net-based netlists usually describe all the instances and their attributes, then describe each net, and say which port they are connected on each instance. This allows for attributes to be associated with nets. EDIF is probably the most famous of the net-based netlists.

6.What Physical timing closure ?

Physical timing closure is the process by which an FPGA or a VLSI design with a physical representation is modified to meet its timing requirements. Most of the modifications are handled by EDA tools based on directives given by a designer. The term is also sometimes used as a characteristic, which is ascribed to an EDA tool, when it provides most of the features required in this process. Physical timing closure became more important with submicrometre technologies, as more and more steps of the design flow had to be made timing-aware. Previously only logic synthesis had to satisfy timing requirements. With present deep submicrometre technologies it is unthinkable to perform any of the design steps of placement, clock-tree synthesis and routing without timing constraints. Logic synthesis with these technologies is becoming less important. It is still required, as it provides the initial netlist of gates for the placement step, but the timing requirements do not need to be strictly satisfied any more. When a physical representation of the circuit is available, the modifications required to achieve timing closure are carried out by using more accurate estimations of the delays.

7.What Physical verification ?

Physical verification of the design, involves DRC(Design rule check), LVS(Layout versus schematic) Check, XOR Checks, ERC (Electrical Rule Check) and Antenna Checks.
XOR Check

This step involves comparing two layout databases/GDS by XOR operation of the layout geometries. This check results a database which has all the mismatching geometries in both the layouts. This check is typically run after a metal spin, where in the re-spin

database/GDS is compared with the previously taped out database/GDS.
Antenna Check

Antenna checks are used to limit the damage of the thin gate oxide during the manufacturing process due to charge accumulation on the interconnect layers (metal, polysilicon) during certain fabrication steps like Plasma etching, which creates highly ionized matter to etch. The antenna basically is a metal interconnect, i.e., a conductor like polysilicon or metal, that is not electrically connected to silicon or grounded, during the processing steps of the wafer. If the connection to silicon does not exist, charges may build up on the interconnect to the point at which rapid discharge does take place and permanent physical damage results to thin transistor gate oxide. This rapid and destructive phenomenon is known as the antenna effect. The Antenna ratio is defined as the ratio between the physical area of the conductors making up the antenna to the total gate oxide area to which the antenna is electrically connected.
ERC (Electrical rule check)

ERC (Electrical rule check) involves checking a design for all well and substrate areas for proper contacts and spacings thereby ensuring correct power and ground connections. ERC steps can also involve checks for unconnected inputs or shorted outputs.

7.What is Stuck-at fault ?

A Stuck-at fault is a particular fault model used by fault simulators and Automatic test pattern generation (ATPG) tools to mimic a manufacturing defect within an integrated circuit. Individual signals and pins are assumed to be stuck at Logical '1', '0' and 'X'. For example, an output is tied to a logical 1 state during test generation to assure that a manufacturing defect with that type of behavior can be found with a specific test pattern. Likewise the output could be tied to a logical 0 to model the behavior of a defective circuit that cannot switch its output pin.

7.What is Different Logic family ?

Listed here in rough chronological order of introduction along with their usual abbreviations of Logic family
* Diode logic (DL)
* Direct-coupled transistor logic (DCTL)
* Complementary transistor logic (CTL)
* Resistor-transistor logic (RTL)
* Resistor-capacitor transistor logic (RCTL)
* Diode-transistor logic (DTL)
* Emitter coupled logic (ECL) also known as Current-mode logic (CML)
* Transistor-transistor logic (TTL) and variants
* P-type Metal Oxide Semiconductor logic (PMOS)
* N-type Metal Oxide Semiconductor logic (NMOS)
* Complementary Metal-Oxide Semiconductor logic (CMOS)
* Bipolar Complementary Metal-Oxide Semiconductor logic (BiCMOS)

* Integrated Injection Logic (I2L)

7.What is Different Types of IC packaging ?

IC are packaged in many types they are: * BGA1
* BGA2
* Ball grid array
* CPGA
* Ceramic ball grid array
* Cerquad
* DIP-8
* Die attachment
* Dual Flat No Lead
* Dual in-line package
* Flat pack
* Flip chip
* Flip-chip pin grid array
* HVQFN
* LQFP
* Land grid array
* Leadless chip carrier
* Low insertion force
* Micro FCBGA
* Micro Leadframe Package
* MicroLeadFrame
* Mini-Cartridge
* Multi-Chip Module
* OPGA
* PQFP
* Package on package
* Pin grid array
* Plastic leaded chip carrier
* QFN
* QFP
* Quadruple in-line package
* ROM cartridge
* Shrink Small-Outline Package
* Single in-line package
* Small-Outline Integrated Circuit
* Staggered Pin Grid Array
* Surface-mount technology
* TO220
* TO3
* TO92
* TQFP
* TSSOP

* Thin small-outline package
* Through-hole technology
* UICC
* Zig-zag in-line package

8.What is Substrate coupling ?

In an integrated circuit, a signal can couple from one node to another via the substrate. This phenomenon is referred to as substrate coupling or substrate noise coupling.
The push for reduced cost, more compact circuit boards, and added customer features has provided incentives for the inclusion of analog functions on primarily digital MOS integrated circuits (ICs) forming mixed-signal ICs. In these systems, the speed of digital circuits is constantly increasing, chips are becoming more densely packed, interconnect layers are added, and analog resolution is increased. In addition, recent increase in wireless applications and its growing market are introducing a new set of aggressive design goals for realizing mixed-signal systems. Here, the designer integrates radio frequency (RF) analog and base band digital circuitry on a single chip. The goal is to make single-chip radio frequency integrated circuits (RFICs) on silicon, where all the blocks are fabricated on the same chip. One of the advantages of this integration is low power dissipation for portability due to a reduction in the number of package pins and associated bond wire capacitance. Another reason that an integrated solution offers lower power consumption is that routing high-frequency signals off-chip often requires a 50O impedance match, which can result in higher power dissipation. Other advantages include improved high-frequency performance due to reduced package interconnect parasitics, higher system reliability, smaller package count, smaller package interconnect parasitics, and higher integration of RF components with VLSI-compatible digital circuits. In fact, the single-chip transceiver is now a reality.

8.What is Latchup ?

A latchup is the inadvertent creation of a low-impedance path between the power supply rails of an electronic component, triggering a parasitic structure, which then acts as a short circuit, disrupting proper functioning of the part and possibly even leading to its destruction due to overcurrent. A power cycle is required to correct this situation. The parasitic structure is usually equivalent to a thyristor (or SCR), a PNPN structure which acts as a PNP and an NPN transistor stacked next to each other. During a latchup when one of the transistors is conducting, the other one begins conducting too. They both keep each other in saturation for as long as the structure is forward-biased and some current flows through it - which usually means until a power-down. The SCR parasitic structure is formed as a part of the totem-pole PMOS and NMOS transistor pair on the output drivers of the gates.

1) What is latch up?

Latch-up pertains to a failure mechanism wherein a parasitic thyristor (such as a parasitic silicon controlled rectifier, or SCR) is inadvertently created within a circuit, causing a high amount of current to continuously flow through it once it is accidentally triggered or turned on. Depending on the circuits involved, the amount of current flow produced by this mechanism can be large enough to result in permanent destruction of the device due to electrical overstress (EOS) .

2)Why is NAND gate preferred over NOR gate for fabrication?

NAND is a better gate for design than NOR because at the transistor level the mobility of electrons is normally three times that of holes compared to NOR and thus the NAND is a faster gate.
Additionally, the gate-leakage in NAND structures is much lower. If you consider t_phl and t_plh delays you will find that it is more symmetric in case of NAND (the delay profile), but for NOR, one delay is much higher than the other(obviously t_plh is higher since the higher resistance p mos's are in series connection which again increases the resistance).

3)What is Noise Margin? Explain the procedure to determine Noise Margin

The minimum amount of noise that can be allowed on the input stage for which the output will not be effected.

4)Explain sizing of the inverter?

In order to drive the desired load capacitance we have to increase the size (width) of the inverters to get an optimized performance.

5) How do you size NMOS and PMOS transistors to increase the threshold voltage?

6) What is Noise Margin? Explain the procedure to determine Noise Margin?

The minimum amount of noise that can be allowed on the input stage for which the output will not be effected.

7) What happens to delay if you increase load capacitance?

delay increases.

8)What happens to delay if we include a resistance at the output of a CMOS circuit?

Increases. (RC delay)

9)What are the limitations in increasing the power supply to reduce delay?

The delay can be reduced by increasing the power supply but if we do so the heating effect comes because of excessive power, to compensate this we have to increase the die size which is not practical.

10)How does Resistance of the metal lines vary with increasing thickness and increasing length?

R = (*l) / A.

11)For CMOS logic, give the various techniques you know to minimize power consumption?

Power dissipation=CV2f ,from this minimize the load capacitance, dc voltage and the operating frequency.

12) What is Charge Sharing? Explain the Charge Sharing problem while sampling data from a Bus?

In the serially connected NMOS logic the input capacitance of each gate shares the charge with the load capacitance by which the logical levels drastically mismatched than that of the desired once. To eliminate this load capacitance must be very high compared to the input capacitance of the gates (approximately 10 times).

13)Why do we gradually increase the size of inverters in buffer design? Why not give the output of a circuit to one large inverter?

Because it can not drive the output load straight away, so we gradually increase the size to get an optimized performance.

14)What is Latch Up? Explain Latch Up with cross section of a CMOS Inverter. How do you avoid Latch Up?

Latch-up is a condition in which the parasitic components give rise to the Establishment of low resistance conducting path between VDD and VSS with Disastrous results.

15) Give the expression for CMOS switching power dissipation?

CV2

16) What is Body Effect?

In general multiple MOS devices are made on a common substrate. As a result, the substrate voltage of all devices is normally equal. However while connecting the devices serially this may result in an increase in source-to-substrate voltage as we proceed

vertically along the series chain (Vsb1=0, Vsb2 0).Which results Vth2>Vth1.

17) Why is the substrate in NMOS connected to Ground and in PMOS to VDD?

we try to reverse bias not the channel and the substrate but we try to maintain the drain,source junctions reverse biased with respect to the substrate so that we dont loose our current into the substrate.

18) What is the fundamental difference between a MOSFET and BJT ?

In MOSFET, current flow is either due to electrons(n-channel MOS) or due to holes(p-channel MOS) - In BJT, we see current due to both the carriers.. electrons and holes. BJT is a current controlled device and MOSFET is a voltage controlled device.

19)Which transistor has higher gain. BJT or MOS and why?

BJT has higher gain because it has higher transconductance.This is because the current in BJT is exponentially dependent on input where as in MOSFET it is square law.

20)Why do we gradually increase the size of inverters in buffer design when trying to drive a high capacitive load? Why not give the output of a circuit to one large inverter?

We cannot use a big inverter to drive a large output capacitance because, who will drive the big inverter? The signal that has to drive the output cap will now see a larger gate capacitance of the BIG inverter.So this results in slow raise or fall times .A unit inverter can drive approximately an inverter thats 4 times bigger in size. So say we need to drive a cap of 64 unit inverter then we try to keep the sizing like say 1,4,16,64 so that each inverter sees a same ratio of output to input cap. This is the prime reason behind going for progressive sizing.

21)In CMOS technology, in digital design, why do we design the size of pmos to be higher than the nmos.What determines the size of pmos wrt nmos. Though this is a simple question try to list all the reasons possible?

In PMOS the carriers are holes whose mobility is less[aprrox half] than the electrons, the carriers in NMOS. That means PMOS is slower than an NMOS. In CMOS technology, nmos helps in pulling down the output to ground ann PMOS helps in pulling up the output to Vdd. If the sizes of PMOS and NMOS are the same, then PMOS takes long time to charge up the output node. If we have a larger PMOS than there will be more carriers to charge the node quickly and overcome the slow nature of PMOS . Basically we do all this to get equal rise and fall times for the output node.

22)Why PMOS and NMOS are sized equally in a Transmission Gates?

In Transmission Gate, PMOS and NMOS aid each other rather competing with each

other. That's the reason why we need not size them like in CMOS. In CMOS design we have NMOS and PMOS competing which is the reason we try to size them proportional to their mobility.

23)All of us know how an inverter works. What happens when the PMOS and NMOS are interchanged with one another in an inverter?

I have seen similar Qs in some of the discussions. If the source & drain also connected properly...it acts as a buffer. But suppose input is logic 1 O/P will be degraded 1 Similarly degraded 0;

24)A good question on Layouts. Give 5 important Design techniques you would follow when doing a Layout for Digital Circuits?

a)In digital design, decide the height of standard cells you want to layout.It depends upon how big your transistors will be.Have reasonable width for VDD and GND metal paths.Maintaining uniform Height for all the cell is very important since this will help you use place route tool easily and also incase you want to do manual connection of all the blocks it saves on lot of area.
b)Use one metal in one direction only, This does not apply for metal 1. Say you are using metal 2 to do horizontal connections, then use metal 3 for vertical connections, metal4 for horizontal, metal 5 vertical etc...
c)Place as many substrate contact as possible in the empty spaces of the layout.
d)Do not use poly over long distances as it has huge resistances unless you have no other choice.
e)Use fingered transistors as and when you feel necessary.
f)Try maintaining symmetry in your design. Try to get the design in BIT Sliced manner.

25)What is metastability? When/why it will occur?Different ways to avoid this?

Metastable state: A un-known state in between the two logical known states.This will happen if the O/P cap is not allowed to charge/discharge fully to the required logical levels.
One of the cases is: If there is a setup time violation, metastability will occur,To avoid this, a series of FFs is used (normally 2 or 3) which will remove the intermediate states.

26)Let A and B be two inputs of the NAND gate. Say signal A arrives at the NAND gate later than signal B. To optimize delay of the two series NMOS inputs A and B which one would you place near to the output?

The late coming signals are to be placed closer to the output node ie A should go to the nmos that is closer to the output.

1)Explain zener breakdown and avalanche breakdown?

A thermally generated carrier (part of reverse saturation current) falls down the junction barrier and acquires energy from the applied potential. This carriers collides with a crystal ion and imparts sufficient energy to disrupt a covalent bond.In addition to the original carrier, a new electron-hole pair has been generated. These carriers may also pick up sufficient energy and creates still another electron-hole pair. This cumulative process is called the Avalanche breakdown.
A reverse electric field at the junction causes a strong force to be applied on a bounded electron by the field to tear it out of its covalent bond. The new hole-electron pair which is created increases the reverse current, called zener breakdown.

2)What is Instrumentation Amplifier(IA) and what are all the advantages?

An instrumentation amplifier is a differential op-amp circuit providing high input impedances with ease of gain adjustment by varying a single resistor

3) What is the fundamental difference between a MOSFET and BJT ?

In MOSFET,current flow is either due to electrons(n-channel MOS) or due to holes(p-channel MOS)
- In BJT, we see current due to both the carriers.. electrons and holes. BJT is a current controlled device and MOSFET is a voltage controlled device.

4) What is the basic difference between Analog and Digital Design?

Digital design is distinct from analog design. In analog circuits we deal with physical signals which are continuous in amplitude and time. Ex: biological data, sesimic signals, sensor output, audio, video etc.

Analog design is quite challenging than digital design as analog circuits are sensitive to noise, operating voltages, loading conditions and other conditions which has severe effects on performance. Even process technology poses certain topological limitations on the circuit. Analog designer has to deal with real time continuous signals and even manipulate them effectively even in harsh environment and in brutal operating conditions.
Digital design on the other hand is easier to process and has great immunity to noise. No room for automation in analog design as every application requires a different design. Where as digital design can be automated. Analog circuits generally deal with instantaneous value of voltage and current(real time). Can take any value within the domain of specifications for the device.consists of passive elements which contribute to the noise(thermal) of the circuit . They are usually more sensitive to external noise more so because for a particular function a analog design
uses lot less transistors providing design challenges over process corners and temperature ranges. deals with a lot of device level physics and the state of the transistor plays a very important role Digital Circuits on the other hand deal with only two logic levels 0 and

1(Is it true that according to quantum mechanics there is a third logic level?) deal with lot more transistors for a particular logic, easier to design complex designs, flexible logic synthesis and greater speed although at the cost of greater power. Less sensitive to noise. design and analysis of such circuits is dependant on the clock. challenge lies in negating the timing and load delays and ensuring there is no set up or hold violation.

5)What is ring oscillator? And derive the freq of operation?

Ring oscillator circuit is a coupled inverter chain with the output being connected to the input as feedback. The number of stages(inverters) is always odd to ensure that there is no single stable state(output value). sometimes one of the stages consists of a logic gate which is used to initialise and control the circuit. The total time period of operation is the product of 2*number of gates and gate(inverter) delay. And frequency of operation will be inverse of time period.
Application: used as prototype circuits for modeling and designing new semiconductor processes due to simplicity in design and ease of use. Also forms a part of clock recovery circuit.

6)What are RTL, Gate, Metal and FIB fixes? What is a "sewing kits"?

There are several ways to fix an ASIC-based design. >From easiest to most extreme:

RTL Fix -> Gate Fix -> Metal Fix -> FIB Fix

First, let's review fundementals. A standard-cell ASIC consists of at least 2 dozen manufactured layers/masks. Lower layers conists of materialsmaking up the actual CMOS transistors and gates of the design. The upper 3-6 layers are metal layers used ti connect everything together. ASICs, of course, are not intended to be flexible like an FPGA, however, important "fixes" can be made during the manufacturing process. The progression of possible fixes in the manufacturing life cycle is as listed above.

An RTL fix means you change the Verilog/VHDL code and you resynthesize. This usually implies a new Plance&Route. RTL fixes would also imply new masks, etc. etc. In other words - start from scratch.

A Gate Fix means that a select number of gates and their interconections may be added or subtracted from the design (e.g. the netlist). This avoids resynthesis. Gate fixes preserve the previous synthesis effort and involve manually editing a gate-level netlist - adding gates, removing gates, etc. Gate level fixes affect ALL layers of the chip and all masks.

A Metal Fix means that only the upper metal interconnect layers are affected. Connections may be broken or made, but new cells may not be added. A Sewing Kit is a means of adding a new gate into the design while only affecting the metal layers. Sewing Kits are typically added into the initial design either at the RTL level or during synthesis by the customer and are part of the netlist. A Metal Fix affects only the top layers of the wafers and does not affect the "base" layers.

Sewing Kits are modules that contain an unused mix of gates, flip-flops or any other cells considered potentially useful for an unforseen metal fix. A Sewing Kit may be specified in RTL by instantiating the literal cells from the vendor library. The cells in the kit are usually connected such that each cell's output is unconnected and the inputs are tied to ground. Clocks and resets may be wired into the larger design's signals, or not.

A FIB Fix (Focussed Ion Beam) Fix is only performed on a completed chip. FIB is a somewhat exotic technology where a particle beam is able to make and break connections on a completed die. FIB fixes are done on individual chips and would only be done as a last resort to repair an otherwise defective prototype chip. Masks are not affected since it is the final chip that is intrusively repaired.

Clearly, these sorts of fixes are tricky and risky. They are available to the ASIC developer, but must be negotiated and coordinated with the foundry. ASIC designers who have been through enough of these fixes appreciate the value of adding test and fault-tolerant design features into the RTL code so that Software Fixes can correct mior silicon problems!

VLSI FAQs

1. What is metastability?

When setup or hold window is violated in an flip flop then signal attains a unpredictable value or state known as metastability.

2. What is MTBF? What it signifies?

- MTBF-Mean Time Before Failure
- Average time to next failure

3. How chance of metastable state failure can be reduced?

- Lowering clock frequency
- Lowering data speed
- Using faster flip flop

4. What are the advantages of using synchronous reset ?

- No metastability problem with synchronous reset (provided recovery and removal time for reset is taken care).
- Simulation of synchronous reset is easy.

5. What are the disadvantages of using synchronous reset ?

- Synchronous reset is slow.
- Implementation of synchronous reset requires more number of gates compared to asynchronous reset design.
- An active clock is essential for a synchronous reset design. Hence you can expect more power consumption.

6. What are the advantages of using asynchronous reset ?

- Implementation of asynchronous reset requires less number of gates compared to synchronous reset design.
- Asynchronous reset is fast.
- Clocking scheme is not necessary for an asynchronous design. Hence design consumes less power. Asynchronous design style is also one of the latest design options to achieve low power. Design community is scrathing their head over asynchronous design possibilities.

7. What are the disadvantages of using asynchronous reset ?

- Metastability problems are main concerns of asynchronous reset scheme (design).
- Static timing analysis and DFT becomes difficult due to asynchronous reset.

8. What are the 3 fundamental operating conditions that determine the delay characteristics of gate? How operating conditions affect gate delay?

- Process
- Voltage
- Temperature

9. Is verilog/VHDL is a concurrent or sequential language?

- Verilog and VHDL both are concurrent languages.
- Any hardware descriptive language is concurrent in nature.

10. In a system with insufficient hold time, will slowing down the clock frequency help?

- No.
- Making data path slower can help hold time but it may result in setup violation.

11. In a system with insufficient setup time, will slowing down the clock frequency help?

- Yes.
- Making data path faster can also help setup time but it may result in hold violation.

Physical Design Objective Type of Questions and Answers

- **1) Chip utilization depends on ___.**

a. Only on standard cells b. Standard cells and macros c. Only on macros d. Standard cells macros and IO pads

- **2) In Soft blockages ____ cells are placed.**

a. Only sequential cells b. No cells c. Only Buffers and Inverters d. Any cells

- **3) Why we have to remove scan chains before placement?**

a. Because scan chains are group of flip flop b. It does not have timing critical path c. It is series of flip flop connected in FIFO d. None

- **4) Delay between shortest path and longest path in the clock is called ____.**

a. Useful skew b. Local skew c. Global skew d. Slack

- **5) Cross talk can be avoided by ___.**

a. Decreasing the spacing between the metal layers b. Shielding the nets c. Using lower metal layers d. Using long nets

- **6) Prerouting means routing of _____.**

a. Clock nets b. Signal nets c. IO nets d. PG nets

- **7) Which of the following metal layer has Maximum resistance?**

a. Metal1 b. Metal2 c. Metal3 d. Metal4

- **8) What is the goal of CTS?**

a. Minimum IR Drop b. Minimum EM c. Minimum Skew d. Minimum Slack

- **9) Usually Hold is fixed ___.**

a. Before Placement b. After Placement c. Before CTS d. After CTS

- **10) To achieve better timing ____ cells are placed in the critical path.**

a. HVT b. LVT c. RVT d. SVT

- **11) Leakage power is inversely proportional to ___.**

a. Frequency b. Load Capacitance c. Supply voltage d. Threshold Voltage

- **12) Filler cells are added ___.**

a. Before Placement of std cells b. After Placement of Std Cells c. Before Floor planning d. Before Detail Routing

- **13) Search and Repair is used for ___.**

a. Reducing IR Drop b. Reducing DRC c. Reducing EM violations d. None

- **14) Maximum current density of a metal is available in ___.**

a. .lib b. .v c. .tf d. .sdc

- **15) More IR drop is due to ___.**

a. Increase in metal width b. Increase in metal length c. Decrease in metal length d. Lot of metal layers

- **16) The minimum height and width a cell can occupy in the design is called as ___.**

a. Unit Tile cell b. Multi heighten cell c. LVT cell d. HVT cell

- **17) CRPR stands for ___.**

a. Cell Convergence Pessimism Removal b. Cell Convergence Preset Removal c. Clock Convergence Pessimism Removal d. Clock Convergence Preset Removal

- **18) In OCV timing check, for setup time, ___.**

a. Max delay is used for launch path and Min delay for capture path b. Min delay is used for launch path and Max delay for capture path c. Both Max delay is used for launch and Capture path d. Both Min delay is used for both Capture and Launch paths

- **19) "Total metal area and(or) perimeter of conducting layer / gate to gate area" is called ___.**

a. Utilization b. Aspect Ratio c. OCV d. Antenna Ratio

- **20) The Solution for Antenna effect is ___.**

a. Diode insertion b. Shielding c. Buffer insertion d. Double spacing

- **21) To avoid cross talk, the shielded net is usually connected to ___.**

a. VDD b. VSS c. Both VDD and VSS d. Clock

- **22) If the data is faster than the clock in Reg to Reg path ___ violation may come.**

a. Setup b. Hold c. Both d. None

- **23) Hold violations are preferred to fix ___.**

a. Before placement b. After placement c. Before CTS d. After CTS

- **24) Which of the following is not present in SDC ___?**

a. Max tran b. Max cap c. Max fanout d. Max current density

- **25) Timing sanity check means (with respect to PD)___.**

a. Checking timing of routed design with out net delays b. Checking Timing of placed design with net delays c. Checking Timing of unplaced design without net delays d. Checking Timing of routed design with net delays

- **26) Which of the following is having highest priority at final stage (post routed) of the design ___?**

a. Setup violation b. Hold violation c. Skew d. None

- **27) Which of the following is best suited for CTS?**

a. CLKBUF b. BUF c. INV d. CLKINV

- **28) Max voltage drop will be there at(with out macros) ___.**

a. Left and Right sides b. Bottom and Top sides c. Middle d. None

- **29) Which of the following is preferred while placing macros ___?**

a. Macros placed center of the die b. Macros placed left and right side of die c. Macros placed bottom and top sides of die d. Macros placed based on connectivity of the I/O

- **30) Routing congestion can be avoided by ___.**

a. placing cells closer b. Placing cells at corners c. Distributing cells d. None

- **31) Pitch of the wire is ___.**

a. Min width b. Min spacing c. Min width - min spacing d. Min width + min spacing

- **32) In Physical Design following step is not there ___.**

a. Floorplaning b. Placement c. Design Synthesis d. CTS

- **33) In technology file if 7 metals are there then which metals you will use for power?**

a. Metal1 and metal2 b. Metal3 and metal4 c. Metal5 and metal6 d. Metal6 and metal7

- **34) If metal6 and metal7 are used for the power in 7 metal layer process design then which metals you will use for clock ?**

a. Metal1 and metal2 b. Metal3 and metal4 c. Metal4 and metal5 d. Metal6 and metal7

- **35) In a reg to reg timing path Tclocktoq delay is 0.5ns and TCombo delay is 5ns and Tsetup is 0.5ns then the clock period should be ___.**

a. 1ns b. 3ns c. 5ns d. 6ns

- **36) Difference between Clock buff/inverters and normal buff/inverters is __.**

a. Clock buff/inverters are faster than normal buff/inverters b. Clock buff/inverters are slower than normal buff/inverters c. Clock buff/inverters are having equal rise and fall times with high drive strengths compare to normal buff/inverters d. Normal buff/inverters are having equal rise and fall times with high drive strengths compare to Clock buff/inverters.

- **37) Which configuration is more preferred during floorplaning ?**

a. Double back with flipped rows b. Double back with non flipped rows c. With channel spacing between rows and no double back d. With channel spacing between rows and double back

- **38) What is the effect of high drive strength buffer when added in long net?**

a. Delay on the net increases b. Capacitance on the net increases c. Delay on the net decreases d. Resistance on the net increases.

- **39) Delay of a cell depends on which factors ?**

a. Output transition and input load b. Input transition and Output load c. Input transition and Output transition d. Input load and Output Load.

- **40) After the final routing the violations in the design ___.**

a. There can be no setup, no hold violations b. There can be only setup violation but no hold c. There can be only hold violation not Setup violation d. There can be both violations.

- **41) Utilisation of the chip after placement optimisation will be ___.**

a. Constant b. Decrease c. Increase d. None of the above

- **42) What is routing congestion in the design?**

a. Ratio of required routing tracks to available routing tracks b. Ratio of available routing tracks to required routing tracks c. Depends on the routing layers available d. None of the above

- **43) What are preroutes in your design?**

a. Power routing b. Signal routing c. Power and Signal routing d. None of the above.

- **44) Clock tree doesn't contain following cell ___.**

a. Clock buffer b. Clock Inverter c. AOI cell d. None of the above

- **Answers:**

1)b 2)c 3)b 4)c 5)b 6)d 7)a 8)c 9)d 10)b 11)d 12)d 13)b 14)c 15)b 16)a 17)c 18)a 19)d 20)a 21)b 22)b 23)d 24)d 25)c 26)b 27)a 28)c 29)d 30)c 31)d 32)c 33)d 34)c 35)d 36)c 37)a 38)c 39)b 40)d 41)c 42)a 43)a 44)c

CMOS Design Interview Questions

Below are the important VLSI CMOS interview questions. This set of interview questions may be updated in future. Answers will be posted one by one as and when i prepare them ! Readers are encouraged to post answers in comment section. Here we go.....

Draw Vds-Ids curve for an MOSFET. How it varies with a) increasing Vgs b)velocity saturation c)Channel length modulation d)W/L ratio.

What is body effect? Write mathematical expression? Is it due to parallel or serial connection of MOSFETs?

What is latch-up in CMOS design and what are the ways to prevent it?

What is Noise Margin? Explain with the help of Inverter.

What happens to delay if you increase load capacitance?

Give the various techniques you know to minimize power consumption for CMOS logic?

What happens when the PMOS and NMOS are interchanged with one another in an inverter?

What is body effect?

Why is NAND gate preferred over NOR gate for fabrication?

What is Noise Margin? Explain the procedure to determine Noise Margin

Explain sizing of the inverter?
How do you size NMOS and PMOS transistors to increase the threshold voltage?
What happens to delay if we include a resistance at the output of a CMOS circuit?
What are the limitations in increasing the power supply to reduce delay?
How does Resistance of the metal lines vary with increasing thickness and increasing length?
What is Charge Sharing? Explain the Charge Sharing problem while sampling data from a Bus?
 Why do we gradually increase the size of inverters in buffer design? Why not give the output of a circuit to one large inverter?
Give the expression for CMOS switching power dissipation?
Why is the substrate in NMOS connected to ground and in PMOS to VDD?
What is the fundamental difference between a MOSFET and BJT ?
Which transistor has higher gain- BJT or MOS and why?
Why PMOS and NMOS are sized equally in a Transmission Gates?
What is metastability? When/why it will occur? What are the different ways to avoid this?
 Explain zener breakdown and avalanche breakdown?

* What happens if Vds is increased over saturation?

In the I-V characteristics curve, why is the saturation curve flat or constant?
What happens if a resistor is added in series with the drain in a CMOS transistor?
What are the different regions of operation in a CMOS transistor?
What are the effects of the output characteristics for a change in the beta (β) value?
What is the effect of body bias?
What is hot electron effect and how can it be eliminated?
What is channel length modulation?
What is the effect of temperature on threshold voltage?
What is the effect of temperature on mobility?
What is the effect of gate voltage on mobility?
What are the different types of scaling?
What is stage ratio?
What is charge sharing on a bus?
What is electron migration and how can it be eliminated?
Can both PMOS and NMOS transistors pass good 1 and good 0? Explain.
Why is only NMOS used in pass transistor logic?
What are the different methodologies used to reduce the charge sharing in dynamic logic?
What are setup and hold time violations? How can they be eliminated?
Explain the operation of basic SRAM and DRAM.
Which ones take more time in SRAM: Read operation or Write operation? Why?
What is meant by clock race?
What is meant by single phase and double phase clocking?
If given a choice between NAND and NOR gates, which one would you pick? Explain.
Explain the origin of the various capacitances in the CMOS transistor and the physical reasoning behind it.
Why should the number of CMOS transistors that are connected in series be reduced?
What is charge sharing between bus and memory element?
What is crosstalk and how can it be avoided?
Realize an XOR gate using NAND gate.
What are the advantages and disadvantages of Bi-CMOS process?
Draw an XOR gate with using minimum number of transistors and explain the operation.
What are the critical parameters in a latch and flip-flop?
What is the significance of sense amplifier in an SRAM?
Explain Domino logic.
What are the advantages of depletion mode devices over the enhancement mode devices?
How can the rise and fall times in an inverter be equated?

What is meant by leakage current?
Realize an OR gate using NAND gate.
Realize an NAND gate using a 2:1 multiplexer.
Realize an NOR gate using a 2:1 multiplexer.
Draw the layout of a simple inverter.
What are the substrates of PMOS and NMOS transistors connected to and explain the results if the connections are interchanged with the other.
What are repeaters?
What is tunneling problem?
What is meant by negative biased instability and how can it be avoided?
What is Elmore delay algorithm?
What is meant by metastability?
What is the effect of Vdd on delay?
What is the effect of delay, rise and fall times with increase in load capacitance?
What is the value of mobility of electrons?
What is value of mobility of holes?
Give insights of an inverter. Draw Layout. Explain the working.

* Give insights of a 2 input NOR gate. Draw Layout. Explain the working.

Give insights of a 2 input NAND gate. Draw layout. Explain the working?
Implement F= not (AB+CD) using CMOS gates.
What is a pass gate. Explain the working?
Why do we need both PMOS and NMOS transistors to implement a pass gate?
What does the above code synthesize to?
Draw cross section of a PMOS transistor.
Draw cross section of an NMOS transistor.
What is a D-latch?
Implement D flip-flop with a couple of latches?
Implement a 2 input AND gate using transmission gate?
Explain various adders and difference between them?
How can you construct both PMOS and NMOS on a single substrate?
What happens when the gate oxide is very thin?
What is SPICE?
What are the differences between IRSIM and SPICE?
What are the differences between netlist of HSPICE and Spectre?
Implement F = AB+C using CMOS gates?
What is hot electron effect?
Define threshold voltage?
List out the factors affecting power consumption on a chip?
What r the phenomenon which come into play when the devices are scaled to the sub-micron lengths?
What is clock feed through?
Implement an Inverter using a single transistor?
What is Fowler-Nordheim Tunneling?
Which gate is normally preferred while implementing circuits using CMOS logic, NAND or NOR? Why?
Draw the Differential Sense Amplifier and explain its working. How to size this circuit?
What happens if we use an Inverter instead of the Differential Sense Amplifier?
Draw the SRAM Write Circuitry
How did you arrive at sizes of transistor in SRAM?
How does the size of PMOS pull up transistors for bit and bitbar lines affect SRAM's performance?
What is the critical path in a SRAM?

Draw the timing diagram for a SRAM Read. What happens if we delay the enabling of Clock signal?

Give a big picture of the entire SRAM layout showing placements of SRAM cells, row decoders, column decoders, read circuit, write circuit and buffers.

In a SRAM layout, which metal layers would you prefer for Word Lines and Bit Lines? Why?

Design For Test-DFT

In scan chains if some flip flops are +ve edge triggered and remaining flip flops are -ve edge triggered how it behaves?

Answer:

For designs with both positive and negative clocked flops, the scan insertion tool will always route the scan chain so that the negative clocked flops come before the positive edge flops in the chain. This avoids the need of lockup latch.

For the same clock domain the negedge flops will always capture the data just captured into the posedge flops on the posedge of the clock.

For the multiple clock domains, it all depends upon how the clock trees are balanced. If the clock domains are completely asynchronous, ATPG has to mask the receiving flops.

What you mean by scan chain reordering?

Answer1:

Based on timing and congestion the tool optimally places standard cells. While doing so, if scan chains are detached, it can break the chain ordering (which is done by a scan insertion tool like DFT compiler from Synopsis and can reorder to optimize it.... it maintains the number of flops in a chain.

Answer2:

During placement, the optimization may make the scan chain difficult to route due to congestion. Hence the tool will re-order the chain to reduce congestion.

This sometimes increases hold time problems in the chain. To overcome these buffers may have to be inserted into the scan path. It may not be able to maintain the scan chain length exactly. It cannot swap cell from different clock domains.

Because of scan chain reordering patterns generated earlier is of no use. But this is not a problem as ATPG can be redone by reading the new net list.

what are the differences between SIMULATION and SYNTHESIS

Simulation <= verify your design.

synthesis <= Check for your timing

Simulation is used to verify the functionality of the circuit.. a)Functional

Simulation:study of ckt's operation independent of timing parameters and gate delays. b) Timing Simulation :study including estimated delays, verify setup,hold and other timing requirements of devices like flip flops are met.
Synthesis:One of the foremost in back end steps where by synthesizing is nothing but converting VHDL or VERILOG description to a set of primitives(equations as in CPLD) or components(as in FPGA'S)to fit into the target technology.Basically the synthesis tools convert the design description into equations or components

Can u tell me the differences between latches & flipflops?

There are 2 types of circuits:
1. Combinational
2. Sequential

Latches and flipflops both come under the category of "sequential circuits", whose output depends not only on the current inputs, but also on previous inputs and outputs. Difference: Latches are level-sensitive, whereas, FF are edge sensitive. By edge sensitive, I mean O/p changes only when there is a clock transition.(from 1 to 0, or from 0 to 1)

Example: In a flipflop, inputs have arrived on the input lines at time= 2 seconds. But, output won't change immediately. At time = 3 seconds, clock transition takes place. After that, O/P will change.
Flip-flops are of 2 types:
1.Positive edge triggered
2. negative edge triggered

1)fllipflops take twice the nymber of gates as latches
2) so automatically delay is more for flipflops
3)power consumption is also more

latch does not have a clock signal, whereas a flip-flop always does.

What is slack?

The slack is the time delay difference from the expected delay(1/clock) to the actual delay in a particular path.
Slack may be +ve or -ve.

Equivalence between VHDL and C?

There is concept of understanding in C there is structure.Based upon requirement structure provide facility to store collection of different data types.

In VHDL we have direct access to memory so instead of using pointer in C (and member of structure) we can write interface store data in memory and access it.

RTL and Behavioral

Register transfer language means there should be data flow between two registers and logic is in between them for end registers data should flow.

Behavioral means how hardware behave determine the exact way it works we write using HDL syntax.For complex projects it is better mixed approach or more behavioral is used.

VHDL QUESTIONS

1. What is the difference between using direct instantiations and component ones except that you need to declare the component?
2. What is the use of BLOCKS?
3. What is the use of PROCEDURES?
4. What is the usage of using more then one architecture in an entity?
5. What is a D-latch? Write the VHDL Code for it?
6. Implement D flip-flop with a couple of latches? Write a VHDL Code for a D flip-flop?
7. Differences between Signals and Variables in VHDL? If the same code is written using Signals and Variables what does it synthesize to?
8. Differences between functions and Procedures in VHDL?
9. Explain the concept of a Clock Divider Circuit? Write a VHDL code for the same?

Digital Design interview questions:

1. Give two ways of converting a two input NAND gate to an inverter
2. Given a circuit, draw its exact timing response. (I was given a Pseudo Random Signal Generator; you can expect any sequential ckt)

3. What are set up time & hold time constraints? What do they signify? Which one is critical for estimating maximum clock frequency of a circuit?
4. Give a circuit to divide frequency of clock cycle by two
5. Design a divide-by-3 sequential circuit with 50% duty circle. (Hint: Double the Clock)
6. Suppose you have a combinational circuit between two registers driven by a clock. What will you do if the delay of the combinational circuit is greater than your clock signal? (You can't resize the combinational circuit transistors)
7. The answer to the above question is breaking the combinational circuit and pipelining it. What will be affected if you do this?
8. What are the different Adder circuits you studied?
9. Give the truth table for a Half Adder. Give a gate level implementation of the same.
10. Draw a Transmission Gate-based D-Latch.
11. Design a Transmission Gate based XOR. Now, how do you convert it to XNOR? (Without inverting the output)
12. How do you detect if two 8-bit signals are same?
13. How do you detect a sequence of "1101" arriving serially from a signal line?
14. Design any FSM in VHDL or Verilog.

Intel interview questions

The following questions are used for screening the candidates during the first interview. The questions apply mostly to fresh college grads pursuing an engineering career at Intel.

1. Have you studied buses? What types?
2. Have you studied pipelining? List the 5 stages of a 5 stage pipeline. Assuming 1 clock per stage, what is the latency of an instruction in a 5 stage machine? What is the throughput of this machine ?
3. How many bit combinations are there in a byte?
4. For a single computer processor computer system, what is the purpose of a processor cache and describe its operation?
5. Explain the operation considering a two processor computer system with a cache for each processor.
6. What are the main issues associated with multiprocessor caches and how might you solve them?

7. Explain the difference between write through and write back cache.
8. Are you familiar with the term MESI?
9. Are you familiar with the term snooping?
10. Describe a finite state machine that will detect three consecutive coin tosses (of one coin) that results in heads.
11. In what cases do you need to double clock a signal before presenting it to a synchronous state machine?
12. You have a driver that drives a long signal & connects to an input device. At the input device there is either overshoot, undershoot or signal threshold violations, what can be done to correct this problem?
13. What are the total number of lines written by you in C/C++? What is the most complicated/valuable program written in C/C++?
14. What compiler was used?
15. What is the difference between = and == in C?
16. Are you familiar with VHDL and/or Verilog?
17. What types of CMOS memories have you designed? What were their size? Speed?
18. What work have you done on full chip Clock and Power distribution? What process technology and budgets were used?
19. What types of I/O have you designed? What were their size? Speed? Configuration? Voltage requirements?
20. Process technology? What package was used and how did you model the package/system? What parasitic effects were considered?
21. What types of high speed CMOS circuits have you designed?
22. What transistor level design tools are you proficient with? What types of designs were they used on?
23. What products have you designed which have entered high volume production?
24. What was your role in the silicon evaluation/product ramp? What tools did you use?
25. If not into production, how far did you follow the design and why did not you see it into production?

VLSI Design Interview questions

1. Explain why & how a MOSFET works

2. Draw Vds-Ids curve for a MOSFET. Now, show how this curve changes (a) with increasing Vgs (b) with increasing transistor width (c) considering Channel Length Modulation
3. Explain the various MOSFET Capacitances & their significance
4. Draw a CMOS Inverter. Explain its transfer characteristics
5. Explain sizing of the inverter
6. How do you size NMOS and PMOS transistors to increase the threshold voltage?
7. What is Noise Margin? Explain the procedure to determine Noise Margin
8. Give the expression for CMOS switching power dissipation
9. What is Body Effect?
10. Describe the various effects of scaling
11. Give the expression for calculating Delay in CMOS circuit
12. What happens to delay if you increase load capacitance?
13. What happens to delay if we include a resistance at the output of a CMOS circuit?
14. What are the limitations in increasing the power supply to reduce delay?
15. How does Resistance of the metal lines vary with increasing thickness and increasing length?
16. You have three adjacent parallel metal lines. Two out of phase signals pass through the outer two metal lines. Draw the waveforms in the center metal line due to interference. Now, draw the signals if the signals in outer metal lines are in phase with each other
17. What happens if we increase the number of contacts or via from one metal layer to the next?
18. Draw a transistor level two input NAND gate. Explain its sizing (a) considering Vth (b) for equal rise and fall times
19. Let A & B be two inputs of the NAND gate. Say signal A arrives at the NAND gate later than signal B. To optimize delay, of the two series NMOS inputs A & B, which one would you place near the output?
20. Draw the stick diagram of a NOR gate. Optimize it
21. For CMOS logic, give the various techniques you know to minimize power consumption
22. What is Charge Sharing? Explain the Charge Sharing problem while sampling data from a Bus
23. Why do we gradually increase the size of inverters in buffer design? Why not give the output of a circuit to one large inverter?

24. In the design of a large inverter, why do we prefer to connect small transistors in parallel (thus increasing effective width) rather than lay out one transistor with large width?
25. Given a layout, draw its transistor level circuit. (I was given a 3 input AND gate and a 2 input Multiplexer. You can expect any simple 2 or 3 input gates)
26. Give the logic expression for an AOI gate. Draw its transistor level equivalent. Draw its stick diagram
27. Why don't we use just one NMOS or PMOS transistor as a transmission gate?
28. For a NMOS transistor acting as a pass transistor, say the gate is connected to VDD, give the output for a square pulse input going from 0 to VDD
29. Draw a 6-T SRAM Cell and explain the Read and Write operations
30. Draw the Differential Sense Amplifier and explain its working. Any idea how to size this circuit? (Consider Channel Length Modulation)
31. What happens if we use an Inverter instead of the Differential Sense Amplifier?
32. Draw the SRAM Write Circuitry
33. Approximately, what were the sizes of your transistors in the SRAM cell? How did you arrive at those sizes?
34. How does the size of PMOS Pull Up transistors (for bit & bit- lines) affect SRAM's performance?
35. What's the critical path in a SRAM?
36. Draw the timing diagram for a SRAM Read. What happens if we delay the enabling of Clock signal?
37. Give a big picture of the entire SRAM Layout showing your placements of SRAM Cells, Row Decoders, Column Decoders, Read Circuit, Write Circuit and Buffers
38. In a SRAM layout, which metal layers would you prefer for Word Lines and Bit Lines? Why?
39. How can you model a SRAM at RTL Level?
40. Whatï¿½s the difference between Testing & Verification?
41. For an AND-OR implementation of a two input Mux, how do you test for Stuck-At-0 and Stuck-At-1 faults at the internal nodes? (You can expect a circuit with some redundant logic)
42. What is Latch Up? Explain Latch Up with cross section of a CMOS Inverter. How do you avoid Latch Up?

Verilog Interview Questions

- What is the difference between $display and $monitor and $write and $strobe?
- What is the difference between code-compiled simulator and normal simulator?

- What is the difference between wire and reg?
- What is the difference between blocking and non-blocking assignments?
- What is the significance Timescale directivbe?
- What is the difference between bit wise, unary and logical operators?
- What is the difference between task and function?
- What is the difference between casex, casez and case statements?
- Which one preferred-casex or casez?
- For what is defparam used?
- What is the difference between “= =” and “= = =” ?
- What is a compiler directive like ‘include’ and ‘ifdef’?
- Write a verilog code to swap contents of two registers with and without a temporary register?
- What is the difference between inter statement and intra statement delay?
- What is delta simulation time?
- What is difference between Verilog full case and parallel case?
- What you mean by inferring latches?
- How to avoid latches in your design?
- Why latches are not preferred in synthesized design?
- How blocking and non blocking statements get executed?
- Which will be updated first: is it variable or signal?
- What is sensitivity list?
- If you miss sensitivity list what happens?
- In a pure combinational circuit is it necessary to mention all the inputs in sensitivity disk? If yes, why? If not, why?
- In a pure sequential circuit is it necessary to mention all the inputs in sensitivity disk? If yes, why? If not, why?
- What is general structure of Verilog code you follow?
- What are the difference between Verilog and VHDL?
- What are system tasks?
- List some of system tasks and what are their purposes?
- What are the enhancements in Verilog 2001?
- Write a Verilog code for synchronous and asynchronous reset?
- What is pli? why is it used?
- What is file I/O?
- What is difference between freeze deposit and force?
- Will case always infer priority register? If yes how? Give an example.
- What are inertial and transport delays ?
- What does `timescale 1 ns/ 1 ps’ signify in a verilog code?
- How to generate sine wav using verilog coding style?
- How do you implement the bi-directional ports in Verilog HDL?
- How to write FSM is verilog?
- What is verilog case (1)?
- What are Different types of Verilog simulators available?
- What is Constrained-Random Verification ?
- How can you model a SRAM at RTL Level?

Physical Design Questions and Answers

- I am getting several emails requesting answers to the questions posted in this blog. But it is very difficult to provide detailed answer to all questions in my available spare time.

Hence i decided to give "short and sweet" one line answers to the questions so that readers can immediately benefited. Detailed answers will be posted in later stage.I have given answers to some of the physical design questions here. Enjoy !

What parameters (or aspects) differentiate Chip Design and Block level design?

- Chip design has I/O pads; block design has pins.
- Chip design uses all metal layes available; block design may not use all metal layers.
- Chip is generally rectangular in shape; blocks can be rectangular, rectilinear.
- Chip design requires several packaging; block design ends in a macro.

How do you place macros in a full chip design?

- First check flylines i.e. check net connections from macro to macro and macro to standard cells.
- If there is more connection from macro to macro place those macros nearer to each other preferably nearer to core boundaries.
- If input pin is connected to macro better to place nearer to that pin or pad.
- If macro has more connection to standard cells spread the macros inside core.
- Avoid criscross placement of macros.
- Use soft or hard blockages to guide placement engine.

Differentiate between a Hierarchical Design and flat design?

- Hierarchial design has blocks, subblocks in an hierarchy; Flattened design has no subblocks and it has only leaf cells.
- Hierarchical design takes more run time; Flattened design takes less run time.

Which is more complicated when u have a 48 MHz and 500 MHz clock design?

- 500 MHz; because it is more constrained (i.e.lesser clock period) than 48 MHz design.

Name few tools which you used for physical verification?

- Herculis from Synopsys, Caliber from Mentor Graphics.

What are the input files will you give for primetime correlation?

- Netlist, Technology library, Constraints, SPEF or SDF file.

If the routing congestion exists between two macros, then what will you do?

- Provide soft or hard blockage

How will you decide the die size?

- By checking the total area of the design you can decide die size.

If lengthy metal layer is connected to diffusion and poly, then which one will affect by antenna problem?

- Poly

If the full chip design is routed by 7 layer metal, why macros are designed using 5LM instead of using 7LM?

- Because top two metal layers are required for global routing in chip design. If top metal layers are also used in block level it will create routing blockage.

In your project what is die size, number of metal layers, technology, foundry, number of clocks?

- Die size: tell in mm eg. 1mm x 1mm ; remeber 1mm=1000micron which is a big size !!
- Metal layers: See your tech file. generally for 90nm it is 7 to 9.
- Technology: Again look into tech files.
- Foundry:Again look into tech files; eg. TSMC, IBM, ARTISAN etc
- Clocks: Look into your design and SDC file !

How many macros in your design?

- You know it well as you have designed it ! A SoC (System On Chip) design may have 100 macros also !!!!

What is each macro size and number of standard cell count?

- Depends on your design.

What are the input needs for your design?

- For synthesis: RTL, Technology library, Standard cell library, Constraints
- For Physical design: Netlist, Technology library, Constraints, Standard cell library

What is SDC constraint file contains?

- Clock definitions
- Timing exception-multicycle path, false path
- Input and Output delays

How did you do power planning?
How to calculate core ring width, macro ring width and strap or trunk width?
How to find number of power pad and IO power pads?
How the width of metal and number of straps calculated for power and ground?

- Get the total core power consumption; get the metal layer current density value from the tech file; Divide total power by number sides of the chip; Divide the obtained value from the current density to get core power ring width. Then calculate number of straps using some more equations. Will be explained in detail later.

How to find total chip power?

- Total chip power=standard cell power consumption,Macro power consumption pad power consumption.

What are the problems faced related to timing?

- Prelayout: Setup, Max transition, max capacitance
- Post layout: Hold

How did you resolve the setup and hold problem?

- Setup: upsize the cells
- Hold: insert buffers

In which layer do you prefer for clock routing and why?

- Next lower layer to the top two metal layers(global routing layers). Because it has less resistance hence less RC delay.

If in your design has reset pin, then it'll affect input pin or output pin or both?

- Output pin.

During power analysis, if you are facing IR drop problem, then how did you avoid?

- Increase power metal layer width.
- Go for higher metal layer.
- Spread macros or standard cells.

- Provide more straps.

Define antenna problem and how did you resolve these problem?

- Increased net length can accumulate more charges while manufacturing of the device due to ionisation process. If this net is connected to gate of the MOSFET it can damage dielectric property of the gate and gate may conduct causing damage to the MOSFET. This is antenna problem.
- Decrease the length of the net by providing more vias and layer jumping.
- Insert antenna diode.

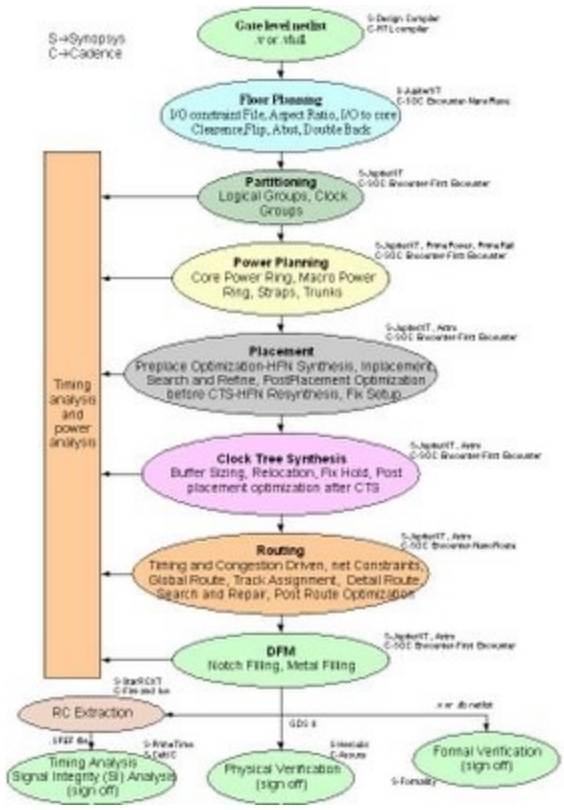
How delays vary with different PVT conditions? Show the graph.

- P increase->dealy increase
- P decrease->delay decrease
- V increase->delay decrease
- V decrease->delay increase
- T increase->delay increase
- T decrease->delay decrease

Explain the flow of physical design and inputs and outputs for each step in flow.

Physical Design Flow

The physical design flow is generally explained in the Figure (1.). In each section of the flow EDA tools available from the two main EDA companies-Synopsys and Cadence is also listed. In each and every step of the flow timing and power analysis can be carried out. If timing and power requirements are not met then either the whole flow has to be re-exercised or going back one or two steps and optimizing the design or incremental optimization may meet the requirements



What is cell delay and net delay?

- **Gate delay**
- Transistors within a gate take a finite time to switch. This means that a change on the input of a gate takes a finite time to cause a change on the output.[Magma]
- Gate delay =function of(i/p transition time, Cnet+Cpin).
- Cell delay is also same as Gate delay.
- **Cell delay**
- For any gate it is measured between 50% of input transition to the corresponding 50% of output transition.
- Intrinsic delay
- Intrinsic delay is the delay internal to the gate. Input pin of the cell to output pin of the cell.
- It is defined as the delay between an input and output pair of a cell, when a near zero slew is applied to the input pin and the output does not see any load condition.It is predominantly caused by the internal capacitance associated with its transistor.

- This delay is largely independent of the size of the transistors forming the gate because increasing size of transistors increase internal capacitors.

- **Net Delay (or wire delay)**

- The difference between the time a signal is first applied to the net and the time it reaches other devices connected to that net.

- It is due to the finite resistance and capacitance of the net.It is also known as wire delay.

- Wire delay =fn(Rnet , Cnet+Cpin)

What are delay models and what is the difference between them?

- Linear Delay Model (LDM)

- Non Linear Delay Model (NLDM)

What is wire load model?

- Wire load model is NLDM which has estimated R and C of the net.

Why higher metal layers are preferred for Vdd and Vss?

- Because it has less resistance and hence leads to less IR drop.

What is logic optimization and give some methods of logic optimization.

- Upsizing

- Downsizing

- Buffer insertion

- Buffer relocation

- Dummy buffer placement

What is the significance of negative slack?

- negative slack==> there is setup voilation==> deisgn can fail

What is signal integrity? How it affects Timing?

- IR drop, Electro Migration (EM), Crosstalk, Ground bounce are signal integrity issues.

- If Idrop is more==>delay increases.

- crosstalk==>there can be setup as well as hold voilation.

What is IR drop? How to avoid? How it affects timing?

- There is a resistance associated with each metal layer. This resistance consumes power causing voltage drop i.e.IR drop.

- If IR drop is more==>delay increases.

What is EM and it effects?

- Due to high current flow in the metal atoms of the metal can displaced from its origial place. When it happens in larger amount the metal can open or bulging of metal layer can happen. This effect is known as Electro Migration.

- Affects: Either short or open of the signal line or power line.

What are types of routing?

- Global Routing

- Track Assignment

- Detail Routing

What is latency? Give the types?

- **Source Latency**

- It is known as source latency also. It is defined as "the delay from the clock origin point to the clock definition point in the design".

- Delay from clock source to beginning of clock tree (i.e. clock definition point).

- The time a clock signal takes to propagate from its ideal waveform origin point to the clock definition point in the design.

- **Network latency**

- It is also known as Insertion delay or Network latency. It is defined as "the delay from the clock definition point to the clock pin of the register".

- The time clock signal (rise or fall) takes to propagate from the clock definition point to a register clock pin.

What is track assignment?

- Second stage of the routing wherein particular metal tracks (or layers) are assigned to the signal nets.

What is congestion?

- If the number of routing tracks available for routing is less than the required tracks then it is known as congestion.

Whether congestion is related to placement or routing?

- Routing

What are clock trees?

- Distribution of clock from the clock source to the sync pin of the registers.

What are clock tree types?

- H tree, Balanced tree, X tree, Clustering tree, Fish bone

What is cloning and buffering?

- Cloning is a method of optimization that decreases the load of a heavily loaded cell by replicating the cell.
- Buffering is a method of optimization that is used to insert beffers in high fanout nets to decrease the dealy.

ASIC

Different Types of Delays in ASIC or VLSI design

- **Source Delay/Latency**
- **Network Delay/Latency**
- **Insertion Delay**
- **Transition Delay/Slew: Rise time, fall time**
- **Path Delay**
- **Net delay, wire delay, interconnect delay**

- **Propagation Delay**
- **Phase Delay**
- **Cell Delay**
- **Intrinsic Delay**
- **Extrinsic Delay**
- **Input Delay**
- **Output Delay**
- **Exit Delay**
- **Latency (Pre/post CTS)**
- **Uncertainty (Pre/Post CTS)**
- **Unateness: Positive unateness, negative unateness**
- **Jitter: PLL jitter, clock jitter**

Gate delay

- Transistors within a gate take a finite time to switch. This means that a change on the input of a gate takes a finite time to cause a change on the output.[Magma]
- Gate delay =function of(i/p transition time, Cnet+Cpin).
- Cell delay is also same as Gate delay.

Source Delay (or Source Latency)

- It is known as source latency also. It is defined as "the delay from the clock origin point to the clock definition point in the design".
- Delay from clock source to beginning of clock tree (i.e. clock definition point).
- The time a clock signal takes to propagate from its ideal waveform origin point to the clock definition point in the design.

Network Delay(latency)

- It is also known as Insertion delay or Network latency. It is defined as "the delay from the clock definition point to the clock pin of the register".

- The time clock signal (rise or fall) takes to propagate from the clock definition point to a register clock pin.

Insertion delay

- The delay from the clock definition point to the clock pin of the register.

Transition delay

- It is also known as "Slew". It is defined as the time taken to change the state of the signal. Time taken for the transition from logic 0 to logic 1 and vice versa . or Time taken by the input signal to rise from 10%(20%) to the 90%(80%) and vice versa.

- Transition is the time it takes for the pin to change state.

Slew

- Rate of change of logic.See Transition delay.

- Slew rate is the speed of transition measured in volt / ns.

Rise Time

- Rise time is the difference between the time when the signal crosses a low threshold to the time when the signal crosses the high threshold. It can be absolute or percent.

- Low and high thresholds are fixed voltage levels around the mid voltage level or it can be either 10% and 90% respectively or 20% and 80% respectively. The percent levels are converted to absolute voltage levels at the time of measurement by calculating percentages from the difference between the starting voltage level and the final settled voltage level.

Fall Time

- Fall time is the difference between the time when the signal crosses a high threshold to the time when the signal crosses the low threshold.

- The low and high thresholds are fixed voltage levels around the mid voltage level or it can be either 10% and 90% respectively or 20% and 80% respectively. The percent levels are converted to absolute voltage levels at the time of measurement by calculating percentages from the difference between the starting voltage level and the final settled voltage level.

- For an ideal square wave with 50% duty cycle, the rise time will be 0.For a symmetric triangular wave, this is reduced to just 50%.

- The rise/fall definition is set on the meter to 10% and 90% based on the linear power in Watts. These points translate into the -10 dB and -0.5 dB points in log mode (10 log 0.1) and (10 log 0.9). The rise/fall time values of 10% and 90% are calculated based on an algorithm, which looks at the mean power above and below the 50% points of the rise/fall times

- **Path delay**

- Path delay is also known as pin to pin delay. It is the delay from the input pin of the cell to the output pin of the cell.

Net Delay (or wire delay)

- The difference between the time a signal is first applied to the net and the time it reaches other devices connected to that net.

- It is due to the finite resistance and capacitance of the net.It is also known as wire delay.

- Wire delay =fn(Rnet , Cnet+Cpin)

Propagation delay

- For any gate it is measured between 50% of input transition to the corresponding 50% of output transition.

- This is the time required for a signal to propagate through a gate or net. For gates it is the time it takes for a event at the gate input to affect the gate output.

- For net it is the delay between the time a signal is first applied to the net and the time it reaches other devices connected to that net.

- It is taken as the average of rise time and fall time i.e. Tpd= (Tphl+Tplh)/2.

Phase delay

- Same as insertion delay

Cell delay

- For any gate it is measured between 50% of input transition to the corresponding 50% of output transition.

Intrinsic delay

- Intrinsic delay is the delay internal to the gate. Input pin of the cell to output pin of the cell.

- It is defined as the delay between an input and output pair of a cell, when a near zero slew is applied to the input pin and the output does not see any load condition.It is predominantly caused by the internal capacitance associated with its transistor.

- This delay is largely independent of the size of the transistors forming the gate because increasing size of transistors increase internal capacitors.

Extrinsic delay

- Same as wire delay, net delay, interconnect delay, flight time.

- Extrinsic delay is the delay effect that associated to with interconnect. output pin of the cell to the input pin of the next cell.

Input delay

- Input delay is the time at which the data arrives at the input pin of the block from external circuit with respect to reference clock.

Output delay

- Output delay is time required by the external circuit before which the data has to arrive at the output pin of the block with respect to reference clock.

Exit delay

- It is defined as the delay in the longest path (critical path) between clock pad input and an output. It determines the maximum operating frequency of the design.

Latency (pre/post cts)

- Latency is the summation of the Source latency and the Network latency. Pre CTS estimated latency will be considered during the synthesis and after CTS propagated latency is considered.

Uncertainty (pre/post cts)

- Uncertainty is the amount of skew and the variation in the arrival clock edge. Pre CTS uncertainty is clock skew and clock Jitter. After CTS we can have some margin of skew + Jitter.

Unateness

- A function is said to be unate if the rise transition on the positive unate input variable causes the ouput to rise or no change and vice versa.

- Negative unateness means cell output logic is inverted version of input logic. eg. In inverter having input A and output Y, Y is -ve unate w.r.to A. Positive unate means cell output logic is same as that of input.

- These +ve ad -ve unateness are constraints defined in library file and are defined for output pin w.r.to some input pin.

- A clock signal is positive unate if a rising edge at the clock source can only cause a rising edge at the register clock pin, and a falling edge at the clock source can only cause a falling edge at the register clock pin.

- A clock signal is negative unate ﾉ if a rising edge at the clock source can only cause a falling edge at the register clock pin, and a falling edge at the clock source can only cause a rising edge at the register clock pin. In other words, the clock signal is inverted.

- A clock signal is not unate if the clock sense is ambiguous as a result of non-unate timing arcs in the clock path. For example, a clock that passes through an XOR gate is not unate because there are nonunate arcs in the gate. The clock sense could be either positive or negative, depending on the state of the other input to the XOR gate.

Jitter

- The short-term variations of a signal with respect to its ideal position in time.
- Jitter is the variation of the clock period from edge to edge. It can varry +/- jitter value.
- From cycle to cycle the period and duty cycle can change slightly due to the clock generation circuitry. This can be modeled by adding uncertainty regions around the rising and falling edges of the clock waveform.

Sources of Jitter Common sources of jitter include:

- Internal circuitry of the phase-locked loop (PLL)
- Random thermal noise from a crystal
- Other resonating devices
- Random mechanical noise from crystal vibration
- Signal transmitters
- Traces and cables
- Connectors
- Receivers

Skew

- The difference in the arrival of clock signal at the clock pin of different flops.
- Two types of skews are defined: Local skew and Global skew.

Local skew

- The difference in the arrival of clock signal at the clock pin of related flops.

Global skew

- The difference in the arrival of clock signal at the clock pin of non related flops.
- Skew can be positive or negative.

- When data and clock are routed in same direction then it is **Positive skew.**

- When data and clock are routed in opposite then it is **negative skew.**

Recovery Time

- Recovery specifies the minimum time that an asynchronous control input pin must be held stable after being de-asserted and before the next clock (active-edge) transition.

- Recovery time specifies the time the inactive edge of the asynchronous signal has to arrive before the closing edge of the clock.

- Recovery time is the minimum length of time an asynchronous control signal (eg.preset) must be stable before the next active clock edge. The recovery slack time calculation is similar to the clock setup slack time calculation, but it applies asynchronous control signals.

Equation 1:

- Recovery Slack Time = Data Required Time â€“ Data Arrival Time

- Data Arrival Time = Launch Edge + Clock Network Delay to Source Register + Tclkq+ Register to Register Delay

- Data Required Time = Latch Edge + Clock Network Delay to Destination Register =Tsetup

If the asynchronous control is not registered, equations shown in Equation 2 is used to calculate the recovery slack time. Equation 2:

- Recovery Slack Time = Data Required Time â€“ Data Arrival Time

- Data Arrival Time = Launch Edge + Maximum Input Delay + Port to Register Delay

- Data Required Time = Latch Edge + Clock Network Delay to Destination Register Delay+Tsetup

- If the asynchronous reset signal is from a port (device I/O), you must make an Input Maximum Delay assignment to the asynchronous reset pin to perform recovery analysis on that path.

Removal Time

- Removal specifies the minimum time that an asynchronous control input pin must be held stable before being de-asserted and after the previous clock (active-edge) transition.

- Removal time specifies the length of time the active phase of the asynchronous signal has to be held after the closing edge of clock.

- Removal time is the minimum length of time an asynchronous control signal must be stable after the active clock edge. Calculation is similar to the clock hold slack calculation, but it applies asynchronous control signals. If the asynchronous control is registered, equations shown in Equation 3 is used to calculate the removal slack time.

- If the recovery or removal minimum time requirement is violated, the output of the sequential cell becomes uncertain. The uncertainty can be caused by the value set by the resetbar signal or the value clocked into the sequential cell from the data input.

Equation 3

- Removal Slack Time = Data Arrival Time â€“ Data Required Time

- Data Arrival Time = Launch Edge + Clock Network Delay to Source Register + Tclkq of Source Register + Register to Register Delay

- Data Required Time = Latch Edge + Clock Network Delay to Destination Register + Thold

- If the asynchronous control is not registered, equations shown in Equation 4 is used to calculate the removal slack time.

Equation 4

- Removal Slack Time = Data Arrival Time â€“ Data Required Time

- Data Arrival Time = Launch Edge + Input Minimum Delay of Pin + Minimum Pin to Register Delay

- Data Required Time = Latch Edge + Clock Network Delay to Destination Register +Thold

- If the asynchronous reset signal is from a device pin, you must specify the Input Minimum Delay constraint to the asynchronous reset pin to perform a removal analysis on this path.

What is the difference between soft macro and hard macro?

- **What is the difference between hard macro, firm macro and soft macro?**

or

- **What are IPs?**

- Hard macro, firm macro and soft macro are all known as IP (Intellectual property). They are optimized for power, area and performance. They can be purchased and used in your ASIC or FPGA design implementation flow. Soft macro is flexible for all type of ASIC implementation. Hard macro can be used in pure ASIC design flow, not in FPGA flow. Before bying any IP it is very important to evaluate its advantages and disadvantages over each other, hardware compatibility such as I/O standards with your design blocks, reusability for other designs.

Soft macros

- Soft macros are in synthesizable RTL.
- Soft macros are more flexible than firm or hard macros.
- Soft macros are not specific to any manufacturing process.
- Soft macros have the disadvantage of being somewhat unpredictable in terms of performance, timing, area, or power.
- Soft macros carry greater IP protection risks because RTL source code is more portable and therefore, less easily protected than either a netlist or physical layout data.
- From the physical design perspective, soft macro is any cell that has been placed and routed in a placement and routing tool such as Astro. (This is the definition given in Astro Rail user manual !)
- Soft macros are editable and can contain standard cells, hard macros, or other soft macros.

Firm macros

- Firm macros are in netlist format.
- Firm macros are optimized for performance/area/power using a specific fabrication technology.
- Firm macros are more flexible and portable than hard macros.
- Firm macros are predictive of performance and area than soft macros.

Hard macro

- Hard macros are generally in the form of hardware IPs (or we termed it as hardwre IPs !).
- Hard macos are targeted for specific IC manufacturing technology.
- Hard macros are block level designs which are silicon tested and proved.
- Hard macros have been optimized for power or area or timing.
- In physical design you can only access pins of hard macros unlike soft macros which allows us to manipulate in different way.
- You have freedom to move, rotate, flip but you can't touch anything inside hard macros.
- Very common example of hard macro is memory. It can be any design which carries dedicated single functionality (in general).. for example it can be a MP4 decoder.

- Be aware of features and characteristics of hard macro before you use it in your design... other than power, timing and area you also should know pin properties like sync pin, I/O standards etc

- LEF, GDS2 file format allows easy usage of macros in different tools.

From the physical design (backend) perspective:

- Hard macro is a block that is generated in a methodology other than place and route (i.e. using full custom design methodology) and is brought into the physical design database (eg. Milkyway in Synopsys; Volcano in Magma) as a GDS2 file.

What is the difference between FPGA and CPLD?

FPGA-Field Programmable Gate Array and CPLD-Complex Programmable Logic Device-- both are programmable logic devices made by the same companies with different characteristics.

"A Complex Programmable Logic Device (CPLD) is a Programmable Logic Device with complexity between that of PALs (Programmable Array Logic) and FPGAs, and architectural features of both. The building block of a CPLD is the macro cell, which contains logic implementing disjunctive normal form expressions and more specialized logic operations".

Architecture

Granularity is the biggest difference between CPLD and FPGA.

FPGA are "fine-grain" devices. That means that they contain hundreds of (up to 100000) of tiny blocks (called as LUT or CLBs etc) of logic with flip-flops, combinational logic and memories.FPGAs offer much higher complexity, up to 150,000 flip-flops and large number of gates available.

CPLDs typically have the equivalent of thousands of logic gates, allowing implementation of moderately complicated data processing devices. PALs typically have a few hundred gate equivalents at most, while FPGAs typically range from tens of thousands to several million.

CPLD are "coarse-grain" devices. They contain relatively few (a few 100's max) large blocks of logic with flip-flops and combinational logic. CPLDs based on AND-OR structure.

CPLD's have a register with associated logic (AND/OR matrix). CPLD's are mostly implemented in control applications and FPGA's in datapath applications. Because of this course grained architecture, the timing is very fixed in CPLDs.

- FPGA are RAM based. They need to be "downloaded" (configured) at each power-up. CPLD are EEPROM based. They are active at power-up i.e. as long as they've been programmed at least once.

FPGA needs boot ROM but CPLD does not. In some systems you might not have enough time to boot up FPGA then you need CPLD+FPGA.

Generally, the CPLD devices are not volatile, because they contain flash or erasable ROM memory in all the cases. The FPGA are volatile in many cases and hence they need a configuration memory for working. There are some FPGAs now which are nonvolatile. This distinction is rapidly becoming less relevant, as several of the latest FPGA products also offer models with embedded configuration memory.

- The characteristic of non-volatility makes the CPLD the device of choice in modern digital designs to perform 'boot loader' functions before handing over control to other devices not having this capability. A good example is where a CPLD is used to load configuration data for an FPGA from non-volatile memory.

- Because of coarse-grain architecture, one block of logic can hold a big equation and hence CPLD have a faster input-to-output timings than FPGA.

Features

- FPGA have special routing resources to implement binary counters,arithmetic functions like adders, comparators and RAM. CPLD don't have special features like this.

- FPGA can contain very large digital designs, while CPLD can contain small designs only.The limited complexity (<500>

- ***Speed***: CPLDs offer a single-chip solution with fast pin-to-pin delays, even for wide input functions. Use CPLDs for small designs, where "instant-on", fast and wide decoding, ultra-low idle power consumption, and design security are important (e.g., in battery-operated equipment).

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- ***Speed...speed...speed....ASICs are faster than FPGA:*** ASIC gives design flexibility. This gives enoromous opportunity for speed optimizations.

- ***Low power....Low power....Low power:*** ASIC can be optimized for required low power. There are several low power techniques such as power gating, clock gating, multi vt cell libraries, pipelining etc are available to achieve the power target. This is where FPGA fails badly !!! Can you think of a cell phone which has to be charged for every call.....never.....low power ASICs helps battery live longer life !!

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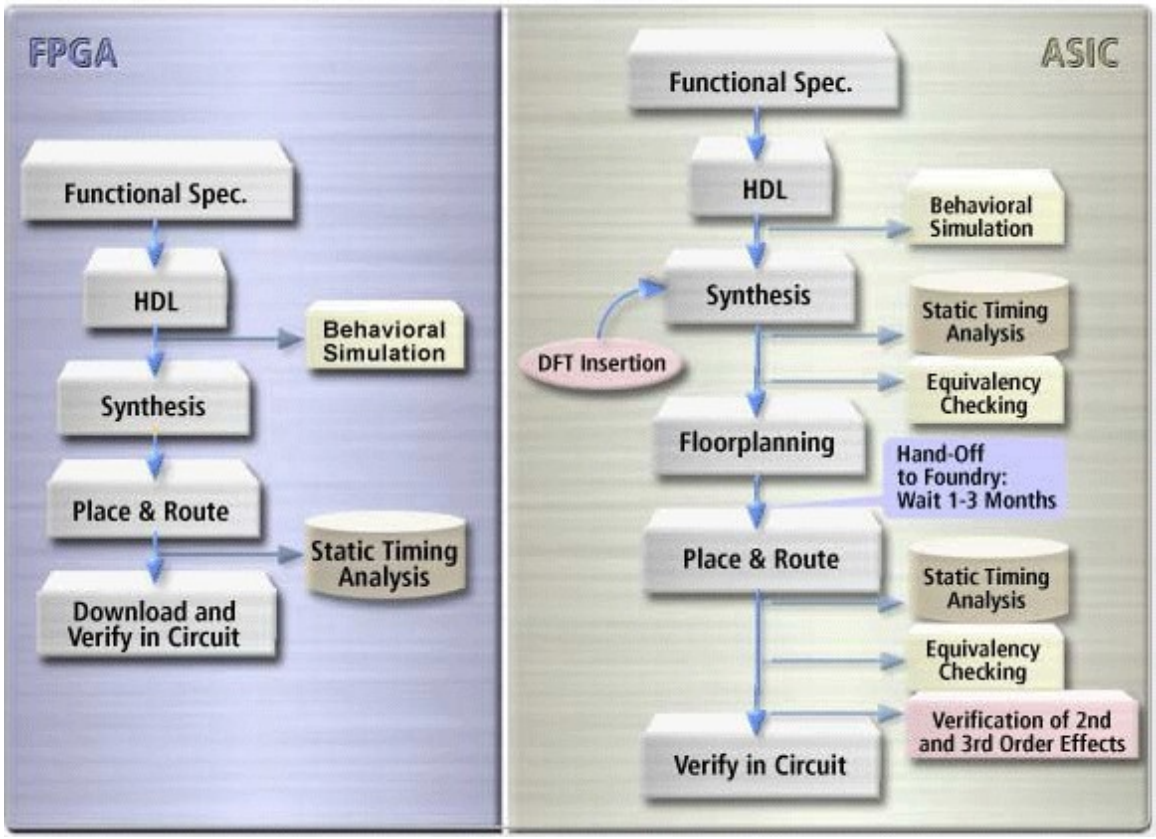
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FPGA vs. ASIC Design Flow Comparison



ASIC Design Check List

Silicon Process and Library Characteristics

- What exact process are you using?
- How many layers can be used for this design?
- Are the Cross talk Noise constraints, Xtalk Analysis configuration, Cell EM & Wire EM available?

Design Characteristics

- What is the design application?
- Number of cells (placeable objects)?
- Is the design Verilog or VHDL?
- Is the netlist flat or hierarchical?
- Is there RTL available?
- Is there any datapath logic using special datapath tools?
- Is the DFT to be considered?
- Can scan chains be reordered?
- Is memory BIST, boundary scan used on this design?
- Are static timing analysis constraints available in SDC format?

Clock Characteristics

- How many clock domains are in the design?

- What are the clock frequencies?
- Is there a target clock skew, latency or other clock requirements?
- Does the design have a PLL?
- If so, is it used to remove clock latency?
- Is there any I/O cell in the feedback path?
- Is the PLL used for frequency multipliers?
- Are there derived clocks or complex clock generation circuitry?
- Are there any gated clocks?
- If yes, do they use simple gating elements?
- Is the gate clock used for timing or power?
- For gated clocks, can the gating elements be sized for timing?
- Are you muxing in a test clock or using a JTAG clock?
- Available cells for clock tree?
- Are there any special clock repeaters in the library?
- Are there any EM, slew or capacitance limits on these repeaters?
- How many drive strengths are available in the standard buffers and inverters?
- Do any of the buffers have balanced rise and fall delays?
- Any there special requirements for clock distribution?
- Will the clock tree be shielded? If so, what are the shielding requirements?

Floorplan and Package Characteristics

- Target die area?
- Does the area estimate include power/signal routing?
- What gates/mm2 has been assumed?
- Number of routing layers?
- Any special power routing requirements?
- Number of digital I/O pins/pads?
- Number of analog signal pins/pads?
- Number of power/ground pins/pads?
- Total number of pins/pads and Location?
- Will this chip use a wire bond package?
- Will this chip use a flip-chip package?
- If Yes, is it I/O bump pitch? Rows of bumps? Bump allocation?Bump pad layout guide?
- Have you already done floorplanning for this design?
- If yes, is conformance to the existing floorplan required?
- What is the target die size?
- What is the expected utilization?
- Please draw the overall floorplan ?
- Is there an existing floorplan available in DEF?
- What are the number and type of macros (memory, PLL, etc.)?
- Are there any analog blocks in the design?
- What kind of packaging is used? Flipchip?
- Are the I/Os periphery I/O or area I/O?
- How many I/Os?
- Is the design pad limited?
- Power planning and Power analysis for this design?
- Are layout databases available for hard macros ?
- Timing analysis and correlatio?

- Physical verification ?

Data Input

- Library information for new library
- .lib for timing information
- GDSII or LEF for library cells including any RAMs
- RTL in Verilog/VHDL format
- Number of logical blocks in the RTL
- Constraints for the block in SDC
- Floorplan information in DEF
- I/O pin location
- Macro locations

ASIC General

General ASIC questions are posted here. More questions related to different catagories of ASICs can be found at respective sections.

- What are the differences between PALs, PLAs, FPGAs, ASICs and PLDs?
- In system with insufficient hold time, will slowing down the clock help?
- In system with insufficient setup time, will slowing down the clock help?
- Why would a testbench not have pins (port) on it?
- When declaring a flip flop, why would not you declare its output value in the port statement?
- Give 2 advantages of using a script to build a chip?
- A “tri state “ bus is directly connected to a set of CMOS input buffers. No other wires or components are attached to the bus wires. Upon observation we can find that under certain conditions, this circuit is consuming considerable power. Why it is so? Is circuit correct? If not, how to correct?
- Is Verilog (or that matter any HDL) is a concurrent or sequential language?
- What is the function of sensitivity list?
- A mealy –type state machine is coded using D-type rising edge flip flops. The reset and clock signals are in the sensitivity list but with one of the next state logic input signals have been left out of the sensitivity list. Explain what happens when the state machine is simulated? Will the state machine be synthesized correctly?
- A moore –type state machine is coded using D-type rising edge flip flops. The reset and clock signals are in the sensitivity list but with one of the next state logic input signals have been left out of the sensitivity list. Explain what happens when the state machine is simulated? Will the state machine be synthesized correctly?
- What type of delay is most like a infinite bandwidth transmission line?
- Define metastability.
- When does metastability occur?
- Give one example of a situation where metastability could occur.
- Give two ways metastability could manifest itself in a state machine.
- What is MTBF?
- Does MTBF give the time until the next failure occurs?
- Give 3 ways in which to reduce the chance of metastable failure.
- Give 2 advantages of using a synchronous reset methodology.
- Give 2 disadvantages of using a synchronous reset methodology.

- Give 2 advantages of using an asynchronous reset methodology.
- Give 2 disadvantages of using an asynchronous reset methodology.
- What are the two most fundamental inputs (files) to the synthesis tool?
- What are two important steps in synthesis? What happens in those steps?
- What are the two major output (files) from the synthesis process?
- Name the fundamental 3 operating consitions that determine (globally) the delay characteristics of CMOS gates. For each how they affect gate delay?
- For a single gate, with global gating conditions held constant , what 3 delay coefficients effect total gate delay? Which is the most sensitive to circuit topology?

FPGA.

What is the difference between FPGA and CPLD?

FPGA-Field Programmable Gate Array and CPLD-Complex Programmable Logic Device-- both are programmable logic devices made by the same companies with different characteristics.

- "A Complex Programmable Logic Device (CPLD) is a Programmable Logic Device with complexity between that of PALs (Programmable Array Logic) and FPGAs, and architectural features of both. The building block of a CPLD is the macro cell, which contains logic implementing disjunctive normal form expressions and more specialized logic operations".

- This is what Wiki defines.....!!

Architecture

- Granularity is the biggest difference between CPLD and FPGA.

- FPGA are "fine-grain" devices. That means that they contain hundreds of (up to 100000) of tiny blocks (called as LUT or CLBs etc) of logic with flip-flops, combinational logic and memories.FPGAs offer much higher complexity, up to 150,000 flip-flops and large number of gates available.

- CPLDs typically have the equivalent of thousands of logic gates, allowing implementation of moderately complicated data processing devices. PALs typically have a few hundred gate equivalents at most, while FPGAs typically range from tens of thousands to several million.

CPLD are "coarse-grain" devices. They contain relatively few (a few 100's max) large blocks of logic with flip-flops and combinational logic. CPLDs based on AND-OR structure.

CPLD's have a register with associated logic (AND/OR matrix). CPLD's are mostly implemented in control applications and FPGA's in datapath applications. Because of this course grained architecture, the timing is very fixed in CPLDs.

FPGA are RAM based. They need to be "downloaded" (configured) at each power-up. CPLD are EEPROM based. They are active at power-up i.e. as long as they've been programmed at least once.

FPGA needs boot ROM but CPLD does not. In some systems you might not have enough time to boot up FPGA then you need CPLD+FPGA.

- Generally, the CPLD devices are not volatile, because they contain flash or erasable ROM memory in all the cases. The FPGA are volatile in many cases and hence they need a configuration memory for working. There are some FPGAs now which are nonvolatile. This distinction is rapidly becoming less relevant, as several of the latest FPGA products also offer models with embedded configuration memory.

- The characteristic of non-volatility makes the CPLD the device of choice in modern digital designs to perform 'boot loader' functions before handing over control to other devices not having this capability. A good example is where a CPLD is used to load configuration data for an FPGA from non-volatile memory.

- Because of coarse-grain architecture, one block of logic can hold a big equation and hence CPLD have a faster input-to-output timings than FPGA.

Features

- FPGA have special routing resources to implement binary counters,arithmetic functions like adders, comparators and RAM. CPLD don't have special features like this.

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- ***Speed***: CPLDs offer a single-chip solution with fast pin-to-pin delays, even for wide input functions. Use CPLDs for small designs, where "instant-on", fast and wide decoding, ultra-low idle power consumption, and design security are important (e.g., in battery-operated equipment).

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FPGA Interview Questions

What is minimum and maximum frequency of DCM in spartan-3 series FPGA?
List some of constraints you used and their purpose during your design?
What is the size of bitmap with changing gate count?
What are different types of FPGA programming modes? How to change from one to another?
List out some important features of FPGA.
List out some of synthesizable and non synthesizable constructs?
Draw general structure of FPGA?
What is the difference between FPGA and CPLD?
What is DCM? Why they are used?
Draw FPGA design flow. Explain each step. What is input and output from each step?
What is slice, CLB, LUT?
Is it possible to configure CLB as RAM?
What is purpose of a constraint file? What is its extension?
How you will choose an FPGA?
How clock is routed through out FPGA?
What are difference between PLL and DLL ?

What is soft processor?
What is hard processor?

Freescale Written test

Q 1. Question on pass transistor logic. Output voltage was asked for various combinations of gate and source voltages.

Q2. A logic expression was given. It's CMOS implementation was asked.

Q3. Simple question on Setup/hold time/clock frequency

Q4. 1 question that combined the concepts of a pass transistor and an op-amp

Q5. Question on an RLC network. Voltage values at t(0) and t (infinity) was asked

Q6. One more question on STA (much tougher than the previous one)

Q7. Question on Fault analysis

Q8. Question on FIFO buffer

Q9. Question on memory mapping and pipelining

Q10. Question on process efficiency.

Note: Some questions may have been missed or reported incorrectly. This is only a rough idea.

Date..... Subject.....

Sig..of invg..... Max. Marks..... Marks Obtd.....

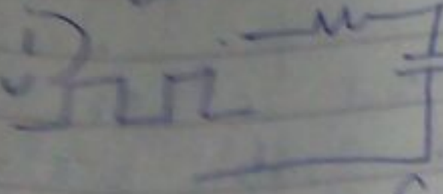
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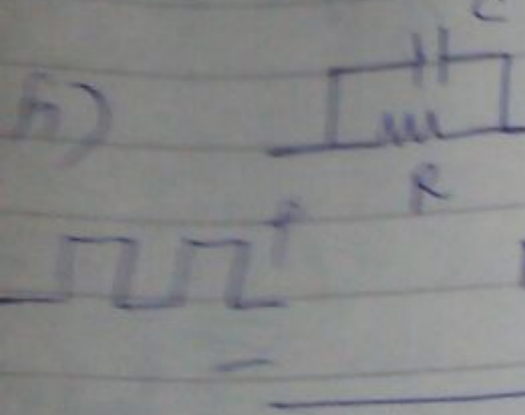
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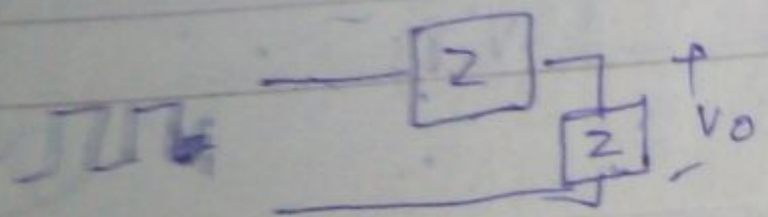
- 20 Q Aptitude → 30 min — 15 att. [data Int, Eq & Series, Eng Comprehension, Bargraph + Passage]
- 20 Q Analog → 45 min — 12 att. [Op-Amp, RLC ckt, Rectifier, diodes]
- 20 Q digital → 45 min — 8 att. [diodes, CMOS Power diss, Sizing of Pmos, Nmos, Kmap, Counter].

Written clear

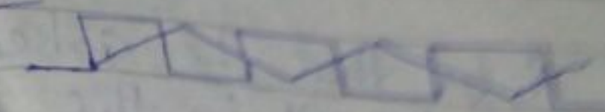
→ Interview →

i) 

ii) 

iii) 

find V_o & Plot My Ans



Now Plot V_o

I was confused & doesn't give satisfactory response. Then he asked me to consider two impedances $Z_1 = Z_2$ Now Plot tell what will be i.e. V_o

I said $V_o = \frac{V_{in}}{2}$

ould I apply +ve
for a stable of P B
erminal.

design a cmos inverte
draw its TC
2 i/p nand

Qualcomm

Written test

1. Aptitude → 20 Q :- 20 min (8-9 attempted) [DI (Bar, Passage), Geometry, Work & Time]
2. C, DS → 20 Q → 25 min. (7-8 attempted) [Stack, LL, DSD, C]
3. Digital → 20 Q → 30 min. (~10 Q) [Counter, K-map, Registers, µP]
   ↳ time consuming questions.

Written clear.

Interview :-

Q.1 Which Phone you are having?
I said HTC 620G.

Q.2 Which processor this phone has? Quad-core/Octa-core or other
I said no idea.

Q.3 Then he asked me. which problem you feel in this phone?
I said Power battery back up. Then he asked how many hours I said 7 AM to 7 PM, so nearly 12 hours. He said power provides 24 hours. I said may be I ~~excess~~ use excess but then I corrected my statement & said that I use it more but not so more that it provides only 12 hours back up.

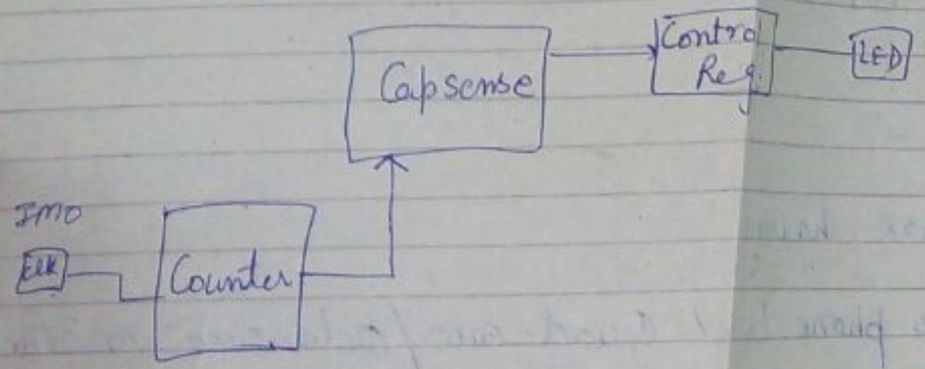
Q.4 So, how do you think we can solve this problem?
I told him about Power dissipation & frequency. So, we have to consider freq. as well as capacitances that are consuming energy. I talked about my b.tech project in which we have worked on Capsense tech[illegible]y in which Capacitance is converted

[illegible] asked me for [illegible] logic [illegible] then I asked any

Roll No. 26
...n & Embedded Systems)

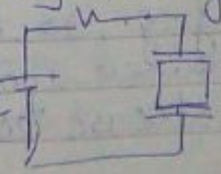
...TIONS

into digital value. He asked me to design the circuit diagram of capsense. But I didn't know that. So, he asked me to draw a schematic which I do in project



Then he asked me how you set the time of 3s for this counter. I explained him about counter concept used in counter. After that he asked to design the clk physically. I said

Clock : 2 clock aplications [VHDL]
He asked me to design physically. I thought a lot. Then I came to answer quartz oscillator. He asked me to design the clk: Then he asked me how to vary the freq. now as if we change V_{DC} & R. there will not be much effect as Quartz crystal is highly stable.



nverter only using NMOS.
& tell the logical effort
for these two.
I could not answer.

the size of PMOS twice only.
he asked me about
. I wrote

nt . I told him Conductanc
its unit is A/V².
I said no cus $\mu_n \frac{W}{L} 2\mu_p$
PMOS has double size.
Si ?

500 cm²/Vsec.

y question?

said whether it effects your
m. Then he explained its the
ays to reduce logical effort
ort book. But I forget that

interview

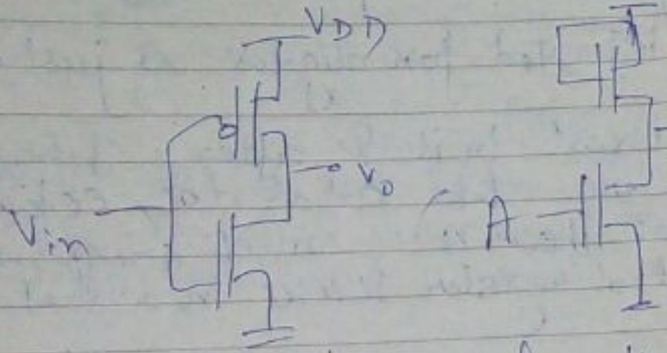
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5]
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age at the
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Draw a CMOS Inverter & an inverter only using NMOS & tell the logical effort for these two.

I could not answer.



Next he asked me why you chose the size of PMOS twice only.
I now forgot the answer then he asked me about process transconductance parameter. I wrote

$$K_n = \mu_n C_{ox}$$

He asked me what it represent. I told him conductance which I think was wrong as its unit is A/V^2.
Then he asked whether $K_n = K_p$. I said no as $\mu_n \approx 2\mu_p$.
Then I realised that that's why PMOS has double size.
Then he asked the μ_n & μ_p of Si ?

$\mu_n = 800\ cm^2/Vsec$. X $500\ cm^2/Vsec$.
$\mu_p = 1300\ cm^2/Vsec$.

Then he asked you wanna ask any question?
I asked him whether any kind of work is going on regarding

logical effort in
desire to do
need for every
He asked me
name too at

Again

Roll No.
...bedded Systems)

Page 8

...XAMINATION

Subject

Marks Obtd

...then Bargraph + Passage

...data int, Seq & Series, Eng Compre...
...Amp, RLC ckt, Rectifier, diode...
...odes, cmos Power diss, Sizing of...
...nos, kmap, Counter].

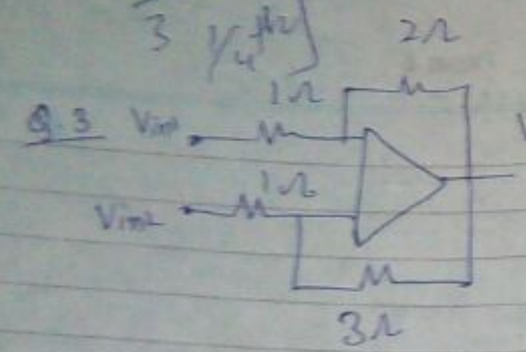
- I took large time & as I thought of an ans, he just skipped the Q.

Q Then he asked me are you interested for digital or just do it?
Then I told him that I have interest in it & after I have done my b.tech project in digital too like adc, dac, eeprom, RTC etc But these he said that these are analog component I told him that I used Counter, control register & Mux in all of these. But he didn't get satisfy.

Q. Then he asked why you wanna come to Banglore?
I told him that its tech city & also most of the research is going on in Banglore Plus I have specific interest in fabrication & cmos domain. So, I am expecting that soon this technology will boost up in banglore.

Q. Then he asked me about mtech project?
I told him that I am working on BCH codes currently, to calculate their minimum distance & then to design this quasi-LDPC decoder with some concept of BCH. He didn't ask any other question regarding this.
Then I told him about logical effort on which I am working He asked me for that. I explained him Then he asked.

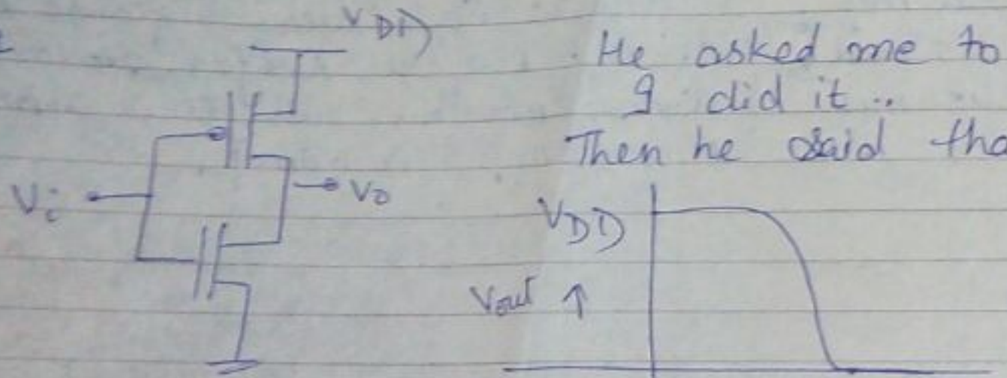
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3 Yr



Q.3 On which terminal should I apply +ve terminal & -ve terminal & for a stable o/p ?

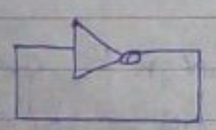
Ans I replied any terminal.

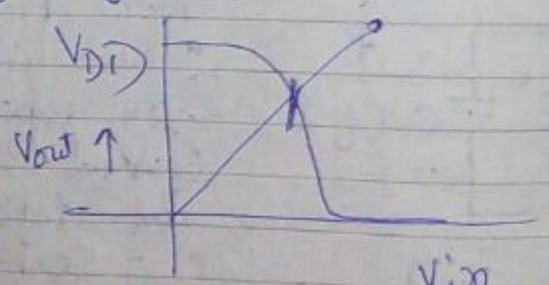
Q.4



He asked me to design a cmos inverter
I did it..
Then he said that let draw its T.C
I did that.

After that, he asked me to short V_{in} & V_{DD} V_{out} & then tell what will be the V_{out}. I plotted V_{in} z V_{out} graph but could not give satisfactory response





Interview over

Not qualified

Written test

1. Aptitude → 20Q. - 2
2. C, DS → 20Q →
3. Digital → 20Q →
   ↳ time consum

Written clear.

Interview :-

Q.1 Which Phone
I said HTC 620G

Q.2 Which processo
I said no idea.

Q.3 Then he asked
I said P~~ower~~
hours I said
mine provides 2
then I corrected
so more that it f

Q.4 So how cl
I told him
to consider fre
energy. I tol
worked on Cap

→ am working

Basic digital system design
- Basic VHDL/Verilog

- For Implementation
- Diffusion
- Y structures

↳ work fx

- Nmos, Pmos
- Clock using cmos & Tx Gate.
- Set up, hold time Numerical

- Skew & its
  - +ve edge skew
  - -ve edge skew
  - I/P Transition of/p load dependancy on delay

- Types of Memory
  - fastest & costliest Ram
  - Expensive
  - Drop drain से होता है gate से होता है।
    - $V_{DD} - V_T$
    - Kam
  - Leakage Power

Paper Pattern

→ Subject to be focussed
→ Criterion
→ Interview aspects

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Date

Subjective

Chakradhar Reddy Pocha

Apm

Paper Pattern →

IP design → RTL → gate

VLSI

Deep Submicron VLSI design

Physical design → Core VLSI

ARM Basics

→ Interview Q

- VLSI delay dependency based on temperature
- based on voltage ?

Digital system basics

$I = I_0 e^{-t/RC}$

V_{GS} ↑

- Never write eq" try to avoid them.

I_D ↑

$I = CR$

V_{GS} ↑ → time taken to charge a capacitor ↓

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STA & SI

Chapter 1: Introduction

Chapter 2: Static Timing Analysis

- 2.1Timing Paths
- 2.2Time Borrowing
- 2.3.a Basic Concept Of Setup and Hold
- 2.3.b Basic Concept of Setup and Hold Violation
- 2.3.c Practical Examples for Setup and Hold Time / Violation
- 2.4.a Delay - Timing Path Delay
- 2.4.b Delay -Interconnect Delay Models
- 2.4.c Delay - Wire Load Model
- 2.5.a Maximum Clock Frequency
- 2.5.b Examples to calculate the “Maximum Clock Frequency” for different circuits
- 2.6.a How to solve Setup and Hold Violation (basic example)
- 2.6.b Continue of... How to solve Setup and Hold Violation (Advance examples)
- 2.6.c Continue of...How to solve Setup and Hold Violation (more advance examples)
- 2.7.a Methods for Increase/Decrease the Delay of Circuit (Effect of Wire Length On the Slew)
- 2.7.b Methods for Increase/Decrease the Delay of Circuit (Effect of Size of the Transistor On the Slew)
- 2.7.c Methods for Increase/Decrease the Delay of Circuit (Effect of Threshold voltage On the Slew)
- 2.8 10 ways to fix Setup and Hold Violation.

Chapter 3: Clock

- 3.1 Sources Of Clock Skew
- 3.2 Types Of Clock Skew

Chapter 4: Advance Static Timing Analysis

- 4.1 Setup And Hold Checks
- 4.2 Setup And Hold Violation

Chapter 5: Signal Integrity

- 5.1 Introduction

Chapter 6: STA using EDA Tool

- 6.1 Static Timing Analysis Using EDA Tool
- 6.2 Static Timing Analysis Using EDA Tool .. (...continue)

Chapter 7: Timing Models

- 7.1.a ETM (Extracted Timing Models) Basics
- 7.1.b ETM (...continue)

Chapter 8: Other topics (Will categorize later)

- 8.1.a How To read SDF (Standard Delay Format)
- 8.1.b How To read SDF (...continue)
- 8.1.c Standard Delay Format (...continue)

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Monday, February 28, 2011

Basic of Timing Analysis in Physical Design

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professional).
Please let me know if anybody thinks that I should add few more things here. It's difficult to put everything in one blog so just consider this as the first part of Timing analysis.

What is Timing Analysis??

Before we start anything at least we should know what exactly we mean by Timing Analysis. Why these days it's so important?

There are a couple of reasons for performing timing analysis.

- We want to verify whether our circuit meet all of its timing requirements (Timing Constraints)
  - There are 3 types of design constraints- timing, power, area. During designing there is a trade-offs between speed, area, power, and runtime according to the constraints set by the designer. However, a chip must meet the timing constraints in order to operate at the intended clock rate, so timing is the most important design constraint.
- We want to make sure that circuit is properly designed and can work properly for all combinations of components over the entire specified operating environment. **"Every Time".**
- Timing analysis can also help with component selection.
  - An example is when you are trying to determine what memory device speed, you should use with a microprocessor. Using a memory device that is too slow may not work in the circuit (or would degrade performance by introducing wait states), and using one that is too fast will likely cost more than it needs to.

So I can say Timing analysis is the methodical analysis of a digital circuit to determine if the timing constraints imposed by components or interfaces are met. Typically, this means that you are trying to prove that all set-up, hold, and pulse-width times are being met.

Note: Timing analysis is integral part of ASIC/VLSI design flow. Anything else can be compromised but not timing!

Types of Timing Analysis:

There are 2 type of Timing Analysis

- Static Timing Analysis
  - Checks static delay requirements of the circuit without any input or output vectors.
- Dynamic Timing Analysis.
  - verifies functionality of the design by applying input vectors and checking for correct output vectors

Basic Of Timing Analysis:

The basis of all timing analysis is the clock and the sequential component (here we will discuss with the help of Flip-flop) . Following are few of the things related to clock and flip-flop which we usually wants to take care during Timing analysis.

Clock related:

- It must be well understood parametrically and glitch-free.
- Timing analysis must ensure that any clocks that are generated by the logic are clean, are of bounded period and duty cycle, and of a known phase relationship to other clock signals of interest.
- The clock must, for both high and low phases, meet the minimum pulse width requirements.

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Delay - "Wire Load

- Certain circuits, such as PLLs, may have other requirements such as maximum jitter. As the clock speeds increase, jitter becomes an increasingly important parameter.
- When "passing" data from one clock edge to the other, ensure that the worst-case duty cycle is used for the calculation. A frequent source of error is the analyst assuming that every clock will have a 50% duty cycle.

Flip-Flop related:

- All of the flip-flops parameters are always met. The only exception to this is when synchronizers are used to synchronize asynchronous signals
- For asynchronous presets and clears, there are two basic parameters that must be met.
- All setup and hold times are met for the earliest/latest arrival times for the clock.
- Setup times are generally calculated by designers and suitable margins can be demonstrated under test. Hold times, however, are frequently not calculated by designers.
- When passing data from one clock domain to another, ensure that there is either known phase relationships which will guarantee meeting setup and hold times or that the circuits are properly synchronized

Now Lets talk about Each type of Timing analysis One by one in the next few blogs.

Next

You might also like:

"Timing Paths" : Static Timing Analysis (STA) basic (Part 1)

Hierarchical Design Flow - part 2

Setup and Hold Check: Advance STA (Static Timing Analysis)

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Anonymous March 27, 2013 at 8:22 PM

useful

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Ravi Gullapalli Gullapally GARV October 8, 2015 at 9:25 PM

I don't know how to address you. Until I know you, will stick to Senior.

Senior, what we loosing by limiting our selfs to static analysis to fix all the timing issues?

Model" : Static Timing Analysis (STA) basic (Part 4c)

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

"Setup and Hold Time" : Static Timing Analysis (STA) basic (Part 3a)

"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

"Examples Of Setup and Hold time" : Static Timing Analysis (STA) basic (Part 3c)

"Time Borrowing" : Static Timing Analysis (STA) basic (Part 2)

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Wednesday, March 9, 2011

"Timing Paths" : Static Timing Analysis (STA) basic (Part 1)

Part1	Part2	Part3a	Part3b	Part3c	Part4a	Part4b	Part4c	Part5a
Part5b	Part6a	Part6b	Part6c	Part7a	Part7b	Part7c	Part8	

Static Timing analysis is divided into several parts:

- Part1 -> Timing Paths
- Part2 -> Time Borrowing
- Part3a -> Basic Concept Of Setup and Hold
- Part3b -> Basic Concept of Setup and Hold Violation
- Part3c -> Practical Examples for Setup and Hold Time / Violation
- Part4a -> Delay - Timing Path Delay
- Part4b -> Delay - Interconnect Delay Models
- Part4c -> Delay - Wire Load Model
- Part5a -> Maximum Clock Frequency
- Part5b -> Examples to calculate the "Maximum Clock Frequency" for different circuits.
- Part 6a -> How to solve Setup and Hold Violation (basic example)
- Part 6b -> Continue of How to solve Setup and Hold Violation (Advance examples)
- Part 6c -> Continue of How to solve Setup and Hold Violation (more advance examples)
- Part 7a -> Methods for Increase/Decrease the Delay of Circuit (Effect of Wire Length On the Slew)
- Part 7b -> Methods for Increase/Decrease the Delay of Circuit (Effect of Size of the Transistor On the Slew)
- Part 7c -> Methods for Increase/Decrease the Delay of Circuit (Effect of Threshold voltage On the Slew)
- Part 8 -> 10 ways to fix Setup and Hold Violation.

As we have discussed in our last blog (about Basic of Timing analysis) that there are 2 types of timing analysis.

- Static Timing Analysis
- Dynamic Timing Analysis.

Note: There is one more type of Timing analysis: "Manual Analysis". But now a days nothing is 100% Manual. Evey thing is more automated and less manual. So that we are not discussing right now.

In this Blog (and few next as a part of this) we will discuss about the Static Timing Analysis. We will discuss Dynamic Timing Analysis later on.
Static Timing analysis is divided into several parts as per the above mentioned list.

Static Timing Analysis:

Static timing analysis is a method of validating the timing performance of a design by checking all possible paths for timing violations under worst-case conditions. *It considers the worst possible delay through each logic element, but not the logical*

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operation of the circuit.

In comparison to circuit simulation, static timing analysis is

- Faster - It is faster because it does not need to simulate multiple test vectors.
- More Thorough - It is more thorough because it checks the worst-case timing for all possible logic conditions, not just those sensitized by a particular set of test vectors.

Once again Note this thing : Static timing analysis checks the design only for proper timing, not for correct logical functionality.

Static timing analysis seeks to answer the question, "Will the correct data be present at the data input of each synchronous device when the clock edge arrives, under all possible conditions?"

In static timing analysis, the word static alludes to the fact that this timing analysis is carried out in an input-independent manner. It locates the worst-case delay of the circuit over all possible input combinations. There are huge numbers of logic paths inside a chip of complex design. The advantage of STA is that it performs timing analysis on all possible paths (whether they are real or potential false paths).
However, it is worth noting that STA is not suitable for all design styles. It has proven efficient only for fully synchronous designs. Since the majority of chip design is synchronous, it has become a mainstay of chip design over the last few decades.

The Way STA is performed on a given Circuit:
To check a design for violations or say to perform STA there are 3 main steps:

- Design is broken down into sets of timing paths,
- Calculates the signal propagation delay along each path
- And checks for violations of timing constraints inside the design and at the input/output interface.

The STA tool analyzes ALL paths from each and every startpoint to each and every endpoint and compares it against the constraint that (should) exist for that path. All paths should be constrained, most paths are constrained by the definition of the period of the clock, and the timing characteristics of the primary inputs and outputs of the circuit.

Before we start all this we should know few key concepts in STA method: timing path, arrive time, required time, slack and critical path.
Let's Talk about these one by one in detail. In this Blog we will mainly Focus over Different Types of Timing Paths.

Timing Paths:

Timing paths can be divided as per the type of signals (e.g clock signal, data signal etc).

Types of Paths for Timing analysis:

- Data Path
- Clock Path
- Clock Gating Path
- Asynchronous Path

Each Timing path has a "Start Point" and an "End Point". Definition of Start Point and End Point vary as per the type of the timing path. E.g for the Data path- The startpoint is a place in the design where data is launched by a clock edge. The data is propagated through combinational logic in the path and then captured at the endpoint by another clock edge.

Start Point and End Point are different for each type of paths. It's very important to understand this clearly to understand and analysing the Timing analysis report and fixing the timing violation.

- Data path
  - Start Point
    - Input port of the design (because the input data can be launched from some external source).
    - Clock pin of the flip-flop/latch/memory (sequential cell)
  - End Point
    - Data input pin of the flip-flop/latch/memory (sequential cell)
    - Output port of the design (because the output data can be captured by some external sink)
- Clock Path
  - Start Point
    - Clock input port
  - End Point
    - Clock pin of the flip-flop/latch/memory (sequential cell)
- Clock Gating Path
  - Start Point
    - Input port of the design
  - End Point
    - Input port of clock-gating element.
- Asynchronous path
  - Start Point
    - Input Port of the design

"Timing Paths" : Static Timing Analysis (STA) basic (Part 1)

Delay - "Wire Load Model" : Static Timing Analysis (STA) basic (Part 4c)

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

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"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

"Examples Of Setup and Hold time" : Static Timing Analysis (STA) basic (Part 3c)

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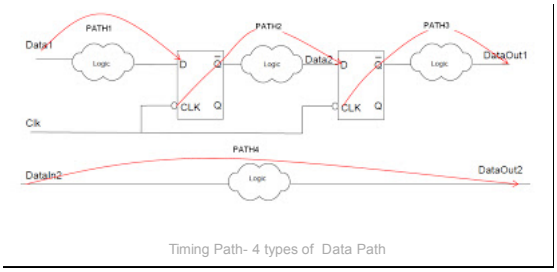
- End Point
  - Set/Reset/Clear pin of the flip-flop/latch/memory (sequential cell)

Data Paths:

If we use all the combination of 2 types of Starting Point and 2 types of End Point, we can say that there are 4 types of Timing Paths on the basis of Start and End point.

- Input pin/port to Register(flip-flop).
- Input pin/port to Output pin/port.
- Register (flip-flop) to Register (flip-flop)
- Register (flip-flop) to Output pin/port

Please see the following fig:

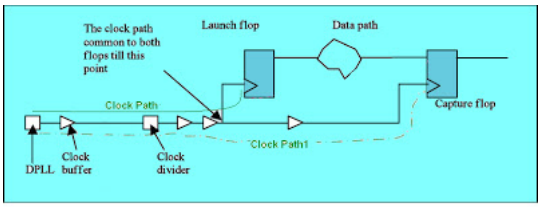


Timing Path- 4 types of Data Path

PATH1- starts at an input port and ends at the data input of a sequential element. (Input port to Register)
PATH2- starts at the clock pin of a sequential element and ends at the data input of a sequential element. (Register to Register)
PATH3- starts at the clock pin of a sequential element and ends at an output port.(Register to Output port).
PATH4- starts at an input port and ends at an output port. (Input port to Output port)

Clock Path:

Please check the following figure



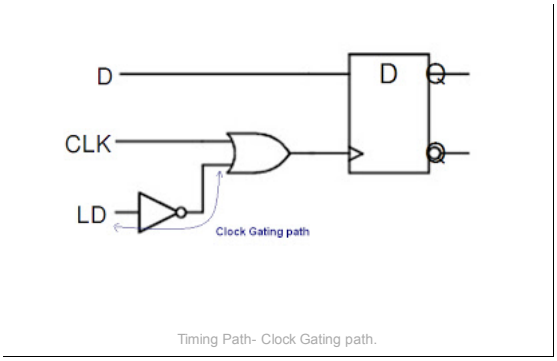
Timing Paths- Clock Paths

In the above fig its very clear that for clock path the starts from the input port/pin of the design which is specific for the Clock input and the end point is the clock pin of a sequential element. In between the Start point and the end point there may be lots of Buffers/Inverters/clock divider.

Clock Gating Path:

Clock path may be passed trough a “gated element” to achieve additional advantages. In this case, characteristics and definitions of the clock change accordingly. We call this type of clock path as “gated clock path”.

As in the following fig you can see that



Timing Path- Clock Gating path.

LD pin is not a part of any clock but it is using for gating the original CLK signal. Such type of paths are neither a part of Clock path nor of Data Path because as per the Start Point and End Point definition of these paths, its different. So such type of paths are part

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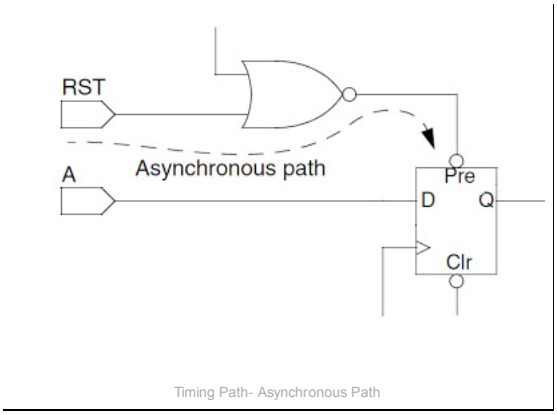
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of Clock gating path.

Asynchronous path:

A path from an input port to an asynchronous set or clear pin of a sequential element.

See the following fig for understanding clearly.



Timing Path- Asynchronous Path

As you know that the functionality of set/reset pin is independent from the clock edge. Its level triggered pins and can start functioning at any time of data. So in other way we can say that this path is not in synchronous with the rest of the circuit and that's the reason we are saying such type of path an Asynchronous path.

Other types of Paths:

There are few more types of path which we usually use during timing analysis reports. Those are subset of above mention paths with some specific characteristics. Since we are discussing about the timing paths, so it will be good if we will discuss those here also.

Few names are

- Critical path
- False Path
- Multi-cycle path
- Single Cycle path
- Launch Path
- Capture Path
- Longest Path (also know as Worst Path , Late Path , Max Path , Maximum Delay Path)
- Shortest Path (Also Know as Best Path , Early Path , Min Path, Minimum Delay Path)

Critical Path:

In short, I can say that the path which creates Longest delay is the critical path.

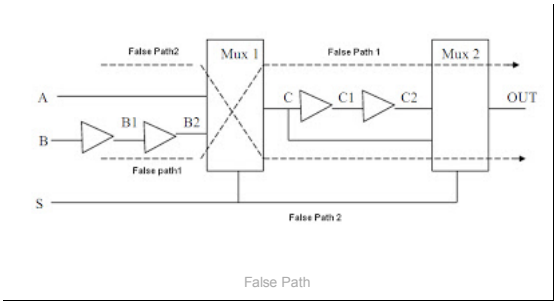
- Critical paths are timing-sensitive functional paths. because of the timing of these paths is critical, no additional gates are allowed to be added to the path, to prevent increasing the delay of the critical path.
- Timing critical path are those path that do not meet your timing. What normally happens is that after synthesis the tool will give you a number of path which have a negative slag. The first thing you would do is to make sure those path are not false or multicycle since it that case you can just ignore them.

Taking a typical example (in a very simpler way), the STA tool will add the delay contributed from all the logic connecting the Q output of one flop to the D input of the next (including the CLK->Q of the first flop), and then compare it against the defined clock period of the CLK pins (assuming both flops are on the same clock, and taking into account the setup time of the second flop and the clock skew). This should be strictly less than the clock period defined for that clock. If the delay is less than the clock period, then the "path meets timing". If it is greater, than the "path fails timing". The "critical path" is the path out of all the possible paths that either exceeds its constraint by the largest amount, or, if all paths pass, then the one that comes closest to failing.

False Path:

- Physically exist in the design but those are logically/functionally incorrect path. Means no data is transferred from Start Point to End Point. There may be several reasons of such path present in the design.
- Some time we have to explicitly define/create few false path with in the design. E.g for setting a relationship between 2 Asynchronous Clocks.
- The goal in static timing analysis is to do timing analysis on all “true” timing paths, these paths are excluded from timing analysis.
- Since false path are not exercised during normal circuit operation, they typically don't meet timing specification,considering false path during timing closure can result into timing violations and the procedure to fix would introduce unnecessary complexities in the design.
- There may be few paths in your design which are not critical for timing or masking other paths which are important for timing optimization, or never occur with in normal situation. In such case , to increase the run time and improving the timing result , sometime we have to declare such path as a False path , so that Timing analysis tool ignore these paths

and so the proper analysis with respect to other paths. Or During optimization don't concentrate over such paths. One example of this. e.g A path between two multiplexed blocks that are never enabled at the same time. You can see the following picture for this.



False Path

Here you can see that False path 1 and False Path 2 can not occur at the same time but during optimization it can effect the timing of another path. So in such scenario, we have to define one of the path as false path.

Same thing I can explain in another way (Note- Took snapshot from one of the forum). As we know that, not all paths that exist in a circuit are "real" timing paths. For example, let us assume that one of the primary inputs to the chip is a configuration input; on the board it must be tied either to VCC or to GND. Since this pin can never change, there are never any timing events on that signal. As a result, all STA paths that start at this particular startpoint are false. The STA tool (and the synthesis tool) cannot know that this pin is going to be tied off, so it needs to be told that these STA paths are false, which the designer can do by telling the tool using a "false_path" directive. When told that the paths are false, the STA tool will not analyze it (and hence will not compare it to a constraint, so this path can not fail), nor will a synthesis tool do any optimizations on that particular path to make it faster; synthesis tools try and improve paths until they "meet timing" - since the path is false, the synthesis tool has no work to do on this path.
Thus, a path should be declared false if the designer KNOWS that the path in question is not a real timing path, even though it looks like one to the STA tool. One must be very careful with declaring a path false. If you declare a path false, and there is ANY situation where it is actually a real path, then you have created the potential for a circuit to fail, and for the most part, you will not catch the error until the chip is on a board, and (not) working. Typically, false paths exists

- from configuration inputs like the one described above
- from "test" inputs; inputs that are only used in the testing of the chip,and are tied off in normal mode (however, there may still be some static timing constraints for the test mode of the chip)
- from asynchronous inputs to the chip (and you must have some form of synchronizing circuit on this input) (this is not an exhaustive list, but covers the majority of legitimate false paths).

So we can say that false paths should NOT be derived from running the STA tool (or synthesis tool); they should be known by the designer as part of the definition of the circuit, and constrained accordingly at the time of initial synthesis.

MultiCycle Path:

- A multicycle path is a timing path that is designed to take more than one clock cycle for the data to propagate from the startpoint to the endpoint.

A multi-cycle path is a path that is allowed multiple clock cycles for propagation. Again, it is a path that starts at a timing startpoint and ends at a timing endpoint. However, for a multi-cycle path, the normal constraint on this path is overridden to allow for the propagation to take multiple clocks.
In the simplest example, the startpoint and endpoint are flops clocked by the same clock. The normal constraint is therefore applied by the definition of the clock; the sum of all delays from the CLK arrival at the first flop to the arrival at the D of the second clock should take no more than 1 clock period minus the setup time of the second flop and adjusted for clock skew.
By defining the path as a multicycle path you can tell the synthesis or STA tool that the path has N clock cycles to propagate; so the timing check becomes "the propagation must be less than N x clock_period, minus the setup time and clock skew". N can be any number greater than 1.

Few examples are

- When you are doing clock crossing from two closely related clocks; ie. from a 30MHz clock to a 60MHz clock,
  - Assuming the two clocks are from the same clock source (i.e. one is the divided clock of the other), and the two clocks are in phase.
  - The normal constraint in this case is from the rising edge of the 30MHz clock to the nearest edge of the 60MHz clock, which is 16ns later. However, if you have a signal in the 60MHz domain that indicates the phase of the 30MHz clock, you can design a circuit that allows for the full 33ns for the clock crossing, then the path from flop30 -> to flop60 is a MCP (again with N=2).
  - The generation of the signal 30MHZ_is_low is not trivial, since it must come from a flop which is clocked by the 60MHz clock, but show the phase of the 30MHz clock.
- Another place would be when you have different parts of the design that run at different, but related frequencies. Again, consider a circuit that has some stuff running at 60MHz and some running on a divided clock at 30MHz.
  - Instead of actually defining 2 clocks, you can use only the faster clock, and have a clock enable that prevents the clocks in the slower domain from updating every other clock,
  - Then all the paths from the "30MHz" flops to the "30MHz" flops can be MCP.
  - This is often done since it is usually a good idea to keep the number of different clock domains to a minimum.

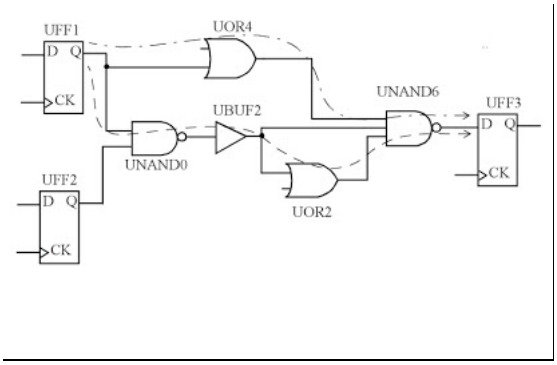
Single Cycle Path:

A Single-cycle path is a timing path that is designed to take only one clock cycle for the data to propagate from the startpoint to the endpoint.

Launch Path and Capture Path:

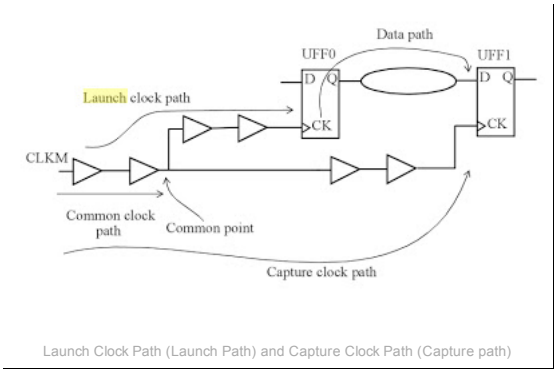
Both are inter-related so I am describing both in one place. When a flip flop to filp-flop path such as UFF1 to UFF3 is considered, one of the flip-flop launches the data and other captures the data. So here UFF1 is referred to "launch Flip-flop" and UFF3 referred to

"capture flip-flop".



These Launch and Capture terminology are always referred to a flip-flop to flip-flop path. Means for this particular path (UFF1->UFF3), UFF1 is launch flip-flop and UFF3 is capture flip-flop. Now if there is any other path starting from UFF3 and ends to some other flip-flop (lets assume UFF4), then for that path UFF3 become launch flip-flop and UFF4 be as capture flip-flop.

The Name "Launch path" referred to a part of clock path. Launch path is launch clock path which is responsible for launching the data at launch flip flop.
And Similarly Capture path is also a part of clock path. Capture path is capture clock path which is responsible for capturing the data at capture flip flop.
This is can be clearly understood by following fig.



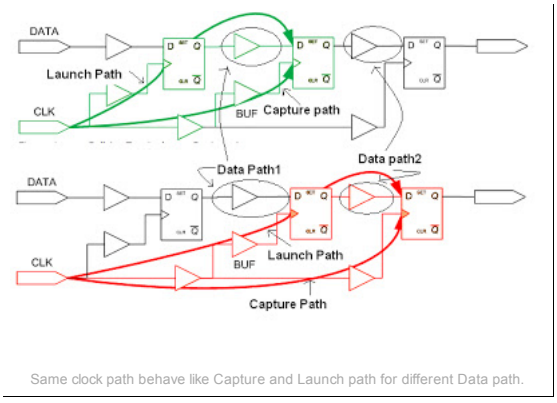
Launch Clock Path (Launch Path) and Capture Clock Path (Capture path)

Here UFF0 is referred to launch flip-flop and UFF1 as capture flip-flop for "Data path" between UFF0 to UFF1.So Start point for this data path is UFF0/CK and end point is UFF1/D.

One thing I want to add here (which I will describe later in my next blog- but its easy to understand here)-

- Launch path and data path together constitute arrival time of data at the input of capture flip-flop.
- Capture clock period and its path delay together constitute required time of data at the input of capture register.

Note: Its very clear that capture and launch paths are correspond to Data path. Means same clock path can be a launch path for one data path and be a capture path for another datapath. Its will be clear by the following fig (source of Fig is From Synopsys).

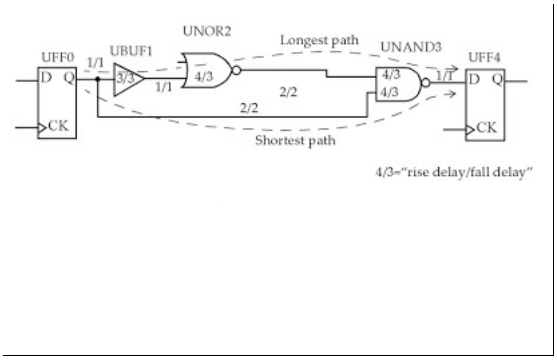


Same clock path behave like Capture and Launch path for different Data path.

Here you can see that for Data path1 the clock path through BUF cell is a capture path but for Data path2 its a Launch Path.

Longest and Shortest Path:

Between any 2 points, there can be many paths.
Longest path is the one that takes longest time, this is also called worst path or late path or a max path.
The shortest path is the one that takes the shortest time; this is also called the best path or early path or a min path.



In the above fig, The longest path between the 2 flip-flop is through the cells UBUF1,UNOR2 and UNAND3. The shortest path between the 2 flip-flops is through the cell UNAND3.

I have tried my best to capture all the important points related to the Timing Paths. Please Let me know If anything is missing here.

Timing Analysis Basis (Previous) | Index | Time Borrowing (Next)

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samiappa sakthikumaran March 10, 2011 at 9:48 PM

Hi sir, if for a certain timing contraint u give for ex: let me say that my clock period is 5ns and if slack is met using synopsys primetime, so can i say my frequency of operation of my whole ckt is 200MHz ??? kindly reply. Thanks

Reply

Replies

Prashanth Anil Mascarenhas February 10, 2014 at 8:01 PM

Yes. Frequency = 1 / Clock Period

Reply

Indu Mathi April 15, 2011 at 4:03 PM

Hi,

Fabulous work u r doing. Reaaly the topics covered in this blog are helping me a lot.

Thanks & Regards,
indu.

Reply

your VLSI April 18, 2011 at 11:19 AM

Hi @Samiappa -- you are right.. if slack is meeting.. then you can say .. but still there is a "can".. :) means its not 100% true. There are other factors also. Please see my other blogs for details...

@Indu -- Thanks a lot for such a appreciation.

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Monday, March 28, 2011

"Time Borrowing" : Static Timing Analysis (STA) basic (Part 2)

Part1	Part2	Part3a	Part3b	Part3c	Part4a	Part4b	Part4c	Part5a
Part5b	Part6a	Part6b	Part6c	Part7a	Part7b	Part7c	Part 8	

Static Timing analysis is divided into several parts:

- Part1 -> Timing Paths
- Part2 -> Time Borrowing
- Part3a -> Basic Concept Of Setup and Hold
- Part3b -> Basic Concept of Setup and Hold Violation
- Part3c -> Practical Examples for Setup and Hold Time / Violation
- Part4a -> Delay - Timing Path Delay
- Part4b -> Delay - Interconnect Delay Models
- Part4c -> Delay - Wire Load Model
- Part5a -> Maximum Clock Frequency
- Part5b -> Examples to calculate the “Maximum Clock Frequency” for different circuits.
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- Part 6b -> Continue of How to solve Setup and Hold Violation (Advance examples)
- Part 6c -> Continue of How to solve Setup and Hold Violation (more advance examples)
- Part 7a -> Methods for Increase/Decrease the Delay of Circuit (Effect of Wire Length On the Slew)
- Part 7b -> Methods for Increase/Decrease the Delay of Circuit (Effect of Size of the Transistor On the Slew)
- Part 7c -> Methods for Increase/Decrease the Delay of Circuit (Effect of Threshold voltage On the Slew)
- Part 8 -> 10 ways to fix Setup and Hold Violation.

In a ASIC there are majorly two types of component. Flip-flop and other is Latches. Basically Here we will discuss about Latched based timing analysis.
Before this we should understand the basic differences between the latch based design and flip-flop based design.

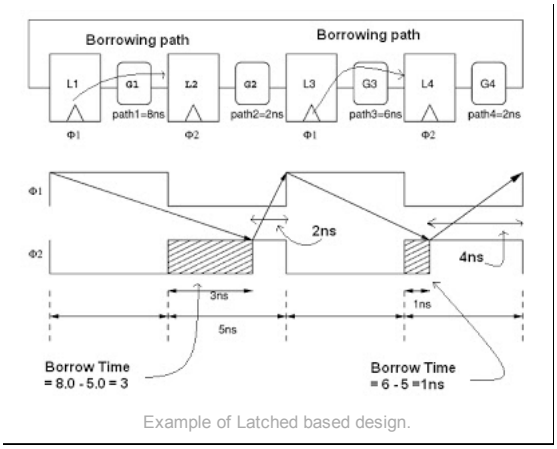
- Edge-triggered flip-flops change states at the clock edges, whereas latches change states as long as the clock pin is enabled.
- The delay of a combinational logic path of a design using edge-triggered flip-flops cannot be longer than the clock period except for those specified as false paths and multiple-cycle paths. So the performance of a circuit is limited by the longest path of a design.
- In latch based design longer combinational path can be compensated by shorter path delays in the sebsequent logic stages.So for higher performance circuits deisgner are turning to latched based design.

Its true that in the latched based design its difficult to control the timing because of multi-phase clockes used and the lack of "hard" clock edges at which events must occur.

The technique of borrowing time from the shorter paths of the subsequent logic stages to the longer path is called ***time borrowing or cycle stealing***.

Lets talk about this. Please See the following figure.

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Basic of Timing Analysis in Physical Design



Example of Latched based design.

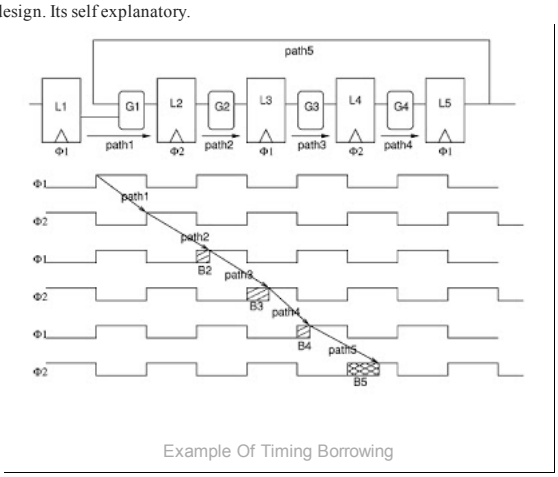
There are 4 latches (positive level sensitive). L1 and L3 are controlled by PH1 and L2 and L4 are controlled by PH2. G1, G2, G3 and G4 are combinational logic paths. For now assume a library setup time is zero for the latches and zero delay in latch data-path in the transparent mode.

Now if assume that if designs using edge-triggered flip-flops, the clock period has to be at least 8 ns because the longest path in G1 is 8 ns. Now as the clock pulse is 5ns , there is a voilation at L2. On the other hand, if the design uses latches , L2 latch is transparent for another 5ns and since the eighth (8th) ns is within the enabled period of L2, the signal along path1 can pass through L2 and continue on path2. Since the delay along path2 is 2 ns, which is short enough to compensate for the overdue delay of path1, this design will work properly. In other word we can say that path1 can borrow sometime (3ns) from the path2. Since the sum of path1 and path2 is 10ns, which is the required time of L3, there will be no voilation in either of the Latches.

For the same reason, path3 can borrow some time (1ns) from path4 without any timing violation.

Note: A latch-based design completes the execution of the four logic stages in 20 ns, whereas an edge-triggered based design needs 32 ns.

Lets see this in a more complex design. Its self explanatory.



Example Of Timing Borrowing

Just wanted to convey here that this Timing borrowing can be multistage. Means we can easily say that for a latched based design, each executing path must start at a time when its driving latch is enabled, and end at a time when its driven latch is enabled.

Few Important things:

- Time borrowing occur with in the same cycle. Means launching and capturing latches be using the same phase of the same clock. when the clocks of the launching and capturing latches are out of phase, time borrowing is not to happen. Usually it was disabled by EDA tools.
- Time borrowing typically only affects setup slack calculation since time borrowing slows data arrival times. Since hold time slack calculation uses fastest data, time-borrowing typically does not affect hold slack calculation.

Few Important terminology:

Maximum Borrow time:

Maximum Borrow time is the clock pulse width minus the library setup time of the latch. Usually to calculate the maximum allowable borrow time, start with clock pulse width and then substract clock latency , clock reconvergence pessimism removal , library setup time of the endpoint latch.

Negative Borrow time:

If the arrival time minus the clock edge is a negative number, the amount of time borrowing is negative (in other way you can say that no borrowing). This amount is know as Negative Borrow time.

"Timing Paths" : Static Timing Analysis (STA) basic (Part 1)

Delay - "Wire Load Model" : Static Timing Analysis (STA) basic (Part 4c)

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

"Setup and Hold Time" : Static Timing Analysis (STA) basic (Part 3a)

"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

"Examples Of Setup and Hold time" : Static Timing Analysis (STA) basic (Part 3c)

"Time Borrowing" : Static Timing Analysis (STA) basic (Part 2)

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"Setup and Hold Time" : Static Timing Analysis (STA) basic (Part 3a)

Part1	Part2	Part3a	Part3b	Part3c	Part4a	Part4b	Part4c	Part5a
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- Part 7c -> Methods for Increase/Decrease the Delay of Circuit (Effect of Threshold voltage On the Slew)
- Part 8 -> 10 ways to fix Setup and Hold Violation.

Its been long time, people are asking about Setup and Hold time blog. Finally time come for that. :)

The way we will discuss this concept in the following manner

1. What is SetUp and Hold time?
2. Definition of Setup and Hold.
3. Setup and Hold Violation.
4. How to calculate the Setup and Hold violation in a design?

I saw that lots of people are confused with respect to this concept. And the reason of this are

1. They know the definition but don't know the origin or say concept behind Setup and Hold timing.
2. They know the formula for calculating setup and hold violation but don't know how this formula come in picture.
3. They become confuse by few of the terminology like capture path delay, launch path delay, previous clock cycle, current clock cycle, data path delay, slew, setup slew, hold slew, min and max concept, slowest path and fastest path, min and max corner, best and worst case etc during the explanation of Setup and Hold Timings/Violation.

I hope I can clarify your confusion. Let me explain this and if you face any problem let me know.

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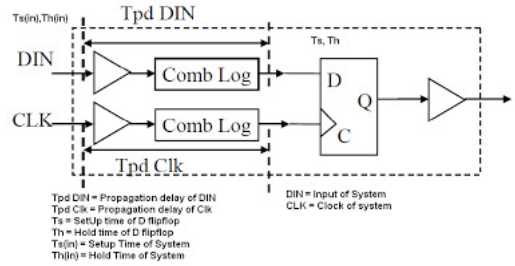
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What is Setup and Hold time?

To understand the origin of the Setup and Hold time concepts first understand it with respect to a System as shown in the fig. An Input DIN and external clock CLK are buffered and passes through combinational logic before they reach a synchronous input and a clock input of a D flipflop (positive edge triggered). Now to capture the data correctly at D flip flop, data should be present at the time of positive edge of clock signal at the C pin (to know the detail just read basis of D flipflop).
Note: here we are assuming D flip flop is ideal so Zero hold and setup time for this.



SetUp and Hold Time of a System

There may be only 2 condition.

- **Tpd DIN > Tpd Clk**
  - For capture the data at the same time when Clock signal (positive clock edge) reaches at pin C, you have to apply the input Data at pin DIN "Ts(in)=(Tpd DIN) - (Tpd Clk)" time before the positive clock edge at pin CLK.
  - In other word, at DIN pin, Data should be stable "Ts(in)" time before the positive clock edge at CLK pin.
  - **This Time "Ts(in)" is know as Setup time of the System.**
- **Tpd DIN < Tpd Clk**
  - For capture the data at the same time when clock signal (positive clock edge) reaches at pin C, input Data at pin DIN should not change before "Th(in)= (Tpd Clk) - (Tpd DIN)" time. If it will change, positive clock edge at pin C will capture the next data.
  - In other word, at DIN pin, Data should be stable "Th(in)" time after the positive clock edge at CLK pin.
  - **This time "Th(in)" is know as Hold Time of the System.**

From the above condition it looks like that both the condition can't exist at the same time and you are right. But we have to consider few more things in this.

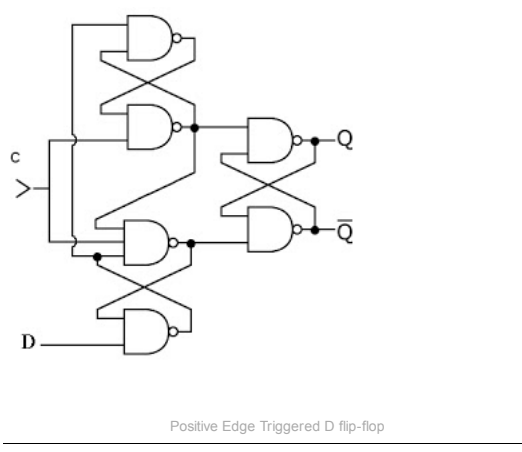
- Worst case and best case (Max delay and min delay)
  - Because of environment condition or because of PVT, we can do this analysis for the worst case (max delay) and best case (min delay) also.
- Shortest Path or Longest path (Min Delay and Max delay)
  - If combinational logic has multiple paths, the we have to do this analysis for the shortest path (min delay) and longest path (max delay) also.

So we can say that above condition can be like this.

- **Tpd DIN (max) > Tpd Clk (min)**
  - **SetUp time == Tpd DIN (max) - Tpd Clk (min)**
- **Tpd DIN (min) < Tpd Clk (max)**
  - **Hold time == Tpd Clk (max) - Tpd DIN (min)**

For example for combinational logic delays are
Data path (max, min) = (5ns, 4 ns)
Clock path (max, min) = (4.5ns, 4.1ns)
Then Setup time= 5-4.1=0.9ns
Hold time is = 4.5-4=0.5ns

Now similar type of explanation we can give for a D flip flop. There is a combinational logic between C and Q , between D and Q of the Flipflop. There are different delays in those conbinational logic and based on there max and min value , a flipflop has Setup and Hold time. One circuitry of the positive edge triggered D flip is shown below.



Positive Edge Triggered D flip-flop

Model" : Static Timing Analysis (STA) basic (Part 4c)

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

"Setup and Hold Time" : Static Timing Analysis (STA) basic (Part 3a)

"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

"Examples Of Setup and Hold time" : Static Timing Analysis (STA) basic (Part 3c)

"Time Borrowing" : Static Timing Analysis (STA) basic (Part 2)

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10 Ways to fix SETUP and HOLD violation: Static Timing Analysis (STA) Basic (Part-8)

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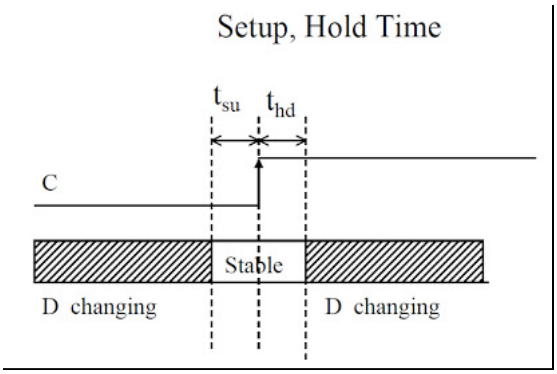
There are different ways for making the D flip flop. Like by JK flipflop, master slave flipflop, Using 2 D type latches etc. Since the internal circuitry is different for each type of Flipflop, the Setup and Hold time is different for every Flipflop.

Definition:
Setup Time:

- **Setup time** is the minimum amount of time the data signal should be held steady **before** the clock event so that the data are reliably sampled by the clock. This applies to synchronous circuits such as the flip-flop.
- Or In short I can say that the amount of time the Synchronous input (D) must be stable **before** the active edge of the Clock.
- The Time when input data is available and stable **before** the clock pulse is applied is called Setup time.

Hold time:

- **Hold time** is the minimum amount of time the data signal should be held steady **after** the clock event so that the data are reliably sampled. This applies to synchronous circuits such as the flip-flop.
- Or in short I can say that the amount of time the synchronous input (D) must be stable **after** the active edge of clock.
- The Time **after** clock pulse where data input is held stable is called hold time.



Setup and Hold Violation:

In simple language-
If Setup time is Ts for a flip-flop and if data is not stable before Ts time from active edge of the clock, there is a Setup violation at that flipflop. So if data is changing in the non-shaded area (in the above figure) before active clock edge, then it's a Setup violation.
And If hold time is Th for a flip flop and if data is not stable after Th time from active edge of the clock , there is a hold violation at that flipflop. So if data is changing in the non-shaded area (in the above figure) after active clock edge, then it's a Hold violation.

How to calculate the setup and hold violation in a design.. please see the next blog.

Time Borrowing (Previous) | Index | Basic Concept of Setup and Hold Violation (Next)

Posted by VLSI EXPERT at 3:09 PM

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68 comments:

anil April 20, 2011 at 5:41 PM

this is really useful... thank you very much...
Really looking forward for your future posts...

Reply

your VLSI April 21, 2011 at 10:27 AM

Thanks for such comments. I will update you once there will be any new post. you can also subscribe to my blog or by twitter account to get a regular update.

Reply

Anonymous April 23, 2011 at 11:26 PM

Nice explanation with useful examples. Thanks!!

Reply

Anonymous July 6, 2011 at 12:25 AM

your work is really very helpful..

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Friday, April 8, 2011

"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

Part1	Part2	Part3a	Part3b	Part3c	Part4a	Part4b	Part4c	Part5a
Part5b	Part6a	Part6b	Part6c	Part7a	Part7b	Part7c	Part 8	

Static Timing analysis is divided into several parts:

- Part1 -> Timing Paths
- Part2 -> Time Borrowing
- Part3a -> Basic Concept Of Setup and Hold
- Part3b -> Basic Concept of Setup and Hold Violation
- Part3c -> Practical Examples for Setup and Hold Time / Violation
- Part4a -> Delay - Timing Path Delay
- Part4b -> Delay - Interconnect Delay Models
- Part4c -> Delay - Wire Load Model
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- Part 7c -> Methods for Increase/Decrease the Delay of Circuit (Effect of Threshold voltage On the Slew)
- Part 8 -> 10 ways to fix Setup and Hold Violation.

Here we will discuss how to calculate the Setup and Hold Violation for a design.

Till now we have discussed setup and hold violation with respect to the single flipflop, now lets extend this to 2 flip flop. In the following fig there are 2 flipflops (FF1 and FF2).

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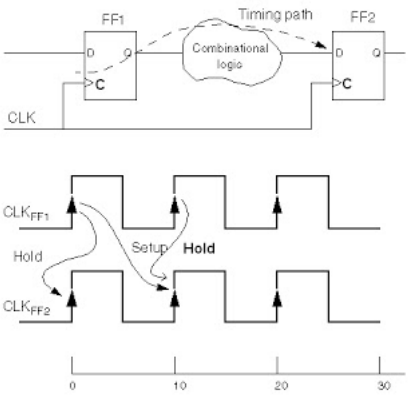
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Single-Cycle Setup and Hold For Flip-Flops

Few important things to note down here-

- Data is launching from FF1/D to FF1/Q at the positive clock edge at FF1/C.
- At FF2/D , input data is coming from FF1/Q through a combinational logic.
- Data is capturing at FF2/D, at the positive clock edge at FF2/C.
- So I can say that Launching Flip-Flop is FF1 and Capturing Flip-Flop is FF2.
- So Data path is FF1/C --> FF1/Q --> FF2/D
- For a single cycle circuit- Signal has to be propagate through Data path in one clock cycle. Means if data is launched at time=0ns from FF1 then it should be captured at time=10ns by FF2.

So for **Setup analysis at FF2**, Data should be stable "Ts" time before the positive edge at FF2/C. Where "Ts" is the Setup time of FF2.

- If Ts=0ns, then , data launched from FF1 at time=0ns should arrive at D of FF2 before or at time=10ns. If data takes too long (greater then 10ns) to arrive (means it is not stable before clock edge at FF2) , it is reported as Setup Violation.
- If Ts=1ns, then, data launched from FF1 at time=0ns should arrive at D of FF2 before or at time=(10ns-1ns)=9ns. If data takes too long (greater then 9ns) to arrive (means it is not stable before 1ns of clock edge at FF2), it is reported as Setup Violation.

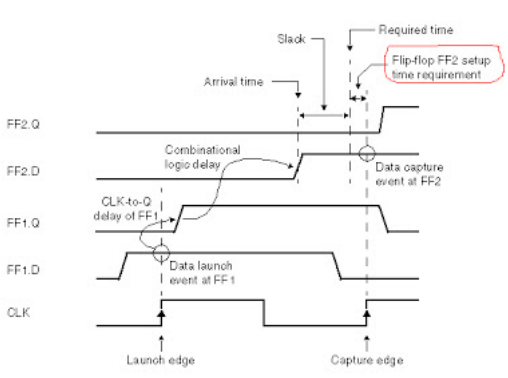
For **Hold Analysis at FF2**, Data should be stable "Th" time after the positive edge at FF2/C. Where "Th" is the Hold time of FF2. Means there should not be any change in the Input data at FF2/D between positive edge of clock at FF2 at Time=10ns and Time=10ns+Th.

- To satisfy the Hold Condition at FF2 for the Data launched by FF1 at 0ns, the data launched by FF1 at 10ns should not reach at FF2/D before 10ns+Th time.
- If Th=0.5ns, then we can say that the data launched from FF1 at time 10ns does not get propagated so soon that it reaches at FF2 before time (10+0.5)=10.5ns (Or say it should reach from FF1 to FF2 with in 0.5ns). If data arrive so soon (means with in 0.5ns from FF1 to FF2, data can't be stable at FF2 for time=0.5ns after the clock edge at FF2), its reported Hold violation.

With the above explanation I can say 2 important points:

1. *Setup is checked at next clock edge.*
2. *Hold is checked at same clock edge.*

Setup Check timing can be more clear for the above Flip-flop combination with the help of following explanation.



Setup Check Timing

In the above fig you can see that the data launched by FF1/D (at launch edge) reaches at FF2/D after a specific delay (CLK-to-Q delay + Conmination Logic Delay) well before the setup time requirement of Flip-Flop FF2, so there is no setup violation.
From the Fig its clear that if Slack= Required Time - Arrival time < 0 (-ive) , then there is a Setup violation at FF2.

Hold Check timing can be more clear with the help of following circuit and explanation.

"Timing Paths" : Static Timing Analysis (STA) basic (Part 1)

Delay - "Wire Load Model" : Static Timing Analysis (STA) basic (Part 4c)

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

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"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

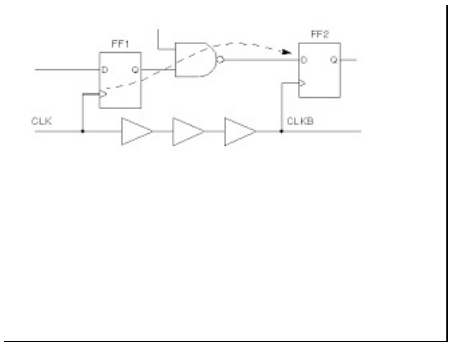
"Examples Of Setup and Hold time" : Static Timing Analysis (STA) basic (Part 3c)

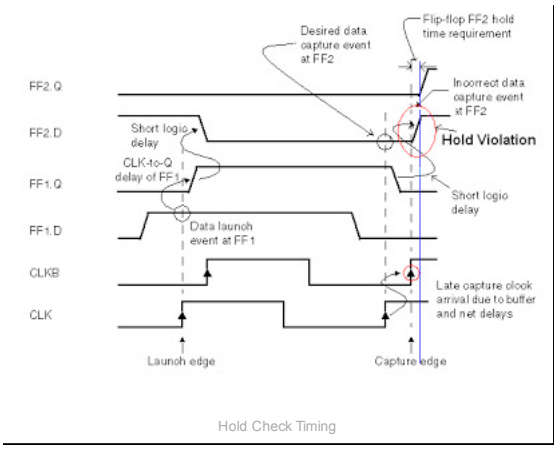
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Hold Check Timing

In the above fig you can see that there is a delay in the CLK and CLKB because of the delay introduced by the series of buffer in the clock path. Now Flip-flop FF2 has a hold requirement and as per that data should be constant after the capture edge of CLKB at Flip-flop FF2.
You can see that desired data which suppose to capture by CLKB at FF2.D should be at Zero (0) logic state and be constant long enough after the CLKB capture edge to meet hold requirement but because of very short logic delay between FF1/Q and FF1/D, the change in the FF1/Q propagates very soon. As a result of that there occurs a Hold violation.
This type of violation (Hold Violation) can be fixed by shortening the delay in the clock line or by increasing the delay in the data path.

Setup and Hold violation calculation for the single clock cycle path is very easy to understand. But the complexity increases in case of multi-cycle path ,Gated clock, Flip-flop using different clocks, Latches in place of Flip-Flop. We will discuss all these later sometime.

Basic Concept of Setup and Hold (Previous) **Index** **Examples Of Setup/Hold Violation (Next)**

Posted by VLSI EXPERT at 12:45 AM

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Reactions: Excellent (9) Good (0) Interesting (1) Need More (3)

33 comments:

Anonymous February 2, 2012 at 2:57 AM

should not the launch edge be the second rising edge of CLK, since hold checks are done for the same clock cycle?

Reply

Replies

your VLSI February 7, 2012 at 10:18 AM

Hi,

I didn't get you question very clearly. Please elaborate it.

Anonymous May 19, 2012 at 9:20 AM

dear clock checks ideally should be done at the rising or capturing edge of clock b only.. but sir has written and shown in the diagram that the data is arriving at the ff2 d before the clock b and it is in the transition region..which is a hold violation

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Tuesday, May 3, 2011

"Examples Of Setup and Hold time" : Static Timing Analysis (STA) basic (Part 3c)

Part1	Part2	Part3a	Part3b	Part3c	Part4a	Part4b	Part4c	Part5a
Part5b	Part6a	Part6b	Part6c	Part7a	Part7b	Part7c	Part 8	

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- Part4c -> Delay - Wire Load Model
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- Part 8 -> 10 ways to fix Setup and Hold Violation.

Till now we have discussed a lot of theory about setup and hold time (with and without Example). Now it's time to discuss the practical implementation of that. Means in a circuit

- How will you calculate the setup and hold values?
- How will you analyze setup and hold violation in a circuit?
- If you have to improve timing of a circuit then what can you do?

There are few formulas to calculate different parameter (Theory of those I already explained in my previous blogs). I am not going to explain those right now. First we will solve few examples which will give you an basic idea about these formulas, then in the last I will summarize all those in one place.

I saw a lot of confusion with respect to setup and hold timing calculation. Actually there are two things.

- Timing Specification of a Block/Circuit/Library:
  - You have a block with input A and output Y. Some combinational logic is there between A and Y. Now you have to calculate following parameters for that block
    - Setup Time Value at input A
    - Hold Time value at input A.

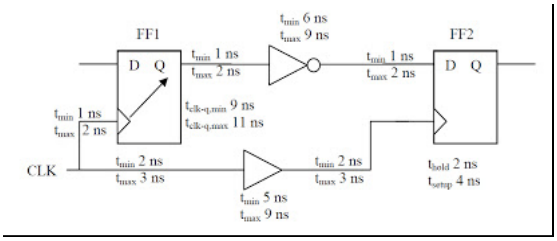
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Basic of Timing Analysis in Physical Design
EDN:

- Maximum operating Clock Frequency or Time Period for that block.
- Clock To Y delay value
- Input A to Output Y delay value.

- Timing Violation of a circuit:
  - You have to operate a circuit at a particular clock frequency and now you have to find out whether this circuit has any setup or Hold Violation.

So in second case all the parameters are given and you have to find out whether this circuit has any violation or not and In first case you have to find out all the parameters keeping in mind that there should not be any violation.
Lets Discuss in the reverse order.
****
****

Problem1: In the following Circuit, Find out whether there is any Setup Or Hold Violation?



Solution:
Hold Analysis:
When a hold check is performed, we have to consider two things-

- Minimum Delay along the data path.
- Maximum Delay along the clock path.

If the difference between the data path and the clock path is negative, then a timing violation has occurred. (Note: there are few Exceptions for this- We will discuss that some other time)

Data path is: CLK->FF1/CLK ->FF1/Q ->Inverter ->FF2/D

Delay in Data path
= min(wire delay to the clock input of FF1) + min(Clk-to-Q delay of FF1) +min(cell delay of inverter) + min(2 wire delay- "Qof FF1-to-inverter" and "inverter-to-D of FF2")
=Td = 1+9+6+(1+1)=18ns

Clock path is: CLK-> buffer -> FF2/CLK

Clock path Delay
= max(wire delay from CLK to Buffer input) + max(cell delay of Buffer) + max(wire delay from Buffer output to FF2/CLK pin) + (hold time of FF2)
=Tclk = 3+9+3+2 = 17 ns

Hold Slack = Td - Tclk = 18ns -17ns = 1ns
Since Hold Slack is positive-> No hold Violation.

Note: If the hold time had been 4 ns instead of 2 ns, then there would have been a hold violation.
Td=18ns and Tclk = 3+9+3+4=19ns
So Hold Slack=Td - Tclk = 18ns - 19ns = -1ns (Violation)

Setup Analysis:
When a setup check is performed, we have to consider two things-

- Maximum Delay along the data path.
- Minimum Delay along the clock path.

If the difference between the clock path and the data path is negative, then a timing violation has occurred. (Note: there are few Exceptions for this- We will discuss that some other time)

Data path is: CLK->FF1/CLK ->FF1/Q ->Inverter ->FF2/D

Delay in Data path
= max(wire delay to the clock input of FF1) + max(Clk-to-Q delay of FF1) +max(cell delay of inverter) + max(2 wire delay- "Qof FF1-to-inverter" and "inverter-to-D of FF2")
=Td = 2+11+9+(2+2) = 26ns

*Note: The first part of the clock path delay (during setup calculation) is the clock period, which has been set to 15 ns. Hope You remember in last blog, I have mentioned very clearly that **Setup is checked at the next clock cycle.** That's the reason for clock path delay we have to include clock period also.*

Clock path is: CLK-> buffer -> FF2/CLK

"Timing Paths" : Static Timing Analysis (STA) basic (Part 1)

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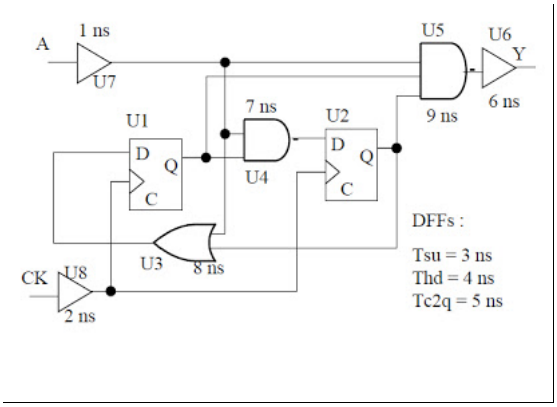
Clock path Delay
= (Clock period) + min(wire delay from CLK to Buffer input) + min(cell delay of Buffer) + min(wire delay from Buffer output to FF2/CLK pin) - (Setup time of FF2)
=Tclk = 15+2+5+2-4=20ns

Setup Slack = Tclk - Td = 20ns - 26ns = -6ns.
Since Setup Slack is negative -> Setup violation.

Note: A bigger clock period or a less maximum delay of the inverter solve this setup violations in the circuit.
E.g
If Clock period is 22ns then
Tclk = 22+2+5+2-4=31-4=27ns AND Td = 26ns
Setup Slack = Tclk - Td = 27-26=1ns (No Violation)

****
****

Problem2: In order to work correctly, what should be the Setup and Hold time at Input A in the following Circuit. Also find out the maximum operating frequency for this circuit. (Note: Ignore Wire delay). Where Tsu- Setup time; Thd-Hold Time; Tc2q- Clock-to-Q delay



Solution:
Step1: Find out the maximum Register to register Delay.

Max Register to Register Delay
= (clk-to-Q delay of U2) + (cell delay of U3) + (all wire delay) + (setup time of U1)
= 5 + 8 + 3 = 16 ns.

Note:

- *There are 2 register to register paths*
  - *U2 -> U3 ->U1 (Delay=5+8+3=16ns)*
  - *U1 -> U4 -> U2 (Delay=5+7+3=15ns)*
- *We have to pick maximum one.*

Step2: Find Out Setup Time:

A setup time = Setup time of Flipflop + Max (Data path Delay) - min(Clock path Delay)
= (Setup time of Flipflop + A2D max delay) - (Clk path min delay)
= Tsu + (Tpd U7 + Tpd U3 + wire delay) - Tpd U8
= 3 + (1+8) - 2 = 10 ns

Note:

- *Here we are not using the Clock period. Because we are not suppose to calculate the Setup violation. We are calculating Setup time. Please refer the part3a for the referance.*
- *All the wire dealy is neglected. If Wire delay present, we have to consider those one.*
- *There are 2 Data path*
  - *A -> U7 -> U4 -> D of U2 (Data path Delay = 1+7 =8ns)*
  - *A -> U7 -> U3 -> D of U1 (Data path Delay = 1+8 =9ns)*
- *Since for Setup calculation we need maximum Data path delay, we have choosen 2nd for our calculation.*

Step3: Find Out Hold Time:
A hold time = Hold time of Flipflop + max(Clock path Delay) - min(Data path delay)
=(Hold time of Flipflop + Clk path max delay) - (A2D max delay)
= Thd + Tpd U8 - (Tpd U7 + Tpd U4+wire delay)
= 4 + 2 - (1+7) = -2 ns

Note: Same explanation as for Setup time. For hold time we need minimum data path , so we have picked first Data path.

Step4: Find out Clock to Out Time:

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Clock to Out
= Cell delay of U8 + Clk-to-Q delay of FlipFlop+ Cell delay of U5+ Cell delay of U6+ (all wire delay)
= Tpd U8+ U2 Tc2q + U5 Tpd + U6 Tpd
= 2 + 5 + 9 + 6 = 22 ns

Note:

- *There are 2 Clock to Out path- one from Flip flop U1 and other from U2.*
- *Since in this case the Clk-to-Q path for both Flipflop is same, we can consider any path. But in some other Circuit where the delay is different for both the paths, we should consider Max delay path.*

Step5: Find Pin to Pine Combinational Delay (A to Y delay)

Pin to Pin Combinational Delay (A to Y)
= U7 Tpd + U5 Tpd + U6 Tpd
= 1 + 9 + 6 = 16 ns

Step5: Find Out Max Clock Frequency:

Max Clock Freq = 1/ Max (Reg2reg, Clk2Out, Pin2Pin)
= 1/ Max (16, 22, 16)
= 45.5 Mhz

So summery is:

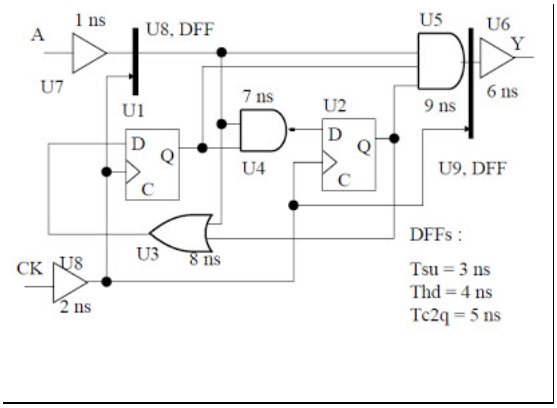
Parameter	Description	Min	Max	Units
Tclk	Clock Period	22		ns
Fclk	Clock Frequency		45.5	Mhz
Atsu	A setup time	10		ns
Athd	A hold time	-2		ns
A2Y	A to Y Tpd		16	ns
Ck2Y	Clock to Y tpd		22	ns

Note: Negative hold times are typically specified as 0 ns.

****
****

Problem3: In the above Circuit, Try to improve the timing by adding any "buffer" or "Register".

Solution:
Best way of doing this is “Register all Input and Output”. We are adding DFF so same specification (as U2 and U1).



Now follow all those 5 Steps onn by one.
Step1:

Max Register to Register Delay
U2 Tc2q + U5 Tpd + U9 Tsu = 5 + 9 + 3 = 17 ns

Note:

- *A lot of Register to Register path*
  - *U8 -> U5 -> U9 (Delay = 5+9+3=17ns)*
  - *U8 -> U4 -> U2 (Delay = 5+7+3=15ns)*
  - *U8 -> U3 -> U1 (Delay = 5+8+3=16ns)*
  - *U1 -> U4 -> U2 (Delay= 5+7+3=15ns)*
  - *U1 -> U5 -> U9 (Delay= 5+9+3=17ns)*
  - *U2 -> U5 -> U9 (Delay = 5+9+3=17ns)*
  - *U2 -> U3 -> U1 (Delay = 5+8+3=16ns)*
- *Maximum delay is 17ns, Just picked anyone.*

Step2:

A setup time = Tsu + A2D Tpd max - Clk Tpd min
= Tsu + (Tpd U7) - Tpd U8
= 3 + (1) - 2 = 2 ns

Note: Only One path between A and D of FF(i.e U8)

Step3:
A hold time = Thd + Clk Tpd max - A2D Tpd min
= Thd + Tpd U8 - (Tpd U7)
= 4 + 2 - (1) = 5 ns

Note: Only One path between A and D of FF(i.e U8)

Step4:
Clock to out:
=Tpd U8+ U9 Tc2q + U6 Tpd
=2+5+6 = 13 ns

Step5:
No direct link between A and Y. So Not Applicable.

Step6:
Max Clock Freq = 1/ Max (Reg2reg, Clk2Out, Pin2Pin)
= 1/ Max (17, 13)
=58.8 Mhz

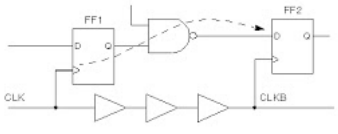
Parameter	Description	Min	Max	Units
Tclk	Clock Period	17		ns
Fclk	Clock Frequency		58.8	Mhz
Atsu	A setup time	2		ns
Athd	A hold time	5		ns
Ck2Y	Clock to Y tpd		13	ns

****
****

I hope This much will help you. Now its the time to summarize all the important things and formulas.

Points to remember:

1. *Setup is checked at next clock edge.*
2. *Hold is checked at same clock edge.*
3. *For Hold Check (Checking of hold Violation)*
   - ***Minimum Delay** along the **data path**.*
   - ***Maximum Delay** along the **clock path**.*
4. *For SetUp Check (Checking of Setup Violation)*
   1. ***Maximum Delay** along the **data path**.*
   2. ***Minimum Delay** along the **clock path.***



Basic 2 FlipFlop circuit.

Calculation of Setup Violation Check: Consider above circuit of 2 FF connected to each other.

Setup Slack = Required time - Arrival time (since we want data to arrive before it is required)

Where:

Arrival time (max) = clock delay FF1 (max) +clock-to-Q delay FF1 (max) + comb. Delay(max)
Required time = clock adjust + clock delay FF2 (min) - set up time FF2
Clock adjust = clock period (since setup is analyzed at next edge)

Calculation of Hold Violation Check: Consider above circuit of 2 FF connected to each other.

Hold Slack = Arrival Time - Required time (since we want data to arrive after it is required)

Where:

Arrival time (min) = clock delay FF1 (min) +clock-to-Q delay FF1 (min) + comb. Delay(min)
Required time = clock adjust + clock delay FF2 (max) + hold time FF2
Clock adjust = 0 (since hold is analyzed at same edge)

Calculation of Maximum Clock Frequency:

Max Clock Freq = 1/ Max (Reg2reg delay, Clk2Out delay, Pin2Pin delay)

Where:

Reg2Reg Delay = Clk-to-Q delay of first FF (max) + conb delay (max) + setup time of 2nd FF.
Clk2Out Delay = Clock delay w.r.t FF (max) + clock-to-Q delay of FF1 (max) + comb. delay (max)
Pin2Pin delay = Comb delay between input pin to output pin (max)

Basic Concept of Setup and Hold Violation (Previous) | Index | Timing Path Delay (Next)

Posted by VLSI EXPERT at 12:38 AM

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Reactions: Excellent (17) Good (1) Interesting (1) Need More (0)

116 comments:

Anonymous May 5, 2011 at 10:54 AM

Thanks... I really appreciate your work.. Nice to see experts like you sharing knowledge in a detailed manner

Reply

Anonymous May 12, 2011 at 1:22 PM

Hi, thanks for such useful blog.
But one comment I wanna say that can you remove all the previous comment (maybe include this), which pointed out the mistake you have been corrected. Its help the audience save much time.
Thanks again for great job!

Reply

Anonymous May 17, 2011 at 6:03 PM

Hi,

Thanks a lot for this blog. Great Job!

Reply

Anonymous May 24, 2011 at 11:53 PM

Hi,

In the 1st problem, for checking hold violation, you calculated hold slack as Td-Tclk, but you mentioned earlier that for calculating hold time we should do Tclk(max)-Td(min), but here I find its opposite, same thing I found in regard to setup time analysis. Can you please clear me that doubt.
By the way, very very nice job. This blog helped me a lot for my interviews. :)
Thanking You

Reply

vlsi.expert May 25, 2011 at 12:34 PM

Hi,

See there are 2 different things.. (i have mentioned very clearly in the starting of this)

1) calculate the Timing Specification of a Block/Circuit/Library:

2) Calculate the Timing voilation in a ckt.

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Thursday, August 4, 2011

"Delay - Timing path Delay" : Static Timing Analysis (STA) basic (Part 4a)

Part1	Part2	Part3a	Part3b	Part3c	Part4a	Part4b	Part4c	Part5a
Part5b	Part6a	Part6b	Part6c	Part7a	Part7b	Part7c	Part 8	

Static Timing analysis is divided into several parts:

- Part1 -> Timing Paths
- Part2 -> Time Borrowing
- Part3a -> Basic Concept Of Setup and Hold
- Part3b -> Basic Concept of Setup and Hold Violation
- Part3c -> Practical Examples for Setup and Hold Time / Violation
- Part4a -> Delay - Timing Path Delay
- Part4b -> Delay - Interconnect Delay Models
- Part4c -> Delay - Wire Load Model
- Part5a -> Maximum Clock Frequency
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- Part 6b -> Continue of How to solve Setup and Hold Violation (Advance examples)
- Part 6c -> Continue of How to solve Setup and Hold Violation (more advance examples)
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- Part 7b -> Methods for Increase/Decrease the Delay of Circuit (Effect of Size of the Transistor On the Slew)
- Part 7c -> Methods for Increase/Decrease the Delay of Circuit (Effect of Threshold voltage On the Slew)
- Part 8 -> 10 ways to fix Setup and Hold Violation.

This particular post is inspired by a question asked by Lalit. And Frankly speaking I am not able to resist myself to write a blog on this. I was thinking to capture all this since long but every time because of work I have to drop my thoughts.. But today after reading his question.. I am not able to control myself. :)

So the Question is: (original question)

I have a doubt regarding how delay is calculated along a path.i think there are two ways
1) to calculate max delay and min delay, we keep adding max delays and min delays of all cells(buffer/inverter/mux) from start point to end point respectively.
2)in other way, we calculate path delay for rising edge and falling edge separately. we apply a rise edge at start point and keep adding cell delay. cell delay depends upon input transition and output fanout. so now we have two path delay values for rise edge and falling edge. greater one is considered as Max delay and smaller one is min delay.
which one is correct ?

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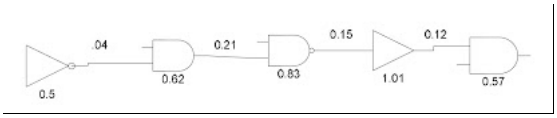
Short Ans is .. both are correct and you have to use both. May be you all become confuse, so let me give you few details.

As I have mention that for Setup and Hold calculation , you have to calculate the Delay of the Timing path (capture path or launch path). Now in a circuit there are 2 major type of Delay.

1. CELL DELAY
    - Timing Delay between an input pin and an output pin of a cell.
    - Cell delay information is contained in the library of the cell. e.g- .lef file
2. NET DELAY.
    - Interconnect delay between a driver pin and a load pin.
    - To calculate the NET delay generally you require 3 most important information.
        - Characteristics of the Driver cell (which is driving the particular net)
        - Load characteristic of the receiver cell. (which is driven by the net)
        - RC (resistance capacitance) value of the net. (It depends on several factor- which we will discuss later)

Both the delay can be calculated by multiple ways. It depends at what stage you require this information with in the design. e.g During pre layout or Post layout or during Signoff timing. As per the stage you are using this, you can use different ways to calculate these Delay. Sometime you require accurate numbers and sometime approximate numbers are also sufficient.

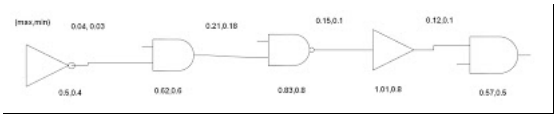
Now lets discuss this with previous background and then we will discuss few new concepts.



Now in the above fig- If I will ask you to calculate the delay of the circuit, then the delay will be

Delay=0.5+0.04+0.62+0.21+0.83+0.15+1.01+0.12+0.57=4.05ns (if all the delay in ns)

Now lets add few more value in this. As we know that every gate and net has max and min value, so in that case we can find out the max delay and min delay. (on what basis these max delay and min delay we are calculating .. we will discuss after that)



So in the above example, first value is max value and 2nd value is min value. So

Delay(max)= 0.5+0.04+0.62+0.21+0.83+0.15+1.01+0.12+0.57=4.05ns
Delay(min)= 0.4+0.03+0.6+0.18+0.8+0.1+0.8+0.1+0.5=3.51ns

Till now every one know the concept. Now lets see what's the meaning of min and max delay.

The delay of a cell or net depends on various parameters. Few of them are listed below.

- Library setup time
- Library delay model
- External delay
- Cell load characteristic
- Cell drive characteristic
- Operating condition (PVT)
- Wire load model
- Effective Cell output load
- Input skew
- Back annotated Delay

If any of these parameter vary , the delay vary accordingly. Few of them are mutually exclusive. and In that case we have to consider the effect of only one parameter at a time. If that's the case , then for STA, we calculated the delay in both the condition and then categorize them in worst (max delay) condition or the best condition (min delay). E.g- if a cell has different delay for rise edge and fall edge. Then we are sure that in delay calculation we have to use only one value. So as per their value , we can categorize fall and rise delay of all the cell in the max and min bucket. And finally we come up with max Delay and min delay.

"Timing Paths" : Static Timing Analysis (STA) basic (Part 1)

Delay - "Wire Load Model" : Static Timing Analysis (STA) basic (Part 4c)

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

"Setup and Hold Time" : Static Timing Analysis (STA) basic (Part 3a)

"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

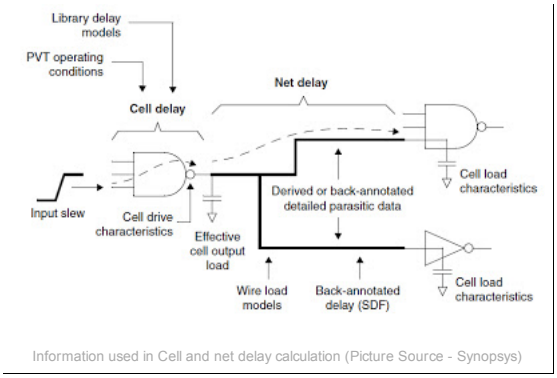
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"Time Borrowing" : Static Timing Analysis (STA) basic (Part 2)

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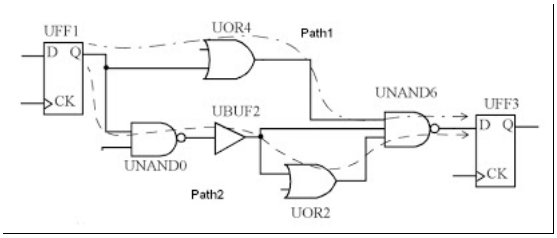
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Information used in Cell and net delay calculation (Picture Source - Synopsys)

The way delay is calculated also depends which tool are you using for STA or delay calculation. Cadence may have different algorithm from Synopsys and same is the case of other vendor tools like mentor,magma and all. But in general the basic or say concepts always remain same.
I will explain about all these parameter in detail in next of few blogs, but right now just one example which can help you to understand the situation when you have a lot of information about the circuit and you want to calculate the delay.



In the above diagram, you have 2 paths between UFF1 and UFF3. So when ever you are doing setup and hold analysis, these path will be the part of launch path (arrival time). So lets assume you want to calculate the max and min value of delay between UFF1 and UFF2.

Information1:

	UOR4	UNAND6	UNAND0	UBUF2	UOR2
DELAY(ns)	5	6	6	2	5

Calculation:
Delay in Path1 : 5+6=11ns,
Delay in Path2: 6+2+5+6=19ns,
So
Max Delay = 19ns - Path2 - Longest Path - Worst Path
Min Delay = 11ns - Path1 - Smallest Path - Best Path

Information2:

	UOR4	UNAND6	UNAND0	UBUF2	UOR2
Rise Delay (ns)	5	6	4	1	1
Fall Delay (ns)	6	7	3	1	1

Calculation:
Delay in Path1 : Rise Delay : 5+6=11ns, Fall Delay: 6+7=13ns
Delay in Path2: Rise Delay : 4+1+1+6=12ns, Fall Delay: 3+1+1+7=12ns
So
Max Delay = 13ns -Path1 (Fall Delay)
Min Delay = 11ns - Path1 (Rise Delay)

Note: here there are lot of more concepts which can impact the delay calculation sequence, like unate. We are not considering all those right now. I will explain later.

Information3:

Library	Delay	UOR4	UNAND6	UNAND0	UBUF2	UOR2
Min	Rise Delay (ns)	5	6	4	1	1
	Fall Delay (ns)	6	7	3	1	1
Max	Rise Delay (ns)	5.5	6.5	4.5	1.5	1.5
	Fall Delay (ns)	5.5	6.5	2.5	0.5	0.5

Calculation:
For Min Library:
Delay in Path1 : Rise Delay : 5+6=11ns, Fall Delay: 6+7=13ns
Delay in Path2: Rise Delay : 4+1+1+6=12ns, Fall Delay: 3+1+1+7=12ns
For Max Library:
Delay in Path1 : Rise Delay : 5.5+6.5=12ns, Fall Delay: 5.5+6.5=14ns
Delay in Path2: Rise Delay : 4.5+1.5+1.5+6.5=14ns, Fall Delay: 2.5+0.5+0.5+6.5=10ns
So

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Max Delay = 14ns- Path1(Fall Delay)/Path2(Rise Delay)
Min Delay = 10ns - Path2(Fall Delay)

As we have calculated above, STA tool also uses similar approach for finding the Max delay and Min Delay. Once Max and Min delay is calculated then during setup and hold calculation, we use corresponding value.

Once again I am mentioning that all these values are picked randomly. So it may be possible that practically the type/amount of variation in value is not possible.

In next part we will discuss these parameter in detail one by one.

Examples Of Setup/Hold Violation (Previous) | Index | Interconnect Delay Models (Next)

Posted by VLSI EXPERT at 9:17 AM

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Reactions: Excellent (2) Good (0) Interesting (0) Need More (0)

22 comments:

Anonymous August 21, 2011 at 2:22 AM

Nice article. Many thanks to the author

Reply



Karadi October 5, 2011 at 7:36 AM

Isn't propagating rise and fall delays without considering the transitions an over or under estimate as some transitions may not even be possible. Is there some way to workaround this without having to use the input vectors.

Reply

Your VLSI October 7, 2011 at 9:15 AM

Hi Karadi,

You are right but usually we do the worst case analysis. Or you can sat most of the STA tool do the calculation based like this. But if you need to check the actual calculation, then there is a methodology - know as PATH BASED ANALYSIS. In that we do delay calcualtion with respect to a particular path (timing path- please cehck the detials of timing path in my previous blogs.) And in that you can avoide all those transitions which is not possible.

Reply

Anonymous December 8, 2011 at 6:03 PM

one correction pls....When you explained Cell Delay you said it is mentioned in .lef file but it is in .lib file right?

Reply



Maninder January 31, 2012 at 6:35 AM

Hi Your VLSI.....

This is the best explanation ever for Setup and hold timing and its violations.....I am waiting for your next blog....

Reply

Replies



your VLSI February 7, 2012 at 9:56 AM

thanks Maninder.
I was busy in last few months.. but soon you will see few more updates from my side.

Reply

Anonymous February 3, 2012 at 11:33 AM

Hi,
I have just started to understand about the STA concepts.
I got great idea about the setup and hold time from your blogs. Thank you for sharing this knowledge.
I want to know how can I apply these concepts in the real design. i.e If I want to add some constraints in the design then how can I add ?

Reply

Replies



your VLSI February 7, 2012 at 10:01 AM

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VLSI Concepts

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Sunday, September 11, 2011

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

Part1	Part2	Part3a	Part3b	Part3c	Part4a	Part4b	Part4c	Part5a
Part5b	Part6a	Part6b	Part6c	Part7a	Part7b	Part7c	Part 8	

Static Timing analysis is divided into several parts:

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- Part 8 -> 10 ways to fix Setup and Hold Violation.

In the previous post we have discussed about the way tool calculate the max and min delay in a circuit. Now we will discuss other basics of the Delay and delay calculation. During your day to day work (in Semiconductor Field) or say in different Books, you come across different terminology related to the delays. There is a long list of that.

- Input Delay
- Output Delay
- Cell Delay
- Net Delay
- Wire Delay
- Slope Delay
- Intrinsic Delay
- Transition Delay
- Connect Delay
- Interconnect Delay

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Basic of Timing Analysis in Physical Design
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- Propagation Delay
- Min/Max Delay
- Rising/Falling Delay
- Gate Delay
- Stage delay

Fortunately or say luckily out of the above mention long list few are just synonym of other and few are interrelated to each other . Like Net delay also know as Wire Delay , Interconnect delay. Broadly we can divide this Long List into 2 type of delay. Net Delay (Wire delay) and Cell Delay. (**Note :** Stage Delay = Net delay + Cell Delay.)
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Now to calculate the Net delay, the wires are modeled in different ways and there are different way to do the calculation. Practically, when you are applying a particular delay model in a design , then you have to apply that to all cells in a particular library. You cannot mix delay models within a single library. There are few recommendations provided by experts or say experienced designer regarding the application of a particular Delay model in a design and that depends on

- Technology of design.
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- How accurately you want to calculate the delay.

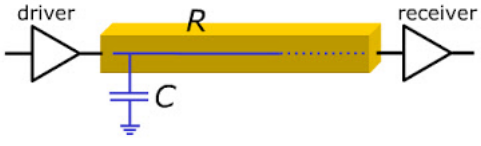
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There are several delay models. Those which can provide more accurate result, takes more runtime to do the calculation and those which are fast provides less accurate value of delay. Lets discuss few of them. Most popular delay models are -

- Lumped Capacitor Model
- Lumped RC model
- Distributed RC model
  - Pi RC network
  - T RC network
- RLC model
- Wire Load model
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Lumped Capacitor Model.

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"Timing Paths" : Static Timing Analysis (STA) basic (Part 1)

Delay - "Wire Load Model" : Static Timing Analysis (STA) basic (Part 4c)

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

"Setup and Hold Time" : Static Timing Analysis (STA) basic (Part 3a)

"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

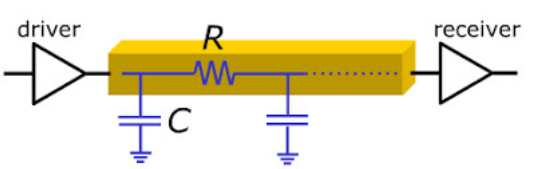
"Examples Of Setup and Hold time" : Static Timing Analysis (STA) basic (Part 3c)

"Time Borrowing" : Static Timing Analysis (STA) basic (Part 2)

Effect of Wire Length On the Slew: Static Timing Analysis (STA) Basic (Part-7a)

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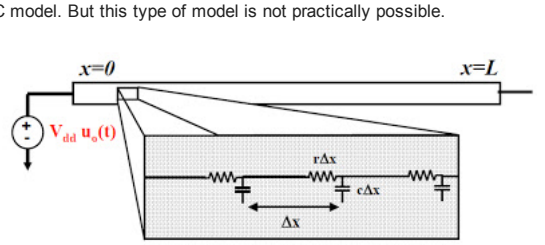
Lumped RC Model

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Where
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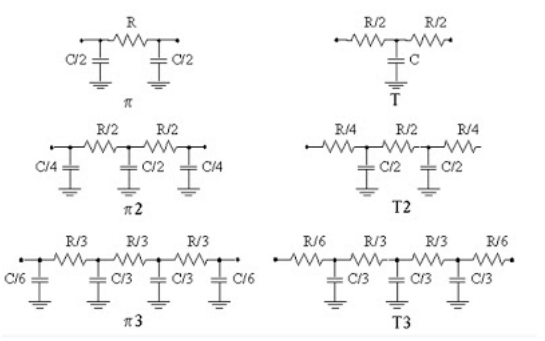
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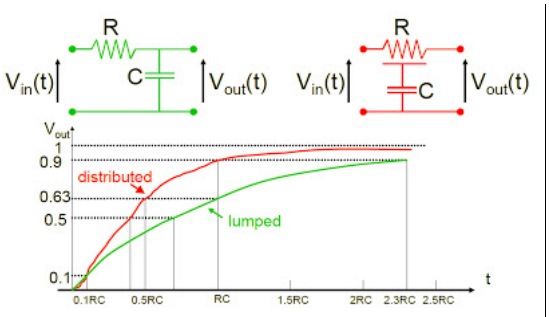
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2 Types of Distributed RC Model (Pi- Model and T- Model)

Note: Lumped Vs Distributed RC wire:

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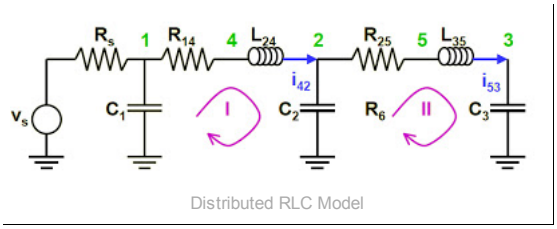
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Distributed RLC Model

In next part we will discuss Wire Load Delay Model...

Timing Path Delay (Previous) | Index | Wire Load Model (Next)

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Shubham September 16, 2011 at 2:24 PM

nice explanation.
I have a question though, when we calculate the total delay of the path we addup cell delay + interconnect delay and cell delay is calculated by taking into account the interconnect loading effect.So, isn't this wrong as we are taking the interconnect effect twice in our delay calculation

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Sunday, September 11, 2011

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

Part1	Part2	Part3a	Part3b	Part3c	Part4a	Part4b	Part4c	Part5a
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In the previous post we have discussed about the way tool calculate the max and min delay in a circuit. Now we will discuss other basics of the Delay and delay calculation. During your day to day work (in Semiconductor Field) or say in different Books, you come across different terminology related to the delays. There is a long list of that.

- Input Delay
- Output Delay
- Cell Delay
- Net Delay
- Wire Delay
- Slope Delay
- Intrinsic Delay
- Transition Delay
- Connect Delay
- Interconnect Delay
- Propagation Delay
- Min/Max Delay
- Rising/Falling Delay

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Basic of Timing Analysis in Physical Design

"Timing Paths" : Static Timing Analysis (STA) basic (Part 1)

Delay - "Wire Load

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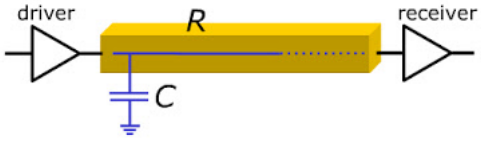
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Model" : Static Timing Analysis (STA) basic (Part 4c)

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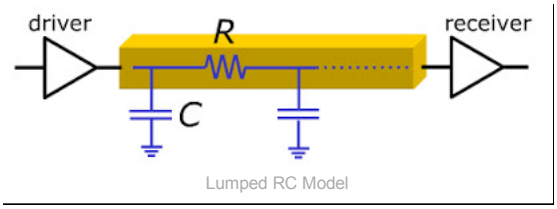
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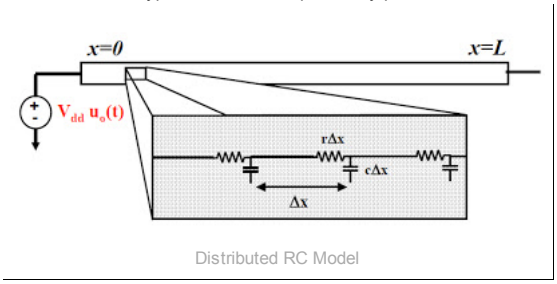
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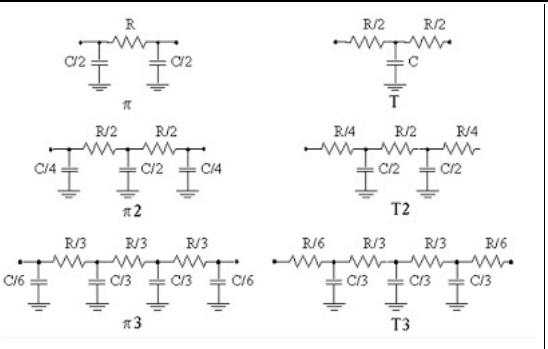
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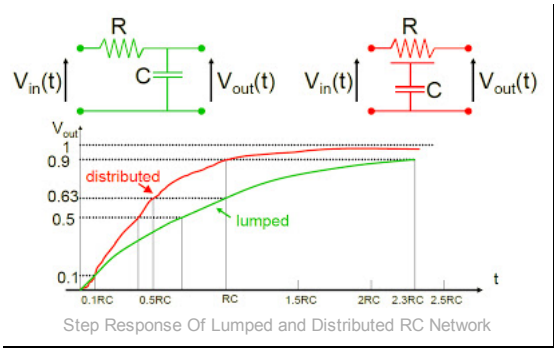
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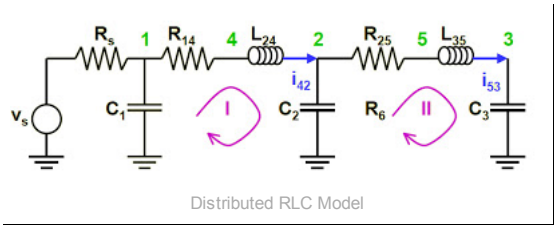
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Distributed RLC Model

In next part we will discuss Wire Load Delay Model...

Timing Path Delay (Previous) | Index | Wire Load Model (Next)

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Wednesday, March 7, 2012

Delay - "Wire Load Model" : Static Timing Analysis (STA) basic (Part 4c)

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Now the question is: What is Wire Load Models (WLM).

Wire loading models

- Used to estimate the interconnect wire delay during pre-layout in a design cycle.
- Wire load information is based on statistics from physical layout parasitic
  - Information from the statistics is used in both conservative and aggressive tables.
  - The conservative tables are based on "mean value" plus 3-sigma; the aggressive tables on "mean value" plus 1-sigma.
- Different for different technology.
  - Wire load models are approximated from one technology to another based on scaling factors. Due to these approximations, the accuracy of these models diminish over multiple technology nodes
- Describes effect of wire length and fanout on

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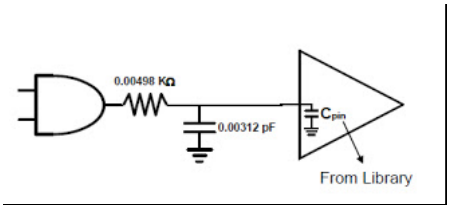
- Resistance
- Capacitance
- Area of the nets.

- All attributes (R, C and Area) are given per unit length wire.
- Slope value is used to characterize linear fanout.
- Basically a set of tables
  - Net fanout vs load
  - Net fanout vs resistance
  - Net fanout vs area

One example of such type of table is:

Net Fanout	Resistance KΩ	Capacitance pF
1	**0.00498**	**0.00312**
2	0.01295	0.00812
3	0.02092	0.01312
4	0.02888	0.01811

As per this



Model" : Static Timing Analysis (STA) basic (Part 4c)

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

"Setup and Hold Time" : Static Timing Analysis (STA) basic (Part 3a)

"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

"Examples Of Setup and Hold time" : Static Timing Analysis (STA) basic (Part 3c)

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```
wire_load("WLM1")     {
 resistance  :       0.0006      ;------>R per unit length
 capacitance :       0.0001      ;------> C per unit length
 area :              0.1         ;------> Area per unit length
 slope       :       1.5         ;------> Used for linear extrapolation
 fanout_length(1,  0.002)        ; ------> at fanout "1" length of the wire is 0.002

fanout_length(2,  0.006);
fanout_length(3,  0.009);
fanout_length(4,  0.015);
fanout_length(5,  0.020);

fanout_length(7,  0.028);          ------> at fanout "7" length of the wire is 0.028
fanout_length(8,  0.030);
fanout_length(9,  0.035);
fanout_length(10, 0.040);
}

wire_load("WLM2") {
      fanout_length(  1, 1 );
      fanout_length(  2, 2 );

       fanout_capacitance( 1, 0.002 );
       fanout_capacitance( 2, 0.004 );
       fanout_capacitance( 3, 0.006 );
       fanout_capacitance( 4, 0.008 );
       fanout_capacitance( 5, 0.010 );
       fanout_capacitance( 6, 0.013 );
       fanout_capacitance( 7, 0.015 );
       fanout_capacitance( 8, 0.019 );
       fanout_capacitance( 9, 0.023 );
       fanout_capacitance( 10, 0.027);


       fanout_resistance( 1, 0.01 );
```

```
        fanout_resistance( 2, 0.015 );
        fanout_resistance( 3, 0.022 );
        fanout_resistance( 4, 0.026 );
        fanout_resistance( 5, 0.030 );
        fanout_resistance( 6, 0.035 );
        fanout_resistance( 7, 0.039 );
        fanout_resistance( 8, 0.048 );
        fanout_resistance( 9, 0.057 );
        fanout_resistance( 10, 0.06 );

      fanout_area(  1, 0.11 );
      fanout_area( 20, 2.20 );
}
```

Here --
Area, Resistance and Capacitance are in per unit length of the interconnect.
The slope is the extrapolation slop to be used for data points that are not specified in the fan-out length table.

In general, not all fanouts are mentioned in a given WLM lookup table. For example, in above WLM1 and WLM2 lookup table, capacitance and resistance values for fanouts 1, 2, 3, 4, 5, 7, 8, 9, 10 is given. If we want to estimate the values at fanouts in the gaps (e.g. from 6) or outside the fanout range specified in the table (e.g Fanout 20), we have to calculated those value using (linear) interpolation and extrapolation.

For WLM1
For Fanout=20

Since its more than the max value of Fanout available in table (i.e 10) , so we have to perform extrapolation.

Net length = <length of net at fanout 10> + (20-10) x Slope
Resistance = <new calculated Net length at fanout 6> x Resistance or Capacitance value per unit length
Capacitance = <new calculated Net length at fanout 6> x Capacitance value per unit length

Net length = 0.040 + 10 x 1.5 (slope) = 15.04 ----------> length of net with fanout of 20
Resistance = 15.04 x 0.0006 = 0.009024 units
Capacitance = 15.04 x 0.0001 = 0.001504 units

For Fanout=6

Since it's between 5 and 7 and corresponding fanout Vs length is available, we can do the interpolation.

Net length = ((net length at fanout 5) + (net length at fanout 7)) / 2
Resistance = <new calculated Net length at fanout 20> x Resistance value per unit length
Capacitance = <new calculated Net length at fanout 20> x Capacitance value per unit length

Net length = (0.0020 + 0.0028)/2=0.0048/2=0.0024 ----------> length of net with fanout of 6
Resistance = 0.0024 x 0.0006 = 0.00000144 units

Capacitance = 0.0024 x 0.0001 = 0.00000024 units
In the similar way we can calculate the WLM for any no of fanout value.

WLMs are often used in pre-placement optimization to drive speedups of critical paths. Since timing-driven placement plausibly makes nets on critical paths shorter than average, some optimism may be incorporated into the WLM. Thus, a WLM may actually consist of more than one lookup table, with each table corresponding to a different optimism level. There are several ways to incorporate the optimism level. If we use the WLMs that come from the (ASIC vendor's) design library, usually there are several tables from which we can select. We can also increase the optimism level of a WLM by multiplying all values in the WLM by some factor less than 1.2 For example, we can use 0.25, 0.5, or 0.75.

WLM Types
For flows that run timing-based logic optimization before placement, there are three basic types of

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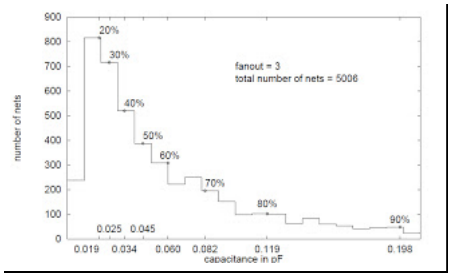
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WLMs that can be used:

1. Statistical WLMs
   - Are based on averages over many similar designs using the same or similar physical libraries.
2. Structural WLMs
   - Use information about neighboring nets, rather than just fanout and module size information.
3. Custom WLMs
   - Are based on the current design after placement and routing, but before the current iteration of preplacement synthesis.

Now the Question is: Where do the wire load models come from?

Normally the semiconductor vendors will develop the models.
ASIC vendors typically develop wireload models based on statistical information taken from a variety of example designs. For all the nets with a particular fanout, the number of nets with a given capacitance is plotted as a histogram. A single capacitance value is picked to represent this fanout value in the wireload model. If a very conservative wireload model is desired, the 90% decile might be picked (i.e. 90% of the nets in the sample have a capacitance smaller than that value).



In this example 90% of nets have a capacitance smaller then 0.198pf. So in the WLM table, you will notice that fanout_capacitance(3, 0.198).
Similar statistics are gathered for resistance and net area.
Usually the vendor supplies a family of wireload models, each to be used for a different size design. This is called *area-based wireload selection*

Few Advance concepts:

Till now we have discussed that for a particular Net you can estimate the RC value as per the WLM. Let me ask you one question. What if your design is hierarchical? Do you think even in that case you can use the same WLM for a particular net which is crossing the hierarchical boundaries? Short ANS is: you can use it but you will lose the accuracy.
Just to solve this problem, Vendors usually supplies multiple WLMs. There are different Modes for WLM analysis- few important are:

WLM analysis has three modes:

1. Top:
   - Consider the design as it has no hierocracy and use the WLM for the top module to calculate delays for all modules.
   - Any low level WLM is ignored.
2. Enclosed:
   - Use the WLM of the module which completely encloses the net to compute delay for that net.
3. Segmented:
   - If a net goes across several WLM, use the WLM that corresponds to that portion of the net which it encloses only.

Interconnect Delay Models (Previous) | Index | Maximum Clock Frequency (Next)

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Monday, September 24, 2012

Maximum Clock Frequency : Static Timing Analysis (STA) basic (Part 5a)

Part1	Part2	Part3a	Part3b	Part3c	Part4a	Part4b	Part4c	Part5a
Part5b	Part6a	Part6b	Part6c	Part7a	Part7b	Part7c	Part 8	

Static Timing analysis is divided into several parts:

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- Part4b -> Delay - Interconnect Delay Models
- Part4c -> Delay - Wire Load Model
- Part5a -> Maximum Clock Frequency
- Part5b -> Examples to calculate the "Maximum Clock Frequency" for different circuits.
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- Part 6c -> Continue of How to solve Setup and Hold Violation (more advance examples)
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- Part 8 -> 10 ways to fix Setup and Hold Violation.

This is a general question in most of the interview, what's the maximum clock frequency for a particular circuit? Or Interviewer will provide some data and they will repeat the same question. Many of us know the direct formula and after applying that we can come across the final "Ans" but if someone twist the question. Some -time we become confuse. I motivation of this blog is the same. Several people asked me how to calculate the max-clock frequency. So I thought that it's best if I can write something over this.

Here I will discuss the same but from basic point of view. It has 3 major sections.

1. In 1st section, we will discuss different definitions with respect to Sequential and combinational Circuits.
2. 2nd Section contains the basics of "Maximum Clock Frequency". I will explain why and how you can calculate the max Clock frequency.
3. I will take few examples and try to solve them. I will make sure that I can capture at least 2-4 examples from easy one to difficult one.

As we know that now a days all the chips has combinational + sequential circuit. So before we move forward, we should know the definition of "Propagation delay" in both types of circuits. Please read it once because it will help you to understand the "Maximum Clock Frequency" concepts.

Propagation Delay in the Combinational circuits:

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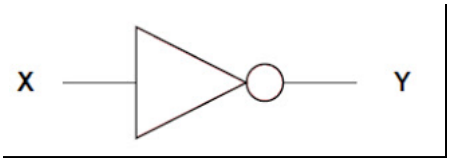
Basic of Timing Analysis in Physical Design

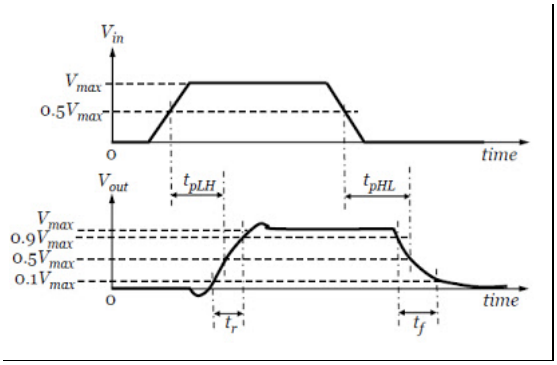
"Timing Paths" : Static Timing Analysis (STA) basic (Part 1)

Delay - "Wire Load

EDN:

Let's consider a "NOT" gate and Input/output waveform as shown in the figure





From the above figure, you can define

- **Rise Time (tr):** The time required for a signal to transition from 10% of its maximum value to 90% of its maximum value.
- **Fall Time (tf):** The time required for a signal to transition from 90% of its maximum value to 10% of its maximum value.
- **Propagation Delay (tpLH, tpHL):** The delay measured from the time the input is at 50% of its full swing value to the time the output reaches its 50% value.

I want to rephrase above mention definition as

- This value indicates the amount of time needed to reflect *a permanent change* at an output, if there is any change in logic of input.
- Combinational logic is guaranteed not to show any further output changes in response to an input change after tpLH or tpHL time units have passed.

So, when an input X change, the output Y is not going to change instantaneous. Inverter output is going to maintain its initial value for some time and then it's going to change from its initial value. After the propagation delay (tpLH or tpHL - depends on what type of change- low to high or high to low), the inverter output is stable and is guaranteed not to change again until another input change (here we are not considering any SI/noise effect).

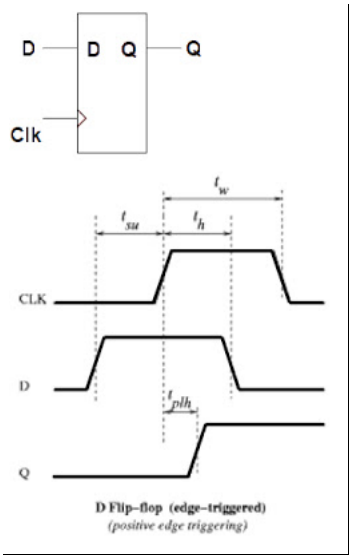
Propagation Delay in the Sequential circuits:

In the sequential circuits, timing characteristics are with respect to the clock input. You can correlate it in this way that in the combinational circuit every timing characteristic/parameter are with respect to the data input change but in the sequential circuits the change In the "data input" is important but change in the clock value has higher precedence. E.g in a positive-edged-triggered Flip-flop, the output value will change only after a presence of positive-edge of clock whether the input data has changed long time ago.

So flip-flops only change value in response to a change in the clock value, timing parameters can be specified in relation to the rising (for positive edge-triggered) or falling (for negative-edge triggered) clock edge.

Note: Setup and hold time we have discussed in detail in the following blogs. Setup and Hold part1; Setup and Hold part2; Setup and Hold part3 . But just to refresh your memories :) , I have captured the definition here along with "propagation delay".

Let's consider the positive-edge flip-flop as shown in figure.



Model" : Static Timing Analysis (STA) basic (Part 4c)

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

"Setup and Hold Time" : Static Timing Analysis (STA) basic (Part 3a)

"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

"Examples Of Setup and Hold time" : Static Timing Analysis (STA) basic (Part 3c)

"Time Borrowing" : Static Timing Analysis (STA) basic (Part 2)

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Propagation delay, tpHL and tpLH , has the same meaning as in combinational circuit – beware propagation delays usually will not be equal for all input to output pairs.

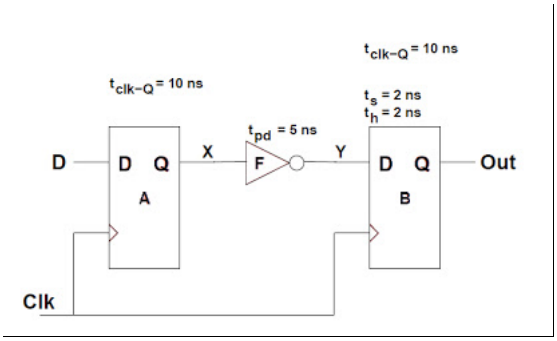
Note: In case of flip-flop there is only one propagation delay i.e tclk-Q (clock→Q delay) but in case of Latches there can be two propagation delays: tClk-Q (clock→Q delay) and tD-Q (data→Q delay). Lation delay we will discuss later.
So again let me rephrase the above mention definition

- This value indicates the amount of time needed for a permanent change at the flip-flop output (Q) with respect to a change in the flip flop-clock input (e.g. rising edge).
- When the clock edge arrives, the D input value is transferred to output Q. After tClk−Q (here which is equivalent to tpLH), the output is guaranteed not to change value again until another clock edge trigger (e.g. rising edge) arrives and corresponding Input also.

Setup time (*tsu*) - This value indicates the amount of time **before** the clock edge that data input *D* **must** be stable.
Hold time (*th*) - This value indicates the amount of time **after** the clock edge that data input *D* **must** be held stable.
The circuit must be designed so that the *D* flip flop input signal arrives at least *"tsu"* time units **before** the clock edge and does not change until at least *"th"* time units **after** the clock edge. If either of these **restrictions are violated for any of the flip-flops in the circuit, the circuit will not operate correctly. These** restrictions limit the maximum clock frequency at which the circuit can operate (that's what I am going to explain in the next section J)

The Maximum Clock Frequency for a circuit:

I hope you may be asking that why there is a need of explaining the combinational circuit propagation delay here. Combinational circuit is always independent of clock, so why combination circuit here. J
Now the point is combinational circuit plays a very important role in deciding the clock frequency of the circuit. Let's first discuss an example and try to calculate the circuit frequency, and then we will discuss rest of the things in details. J
Note: Following diagram and numbers, I have copied from one of the pdf downloaded by me long time back.



Now let's understand the flow of data across these Flip-flops.

- Let's assume data is already present at input D of flip-flop A and it's in the stable form.
- Now Clock pin of FF (Flip-Flop) A i.e Clk has been triggered with a positive clock edge (Low to high) at time "0ns".
- As per the propagation delay of the sequential circuit (tclk-Q), it will take at least 10ns for a valid output data at the pin X.
  - Remember- If you will capture the output before 10ns, then no one can give you the guarantee for the accurate/valid value at the pint X.
- This data is going to transfer through the inverter F. Since the propagation delay of "F" is 5ns, it means, you can notice the valid output at the pin Y only after 10ns+5ns=15ns (with reference to the positive clock edge- 10ns of FF A and 5 ns of inverter)
  - Practically this is the place where a more complex combinational circuit are present between 2 FFs. So in a more complex design, if a single path is present between X and Y, then the total time taken by the data to travel from X to Y is equal to the sum of the propagation delay of all the combinational circuits/devices. (I will explain this in more detail in the next section with more example)
- Now once valid data reaches at the pin Y, then this data supposed to capture by FF B at the next clock positive edge (in a single cycle circuit).
  - We generally try to design all the circuit in such a way that it operates in a single clock cycle. Multiple clock cycle circuit are special case and we are not going to discuss that right now (as someone says – it's out of scope of this blog J)
- For properly capturing the data at FF B, data should be present and stable 2ns (setup time) before the next clock edge as part of setup definition).

So it means between 2 consecutive positive clock edge, there should be minimum time difference of 10ns +5ns +2ns = 17ns. And we can say that for this circuit the minimum clock period should be 17ns (if we want to operate the circuit in single clock cycle and accurately).
Now we can generalize this
Minimum Clock Period = tclk-Q (A) + tpd (F) + ts (B)
And "**Maximum Clock Frequency = 1/(Min Clock Period)"**

Now at least we have some idea how to calculate the Max clock frequency or Min Clock Period. So even if we will forget the formula then we can calculate our self and we can also prove the logic behind that. Let me use the same concept in few of the more complex design circuit or you can say the practical circuit.

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Maximum Clock Frequency : Static Timing Analysis (STA) basic (Part 5b)

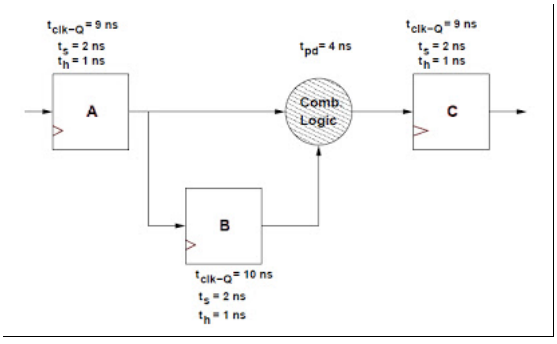
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- Part4c -> Delay - Wire Load Model
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- Part 7c -> Methods for Increase/Decrease the Delay of Circuit (Effect of Threshold voltage On the Slew)
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Example 1: Multiple FF's Sequential Circuit

In a typical sequential circuit design there are often millions of flip-flop to flip-flop paths that need to be considered in calculating the maximum clock frequency. This frequency must be determined by locating the longest path among all the flip-flop paths in the circuit. Consider the following circuit.



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Basic of Timing Analysis in Physical Design

"Timing Paths" : Static Timing Analysis (STA) basic (Part 1)

Delay - "Wire Load

There are three flip-flop to flip-flop paths (flop A to flop B, flop A to flop C, flop B to flop C). Using an approach similar to whatever I have explained in the last section, the delay along all three paths are:

- TAB = tClk−Q(A) + ts(B) = 9 ns + 2 ns = 11 ns
- TAC = tClk−Q(A) + tpd(Z) + ts(C) = 9 ns + 4 ns + 2 ns = 15 ns
- TBC = tClk−Q(B) + tpd(Z) + ts(C) = 10 ns + 4 ns + 2 ns = 16 ns

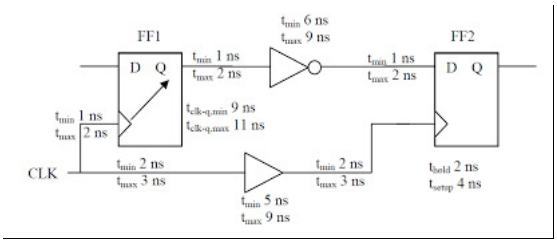
Since the TBC is the largest of the path delays, the minimum clock period for the circuit is Tmin = 16ns and the maximum clock frequency is 1/Tmin = 62.5 MHz.

Example 2: Circuit with min and max delay Specification

Let's consider following circuit. Now this circuit is similar to the normal FF circuitry, only differences are

- Every specification has 2 values (Min and Max).
- There is a combinational circuit in the clock path also.

Note: if you are wondering why there are min and max value (or like from where these values are coming, then you have to refer another blog).



Now let's understand the flow/circuit once again.

- Every interconnect wire also has some delay, so you can see clock CLK will take some time to reach the clock pin of the FF1.
- That's means with reference to original clock edge (let's assume at 0ns), clock edge will take minimum 1ns and maximum 2ns to reach the clock pin of the FF1.
- So in the similar fashion, if we will calculate the total minimum delay and maximum delay.
  - In data path : max delay = (2+11+2+9+2)ns=26ns
  - In data path : min delay = (1+9+1+6+1)ns=18ns
  - In clock path: max delay= (3+9+3)ns=15ns
  - In clock path : min delay = (2+5+2)ns=9ns
- In the last 2 example, there were no delays in the clock path, so it was easy to figure out the minimum clock period. But in this example we have to consider the delay in the clock path also.
- So for minimum clock period, we just want to make sure that at FF2, data should be present at least "tsetup" time before positive clock edge (if it's a positive edged triggered flipflop) at the FF2.
  - So Clock edge can reach at the FF2 after 9ns/15ns (min/max) with the reference of original clock edge.
  - And data will take time 18ns/26ns (min/max) with the reference of original clock edge.
  - So clock period in all the 4 combinations are
    - Clock period (T1)= (Max data path delay)-(max clock path delay)+tsetup=26-15+4=15ns
    - Clock period (T2)= (Min data path delay)-(max clock path delay)+tsetup=18-15+4=7ns
    - Clock period (T3)= (Max data path delay)-(min clock path delay)+tsetup=26-9+4=21ns
    - Clock period (T4)= (Min data path delay)-(min clock path delay)+tsetup=18-9+4=11ns
- Since we want that this circuit should work in the entire scenario (all combination of data and clock path delay), so we have to calculate the period on the basis of that.
  - Now if you will see all the above clock period, you can easily figure out that if the clock period is less than 21ns, then either one or all of the scenarios/cases/combinations fail.
  - **So we can easily conclude that for working of the entire circuit properly**
    - **Minimum Clock Period = Clock period (T3) = (Max data path delay)-(min clock path delay)+tsetup=26-9+4=21ns**

So in general:
Minimum Clock Period = (Max data path delay)-(min clock path delay) + tsetup

And **"Maximum Clock Frequency = 1/(Min Clock Period)"**

Example 3: Circuit with multiple Combinational paths between 2 FFs:

Model" : Static Timing Analysis (STA) basic (Part 4c)

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

"Setup and Hold Time" : Static Timing Analysis (STA) basic (Part 3a)

"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

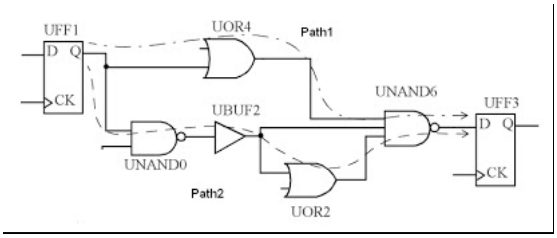
"Examples Of Setup and Hold time" : Static Timing Analysis (STA) basic (Part 3c)

"Time Borrowing" : Static Timing Analysis (STA) basic (Part 2)

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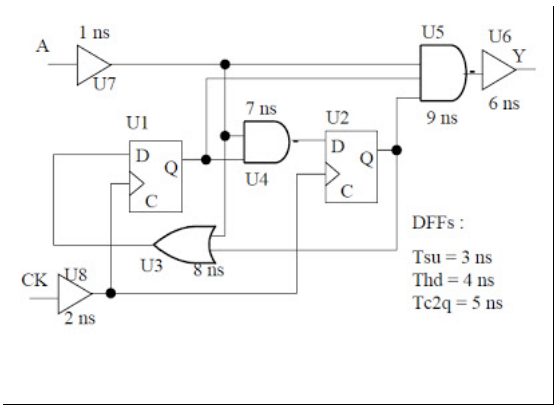
10 Ways to fix SETUP and HOLD violation: Static Timing Analysis (STA) Basic (Part-8)

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Now same scenario is with this example. I am not going to explain much in detail. Just it's like that if you have multiple paths in between the 2-flipflops, then as we have done in previous examples, please calculate the delays.
Then calculate the time period and see which one is satisfying all the condition. Or directly I can say that we can calculate the Clock period on the bases of the delay of that path which has big number.
Min Clock Time Period = Tclk-q (of UFF1) + max(delay of Path1,delay of Path2) +Tsetup (of UFF3)

Example 4: Circuit with Different kind of Timing paths:



Since I have mentioned that it has different kind of timing path, so you should know about the timing paths. For that you can refer the (Post link) post. After reading the Timing path, you can easily figure out that in the above circuit there are 4 types of data paths and 2 clock paths

Data path:

1. *Register to register Path*
   - *U2 -> U3 ->U1 (Delay=5+8=13ns)*
   - *U1 -> U4 -> U2 (Delay=5+7=12ns)*
2. *Input pin/port to Register(flip-flop)*
   - *U7 -> U4 -> U2 (Delay=1+7=8ns)*
   - *U7 -> U3 -> U1 (Delay=1+8=9ns)*
3. *Input pin/port to Output pin/port*
   - *U7 -> U5 -> U6 (Delay=1+9+6=16ns)*
4. *Register (flip-flop) to Output pin/port*
   - *U1 -> U5 -> U6 (Delay=5+9+6=20ns)*
   - *U2 -> U5 -> U6 (Delay=5+9+6=20ns)*

Clock path:

- *U8 -> U1 (Delay = 2ns)*
- *U8 -> U2 (Delay =2ns)*

Now few important points- This is not a full chip circuit. In general, recommendation is that you use registers at every input and output port. But for the time being, we will discuss this circuit, considering this as full chip circuit. And you will how much analysis you have to do in this case. Next example, I will add the FFs (registers) at input and output port and then you come to know the difference.

Now let's Study this circuit in more details.

- In this circuit, we have to do the analysis in such a way that if we will apply an input at Port A, then how much time it will take to reach at output Port Y. It will help us to find out the time period of clock.

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- Output pin Y is connected with a 3input NAND gate. So if we want a stable out at Y, we have to make sure that all 3 Inputs of NAND gate should have stable data.
- One input of NAND gate is connected with Input pin A with the help of U7.
  - Time take by data to reach NAND gate is 1ns (gate delay of U7)
- Second input pin of NAND gate is connected with output pin Q of Flip flop U2.
  - Time take by data which is present at input D of FF –U2 to reach NAND gate:
    - 2ns(delay of U8)+5ns(Tc2q of FF U2)=7ns
- Third input pin of NAND gate is connected with the output pin Q of Flip Flop U1.
  - Time take by data which is present at input D of FF –U2 to reach NAND gate:
    - 2ns(delay of U8)+5ns(Tc2q of FF U1)=7ns

Note:

- I know you may have doubt that why delay of U8 comes in picture.
  - With reference to the clock edge at CLK pin, we can receive the data at NAND pin after 7ns only (Don't ask me- why we can't take reference in negative?)
- May be you can ask why we haven't consider the setup time of FF in this calculation.
  - If in place of NAND gate, any FF would there then we will consider the setup. We never consider the setup and Tc2q (Clk-2-Q) values of same FF in the delay calculation at the same time. Because when we are considering Clk-2-Q delay, we assume that Data is already present at input Pin D of the FF.

So Time required for the data to transfer from input (A) to output (Y) Pin is the maximum of:

Pin2Pin Delay = U7+U5+U6 = 1+9+6=16ns
Clk2Out (through U1) delay = U8 +U1+U5+U6=2+5+9+6=22ns
Clk2Out (through U2) delay = U8 +U2+U5+U6=2+5+9+6=22ns.

So out of this Clk2Out Delay is Maximum.

From the above Study, you can conclude that data can be stable after 7ns at the NAND gate and maximum delay is 22ns. And you can also assume that this much data is sufficient for calculating the Max Clock Frequency or Minimum Time Period. But that's not the case. Still our analysis is half done in calculating the Max-clock-frequency.

As we have done in our previous example, we have to consider the path between 2 flip-flops also. So the paths are:

- From U1 to U2 (Reg1Reg2)
  - Path delay= 2ns (Delay of U8) + 5ns (Tclk2Q of U1)+7ns (Delay of U4)+3ns (Setup of U2) – 2ns (Delay of U8)=17ns-2ns=15ns
- From U2 to U1 (Reg2Reg1)
  - Path delay = 2ns (Delay of U8) + Tclk2Q of U2 (5ns) + Delay of U3 (8ns) + setup of U1 (3ns) – Delay of U8 (2ns) =18ns -2ns = 16ns.

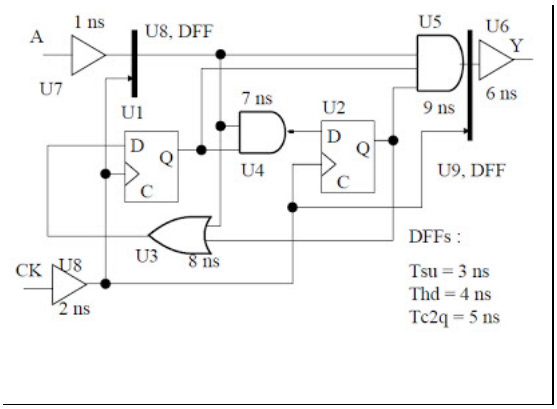
Note:

- I am sure you will ask why did I subtract "Delay of U8" from the above calculation :) because Delay of U8 is common to both the launch and capture path (In case you want to know what's Launch and capture path please follow this post). So we are not supposed to add this delay in our calculation. But just to make it clear, I have added as per the previous logic and then subtracted it to make it clear.

So now if you want to calculate the maximum clock frequency then you have to consider all the delay which we have discussed above.

So
Max Clock Freq = 1/ Max (Reg1Reg2, Reg2Reg1, Clk2Out_1, Clk2Out_2, Pin2Pin)
= 1/ Max (15, 16, 22, 22, 16)
=1/22 =45.5MHz

Example 5: Circuit with Different kind of Timing paths with Register at Input and output ports:



In this example, we have just added 2 FFs U8 at Input pin and U9 at output pin. Now for this circuit, if we want to calculate the max clock frequency then it's similar to example 1.
There are 7 Flip flop to flipflop paths

1. *U8 -> U4 -> U2*
   - *Delay = 5ns+7ns+3ns=15ns*
2. *U8 -> U3 -> U1*
   - *Delay = 5ns+8ns+3ns=16ns*
3. *U8 -> U5 -> U9*
   - *Delay = 5ns+9ns+3ns=17ns*
4. *U1 -> U4 -> U2*
   - *Delay = 5ns +7ns +3ns = 15ns*
5. *U1 -> U5 -> U9*
   - *Delay= 5ns+9ns+3ns=17ns*
6. *U2 -> U5 -> U9*
   - *Delay=5ns+9ns+3ns=17ns*
7. *U2 -> U3 -> U1*
   - *Delay=5ns+8ns+3ns=16ns*

Since the maximum path delay is 17ns,
The Minimum clock period for the circuit should be Tmin = 17 ns
And the Maximum clock frequency is 1/Tmin = 58.8 MHz.

Maximum Clock Frequency (Previous) **Index** **Examples: Solve Setup and Hold Violation (a) (Next)**

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Monday, November 19, 2012

Fixing Setup and Hold Violation : Static Timing Analysis (STA) Basic (Part 6a)

How To Solve Setup and Hold Violation

Part1	Part2	Part3a	Part3b	Part3c	Part4a	Part4b	Part4c	Part5a
Part5b	Part6a	Part6b	Part6c	Part7a	Part7b	Part7c	Part 8	

Static Timing analysis is divided into several parts:

- Part1 -> Timing Paths
- Part2 -> Time Borrowing
- Part3a -> Basic Concept Of Setup and Hold
- Part3b -> Basic Concept of Setup and Hold Violation
- Part3c -> Practical Examples for Setup and Hold Time / Violation
- Part4a -> Delay - Timing Path Delay
- Part4b -> Delay - Interconnect Delay Models
- Part4c -> Delay - Wire Load Model
- Part5a -> Maximum Clock Frequency
- Part5b -> Examples to calculate the "Maximum Clock Frequency" for different circuits.
- Part 6a -> How to solve Setup and Hold Violation (basic example)
- Part 6b -> Continue of How to solve Setup and Hold Violation (Advance examples)
- Part 6c -> Continue of How to solve Setup and Hold Violation (more advance examples)
- Part 7a -> Methods for Increase/Decrease the Delay of Circuit (Effect of Wire Length On the Slew)
- Part 7b -> Methods for Increase/Decrease the Delay of Circuit (Effect of Size of the Transistor On the Slew)
- Part 7c -> Methods for Increase/Decrease the Delay of Circuit (Effect of Threshold voltage On the Slew)
- Part 8 -> 10 ways to fix Setup and Hold Violation.

We have discussed few basics about the "Setup and Hold violation" in last few posts. Once designer's figured out the number of setup and hold violation then their next challenge is: "How to fix these violations".

EDA tools usually take care but still you have to provide the input (or say proper switch) to fix these violations. That means I can say that "Timing/Routing Tools are enough intelligent to solve most of the timing violation, but still Tools never be more intelligent than the human brain". ☺
There are different ways to fix these issues and every way has the reason for that. So designers should know what exactly the reason of Issue and what are the different methods (priority wise) or at least different EDA's switch for fixing those violation.

In this series we will discuss the following things one by one.

- Basic of Fixing the SETUP and HOLD violations.

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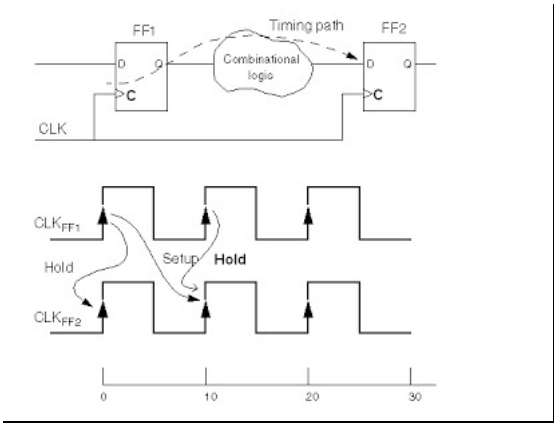
EDN:

- More Examples here. Very less theory.
- Few shortcuts/formula/tricks to find out whether these violations are fixable or not. And If fixable, then a rough idea where and how.

- Different ways to fix.
  - Their basics or say physics/Engineering of using that method for fixing.
  - Which method is good and in what scenario you can use them.

Before that- If still you have any doubt regarding the Setup and Hold then please refer following post.
What exactly is the setup and hold – please refer the previous blog.
What are setup and hold violation – please refer the previous blog.

Basics of Fixing the "SETUP and HOLD" violations.
Let's start with following Diagram and consider this as common for next few examples.



In the following examples we will pick different values of Setup/Hold values of Capture FF and Combinational Path delay. Through these example we will study - How the setup and hold violations are dependent to each other and on the delay of the circuit. If these things are clear then it's very easy for you to understand -- how can we fix the violations and if we are using any particular methods, then why?

Example 1:

Specification of the FF Circuit					
Setup	Hold	Clock period	Tck2q delay	Net Delay	Combinational Logic Delay
2ns	1ns	10ns	0ns (Ideal)	0ns (Ideal)	0.5ns

Let's discuss the flow of the data from FF1 to FF2

- Data is going to launch from FF1 at +ive Clock Edge at 0ns and it will reach to FF2 after 0.5ns (combinational logic delay only).
- This data is going to capture at FF2 at +ive Clock Edge at 10ns.
- **As per the Setup definition**, data should be stable 2ns (Setup time of FF2) at FF2 before the +ive Clock Edge (which is at 10ns)
  - In the above case – data become stable 9.5ns before the Clock edge at 10ns (10ns – 0.5ns). That means it satisfy the Setup condition. **NO SETUP VIOLATION.**
- At the FF1 – second set of data is going to launch at t=10ns and it will reach the FF2 in another 0.5ns, means at t=10.5ns.
- This second set of data is going to update/override the first set of data.

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after 0.5ns of the +ive Clock edge at FF2). This means it is not satisfying the hold condition.
HOLD VIOLATION.

To fix this Hold violation – we have to increase the delay of the Data path so that the second set of data should not reach before t=11ns (10ns+1ns). That means the minimum delay of the Combinational Logic Path should be 1ns for NO HOLD VIOLATION.
That means if you want to fix the HOLD violation, you can increase the Delay of the Data path by any method (we will discuss all those methods in detail – Just keep small patience 💻🕐).

But it doesn't mean that you can increase the Delay by any Value. Let's assume that you have increased the delay of combinational path by adding extra buffer (with delay of 8.5ns). Now new specifications are

Specification of the FF Circuit					
Setup	Hold	Clock period	Tck2q delay	Net Delay	Combinational Logic Delay

Model" : Static Timing Analysis (STA) basic (Part 4c)

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

"Setup and Hold Time" : Static Timing Analysis (STA) basic (Part 3a)

"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

"Examples Of Setup and Hold time" : Static Timing Analysis (STA) basic (Part 3c)

"Time Borrowing" : Static Timing Analysis (STA) basic (Part 2)

Effect of Wire Length On the Slew: Static Timing Analysis (STA) Basic (Part-7a)

10 Ways to fix SETUP and HOLD violation: Static Timing Analysis (STA) Basic (Part-8)

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2ns	1ns	10ns	0ns (Ideal)	0ns (Ideal)	=0.5ns+8.5ns=9ns

As per the Setup definition, data should be stable 2ns (Setup time of FF2) before the Clock Edge (at FF2 which is at 10ns) and with the updated specification – data will be stable at t=9ns, just 1ns before the Clock edge at t=10ns at FF2. That means it is not satisfying the Setup condition. **SETUP VIOLATION**.

Since Data path delay is more than 1ns, there is **NO HOLD VIOLATION** (just we have discussed few paragraph above)

So it means that if we want to fix the setup violation, the Delay of the combinational path should not be more then 8ns (10ns – 2ns). Means 8ns is the maximum value of the Delay of the Combinational Logic path for NO SETUP VIOLATION.

So we can generalize this –
For NO HOLD and SETUP VIOLATION, the delay of the path should be in between 1ns and 8ns.
OR

For Violation free Circuit:

Min delay of Combinational path > Hold time of Capture FF.
Max delay of Combinational path < Clock Period - Setup time of Capture FF.

Example 2:

Specification of the FF Circuit					
Setup	Hold	Clock period	Tck2q delay	Net Delay	Combinational Logic Delay
6ns	5ns	10ns	0ns (Ideal)	0ns (Ideal)	0.5ns

Flow of the data from FF1 to FF2:

- Data is going to launch from FF1 at Clock Edge at 0ns and it will reach to FF2 after 0.5ns (combinational logic delay only).
- This data is going to capture at FF2 at Clock Edge at 10ns.
- **As per the Setup definition**, data should be stable 6ns (Setup time of FF2) before the Clock Edge (which is at 10ns)
  - In the above case – data become stable 9.5ns before the Clock edge at 10ns (10ns – 0.5ns). That means it satisfy the Setup condition. **NO SETUP VIOLATION**.
- At the FF1 – second set of data is going to launch at t=10ns and it will reach the FF1 in another 0.5ns, means at t=10.5ns.
- This second set of data is going to update/override the first set of data.
- **As per the Hold Definition**, data should be stable till 5ns (Hold time of FF2) after the clock edge (which is at t=10ns) at FF2
  - In the above case – first set of data is going to override by second set of data at 10.5ns (means just after 0.5ns of the Clock edge at FF2). This means it is not satisfying the hold condition. **HOLD VIOLATION**.

To fix this Hold violation – (As per the previous example) we may increase the delay of the Data path, so that the second set of data should not reach before t=15ns (10ns+5ns). That means the minimum delay of the Combinational Logic Path should be 5ns for NO HOLD VIOLATION.

But Now if you will verify the Setup condition once again (with combination delay of 5ns- which we have assumed for fixing the hold violation) then you come to know that data is going to stable only after 5ns (means 10ns-5ns = 5ns before the clock edge at t-10ns). But as per the setup condition data should be stable before 6ns. So it means now it's not satisfying Setup Condition. Means SETUP VIOLATION.

So in this scenario, we can't fix the setup and hold violation at the same time by adjusting the delay in the combinational logic.

You can also see it directly with the help of minimum and maximum value of combinational delay.

Min delay > Hold time of Capture FF (means 5ns)
Max Delay < Clock Period – Setup time of capture FF (Means 10ns – 6ns = 4ns)

So Min delay > 5ns and Max Delay < 4ns which is not possible.

Now the point is how to fix these violations? Actually this is a non-fixable issue until you just change the clock frequency or replace the FF with lesser setup/hold value. ☺

Let me explain this.
Min delay has dependence only on Hold time, which is fixed for a particular FF.
Max delay has dependence on 2 parameters – Clock Period and Setup time - where Setup time is fixed for a particular FF.
So if you can change the FF with lower setup/hold violation, then you can fix this issue. But in case if that's not possible then we have to change the Clock period.

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In case we are changing the Clock period:

Keep -- Min delay >= 5ns (No HOLD Violation)

Setup violation is by 6ns-5ns =1ns (6ns= Setup time and 5ns = combinational delay). What if we will increase the Clock period by 1ns. Means New clock period should be > 11ns.

So for Clock Period 11ns:

Max delay <= Clock period (11ns) – Setup time (6ns) =5ns.

Now - Max Delay=Min Delay = 5ns. (Neither Hold nor Setup Violation.)

We can generalize-

For Violation Free Circuit

Clock Period >= Setup time + Hold time.

Summary of this Post:

Min delay of Combinational path > Hold time of Capture FF.
Max delay of Combinational path < Clock Period - Setup time of Capture FF.
Clock Period >= Setup time + Hold time.

In the next part we will discuss few more examples with more restrictions. Like-

- What if we can't reduce the Delay in the Data path?

Examples: Calculating Maximum Clock Frequency (Previous) | Index
Examples: Solve Setup and Hold Violation (b) (Next)

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- Part 3a -> Basic Concept Of Setup and Hold
- Part 3b -> Basic Concept of Setup and Hold Violation
- Part 3c -> Practical Examples for Setup and Hold Time / Violation
- Part 4a -> Delay - Timing Path Delay
- Part 4b -> Delay - Interconnect Delay Models
- Part 4c -> Delay - Wire Load Model
- Part 5a -> Maximum Clock Frequency
- Part 5b -> Examples to calculate the "Maximum Clock Frequency" for different circuits.
- Part 6a -> How to solve Setup and Hold Violation (basic example)
- Part 6b -> Continue of How to solve Setup and Hold Violation (Advance examples)
- Part 6c -> Continue of How to solve Setup and Hold Violation (more advance examples)
- Part 7a -> Methods for Increase/Decrease the Delay of Circuit (Effect of Wire Length On the Slew)
- Part 7b -> Methods for Increase/Decrease the Delay of Circuit (Effect of Size of the Transistor On the Slew)
- Part 7c -> Methods for Increase/Decrease the Delay of Circuit (Effect of Threshold voltage On the Slew)
- Part 8 -> 10 ways to fix Setup and Hold Violation.

In the last part/post we have discussed 2 examples with different specifications (Both net delay and Tck2Q were ideal means 0ns) and come to know that for Violation free Circuit, following conditions should be satisfied.

Min delay of Combinational path > Hold time of Capture FF.
Max delay of Combinational path < Clock Period - Setup time of Capture FF.

Clock Period >= Setup time + Hold time.

In this post we will discuss few more examples with more restrictions. Like

- What if we can't reduce the Delay of Data path?

Let's consider the following figure common to all examples until unless it's specified

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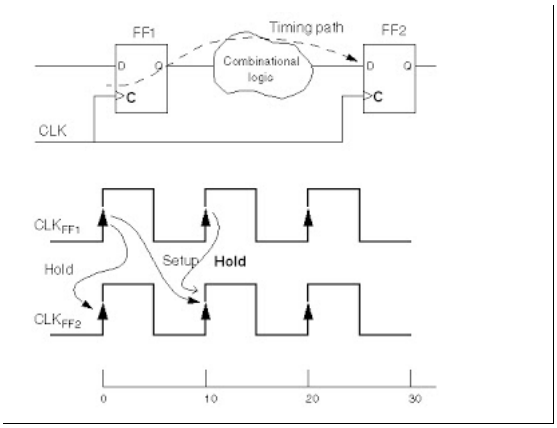
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Example 3:

Specification of the FF Circuit					
Setup	**Hold**	**Clock period**	**Tck2q delay**	**Net Delay**	**Combinational Logic Delay**
3ns	**2ns**	**10ns**	**0ns (Ideal)**	**0ns (Ideal)**	**5ns (can't be further reduced).**

On the basic of last post …let's start with checking few conditions directly.
Clock Period Condition: (Satisfied)
Setup time +Hold time = 5ns
Clock period = 10ns
Clock Period > Setup time +Hold time (10> 5)

Min delay / Hold Condition: (Satisfied)
Combinational Delay (5ns) > Hold time.
Means - NO HOLD VIOLATION

Max Delay / Setup Condition: (Satisfied)
Combinational delay (5ns) < Clock period (10ns) – Setup (3ns)
Means - NO SETUP VIOLATION.

This example is just to refresh your memories about the previous post.

Example 4:

Specification of the FF Circuit					
Setup	Hold	Clock period	Tck2q delay	Net Delay	Combinational Logic Delay
4ns	3ns	10ns	0ns (Ideal)	0ns (Ideal)	8ns (can't be further reduced).

Clock Period Condition: (Satisfied)
Setup time +Hold time = 4ns+3ns = 7ns
Clock period = 10ns
Clock Period > Setup time + Hold time (10 > 7)

Min delay / Hold Condition: (Satisfied)
Combinational Delay (8ns) > Hold time (3ns)
Means - NO HOLD VIOLATION

Max Delay / Setup Condition: (Not Satisfied)
Combinational delay (8ns) **Is Not Less Than** "Clock period (10ns) – Setup (4ns)"
Means - SETUP VIOLATION.

Since we can't change this Combinational delay and also Setup time for the FF, so we have to think something else. J. Since we can't touch the data path, we can try with clock path.

Flow of the data from FF1 to FF2:

- Let's assume that you have added one buffer of T_capture delay in the clock path between the FF1 and FF2.
- Data is going to launch from FF1 at Clock Edge at 0ns and it will reach to FF2 after 8ns (combinational logic delay only).
- This data is going to capture at FF2 at Clock Edge at 10ns+T_capture. (because of Delay added by Buffer).
- **As per the Setup definition**, data should be stable 4ns (Setup time of FF2) before the Clock Edge at FF2 and in the above case clock edge is at t=T_capture+10ns.

So, for No Setup violation:

Model" : Static Timing Analysis (STA) basic (Part 4c)

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

"Setup and Hold Time" : Static Timing Analysis (STA) basic (Part 3a)

"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

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"Time Borrowing" : Static Timing Analysis (STA) basic (Part 2)

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10 Ways to fix SETUP and HOLD violation: Static Timing Analysis (STA) Basic (Part-8)

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=> 8ns (Combinational Delay) < T_capture+10ns (clock period) – 4ns (Setup Time of FF2)
=> 12ns – 10ns < T_capture
=> **T_capture > 2ns**.

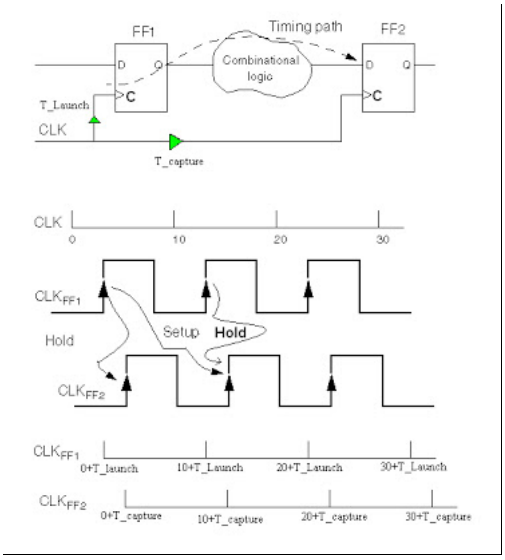
Let's assume if my T_capture = 3ns. Then NO SETUP VIOLATION.

Now, recheck the Hold violation.

- At the FF1 – second set of data is going to launch at t=10ns and it will reach the FF2 in another 8ns, means at t=18ns.
- This second set of data is going to update/override the first set of data present at FF2.
- **As per the Hold Definition**, data should be stable till 3ns (Hold time of FF2) after the clock edge at FF2 (Which is at t=10ns+3ns=13ns – where 3ns is the T_capture).
- That means Data should be remain stable till t=13ns+3ns=16ns.
    - In the above case the second set of data is going to override only after t=18ns. That means first set of data remain Stable till 16ns. Means **NO HOLD VIOLATION.**

Let me Generalize this concept:

I am sure, few people may have question that what will happen if we will add the buffer in the Launch path. Let's discuss that. Please consider the following Diagram for this. In this Launch Clock path has a buffer with a delay of "T_launch" and Capture clock path has another buffer of delay "T_capture".



Specification of the FF Circuit	
Setup	T_setup
Hold	T_hold
Clock Period	Clk_period
Tck2q Delay	0 (Ideal)
Net Delay	0 (Ideal)
Combinational Logic Delay (b/w 2FFs)	Td
Launch Clock path Delay	T_launch
Capture Clock path Delay	T_capture

Let's understand the data flow from FF1 to FF2

- Data is going to launch from FF1 at Clock Edge at T_launch and it will reach to FF2 after Td (combinational logic delay only) that means t= "Td + T_launch".
- This data is going to capture at FF2 at Clock Edge at "Clk_period + T_capture"
- **As per the Setup definition**, data should be stable "T_setup" (Setup time of FF2) time before the Clock Edge at FF2
    - Means data should reach at FF2 before t= "Clk_period + T_capture – T_setup".

So For NO SETUP VIOLATION:

=> T_launch + Td < Clk_period + T_capture – T_setup
=> **Td < Clk_Period + (T_capture - T_launch) – T_setup**

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- At the FF1 – second set of data is going to launch at t= “Clk_Period + T_launch” and it will reach the FF2 in another Td, means at t=” Clk_Period + Td + T_launch”.
- This second set of data is going to update/override the first set of data present at FF2.
- **As per the Hold Definition**, data should be stable till “T_hold” (Hold time of FF2) time after the Clock edge (which is at t= “Clk_Period + T_capture”).
  - Means Next set of data should not reach FF2 before t= “Clk_Period + T_capture + T_hold”

So For NO HOLD VIOLATION:

=> Clk_Period + Td + T_launch > Clk_Period + T_capture + T_hold
=> **Td > (T_capture - T_launch) + T_hold**

Summary of this post:

Clock Period Condition:
Clock period > Setup time + Hold Time

Max Delay/ Setup Condition:
Td < Clk_Period + (T_capture - T_launch) – T_setup

Min Delay / Hold Condition:
Td > (T_capture - T_launch) + T_hold

In the next part we will discuss

- More examples which will explain the above conditions in more details.
- How to fix the Setup and hold violation, if we can neither decrease nor increase the Delay in the Data path?

Examples: Solve Setup and Hold Violation (a) (Previous) | Index
Examples: Solve Setup and Hold Violation (c) (Next)

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Awesome :) crystal clear..

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Wednesday, November 21, 2012

Fixing Setup and Hold Violation : Static Timing Analysis (STA) Basic (Part 6c)

How To Solve Setup and Hold Violation.. continue...

Part1	Part2	Part3a	Part3b	Part3c	Part4a	Part4b	Part4c	Part5a
Part5b	Part6a	Part6b	Part6c	Part7a	Part7b	Part7c	Part 8	

Static Timing analysis is divided into several parts:

- Part1 -> Timing Paths
- Part2 -> Time Borrowing
- Part3a -> Basic Concept Of Setup and Hold
- Part3b -> Basic Concept of Setup and Hold Violation
- Part3c -> Practical Examples for Setup and Hold Time / Violation
- Part4a -> Delay - Timing Path Delay
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- Part 7c -> Methods for Increase/Decrease the Delay of Circuit (Effect of Threshold voltage On the Slew)
- Part 8 -> 10 ways to fix Setup and Hold Violation.

In the last part/post we have discussed 2 more examples with different specifications with more restrictions (Both net delay and Tck2Q were ideal means 0ns) and figure out that if you want to fix the violation by increasing/decreasing the delay in the data path then following condition should be satisfied.

Min delay of Combinational path > Hold time of Capture FF.
Max delay of Combinational path < Clock Period - Setup time of Capture FF.

Clock Period >= Setup time + Hold time.

But in case if you can't touch the data path and you have to increase/decrease the delay in the clock path (means between "Clk pin to Launch FF clock pin" Or between "Clk pin and capture FF clock pin"), then following conditions should satisfied.

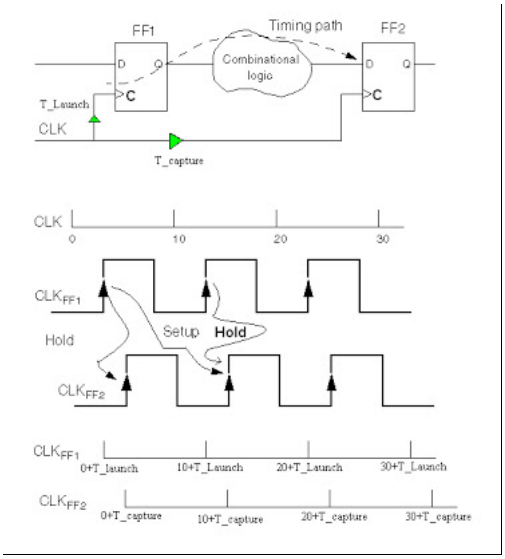
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Max Delay/ Setup Condition:
Td < Clk_Period + (T_capture - T_launch) – T_setup

Min Delay / Hold Condition:
Td > (T_capture - T_launch) + T_hold

Where:
Td -> Combinational path delay (between the 2 FFs)
T_capture -> Delay of circuit present between "Clk pin and capture FF clock pin"
T_launch -> Delay of circuit present between "Clk pin to Launch FF clock pin"

In this post we will discuss few more examples with more restrictions.
Let's consider the following figure common to all examples until unless it's specified.



Example 5:

Specification of the FF Circuit					
Setup	Hold	Clock period	Tck2q delay	Net Delay	Combinational Logic Delay
3ns	2ns	10ns	0ns (Ideal)	0ns (Ideal)	11ns (can't be further reduced)

On the basic of last post …let's start with checking few conditions directly.

Clock Period Condition: (Satisfied)
Setup time +Hold time = 5ns
Clock period = 10ns
Clock Period > Setup time +Hold time (10> 5)

Min delay / Hold Condition: (Satisfied)
Combinational Delay (11ns) > Hold time.
Means - NO HOLD VIOLATION

Max Delay / Setup Condition:
Combinational delay (11ns) **Is Not Less Than** "Clock period (10ns) – Setup (3ns)"
Means - SETUP VIOLATION.

Since adding delay in the data path is not going to fix this violation and we can't reduce the combinational delay. So as we have discussed in our last post, we will try with Clock path.

From the last post, if T_capture is the delay of buffer which is inserted between the CLK and Capture's FF and T_launch is the delay of buffer which is inserted between the CLK and Launch's FF, then

Max Delay /Setup condition is :
Td < Clock Period + (T_capture - T_launch) – T_setup
=> 11ns < 10ns – 3ns + (T_capture - T_launch)
=> 11ns < 7ns + (T_capture - T_launch)
=> 4ns < (T_capture - T_launch)

Now we can choose any combination of T_capture and T_launch such that their difference should be less than 4ns.
Note: Remember in the design if you are fixing the violation by increasing or decreasing the delay in the Clock path then always prefer not to play too much with this path.

I never prefer to use T_launch in this case (For setup fixing, I ignore to use T_launch).
So let's assume T_launch =0ns and T_capture = 5ns

Then

"Timing Paths" : Static Timing Analysis (STA) basic (Part 1)

Delay - "Wire Load Model" : Static Timing Analysis (STA) basic (Part 4c)

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"Examples Of Setup and Hold time" : Static Timing Analysis (STA) basic (Part 3c)

"Time Borrowing" : Static Timing Analysis (STA) basic (Part 2)

Effect of Wire Length On the Slew: Static Timing Analysis (STA) Basic (Part-7a)

10 Ways to fix SETUP and HOLD violation: Static Timing Analysis (STA) Basic (Part-8)

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11ns < 7ns + 5ns **means no Setup Violation.**

Check once again the Hold condition.
Min delay / Hold Condition:
Td > (T_capture - T_launch) + T_hold
=> 11ns > (T_capture - T_launch) + T_hold
=> 11ns > 5ns + 2ns
=> 11ns > 7ns – **Means No Hold Violation.**

Example 6:

Specification of the FF Circuit					
Setup	Hold	Clock period	Tck2q delay	Net Delay	Combinational Logic Delay
3ns	5ns	10ns	0ns (Ideal)	0ns (Ideal)	2ns (can't be further reduced and we can't increase the delay in the data path by any methods)

Let's check the conditions directly.

Clock Period Condition (Satisfied):
Setup time +Hold time = 8ns
Clock period = 10ns
Clock Period > Setup time +Hold time (10ns > 8ns)
Means we can fix violations, if there is any.

Max Delay/ Setup Condition (Satisfied):
Td < Clk_Period + (T_capture - T_launch) – T_setup
Combinational Delay = 2ns
There is no delay in the clock path till now, so T_capture=T_launch=0ns
=> Td (2ns) < Clk_period (10ns) + 0ns – T_setup (3ns)
=> 2ns < 7ns – **Means NO SETUP Violations**

Min Delay / Hold Condition (Not Satisfied):
Td > (T_capture - T_launch) + T_hold
Combinational Delay = 2ns
There is no delay in the clock path till now, so T_capture=T_launch=0ns
=> Td (2ns) **is not greater than** 0ns + T_hold (5ns)
Means HOLD VIOLATION

Since we can't make change in the delay path, so we have to touch the clock path.
For Hold fixing -
=> Td > (T_capture - T_launch) + T_hold
=> 2ns > (T_capture - T_launch) + 5ns
=> -3ns > (T_capture - T_launch)

For Satisfying the above equation T_launch should have more value in comparison to T_capture.
We can choose any combination of T_capture and T_launch.

Note: Remember in the design if you are fixing the violation by increasing or decreasing the delay in the Clock path then always prefer not to play too much with this path.

I will never prefer to use T_capture in this case (For Hold fixing, I ignore to use T_capture).
So let's assume T_capture =0ns and T_launch = 4ns

Then

T_launch + Td > 5ns (hold time)
=> 4ns +2ns > 5ns **NO HOLD Violation.**
Check once again the Setup Condition:
Td < Clock Period + (T_capture - T_launch) – T_setup
=> 2ns < 10ns + 0ns -4ns – 3ns
=> 2ns < 3ns **Means No Setup Violation.**

Note: (T_capture - T_launch) also known as CLOCK SKEW. I will explain this later in this blog. Right now, it's Just for your info.

Example 7:

Specification of the FF Circuit					
Setup	Hold	Clock period	Tck2q delay	Net Delay	Combinational Logic Delay
6ns	5ns	10ns	0ns (Ideal)	0ns (Ideal)	0.5ns

Note: this is the same example which we have discussed in the part-6a. Let's check all the conditions one by one.

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Clock Period Condition (Not Satisfied):
Setup time +Hold time = 11ns
Clock period = 10ns
Clock Period **is not greater than** Setup time +Hold time
Means we can't fix violations, if there is any.

But still we will try once again with all other conditions, just to prove that above mention condition should be true for fixing the violations.

Max Delay/ Setup Condition (Satisfied):
Td < Clk_Period + (T_capture - T_launch) – T_setup
Combinational Delay = 0.5ns
There is no delay in the clock path till now, so T_capture=T_launch=0ns
=> Td (0.5ns) < Clk_period (10ns) + 0ns – T_setup (6ns)
=> 0.5ns < 4ns – **Means NO SETUP Violations**

Min Delay / Hold Condition (Not Satisfied):
Td > (T_capture - T_launch) + T_hold
Combinational Delay = 0.5ns
There is no delay in the clock path till now, so T_capture=T_launch=0ns
=> Td (0.5ns) **is not greater than** 0ns + T_hold (5ns)
Means HOLD VIOLATION

If you want to fix the Hold violation, then we have already seen that by increasing/decreasing the delay in the data path it can't be fixed. Even if this will fixed, then Setup violation will occur.
Let's Try with T_capture or T_launch. Means by adding delay in the clock circuit.

As per the above equations/conditions and corresponding values:
Max Delay/ Setup Condition :
Td < Clock Period + (T_capture - T_launch) – T_setup
=> Td < 10ns -6ns + (T_capture - T_launch)
=> Td < 4ns + (T_capture - T_launch)

Min Delay / Hold Condition:
Td > (T_capture - T_launch) + T_hold
=> Td > (T_capture - T_launch) + 5ns

Remember all 3 variable Td,T_capture,T_launch are positive number.
Possible values of (T_capture - T_launch) = +/-A (where A is a positive number)

Case 1: (T_capture - T_launch) = +A

=> Td < 4ns+A - Condition (a)
=> Td> 5ns+A – Condition (b)
Satisfying both the conditions ("a" and "b") not possible for any +ive value of A.

Case 1: (T_capture - T_launch) = -A

=> Td< 4ns-A => Td+A < 4ns - Condition (a)
=> Td> 5ns-A => Td +A > 5ns - Condition (a)
Satisfying both the conditions ("a" and "b") not possible for any +ive value of A.

That means, I am successfully able to prove that if following condition is not satisfied then you can't fix any type of violation by increasing/decreasing delay in either data_path or clock_path.

Clock Period > Setup time + Hold time.

Summary of this post:

Clock Period Condition:
Clock period > Setup time + Hold Time
For fixing any type of violation (without changing Clock period) - This condition should be satisfied.

Max Delay/ Setup Condition:
Td < Clk_Period + (T_capture - T_launch) – T_setup
For Fixing the Setup Violation – Always prefer T_capture over T_launch

Min Delay / Hold Condition:
Td > (T_capture - T_launch) + T_hold
For Fixing the hold Violation – Always prefer T_launch over T_capture.

Till now we have discussed almost all the necessary basic of fixing the violation of Setup and Hold time. You have been noticed that everywhere I have talked about the increasing/decreasing the delay. If I have mentioned anywhere adding/removing the buffer, that also mean increasing/decreasing the delay.

There are several other ways through which you can increase/decreasing the delay of the circuit. In the next post we will discuss

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Saturday, October 5, 2013

Effect of Wire Length On the Slew: Static Timing Analysis (STA) Basic (Part-7a)

(Effect of Wire Length On the Slew)

Part1	Part2	Part3a	Part3b	Part3c	Part4a	Part4b	Part4c	Part5a
Part5b	Part6a	Part6b	Part6c	Part7a	Part7b	Part7c	Part 8	

Static Timing analysis is divided into several parts:

- Part1 -> Timing Paths
- Part2 -> Time Borrowing
- Part3a -> Basic Concept Of Setup and Hold
- Part3b -> Basic Concept of Setup and Hold Violation
- Part3c -> Practical Examples for Setup and Hold Time / Violation

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- Part5a -> Maximum Clock Frequency
- Part5b -> Examples to calculate the "Maximum Clock Frequency" for different circuits.
- Part 6a -> How to solve Setup and Hold Violation (basic example)
- Part 6b -> Continue of How to solve Setup and Hold Violation (Advance examples)
- Part 6c -> Continue of How to solve Setup and Hold Violation (more advance examples)
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- Part 7b -> Methods for Increase/Decrease the Delay of Circuit (Effect of Size of the Transistor On the Slew)
- Part 7c -> Methods for Increase/Decrease the Delay of Circuit (Effect of Threshold voltage On the Slew)
- Part 8 -> 10 ways to fix Setup and Hold Violation.

Till now we have discussed the Ideal scenario for few of the cases. Like No Clock-to-Q delay, No Net Delay. But now we will discuss about those parameter also.

First understand/revise what are the different types or forms of Delay into a circuit.
In FFs:

- Clock to Q delay
  - Propagation delay of sequential flip flop
- Time taken to charge and discharge the output load (capacitance) at Pin Q.
  - Rise time and Fall time delay

Combinational Circuit:

- Cell delay
  - Delay contributed by Gate itself.

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Basic of Timing Analysis in Physical Design
"Timing Paths" : Static Timing Analysis (STA) basic (Part 1)
Delay - "Wire Load

EDN:

- Typically defined as 50% input pin voltage to 50% output voltage.
- Usually a function of Both Output Loading and Input Transition time.
- Can be divide into propagation delay and transition delay.
- Propagation delay is the time from input transition to completion of a specific % (e.g 10%) of the output transition.
  - Propagation delay is function of output loading and input transition time.
- Transition Delay is the time for an output pin to change the stage.
  - Transition delay is function of capacitance at the output pin and can also be a function of input transition time.
- Time taken to charge and discharge the output load (capacitance) of the Cell output.

Net Delay:

- RC delay.
  - Long wire has more delay in comparison to short wire.
  - More coupling means more delay.

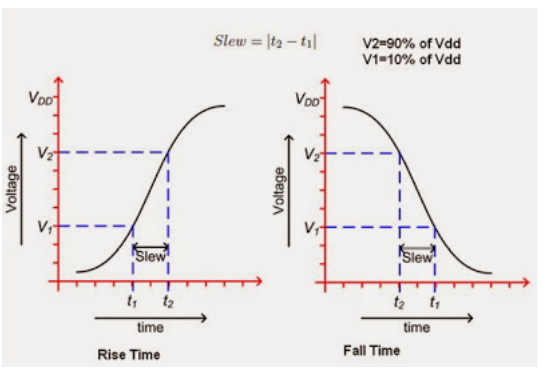
Now we will discuss different techniques to increase or decrease the delay in the design. We will also discuss the basics of different techniques, which will help us to understand why we are using any particular technique.

Now we have to see what best we can do to remove these violations or as explained earlier – How can we increase or decrease the delay of the clock or data path in the design. If I will ask you, then might be you can tell me 10 ways to do so. But I don't want to explain in that way. Let's start one by one with basics and then in the last I will brief all those points.

Let's talk about the Transition delay first. There are 2 types of transition delays. Rise Delay and Fall delay. In terms of definition

- **Rise Time Delay (tr):** The time required for a signal to transition from 10% of its maximum value to 90% of its maximum value.
- **Fall Time Delay (tf):** The time required for a signal to transition from 90% of its maximum value to 10% of its maximum value.

Basically these times (rise time and fall time) are related to the Capacitance Charging and Discharging time.
So when capacitance is charging just because of any change in the input voltage then time taken by capacitance to reach from 10% to 90% of maximum value is known as rise time. Since this time (rise time) is going to introduce the delay in the circuit in comparison to the Ideal scenario (Capacitance charging time is Zero – It can charge instantly), it's known as Rise Time Delay also.
Similarly, during the discharging of the capacitance from 90% to 10% of its maximum value, it's going to add one more delay – known as Fall Time Delay.
Following figure is just an example of rise time and fall time.
Note: Transition time is also known as Slew.



So we can say that Capacitance (and the associated Resistance) is the culprit. ☺ And if we can play with capacitance/resistance, we can increase and decrease Transition Delay.

Now, whenever we are talking about any signal which is changing its state from "0" to "1" or from "1" to "0", we are sure that it can't be ideal (Ideal means its changing its state in Zero "0" time). If you have any doubt on this statement then defiantly I have to ask you to read some very basic books once again. ☺
Every "state changing signal" has a Slew Number (common name of Rise time and Fall time) associated with itself at any given point of time.

Effect of Wire length on the Slew (transition time):

Model" : Static Timing Analysis (STA) basic (Part 4c)

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

"Setup and Hold Time" : Static Timing Analysis (STA) basic (Part 3a)

"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

"Examples Of Setup and Hold time" : Static Timing Analysis (STA) basic (Part 3c)

"Time Borrowing" : Static Timing Analysis (STA) basic (Part 2)

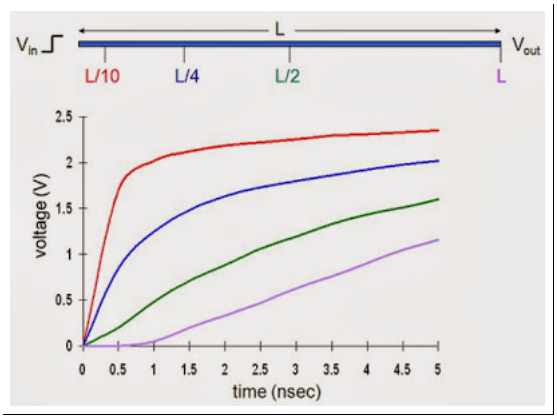
Effect of Wire Length On the Slew: Static Timing Analysis (STA) Basic (Part-7a)

10 Ways to fix SETUP and HOLD violation: Static Timing Analysis (STA) Basic (Part-8)

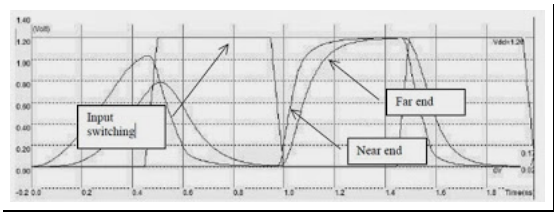
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In the below figure you can observe, how the step waveform (consider this as ideal one) degrades from the start to the end of the wire (color coding can help you to understand) and this is resulting a considerable amount of delay for long wires. That means if wire length is less, then degradation of waveform be less, means less effective delay and Vice-versa. We can conclude from this-

"If we want to increase the delay- we can increase the wire length and vice versa"



More simulation results you can see from this picture… (Following picture I have copied from book "DEEP SUBMICRON CMOS DESIGN" written by E.Sicard, S. Delmas-Bendhia)



I am sure you can cross question me that why this degradation is happing. Simple Ans is - you can model a wire into a series of Resistance and Capacitance network. For more detail please refer following post Interconnect Delay Models.

Note: This delay is also known as Net delay/Wire Delay/Interconnect Delay.

In the next post we will discuss about the effect of Size of the Transistor on the "Transition Delay" and "Propagation Delay".

Examples: Solve Setup and Hold Violation (c) (Previous) | Index
Effect of Size of the Transistor On the Slew (Next)

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Wednesday, October 16, 2013

Effect of Transistor's Size On the Slew: Static Timing Analysis (STA) Basic (Part-7b)

Methods for Increase / Decrease the Delay in Clock / Data path.

(Effect of Transistor's Size On the Slew)

Part1	Part2	Part3a	Part3b	Part3c	Part4a	Part4b	Part4c	Part5a
Part5b	Part6a	Part6b	Part6c	Part7a	Part7b	Part7c	Part 8	

Static Timing analysis is divided into several parts:

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- Part2 -> Time Borrowing
- Part3a -> Basic Concept Of Setup and Hold
- Part3b -> Basic Concept of Setup and Hold Violation
- Part3c -> Practical Examples for Setup and Hold Time / Violation
- Part4a -> Delay - Timing Path Delay
- Part4b -> Delay - Interconnect Delay Models
- Part4c -> Delay - Wire Load Model
- Part5a -> Maximum Clock Frequency
- Part5b -> Examples to calculate the "Maximum Clock Frequency" for different circuits.
- Part 6a -> How to solve Setup and Hold Violation (basic example)
- Part 6b -> Continue of How to solve Setup and Hold Violation (Advance examples)
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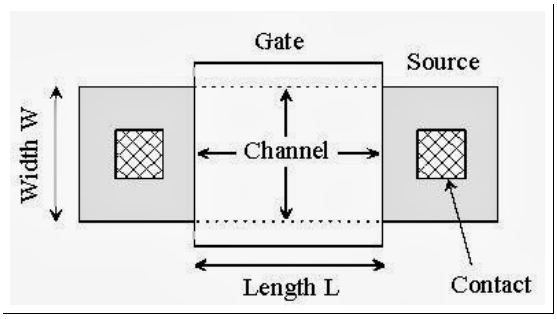
In the Last post we have discussed - how the wire length effects the slew? Now lets discuss about the effect of size of the transistors. Also before that let's discuss few basics also.

Size of transistor:

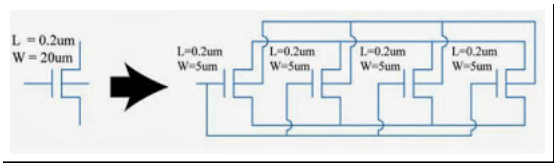
There are 2 parameters – Width and Length, by which you can decide the size of the transistor. For a particular technology – Channel - length is almost constant. So it means Width is going to decide the size of the transistor. Below

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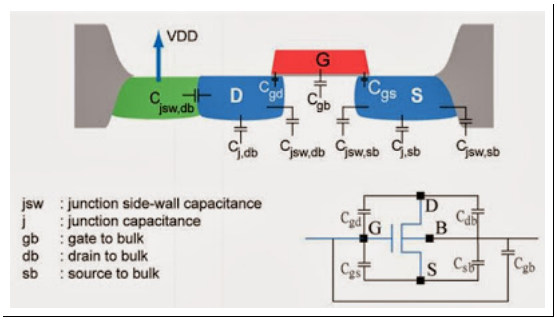
figure will refresh your memory - about which, parameter I am talking.



If you want to increase the width of the transistor, then you have 2 options. One – Just increase the Width directly, Second -connect multiple transistors in parallel in such a way that their effective impact remains same. For example – if you want to manufacture a transistor with a width of 20um and a length of 0.2um then it's similar (not exactly the same) to having four transistors connected in parallel, each with a width of 5um and a length of 0.2um. Here I am not going to discuss the difference in both the way of representation of Layout. If you are interested then you can check any basic book of CMOS design. Below figure will refresh your memory (Note: Below figure I have copied from www.eda-utilities.com)



Now since we are talking about the transition time /transition delay /slew, we know that it depend on the capacitance and resistance. So before we start to discuss how width (means size of the transistor) impact on the transition delay, we should know what all are the capacitance associated with the transistor. Below diagram help you in that. (Note: Below figure I have copied from www.eda-utilities.com)



How the capacitance are calculated (means whole derivation and explanation), I will discuss some other time, right now I am writing/copying the value of these capacitance directly.

"Timing Paths" : Static Timing Analysis (STA) basic (Part 1)

Delay - "Wire Load Model" : Static Timing Analysis (STA) basic (Part 4c)

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

"Setup and Hold Time" : Static Timing Analysis (STA) basic (Part 3a)

"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

"Examples Of Setup and Hold time" : Static Timing Analysis (STA) basic (Part 3c)

"Time Borrowing" : Static Timing Analysis (STA) basic (Part 2)

Effect of Wire Length On the Slew: Static Timing Analysis (STA) Basic (Part-7a)

10 Ways to fix SETUP and HOLD violation: Static Timing Analysis (STA) Basic (Part-8)

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$$C_G = C_{ox}LW + 2C_o$$

$$C_{GD} = G_{GS} = \frac{1}{2}C_{ox}(WL) + C_o$$

$$C_{DB,bot} = K(V_l)C_{j0}A_D$$

$$C_{DB,sw} = K_{1/3}(V_l)C_{jsw}l_D$$

$$C_{DB} = C_{DB,bot} + C_{DB,sw}$$

$$C_{SB,bot} = K(V_l)C_{j0}A_S$$

$$C_{SB,sw} = K_{1/3}(V_l)C_{jsw}l_S$$

$$C_{SB} = C_{SB,bot} + C_{SB,sw}$$

where:
L = channel length
W = channel width
C_o = overlap capacitance
C_{j0} = zero bias capacitance per unit area
C_{jsw} = zero-bias sidewall capacitance per unit perimeter
A_D = Area of Drain
l_D = perimeter of Drain
A_S = Area of source
L_S = perimeter of source

$$C_{ox} = \frac{\varepsilon_{ox}}{x_{ox}}$$

x_{ox} = Gate oxide thickness

$$\frac{K(V_l)}{K_{1/3}(V_l)} = \text{function of Voltages applied across Source and Drain}$$

Note: the reference of above formulas is from the book written by “J.P.uyemura” - Cmos Logic Circuit Design Edition -2002.

Now from the above, you can see that Gate Capacitance (this gate capacitance has 3 component – Gate to Base, Gate to Source and Gate to Drain) has dependence on the Width of the Channel (W). So it means, if you increase the width, Gate Capacitance will increase and Vice-Versa.
Source and Drain Capacitance has a multiplying factor As and Ad (which is equavilant to WxLs or WxLd). It means source and drain capacitance also increases with Width of the Channel and Vice-Versa.

Now let’s talk about the Resistance. Below Resistance formula is with respect to NMOS. You can derive similar formula for PMOS also (Just replace subscript “n” with “p” ☺).

$$R_n = \frac{1}{\beta_n(V_{DD} - V_{Tn})} = \frac{1}{k'_n\left(\frac{W}{L}\right)(V_{DD} - V_{Tn})}$$

where:
V_{DD} = Power Supply voltage applied to circuit
V_{Tn} = Threshold Voltage for Nmos
$k'_n = \mu_n C_{ox}$
R_n = On Resisitance of Nmos

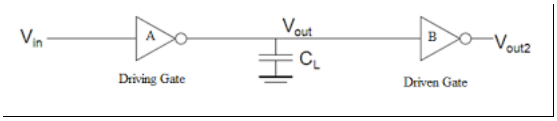
Now here the Resistance is inversely proportional to Width of the Transistor.

Effect of Device Size on the Slew (transition time) and Propogation Delay.:

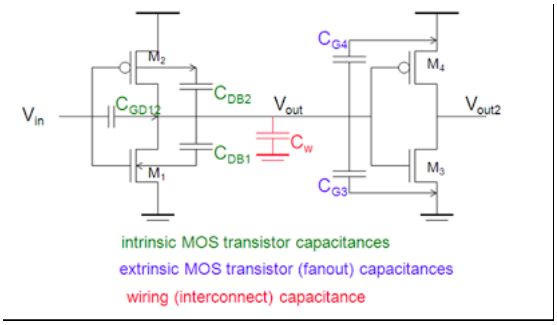
I can’t write in a single line the effect of size of transistor on the slew because it’s not straight forward (I know you might have doubt on my statement☹). There are some other factors which we have to consider. I hope, below paragraph helps you to understand the same.

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Consider the above circuit. Gate "A" is the Driving Gate and Gate "B" is the Driven Gate. If we will expand this with the actual capacitance, it will be something similar to…



- Capacitance Cgd12 is the Gate Capacitance of Driving Gate A due to overlap in M1 and M2.
- Cdb1 and Cdb2 are the diffusion capacitances due to the reverse-biased pn-junction
- Cw is the wiring capacitance (pp, fringe, and interwire) that depends on the length and width of the connecting wire. It is a function of the fanout of the gate and the distance to those gates.
- Cg3 and Cg4 are the gate capacitance of the fanout gate (Driven gate).

If we increase the size of the transistor (Width of the Transistor) it's current carrying capability increase. Means "larger is the size of a transistor, the larger is the driving capability (the ability to source or sink current) of a transistor". Thus a larger transistor would normally make its output transition faster (when output load is constant). The output load of a driving gate consists of the source/drain capacitance of the driving gate, the routing capacitance of wire, and the gate capacitance of the driven gate.
The larger is the output load, the longer is the time to charge or discharge it. This would increase the transition (rise or fall) time and propagation delay.

Let me summarize few important points.

- On increasing the Size of Gate A –
  - On Resistance Decreases (R - inversely proportional to W)
  - Means large Driving capability (Ability to source or sink current)
  - Decrease the time to charge the output load (capacitance) (Consists of source/drain capacitance of the driving gate, the routing capacitance of wire, and the gate capacitance of the driven gate) **
  - Means - Output Transition time of Gate A and Input Transition time for Gate B decreases.

I am sure you have noticed that I have marked point 3 with ** because there are terms and conditions. :)

On increasing the Size of the Gate A – Source/Drain Capacitance also increases which are the part of output load of Gate A. Means it's going to increase the output load. That means as I have mentioned in my point no 3 – that can be possible only when S/D Capacitance of Driving gate are not dominating the rest of the Capacitance. Which is only possible when either "Net capacitance is large" (length of wire is large) or "Size of the driven gate (Gate B) is large" (which increase the Gate capacitance of GateB) or "Both should be true".
So for Minimizing Propagation Delay, A fast Gate/Cell is required, which is only possible by

1. Keeping the output capacitance CL small (it decreases the charging and discharging time). And for this
   - Minimize the area of drain pn junctions. (Decrease W)
   - Minimize Interconnect capacitance. (Decrease wire/net Length)
   - Avoid large fan-out. Means Minimize gate capacitance of Driven Cell. (Decrease W of Driven cell)
2. Decreasing the equivalent Resistance of the transistors
   - Decrease L (For a particular technology Node It's almost constant)
   - Increase W
     - But this increases pn junction area and hence CL.

So if we want to use the size of the transistor as one of the parameter to increase/decrease of the propagation/transition delay, then we should have understanding of the design and also it depends on the property of Driven Cell and Net length also.

Few last points:

1. **"Delay reduces with increase in input transition and constant load capacitance".**
2. **"Delay increases with increase in output capacitance and constant input transition"**
   - **Because on increasing the output capacitance – charging and discharging time will increase.**

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Wednesday, January 8, 2014

Effect of Threshold voltage: Static Timing Analysis (STA) Basic (Part-7c)

Part1	Part2	Part3a	Part3b	Part3c	Part4a	Part4b	Part4c	Part5a
Part5b	Part6a	Part6b	Part6c	Part7a	Part7b	Part7c	Part 8	

Static Timing analysis is divided into several parts:

- Part1 -> Timing Paths
- Part2 -> Time Borrowing
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- Part3b -> Basic Concept of Setup and Hold Violation
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- Part4a -> Delay - Timing Path Delay
- Part4b -> Delay - Interconnect Delay Models

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- Part6b -> Examples to calculate the Maximum Clock Frequency for different circuits.
- Part 6a -> How to solve Setup and Hold Violation (basic example)
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- Part 7c -> Methods for Increase/Decrease the Delay of Circuit (Effect of Threshold voltage On the Slew)
- Part 8 -> 10 ways to fix Setup and Hold Violation.

Effect of Threshold voltage on the propagation delay and transition delay:

If you will see the below equations – I am sure you can easily figure out how threshold voltage effect the cell delay. (Note: Below Resistance formula is with respect to NMOS. You can derive similar formula for PMOS also (Just replace subscript "n" with "p" ☺).

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Delay - "Wire Load

EDN:

$$R_n = \frac{1}{\beta_n (V_{DD} - V_{Tn})} = \frac{1}{k'_n \left(\frac{W}{L}\right)(V_{DD} - V_{Tn})}$$

where:

V_{DD} = Power Supply voltage applied to circuit

V_{Tn} = Threshold Voltage for Nmos

$k'_n = \mu_n C_{ox}$

R_n = On Resisitance of Nmos

From above equation we have following points

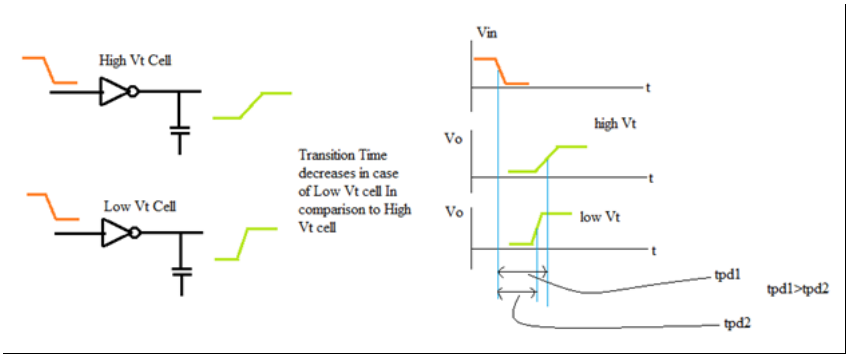
- On Resistance of MOS is inversely proportional to the "V_{DD}-V_{Tn}" (where V_{Tn} is Threshold Voltage).
- Decreasing the threshold voltage (LOW V_{Tn}) increases "V_{DD}-V_{Tn}" for constant V_{DD}.
- Increasing "V_{DD}-V_{Tn}" means decreasing "On Resistance" R_n.
- Decreasing R_n → RC decreases.
    - Means large Driving capability (Ability to source or sink current)
    - Decrease the time to charge the output load (capacitance) (Consists of source/drain capacitance of the driving gate, the routing capacitance of wire, and the gate capacitance of the driven gate) **
    - Means "Output Transition time of Gate A" and "Input Transition time for Gate B" decreases.
- Decreasing the transition time means decreases the propagation time.

So we can say that…

"Delay can be reduced by using low Vt cells, but the cost paid is high leakage power"

Direct effect is that low Vt cells are often more leaky i.e. leakage power increases.

If still you have any confusion below diagram should clarify your doubts.



I hope above diagram should clear your doubts about the effect of Threshold voltage on Delay.

In the next post we will summarize/list down all the methods of fixing the setup and hold violations.

Effect of Size of the Transistor On the Slew (Previous) | Index | 10 ways to fix Setup and Hold Violation (Next)

Posted by VLSI EXPERT at 2:07 PM

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4 comments:

Anonymous January 14, 2014 at 7:55 AM

Decreasing "VDD-VTn" means decreasing "On Resistance" Rn.
Doesnt decreasing this increase On resistance since its in the denominator?

Reply

Model" : Static Timing Analysis (STA) basic (Part 4c)

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

"Setup and Hold Time" : Static Timing Analysis (STA) basic (Part 3a)

"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

"Examples Of Setup and Hold time" : Static Timing Analysis (STA) basic (Part 3c)

"Time Borrowing" : Static Timing Analysis (STA) basic (Part 2)

Effect of Wire Length On the Slew: Static Timing Analysis (STA) Basic (Part-7a)

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Friday, January 10, 2014

10 Ways to fix SETUP and HOLD violation: Static Timing Analysis (STA) Basic (Part-8)

Part1	Part2	Part3a	Part3b	Part3c	Part4a	Part4b	Part4c	Part5a
Part5b	Part6a	Part6b	Part6c	Part7a	Part7b	Part7c	Part 8	

Static Timing analysis is divided into several parts:

- Part1 -> Timing Paths
- Part2 -> Time Borrowing
- Part3a -> Basic Concept Of Setup and Hold
- Part3b -> Basic Concept of Setup and Hold Violation
- Part3c -> Practical Examples for Setup and Hold Time / Violation
- Part4a -> Delay - Timing Path Delay
- Part4b -> Delay - Interconnect Delay Models
- Part4c -> Delay - Wire Load Model
- Part5a -> Maximum Clock Frequency
- Part5b -> Examples to calculate the "Maximum Clock Frequency" for different circuits.
- Part 6a -> How to solve Setup and Hold Violation (basic example)
- Part 6b -> Continue of How to solve Setup and Hold Violation (Advance examples)
- Part 6c -> Continue of How to solve Setup and Hold Violation (more advance examples)
- Part 7a -> Methods for Increase/Decrease the Delay of Circuit (Effect of Wire Length On the Slew)
- Part 7b -> Methods for Increase/Decrease the Delay of Circuit (Effect of Size of the Transistor On the Slew)
- Part 7c -> Methods for Increase/Decrease the Delay of Circuit (Effect of Threshold voltage On the Slew)
- Part 8 -> 10 ways to fix Setup and Hold Violation.

10 Ways to fix SETUP and HOLD violation:

Till now, We have discussed basic concepts of fixing the Setup and Hold violation which include

- Different formulas + explanation to identify the type of violation in design.
- How to fix those violations?
- Different methods of Increasing and Decreasing the Delay in the circuit to fix these type of violations?

And Now it's the time to list down different methods to fix these violations. I have also explained in brief each and every method, which also referring previous post for reference. One point to remember here that Fixing the Setup and Hold Violation are reverse in nature. All the methods which are applicable to fix one type of methods , hold true and can be apply to fix other type of if we will do the opposite thing. E.g - if setup can be fix by adding 1 buffer in some path then Hold can be fix by removing buffer in that path. (You will see these things below in the post)

In the last you will also find DOs and DON'Ts and recommended approach to fix these violations. These Recommendations helps designer in reducing iteration and fix the violations fast.

8 Ways To Fix Setup violation:

Setup violations are essentially where the data path is too slow compared to the clock speed at the capture

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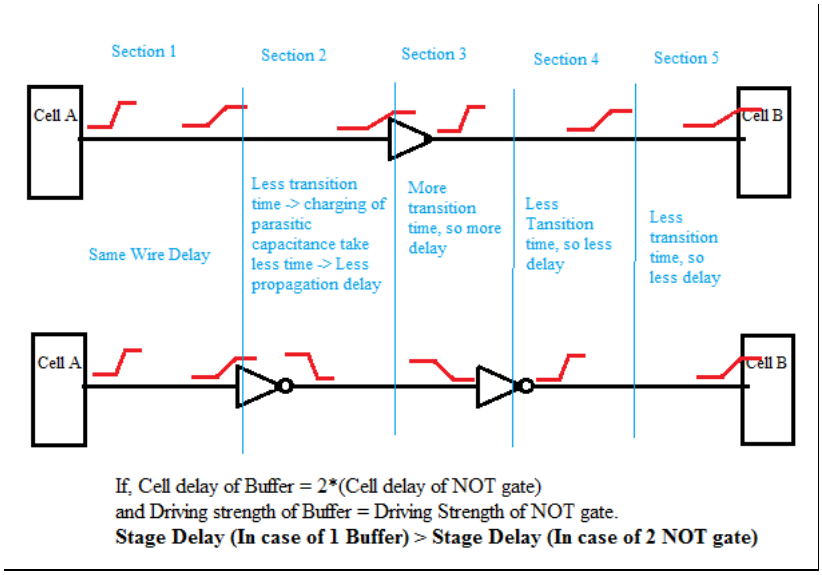
flip-flop. With that in mind there are several things a designer can do to fix the setup violations.

Method 1 : Reduce the amount of buffering in the path.

- It will reduce the cell delay but increase the wire delay. So if we can reduce more cell delay in comparison to wire delay, the effective stage delay decreases.

Method 2 : Replace buffers with 2 Inverters place farther apart

- Adding 2 inverters in place of 1 buffer, reducing the overall stage delay.
  - Adding inverter decreases the transition time 2 times then the existing buffer gate. Due to that, the RC delay of the wire (interconnect delay) decreases.
  - As such cell delay of 1 buffer gate ≈ cell delay of 2 Inverter gate
  - So stage delay (cell delay + wire delay) in case of single buffer < stage delay in case of 2 inverter in the same path.
  - You will get the clear understanding by following figure and you can refer the first post to understand how transition time varies across the wire.



Method 3 : HVT swap. Means change HVT cells into SVT/RVT or into LVT.

- Low Vt decrease the transition time and so propagation delay decreases.
- HVT/NVT/LVT type cells have same size and pin position. In both leakage current and speed, LVT>NVT>HVT. So replace HVT with NVT or LVT will speed up the timing without disturb layout.
- **Negative effect**: Leakage current/power also increases.

Method 4 : Increase Driver Size or say increase Driver strength (also known as upsize the cell)

- Explained the basic and details in the previous post
- **Note:** Normally larger cell has higher speed. But some special cell may have larger cell slower than normal cell. Check the technology library timing table to find out these special cells. Increasing driver is very commonly used in setup fix.
- **Negative effect:** Higher power consumption and more area used in the layout.
- I have notice one explanation in book <book name>. I am copying and pasting (not 100%) that here because I like that one. J Marked the important part by Bold.
  - The basic layout technique for reducing the gate delay consists in connecting MOS devices in parallel.
  - The equivalent width of the resulting MOS device is the sum of each elementary gate width. Both nMOS and pMOS devices are designed using parallel elementary devices.
  - Most cell libraries include so-called x1, x2, x4, x8 inverters.
  - The x1 inverter has the minimum size, and is targeted for low speed, low power operations.
  - **The x2 inverter uses two devices x1 inverters, in parallel. The resulting circuit is an inverter with twice the current capabilities. The output capacitance may be charge and discharged twice as fast as for the basic inverter (see below figure), because the Ron resistance of the MOS device is divided by two. The price to pay is a higher power consumption.**

Model" : Static Timing Analysis (STA) basic (Part 4c)

Delay - "Interconnect Delay Models" : Static Timing Analysis (STA) basic (Part 4b)

"Setup and Hold Time" : Static Timing Analysis (STA) basic (Part 3a)

"Setup and Hold Time Violation" : Static Timing Analysis (STA) basic (Part 3b)

"Examples Of Setup and Hold time" : Static Timing Analysis (STA) basic (Part 3c)

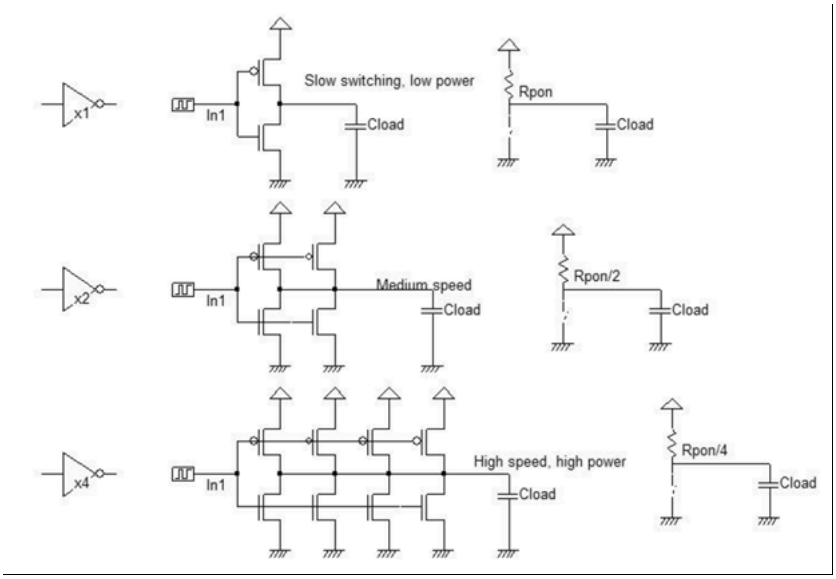
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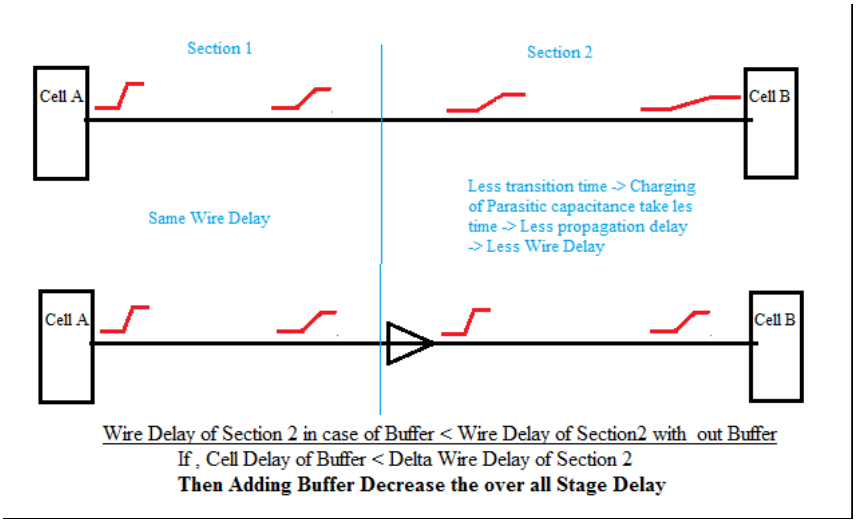
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- The equivalent Ron resistance of the x4 inverter is divided by four.
- The clock signals, bus, ports and long wires with severe time constraints use such high drive circuits.



Method 5 : Insert Buffers

- Some time we insert the buffer to decrease over all delay in case of log wire.
- Inserting buffer decreases the transition time, which decreases the wire delay.
- If, the amount of wire delay decreases due to decreasing of transition time > Cell delay of buffer, over all delay decreases.
- **Negative Effect**: Area will increase and increase in the power consumption.

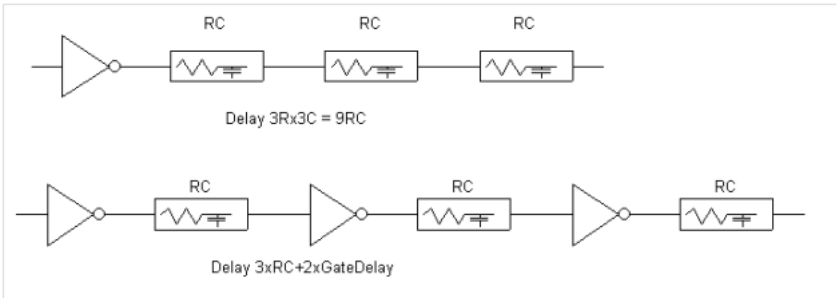


Method 6 : Inserting repeaters:

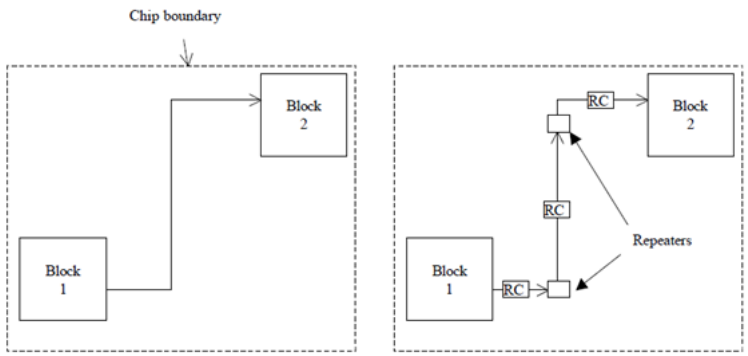
- Concepts of Repeaters are same as I have discussed in “Inserting the Buffer” (above point). Just I am trying to explain this in a different way but the over concept are same.
- Long distance routing means a huge RC loading due to a series of RC delays, as shown in figure. A good alternative is to use repeaters, by splitting the line into several pieces. **Why can this solution be better in terms of delay? Because the gate delay is quite small compared to the RC delay.**

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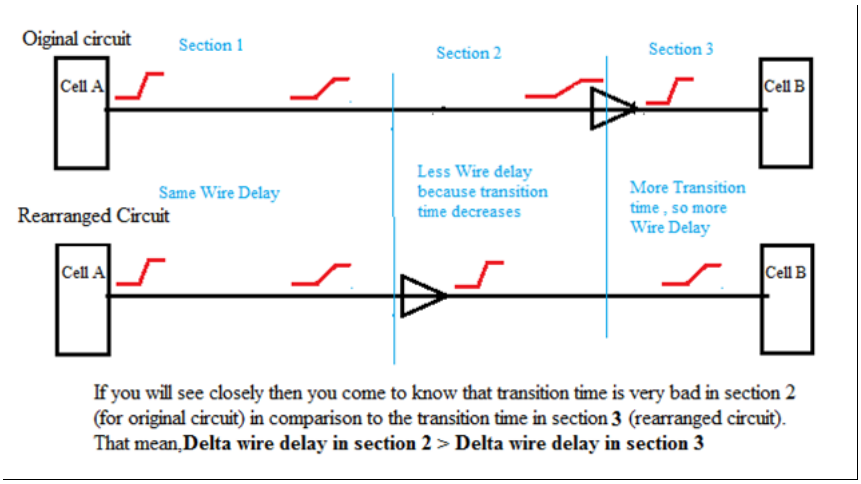
- In case of Interconnect driven by a single inverter, the propagation delay become
  - Tdelay= tgate+ nR.nC = tgate + n^2RC
- If two repeaters are inserted, the delay becomes:
  - Tdelay=tgate (delay of inverter) + 2tgate (delay of repeater) +3RC = 3tgate + 3RC
- So you can see how RC delay is impacting in case of non-repeater in the circuit.
- Consequently, if the gate delay is much smaller than the RC delay, repeaters improve the switching speed performances, at the price of higher power consumption.
- Below figure helps you to understand the practical use of this.



Inserting repeaters in long lines

Method 7 : Adjust cell position in layout.

- Let's assume there are 2 gate (GATE A and GATE B) separated by 1000um. There is another GATE C placed at the distance of 900um from GATE A.
- If we re-position the GATE C at 500um from GATE A (center of GATE A and B), overall delay between GATE A and B decreases.
- You will get the clear understanding by first post and the following diagram.
- **Note:** The placement in layout may prevent such movement. Always use layout viewer to check if there are any spare space to move the critical cell to an optimal location.



If you will see closely then you come to know that transition time is very bad in section 2 (for original circuit) in comparison to the transition time in section **3** (rearranged circuit). That mean,**Delta wire delay in section 2 > Delta wire delay in section 3**

Method 8 : Clock skew:

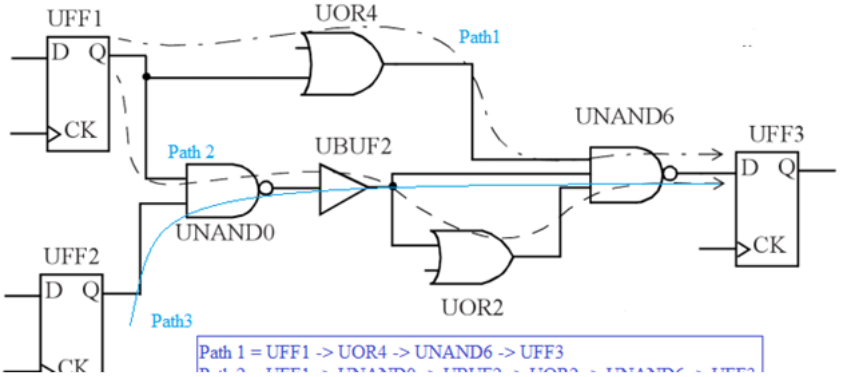
- By delaying the clock to the end point can relax the timing of the path, but you have to make sure the downstream paths are not critical paths.
- Related to clock skew basic – I will discuss that in SI section.

2 Ways to Fix Hold Violations:

Hold violation is the opposite of setup violation. Hold violation happen when data is too fast compared to the clock speed. For fixing the hold violation, delay should be increases in the data path.
Note: Hold violations is critical and on priority basis in comparison are not fixed before the chip is made, more there is nothing that can be done post fabrication to fix hold problems unlike setup violation where the clock speed can be reduced.
The designer needs to simply add more delay to the data path. This can be done by

Method 9 : By Adding delays.

- Adding buffer / Inverter pairs /delay cells to the data path helps to fix the hold violation.
- **Note:** The hold violation path may have its start point or end point in other setup violation paths. So we have to take extra care before adding the buffer/delay.
  - E.G. if the endpoint of hold violation path has setup violation with respect to some other path, insert the buffer/delay nearer to start point of hold violation path. Else the setup violation increases in other path.
  - if the start point of hold violation path has setup violation with respect to some other path, insert the buffer/delay nearer to end point of hold violation path. Else the setup violation increases in other path.
- I am sure you may be asking what is this and why?
- Below figure and explanation can help you to understand this.
- **From below figure, you can also conclude that don't add buffer/delay in the common segment of 2 paths (where one path has hold violation and other setup violation).**



Method 10 : Decreasing the size of certain cells in the data path.

- It is better to reduce the cells closer to the capture flip flop because there is less likely hood of affecting other paths and causing new errors.

Note: Following points are recommended while fixing setup and hold violations.

- Make modification to the data path only.
  - Adjusting register location or removing/adding buffers to the clock path will fix the violation that but it may cause more violations for some other paths which may not present before.
- First try to fix setup violation as much as possible. Then later on start fixing hold violation.
- In general, hold time will be fixed during back-end work (during PNR) while building clock tree. If u r a front-end designer, concentrate on fixing setup time violations rather than hold violations.
- Fix all the hold violation, if you have to choose between setup and hold.
  - If a chip is done with some setup violations it can work by reducing the frequency.
  - If a chip is done with hold violations, we have “JUST DUMP” the chip.

Previous Index Next

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"Examples Of Setup and Hold time" : Static Timing ...

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Posted by VLSI EXPERT at 5:17 PM

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30 comments:

Prashanth Anil Mascarenhas February 16, 2014 at 2:00 PM

Awesome Learning Experience. It is a very interesting blog to refresh the concepts on STA. Thank you:)

Reply

Anonymous March 7, 2014 at 2:24 AM

Kudos to you!! This is one great blog for learners like me! Thank you loads for all your effort! May god bless you!

Reply

Anonymous April 20, 2014 at 7:31 PM

really nice to learn...

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Anonymous June 5, 2014 at 2:51 PM

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Anonymous June 28, 2014 at 3:42 AM

thank you very much.
interesting.....

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Anonymous August 20, 2014 at 3:01 PM

The explanation is great. Thanks for sharing the post :)

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Anonymous August 24, 2014 at 6:05 AM

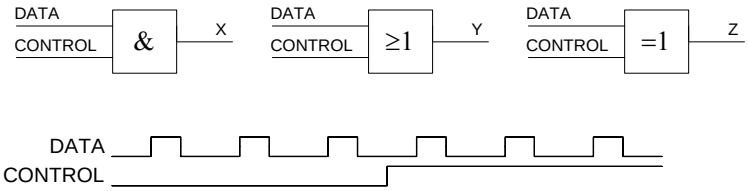
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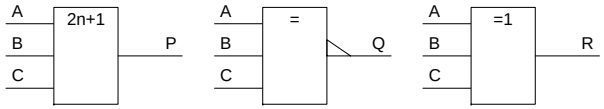
Problem Sheet 1

(Question ratings: A=Easy, …, E=Hard. All students should do questions rated A, B or C as a minimum)

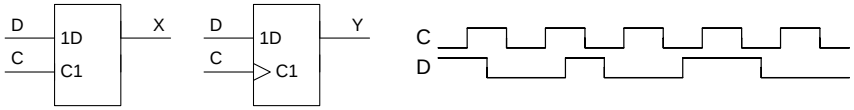
1A. The diagram shows three gates in which one input (CONTROL) is being used to modify a signal at the other input (DATA). Complete the timing diagram by drawing the waveforms of X, Y and Z. Describe in words the effect each of the gates has on DATA when CONTROL is low and when it is high.



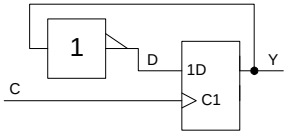
2B. The symbol in a gate generally indicates how many of the inputs need to be high to make the output high. Guess the truth tables of the following gates from their symbols. Explain why any one of them could be considered as a 3-input XOR gate.



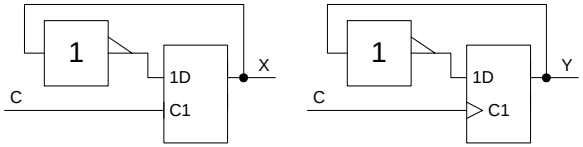
3A. The circuits below are a D-latch and a D-flipflop. Complete the timing diagram by drawing the waveforms of X and Y assuming that they are both low initially.



4B. The circuit below forms a ÷2 counter. If the inverter has a propagation delay of 5 ns and the propagation delay, setup time and hold time of the flipflop are 8 ns, 4 ns and 2 ns respectively, calculate the highest clock frequency for reliable operation.

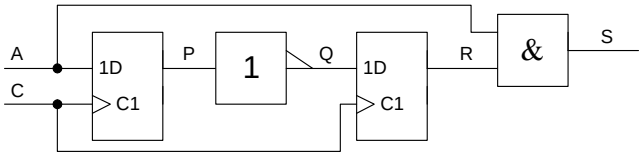


5B. The circuits below are a D-latch and a D-flipflop with their outputs connected to their inputs via an inverter. Draw the waveforms of X and Y assuming that they are both low initially and that C is a uniform square wave. (One of these circuits is a disaster and should never be used)



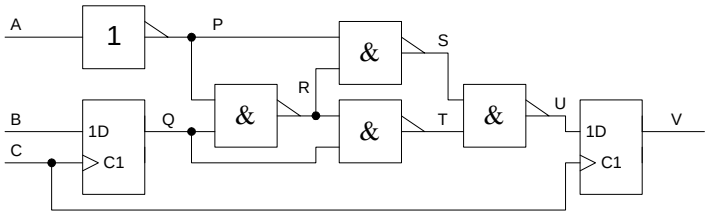
6B. In the circuit below the propagation delay of the flipflops may vary between 4 and 7 ns while the propagation delay of the gates may vary between 2 and 6 ns.

Calculate the minimum and the maximum propagation delays from each of A and C to each of P, Q and R and S.



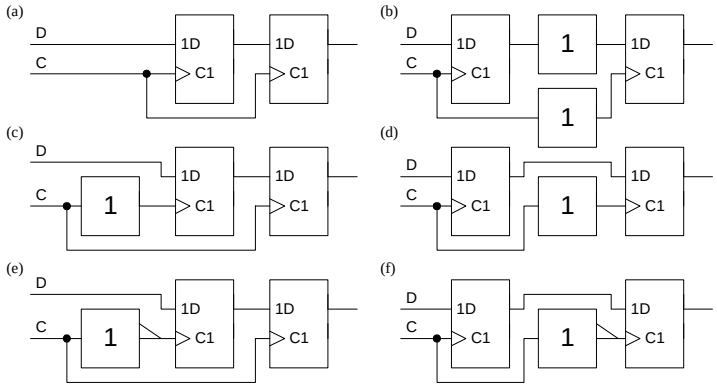
7C. In the circuit below the setup and hold times of the flipflops are 5 ns and 1 ns respectively. The propagation delay of the flipflops may vary between 4 and 7 ns while the propagation delay of the gates may vary between 2 and 6 ns.

Calculate the minimum and the maximum propagation delays between C and U. Hence calculate the maximum frequency of the clock, C.



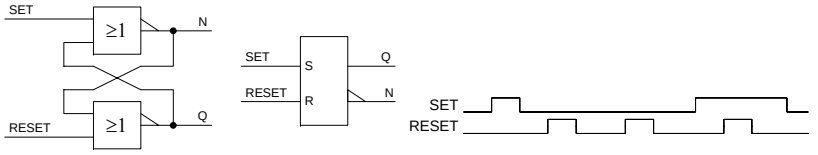
8C. In the six circuits below the setup and hold times of the flipflops are 5 ns and 1 ns respectively. The propagation delay of the flipflops may vary between 4 and 7 ns while the propagation delay of the gates may vary between 2 and 6 ns. The signal C is a symmetrical square wave.

Write down the setup and hold inequalities that relate to the ***second*** flipflop in each circuit. You should measure all times from the rising edge of CLOCK. Identify which of the circuits will not work reliably and determine the maximum clock frequency for each of the others.

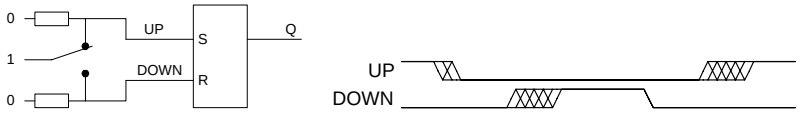


9B. The dual NOR-gate circuit shown below is called a Set-Reset latch and has the symbol shown at right. Complete the timing diagram by showing the waveforms of Q and N assuming that Q is initially low.

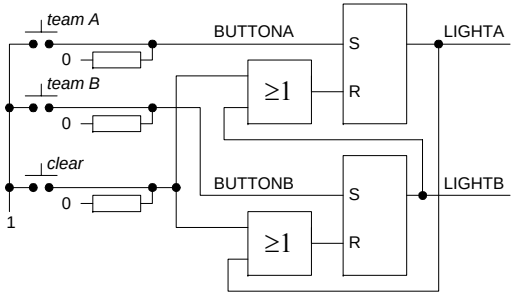
If SET and RESET are both high, say which one of these inputs dominates as far as Q is concerned and as far as N is concerned.



10A. The springing contacts in switches always bounce when they close and sometimes do so when they open as well. This contact bounce can last for several milliseconds. An SR-latch can be used to debounce switch signals in the following circuit. Complete the timing diagram by drawing the waveform of Q.



11C. The circuit shows a circuit to indicate who pressed their button first in a 2-contestant game show. Design a similar circuit for a 3-contestant game show. The SR-latches use the circuit from question 9.



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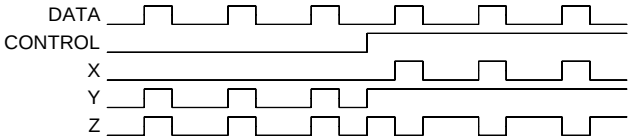
Solution Sheet 1

(Question ratings: A=Easy, …, E=Hard. All students should do questions rated A, B or C as a minimum)

1A. AND gate: 0 forces output low, 1 allows DATA through

OR gate: 0 allows DATA through, 1 forces output high

XOR gate: 0 allows DATA through, 1 inverts DATA



It is often useful to think of a gate like this: one input a signal, the others controlling it.

2B. P is high when an odd number of its inputs are high (an odd parity gate)

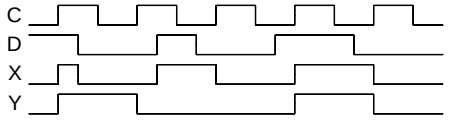
Q is low when all its inputs are the same

R is high when exactly one of its inputs is high

All of these properties are true of a 2-input XOR gate. Talking about a 3-input XOR gate (or larger) is ambiguous because no one can tell which of these three gates you mean.

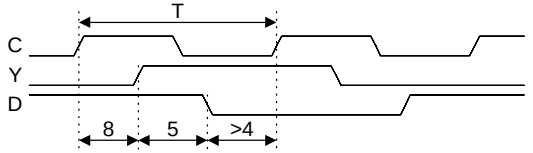
A	B	C	P	Q	R
0	0	0	0	0	0
0	0	1	1	1	1
0	1	0	1	1	1
0	1	1	0	1	0
1	0	0	1	1	1
1	0	1	0	1	0
1	1	0	0	1	0
1	1	1	1	0	0

3A. The latch output, X, follows D whenever C is high and freezes in its current state when C goes low. The flipflop output Y, only ever changes on the rising edge of C when it changes to the value that D has just prior to the edge.



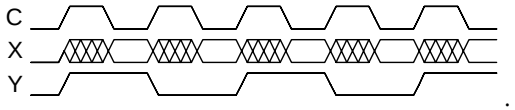
4B. From the diagram we obtain:

$$T-(5+8)>4 \;\Rightarrow\; T>17\,\text{ns} \;\Rightarrow\; f<58.8\,\text{MHz}$$



5B. Whenever C is high, the latch output, X, will follow its input: this means that we get a feedback loop containing an odd number of inverters. Such a loop will oscillate (indeed the oscillation frequency of such a loop is the standard way of measuring the propagation delay of a logic circuit). The only real use for this circuit is as a random number generator.

The flipflop circuit, Y, has no such problems because it only looks at its input for an instant and then effectively disconnects it until the next rising clock edge. It forms a ÷2 counter.



6B. This is a trick question: there is no propagation delay between A and any of P, Q or R since a transition in A does not directly cause any of these other signals to change. The min and max delays from A to S are 2 and 6 ns. Of course if R happens to be low, there is no propagation delay between A and S either.

The min and max delays from C to P, Q, R and S are 4 and 7, 6 and 13, 4 and 7, and 6 and 13 ns respectively. The important point to realise is that since a transition at Q does

not directly cause R to change, it follows that there is no delay path through both flipflops. The expression for a propagation delay **never** involves more than one flipflop delay.

7C. The shortest path from C to U passes through the flipflop and then through two gates: this gives a minimum propagation delay of 8 ns. This happens when A=1⇒P=0⇒R=S=1⇒U=!T=Q. We therefore use $2t_g$ in the hold inequality below.

The longest path from C to U passes through the flipflop and then through three gates: this gives a maximum propagation delay of 25 ns. This happens when A=0⇒P=1⇒R=!Q⇒T=1⇒U=!S=R=!Q. We therefore use $3t_g$ in the setup inequality below.

Setup: $t_p + 3t_g + t_s < T \Rightarrow T > 7 + 3\times 6 + 5 = 30\,\text{ns} \Rightarrow f < 33\,\text{MHz}$

Hold: $t_h < t_p + 2t_g \quad \Rightarrow \quad 1 < 4 + 2\times 2 = 8$ ✓

The hold inequality is always satisfied and the setup inequality gives a maximum clock frequency of 33 MHz.

8C. The setup inequality is given by

maximum delay to flipflop data input + setup time < minimum delay to flipflop clock ↑

Both delays must of course be measured from the same reference point: in this question, we are told to use the rising edge of C as our reference. We have to be a bit careful about which clock edge we are talking about. In parts (a), (b), (c) and (d) the first rising edge of C (at time 0) causes the output of the first flipflop to change and the **next** rising edge of C clocks the new data into the second flipflop. Thus, assuming the clock period to be *T*, the setup inequalities for these circuits are:

(a) $7+5 < T \Rightarrow T > 12 \Rightarrow f < 83$ MHz

(b) $7+6+5 < T+2 \Rightarrow T > 16 \Rightarrow f < 62.5$ MHz

(c) $6+7+5 < T \Rightarrow T > 18 \Rightarrow f < 55$ MHz

(d) $7+5 > T+2 \Rightarrow T > 10 \Rightarrow f < 100$ MHz

For part (e), the **falling** edge of C clocks the first flipflop and the following **rising** edge of C clocks the second one while for part (f) these rôles are reversed. This gives a ½*T* term on one side of the inequality and substantially slower clock speeds since the data must now reach the second flipflop in half a clock cycle rather than a whole one:

(e) $\frac{1}{2}T+6+7+5 < T \Rightarrow T > 36 \Rightarrow f < 28$ MHz

(f) $7+5 < \frac{1}{2}T+2 \Rightarrow T > 20 \Rightarrow f < 50$ MHz

The hold inequality is given by

maximum delay to to flipflop clock ↑ + hold time < minimum delay flipflop data input

This time though we are concerned with the clock edge that is meant to be clocking data into the second flipflop from the previous cycle. This means that for the normal shift register circuits of (a), (b), (c) and (d), the hold inequalities will not involve *T* at all:

(a) $0+1 < 4 \Rightarrow 1 < 4$ ☑

(b) $6+1 < 4+2 \Rightarrow 1 < 0$ ☒ ⇒ **Won't work**

(c) $0+1 < 2+4 \Rightarrow 1 < 6$ ☑

(d) $6+1 < 4 \Rightarrow 7 < 4$ ☒ ⇒ **Won't work**

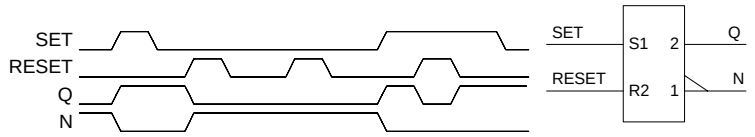
The moral is that if you clock both flipflops with the same clock edge you mustn't have any delay in the second flipflop's clock signal. Life is much easier with circuits (e) and (f) because the ½*T* that we lost from the setup equation reappears:

(e) $1 < \frac{1}{2}T+2+4 \Rightarrow \frac{1}{2}T > -5$ ☑

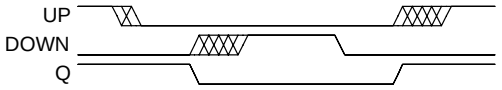
(f) $\frac{1}{2}T+6+1 < T+4 \Rightarrow \frac{1}{2}T > 3 \Rightarrow f < 167$ MHz (but also needs to be < 50 MHz above)

These circuits are used when transmitting information between circuit boards and in other situations where clock signal delays might arise.

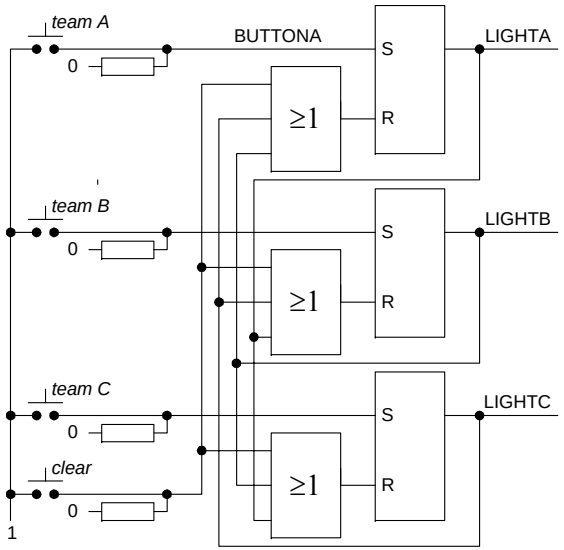
9B. If SET and RESET are both high then both outputs will be forced low. This means that SET wins as far as N is concerned and RESET wins as far as Q is concerned. This can be indicated in the logic symbol by labelling the inputs S1 and R2 and labelling each output with the identification number of the dominant input:



10A. Note that it is essential for the 2-way switch to be of the *break-before-make* variety to ensure that the latch inputs are never high simultaneously.



11C. The extension to an arbitrary number of contestants is easy: each latch must be held reset if any of the other contestants have their light on or if the CLEAR button is pressed. This circuit relies on the dominance of the RESET input referred to in question 9. In fact the OR gates that feed the reset inputs of the latches can be absorbed into the latch itself so we only need two gates per contestant.

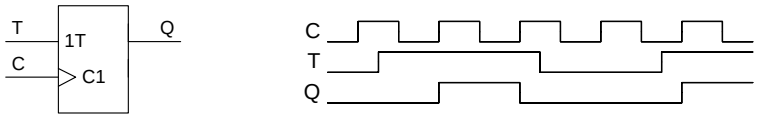


E2.11/ISE2.22 – Digital Electronics II

Problem Sheet 2

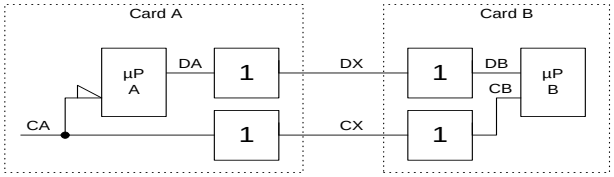
(Question ratings: A=Easy, …, E=Hard. All students should do questions rated A, B or C as a minimum)

1B. A toggle flipflop (T-flipflop) changes state whenever its T input is high on the CLOCK ↑ edge as shown in the timing diagram.

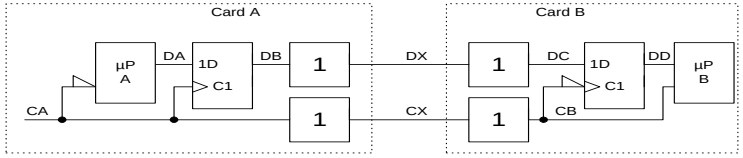


Show how a T-flipflop can be made by combining an XOR gate with a D-flipflop.

2C. A multi-processor system contains two microprocessors which are mounted on separate printed circuit cards. The clock and data signals pass through a line driver when they leave one card and a line receiver when they pass onto the next. The combined delay of the driver+receiver may vary between 13 ns and 22 ns. New data values appear at DA on the falling edge of CA with a propagation delay of 5 to 50 ns. Data is clocked into µP B on the rising edge of CB with a setup time of 12 ns and a hold time of 27 ns. If the clock, CA, is a symmetrical squarewave, calculate its maximum frequency.

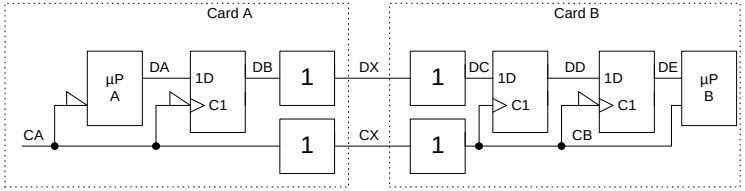


3C. We can speed up the circuit from the previous question by using high-speed flipflops with shorter propagation delays and setup times. The flipflops in the revised circuit have setup and hold times of 5 ns and 3 ns and propagation delays in the range 2 to 10 ns. Note that the second flipflop has an inverted clock. Calculate the new maximum clock frequency by considering its minimum period for each of µPA→flipflop, flipflop→flipflop and flipflop→µPB.



4C. By considering the Hold requirements, explain why the circuit in question 3 would not work if the two flipflops were interchanged.

5D. If we add a third flipflop, we can improve the speed further. Calculate the maximum clock frequency for the following circuit and explain in words how it has achieved the performance increase when compared with the original circuit of question 2.



6A. Explain why most memory integrated circuits have “tri-state” data output pins.

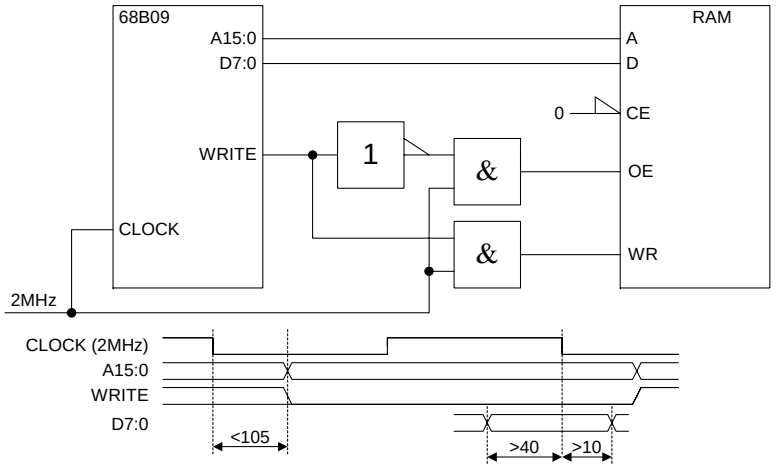
7B. In an 8-bit microprocessor system, addresses 0000 to 9FFF are occupied by RAM and addresses A000 to DFFF are occupied by ROM. The system also contains two peripheral devices: a serial port occupying addresses E100 to E107 and a parallel port occupying addresses E200 to E201. You have a supply of 8k×8 RAM integrated circuits and a supply of 16k×8 ROM integrated circuits.

a) State how many input address pins you would expect to find on each of the RAM integrated circuits, ecah of the ROM integrated circuits and on each of the peripheral device integrated circuits.

b) Derive Boolean expressions for the CE inputs of each memory and peripheral integrated circuit.

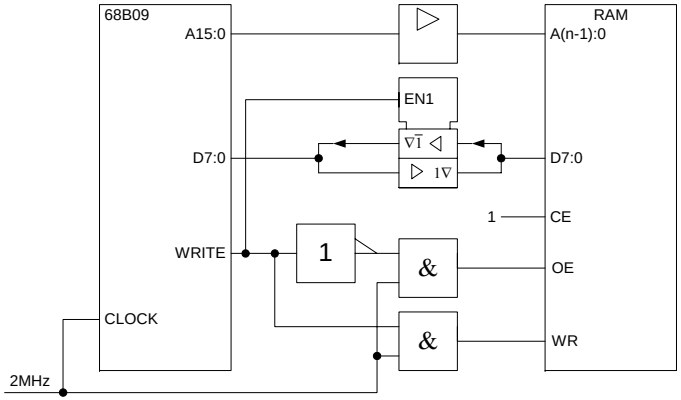
c) Say what is unusual about the byte ordering within the ROM.

8B. The diagram shows a Motorola 68B09 microprocessor connected to a memory circuit together with the timing diagram for a microprocessor read cycle..



Each logic gate has a propagation delay that may vary independently in the range 5 to 10 ns. Calculate the maximum permissible access times of the memory from (a) its address inputs, and (b) its OE input.

9C. The circuit of question 8 is altered by the introduction of buffers in the address lines and bi-directional buffers in the data lines.



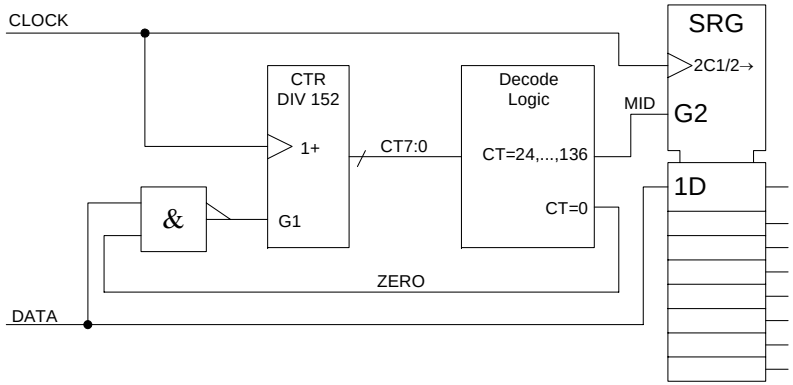
The address line buffers have a propagation delay of <18 ns while the data line buffers have a propagation delay of <12 ns, an enable time of <40 ns and a disable time of <25ns. The enable time applies when an output changes from a high impedance state to a driven state, the disable time applies when an output changes from a driven state to a high impedance state.

Calculate the new values of the maximum permissible access times of the memory from (a) its address inputs, and (b) its OE input.

10A. Explain why a bi-directional buffer is normally designed to have a longer enable time than disable time.

11D. Buffers for address and data lines can be made faster if they have inverted outputs. Say how the operation of a microprocessor is affected if the address and data lines pass through inverting buffers between the microprocessor and (a) read-write RAM memory and (b) read-only ROM memory.

12B. Asynchronous bit serial data consisting of a start bit (logical 0), 8 data bits and a stop bit (logical 1) is received by the circuit below. The CLOCK frequency is 16× the transmission bit rate. Give a Boolean expression for the signal MID; simplify it where possible.



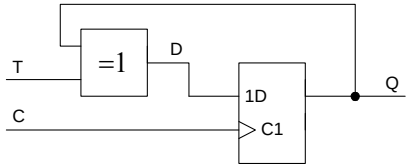
13C. In question 12, the CLOCK frequency is changed to 4× the transmission bit rate. Determine the appropriate counter division ratio and the counter values for which MID should now be high. Determine the clock accuracy required by the transmitter and receiver to ensure that the data is received correctly.

E2.11/ISE2.22 – Digital Electronics II

Solution Sheet 2

(Question ratings: A=Easy, …, E=Hard. All students should do questions rated A, B or C as a minimum)

1B. As seen in problem sheet 1, an XOR gate can be used to invert a signal or pass it through unchanged according to whether a control input is high or low.



2C. We define t=0 as the falling edge of CA.

Setup requirement: $\max(DB\uparrow\downarrow)+12 < \min(CB\uparrow)$
$50+22+12 < (13 + \frac{1}{2}T)$
$\frac{1}{2}T > 71 \Rightarrow f < 7$ MHz

Hold requirement: $\max(CB\uparrow) + 27 > \min(T + DB\uparrow\downarrow)$
$\frac{1}{2}T+22 + 27 > T + 5+13$
$\frac{1}{2}T > 31$ (less severe restriction than above)

Note the extra T term in the hold requirement: this is because we want the *second* transition of DB to occur >27 ns after CB↑. The Hold requirement is so easily satisfied that it wouldn't normally be necessary to calculate it exactly.

3C.

µPA → flipflop CA↓=0	Setup:	$\max(DA\uparrow\downarrow) +5 < \min(CA\uparrow)$ $50+5 < \frac{1}{2}T$ $\frac{1}{2}T > 55 \Rightarrow$ **f < 9 MHz**	Hold:	$\max(CA\uparrow)+3< \min(T+DA\uparrow\downarrow)$ $\frac{1}{2}T+3< T + 5$ $\frac{1}{2}T > -2$ ☑
flipflop → flipflop CA↑=0	Setup:	$\max(DB\uparrow\downarrow)+5 < \min(CB\downarrow)$ $10+22+5 < \frac{1}{2}T+13$ $\frac{1}{2}T > 24 \Rightarrow f < 21$ MHz	Hold:	$\max(CB\downarrow)+3< \min(T+DB\uparrow\downarrow)$ $\frac{1}{2}T+22+3< T +2+13$ $\frac{1}{2}T > 10 \Rightarrow f < 50$ MHz
flipflop → µPB CB↓=0	Setup:	$\max(FB\uparrow\downarrow)+12<\min(CB\uparrow)$ $10+12<\frac{1}{2}T$ $\frac{1}{2}T > 22 \Rightarrow f < 23$ MHz	Hold:	$\max(CB\uparrow)+27<\min(T+FB\uparrow\downarrow)$ $\frac{1}{2}T+27<T + 2$ $\frac{1}{2}T > 25 \Rightarrow f < 20$ MHz

It can be seen that the critical figure is the setup time for the first flipflop: this is because the microprocessor takes such a long time (up to 50 ns) to output its data. Question 4 (which doesn't work) and question 5 (which does) show how to relax this constraint. In the third row of the previous table, I have cancelled out the delay of the clock line driver/receiver from the two sides of the inequality. This is only valid if we can assume that the propagation delays for rising and falling edges are the same (not generally true).

4C. The first flipflop now responds to a falling clock edge: this means that µPA now has a full clock cycle to output its data rather than only a half cycle. We have therefore doubled maximum clock frequency of the circuit. (Note that the middle row of this table is unchanged from the previous question).

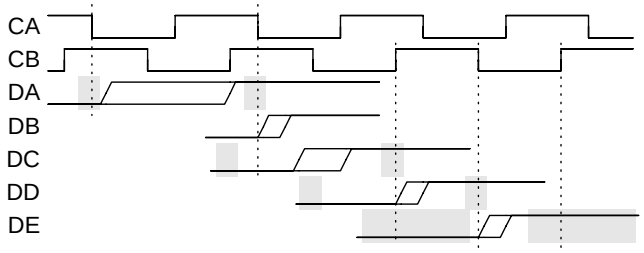
The problem is that the output from the second flipflop now changes on the rising clock edge and therefore fails to meet the hold time of µPB.

µPA → flipflop CA↓=0	Setup:	$\max(DA\uparrow\downarrow)+5<\min(T+CA\downarrow)$ $50+5<T$ $T > 55 \Rightarrow$ **f < 18 MHz**	Hold:	$\max(CA\downarrow)+3<\min(DA\uparrow\downarrow)$ $0+3<5$ ☑
flipflop → flipflop CA↓=0	Setup:	$\max(DC\uparrow\downarrow)+5<\min(CB\uparrow)$ $10+22+5 < \frac{1}{2}T+13$ $\frac{1}{2}T > 24 \Rightarrow f < 21$ MHz	Hold:	$\max(CB\uparrow)+3<\min(T+DC\uparrow\downarrow)$ $\frac{1}{2}T+22+3<T + 2+13$ $\frac{1}{2}T > 10 \Rightarrow f < 50$ MHz
flipflop → µPB CB↑=0	Setup:	$\max(DD\uparrow\downarrow)+12<\min(T+CB\uparrow)$ $10+12<T$ $T > 22 \Rightarrow f < 46$ MHz	Hold:	$\max(CB\uparrow)+27<\min(DD\uparrow\downarrow)$ **2 > 27** ☒

5D. We can fix the hold problem by adding a third flipflop. The last row of the previous table is now replaced by the two rows below and the maximum frequency is now 18 MHz.

flipflop → flipflop CB↑=0	Setup:	$\max(DD\uparrow\downarrow)+5<\min(CB\downarrow)$ $10+5<\frac{1}{2}T$ $\frac{1}{2}T > 15 \Rightarrow f < 33$ MHz	Hold:	$\max(CB\downarrow)+3<\min(T+DD\uparrow\downarrow)$ $\frac{1}{2}T+3<T + 2$ $\frac{1}{2}T > 1 \Rightarrow f < 500$ MHz
flipflop → µPB CB↓=0	Setup:	$\max(DE\uparrow\downarrow)+12<\min(CB\uparrow)$ $10+12<\frac{1}{2}T$ $\frac{1}{2}T > 22 \Rightarrow f < 23$ MHz	Hold:	$\max(CB\uparrow)+27<\min(T+DE\uparrow\downarrow)$ $\frac{1}{2}T+27<T + 2$ $\frac{1}{2}T > 25 \Rightarrow f < 20$ MHz

The timing of this circuit with a clock period of about 60 ns (16.7 MHz) is shown below with setup/hold windows shaded:



6A. Two reasons. Firstly many memories, though not all, use the same pins for data input as for data output: the outputs must therefore be turned off (or "tristated") to allow new data to be written into a memory location. Secondly, a large memory system contains several memory integrated circuits which are enabled one at a time according to the address range selected (as in question 7 below). The use of tri-state outputs allows all memory data lines to be connected together without the need for an external multiplexer to switch between them.

7B. a) RAM has 13 address inputs, ROM has 14, Serial Port has 3 and Parallel Port has 1.

b) We need 5 RAM chips and one ROM, serial and parallel chips. The CE inputs are given in the following table:

Chip	Address Range	CE
RAM	0000 – 1FFF	$\overline{A15}\cdot\overline{A14}\cdot\overline{A13}$
	2000 – 3FFF	$\overline{A15}\cdot\overline{A14}\cdot A13$
	4000 – 5FFF	$\overline{A15}\cdot A14\cdot\overline{A13}$
	6000 – 7FFF	$\overline{A15}\cdot A14\cdot A13$
	8000 – 9FFF	$A15\cdot\overline{A14}\cdot\overline{A13}$
ROM	A000 – DFFF	$A15\cdot\left(\overline{A14}\cdot A13+A14\cdot\overline{A13}\right)$
Serial	E100 – E107	$A15\cdot A14\cdot A13\cdot A8$
Parallel	E200 – E201	$A15\cdot A14\cdot A13\cdot A9$

Note that I have not included all the address lines needed to fully decode the Serial and Parallel port adress ranges. The microprocessor should never access the undefined memory locations in the range E000 to FFFF so it does not matter if the peripheral ports respond to several of them. As defined above, the following addresses will all refer to the serial port's lowest location: E100, E108, E110, E118, …, FFF8. Of the 16 address lines, 3 are direct inputs to the serial port, 4 are used in forming its CE and 9 are unused. The 9 unused lines can take on 512 possible values and so the serial port will appear 512 times in the memory map. If a PAL is being used to generate the CE signals, then the number of PAL inputs required may be reduced by not decoding peripheral address ranges fully.

c) The bytes are in the wrong order in the ROM. The ROM has 14 address inputs, namely A13:0. Addresses A000 to BFFF have A13=1 and will therefore be mapped to the second half of the ROM; addresses C000 to DFFF have A13=0 and will be mapped to the first half of the ROM. This situation could be corrected by inverting A13 before connecting it to the ROM but this would add delay: a neater solution is to use A14 as the most significant address bit rather than A13.

8B. Note that only the setup time matters for this question. Let A be the access time (or propagation delay) from the address inputs and E be the access time from the OE input. The clock period is 500 ns. P is the propagation delay of a gate (i.e. 5 to 10 ns)

$$\max(105+A)+40<500 \Rightarrow \underline{\mathbf{A < 355\ ns}}$$
$$\max(P+E)+40<250 \Rightarrow \underline{\mathbf{E < 200\ ns}} \text{ (taking P=10, its maximum value)}$$

9C. We now have three possible paths to consider:

A15:0→Mem→D7:0	$\max(105+18+A+12)+40<500 \Rightarrow \underline{\mathbf{A < 325\ ns}}$
CLOCK↑→Mem:OE→D7:0	$\max(P+E+12)+40<250 \Rightarrow \underline{\mathbf{E < 188\ ns}}$
WRITE→Buff:EN1→D7:0	$\max(105+40)+40<500 \Rightarrow \underline{\mathbf{185>500}}$ ☑

Note that in the symbol for the bidirectional buffer: the ∇ denotes tristate outputs: the output marked with a 1 is enabled when EN1 is high while the output marked with a $\overline{1}$ is enabled when EN1 is low. The ∇ always goes immediately next to the output pin. The symbol ▷ denotes a buffer: a gate with a higher than normal output current capability. Any signals that flow right to left instead of the more normal left to right must be marked with a ←.

10A. If the same wire is driven high by the output of one device and low by that of another, a high current will flow which will waste power and which may even damage the integrated circuits involved. To reduce the possibility of two devices trying to drive the same wire simultaneously, tristate outputs are almost always designed to turn off more quickly than they turn on.

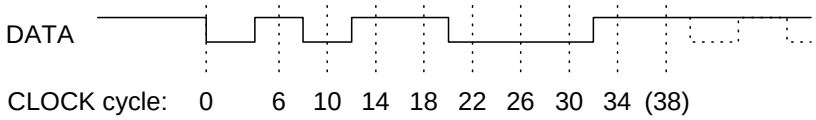
11D. The effect of inverting the address lines is to subtract them from FFFF. Thus what were memory locations 0000, 0001 and 0002 now become FFFF, FFFE and FFFD. Providing the chip enables are generated correctly, this reversal does not matter at all for a RAM: as long as each distinct address refers to a unique memory location the microprocessor does not care where it is inside the chip. For a ROM, the contents must be preprogrammed in the correct locations: thus the program would need to be stored backwards.

12B. The counter counts up from 0 to 151.

MID=CT3•!CT2•!CT1•!CT0 will get all odd multiples of 8, i.e. 8, 24, …, 136.

To eliminate the first of these we make MID=(CT7+CT6+CT5+CT4)•CT3•!CT2•!CT1•!CT0

13C. The counter should now divide by 38:



The eighth bit will now be clocked in at the end of the clock cycle in which CT5:0 equals 34. This is at time 34P+t from the beginning of the START bit where 0<t<P and P is the receiver clock period. This time must be in the range 8T to 9T where T is the duration of a bit cell. Thus

$$8T<34P+0 \Rightarrow T/P<4.25$$
$$9T>34P+P \Rightarrow T/P>3.89$$

If we assume that T and P have nominal values of T_0 and P_0 with a fractional tolerance of $\pm x$, we have

$$T_0(1-x) < T < T_0(1+x) \quad \text{and} \quad P_0(1-x) < P < P_0(1+x)$$

By considering the maximum and minimum possible values of the ratio T/P, we get:

$$\frac{T_0(1+x)}{P_0(1-x)} = 4.25 \quad \text{and} \quad \frac{T_0(1-x)}{P_0(1+x)} = 3.89$$

From which

$$\frac{(1+x)^2}{(1-x)^2} = \frac{4.25}{3.89} \quad \Rightarrow \quad (1+x) = 1.045(1-x) \quad \Rightarrow \quad x = \frac{0.045}{2.045} = 2.2\%$$

Substituting this value for x into the original equations gives $T_0/P_0 = 4.07$

Hence both the transmit and receive clocks must have a ratio of 4.07 and a tolerance of ±2.2%.This requirement is slightly more stringent than for the more usual 16× clock.

Companywise ASIC/VLSI Interview Questions

Below interview questions are contributed by ASIC_diehard (Thanks a lot !). Below questions are asked for senior position in Physical Design domain. The questions ar also related to Static Timing Analysis and Synthesis. Answers to some questions are given as link. Remaining questions will be answered in coming blogs.

Common introductory questions every interviewer asks are:

- Discuss about the projects worked in the previous company.
- What are physical design flows, various activities you are involved?
- Design complexity, capacity, frequency, process technologies, block size you handled.

Intel

- **Why power stripes routed in the top metal layers?**

The resistivity of top metal layers are less and hence less IR drop is seen in power distribution network. If power stripes are routed in lower metal layers this will use goo amount of lower routing resources and therefore it can create routing congestion.

- **Why do you use alternate routing approach HVH/VHV (Horizontal-Vertical-Horizontal/ Vertical-Horizontal-Vertical)?**

Answer:

This approach allows routability of the design and better usage of routing resources.

- **What are several factors to improve propagation delay of standard cell?**

Answer:

Improve the input transition to the cell under consideration by up sizing the driver.

Reduce the load seen by the cell under consideration, either by placement refinemei buffering.

If allowed increase the drive strength or replace with LVT (low threshold voltage) cell.

- How do you compute net delay (interconnect delay) / decode RC values present in tech file?
- **What are various ways of timing optimization in synthesis tools?**

Answer:

Logic optimization: buffer sizing, cell sizing, level adjustment, dummy buffering etc.

Less number of logics between Flip Flops speedup the design.

Optimize drive strength of the cell , so it is capable of driving more load and hence reducing the cell delay.

Better selection of design ware component (select timing optimized design ware components).

Use LVT (Low threshold voltage) and SVT (standard threshold voltage) cells if allowe

- **What would you do in order to not use certain cells from the library?**

Answer:

Set don't use attribute on those library cells.

- **How delays are characterized using WLM (Wire Load Model)?**

Answer:

For a given wireload model the delay are estimated based on the number of fanout c cell driving the net.

Fanout vs net length is tabulated in WLMs.

Values of unit resistance R and unit capacitance C are given in technology file.

Net length varies based on the fanout number.

Once the net length is known delay can be calculated; Sometimes it is again tabulate

- **What are various techniques to resolve congestion/noise?**

Answer:

Routing and placement congestion all depend upon the connectivity in the netlist , a better floor plan can reduce the congestion.

Noise can be reduced by optimizing the overlap of nets in the design.

- **Let's say there enough routing resources available, timing is fine, can you increase clock buffers in clock network? If so will there be any impact on other parameters?**

Answer:

No. You should not increase clock buffers in the clock network. Increase in clock buff cause more area , more power. When everything is fine why you want to touch clock tree??

- **How do you optimize skew/insertion delays in CTS (Clock Tree Synthesis)?**

Answer:

Better skew targets and insertion delay values provided while building the clocks.

Choose appropriate tree structure – either based on clock buffers or clock inverters c mix of clock buffers or clock inverters.

For multi clock domain, group the clocks while building the clock tree so that skew is balanced across the clocks. (Inter clock skew analysis).

- **What are pros/cons of latch/FF (Flip Flop)?**

Answer: Pros and cons of latch and flip flop

- How you go about fixing timing violations for latch- latch paths?
- As an engineer, let's say your manager comes to you and asks for next project die size estimation/projection, giving data on RTL size, performance requirements. How do you go about the figuring out and come up with die size considering physical aspects?
- How will you design inserting voltage island scheme between macro pins crossing core and are at different power wells? What is the optimal resource solution?
- What are various formal verification issues you faced and how did you resolve?
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- **What are effects of metastability?**

Answer: Metastability

- Consider a timing path crossing from fast clock domain to slow clock domain. How do you design synchronizer circuit without knowing the source clock frequency?
- How to solve cross clock timing path?
- **How to determine the depth of FIFO/ size of the FIFO?**

Answer: FIFO Depth

STmicroelectronics

- What are the challenges you faced in place and route, FV (Formal Verification), ECO (Engineering Change Order) areas?
- How long the design cycle for your designs?
- What part are your areas of interest in physical design?
- Explain ECO (Engineering Change Order) methodology.
- **Explain CTS (Clock Tree Synthesis) flow.**

Answer: Clock Tree Synthesis

- What kind of routing issues you faced?
- **How does STA (Static Timing Analysis) in OCV (On Chip Variation) conditions done? How do you set OCV (On Chip Variation) in IC compiler? How is timing correlation done before and after place and route?**

Answer: Process-Voltage-Temperature (PVT) Variations and Static Timing Analysis (S

- If there are too many pins of the logic cells in one place within core, what kind of issues would you face and how will you resolve?

- Define hash/ @array in perl.
- Using TCL (Tool Command Language, Tickle) how do you set variables?
- What is ICC (IC Compiler) command for setting derate factor/ command to perform physical synthesis?
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- If hold violation exists in design, is it OK to sign off design? If not, why?

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- How are timing constraints developed?
- Explain timing closure flow/methodology/issues/fixes.
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- Explain ECO (Engineering Change Order) implementation flow. Given post routed database and functional fixes, how will you take it to implement ECO (Engineering Change Order) and what physical and functional checks you need to perform?

Qualcomm

- In building the timing constraints, do you need to constrain all IO (Input-Output) ports?
- Can a single port have multi-clocked? How do you set delays for such ports?
- How is scan DEF (Design Exchange Format) generated?
- What is purpose of lockup latch in scan chain?
- **Explain short circuit current.**

Answer: Short Circuit Power

- **What are pros/cons of using low Vt, high Vt cells?**

Answer:

Multi Threshold Voltage Technique

Issues With Multi Height Cell Placement in Multi Vt Flow

- **How do you set inter clock uncertainty?**

Answer:

set_clock_uncertainty –from clock1 -to clock2

- In DC (Design Compiler), how do you constrain clocks, IO (Input-Output) ports, maxcap, max tran?
- **What are differences in clock constraints from pre CTS (Clock Tree Synthesis) to post CTS (Clock Tree Synthesis)?**

Answer:

Difference in clock uncertainty values; Clocks are propagated in post CTS.

In post CTS clock latency constraint is modified to model clock jitter.

- **How is clock gating done?**

Answer: Clock Gating

- **What constraints you add in CTS (Clock Tree Synthesis) for clock gates?**

Answer:

Make the clock gating cells as through pins.

- **What is trade off between dynamic power (current) and leakage power (current)?**

Answer:

Leakage Power Trends

Dynamic Power

- **How do you reduce standby (leakage) power?**

Answer: Low Power Design Techniques

- Explain top level pin placement flow? What are parameters to decide?
- Given block level netlists, timing constraints, libraries, macro LEFs (Layout Exchange Format/Library Exchange Format), how will you start floor planning?
- With net length of 1000um how will you compute RC values, using equations/tech file info?
- What do noise reports represent?
- What does glitch reports contain?
- What are CTS (Clock Tree Synthesis) steps in IC compiler?
- What do clock constraints file contain?
- How to analyze clock tree reports?
- What do IR drop Voltagestorm reports represent?
- Where /when do you use DCAP (Decoupling Capacitor) cells?
- **What are various power reduction techniques?**

Answer: Low Power Design Techniques

Hughes Networks

- What is setup/hold? What are setup and hold time impacts on timing? How will you fix setup and hold violations?
- Explain function of Muxed FF (Multiplexed Flip Flop) /scan FF (Scal Flip Flop).
- What are tested in DFT (Design for Testability)?
- In equivalence checking, how do you handle scanen signal?
- In terms of CMOS (Complimentary Metal Oxide Semiconductor), explain physical parameters that affect the propagation delay?
- **What are power dissipation components? How do you reduce them?**

Answer:

Short Circuit Power

Leakage Power Trends

Dynamic Power

Low Power Design Techniques

- **How delay affected by PVT (Process-Voltage-Temperature)?**

Answer: Process-Voltage-Temperature (PVT) Variations and Static Timing Analysis (S

- Why is power signal routed in top metal layers?

Avago Technologies (former HP group)

- How do you minimize clock skew/ balance clock tree?
- Given 11 minterms and asked to derive the logic function.
- Given C1= 10pf, C2=1pf connected in series with a switch in between, at t=0 switch is open and one end having 5v and other end zero voltage; compute the voltage across C2 when the switch is closed?
- Explain the modes of operation of CMOS (Complimentary Metal Oxide Semiconductor) inverter? Show IO (Input-Output) characteristics curve.
- Implement a ring oscillator.
- How to slow down ring oscillator?

Hynix Semiconductor

- How do you optimize power at various stages in the physical design flow?
- What timing optimization strategies you employ in pre-layout /post-layout stages?
- What are process technology challenges in physical design?
- Design divide by 2, divide by 3, and divide by 1.5 counters. Draw timing diagrams.
- What are multi-cycle paths, false paths? How to resolve multi-cycle and false paths?
- Given a flop to flop path with combo delay in between and output of the second flop fed back to combo logic. Which path is fastest path to have hold violation and how will you resolve?
- What are RTL (Register Transfer Level) coding styles to adapt to yield optimal backend design?
- Draw timing diagrams to represent the propagation delay, set up, hold, recovery, removal, minimum pulse width.

About Contributor

ASIC_diehard has more than 5 years of experience in physical design, timing, netlist GDS flows of Integrated Circuit development. ASIC_diehard's fields of interest are backend design, place and route, timing closure, process technologies.

Readers are encouraged to discuss answers to these questions. Just click on the 'pc comment' option below and put your comments there. Alternatively you can send yo answers/discussions to my mail id: shavakmm@gmail.com

Tags: ASIC, Physical Design, Static Timing Analysis (STA), Synthesis, Timing Analysis, VLSI

5 comments:

Anonymous November 4, 2008 at 11:02 PM

1. You might want to include things like what are the different ways of fixing an violations
-- there are about 4 methods

2. Why non-leaf clock nets are routed on top-most layers

3. Does jitter effect setup/hold paths?

Reply

valladolid43 October 2, 2010 at 7:43 PM

Hi

Tks very much for post:

I like it and hope that you continue posting.

Let me show other source that may be good for community.

Source: Interview questions and answers

Best rgs
David

Reply

Raj August 20, 2013 at 4:53 PM

Hey, great post man...Thanks for sharing! I also found an article for Qualcomm, Embedded Systems and other electronics companies here..... Electronics comp interview questions and tips

Hope this helps!

Reply

Replies

Anonymous September 6, 2013 at 1:15 PM

cant open the doc..

Thanks.

Reply

Anonymous May 31, 2015 at 10:19 AM

Hi sir...Can you post the question for AsicnTechnology, Banglore,,,

Reply

Enter your comment...

Comment as: Google Accour

Publish Preview

Your Comments... (comments are moderated)

Newer Post Home Older

Subscribe to: Post Comments (Atom)

About Me

Murali

M.Sc in Electronics;M.S in VLSI System Design;worked 3 years as design engineer in embedded system domain; experience of PIC and 8051 based microcontroller applications;working as VLSI physical design engineer; photography,travel and literature are my hobbies.

View my complete profile

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Thank You for reading this blog !

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Backend physical design Interview Questions

I have listed below a set of common interview questions asked mainly in interviews related to physical design or backend activities in ASIC or VLSI chip design process. Typically these interviews start with questions on physical design(PD) flow and goes on to deeper details.

* What is signal integrity? How it affects Timing?

* What is IR drop? How to avoid .how it affects timing?

* What is EM and it effects?

* What is floor plan and power plan?

* What are types of routing?

* What is a grid .why we need and different types of grids?

* What is core and how u will decide w/h ratio for core?

* What is effective utilization and chip utilization?

* What is latency? Give the types?

* What is LEF?

* What is DEF?

* What are the steps involved in designing an optimal pad ring?

* What are the steps that you have done in the design flow?

* What are the issues in floor plan?

* How can you estimate area of block?

* How much aspect ratio should be kept (or have you kept) and what is the utilization?

* How to calculate core ring and stripe widths?

* What if hot spot found in some area of block? How you tackle this?

* After adding stripes also if you have hot spot what to do?

* What is threshold voltage? How it affect timing?

* What is content of lib, lef, sdc?

* What is meant my 9 track, 12 track standard cells?

* What is scan chain? What if scan chain not detached and reordered? Is it compulsory?

* What is setup and hold? Why there are ? What if setup and hold violates?

* In a circuit, for reg to reg path ...Tclktoq is 50 ps, Tcombo 50ps, Tsetup 50ps, tskew is 100ps. Then what is the maximum operating frequency?

* How R and C values are affecting time?

* How ohm (R), fared (C) is related to second (T)?

* What is transition? What if transition time is more?

* What is difference between normal buffer and clock buffer?

* What is antenna effect? How it is avoided?

* What is ESD?

* What is cross talk? How can you avoid?

* How double spacing will avoid cross talk?

* What is difference between HFN synthesis and CTS?

* What is hold problem? How can you avoid it?

* For an iteration we have 0.5ns of insertion delay and 0.1 skew and for other iteration 0.29ns insertion delay and 0.25 skew for the same circuit then which one you will select? Why?

* What is partial floor plan?

* What parameters (or aspects) differentiate Chip Design & Block level design??

* How do you place macros in a full chip design?

* Differentiate between a Hierarchical Design and flat design?

* Which is more complicated when u have a 48 MHz and 500 MHz clock design?

* Name few tools which you used for physical verification?

* What are the input files will you give for primetime correlation?

* What are the algorithms used while routing? Will it optimize wire length?

* How will you decide the Pin location in block level design?

* If the routing congestion exists between two macros, then what will you do?

* How will you place the macros?

* How will you decide the die size?

* If lengthy metal layer is connected to diffusion and poly, then which one will affect by antenna problem?

* If the full chip design is routed by 7 layer metal, why macros are designed using 5LM instead of using 7LM?

* In your project what is die size, number of metal layers, technology, foundry, number of clocks?

* How many macros in your design?

* What is each macro size and no. of standard cell count?

* How did u handle the Clock in your design?

* What are the Input needs for your design?

* What is SDC constraint file contains?

* How did you do power planning?

* How to find total chip power?

* How to calculate core ring width, macro ring width and strap or trunk width?

* How to find number of power pad and IO power pads?

* What are the problems faced related to timing?

* How did u resolve the setup and hold problem?

* If in your design 10000 and more numbers of problems come, then what you will do?

* In which layer do you prefer for clock routing and why?

* If in your design has reset pin, then it'll affect input pin or output pin or both?

* During power analysis, if you are facing IR drop problem, then how did u avoid?

* Define antenna problem and how did u resolve these problem?

* How delays vary with different PVT conditions? Show the graph.

* Explain the flow of physical design and inputs and outputs for each step in flow.

* What is cell delay and net delay?

* What are delay models and what is the difference between them?

* What is wire load model?

* What does SDC constraints has?

* Why higher metal layers are preferred for Vdd and Vss?

* What is logic optimization and give some methods of logic optimization.

* What is the significance of negative slack?

* How the width of metal and number of straps calculated for power and ground?

* What is negative slack ? How it affects timing?

* What is track assignment?

* What is grided and gridless routing?

* What is a macro and standard cell?

* What is congestion?

* Whether congestion is related to placement or routing?

* What are clock trees?

* What are clock tree types?

* Which layer is used for clock routing and why?

* What is cloning and buffering?

* What are placement blockages?

* How slow and fast transition at inputs effect timing for gates?

* What is antenna effect?

* What are DFM issues?

* What is .lib, LEF, DEF, .tf?

* What is the difference between synthesis and simulation?

* What is metal density, metal slotting rule?

* What is OPC, PSM?

* Why clock is not synthesized in DC?

* What are high-Vt and low-Vt cells?

* What corner cells contains?

* What is the difference between core filler cells and metal fillers?

* How to decide number of pads in chip level design?

* What is tie-high and tie-low cells and where it is used

Common introductory questions every interviewer asks are:

Discuss about the projects worked in the previous company.

What are physical design flows, various activities you are involved?

Design complexity, capacity, frequency, process technologies, block size you handled.

Intel

Why power stripes routed in the top metal layers?

The resistivity of top metal layers are less and hence less IR drop is seen in power distribution network. If power stripes are routed in lower metal layers this will use good amount of lower routing resources and therefore it can create routing congestion.

Why do you use alternate routing approach HVH/VHV (Horizontal-Vertical-Horizontal/ Vertical-Horizontal-Vertical)?

Answer:

This approach allows routability of the design and better usage of routing resources.

What are several factors to improve propagation delay of standard cell?

Answer:

Improve the input transition to the cell under consideration by up sizing the driver.

Reduce the load seen by the cell under consideration, either by placement refinement or buffering.

If allowed increase the drive strength or replace with LVT (low threshold voltage) cell.

How do you compute net delay (interconnect delay) / decode RC values present in tech file?

What are various ways of timing optimization in synthesis tools?

Answer:

Logic optimization: buffer sizing, cell sizing, level adjustment, dummy buffering etc.

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Set don't use attribute on those library cells.

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Answer:

Routing and placement congestion all depend upon the connectivity in the netlist , a better floor plan can reduce the congestion.

Noise can be reduced by optimizing the overlap of nets in the design.

Let's say there enough routing resources available, timing is fine, can you increase clock buffers in clock network? If so will there be any impact on other parameters?

Answer:

No. You should not increase clock buffers in the clock network. Increase in clock buffers cause more area , more power. When everything is fine why you want to touch clock tree??

How do you optimize skew/insertion delays in CTS (Clock Tree Synthesis)?

Answer:

Better skew targets and insertion delay values provided while building the clocks.

Choose appropriate tree structure – either based on clock buffers or clock inverters or mix of clock buffers or clock inverters.

For multi clock domain, group the clocks while building the clock tree so that skew is balanced across the clocks. (Inter clock skew analysis).

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Answer: Pros and cons of latch and flip flop

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As an engineer, let's say your manager comes to you and asks for next project die size estimation/projection, giving data on RTL size, performance requirements. How do you go about the figuring out and come up with die size considering physical aspects?

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How to determine the depth of FIFO/ size of the FIFO?

Answer: FIFO Depth

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What are the challenges you faced in place and route, FV (Formal Verification), ECO (Engineering Change Order) areas?

How long the design cycle for your designs?

What part are your areas of interest in physical design?

Explain ECO (Engineering Change Order) methodology.

Explain CTS (Clock Tree Synthesis) flow.

Answer: Clock Tree Synthesis

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What parameters (or aspects) differentiate Chip Design and Block level design?

Chip design has I/O pads; block design has pins.

Chip design uses all metal layes available; block design may not use all metal layers.

Chip is generally rectangular in shape; blocks can be rectangular, rectilinear.

Chip design requires several packaging; block design ends in a macro.

How do you place macros in a full chip design?

First check flylines i.e. check net connections from macro to macro and macro to standard cells.

If there is more connection from macro to macro place those macros nearer to each other preferably nearer to core boundaries.

If input pin is connected to macro better to place nearer to that pin or pad.

If macro has more connection to standard cells spread the macros inside core.

Avoid criscross placement of macros.

Use soft or hard blockages to guide placement engine.

Differentiate between a Hierarchical Design and flat design?

Hierarchial design has blocks, subblocks in an hierarchy; Flattened design has no subblocks and it has only leaf cells.

Hierarchical design takes more run time; Flattened design takes less run time.

Which is more complicated when u have a 48 MHz and 500 MHz clock design?

500 MHz; because it is more constrained (i.e.lesser clock period) than 48 MHz design.

Name few tools which you used for physical verification?

Herculis from Synopsys, Caliber from Mentor Graphics.

What are the input files will you give for primetime correlation?

Netlist, Technology library, Constraints, SPEF or SDF file.

If the routing congestion exists between two macros, then what will you do?

Provide soft or hard blockage

How will you decide the die size?

By checking the total area of the design you can decide die size.

If lengthy metal layer is connected to diffusion and poly, then which one will affect by antenna problem?

Poly

If the full chip design is routed by 7 layer metal, why macros are designed using 5LM instead of using 7LM?

Because top two metal layers are required for global routing in chip design. If top metal layers are also used in block level it will create routing blockage.

In your project what is die size, number of metal layers, technology, foundry, number of clocks?

Die size: tell in mm eg. 1mm x 1mm ; remeber 1mm=1000micron which is a big size !!

Metal layers: See your tech file. generally for 90nm it is 7 to 9.

Technology: Again look into tech files.

Foundry:Again look into tech files; eg. TSMC, IBM, ARTISAN etc

Clocks: Look into your design and SDC file !

How many macros in your design?

You know it well as you have designed it ! A SoC (System On Chip) design may have 100 macros also !!!!

What is each macro size and number of standard cell count?

Depends on your design.

What are the input needs for your design?

For synthesis: RTL, Technology library, Standard cell library, Constraints

For Physical design: Netlist, Technology library, Constraints, Standard cell library

What is SDC constraint file contains?

Clock definitions

Timing exception-multicycle path, false path

Input and Output delays

How did you do power planning? How to calculate core ring width, macro ring width and strap or trunk width? How to find number of power pad and IO power pads? How the width of metal and number of straps calculated for power and ground?

Get the total core power consumption; get the metal layer current density value from the tech file; Divide total power by number sides of the chip; Divide the obtained value from the current density to get core power ring width. Then calculate number of straps using some more equations. Will be explained in detail later.

How to find total chip power?

Total chip power=standard cell power consumption,Macro power consumption pad power consumption.

What are the problems faced related to timing?

Prelayout: Setup, Max transition, max capacitance

Post layout: Hold

How did you resolve the setup and hold problem?

Setup: upsize the cells

Hold: insert buffers

In which layer do you prefer for clock routing and why?

Next lower layer to the top two metal layers(global routing layers). Because it has less resistance hence less RC delay.

If in your design has reset pin, then it'll affect input pin or output pin or both?

Output pin.

During power analysis, if you are facing IR drop problem, then how did you avoid?

Increase power metal layer width.

Go for higher metal layer.

Spread macros or standard cells.

Provide more straps.

Define antenna problem and how did you resolve these problem?

Increased net length can accumulate more charges while manufacturing of the device due to ionisation process. If this net is connected to gate of the MOSFET it can damage dielectric property of the gate and gate may conduct causing damage to the MOSFET. This is antenna problem.

Decrease the length of the net by providing more vias and layer jumping.

Insert antenna diode.

How delays vary with different PVT conditions? Show the graph.

P increase->dealy increase

P decrease->delay decrease

V increase->delay decrease

V decrease->delay increase

T increase->delay increase

T decrease->delay decrease

Explain the flow of physical design and inputs and outputs for each step in flow.

Click here to see the flow diagram

What is cell delay and net delay?

Gate delay

Transistors within a gate take a finite time to switch. This means that a change on the input of a gate takes a finite time to cause a change on the output.[Magma]

Gate delay =function of(i/p transition time, Cnet+Cpin).

Cell delay is also same as Gate delay.

Cell delay

For any gate it is measured between 50% of input transition to the corresponding 50% of output transition.

Intrinsic delay

Intrinsic delay is the delay internal to the gate. Input pin of the cell to output pin of the cell.

It is defined as the delay between an input and output pair of a cell, when a near zero slew is applied to the input pin and the output does not see any load condition.It is predominantly caused by the internal capacitance associated with its transistor.

This delay is largely independent of the size of the transistors forming the gate because increasing size of transistors increase internal capacitors.

Net Delay (or wire delay)

The difference between the time a signal is first applied to the net and the time it reaches other devices connected to that net.

It is due to the finite resistance and capacitance of the net.It is also known as wire delay.

Wire delay =fn(Rnet , Cnet+Cpin)

What are delay models and what is the difference between them?

Linear Delay Model (LDM)

Non Linear Delay Model (NLDM)

What is wire load model?

Wire load model is NLDM which has estimated R and C of the net.

Why higher metal layers are preferred for Vdd and Vss?

Because it has less resistance and hence leads to less IR drop.

What is logic optimization and give some methods of logic optimization.

Upsizing

Downsizing

Buffer insertion

Buffer relocation

Dummy buffer placement

What is the significance of negative slack?

negative slack==> there is setup voilation==> deisgn can fail

What is signal integrity? How it affects Timing?

IR drop, Electro Migration (EM), Crosstalk, Ground bounce are signal integrity issues.

If Idrop is more==>delay increases.

crosstalk==>there can be setup as well as hold voilation.

What is IR drop? How to avoid? How it affects timing?

There is a resistance associated with each metal layer. This resistance consumes power causing voltage drop i.e.IR drop.

If IR drop is more==>delay increases.

What is EM and it effects?

Due to high current flow in the metal atoms of the metal can displaced from its origial place. When it happens in larger amount the metal can open or bulging of metal layer can happen. This effect is known as Electro Migration.

Affects: Either short or open of the signal line or power line.

What are types of routing?

Global Routing

Track Assignment

Detail Routing

What is latency? Give the types?

Source Latency

It is known as source latency also. It is defined as "the delay from the clock origin point to the clock definition point in the design".

Delay from clock source to beginning of clock tree (i.e. clock definition point).

The time a clock signal takes to propagate from its ideal waveform origin point to the clock definition point in the design.

Network latency

It is also known as Insertion delay or Network latency. It is defined as "the delay from the clock definition point to the clock pin of the register".

The time clock signal (rise or fall) takes to propagate from the clock definition point to a register clock pin.

What is track assignment?

Second stage of the routing wherein particular metal tracks (or layers) are assigned to the signal nets.

What is congestion?

If the number of routing tracks available for routing is less than the required tracks then it is known as congestion.

Whether congestion is related to placement or routing?

Routing

What are clock trees?

Distribution of clock from the clock source to the sync pin of the registers.

What are clock tree types?

H tree, Balanced tree, X tree, Clustering tree, Fish bone

What is cloning and buffering?

Cloning is a method of optimization that decreases the load of a heavily loaded cell by replicating the cell.

Buffering is a method of optimization that is used to insert beffers in high fanout nets to decrease the dealy.

What parameters (or aspects) differentiate Chip Design & Block level design??

How do you place macros in a full chip design?

Differentiate between a Hierarchical Design and flat design?

Which is more complicated when u have a 48 MHz and 500 MHz clock design?

Name few tools which you used for physical verification?

What are the input files will you give for primetime correlation?

What are the algorithms used while routing? Will it optimize wire length?

How will you decide the Pin location in block level design?

If the routing congestion exists between two macros, then what will you do?

How will you place the macros?

How will you decide the die size?

If lengthy metal layer is connected to diffusion and poly, then which one will affect by antenna problem?

If the full chip design is routed by 7 layer metal, why macros are designed using 5LM instead of using 7LM?

In your project what is die size, number of metal layers, technology, foundry, number of clocks?

How many macros in your design?

What is each macro size and no. of standard cell count?

How did u handle the Clock in your design?

What are the Input needs for your design?

What is SDC constraint file contains?

How did you do power planning?

How to find total chip power?

How to calculate core ring width, macro ring width and strap or trunk width?

How to find number of power pad and IO power pads?

What are the problems faced related to timing?

How did u resolve the setup and hold problem?

If in your design 10000 and more numbers of problems come, then what you will do?

In which layer do you prefer for clock routing and why?

If in your design has reset pin, then it'll affect input pin or output pin or both?

During power analysis, if you are facing IR drop problem, then how did u avoid?

Define antenna problem and how did u resolve these problem?

How delays vary with different PVT conditions? Show the graph.

Explain the flow of physical design and inputs and outputs for each step in flow.

What is cell delay and net delay?

What are delay models and what is the difference between them?

What is wire load model?

What does SDC constraints has?

Why higher metal layers are preferred for Vdd and Vss?

What is logic optimization and give some methods of logic optimization.

What is the significance of negative slack?

What is signal integrity? How it affects Timing?

What is IR drop? How to avoid .how it affects timing?

What is EM and it effects?

What is floor plan and power plan?

What are types of routing?

What is a grid .why we need and different types of grids?

What is core and how u will decide w/h ratio for core?

What is effective utilization and chip utilization?

What is latency? Give the types?

How the width of metal and number of straps calculated for power and ground?

What is negative slack ? How it affects timing?

What is track assignment?

What is grided and gridless routing?

What is a macro and standard cell?

What is congestion?

Whether congestion is related to placement or routing?

What are clock trees?

What are clock tree types?

Which layer is used for clock routing and why?

What is cloning and buffering?

What are placement blockages?

How slow and fast transition at inputs effect timing for gates?

What is antenna effect?

What are DFM issues?

What is .lib, LEF, DEF, .tf?

What is the difference between synthesis and simulation?

What is metal density, metal slotting rule?

What is OPC, PSM?

Why clock is not synthesized in DC?

What are high-Vt and low-Vt cells?

What corner cells contains?

What is the difference between core filler cells and metal fillers?

How to decide number of pads in chip level design?

What is tie-high and tie-low cells and where it is used

What is LEF?

What is DEF?

What are the steps involved in designing an optimal pad ring?

What are the steps that you have done in the design flow?

What are the issues in floor plan?

How can you estimate area of block?

How much aspect ratio should be kept (or have you kept) and what is the utilization?

How to calculate core ring and stripe widths?

What if hot spot found in some area of block? How you tackle this?

After adding stripes also if you have hot spot what to do?

What is threshold voltage? How it affect timing?

What is content of lib, lef, sdc?

What is meant my 9 track, 12 track standard cells?

What is scan chain? What if scan chain not detached and reordered? Is it compulsory?

What is setup and hold? Why there are ? What if setup and hold violates?

In a circuit, for reg to reg path ...Tclktoq is 50 ps, Tcombo 50ps, Tsetup 50ps, tskew is 100ps. Then what is the maximum operating frequency?

How R and C values are affecting time?

How ohm (R), fared (C) is related to second (T)?

What is transition? What if transition time is more?

What is difference between normal buffer and clock buffer?

What is antenna effect? How it is avoided?

What is ESD?

What is cross talk? How can you avoid?

How double spacing will avoid cross talk?

What is difference between HFN synthesis and CTS?

What is hold problem? How can you avoid it?

For an iteration we have 0.5ns of insertion delay and 0.1 skew and for other iteration 0.29ns insertion delay and 0.25 skew for the same circuit then which one you will select? Why?

What is partial floor plan?

VLSI Questions

Most frequently Asked in Interview.

Wednesday, September 22, 2010

High and Low Vt Cells and 5 important Design techniques

1) What are High-Vt and Low-Vt cells?

Ans: Hvt cells are MOS devices with less leakage due to high Vt but they have higher delay than low VT, where as the low Vt cells are devices, which have less delay, but leakage is high. The threshold (t) voltage dictates the transistor switching speed, it matters how much minimum threshold voltage applied can make the transistor switching to active state, which results to how fast we can switch the transistor. Disadvantage is it needs to maintain the transistor in a minimum sub threshold voltage level to make it switch fast so it leads to leakage of current in turn loss of power.

2) Give 5 important Design techniques you would follow when doing a Layout for Digital Circuits
Ans:
1) In digital design, decide the height of standard cells you want to layout. It depends upon how big your transistors will be. Have reasonable width for VDD and GND metal paths. Maintaining uniform Height for all the cell is very important since this will help you use place route tool easily and also incase you want to do manual connection of all the blocks it saves on lot of area.
2) Use one metal in one direction only; this does not apply for metal 1. Say you are using metal 2 to do horizontal connections, and then use metal 3 for vertical connections, metal4 for horizontal, metal 5 vertical etc...
3) Place as much substrate contact as possible in the empty spaces of the layout.
4) Do not use poly over long distances as it has huge resistances unless you have no other choice.
5) Use fingered transistors as and when you feel necessary.
6) Try maintaining symmetry in your design. Try to get the design in BIT Sliced manner.

You might also like:

Metal Width Variation (Summary)

DIGITAL BASIC - 1.4 : Combinational Circuits

DIGITAL BASIC - 1.3 : LOGIC GATES (Part - b)

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VLSI placement

This is a blog for people preparing for placement and company interviews in the field of vlsi and electronics. this blog contains important questions which are generally asked in company written and interview exams

Tuesday, July 5, 2011

Verilog questions part 1

1) Write a verilog code to swap contents of two registers with and without a temporary register?

With temp reg ;
always @ (posedge clock)
begin
temp=b;
b=a;
a=temp;
end
Without temp reg;
always @ (posedge clock)
begin
a <= b;
b <= a;
end

2) Difference between blocking and non-blocking?
(Verilog interview questions that is most commonly asked)

The Verilog language has two forms of the procedural assignment statement: blocking and non-blocking. The two are distinguished by the = and <= assignment operators. The blocking assignment statement (= operator) acts much like in
traditional programming languages. The whole statement is done before control passes on to the next statement. The non-blocking (<= operator) evaluates all the right-hand sides for the current time unit and assigns the left-hand sides at the end of the time unit. For example, the following Verilog program

```
// testing blocking and non-blocking assignment
module blocking;
reg [0:7] A, B;
initial begin: init1
A = 3;
#1 A = A + 1; // blocking procedural assignment
B = A + 1;
$display("Blocking: A= %b B= %b", A, B ); A = 3;
#1 A <= A + 1; // non-blocking procedural assignment
B <= A + 1;
#1 $display("Non-blocking: A= %b B= %b", A, B );
end
endmodule
```

produces the following output:

```
Blocking: A= 00000100 B= 00000101
Non-blocking: A= 00000100 B= 00000100
```

The effect is for all the non-blocking assignments to use the old values of the variables at the beginning of the current time unit and to assign the registers new values at the end of the current time unit. This reflects how register transfers
occur in some hardware systems.
blocking procedural assignment is used for combinational logic and non-blocking procedural assignment for sequential.

3) Tell me about verilog file I/O?

OPEN A FILE

```
integer file;
file = $fopenr("filename");
file = $fopenw("filename");
```

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There was an error in this gadget

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file = $fopena("filename");
The function $fopenr opens an existing file for reading. $fopenw opens a new file for writing, and $fopena opens a new
file for writing where any data will be appended to the end of the file. The file name can be either a quoted string or a reg holding the file name. If the file was successfully opened, it returns an integer containing the file number (1..MAX_FILES) or NULL (0) if there was an error. Note that these functions are not the same as the built-in system function $fopen which opens a file for writing by $fdisplay. The files are opened in C with 'rb', 'wb', and 'ab' which allows
reading and writing binary data on the PC. The 'b' is ignored on Unix.
CLOSE A FILE
integer file, r;
r = $fcloser(file);
r = $fclosew(file);
The function $fcloser closes a file for input. $fclosew closes a file for output. It returns EOF if there was an error, otherwise 0. Note that these are not the same as $fclose which closes files for writing.

4) Difference between task and function?
Function:
A function is unable to enable a task however functions can enable other functions.
A function will carry out its required duty in zero simulation time. (The program time will not be incremented during the function routine)
Within a function, no event, delay or timing control statements are permitted
In the invocation of a function their must be at least one argument to be passed.
Functions will only return a single value and can not use either output or inout statements.
Tasks:
Tasks are capable of enabling a function as well as enabling other versions of a Task
Tasks also run with a zero simulation however they can if required be executed in a non zero simulation time.
Tasks are allowed to contain any of these statements.
A task is allowed to use zero or more arguments which are of type output, input or inout.
A Task is unable to return a value but has the facility to pass multiple values via the output and inout statements .

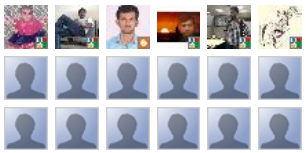
5) Difference between inter statement and intra statement delay?

```
//define register variables
reg a, b, c;
//intra assignment delays
initial
begin
a = 0; c = 0;
b = #5 a + c; //Take value of a and c at the time=0, evaluate
//a + c and then wait 5 time units to assign value
//to b.
end
//Equivalent method with temporary variables and regular delay control
initial
begin
a = 0; c = 0;
temp_ac = a + c;
#5 b = temp_ac; //Take value of a + c at the current time and
//store it in a temporary variable. Even though a and c
//might change between 0 and 5,
//the value assigned to b at time 5 is unaffected.
end
```

5) What is delta simulation time?

6) Difference between $monitor,$display & $strobe?
These commands have the same syntax, and display text on the screen during simulation. They are much less convenient than waveform display tools like cwaves?. $display and $strobe display once every time they are executed,
whereas $monitor displays every time one of its parameters changes.
The difference between $display and $strobe is that $strobe displays the parameters at the very end of the current simulation time unit rather than exactly where it is executed. The format string is like that in C/C++, and may contain format characters. Format characters include %d (decimal), %h (hexadecimal), %b (binary), %c (character), %s (string)
and %t (time), %m (hierarchy level). %5d, %5b etc. would give exactly 5 spaces for the number instead of the space needed. Append b, h, o to the task name to change default format to binary, octal or hexadecimal.
Syntax:
$display ("format_string", par_1, par_2, ...);
$strobe ("format_string", par_1, par_2, ...);
$monitor ("format_string", par_1, par_2, ...);

7) What is difference between Verilog full case and parallel case?
A "full" case statement is a case statement in which all possible case-expression binary patterns can be matched to a case item or to a case default. If a case statement does not include a case default and if it is possible to find a binary case

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expression that does not match any of the defined case items, the case statement is not “full.”
A “parallel” case statement is a case statement in which it is only possible to match a case expression to one and only one case item. If it is possible to find a case expression that would match more than one case item, the matching case items are called “overlapping” case items and the case statement is not “parallel.”

8) What is meant by inferring latches, how to avoid it?
Consider the following :
always @(s1 or s0 or i0 or i1 or i2 or i3)
case ({s1, s0})
2′d0 : out = i0;
2′d1 : out = i1;
2′d2 : out = i2;
endcase
in a case statement if all the possible combinations are not compared and default is also not specified like in example above a latch will be inferred ,a latch is inferred because to reproduce the previous value when unknown branch is specified.
For example in above case if {s1,s0}=3 , the previous stored value is reproduced for this storing a latch is inferred.
The same may be observed in IF statement in case an ELSE IF is not specified.
To avoid inferring latches make sure that all the cases are mentioned if not default condition is provided.

9) Tell me how blocking and non blocking statements get executed?
Execution of blocking assignments can be viewed as a one-step process:
1. Evaluate the RHS (right-hand side equation) and update the LHS (left-hand side expression) of the blocking assignment
without interruption from any other Verilog statement. A blocking assignment “blocks” trailing assignments in the same
always block from occurring until after the current assignment has been completed
Execution of nonblocking assignments can be viewed as a two-step process:
1. Evaluate the RHS of nonblocking statements at the beginning of the time step. 2. Update the LHS of nonblocking statements at the end of the time step.

10) Variable and signal which will be Updated first?
Signals

Basic of verilog/vhdl

Compilation
VHDL. Multiple design-units (entity/architecture pairs), that reside in the same system file, may be separately compiled if so desired. However, it is good design practice to keep each design unit in it's own system file in which case separate compilation should not be an issue.

Verilog. The Verilog language is still rooted in it's native interpretative mode. Compilation is a means of speeding up simulation, but has not changed the original nature of the language. As a result care must be taken with both the compilation order of code written in a single file and the compilation order of multiple files. Simulation results can change by simply changing the order of compilation.
Data types
VHDL. A multitude of language or user defined data types can be used. This may m ean dedicated conversion functions are needed to convert objects from one type to another. The choice of which data types to use should be considered wisely, especially enumerated (abstract) data types. This will make models easier to write, clearer to read and avoid unnecessary conversion functions that can clutter the code. VHDL may be preferred because it allows a multitude of language or user defined data types to be used.

Verilog. Compared to VHDL, Verilog data types a re very simple, easy to use and very much geared towards modeling hardware structure as opposed to abstract hardware modeling. Unlike VHDL, all data types used in a Verilog model are defined by the Verilog language and not by the user. There are net data types, for example wire, and a register data type called reg. A model with a signal whose type is one of the net data types has a corresponding electrical wire in the implied modeled circuit. Objects, that is signals, of type reg hold their value over simulation delta cycles and should not be confused with the modeling of a hardware register. Verilog may be preferred because of it's simplicity.
Design reusability
VHDL. Procedures and functions may be placed in a package so that they are avail able to any design-unit that wishes to use them.

Verilog. There is no concept of packages in Verilog. Functions and procedures used within a model must be defined in the module. To make functions and procedures generally accessible from different module statements the functions and procedures must be placed in a separate system file and included using the `include compiler directive.
Easiest to Learn
Starting with zero knowledge of either language, Verilog is probably the easiest to grasp and understand. This assumes the Verilog compiler directive language for simulation and the PLI language is not included. If these languages are included they can be looked upon as two additional languages that need to be learned. VHDL may seem less intuitive at first for two primary reasons. First, it is very strongly typed; a feature that makes it robust and powerful for the advanced user after a longer learning phase. Second, there are many ways to model the same circuit, specially those with large hierarchical structures.
Forward and back annotation
A spin-off from Verilog is the Standard Delay Format (SDF). This is a general purpose format used to define the timing delays in a circuit. The format provides a bidirectional link between, chip layout tools, and either synthesis or simulation tools, in order to provide more accurate timing representations. The SDF format is now an industry standard in it's own right.

High level constructs
VHDL. There are more constructs and features for high-level modeling in VHDL than there are in Verilog. Abstract data types can be used along with the following statements:

* package statements for model reuse,

* configuration statements for configuring design structure,

* generate statements for replicating structure,

* generic statements for generic models that can be individually characterized, for example, bit width.

All these language statements are useful in synthesizable models.

Verilog. Except for being able to parameterize models by overloading parameter constants, there is no equivalent to the high-level VHDL modeling statements in Verilog.
Language Extensions
The use of language extensions will make a model non standard and most likely not portable across other design tools. However, sometimes they are necessary in order to achieve the desired results.

VHDL. Has an attribute called 'foreign that allows architectures and subprograms to be modeled in another language.

Verilog. The Programming Language Interface (PLI) is an interface mechanism between Verilog models and Verilog software tools. For example, a designer, or more likely, a Verilog tool vendor, can specify user defined tasks or functions in the C programming language, and then call them from the Verilog source description. Use of such tasks or functions make a Verilog model nonstandard and so may not be usable by other Verilog tools. Their use is not recommended.
Libraries
VHDL. A library is a store for compiled entities, architectures, packages and configurations. Useful for managing multiple design projects.

Verilog. There is no concept of a library in Verilog. This is due to it's origins as an interpretive language.
Low Level Constructs
VHDL. Simple two input logical operators are built into the language, they are: NOT, AND, OR, NAND, NOR, XOR and XNOR. Any timing must be separately specified using the after clause. Separate constructs defined under the VITAL language must be used to define the cell primitives of ASIC and FPGA libraries.

Verilog. The Verilog language was originally developed with gate level modeling in mind, and so has very good constructs for modeling at this level and for modeling the cell primitives of ASIC and FPGA libraries. Examples include User Defined Primitive s (UDP), truth tables and the specify block for specifying timing delays across a module.
Managing large designs
VHDL. Configuration, generate, generic and package statements all help manage large design structures.

Verilog. There are no statements in Verilog that help manage large designs.
Operators
The majority of operators are the same between the two languages. Verilog does have very useful unary reduction operators that are not in VHDL. A loop statement can be used in VHDL to perform the same operation as a Verilog unary reduction operator. VHDL has the mod operator that is not found in Verilog.
Parameterizable models
VHDL. A specific bit width model can be instantiated from a generic n-bit model using the generic statement. The generic model will not synthesize until it is instantiated and the value of the generic given.

Verilog. A specific width model can be instantiated from a generic n-bit model using overloaded parameter values. The generic model must have a default parameter value defined. This means two things. In the absence of an overloaded value being specified, it will still synthesize, but will use the specified default parameter value. Also, it does not need to be instantiated with an overloaded parameter value specified, before it will synthesize.
Procedures and tasks
VHDL allows concurrent procedure calls; Verilog does not allow concurrent task calls.
Readability
This is more a matter of coding style and experience than language feature. VHDL is a concise and verbose language; its roots are based on Ada. Verilog is more like C because it's constructs are based approximately 50% on C and 50% on Ada. For this reason an existing C programmer may prefer Verilog over VHDL. Although an existing programmer of both C and Ada may find the mix of constructs somewhat confusing at first. Whatever HDL is used, when writing or reading an HDL model to be synthesized it is important to think about hardware intent.
Structural replication
VHDL. The generate statement replicates a number of instances of the same design-unit or some sub part of a design, and connects it appropriately.

Verilog. There is no equivalent to the generate statement in Verilog.
Test harnesses
Designers typically spend about 50% of their time writing synthesizable models and the other 50% writing a test harness to verify the synthesizable models. Test harnesses are not restricted to the synthesizable subset and so are free to use the full potential of the language. VHDL has generic and configuration statements that are useful in test harnesses, that are not found in Verilog.
Verboseness
VHDL. Because VHDL is a very strongly typed language models must be coded precisely with defined and matching data types. This may be considered an advantage or disadvantage. However, it does mean models are often more verbose, and the code often longer, than it's Verilog equivalent.

Verilog. Signals representing objects of different bits widths may be assigned to each other. The signal representing the smaller number of bits is automatically padded out to that of the larger number of bits, and is independent of whether it is the assigned signal or not. Unused bits will be automatically optimized away during the synthesis process. This has the

advantage of not needing to model quite so explicitly as in VHDL, but does mean unintended modeling errors will not be identified by an analyzer.

Posted by RaJ at 11:50 AM

Labels: Verilog and VHDL

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The Conditional Statement if-else

The if - else statement controls the execution of other statements. In programming language like c, if - else controls the flow of program. When more than one statement needs to be executed for an if condition, then we need to use begin and end as seen in earlier examples.

Syntax : if
if (condition)
statements;

Syntax : if-else
if (condition)
statements;
else
statements;

Syntax : nested if-else-if
if (condition)
statements;
else if (condition)
statements;
.....
.....
else
statements;

Example- simple if

```
1 module simple_if();
2
3 reg latch;
4 wire enable,din;
5
6 always @ (enable or din)
7 if (enable) begin
8   latch <= din;
9 end
10
11 endmodule
```

You could download file simple_if.v here

Example- if-else

```
1 module if_else();
2
3 reg dff;
4 wire clk,din,reset;
5
6 always @ (posedge clk)
7 if (reset) begin
8   dff <= 0;
9 end else  begin
```

```
10   dff <= din;
11 end
12
13 endmodule
```

You could download file if_else.v here

Example- nested-if-else-if

```
1  module nested_if();
2
3  reg [3:0] counter;
4  reg clk,reset,enable, up_en, down_en;
5
6  always @ (posedge clk)
7  // If reset is asserted
8  if (reset == 1'b0) begin
9     counter <= 4'b0000;
10 // If counter is enable and up count is asserted
11 end else if (enable == 1'b1 && up_en == 1'b1) begin
12    counter <= counter + 1'b1;
13 // If counter is enable and down count is asserted
14 end else if (enable == 1'b1 && down_en == 1'b1) begin
15    counter <= counter - 1'b1;
16 // If counting is disabled
17 end else begin
18    counter <= counter; // Redundant code
19 end
20
21 // Testbench code
22 initial begin
23   $monitor ("@%0dns reset=%b enable=%b up=%b down=%b count=%b",
24             $time, reset, enable, up_en, down_en,counter);
25   $display("@%0dns Driving all inputs to know state",$time);
26   clk = 0;
27   reset = 0;
28   enable = 0;
29   up_en = 0;
30   down_en = 0;
31   #3  reset = 1;
32   $display("@%0dns De-Asserting reset",$time);
33   #4  enable = 1;
34   $display("@%0dns De-Asserting reset",$time);
35   #4  up_en = 1;
36   $display("@%0dns Putting counter in up count mode",$time);
37   #10  up_en = 0;
38   down_en = 1;
39   $display("@%0dns Putting counter in down count mode",$time);
40   #8  $finish;
41 end
42
43 always  #1  clk = ~clk;
44
45 endmodule
```

You could download file nested_if.v here

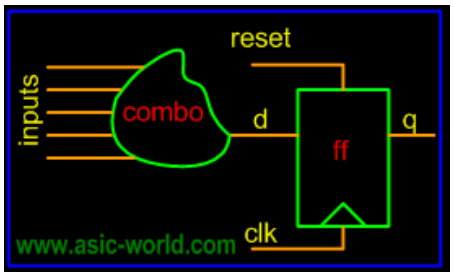
Simulation Log- nested-if-else-if

```
@0ns Driving all inputs to know state
@0ns reset=0 enable=0 up=0 down=0 count=xxxx
@1ns reset=0 enable=0 up=0 down=0 count=0000
@3ns De-Asserting reset
@3ns reset=1 enable=0 up=0 down=0 count=0000
@7ns De-Asserting reset
@7ns reset=1 enable=1 up=0 down=0 count=0000
@11ns Putting counter in up count mode
@11ns reset=1 enable=1 up=1 down=0 count=0001
@13ns reset=1 enable=1 up=1 down=0 count=0010
```

```
@15ns reset=1 enable=1 up=1 down=0 count=0011
@17ns reset=1 enable=1 up=1 down=0 count=0100
@19ns reset=1 enable=1 up=1 down=0 count=0101
@21ns Putting counter in down count mode
@21ns reset=1 enable=1 up=0 down=1 count=0100
@23ns reset=1 enable=1 up=0 down=1 count=0011
@25ns reset=1 enable=1 up=0 down=1 count=0010
@27ns reset=1 enable=1 up=0 down=1 count=0001
```

Parallel if-else

In the above example, the (enable == 1'b1 && up_en == 1'b1) is given highest priority and condition (enable == 1'b1 && down_en == 1'b1) is given lowest priority. We normally don't include reset checking in priority as this does not fall in the combo logic input to the flip-flop as shown in the figure below.



So when we need priority logic, we use nested if-else statements. On the other hand if we don't want to implement priority logic, knowing that only one input is active at a time (i.e. all inputs are mutually exclusive), then we can write the code as shown below.

It's known fact that priority implementation takes more logic to implement than parallel implementation. So if you know the inputs are mutually exclusive, then you can code the logic in parallel if.

```
1  module parallel_if();
2
3  reg [3:0] counter;
4  wire clk,reset,enable, up_en, down_en;
5
6  always @ (posedge clk)
7  // If reset is asserted
8  if (reset == 1'b0) begin
9     counter <= 4'b0000;
10 end else begin
11   // If counter is enable and up count is mode
12   if (enable == 1'b1 && up_en == 1'b1) begin
13     counter <= counter + 1'b1;
14   end
15   // If counter is enable and down count is mode
16   if (enable == 1'b1 && down_en == 1'b1) begin
17     counter <= counter - 1'b1;
18   end
19 end
20
21 endmodule
```

You could download file parallel_if.v here

The Case Statement

The case statement compares an expression to a series of cases and executes the statement or statement group associated with the first matching case:

- case statement supports single or multiple statements.
- Group multiple statements using begin and end keywords.

Syntax of a case statement look as shown below.
case ()
< case1 > : < statement >

< case2 > : < statement >
.....
default : < statement >
endcase

Normal Case

Example- case

```
1  module mux (a,b,c,d,sel,y);
2  input a, b, c, d;
3  input [1:0] sel;
4  output y;
5
6  reg y;
7
8  always @ (a or b or c or d or sel)
9  case (sel)
10   0 : y = a;
11   1 : y = b;
12   2 : y = c;
13   3 : y = d;
14   default : $display("Error in SEL");
15 endcase
16
17 endmodule
```

You could download file mux.v here

Example- case without default

```
1  module mux_without_default (a,b,c,d,sel,y);
2  input a, b, c, d;
3  input [1:0] sel;
4  output y;
5
6  reg y;
7
8  always @ (a or b or c or d or sel)
9  case (sel)
10   0 : y = a;
11   1 : y = b;
12   2 : y = c;
13   3 : y = d;
14   2'bxx,2'bx0,2'bx1,2'b0x,2'b1x,
15   2'bzz,2'bz0,2'bz1,2'b0z,2'b1z : $display("Error in SEL");
16 endcase
17
18 endmodule
```

You could download file mux_without_default.v here

The example above shows how to specify multiple case items as a single case item.

The Verilog case statement does an identity comparison (like the === operator); one can use the case statement to check for logic x and z values as shown in the example below.

Example- case with x and z

```
1  module case_xz(enable);
2  input enable;
3
4  always @ (enable)
5  case(enable)
```

```
  1'bz : $display ("enable is floating");
  1'bx : $display ("enable is unknown");
   default : $display ("enable is %b",enable);
endcase

endmodule
```

You could download file case_xz.v here

The casez and casex statement

Special versions of the case statement allow the x ad z logic values to be used as "don't care":

- casez : Treats z as don't care.
- casex : Treats x and z as don't care.

Example- casez

```
module casez_example();
reg [3:0] opcode;
reg [1:0] a,b,c;
reg [1:0] out;

always @ (opcode or a or b or c)
casez(opcode)
  4'b1zzx : begin  // Don't care about lower 2:1 bit, bit 0 match with x
              out = a;
              $display("@%0dns 4'b1zzx is selected, opcode %b",$time,opcode);
            end
  4'b01?? : begin
              out = b;  // bit 1:0 is don't care
              $display("@%0dns 4'b01?? is selected, opcode %b",$time,opcode);
            end
  4'b001? : begin   // bit 0 is don't care
              out = c;
              $display("@%0dns 4'b001? is selected, opcode %b",$time,opcode);
            end
  default : begin
              $display("@%0dns default is selected, opcode %b",$time,opcode);
            end
endcase

// Testbench code goes here
always  #2  a = $random;
always  #2  b = $random;
always  #2  c = $random;

initial  begin
  opcode = 0;
   #2  opcode = 4'b101x;
   #2  opcode = 4'b0101;
   #2  opcode = 4'b0010;
   #2  opcode = 4'b0000;
   #2  $finish;
end

endmodule
```

You could download file casez_example.v here

Simulation Output - casez

```
@0ns default is selected, opcode 0000
@2ns 4'b1zzx is selected, opcode 101x
@4ns 4'b01?? is selected, opcode 0101
@6ns 4'b001? is selected, opcode 0010
@8ns default is selected, opcode 0000
```

Example- casex

```
1  module casex_example();
2  reg [3:0] opcode;
3  reg [1:0] a,b,c;
4  reg [1:0] out;
5
6  always @ (opcode or a or b or c)
7  casex(opcode)
8    4'b1zzx : begin // Don't care  2:0 bits
9                out = a;
10               $display("@%0dns 4'b1zzx is selected, opcode %b",$time,opcode);
11             end
12   4'b01?? : begin // bit 1:0 is don't care
13               out = b;
14               $display("@%0dns 4'b01?? is selected, opcode %b",$time,opcode);
15             end
16   4'b001? : begin // bit 0 is don't care
17               out = c;
18               $display("@%0dns 4'b001? is selected, opcode %b",$time,opcode);
19             end
20   default : begin
21               $display("@%0dns default is selected, opcode %b",$time,opcode);
22             end
23 endcase
24
25 // Testbench code goes here
26 always  #2  a = $random;
27 always  #2  b = $random;
28 always  #2  c = $random;
29
30 initial begin
31   opcode = 0;
32   #2  opcode = 4'b101x;
33   #2  opcode = 4'b0101;
34   #2  opcode = 4'b0010;
35   #2  opcode = 4'b0000;
36   #2  $finish;
37 end
38
39 endmodule
```

You could download file casex_example.v here

Simulation Output - casex

@0ns default is selected, opcode 0000
@2ns 4'b1zzx is selected, opcode 101x
@4ns 4'b01?? is selected, opcode 0101
@6ns 4'b001? is selected, opcode 0010
@8ns default is selected, opcode 0000

Example- Comparing case, casex, casez

```
1  module case_compare;
2
3  reg sel;
4
5  initial begin
6    #1  $display ("\n    Driving 0");
7    sel = 0;
8    #1  $display ("\n    Driving 1");
9    sel = 1;
10   #1  $display ("\n    Driving x");
11   sel = 1'bx;
12   #1  $display ("\n    Driving z");
```

```
13    sel = 1'bz;
14    #1  $finish;
15  end
16
17  always @ (sel)
18  case (sel)
19    1'b0 : $display("Normal : Logic 0 on sel");
20    1'b1 : $display("Normal : Logic 1 on sel");
21    1'bx : $display("Normal : Logic x on sel");
22    1'bz : $display("Normal : Logic z on sel");
23  endcase
24
25  always @ (sel)
26  casex (sel)
27    1'b0 : $display("CASEX  : Logic 0 on sel");
28    1'b1 : $display("CASEX  : Logic 1 on sel");
29    1'bx : $display("CASEX  : Logic x on sel");
30    1'bz : $display("CASEX  : Logic z on sel");
31  endcase
32
33  always @ (sel)
34  casez (sel)
35    1'b0 : $display("CASEZ  : Logic 0 on sel");
36    1'b1 : $display("CASEZ  : Logic 1 on sel");
37    1'bx : $display("CASEZ  : Logic x on sel");
38    1'bz : $display("CASEZ  : Logic z on sel");
39  endcase
40
41  endmodule
```

You could download file case_compare.v here

Simulation Output

```
    Driving 0
Normal : Logic 0 on sel
CASEX  : Logic 0 on sel
CASEZ  : Logic 0 on sel

    Driving 1
Normal : Logic 1 on sel
CASEX  : Logic 1 on sel
CASEZ  : Logic 1 on sel

    Driving x
Normal : Logic x on sel
CASEX  : Logic 0 on sel
CASEZ  : Logic x on sel

    Driving z
Normal : Logic z on sel
CASEX  : Logic 0 on sel
CASEZ  : Logic 0 on sel
```

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Verilog and VHDL

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To work in vlsi industry you have to know one of these two languages : VHDL and verilog. Doing project in these languages definitely will increase wight of your resume. This portion will cover almost all the questions most frequently asked in interview related to verilog and VHDL. In general 70-80% questions are asked on verilog, but while interview you can tell that you know VHDL. My suggestion is its always better to read verilog basics too, atleast that will help you in written exam. but don't get confuse between two language in order to mastery both the languages.

Part 1 :
http://vlsiplacementindia.blogspot.in/2011/07/verilog-questions-part-1.html

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http://vlsiplacementindia.blogspot.in/2012/09/verilog-part-6.html

Verilog important questions :
http://www.vlsiplacementindia.blogspot.com/2012/10/verilog-important-questions.html

Part I

1) Write a verilog code to swap contents of two registers with and without a temporary register?

With temp reg ;
always @ (posedge clock)
begin
temp=b;
b=a;
a=temp;
end
Without temp reg;
always @ (posedge clock)
begin
a <= b;
b <= a;
end

2) Difference between blocking and non-blocking?

(Verilog interview questions that is most commonly asked)
The Verilog language has two forms of the procedural assignment statement: blocking and non-blocking. The two are

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distinguished by the = and <= assignment operators. The blocking assignment statement (= operator) acts much like in traditional programming languages. The whole statement is done before control passes on to the next statement. The non-blocking (<= operator) evaluates all the right-hand sides for the current time unit and assigns the left-hand sides at the end of the time unit. For example, the following Verilog program

```
// testing blocking and non-blocking assignment
module blocking;
reg [0:7] A, B;
initial begin: init1
A = 3;
#1 A = A + 1; // blocking procedural assignment
B = A + 1;
$display("Blocking: A= %b B= %b", A, B ); A = 3;
#1 A <= A + 1; // non-blocking procedural assignment
B <= A + 1;
#1 $display("Non-blocking: A= %b B= %b", A, B );
end

endmodule
```

produces the following output:

```
Blocking: A= 00000100 B= 00000101
Non-blocking: A= 00000100 B= 00000100
```

The effect is for all the non-blocking assignments to use the old values of the variables at the beginning of the current time unit and to assign the registers new values at the end of the current time unit. This reflects how register transfers occur in some hardware systems.
blocking procedural assignment is used for combinational logic and non-blocking procedural assignment for sequential.

3) Tell me about verilog file I/O?
OPEN A FILE

```
integer file;
file = $fopenr("filename");
file = $fopenw("filename");
file = $fopena("filename");
```

The function $fopenr opens an existing file for reading. $fopenw opens a new file for writing, and $fopena opens a new file for writing where any data will be appended to the end of the file. The file name can be either a quoted string or a reg holding the file name. If the file was successfully opened, it returns an integer containing the file number (1..MAX_FILES) or NULL (0) if there was an error. Note that these functions are not the same as the built-in system function $fopen which opens a file for writing by $fdisplay. The files are opened in C with 'rb', 'wb', and 'ab' which allows reading and writing binary data on the PC. The 'b' is ignored on Unix.
CLOSE A FILE

```
integer file, r;
r = $fcloser(file);
r = $fclosew(file);
```

The function $fcloser closes a file for input. $fclosew closes a file for output. It returns EOF if there was an error, otherwise 0. Note that these are not the same as $fclose which closes files for writing.

4) Difference between task and function?

Function:
A function is unable to enable a task however functions can enable other functions.
A function will carry out its required duty in zero simulation time. (The program time will not be incremented during the function routine)
Within a function, no event, delay or timing control statements are permitted
In the invocation of a function their must be at least one argument to be passed.
Functions will only return a single value and can not use either output or inout statements.

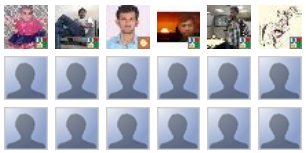
Tasks:
Tasks are capable of enabling a function as well as enabling other versions of a Task
Tasks also run with a zero simulation however they can if required be executed in a non zero simulation time.
Tasks are allowed to contain any of these statements.
A task is allowed to use zero or more arguments which are of type output, input or inout.
A Task is unable to return a value but has the facility to pass multiple values via the output and inout statements .

5) Difference between inter statement and intra statement delay?

```
//define register variables
reg a, b, c;
//intra assignment delays
initial
begin
a = 0; c = 0;
b = #5 a + c; //Take value of a and c at the time=0, evaluate
//a + c and then wait 5 time units to assign value
//to b.
end
//Equivalent method with temporary variables and regular delay control
initial
begin
a = 0; c = 0;
temp_ac = a + c;
#5 b = temp_ac; //Take value of a + c at the current time and
//store it in a temporary variable. Even though a and c
```

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```
//might change between 0 and 5,
//the value assigned to b at time 5 is unaffected.
end
```

5) What is delta simulation time?

6) Difference between $monitor,$display & $strobe?

These commands have the same syntax, and display text on the screen during simulation. They are much less convenient than waveform display tools like cwaves?. $display and $strobe display once every time they are executed, whereas $monitor displays every time one of its parameters changes.
The difference between $display and $strobe is that $strobe displays the parameters at the very end of the current simulation time unit rather than exactly where it is executed. The format string is like that in C/C++, and may contain format characters. Format characters include %d (decimal), %h (hexadecimal), %b (binary), %c (character), %s (string) and %t (time), %m (hierarchy level). %5d, %5b etc. would give exactly 5 spaces for the number instead of the space needed. Append b, h, o to the task name to change default format to binary, octal or hexadecimal.
Syntax:
$display ("format_string", par_1, par_2, …);
$strobe ("format_string", par_1, par_2, …);
$monitor ("format_string", par_1, par_2, …);

7) What is difference between Verilog full case and parallel case?

A "full" case statement is a case statement in which all possible case-expression binary patterns can be matched to a case item or to a case default. If a case statement does not include a case default and if it is possible to find a binary case expression that does not match any of the defined case items, the case statement is not "full."
A "parallel" case statement is a case statement in which it is only possible to match a case expression to one and only one case item. If it is possible to find a case expression that would match more than one case item, the matching case items are called "overlapping" case items and the case statement is not "parallel."

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8) What is meant by inferring latches, how to avoid it?
Consider the following :

```
always @(s1 or s0 or i0 or i1 or i2 or i3)
case ({s1, s0})
2'd0 : out = i0;
2'd1 : out = i1;
2'd2 : out = i2;
endcase
```

in a case statement if all the possible combinations are not compared and default is also not specified like in example above a latch will be inferred ,a latch is inferred because to reproduce the previous value when unknown branch is specified.
For example in above case if {s1,s0}=3 , the previous stored value is reproduced for this storing a latch is inferred.
The same may be observed in IF statement in case an ELSE IF is not specified.
To avoid inferring latches make sure that all the cases are mentioned if not default condition is provided.

9) Tell me how blocking and non blocking statements get executed?
Execution of blocking assignments can be viewed as a one-step process:
1. Evaluate the RHS (right-hand side equation) and update the LHS (left-hand side expression) of the blocking assignment without interruption from any other Verilog statement. A blocking assignment "blocks" trailing assignments in the same always block from occurring until after the current assignment has been completed
Execution of nonblocking assignments can be viewed as a two-step process:
1. Evaluate the RHS of nonblocking statements at the beginning of the time step. 2. Update the LHS of nonblocking statements at the end of the time step.

10) Variable and signal which will be Updated first?
Signals

11) What is sensitivity list?

The sensitivity list indicates that when a change occurs to any one of elements in the list change, begin…end statement inside that always block will get executed.

12) In a pure combinational circuit is it necessary to mention all the inputs in sensitivity disk? if yes, why?

Yes in a pure combinational circuit is it necessary to mention all the inputs in sensitivity disk other wise it will result in pre and post synthesis mismatch.

14) Difference between Verilog and vhdl?

Compilation
VHDL. Multiple design-units (entity/architecture pairs), that reside in the same system file, may be separately compiled if so desired. However, it is good design practice to keep each design unit in it's own system file in which case separate compilation should not be an issue.
Verilog. The Verilog language is still rooted in it's native interpretative mode. Compilation is a means of speeding up simulation, but has not changed the original nature of the language. As a result care must be taken with both the

compilation order of code written in a single file and the compilation order of multiple files. Simulation results can change by simply changing the order of compilation.

Data types
VHDL. A multitude of language or user defined data types can be used. This may mean dedicated conversion functions are needed to convert objects from one type to another. The choice of which data types to use should be considered wisely, especially enumerated (abstract) data types. This will make models easier to write, clearer to read and avoid unnecessary conversion functions that can clutter the code. VHDL may be preferred because it allows a multitude of language or user defined data types to be used.
Verilog. Compared to VHDL, Verilog data types a re very simple, easy to use and very much geared towards modeling hardware structure as opposed to abstract hardware modeling. Unlike VHDL, all data types used in a Verilog model are defined by the Verilog language and not by the user. There are net data types, for example wire, and a register data type called reg. A model with a signal whose type is one of the net data types has a corresponding electrical wire in the implied modeled circuit. Objects, that is signals, of type reg hold their value over simulation delta cycles and should not be confused with the modeling of a hardware register. Verilog may be preferred because of it's simplicity.

Design reusability
VHDL. Procedures and functions may be placed in a package so that they are avail able to any design-unit that wishes to use them. Verilog. There is no concept of packages in Verilog. Functions and procedures used within a model must be defined in the module. To make functions and procedures generally accessible from different module statements the functions and procedures must be placed in a separate system file and included using the `include compiler directive.

15) What are different styles of Verilog coding I mean gate-level,continuous level and others explain in detail?

---

16) Can you tell me some of system tasks and their purpose?

$display, $displayb, $displayh, $displayo, $write, $writeb, $writeh, $writeo.
The most useful of these is $display.This can be used for displaying strings, expression or values of variables. Here are some examples of usage.
$display("Hello oni");
--- output: Hello oni
$display($time) // current simulation time.
--- output: 460
counter = 4'b10;
$display(" The count is %b", counter);
--- output: The count is 0010
$reset resets the simulation back to time 0;
$stop halts the simulator and puts it in interactive mode where the user can enter commands; $finish exits the simulator back to the operating system

18) Write a Verilog code for synchronous and asynchronous reset?

Synchronous reset, synchronous means clock dependent so reset must not be present in sensitivity disk eg:
always @ (posedge clk)
begin if (reset)
. . . end
Asynchronous means clock independent so reset must be present in sensitivity list.
Eg
Always @(posedge clock or posedge reset)
begin
if (reset)
. . . end

19) What is pli?why is it used?

Programming Language Interface (PLI) of Verilog HDL is a mechanism to interface Verilog programs with programs written in C language. It also provides mechanism to access internal databases of the simulator from the C program.
PLI is used for implementing system calls which would have been hard to do otherwise (or impossible) using Verilog syntax. Or, in other words, you can take advantage of both the paradigms - parallel and hardware related features of Verilog and sequential flow of C - using PLI.

21) How to write FSM in verilog?

There are mainly 4 ways 2 write fsm code
1) using 1 process where all input decoder, present state, and output decoder r combine in one process.
2) using 2 process where all comb ckt and sequential ckt separated in different process.
3) using 2 process where input decoder and persent state r combine and output decoder seperated in other process
4) using 3 process where all three, input decoder, present state and output decoder r separated in 3 process.

22) Will case infer priority register if yes how give an example?

yes case can infer priority register depending on coding style
reg r;

```
// Priority encoded mux,
always @ (a or b or c or select2)
begin
r = c;
case (select2)
2'b00: r = a;
2'b01: r = b;
endcase
end
```

23) Casex,z difference,which is preferable,why?

CASEZ : Special version of the case statement which uses a Z logic value to represent don't-care bits.
CASEX : Special version of the case statement which uses Z or X logic values to represent don't-care bits.
CASEZ should be used for case statements with wildcard don't cares, otherwise use of CASE is required;
CASEX should never be used. This is because: Don't cares are not allowed in the "case" statement. Therefore casex or casez are required. Casex will automatically match any x or z with anything in the case statement. Casez will only match z's -- x's require an absolute match.

24) Given the following Verilog code, what value of "a" is displayed?

```
always @(clk) begin
a = 0;
a <= 1;
$display(a);
end
```

This is a tricky one! Verilog scheduling semantics basically imply a
four-level deep queue for the current simulation time:
1: Active Events (blocking statements)
2: Inactive Events (#0 delays, etc)
3: Non-Blocking Assign Updates (non-blocking statements)
4: Monitor Events ($display, $monitor, etc).
Since the "a = 0" is an active event, it is scheduled into the 1st "queue".
The "a <= 1" is a non-blocking event, so it's placed into the 3rd queue.
Finally, the display statement is placed into the 4th queue. Only events in the active queue are completed this sim cycle, so the "a = 0" happens, and then the display shows a = 0. If we were to look at the value of a in the next sim cycle, it would show 1.

25) What is the difference between the following two lines of Verilog code?

```
#5 a = b;
a = #5 b;
```

#5 a = b; Wait five time units before doing the action for "a = b;".
a = #5 b; The value of b is calculated and stored in an internal temp register,After five time units, assign this stored value to a.

27) What are Intertial and Transport Delays ??

---

28) What does `timescale 1 ns/ 1 ps signify in a verilog code?

'timescale directive is a compiler directive.It is used to measure simulation time or delay time. Usage : `timescale / reference_time_unit : Specifies the unit of measurement for times and delays. time_precision: specifies the precision to which the delays are rounded off.

29) What is the difference between === and == ?

output of "==" can be 1, 0 or X.
output of "===" can only be 0 or 1.
When you are comparing 2 nos using "==" and if one/both the numbers have one or more bits as "x" then the output would be "X" . But if use "===" outpout would be 0 or 1.
e.g A = 3'b1x0
B = 3'b10x
A == B will give X as output.
A === B will give 0 as output.
"==" is used for comparison of only 1's and 0's .It can't compare Xs. If any bit of the input is X output will be X
"===" is used for comparison of X also.

30) How to generate sine wav using verilog coding style?
A: The easiest and efficient way to generate sine wave is using CORDIC Algorithm.

31) What is the difference between wire and reg?

Net types: (wire,tri)Physical connection between structural elements. Value assigned by a continuous assignment or a gate output.
Register type: (reg, integer, time, real, real time) represents abstract data storage element. Assigned values only within an always statement or an initial statement. The main difference between wire and reg is wire cannot hold (store) the value when there no connection between a and b like a->b, if there is no connection in a and b, wire loose value. But reg can hold the value even if there in no connection. Default values:wire is Z,reg is x.

32) How do you implement the bi-directional ports in Verilog HDL?

```
module bidirec (oe, clk, inp, outp, bidir);
// Port Declaration
input oe;
input clk;
input [7:0] inp;
output [7:0] outp;
inout [7:0] bidir;
reg [7:0] a;
reg [7:0] b;
assign bidir = oe ? a : 8'bZ ;
assign outp = b;
// Always Construct
always @ (posedge clk)
begin
b <= bidir;
a <= inp;
end
endmodule
```

34) what is verilog case (1) ?

```
wire [3:0] x;
always @(...) begin
case (1'b1)
x[0]: SOMETHING1;
x[1]: SOMETHING2;
x[2]: SOMETHING3;
x[3]: SOMETHING4;
endcase
end
```

The case statement walks down the list of cases and executes the first one that matches. So here, if the lowest 1-bit of x is bit 2, then something3 is the statement that will get executed (or selected by the logic).

35) Why is it that "if (2'b01 & 2'b10)..." doesn't run the true case?

This is a popular coding error. You used the bit wise AND operator (&) where you meant to use the logical AND operator (&&).

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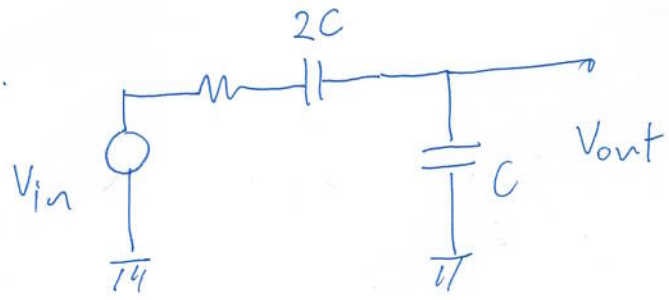
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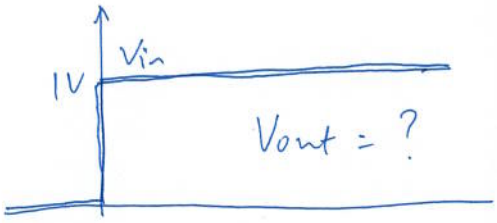
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1.

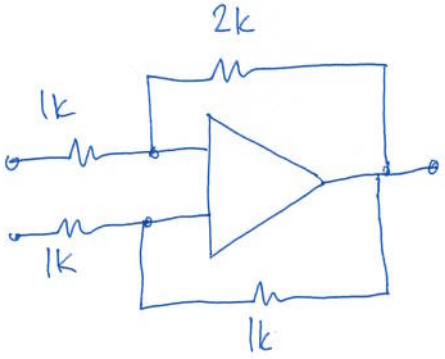


Draw step response:



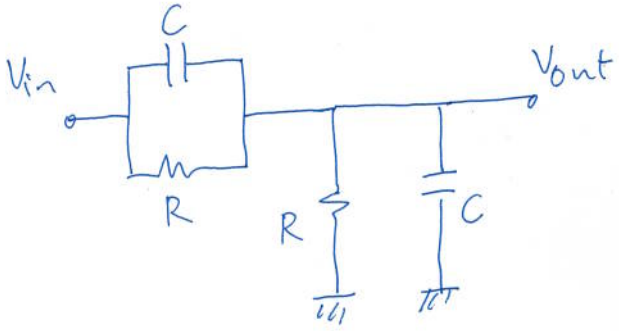
Time constant & final value of Vout = ?

2.

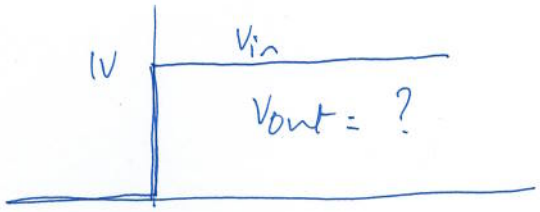


Mark polarity of the opamp such that the system is stable.

3.

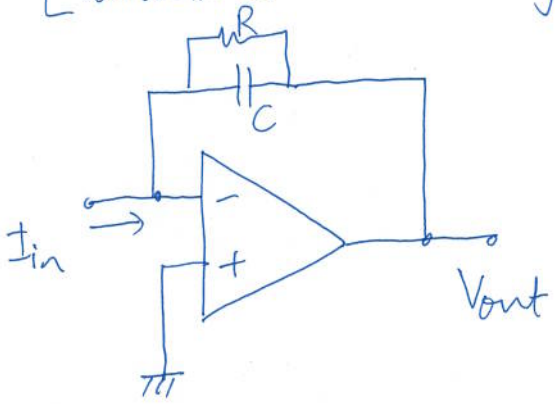


Draw step response

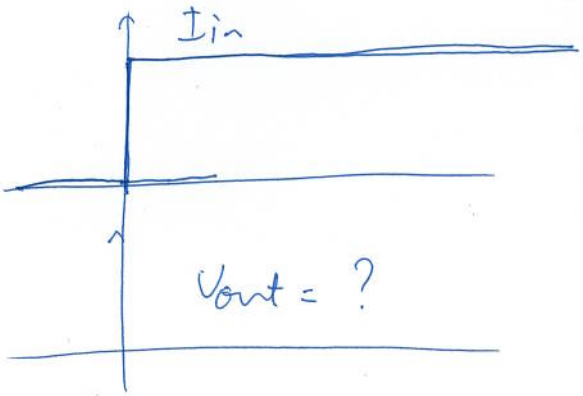


[Variations : change one of the resistors/capacitors]

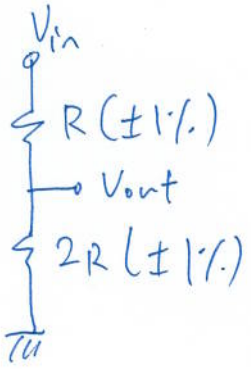
4.



Draw step response:

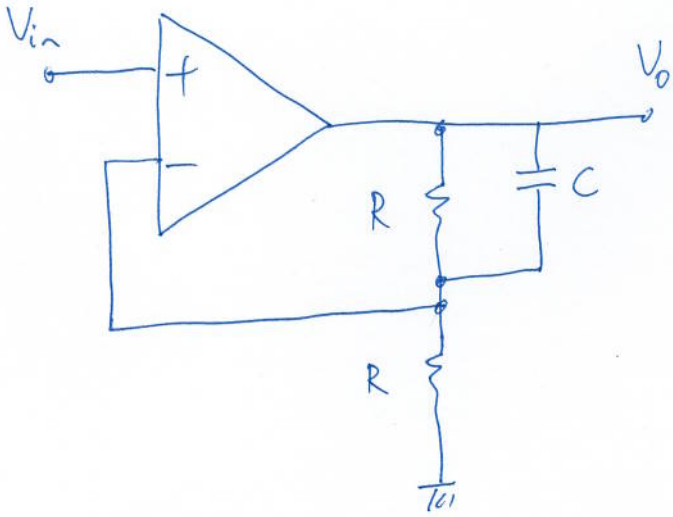


5.



Percentage accuracy of Vout ?

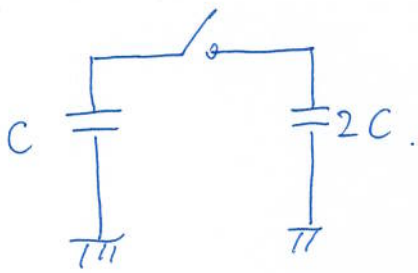
6.



6.1. Draw step response

6.2 Draw bode plot of transfer function.

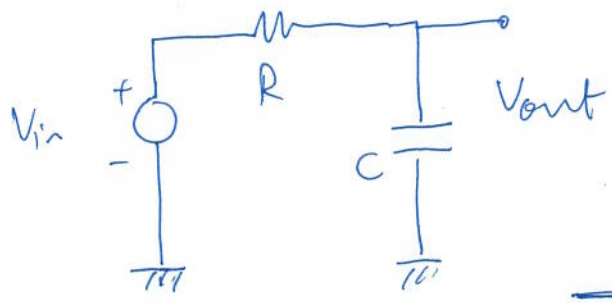
7.

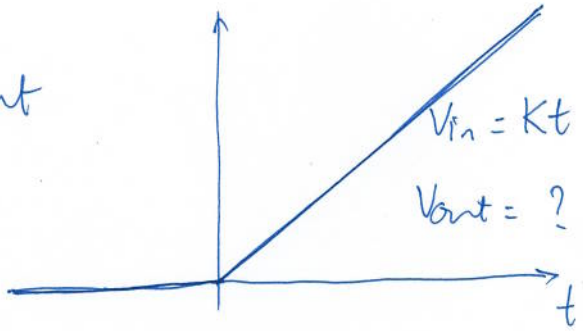


Initial charge on both capacitors = Q.

What is the charge on each capacitor after the switch is closed ?

8.





(Draw ramp response)

9.

